Operating Benchmark Source Compendium

Complete provenance, evidence audit, and final briefs

SIX SUBVERTICALS

- Home Comfort
- Point-of-Use Water
- Submetering
- Liquid Environmental Services
- Transportation Solutions
- Specialty Rental

Compiled July 18, 2026

Evidence policy: comp inputs repeat on company pages; unsupported estimates are omitted.

Document Control and Reading Guide

Canonical artifacts reproduced	**31**
Deep Research pages	**198**
Final brief pages	**12**
Duplicate derivative files listed	**62**

Evidence policy

The concise briefs omit inline source callouts. Every comparable-company input used for a Page 1 estimate is repeated in that company's Page 2 Reference data section; the evidence matrix and embedded Deep Research reports preserve page mapping and source hyperlinks.

Estimate policy

Numeric or qualitative estimates rated Low are removed from the final brief. Each affected row or archetype remains visible as 'Not estimated' with a first-principles directional framework.

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Normalization

Unicode dash variants in rendered text appendices are normalized to ASCII hyphens; substantive source wording is otherwise preserved.

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```
Prepare Six Independent GPT-5.6 Pro Research Prompts

Your role

You are Codex acting as the preparation and quality-control agent for a six-part industry benchmarking study.

Your task is to create six complete, self-contained research prompts, one for each subvertical listed below.

Each prompt will be pasted by the user into:

* a separate, newly created standard ChatGPT conversation;
* with Pro selected in the model picker;
* so that the research is conducted by GPT-5.6 Sol Pro;
* with no reliance on context from any other conversation.

Do not conduct the underlying company research yourself.

Do not combine the six subverticals into one prompt.

Do not use Codex subagents as a substitute for the six standard ChatGPT conversations.

Do not attempt to automate or control the ChatGPT interface.

Your role is to prepare, validate, and organize the six prompts so they are ready to paste into six independent
  ChatGPT Pro conversations.

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1. Required deliverables

Create the following files:

/research_prompts/
    00_launch_manifest.md
    01_home_comfort_prompt.md
    02_point_of_use_water_prompt.md
    03_submetering_prompt.md
    04_liquid_environmental_services_prompt.md
    05_transportation_solutions_prompt.md
    06_specialty_rental_prompt.md
    07_quality_control_checklist.md

Each of the six research prompts must be:

* complete and self-contained;
* written for use in a fresh ChatGPT conversation;
* specific to one subvertical;
* inclusive of every designated company in that subvertical;
* free of references to files or conversations unavailable in the new chat;
* ready to paste without further editing;
* sufficiently detailed to drive rigorous first-principles research;
* structured consistently so the six final reports can be compared.

Do not leave placeholders in the final prompts.

Use the current date as the analysis date.

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2. Launch manifest

Create 00_launch_manifest.md containing the following table:

Sequence    New Chat Title  Prompt File Required Model  Status
1   Operating Benchmark - Home Comfort  01_home_comfort_prompt.md   Pro Ready
2   Operating Benchmark - Point-of-Use Water    02_point_of_use_water_prompt.md Pro Ready
3   Operating Benchmark - Submetering   03_submetering_prompt.md    Pro Ready
4   Operating Benchmark - Liquid Environmental Services 04_liquid_environmental_services_prompt.md  Pro Ready
5   Operating Benchmark - Transportation Solutions  05_transportation_solutions_prompt.md   Pro Ready
6   Operating Benchmark - Specialty Rental  06_specialty_rental_prompt.md   Pro Ready

Also include these user instructions:

1. Open a new standard ChatGPT conversation.
2. Select Pro in the model picker.
3. Confirm the chat is not continuing from an earlier conversation.
4. Paste the full contents of the applicable prompt file.
5. Submit the prompt once.
6. Allow that chat to complete the full subvertical report.
7. Repeat the process in a new chat for the next subvertical.
8. Do not paste multiple subvertical prompts into the same conversation.
```

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3. Common research prompt requirements

Every subvertical prompt must contain the following instructions.

Role

The prompt must begin substantially as follows:

You are ChatGPT 5.6 Pro acting as an expert infrastructure, business-services, asset-leasing, and M&A research analyst. Prepare an independent, first-principles operating benchmark study of the designated subvertical and every company listed in the research universe below.

It must also state:

* this is a fresh analysis;
* no predetermined company or benchmark should influence the conclusions;
* the analyst must not compare the findings with CRA or Compactor Rentals of America;
* every listed company must receive an individual profile;
* company analysis must be completed before the subvertical benchmark is developed;
* the analyst should proceed without requesting clarification;
* reasonable assumptions and unresolved uncertainties should be documented.

Research objective

The report must develop a fundamental understanding of:

1. Revenue growth
2. EBITDA margin
3. Maintenance capex as a percentage of revenue
4. Post-maintenance-capex conversion
5. Initial contract tenor
6. Effective customer tenure
7. Utilization exposure
8. New-unit or new-customer payback
9. B2B versus B2C customer profile

The analysis must explain both:

* what the most defensible normalized assumption is; and
* why the underlying business model naturally produces that result.

Governing analytical method

Require the analyst to use this sequence for every company and metric:

1. Observed facts

Identify the most relevant company-specific disclosures.

2. Root business mechanism

Explain the actual customer, asset, contract, operating-cost, or capital mechanism that drives the metric.

3. Normalized assumption

Develop low, central, and high estimates under normal operating conditions.

4. Sensitivities

Identify the principal factors that could move the result above or below the central estimate.

5. Confidence

Assign High, Medium, or Low confidence and explain why.

Sources must establish facts. They must not replace the analysis.

The prompt must expressly prohibit:

* simply averaging disconnected source estimates;
* repeating management language without analyzing it;
* assuming recurring revenue automatically means low churn;
* assuming high EBITDA margins automatically imply strong cash conversion;
* treating acquisition-driven growth as organic growth;
* treating contract tenor as customer tenure;
* treating total capex as maintenance capex;
* calculating payback using revenue rather than unit-level cash contribution;
* forcing companies with materially different operating models into one average.

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4. Required first-principles company profile

Every prompt must require the following analysis for each listed company.

A. Resolve the company and segment

Confirm:

* exact legal and operating identity;
* current and former names;
* current ownership;
* relevant transaction history;
* geographic footprint;
* relevant operating segment;
* latest reliable standalone reporting period.

For acquired companies, use the last reliable standalone information and disclose the data period.

For diversified companies, isolate the relevant business segment wherever possible.

B. Define the economic unit

Identify the operating unit that most directly creates revenue.

Examples include:

* one installed asset;
* one customer location;
* one meter or metered dwelling;
* one route customer;
* one railcar;
* one locomotive;
* one airport equipment unit;
* one storage container;
* one modular unit.

Use that unit consistently when analyzing growth, utilization, capex, and payback.

C. Explain the customer problem

Explain:

* what the customer needs;
* how essential or discretionary the need is;
* what happens if it is not addressed;
* whether the need is permanent, recurring, temporary, or project based;
* whether demand is driven by cost savings, convenience, regulation, safety, outsourcing, or capacity.

D. Explain customer alternatives

Determine whether the customer can:

* purchase the asset;
* use another rental or service provider;
* perform the activity internally;
* use a manual substitute;
* defer the decision;
* eliminate the need;
* switch providers easily.

Explain why the customer selects and retains the company's solution.

E. Map the revenue model

Identify whether revenue comes from:

* recurring equipment rental;
* fixed subscription;
* asset leasing;
* service fees;
* usage or volume;
* product and consumable sales;
* installation;
* delivery;
* monitoring or data;
* processing or disposal;
* ancillary fees.

Separate recurring and transactional revenue.

F. Map the operating model

Explain what the company must continue doing after acquiring the customer.

Consider:

* technician visits;
* preventive maintenance;
* emergency repairs;
* route service;
* deliveries;
* consumables;
* inspections;

* monitoring;
* equipment removal;
* refurbishment;
* facility operations;
* processing;
* customer support;
* billing and data services.

G. Explain customer retention and churn

Identify why customers remain, including:

* physical installation;
* operational integration;
* switching disruption;
* cost savings;
* essentiality;
* service reliability;
* regulation;
* proprietary systems;
* contract restrictions;
* limited alternatives.

Identify why customers leave, including:

* competitive switching;
* site closure;
* bankruptcy;
* insourcing;
* equipment purchase;
* poor service;
* repricing;
* technology replacement;
* loss of concessions or contracts.

H. Map the capital model

Explain:

* upfront asset investment;
* installation cost;
* customer-acquisition cost;
* asset ownership;
* useful life;
* replacement cycle;
* refurbishment requirements;
* redeployment potential;
* residual value;
* whether the initial contract recovers the investment;
* whether renewals are required to generate the expected return.

The metric assumptions must follow from this operating analysis.

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5. Required treatment of the core metrics

Revenue growth

Require the analyst to decompose growth into the relevant drivers:

* price;
* new customers;
* new installed units;
* net customer retention;
* service frequency;
* customer usage or volumes;
* fleet additions;
* utilization;
* cross-selling;
* acquisitions;
* foreign exchange.

Require:

* latest reported growth;
* historical growth where available;
* organic growth;
* acquisition contribution;
* normalized low, central, and high assumptions;
* explanation of structural tailwinds and constraints.

The analyst must explain why growth is structurally faster or slower than that of the other business models, without relying only on reported historical figures.

EBITDA margin

Require the analyst to construct the margin from:

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Revenue
less product and consumable costs
less service and field labor
less delivery, route, and fuel costs
less repairs and maintenance
less facilities or processing costs
less technology and monitoring expenses
less sales and customer support
less corporate overhead
equals EBITDA

The analyst must explain:

* fixed versus variable costs;
* installation-only versus recurring costs;
* impact of technician and route frequency;
* consumables burden;
* route or installed-base density;
* operating leverage;
* capitalized operating costs;
* recurring adjustments excluded from reported EBITDA.

Require reported and normalized margins to be shown separately.

Maintenance capex

Define maintenance capex as physical investment required to preserve existing revenue and operating capacity.

Potential components include:

* replacement equipment;
* major refurbishment;
* replacement fleet;
* service vehicles;
* mandatory technology upgrades;
* meter replacements;
* processing equipment;
* facility maintenance;
* equipment removal and reinstallation;
* regulatory and safety capital.

Require maintenance capex to be separated from:

* new growth units;
* expansion facilities;
* acquisition capex;
* development projects;
* discretionary growth investments.

Require at least two estimation approaches where possible:

1. Direct disclosure
2. Replacement-cycle analysis
3. Total capex less identifiable growth capex
4. Asset-life and depreciation cross-check

For mobile-fleet businesses, distinguish gross replacement spending from net spending after normal disposal
  proceeds.

Post-maintenance-capex conversion

Require:

Post-Maintenance-Capex Conversion
=
(EBITDA - Maintenance Capex) / EBITDA

State that this is not full free-cash-flow conversion because it excludes interest, taxes, working capital,
  lease principal, and growth capex.

Where capitalized customer acquisition, installation, or technology spending is necessary merely to preserve
  current revenue, require a separate economic sustaining-spend adjustment.

Contract tenor and customer tenure

Require separate analysis of:

1. Initial legal term
2. Renewal term
3. Effective customer tenure
4. Expected customer life

Consider:

* automatic renewal;
* cancellation rights;
* termination penalties;
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* purchase options;
* minimum-volume commitments;
* price escalators;
* transferability;
* rebid frequency;
* actual churn and retention.

The analyst must explain why the legal term may understate or overstate actual revenue durability.

Utilization exposure

Require separate analysis of:

Asset utilization

Whether an owned asset is under a paying contract.

Customer usage

How intensively the customer uses the asset.

Service-capacity utilization

How productively routes, technicians, vehicles, or facilities are employed.

Require the analyst to address:

* whether idle assets earn revenue;
* asset returns and remarketing;
* customer resizing;
* exposure to project, freight, passenger, office-occupancy, waste-volume, or other activity;
* redeployment time and expense;
* residual-value protection;
* the most relevant operating KPI.

Classify utilization exposure as Low, Moderate, or High and explain the classification.

New-unit or new-customer payback

Require fully burdened investment to include:

* equipment;
* installation;
* site work;
* commissioning;
* delivery;
* sales commission;
* customer-acquisition cost;
* other startup investment.

Require annual unit-level cash contribution to equal:

Annual revenue
less direct product cost
less service labor
less transportation and route expense
less consumables
less direct repairs
less maintenance-capex reserve

Require:

Payback Period
=
Fully Burdened Initial Investment
/ Annual Unit-Level Cash Contribution

Require low, central, and high estimates, initial contract coverage, customer life relative to payback, and
  residual or redeployment value.

B2B versus B2C profile

Require the analyst to identify:

* commercial and industrial exposure;
* municipal and utility exposure;
* residential exposure;
* institutional exposure;
* national-account concentration;
* customer concentration.

Classify each company as:

* predominantly B2B;
* predominantly B2C;
* mixed B2B/B2C;
* B2B contract with residential end users.
```

Original request

/Users/mikeberry6/.codex/attachments/3a6e9158-9f45-4701-a9f6-39d4ecedcb4e/pasted-text.txt

Explain implications for:

* bargaining power;
* procurement sophistication;
* sales cycles;
* bid frequency;
* concentration;
* pricing;
* customer-acquisition costs;
* churn;
* regulation.

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6. Required company output

Every company profile must include this table:

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Normalized revenue growth						
Normalized EBITDA margin						
Maintenance capex % of revenue						
Post-maintenance-capex conversion						
Initial contract tenor						
Effective customer tenure						
Utilization exposure						
New-unit or customer payback						
B2B/B2C profile						

Evidence Status must be one of:

* Disclosed
* Calculated
* Inferred
* Triangulated
* Unavailable

For each metric, require a separate explanation covering:

1. Conclusion
2. Observed facts
3. Root business mechanism
4. Calculation or driver bridge
5. Sensitivities
6. Confidence

Each profile must conclude with:

* a plain-English business description;
* principal business strengths;
* principal risks;
* key unavailable information;
* most important diligence questions.

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7. Required subvertical synthesis

After all company profiles are complete, require the analyst to:

Identify operating archetypes

Determine whether companies in the subvertical have materially different operating models.

Do not force a blended benchmark when distinct archetypes should be presented separately.

Classify companies

Classify every company as:

* Core representative
* Secondary reference
* Mixed-business company
* Outlier
* Insufficient disclosure

Develop the benchmark

Require:

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
Revenue growth						
EBITDA margin						
Maintenance capex % revenue						
Post-maintenance conversion						
Initial contract tenor						
Effective customer tenure						
Utilization exposure						

New-unit/customer payback
B2B/B2C profile

For every metric, require an explanation of:

* why the central range is appropriate;
* which companies support it;
* what operating mechanism drives it;
* why certain companies fall above or below it;
* whether the differences are structural or temporary;
* what evidence remains uncertain.

Do not use a simple average unless equal weighting is demonstrably appropriate.

Final subvertical conclusion

Require:

1. Typical customer problem
2. Typical revenue model
3. Primary growth drivers
4. Primary margin drivers
5. Principal maintenance-capex requirements
6. Typical contract and customer duration
7. Typical utilization dynamics
8. Typical new-unit payback
9. Typical B2B/B2C profile
10. Largest source of variation among companies
11. Lowest-confidence conclusion
12. A 25-30-word slide-ready business-model description

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8. Source standards

Every prompt must instruct the analyst to prioritize:

1. Audited filings and statutory accounts
2. Official earnings releases and investor materials
3. Official company operating information
4. Debt, securitization, and ratings materials
5. Official transaction and sponsor materials
6. Government and regulatory filings
7. Reputable financial and industry publications

The analyst must:

* use company-specific evidence wherever possible;
* cite every load-bearing factual assertion;
* avoid using search-result snippets;
* avoid citing multiple sources that merely repeat the same original information;
* use sources to establish facts, not replace reasoning;
* distinguish disclosed facts from estimates;
* state when information is unavailable;
* avoid false precision.

A smaller number of authoritative sources is preferable to a large number of weak or repetitive sources.

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9. Subvertical-specific prompt content

Prompt 1: Home Comfort

Companies:

* Reliance Home Comfort
* Enercare

Special instructions:

* isolate recurring equipment-rental and home-services economics;
* separate rental revenue from one-time HVAC and equipment sales;
* distinguish water heaters, HVAC, and protection plans where possible;
* analyze installation and equipment replacement requirements;
* distinguish residential inertia from enforceable contractual protection;
* distinguish customer tenure from equipment life.

Do not include Vista.

Prompt 2: Point-of-Use Water

Companies:

* AquaVenture Holdings / Quench
* Primo Water

Special instructions:

```
* analyze Quench as the relevant point-of-use operation;
* exclude AquaVenture's desalination and unrelated infrastructure operations;
* isolate point-of-use filtration and recurring service economics;
* separate point-of-use systems from bottled-water delivery, retail refill, dispenser sales, and unrelated
  products;
* analyze filter replacement, sanitation, technician visits, route density, and office-occupancy exposure;
* distinguish equipment payback from customer-acquisition payback.

Do not include Waterlogic.

Prompt 3: Submetering

Companies:

* Energy Assets
* ista
* Techem
* Smart Metering Systems
* Calisen
* CARMA

Special instructions:

* separate meter ownership and rental from installation activity;
* separate physical meter economics from billing, monitoring, software, and data;
* identify regulatory or policy-supported demand;
* analyze meter replacement, communications upgrades, and technology refresh;
* distinguish mature installed-base economics from rollout economics;
* isolate battery storage, EV charging, and unrelated energy-transition activities;
* distinguish contractual customers from residential end users.

Prompt 4: Liquid Environmental Services

Companies:

* Restaurant Technologies
* Liquid Environmental Solutions
* Waste Resources Management
* Wind River Environmental

The prompt must require at least two operating archetypes.

Installed onsite-system model

Restaurant Technologies

Analyze:

* installed fresh- and used-oil equipment;
* installation economics;
* equipment ownership;
* consumable and cooking-oil economics;
* route service;
* used-oil recovery value;
* customer-site payback;
* customer retention.

Route, collection, treatment, and processing model

* Liquid Environmental Solutions
* Waste Resources Management
* Wind River Environmental

Analyze:

* route density;
* service frequency;
* waste volumes;
* truck and technician productivity;
* treatment and processing assets;
* permits;
* disposal access;
* customer-site equipment;
* recovered-material value.

Do not force the two archetypes into one average unless the economics support it.

Prompt 5: Transportation Solutions

Companies:

Railcar and locomotive leasing

* VTG
* Akiem

Airport ground-support-equipment leasing
```

```
* Air Rail
* TCR

Outsourced yard logistics

* Lazer Logistics

Require separate analysis of all three archetypes.

Rail and locomotive leasing must address:

* fleet on lease;
* lease terms;
* off-lease periods;
* asset specialization;
* residual values;
* overhauls;
* fleet replacement;
* freight and customer demand.

Airport equipment leasing must address:

* full-service leasing;
* airline and airport contracts;
* flight activity;
* fleet utilization;
* maintenance intensity;
* electrification;
* airport-specific deployment and redeployment.

Yard logistics must address:

* labor intensity;
* service-hour economics;
* customer contracts;
* fleet ownership;
* site dedication;
* employee productivity;
* whether the business is primarily asset leasing or outsourced operations.

Require the analyst to confirm the exact legal entity represented by Air Rail before conducting the analysis.

Do not allow Lazer Logistics to determine the asset-leasing benchmark without explicit justification.

Prompt 6: Specialty Rental

Companies:

* PODS
* Mobile Mini
* Trench Plate Rental
* General Finance Corporation
* Modulaire Group
* McGrath RentCorp
* Reliant Rental

Require confirmation of the exact Reliant Rental entity using the relevant transaction context.

Distinguish:

* portable storage;
* mobile storage;
* modular space;
* temporary offices and classrooms;
* trench-safety equipment;
* other specialty rental assets.

Analyze:

* physical utilization;
* dollar utilization;
* rental rates;
* rental duration;
* delivery and branch costs;
* repair and refurbishment;
* gross and net replacement capex;
* residual value;
* project and construction cyclicality;
* customer concentration;
* asset redeployment;
* new-fleet payback.

For acquired companies, use the last reliable standalone reporting period and clearly disclose its date.

Do not assume all specialty-rental companies have the same utilization, contract duration, margin, or capital
  requirements.
```

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10. Prompt-level completion criteria

Each generated prompt is complete only if:

1. It includes every designated company.
2. It includes no company from another subvertical.
3. It contains all common research instructions.
4. It contains the applicable subvertical-specific instructions.
5. It is fully self-contained.
6. It contains no unexplained placeholders.
7. It requires individual company profiles before synthesis.
8. It requires root-cause analysis for every metric.
9. It requires low, central, and high estimates.
10. It distinguishes facts, calculations, inferences, triangulations, and unavailable information.
11. It separates organic growth from acquisitions.
12. It separates maintenance capex from growth capex.
13. It separates contract tenor from customer tenure.
14. It defines utilization according to the actual operating model.
15. It calculates payback using cash contribution.
16. It identifies distinct archetypes where necessary.
17. It prohibits unjustified simple averages.
18. It prohibits comparison with CRA.
19. It requires citations and source discipline.
20. It instructs the analyst to complete the entire designated subvertical in that single conversation.

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11. Quality-control checklist

Create 07_quality_control_checklist.md.

For each generated prompt, verify:

* correct file name;
* correct chat title;
* correct company list;
* correct special instructions;
* no Vista;
* no Waterlogic;
* no CRA comparison;
* no unresolved placeholders;
* no dependency on external local files;
* complete metric definitions;
* complete company-profile requirements;
* complete subvertical-synthesis requirements;
* explicit first-principles analytical requirement;
* explicit citation and evidence rules;
* explicit instruction to analyze all companies;
* explicit instruction to complete the full report in the current chat.

Correct every issue before marking the prompt ready.

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12. Final Codex response

After creating and reviewing the files, report:

* the six file names;
* the corresponding new-chat titles;
* confirmation that all prompts are self-contained;
* confirmation that every designated company is included;
* confirmation that Vista and Waterlogic are excluded;
* confirmation that no CRA comparison is included;
* confirmation that each prompt requires first-principles analysis;
* any unresolved company-identity questions, particularly Air Rail and Reliant Rental.

Do not summarize or conduct the underlying company research.

00 Launch Manifest

research_prompts/00_launch_manifest.md

SHA-256: 61a25a624fc4f19f3f683ab16c7df9bcecb11da8bff085605075dc1da39fa535

```
# Six-Part Operating Benchmark - Launch Manifest

Analysis date for every prompt: **July 18, 2026**

| Sequence | New Chat Title | Prompt File | Required Model | Status |
|---:|---|---|---|---|
| 1 | Operating Benchmark - Home Comfort | `01_home_comfort_prompt.md` | Pro | Ready |
| 2 | Operating Benchmark - Point-of-Use Water | `02_point_of_use_water_prompt.md` | Pro | Ready |
| 3 | Operating Benchmark - Submetering | `03_submetering_prompt.md` | Pro | Ready |
| 4 | Operating Benchmark - Liquid Environmental Services | `04_liquid_environmental_services_prompt.md` | Pro |
   Ready |
| 5 | Operating Benchmark - Transportation Solutions | `05_transportation_solutions_prompt.md` | Pro | Ready |
| 6 | Operating Benchmark - Specialty Rental | `06_specialty_rental_prompt.md` | Pro | Ready |

## Launch instructions

1. Open a new standard ChatGPT conversation.
2. Select Pro in the model picker.
3. Confirm the chat is not continuing from an earlier conversation.
4. Paste the full contents of the applicable prompt file.
5. Submit the prompt once.
6. Allow that chat to complete the full subvertical report.
7. Repeat the process in a new chat for the next subvertical.
8. Do not paste multiple subvertical prompts into the same conversation.

Each prompt is independent. Do not provide a new chat with any of the other prompt files, prior reports, or
  conclusions unless you intentionally want to contaminate the independent first-principles analysis.
```

01 Home Comfort Prompt

research_prompts/01_home_comfort_prompt.md

SHA-256: 68528cd6fccd6342388b03c7fb14343ec82d18e43124afaebadf6643a6fa8a79

```
You are ChatGPT 5.6 Pro acting as an expert infrastructure, business-services, asset-leasing, and M&A research analyst. Prepare an independent, first-principles operating benchmark study of the designated subvertical and every company listed in the research universe below.

# Operating Benchmark - Home Comfort

## Mandate and analysis date

Conduct a fresh analysis of the Home Comfort subvertical as of July 18, 2026. Complete the entire designated subvertical report in this single conversation. Do not stop after producing a plan, preliminary findings, or only some of the company profiles.

No predetermined company, prior conclusion, or predetermined benchmark should influence your conclusions. Do not compare the findings with CRA or Compactor Rentals of America under any circumstances. Do not use either one as a benchmark, analogy, cross-check, reference company, or point of comparison. Analyze every company in the research universe individually before developing the subvertical benchmark. Do not add companies to the research universe or substitute other companies for those listed.

Proceed without requesting clarification. Make reasonable assumptions where necessary, label them clearly, document every unresolved uncertainty, and avoid false precision.

## Research universe

Analyze each of the following companies and no additional company as a research-universe profile:

1. Reliance Home Comfort
2. Enercare

Do not include Vista in the research universe, the company profiles, or the benchmark population.

## Home Comfort scope and special instructions

Apply all of the following throughout the company profiles and the subvertical synthesis:

- Isolate recurring equipment-rental and home-services economics.
- Separate rental revenue from one-time HVAC and equipment sales.
- Distinguish water heaters, HVAC, and protection plans where possible.
- Analyze installation and equipment replacement requirements.
- Distinguish residential inertia from enforceable contractual protection.
- Distinguish customer tenure from equipment life.

If a company has activities outside this scope, explain the boundary and isolate the relevant Home Comfort operation wherever reliable evidence permits. Do not allow unrelated operations to distort the benchmark.

## Research objective

Develop a fundamental understanding of these nine core metrics for every company and for the Home Comfort subvertical:

1. Revenue growth
2. EBITDA margin
3. Maintenance capex as a percentage of revenue
4. Post-maintenance-capex conversion
5. Initial contract tenor
6. Effective customer tenure
7. Utilization exposure
8. New-unit or new-customer payback
9. B2B versus B2C customer profile

For every metric, explain both:

- the most defensible normalized assumption; and
- why the underlying business model naturally produces that result.

## Governing analytical method

Use the following sequence for every company and every metric. Complete the company analysis before developing any subvertical benchmark.

1. **Observed facts** - Identify the most relevant company-specific disclosures.
2. **Root business mechanism** - Explain the actual customer, asset, contract, operating-cost, or capital mechanism that drives the metric.
3. **Normalized assumption** - Develop low, central, and high estimates under normal operating conditions.
4. **Sensitivities** - Identify the principal factors that could move the result above or below the central estimate.
5. **Confidence** - Assign High, Medium, or Low confidence and explain why.

Sources must establish facts. They must not replace the analysis.

Do not:

- simply average disconnected source estimates;
```

- repeat management language without analyzing it;
- assume recurring revenue automatically means low churn;
- assume high EBITDA margins automatically imply strong cash conversion;
- treat acquisition-driven growth as organic growth;
- treat contract tenor as customer tenure;
- treat total capex as maintenance capex;
- calculate payback using revenue rather than unit-level cash contribution;
- force companies with materially different operating models into one average.

Required first-principles company profile

Prepare a complete, standalone profile for Reliance Home Comfort and then a complete, standalone profile for Enercare. Each profile must contain all of sections A through H below, followed by the required metric table, metric explanations, and profile conclusion.

A. Resolve the company and segment

Confirm:

- exact legal and operating identity;
- current and former names;
- current ownership;
- relevant transaction history;
- geographic footprint;
- relevant operating segment;
- latest reliable standalone reporting period.

For an acquired company, use the last reliable standalone information and disclose the data period. For a diversified company, isolate the relevant business segment wherever possible. Clearly separate facts belonging to the company at the applicable reporting date from facts belonging to a parent, affiliate, former owner, or unrelated segment.

B. Define the economic unit

Identify the operating unit that most directly creates revenue. Examples may include one installed asset or one customer location. Choose the unit that best represents the actual economics and use it consistently when analyzing growth, utilization, capex, and payback. If materially different activities require different economic units, define each unit explicitly and do not combine unlike units without a bridge.

C. Explain the customer problem

Explain:

- what the customer needs;
- how essential or discretionary the need is;
- what happens if it is not addressed;
- whether the need is permanent, recurring, temporary, or project based;
- whether demand is driven by cost savings, convenience, regulation, safety, outsourcing, or capacity.

D. Explain customer alternatives

Determine whether the customer can:

- purchase the asset;
- use another rental or service provider;
- perform the activity internally;
- use a manual substitute;
- defer the decision;
- eliminate the need;
- switch providers easily.

Explain why the customer initially selects and subsequently retains the company's solution.

E. Map the revenue model

Identify whether revenue comes from:

- recurring equipment rental;
- fixed subscription;
- asset leasing;
- service fees;
- usage or volume;
- product and consumable sales;
- installation;
- delivery;
- monitoring or data;
- processing or disposal;
- ancillary fees.

Separate recurring and transactional revenue. For this subvertical, explicitly separate recurring equipment-rental and home-services revenue from one-time HVAC and equipment sales, and distinguish water-heater, HVAC, and protection-plan economics wherever possible.

F. Map the operating model

Explain what the company must continue doing after acquiring the customer. Consider:

- technician visits;

```
- preventive maintenance;
- emergency repairs;
- route service;
- deliveries;
- consumables;
- inspections;
- monitoring;
- equipment removal;
- refurbishment;
- facility operations;
- processing;
- customer support;
- billing and data services.

Identify which activities are installation-only, which recur predictably, which arise only after a service
  event, and which can be spread over a dense installed base.

### G. Explain customer retention and churn

Identify why customers remain, including:

- physical installation;
- operational integration;
- switching disruption;
- cost savings;
- essentiality;
- service reliability;
- regulation;
- proprietary systems;
- contract restrictions;
- limited alternatives.

Identify why customers leave, including:

- competitive switching;
- site closure;
- bankruptcy;
- insourcing;
- equipment purchase;
- poor service;
- repricing;
- technology replacement;
- loss of concessions or contracts.

For Home Comfort specifically, distinguish retention produced by residential inertia, moving friction, and
  installed-equipment convenience from retention produced by enforceable contractual protection. Do not infer
  legal protection merely from observed customer longevity.

### H. Map the capital model

Explain:

- upfront asset investment;
- installation cost;
- customer-acquisition cost;
- asset ownership;
- useful life;
- replacement cycle;
- refurbishment requirements;
- redeployment potential;
- residual value;
- whether the initial contract recovers the investment;
- whether renewals are required to generate the expected return.

Explicitly analyze equipment installation and replacement requirements and distinguish customer tenure from
  equipment life. The metric assumptions must follow from this operating analysis.

## Required treatment of the core metrics

Apply every instruction in this section separately to each company and again in the subvertical synthesis.

### 1. Revenue growth

Decompose growth into the relevant drivers:

- price;
- new customers;
- new installed units;
- net customer retention;
- service frequency;
- customer usage or volumes;
- fleet additions;
- utilization;
- cross-selling;
- acquisitions;
- foreign exchange.

Show, where evidence permits:
```

- latest reported growth;
- historical growth;
- organic growth;
- acquisition contribution;
- normalized low, central, and high assumptions;
- structural tailwinds and constraints.

Do not treat acquisition-driven growth as organic growth. Explain why growth is structurally faster or slower than that of the other operating models inside this research universe, rather than relying only on reported historical figures.

2. EBITDA margin

Construct the margin using this operating bridge:

- Revenue
- less product and consumable costs
- less service and field labor
- less delivery, route, and fuel costs
- less repairs and maintenance
- less facilities or processing costs
- less technology and monitoring expenses
- less sales and customer support
- less corporate overhead
- equals EBITDA

Explain:

- fixed versus variable costs;
- installation-only versus recurring costs;
- impact of technician and route frequency;
- consumables burden;
- route or installed-base density;
- operating leverage;
- capitalized operating costs;
- recurring adjustments excluded from reported EBITDA.

Show reported and normalized margins separately. Reconcile important differences in perimeter, accounting, and adjustments before making comparisons.

3. Maintenance capex

Define maintenance capex as physical investment required to preserve existing revenue and operating capacity.

Potential components include:

- replacement equipment;
- major refurbishment;
- replacement fleet;
- service vehicles;
- mandatory technology upgrades;
- meter replacements;
- processing equipment;
- facility maintenance;
- equipment removal and reinstallation;
- regulatory and safety capital.

Separate maintenance capex from:

- new growth units;
- expansion facilities;
- acquisition capex;
- development projects;
- discretionary growth investments.

Use at least two estimation approaches where possible:

1. Direct disclosure
2. Replacement-cycle analysis
3. Total capex less identifiable growth capex
4. Asset-life and depreciation cross-check

For mobile-fleet businesses, distinguish gross replacement spending from net spending after normal disposal proceeds. For Home Comfort, explicitly address whether replacement equipment and removal or reinstallation spending are sustaining or growth investments.

4. Post-maintenance-capex conversion

Calculate:

Post-Maintenance-Capex Conversion = (EBITDA - Maintenance Capex) / EBITDA

State clearly that this is not full free-cash-flow conversion because it excludes interest, taxes, working capital, lease principal, and growth capex.

Where capitalized customer acquisition, installation, or technology spending is necessary merely to preserve current revenue, provide a separate economic sustaining-spend adjustment. Show the reported-accounting view and the economically adjusted view when they differ materially.

5. Contract tenor and customer tenure

Analyze separately:

1. Initial legal term
2. Renewal term
3. Effective customer tenure
4. Expected customer life

Consider:

- automatic renewal;
- cancellation rights;
- termination penalties;
- purchase options;
- minimum-volume commitments;
- price escalators;
- transferability;
- rebid frequency;
- actual churn and retention.

Explain why the legal term may understate or overstate actual revenue durability. For this subvertical, explicitly distinguish residential inertia from enforceable contractual protection and customer tenure from equipment life.

6. Utilization exposure

Analyze separately:

Asset utilization - Whether an owned asset is under a paying contract.

Customer usage - How intensively the customer uses the asset.

Service-capacity utilization - How productively routes, technicians, vehicles, or facilities are employed.

Address:

- whether idle assets earn revenue;
- asset returns and remarketing;
- customer resizing;
- exposure to project, freight, passenger, office-occupancy, waste-volume, or other relevant activity;
- redeployment time and expense;
- residual-value protection;
- the most relevant operating KPI.

Classify utilization exposure as Low, Moderate, or High and explain the classification. Do not use a generic rental-industry definition if the installed Home Comfort asset and service model requires a different one.

7. New-unit or new-customer payback

Include all fully burdened investment required to put a new economic unit or customer into service:

- equipment;
- installation;
- site work;
- commissioning;
- delivery;
- sales commission;
- customer-acquisition cost;
- other startup investment.

Calculate annual unit-level cash contribution as:

- Annual revenue
- less direct product cost
- less service labor
- less transportation and route expense
- less consumables
- less direct repairs
- less maintenance-capex reserve

Calculate:

Payback Period = Fully Burdened Initial Investment / Annual Unit-Level Cash Contribution

Do not calculate payback using revenue. Provide low, central, and high payback estimates and analyze initial-contract coverage, customer life relative to payback, and residual or redeployment value. State whether the estimate is for a new installed unit, a new customer relationship, or both, and keep those payback concepts separate when their economics differ.

8. B2B versus B2C profile

Identify:

- commercial and industrial exposure;
- municipal and utility exposure;

- residential exposure;
- institutional exposure;
- national-account concentration;
- customer concentration.

Classify each company as one of:

- predominantly B2B;
- predominantly B2C;
- mixed B2B/B2C;
- B2B contract with residential end users.

Explain implications for:

- bargaining power;
- procurement sophistication;
- sales cycles;
- bid frequency;
- concentration;
- pricing;
- customer-acquisition costs;
- churn;
- regulation.

Required company output

For each company, present this table after the first-principles operating profile. Use percentages, years, months, or other units explicitly. For qualitative rows, use the Low, Central, and High columns to show the defensible range of classifications or scenarios rather than inventing numeric precision.

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Normalized revenue growth						
Normalized EBITDA margin						
Maintenance capex % of revenue						
Post-maintenance-capex conversion						
Initial contract tenor						
Effective customer tenure						
Utilization exposure						
New-unit or customer payback						
B2B/B2C profile						

Evidence Status must be exactly one of:

- Disclosed
- Calculated
- Inferred
- Triangulated
- Unavailable

Confidence must be High, Medium, or Low and must be explained. If evidence status differs across the Low, Central, and High cases, identify the status supporting the central estimate in the table and explain the differences in the accompanying text.

For each metric, provide a separate explanation with these six labeled elements:

1. Conclusion
2. Observed facts
3. Root business mechanism
4. Calculation or driver bridge
5. Sensitivities
6. Confidence

Each company profile must conclude with:

- a plain-English business description;
- principal business strengths;
- principal risks;
- key unavailable information;
- most important diligence questions.

Do not begin the subvertical synthesis until both individual company profiles are complete.

Required Home Comfort subvertical synthesis

After all company profiles are complete, perform the following synthesis.

Identify operating archetypes

Determine whether the companies have materially different operating models. Do not force a blended benchmark when distinct archetypes should be presented separately. Explain whether rental, home-services, one-time equipment sales, water-heater, HVAC, and protection-plan mixes create distinct economics. If archetypes are required, produce a benchmark for each relevant archetype and then explain what, if anything, can defensibly be said for the subvertical as a whole.

Classify every company

Classify each company as exactly one of:

- Core representative
- Secondary reference
- Mixed-business company
- Outlier
- Insufficient disclosure

Explain each classification and how it affects benchmark weight.

Develop the benchmark

Produce this benchmark table. If distinct archetypes are necessary, repeat the table for each archetype rather than forcing one blended range.

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
Revenue growth						
EBITDA margin						
Maintenance capex % revenue						
Post-maintenance conversion						
Initial contract tenor						
Effective customer tenure						
Utilization exposure						
New-unit/customer payback						
B2B/B2C profile						

For every metric, explain:

- why the central range is appropriate;
- which companies support it;
- what operating mechanism drives it;
- why certain companies fall above or below it;
- whether the differences are structural or temporary;
- what evidence remains uncertain.

Do not use a simple average unless equal weighting is demonstrably appropriate. Explain any weighting or exclusion. Never let a larger or more disclosed company mechanically determine the benchmark if its operating model is not representative.

Final subvertical conclusion

Conclude with all of the following:

1. Typical customer problem
2. Typical revenue model
3. Primary growth drivers
4. Primary margin drivers
5. Principal maintenance-capex requirements
6. Typical contract and customer duration
7. Typical utilization dynamics
8. Typical new-unit payback
9. Typical B2B/B2C profile
10. Largest source of variation among companies
11. Lowest-confidence conclusion
12. A slide-ready business-model description of exactly 25-30 words

Source standards

Prioritize sources in this order:

1. Audited filings and statutory accounts
2. Official earnings releases and investor materials
3. Official company operating information
4. Debt, securitization, and ratings materials
5. Official transaction and sponsor materials
6. Government and regulatory filings
7. Reputable financial and industry publications

Use company-specific evidence wherever possible. Cite every load-bearing factual assertion at the point where it is used. Provide enough citation detail or a direct link for the reader to identify the underlying source and its date. Do not use search-result snippets as evidence. Do not cite multiple sources that merely repeat the same original information. Trace repeated claims to the authoritative original source wherever possible.

Use sources to establish facts, not to replace reasoning. Distinguish disclosed facts from calculations, inferences, triangulations, and unavailable information. State when information is unavailable. Avoid false precision. A smaller number of authoritative sources is preferable to a large number of weak or repetitive sources.

When a source provides a group-wide figure rather than the relevant operation, label that limitation and do not present the figure as segment-specific. When evidence dates from before an acquisition or restructuring, state the applicable period and explain whether it remains informative.

Required report order

Deliver the final report in this order:

1. Scope, analysis date, and concise methodology
2. Reliance Home Comfort individual company profile

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3. Enercare individual company profile
4. Operating-archetype assessment and company classifications
5. Home Comfort benchmark table or archetype-specific benchmark tables
6. Metric-by-metric benchmark explanations
7. Final Home Comfort conclusion, including the exactly 25-30-word slide-ready description
8. Consolidated source list grouped by company and synthesis topic

The methodology section may be concise, but no required analysis may be omitted. Complete the full report in
  this conversation.

## Completion requirements

Before submitting the report, silently verify and correct it against every requirement below:

1. Both designated companies-Reliance Home Comfort and Enercare-receive complete individual profiles.
2. No company from another subvertical is included in the research universe or benchmark population.
3. Vista is not included.
4. The analysis is fresh and first-principles, with no predetermined company or benchmark influencing the
  conclusions.
5. The findings are not compared with CRA or Compactor Rentals of America in any form.
6. Company profiles are completed before the subvertical synthesis.
7. Every metric includes observed facts, a root business mechanism, normalized low/central/high estimates,
  sensitivities, and confidence.
8. Every company table distinguishes Disclosed, Calculated, Inferred, Triangulated, and Unavailable evidence
  using the required Evidence Status field.
9. Organic growth is separated from acquisitions.
10. Maintenance capex is separated from growth capex.
11. Contract tenor is separated from customer tenure, and customer tenure is separated from equipment life.
12. Utilization is defined according to the actual operating model and covers asset, customer-usage, and
  service-capacity utilization.
13. Payback uses fully burdened investment divided by unit-level cash contribution, not revenue.
14. Recurring rental and home-services economics are isolated from one-time HVAC and equipment sales.
15. Water heaters, HVAC, and protection plans are distinguished where possible.
16. Installation and equipment replacement requirements are analyzed.
17. Residential inertia is distinguished from enforceable contractual protection.
18. Distinct operating archetypes are identified where necessary and materially different models are not forced
  into one average.
19. No simple average is used without demonstrating that equal weighting is appropriate.
20. Every load-bearing factual assertion is cited under the stated source discipline.
21. Reasonable assumptions and unresolved uncertainties are documented, with no unexplained placeholders.
22. The entire designated Home Comfort report is completed in this single conversation.

Proceed now and deliver the complete report without asking for clarification.
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02 Point Of Use Water Prompt

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SHA-256: dddc332863a7c7a2069a06a2f862a6ef3df8d64636556f308acc394fed4f0bde

You are ChatGPT 5.6 Pro acting as an expert infrastructure, business-services, asset-leasing, and M&A research analyst. Prepare an independent, first-principles operating benchmark study of the designated subvertical and every company listed in the research universe below.

Operating Benchmark - Point-of-Use Water

Mandate and analysis date

Conduct a fresh analysis of the Point-of-Use Water subvertical as of July 18, 2026. Complete the entire designated subvertical report in this single conversation. Do not stop after producing a plan, preliminary findings, or only some of the company profiles.

No predetermined company, prior conclusion, or predetermined benchmark should influence your conclusions. Do not compare the findings with CRA or Compactor Rentals of America under any circumstances. Do not use either one as a benchmark, analogy, cross-check, reference company, or point of comparison. Analyze every company in the research universe individually before developing the subvertical benchmark. Do not add companies to the research universe or substitute other companies for those listed.

Proceed without requesting clarification. Make reasonable assumptions where necessary, label them clearly, document every unresolved uncertainty, and avoid false precision.

Research universe

Analyze each of the following companies and no additional company as a research-universe profile:

1. AquaVenture Holdings / Quench
2. Primo Water

Treat AquaVenture Holdings / Quench as one designated company profile focused on Quench as the relevant point-of-use operation. Do not include Waterlogic in the research universe, the company profiles, or the benchmark population.

Point-of-Use Water scope and special instructions

Apply all of the following throughout the company profiles and the subvertical synthesis:

- Analyze Quench as the relevant point-of-use operation.
- Exclude AquaVenture's desalination and unrelated infrastructure operations.
- Isolate point-of-use filtration and recurring service economics.
- Separate point-of-use systems from bottled-water delivery, retail refill, dispenser sales, and unrelated products.
- Analyze filter replacement, sanitation, technician visits, route density, and office-occupancy exposure.
- Distinguish equipment payback from customer-acquisition payback.

If a company has activities outside this scope, explain the boundary and isolate the relevant Point-of-Use Water operation wherever reliable evidence permits. Do not allow unrelated operations to distort the benchmark.

Research objective

Develop a fundamental understanding of these nine core metrics for every company and for the Point-of-Use Water subvertical:

1. Revenue growth
2. EBITDA margin
3. Maintenance capex as a percentage of revenue
4. Post-maintenance-capex conversion
5. Initial contract tenor
6. Effective customer tenure
7. Utilization exposure
8. New-unit or new-customer payback
9. B2B versus B2C customer profile

For every metric, explain both:

- the most defensible normalized assumption; and
- why the underlying business model naturally produces that result.

Governing analytical method

Use the following sequence for every company and every metric. Complete the company analysis before developing any subvertical benchmark.

1. **Observed facts** - Identify the most relevant company-specific disclosures.
2. **Root business mechanism** - Explain the actual customer, asset, contract, operating-cost, or capital mechanism that drives the metric.
3. **Normalized assumption** - Develop low, central, and high estimates under normal operating conditions.
4. **Sensitivities** - Identify the principal factors that could move the result above or below the central estimate.
5. **Confidence** - Assign High, Medium, or Low confidence and explain why.

Sources must establish facts. They must not replace the analysis.

Do not:

- simply average disconnected source estimates;
- repeat management language without analyzing it;
- assume recurring revenue automatically means low churn;
- assume high EBITDA margins automatically imply strong cash conversion;
- treat acquisition-driven growth as organic growth;
- treat contract tenor as customer tenure;
- treat total capex as maintenance capex;
- calculate payback using revenue rather than unit-level cash contribution;
- force companies with materially different operating models into one average.

Required first-principles company profile

Prepare a complete, standalone profile for AquaVenture Holdings / Quench and then a complete, standalone profile for Primo Water. Each profile must contain all of sections A through H below, followed by the required metric table, metric explanations, and profile conclusion.

A. Resolve the company and segment

Confirm:

- exact legal and operating identity;
- current and former names;
- current ownership;
- relevant transaction history;
- geographic footprint;
- relevant operating segment;
- latest reliable standalone reporting period.

For an acquired company, use the last reliable standalone information and disclose the data period. For a diversified company, isolate the relevant business segment wherever possible. Clearly separate facts belonging to the company at the applicable reporting date from facts belonging to a parent, affiliate, former owner, or unrelated segment. In the AquaVenture Holdings / Quench profile, focus on Quench and exclude desalination and unrelated infrastructure operations from the operating assumptions.

B. Define the economic unit

Identify the operating unit that most directly creates revenue. Examples may include one installed asset or one customer location. Choose the unit that best represents the actual economics and use it consistently when analyzing growth, utilization, capex, and payback. If materially different activities require different economic units, define each unit explicitly and do not combine unlike units without a bridge.

For Point-of-Use Water, test whether the most useful unit is an installed point-of-use system, a customer location, or a customer relationship with multiple installed systems. State which unit supports each calculation.

C. Explain the customer problem

Explain:

- what the customer needs;
- how essential or discretionary the need is;
- what happens if it is not addressed;
- whether the need is permanent, recurring, temporary, or project based;
- whether demand is driven by cost savings, convenience, regulation, safety, outsourcing, or capacity.

D. Explain customer alternatives

Determine whether the customer can:

- purchase the asset;
- use another rental or service provider;
- perform the activity internally;
- use a manual substitute;
- defer the decision;
- eliminate the need;
- switch providers easily.

Explain why the customer initially selects and subsequently retains the company's solution. Compare the operating implications of a point-of-use system with relevant alternatives such as bottled-water delivery only to explain the customer decision; do not add the alternative provider as a benchmark company.

E. Map the revenue model

Identify whether revenue comes from:

- recurring equipment rental;
- fixed subscription;
- asset leasing;
- service fees;
- usage or volume;
- product and consumable sales;
- installation;
- delivery;
- monitoring or data;
- processing or disposal;
- ancillary fees.

Separate recurring and transactional revenue. For this subvertical, isolate point-of-use filtration and recurring service economics. Separate point-of-use systems from bottled-water delivery, retail refill, dispenser sales, and unrelated products. Do not use blended company revenue or margins as if they were point-of-use-specific without a defensible segment bridge.

F. Map the operating model

Explain what the company must continue doing after acquiring the customer. Consider:

- technician visits;
- preventive maintenance;
- emergency repairs;
- route service;
- deliveries;
- consumables;
- inspections;
- monitoring;
- equipment removal;
- refurbishment;
- facility operations;
- processing;
- customer support;
- billing and data services.

Identify which activities are installation-only, which recur predictably, which arise only after a service event, and which can be spread over a dense installed base. Explicitly analyze filter replacement, sanitation, technician visits, and route density.

G. Explain customer retention and churn

Identify why customers remain, including:

- physical installation;
- operational integration;
- switching disruption;
- cost savings;
- essentiality;
- service reliability;
- regulation;
- proprietary systems;
- contract restrictions;
- limited alternatives.

Identify why customers leave, including:

- competitive switching;
- site closure;
- bankruptcy;
- insourcing;
- equipment purchase;
- poor service;
- repricing;
- technology replacement;
- loss of concessions or contracts.

For Point-of-Use Water, assess how office closure, hybrid work, reduced office occupancy, site downsizing, and customer-location changes affect system counts, revenue retention, and churn. Do not infer low churn merely from recurring billing or installed equipment.

H. Map the capital model

Explain:

- upfront asset investment;
- installation cost;
- customer-acquisition cost;
- asset ownership;
- useful life;
- replacement cycle;
- refurbishment requirements;
- redeployment potential;
- residual value;
- whether the initial contract recovers the investment;
- whether renewals are required to generate the expected return.

Explicitly distinguish equipment payback from customer-acquisition payback. Identify costs shared by both calculations and prevent double counting. The metric assumptions must follow from this operating analysis.

Required treatment of the core metrics

Apply every instruction in this section separately to each company and again in the subvertical synthesis.

1. Revenue growth

Decompose growth into the relevant drivers:

- price;

- new customers;
- new installed units;
- net customer retention;
- service frequency;
- customer usage or volumes;
- fleet additions;
- utilization;
- cross-selling;
- acquisitions;
- foreign exchange.

Show, where evidence permits:

- latest reported growth;
- historical growth;
- organic growth;
- acquisition contribution;
- normalized low, central, and high assumptions;
- structural tailwinds and constraints.

Do not treat acquisition-driven growth as organic growth. Explain why growth is structurally faster or slower than that of the other operating models inside this research universe, rather than relying only on reported historical figures. Isolate the effect of point-of-use system placements, net customer retention, pricing, acquisitions, and office occupancy wherever possible.

2. EBITDA margin

Construct the margin using this operating bridge:

- Revenue
- less product and consumable costs
- less service and field labor
- less delivery, route, and fuel costs
- less repairs and maintenance
- less facilities or processing costs
- less technology and monitoring expenses
- less sales and customer support
- less corporate overhead
- equals EBITDA

Explain:

- fixed versus variable costs;
- installation-only versus recurring costs;
- impact of technician and route frequency;
- consumables burden;
- route or installed-base density;
- operating leverage;
- capitalized operating costs;
- recurring adjustments excluded from reported EBITDA.

Show reported and normalized margins separately. Reconcile important differences in perimeter, accounting, and adjustments before making comparisons. Do not use a margin blended with bottled-water delivery, retail refill, dispenser sales, desalination, or other unrelated operations as a point-of-use margin without a defensible analytical bridge.

3. Maintenance capex

Define maintenance capex as physical investment required to preserve existing revenue and operating capacity.

Potential components include:

- replacement equipment;
- major refurbishment;
- replacement fleet;
- service vehicles;
- mandatory technology upgrades;
- meter replacements;
- processing equipment;
- facility maintenance;
- equipment removal and reinstallation;
- regulatory and safety capital.

Separate maintenance capex from:

- new growth units;
- expansion facilities;
- acquisition capex;
- development projects;
- discretionary growth investments.

Use at least two estimation approaches where possible:

1. Direct disclosure
2. Replacement-cycle analysis
3. Total capex less identifiable growth capex
4. Asset-life and depreciation cross-check

For mobile-fleet businesses, distinguish gross replacement spending from net spending after normal disposal proceeds. For Point-of-Use Water, explicitly consider replacement systems, refurbishment, filter and sanitation equipment, service vehicles, removal, and reinstallation, while distinguishing consumable operating expense from capital investment.

4. Post-maintenance-capex conversion

Calculate:

Post-Maintenance-Capex Conversion = (EBITDA - Maintenance Capex) / EBITDA

State clearly that this is not full free-cash-flow conversion because it excludes interest, taxes, working capital, lease principal, and growth capex.

Where capitalized customer acquisition, installation, or technology spending is necessary merely to preserve current revenue, provide a separate economic sustaining-spend adjustment. Show the reported-accounting view and the economically adjusted view when they differ materially.

5. Contract tenor and customer tenure

Analyze separately:

1. Initial legal term
2. Renewal term
3. Effective customer tenure
4. Expected customer life

Consider:

- automatic renewal;
- cancellation rights;
- termination penalties;
- purchase options;
- minimum-volume commitments;
- price escalators;
- transferability;
- rebid frequency;
- actual churn and retention.

Explain why the legal term may understate or overstate actual revenue durability. Assess whether installation, service continuity, route integration, and switching effort extend customer tenure beyond the initial term, and whether office closure or occupancy changes shorten it.

6. Utilization exposure

Analyze separately:

Asset utilization - Whether an owned asset is under a paying contract.

Customer usage - How intensively the customer uses the asset.

Service-capacity utilization - How productively routes, technicians, vehicles, or facilities are employed.

Address:

- whether idle assets earn revenue;
- asset returns and remarketing;
- customer resizing;
- exposure to project, freight, passenger, office-occupancy, waste-volume, or other relevant activity;
- redeployment time and expense;
- residual-value protection;
- the most relevant operating KPI.

Classify utilization exposure as Low, Moderate, or High and explain the classification. Explicitly distinguish a system that remains under a paying contract despite low end-user draw from a returned or idle system that earns no revenue, and distinguish both from technician and route-capacity utilization. Analyze office-occupancy exposure directly.

7. New-unit or new-customer payback

Include all fully burdened investment required to put a new economic unit or customer into service:

- equipment;
- installation;
- site work;
- commissioning;
- delivery;
- sales commission;
- customer-acquisition cost;
- other startup investment.

Calculate annual unit-level cash contribution as:

- Annual revenue
- less direct product cost
- less service labor
- less transportation and route expense
- less consumables

- less direct repairs
- less maintenance-capex reserve

Calculate:

Payback Period = Fully Burdened Initial Investment / Annual Unit-Level Cash Contribution

Do not calculate payback using revenue. Provide low, central, and high payback estimates and analyze initial-contract coverage, customer life relative to payback, and residual or redeployment value.

Calculate and explain separately, where evidence permits:

1. **Equipment payback** - The fully burdened installed-system investment divided by annual cash contribution attributable to that system.
2. **Customer-acquisition payback** - The relevant customer-acquisition and startup investment divided by annual cash contribution attributable to the new customer relationship.

State the unit used, allocate shared installation and acquisition costs transparently, and do not blur or double count the two payback concepts.

8. B2B versus B2C profile

Identify:

- commercial and industrial exposure;
- municipal and utility exposure;
- residential exposure;
- institutional exposure;
- national-account concentration;
- customer concentration.

Classify each company as one of:

- predominantly B2B;
- predominantly B2C;
- mixed B2B/B2C;
- B2B contract with residential end users.

Explain implications for:

- bargaining power;
- procurement sophistication;
- sales cycles;
- bid frequency;
- concentration;
- pricing;
- customer-acquisition costs;
- churn;
- regulation.

Do not infer the customer profile from the end user alone; identify the contractual customer that pays for the installed point-of-use system.

Required company output

For each company, present this table after the first-principles operating profile. Use percentages, years, months, or other units explicitly. For qualitative rows, use the Low, Central, and High columns to show the defensible range of classifications or scenarios rather than inventing numeric precision.

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Normalized revenue growth						
Normalized EBITDA margin						
Maintenance capex % of revenue						
Post-maintenance-capex conversion						
Initial contract tenor						
Effective customer tenure						
Utilization exposure						
New-unit or customer payback						
B2B/B2C profile						

Evidence Status must be exactly one of:

- Disclosed
- Calculated
- Inferred
- Triangulated
- Unavailable

Confidence must be High, Medium, or Low and must be explained. If evidence status differs across the Low, Central, and High cases, identify the status supporting the central estimate in the table and explain the differences in the accompanying text.

For each metric, provide a separate explanation with these six labeled elements:

1. Conclusion
2. Observed facts
3. Root business mechanism

4. Calculation or driver bridge
5. Sensitivities
6. Confidence

Each company profile must conclude with:

- a plain-English business description;
- principal business strengths;
- principal risks;
- key unavailable information;
- most important diligence questions.

Do not begin the subvertical synthesis until both individual company profiles are complete.

Required Point-of-Use Water subvertical synthesis

After all company profiles are complete, perform the following synthesis.

Identify operating archetypes

Determine whether the companies have materially different operating models. Do not force a blended benchmark when distinct archetypes should be presented separately. Explain whether differences in point-of-use filtration, service delivery, customer mix, route density, and exposure to bottled-water delivery, retail refill, dispenser sales, or other activities require separate archetypes or limit comparability. If archetypes are required, produce a benchmark for each relevant archetype and then explain what, if anything, can defensibly be said for the subvertical as a whole.

Classify every company

Classify each company as exactly one of:

- Core representative
- Secondary reference
- Mixed-business company
- Outlier
- Insufficient disclosure

Explain each classification and how it affects benchmark weight.

Develop the benchmark

Produce this benchmark table. If distinct archetypes are necessary, repeat the table for each archetype rather than forcing one blended range.

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
Revenue growth						
EBITDA margin						
Maintenance capex % revenue						
Post-maintenance conversion						
Initial contract tenor						
Effective customer tenure						
Utilization exposure						
New-unit/customer payback						
B2B/B2C profile						

For every metric, explain:

- why the central range is appropriate;
- which companies support it;
- what operating mechanism drives it;
- why certain companies fall above or below it;
- whether the differences are structural or temporary;
- what evidence remains uncertain.

Do not use a simple average unless equal weighting is demonstrably appropriate. Explain any weighting or exclusion. Never let a larger or more disclosed company mechanically determine the benchmark if its operating model is not representative.

Final subvertical conclusion

Conclude with all of the following:

1. Typical customer problem
2. Typical revenue model
3. Primary growth drivers
4. Primary margin drivers
5. Principal maintenance-capex requirements
6. Typical contract and customer duration
7. Typical utilization dynamics
8. Typical new-unit payback
9. Typical B2B/B2C profile
10. Largest source of variation among companies
11. Lowest-confidence conclusion
12. A slide-ready business-model description of exactly 25-30 words

Source standards

Prioritize sources in this order:

1. Audited filings and statutory accounts
2. Official earnings releases and investor materials
3. Official company operating information
4. Debt, securitization, and ratings materials
5. Official transaction and sponsor materials
6. Government and regulatory filings
7. Reputable financial and industry publications

Use company-specific evidence wherever possible. Cite every load-bearing factual assertion at the point where it is used. Provide enough citation detail or a direct link for the reader to identify the underlying source and its date. Do not use search-result snippets as evidence. Do not cite multiple sources that merely repeat the same original information. Trace repeated claims to the authoritative original source wherever possible.

Use sources to establish facts, not to replace reasoning. Distinguish disclosed facts from calculations, inferences, triangulations, and unavailable information. State when information is unavailable. Avoid false precision. A smaller number of authoritative sources is preferable to a large number of weak or repetitive sources.

When a source provides a group-wide figure rather than the relevant operation, label that limitation and do not present the figure as segment-specific. When evidence dates from before an acquisition or restructuring, state the applicable period and explain whether it remains informative.

Required report order

Deliver the final report in this order:

1. Scope, analysis date, and concise methodology
2. AquaVenture Holdings / Quench individual company profile, focused on Quench's relevant point-of-use operation
3. Primo Water individual company profile, isolating point-of-use operations wherever possible
4. Operating-archetype assessment and company classifications
5. Point-of-Use Water benchmark table or archetype-specific benchmark tables
6. Metric-by-metric benchmark explanations
7. Final Point-of-Use Water conclusion, including the exactly 25-30-word slide-ready description
8. Consolidated source list grouped by company and synthesis topic

The methodology section may be concise, but no required analysis may be omitted. Complete the full report in this conversation.

Completion requirements

Before submitting the report, silently verify and correct it against every requirement below:

1. Both designated companies-AquaVenture Holdings / Quench and Primo Water-receive complete individual profiles.
2. The AquaVenture Holdings / Quench profile analyzes Quench as the relevant point-of-use operation.
3. AquaVenture's desalination and unrelated infrastructure operations are excluded from the operating benchmark.
4. No company from another subvertical is included in the research universe or benchmark population.
5. Waterlogic is not included.
6. The analysis is fresh and first-principles, with no predetermined company or benchmark influencing the conclusions.
7. The findings are not compared with CRA or Compactor Rentals of America in any form.
8. Company profiles are completed before the subvertical synthesis.
9. Every metric includes observed facts, a root business mechanism, normalized low/central/high estimates, sensitivities, and confidence.
10. Every company table distinguishes Disclosed, Calculated, Inferred, Triangulated, and Unavailable evidence using the required Evidence Status field.
11. Organic growth is separated from acquisitions.
12. Maintenance capex is separated from growth capex.
13. Contract tenor is separated from customer tenure.
14. Utilization is defined according to the actual operating model and covers asset, customer-usage, and service-capacity utilization, including office-occupancy exposure.
15. Payback uses fully burdened investment divided by unit-level cash contribution, not revenue.
16. Equipment payback is distinguished from customer-acquisition payback.
17. Point-of-use filtration and recurring service economics are isolated.
18. Point-of-use systems are separated from bottled-water delivery, retail refill, dispenser sales, and unrelated products.
19. Filter replacement, sanitation, technician visits, route density, and office-occupancy exposure are analyzed.
20. Distinct operating archetypes are identified where necessary and materially different models are not forced into one average.
21. No simple average is used without demonstrating that equal weighting is appropriate.
22. Every load-bearing factual assertion is cited under the stated source discipline.
23. Reasonable assumptions and unresolved uncertainties are documented, with no unexplained placeholders.
24. The entire designated Point-of-Use Water report is completed in this single conversation.

Proceed now and deliver the complete report without asking for clarification.

03 Submetering Prompt

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SHA-256: ec69c6a9681d913648399fc2237d50406a4a79fc24194c0f4b998a012009baf6

You are ChatGPT 5.6 Pro acting as an expert infrastructure, business-services, asset-leasing, and M&A research analyst. Prepare an independent, first-principles operating benchmark study of the submetering subvertical and every company listed in the research universe below.

Operating Benchmark - Submetering

Assignment, analysis date, and non-negotiable guardrails

Use **July 18, 2026** as the analysis date. Use information available on or before that date, and state the reporting date or period for every material datapoint.

This is a fresh analysis in a newly created conversation. Treat this prompt as the complete instruction set. Do not rely on any prior conversation, local file, unstated attachment, prior analysis, inherited benchmark, or presumed user preference. No predetermined company, result, or benchmark may influence the conclusions. Develop the findings from company-specific evidence and the root economics of each business model.

Complete the full report in this conversation. Proceed without requesting clarification. When an ambiguity cannot be resolved, make the most reasonable explicitly stated assumption, document the alternatives considered, identify the resulting uncertainty, and continue. Do not stop after an outline, research plan, partial set of profiles, or preliminary findings.

Every company in the research universe must receive a complete individual profile. Finish all individual company analyses before developing the subvertical benchmark or drawing subvertical-wide conclusions. Do not use one company as an assumed benchmark for another before both have been analyzed independently.

Do **not** compare any finding, company, metric, range, archetype, or conclusion with **CRA or Compactor Rentals of America**. Do not use CRA or Compactor Rentals of America as a benchmark, analogue, validation point, reference company, or source of assumptions, and do not create a profile for it. **CARMA is a designated company in this submetering research universe; analyze CARMA fully, but do not confuse CARMA with the prohibited comparator CRA / Compactor Rentals of America.**

Research universe

Analyze all and only these designated companies as company profiles:

1. Energy Assets
2. ista
3. Techem
4. Smart Metering Systems
5. Calisen
6. CARMA

Other entities may be mentioned only when necessary to explain a designated company's ownership, transaction history, customers, regulation, or evidence. Do not add any non-designated company profile or substitute a different peer.

Research objective

Develop a fundamental understanding of each designated company and the subvertical across these nine core metrics:

1. Revenue growth
2. EBITDA margin
3. Maintenance capex as a percentage of revenue
4. Post-maintenance-capex conversion
5. Initial contract tenor
6. Effective customer tenure
7. Utilization exposure
8. New-unit or new-customer payback
9. B2B versus B2C customer profile

For every company, every metric, every operating archetype, and the final benchmark, explain both:

* the most defensible normalized assumption or range; and
* why the underlying customer, asset, contract, operating-cost, and capital model naturally produces that result.

The purpose is not merely to collect reported figures. Reconstruct the business model, identify the causal mechanisms, distinguish reported performance from normalized economics, and show how the evidence supports the assumptions.

Submetering-specific analytical requirements

Apply all of the following requirements to every relevant company and explicitly reconcile differences among the designated companies:

* Separate meter ownership and meter-rental economics from installation activity.
* Separate physical meter economics from billing, monitoring, software, and data-service economics.
* Identify demand supported by regulation or public policy, distinguishing mandatory, incentivized, and merely favorable demand where the evidence permits.
* Analyze meter replacement, communications upgrades, and technology refresh, including who funds them, their timing, whether they are sustaining or growth investment, and the risk of obsolescence.

* Distinguish mature installed-base economics from rollout economics. Do not infer steady-state margins, capital intensity, or payback from a rollout phase without adjustment.
* Isolate battery storage, electric-vehicle charging, and unrelated energy-transition activities from the relevant submetering or metering operation wherever possible. Do not allow those activities to determine the submetering benchmark.
* Distinguish contractual customers from residential end users. Identify who signs, pays, procures, renews, can terminate, and ultimately consumes the service.

Where a company spans regulated meter-asset ownership, installation, billing, data, monitoring, or adjacent services, define the relevant segment and economic unit before estimating any metric. If distinct submetering operating models emerge, present separate archetypes rather than forcing all six companies into one blended benchmark.

Governing analytical method

Use the following sequence for **every company and every core metric**:

1. Observed facts

Identify the most relevant company-specific disclosures. State exactly what is disclosed, by whom, for what entity or segment, for which period, and in what units. Distinguish standalone company facts from parent-level or market-level context.

2. Root business mechanism

Explain the actual customer, asset, contract, operating-cost, or capital mechanism that drives the metric. Trace the causal chain rather than merely describing correlation.

3. Normalized assumption

Develop low, central, and high estimates under normal operating conditions. Define what "normal" means for the company, adjust for rollout phases, unusual pricing, acquisitions, disposals, accounting classifications, one-time costs, and cycle effects, and show the bridge from evidence to the range.

4. Sensitivities

Identify the principal factors that could move the result above or below the central estimate. Where practical, quantify the direction or approximate magnitude of the sensitivity.

5. Confidence

Assign **High**, **Medium**, or **Low** confidence and explain why, based on source quality, segment purity, recency, consistency, and the amount of inference required.

Sources must establish facts; they must not replace the analysis. Explicitly distinguish a disclosed fact, a calculation from disclosed facts, an inference, a triangulation from multiple facts, and unavailable information.

Do not:

* simply average disconnected source estimates;
* repeat management language without analyzing it;
* assume recurring revenue automatically means low churn;
* assume high EBITDA margins automatically imply strong cash conversion;
* treat acquisition-driven growth as organic growth;
* treat contract tenor as customer tenure;
* treat total capex as maintenance capex;
* calculate payback using revenue rather than unit-level cash contribution;
* force companies with materially different operating models into one average; or
* use a simple average unless equal weighting is demonstrably appropriate and explicitly justified.

Required first-principles company profile

Complete the following analysis for **each** designated company, in the order listed in the research universe. Do not abbreviate later profiles or refer the reader back to an earlier profile in place of company-specific work.

A. Resolve the company and segment

Confirm:

* exact legal and operating identity;
* current and former names;
* current ownership;
* relevant transaction history;
* geographic footprint;
* relevant operating segment; and
* latest reliable standalone reporting period.

For an acquired or formerly public company, use the last reliable standalone information and clearly disclose the data period. Then use later official information, when available, to update operating facts without mixing incompatible periods. For a diversified company, isolate the relevant submetering or metering business segment wherever possible. If exact isolation is impossible, state what is included in the reported segment, identify contamination from other activities, and reduce confidence rather than implying false precision.

B. Define the economic unit

Identify the operating unit that most directly creates revenue. Depending on the actual model, this may be one meter, one metered dwelling, one installed asset, one customer location, one building, one connection, one contract, or another specifically justified unit.

Use the selected unit consistently when analyzing growth, utilization, capex, and payback. If separate economic units are required for meter assets and billing/data services, define both and reconcile them. State who owns the unit, who pays for it, when it begins earning, and what happens at contract termination.

C. Explain the customer problem

Explain:

* what the customer needs;
* how essential or discretionary the need is;
* what happens if it is not addressed;
* whether the need is permanent, recurring, temporary, or project based; and
* whether demand is driven by cost savings, convenience, regulation, safety, outsourcing, capacity, data requirements, or another evidenced mechanism.

Identify the economic buyer, contractual counterparty, property owner or manager, utility or institutional counterparty where relevant, and residential end user. Explain how regulation or policy changes the problem, timing, and willingness to pay.

D. Explain customer alternatives

Determine whether the customer can:

* purchase the asset;
* use another rental or service provider;
* perform the activity internally;
* use a manual substitute;
* defer the decision;
* eliminate the need; or
* switch providers easily.

Explain why the customer selects and retains the company's solution. Address interoperability, data portability, procurement cycles, physical replacement, access to properties, tenant disruption, system integration, regulation, billing complexity, and switching cost where relevant.

E. Map the revenue model

Identify whether revenue comes from:

* recurring equipment or meter rental;
* fixed subscription;
* asset leasing;
* service fees;
* usage or volume;
* product and consumable sales;
* installation;
* delivery;
* monitoring or data;
* billing services;
* processing or disposal; or
* ancillary fees.

Separate recurring and transactional revenue. Specifically separate meter ownership/rental, installation, billing, monitoring, software, and data where possible. Identify the pricing unit, escalators, pass-throughs, reimbursable items, and any link between revenue and energy consumption or meter reads.

F. Map the operating model

Explain what the company must continue doing after acquiring the customer. Consider:

* technician visits;
* preventive maintenance;
* emergency repairs;
* route service;
* deliveries;
* consumables;
* inspections;
* monitoring;
* meter reading and communications;
* billing and data services;
* equipment removal;
* refurbishment;
* facility operations;
* processing;
* customer support; and
* technology platform maintenance.

Identify which activities occur once at installation and which recur. Explain service frequency, remote versus field work, route or installed-base density, labor requirements, subcontracting, and the operating consequences of communications failures or obsolete technology.

G. Explain customer retention and churn

Identify why customers remain, including as relevant:

* physical installation;
* operational integration;
* switching disruption;
* cost savings;
* essentiality;
* service reliability;
* regulation;
* proprietary systems;
* contract restrictions; and
* limited alternatives.

Identify why customers leave, including as relevant:

* competitive switching;
* site closure;
* bankruptcy;
* insourcing;
* equipment purchase;
* poor service;
* repricing;
* technology replacement; and
* loss of concessions or contracts.

Separate legal protection from practical inertia. Distinguish loss of a contractual counterparty from churn among residential end users. Explain how property ownership changes, procurement rebids, regulatory changes, meter replacement cycles, and data migration affect effective tenure.

H. Map the capital model

Explain:

* upfront asset investment;
* installation cost;
* customer-acquisition cost;
* asset ownership;
* useful life;
* replacement cycle;
* refurbishment requirements;
* redeployment potential;
* residual value;
* whether the initial contract recovers the investment; and
* whether renewals are required to generate the expected return.

Include communications hardware, mandatory upgrades, technology refresh, meter removal and reinstallation, service vehicles, and capitalized software or customer onboarding where economically relevant. Distinguish the capital needed to preserve an existing installed base from capital required for rollout growth. The metric assumptions must follow from this operating analysis.

Required treatment of the core metrics

1. Revenue growth

Decompose growth into every relevant driver:

* price;
* new customers;
* new installed units;
* net customer retention;
* service frequency;
* customer usage or volumes;
* fleet or meter additions;
* utilization;
* cross-selling;
* acquisitions; and
* foreign exchange.

Show latest reported growth and historical growth where available. Separate organic growth, acquisition contribution, disposal or perimeter effects, and foreign exchange. Distinguish mature installed-base growth from rollout-related additions and separate asset additions from billing, monitoring, software, or data revenue growth.

Provide normalized low, central, and high assumptions and explain structural tailwinds and constraints. Explain why growth is structurally faster or slower than the other operating models within this subvertical, based on mechanisms rather than reported history alone. Do not treat acquisition-driven or rollout-funded growth as recurring organic growth.

2. EBITDA margin

Construct the margin from the economics below, adapting the labels to the actual company while retaining the full bridge:

* **Revenue**
* less product and consumable costs
* less service and field labor
* less delivery, route, and fuel costs
* less repairs and maintenance
* less facilities or processing costs
* less technology, communications, monitoring, billing, and data expenses

* less sales and customer support
* less corporate overhead
* **equals EBITDA**

Explain:

* fixed versus variable costs;
* installation-only versus recurring costs;
* the impact of technician and route frequency;
* consumables burden;
* route or installed-base density;
* operating leverage;
* capitalized operating costs; and
* recurring adjustments excluded from reported EBITDA.

Show reported and normalized margins separately. Identify the EBITDA definition used, reconcile material adjustments, avoid mixing accounting standards or segment scopes without explanation, and explain how rollout activity, installation mix, software/data mix, and mature meter-rental economics affect margin.

3. Maintenance capex as a percentage of revenue

Define maintenance capex as physical investment required to preserve existing revenue and operating capacity.

Potential components include:

* replacement equipment;
* major refurbishment;
* replacement fleet;
* service vehicles;
* mandatory technology upgrades;
* meter replacements;
* communications upgrades;
* processing equipment;
* facility maintenance;
* equipment removal and reinstallation; and
* regulatory and safety capital.

Separate maintenance capex from:

* new growth units;
* rollout capital for new installations;
* expansion facilities;
* acquisition capex;
* development projects; and
* discretionary growth investments.

Use at least two of the following estimation approaches wherever the available evidence permits, and explain why each approach is or is not available:

1. Direct disclosure
2. Replacement-cycle analysis
3. Total capex less identifiable growth capex
4. Asset-life and depreciation cross-check

For any mobile fleet, distinguish gross replacement spending from net spending after normal disposal proceeds. For meter assets, reflect physical life, regulatory replacement requirements, communications obsolescence, access and installation labor, failure rates, and who bears replacement expense. Do not classify growth rollout capital as maintenance solely because it recurs annually.

4. Post-maintenance-capex conversion

Calculate:

Post-Maintenance-Capex Conversion = (EBITDA - Maintenance Capex) / EBITDA

State clearly that this is not full free-cash-flow conversion because it excludes interest, taxes, working capital, lease principal, and growth capex.

Show low, central, and high conversion consistent with the EBITDA-margin and maintenance-capex assumptions. Where capitalized customer acquisition, installation, communications, software, or technology spending is necessary merely to preserve current revenue, calculate and discuss a separate economic sustaining-spend adjustment. Explain any mismatch between accounting maintenance capex and economic sustaining spend.

5. Initial contract tenor and effective customer tenure

Analyze separately:

1. Initial legal term
2. Renewal term
3. Effective customer tenure
4. Expected customer life

Consider:

* automatic renewal;
* cancellation rights;
* termination penalties;
* purchase options;

* minimum-volume commitments;
* price escalators;
* transferability;
* rebid frequency; and
* actual churn and retention.

Identify the relevant contractual customer and do not substitute residential end-user occupancy for contract tenure. Explain why the legal term may understate or overstate actual revenue durability. Distinguish meter life, contract life, property tenure, resident tenure, and expected revenue life. If different revenue streams have different terms, show them separately before selecting a normalized company range.

6. Utilization exposure

Analyze three distinct concepts:

* **Asset utilization:** whether an owned asset is under a paying contract.
* **Customer usage:** how intensively the customer uses the asset or service.
* **Service-capacity utilization:** how productively routes, technicians, vehicles, communications systems, or facilities are employed.

Address:

* whether idle assets earn revenue;
* asset returns and remarketing;
* customer resizing;
* exposure to property occupancy, construction or rollout pace, energy consumption, meter reads, waste volume, or other relevant activity;
* redeployment time and expense;
* residual-value protection; and
* the most relevant operating KPI.

Classify overall utilization exposure as **Low**, **Moderate**, or **High** and explain the classification. Where feasible, provide low, central, and high values for the most relevant quantitative KPI as well. Do not equate a contracted installed meter with fully utilized field-service or technology capacity.

7. New-unit or new-customer payback

Define fully burdened initial investment to include:

* equipment;
* installation;
* site work;
* commissioning;
* delivery;
* sales commission;
* customer-acquisition cost; and
* other startup investment.

Calculate annual unit-level cash contribution as:

* **Annual revenue**
* less direct product cost
* less service labor
* less transportation and route expense
* less consumables
* less direct repairs
* less maintenance-capex reserve
* **equals annual unit-level cash contribution**

Then calculate:

Payback Period = Fully Burdened Initial Investment / Annual Unit-Level Cash Contribution

Provide low, central, and high estimates. Show the numerator, denominator, units, and assumptions; do not calculate payback from revenue. Address whether the initial contract covers payback, how customer life compares with payback, and how residual or redeployment value affects downside protection. When equipment and customer-acquisition payback differ, show both and reconcile them. For a multi-service contract, avoid attributing all contribution to the meter without support.

8. B2B versus B2C customer profile

Identify:

* commercial and industrial exposure;
* municipal and utility exposure;
* residential exposure;
* institutional exposure;
* national-account concentration; and
* customer concentration.

Classify each company as one of:

* predominantly B2B;
* predominantly B2C;
* mixed B2B/B2C; or
* B2B contract with residential end users.

Explain implications for:

* bargaining power;
* procurement sophistication;
* sales cycles;
* bid frequency;
* concentration;
* pricing;
* customer-acquisition costs;
* churn; and
* regulation.

Do not classify a business as B2C merely because residents use the meters. Identify the legal buyer, invoice recipient, decision-maker, and end user.

Required output for every company

For each company, first provide the full first-principles profile and metric analysis above. Then include one populated summary table with exactly these columns and metric rows:

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Normalized revenue growth						
Normalized EBITDA margin						
Maintenance capex % of revenue						
Post-maintenance-capex conversion						
Initial contract tenor						
Effective customer tenure						
Utilization exposure						
New-unit or customer payback						
B2B/B2C profile						

Complete every required cell in the delivered report with a company-specific conclusion; the blank cells above define the output schema and are not permission to omit a conclusion. Use percentages for growth, margin, maintenance capex, and conversion; years or months for tenor, tenure, and payback; and explicit qualitative classifications, supplemented by quantitative KPIs where available, for utilization and customer profile. If a value truly cannot be estimated, state **Unavailable**, explain why, and do not fabricate a range.

Evidence Status must be one of:

* Disclosed
* Calculated
* Inferred
* Triangulated
* Unavailable

If evidence status differs across the low, central, and high estimate, use the status that best describes the central conclusion and explain the difference in the metric narrative. **Confidence** must be High, Medium, or Low.

For each metric, provide a separate explanation covering all six items:

1. Conclusion
2. Observed facts
3. Root business mechanism
4. Calculation or driver bridge
5. Sensitivities
6. Confidence

Conclude each company profile with:

* a plain-English business description;
* principal business strengths;
* principal risks;
* key unavailable information; and
* the most important diligence questions.

Required subvertical synthesis

Begin synthesis only after all six individual profiles are complete.

1. Identify operating archetypes

Determine whether the companies have materially different operating models, including differences between physical meter ownership/rental, installation-led activity, billing/data services, and mixed platforms. Name and define each economically distinct archetype. Do not force a blended benchmark when distinct archetypes should be presented separately.

2. Classify every company

Classify each designated company as exactly one of:

* Core representative
* Secondary reference
* Mixed-business company
* Outlier
* Insufficient disclosure

Explain the classification. Base "Core representative" status on business-model relevance and evidence quality, not company size or familiarity.

3. Develop the benchmark

Provide a populated benchmark table with exactly these columns and rows:

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
Revenue growth						
EBITDA margin						
Maintenance capex % revenue						
Post-maintenance conversion						
Initial contract tenor						
Effective customer tenure						
Utilization exposure						
New-unit/customer payback						
B2B/B2C profile						

Complete every required cell in the delivered report with a supported conclusion. If one consolidated table would conceal structural differences, provide a separate fully completed version of this table for each archetype and then a clearly labeled subvertical reference range that does not imply false comparability.

For every benchmark metric, explain:

* why the central range is appropriate;
* which designated companies support it;
* what operating mechanism drives it;
* why certain designated companies fall above or below it;
* whether the differences are structural or temporary; and
* what evidence remains uncertain.

Do not use a simple average unless equal weighting is demonstrably appropriate. If weighting is used, explain the basis and show why it reflects the intended benchmark. Prefer mechanism-based ranges and representative-company evidence over mathematically precise but economically incoherent aggregation.

4. Final subvertical conclusion

Conclude with all of the following:

1. Typical customer problem
2. Typical revenue model
3. Primary growth drivers
4. Primary margin drivers
5. Principal maintenance-capex requirements
6. Typical contract and customer duration
7. Typical utilization dynamics
8. Typical new-unit payback
9. Typical B2B/B2C profile
10. Largest source of variation among companies
11. Lowest-confidence conclusion
12. A **25-30-word** slide-ready business-model description

Count the words in the slide-ready description and ensure it is between 25 and 30 words inclusive.

Source and evidence standards

Prioritize sources in this order:

1. Audited filings and statutory accounts
2. Official earnings releases and investor materials
3. Official company operating information
4. Debt, securitization, and ratings materials
5. Official transaction and sponsor materials
6. Government and regulatory filings
7. Reputable financial and industry publications

Use company-specific evidence wherever possible. Cite every load-bearing factual assertion close to the claim it supports, using direct links or precise footnotes that identify the source, date, and relevant page or section when available. Cite the original source rather than search-result snippets, aggregators, or articles that merely repeat it. Do not cite multiple sources that only repeat the same originating disclosure.

Use sources to establish facts, not to replace reasoning. Clearly label disclosed facts, calculations, inferences, triangulations, and unavailable information. State when information is unavailable. Do not convert weak evidence into precise figures; avoid false precision. A smaller number of authoritative sources is preferable to a large number of weak or repetitive sources.

When a source covers a parent, mixed segment, different geography, or different period, disclose that limitation before using it. Keep currencies, percentages, periods, numerator definitions, and denominator definitions explicit. Do not silently combine inconsistent EBITDA, capex, contract, or customer definitions.

Required report structure

Use this order:

1. Executive summary and concise description of the subvertical
2. Scope, analysis date, definitions, and methodology
3. Individual company profile: Energy Assets
4. Individual company profile: ista

```
5. Individual company profile: Techem
6. Individual company profile: Smart Metering Systems
7. Individual company profile: Calisen
8. Individual company profile: CARMA
9. Operating-archetype identification and company classification
10. Subvertical or archetype benchmark table and metric-by-metric synthesis
11. Final subvertical conclusion, including the 25-30-word slide-ready description
12. Key evidence gaps and consolidated diligence priorities
13. Source list

The executive summary may preview conclusions, but it must not replace or pre-empt the independent company work.
   The final benchmark must be derived only after the six profiles have been completed.

## Completion requirements and final self-audit

Before ending the response, verify and correct the report against every requirement below:

1. Every designated company-Energy Assets, ista, Techem, Smart Metering Systems, Calisen, and CARMA-has a
  complete individual profile.
2. No non-designated company has been added as a research-universe profile.
3. All common research instructions in this prompt have been followed.
4. Every submetering-specific instruction has been addressed explicitly.
5. The report is fully self-contained and does not depend on unavailable files or other conversations.
6. No unexplained placeholder, "TBD," empty required table cell, or promised later work remains.
7. All individual company profiles are complete before the synthesis and benchmark.
8. Every metric includes root-cause analysis, not just a sourced figure.
9. Every estimable metric has low, central, and high estimates.
10. Facts, calculations, inferences, triangulations, and unavailable information are distinguished through the
  required Evidence Status and narrative.
11. Organic growth is separated from acquisitions and other perimeter effects.
12. Maintenance capex is separated from growth capex.
13. Initial legal contract tenor is separated from effective customer tenure and expected customer life.
14. Utilization is defined according to the actual asset, customer-usage, and service-capacity model.
15. Payback is calculated using fully burdened investment and unit-level cash contribution, not revenue.
16. Distinct operating archetypes are identified and separately benchmarked where necessary.
17. No unjustified simple average is used.
18. No comparison with CRA or Compactor Rentals of America appears; CARMA is correctly treated as the designated
   submetering company and is not confused with the prohibited comparator.
19. Load-bearing facts are cited and the source hierarchy and evidence discipline are followed.
20. The entire designated subvertical report is completed in this single response and conversation.

Do not end with an offer to continue later. Deliver the complete, citation-supported report now.
```

04 Liquid Environmental Services Prompt

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SHA-256: 439a5cda4120d5564675f10110aa1a3753d2b7186132479b3d38491bc4494069

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You are ChatGPT 5.6 Pro acting as an expert infrastructure, business-services, asset-leasing, and M&A research
  analyst. Prepare an independent, first-principles operating benchmark study of the liquid environmental
  services subvertical and every company listed in the research universe below.

# Operating Benchmark - Liquid Environmental Services

## Assignment, analysis date, and non-negotiable guardrails

Use **July 18, 2026** as the analysis date. Use information available on or before that date, and state the
  reporting date or period for every material datapoint.

This is a fresh analysis in a newly created conversation. Treat this prompt as the complete instruction set. Do
  not rely on any prior conversation, local file, unstated attachment, prior analysis, inherited benchmark, or
  presumed user preference. No predetermined company, result, or benchmark may influence the conclusions.
  Develop the findings from company-specific evidence and the root economics of each business model.

Complete the full report in this conversation. Proceed without requesting clarification. When an ambiguity
  cannot be resolved, make the most reasonable explicitly stated assumption, document the alternatives
  considered, identify the resulting uncertainty, and continue. Do not stop after an outline, research plan,
  partial set of profiles, or preliminary findings.

Every company in the research universe must receive a complete individual profile. Finish all individual company
   analyses before developing an archetype or subvertical benchmark or drawing subvertical-wide conclusions. Do
  not use one company as an assumed benchmark for another before both have been analyzed independently.

Do **not** compare any finding, company, metric, range, archetype, or conclusion with **CRA or Compactor Rentals
   of America**. Do not use CRA or Compactor Rentals of America as a benchmark, analogue, validation point,
  reference company, or source of assumptions, and do not create a profile for it.

## Research universe

Analyze all and only these designated companies as company profiles:

1. Restaurant Technologies
2. Liquid Environmental Solutions
3. Waste Resources Management
4. Wind River Environmental

Other entities may be mentioned only when necessary to explain a designated company's ownership, transaction
  history, customers, regulation, or evidence. Do not add any non-designated company profile or substitute a
  different peer.

## Research objective

Develop a fundamental understanding of each designated company, each required operating archetype, and the
  subvertical across these nine core metrics:

1. Revenue growth
2. EBITDA margin
3. Maintenance capex as a percentage of revenue
4. Post-maintenance-capex conversion
5. Initial contract tenor
6. Effective customer tenure
7. Utilization exposure
8. New-unit or new-customer payback
9. B2B versus B2C customer profile

For every company, every metric, both required operating archetypes, and any final benchmark, explain both:

* the most defensible normalized assumption or range; and
* why the underlying customer, asset, contract, operating-cost, and capital model naturally produces that
  result.

The purpose is not merely to collect reported figures. Reconstruct the business model, identify the causal
  mechanisms, distinguish reported performance from normalized economics, and show how the evidence supports the
   assumptions.

## Mandatory liquid-environmental-services operating archetypes

Analyze **at least the following two operating archetypes separately**. These two named archetypes are mandatory
   even if further sub-archetypes are warranted by the evidence.

### Archetype 1 - Installed onsite-system model

Designated company:

* Restaurant Technologies

Analyze explicitly:

* installed fresh- and used-oil equipment;
* installation economics;
```

* equipment ownership;
* consumable and cooking-oil economics;
* route service;
* used-oil recovery value;
* customer-site payback; and
* customer retention.

Separate the economics of owned onsite equipment, fresh cooking-oil supply, used-oil collection or recovery, service, delivery, and other ancillary revenue wherever possible. Identify who funds installation, how oil commodity prices and pass-through mechanisms affect reported growth and margin, whether recovered-oil proceeds are a revenue source or cost offset, and whether the equipment investment or total customer-acquisition investment is the relevant payback unit.

Archetype 2 - Route, collection, treatment, and processing model

Designated companies:

* Liquid Environmental Solutions
* Waste Resources Management
* Wind River Environmental

Analyze explicitly:

* route density;
* service frequency;
* waste volumes;
* truck and technician productivity;
* treatment and processing assets;
* permits;
* disposal access;
* customer-site equipment; and
* recovered-material value.

Separate collection, pumping, hauling, treatment, processing, disposal, onsite service, equipment, and recovered-material economics wherever possible. Identify whether the company owns treatment or processing infrastructure, relies on third-party disposal, or combines both; how permits and disposal access affect competitive advantage, cost, and capacity; and whether volume changes affect revenue, route efficiency, processing utilization, or all three.

Do **not** force these two archetypes into one average unless the economics affirmatively support it. Develop separate archetype conclusions and benchmark tables when structural differences in revenue mix, gross margin, capital intensity, contract structure, utilization, or payback would make a blended benchmark misleading. If you also present a subvertical-wide range, label it as a reference range and explain why the underlying values are or are not comparable.

Governing analytical method

Use the following sequence for **every company and every core metric**:

1. Observed facts

Identify the most relevant company-specific disclosures. State exactly what is disclosed, by whom, for what entity or segment, for which period, and in what units. Distinguish standalone company facts from parent-level or market-level context.

2. Root business mechanism

Explain the actual customer, asset, contract, operating-cost, or capital mechanism that drives the metric. Trace the causal chain rather than merely describing correlation.

3. Normalized assumption

Develop low, central, and high estimates under normal operating conditions. Define what "normal" means for the company, adjust for commodity-price movements, acquisitions, route buildout, facility ramp-up, unusual volume or pricing, disposals, accounting classifications, one-time costs, and cycle effects, and show the bridge from evidence to the range.

4. Sensitivities

Identify the principal factors that could move the result above or below the central estimate. Where practical, quantify the direction or approximate magnitude of the sensitivity.

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Sources must establish facts; they must not replace the analysis. Explicitly distinguish a disclosed fact, a calculation from disclosed facts, an inference, a triangulation from multiple facts, and unavailable information.

Do not:

* simply average disconnected source estimates;
* repeat management language without analyzing it;
* assume recurring revenue automatically means low churn;
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* treat acquisition-driven growth as organic growth;

* treat contract tenor as customer tenure;
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* calculate payback using revenue rather than unit-level cash contribution;
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* use a simple average unless equal weighting is demonstrably appropriate and explicitly justified.

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Complete the following analysis for **each** designated company, in the order listed in the research universe. Do not abbreviate later profiles or refer the reader back to an earlier profile in place of company-specific work.

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Confirm:

* exact legal and operating identity;
* current and former names;
* current ownership;
* relevant transaction history;
* geographic footprint;
* relevant operating segment; and
* latest reliable standalone reporting period.

For an acquired company, use the last reliable standalone information and clearly disclose the data period. Then use later official information, when available, to update operating facts without mixing incompatible periods. For a diversified company, isolate the relevant liquid environmental services segment wherever possible. If exact isolation is impossible, state what is included in the reported segment, identify contamination from other activities, and reduce confidence rather than implying false precision.

B. Define the economic unit

Identify the operating unit that most directly creates revenue. Depending on the actual model, this may be one installed asset, one customer location, one serviced grease trap or tank, one route customer, one pickup, one gallon collected or processed, one truck, one route, one treatment facility, or another specifically justified unit.

Use the selected unit consistently when analyzing growth, utilization, capex, and payback. If the company requires separate economic units for customer-site equipment, route operations, and processing capacity, define each and reconcile them. State who owns the unit, who pays, when it begins earning, what ongoing work it requires, and what happens at contract termination.

C. Explain the customer problem

Explain:

* what the customer needs;
* how essential or discretionary the need is;
* what happens if it is not addressed;
* whether the need is permanent, recurring, temporary, or project based; and
* whether demand is driven by cost savings, convenience, regulation, safety, outsourcing, capacity, sanitation, environmental compliance, or another evidenced mechanism.

Identify the economic buyer, contractual counterparty, site operator, and any other end user. Explain the operational, legal, environmental, safety, or downtime consequences of non-service.

D. Explain customer alternatives

Determine whether the customer can:

* purchase the asset;
* use another rental or service provider;
* perform the activity internally;
* use a manual substitute;
* defer the decision;
* eliminate the need; or
* switch providers easily.

Explain why the customer selects and retains the company's solution. Address installed equipment, operational workflow, safety benefits, compliance requirements, route reliability, emergency response, site access, procurement, switching disruption, service quality, disposal documentation, and alternative disposal or processing access where relevant.

E. Map the revenue model

Identify whether revenue comes from:

* recurring equipment rental;
* fixed subscription;
* asset leasing;
* service fees;
* usage or volume;
* product and consumable sales;
* installation;
* delivery;
* monitoring or data;
* collection;
* processing or disposal;

* recovered-material value; or
* ancillary fees.

Separate recurring and transactional revenue. For the installed onsite-system model, separate equipment, installation, fresh-oil or consumable supply, used-oil recovery, delivery, and service economics. For the route, collection, treatment, and processing model, separate pumping or collection, transport, treatment, disposal, onsite services, customer-site equipment, and recovered-material revenue or credits. Identify pricing units, minimums, escalators, fuel or environmental surcharges, pass-through commodity pricing, and exposure to waste or oil volumes.

F. Map the operating model

Explain what the company must continue doing after acquiring the customer. Consider:

* technician visits;
* preventive maintenance;
* emergency repairs;
* route service;
* deliveries;
* consumables;
* inspections;
* monitoring;
* equipment removal;
* refurbishment;
* facility operations;
* treatment and processing;
* disposal;
* customer support; and
* billing and data services.

Identify which activities occur once at installation and which recur. Explain service frequency, route planning, field labor, truck requirements, density, facility throughput, third-party disposal dependence, permit compliance, emergency or unscheduled work, recovered-material handling, and how the model changes as a local market scales.

G. Explain customer retention and churn

Identify why customers remain, including as relevant:

* physical installation;
* operational integration;
* switching disruption;
* cost savings;
* essentiality;
* service reliability;
* regulation;
* proprietary systems;
* contract restrictions; and
* limited alternatives.

Identify why customers leave, including as relevant:

* competitive switching;
* site closure;
* bankruptcy;
* insourcing;
* equipment purchase;
* poor service;
* repricing;
* technology replacement; and
* loss of concessions or contracts.

Separate enforceable contract protection from practical customer inertia. Address site turnover, service failures, missed pickups, price or surcharge disputes, route exits, changes in waste volume, permitting issues, and the availability of alternative collectors, processors, disposal outlets, or onsite solutions.

H. Map the capital model

Explain:

* upfront asset investment;
* installation cost;
* customer-acquisition cost;
* asset ownership;
* useful life;
* replacement cycle;
* refurbishment requirements;
* redeployment potential;
* residual value;
* whether the initial contract recovers the investment; and
* whether renewals are required to generate the expected return.

Include installed customer-site equipment, trucks and service vehicles, tanks and containers, installation or site work, treatment and processing assets, facility maintenance, regulatory and safety capital, permits or access rights where economically relevant, and capitalized customer acquisition or startup spending. Explain which assets are dedicated to one customer and which serve a route or network. The metric assumptions must follow from this operating analysis.

Required treatment of the core metrics

1. Revenue growth

Decompose growth into every relevant driver:

* price;
* new customers;
* new installed units;
* net customer retention;
* service frequency;
* customer usage or volumes;
* fleet additions;
* utilization;
* cross-selling;
* acquisitions; and
* foreign exchange.

Show latest reported growth and historical growth where available. Separate organic growth, acquisition contribution, disposal or perimeter effects, and foreign exchange. For cooking-oil or recovered-material activities, distinguish physical volume, service pricing, commodity-price pass-through, and recovered-value effects. For collection and processing businesses, distinguish new routes, route density, pickup frequency, waste volume, price, acquired revenue, facility additions, and cross-selling.

Provide normalized low, central, and high assumptions and explain structural tailwinds and constraints. Explain why growth is structurally faster or slower than the other operating model in this subvertical, based on mechanisms rather than reported history alone. Do not treat acquisition-driven growth or commodity inflation as organic volume growth.

2. EBITDA margin

Construct the margin from the economics below, adapting the labels to the actual company while retaining the full bridge:

* **Revenue**
* less product and consumable costs
* less service and field labor
* less delivery, route, and fuel costs
* less repairs and maintenance
* less facilities, treatment, processing, and disposal costs
* less technology and monitoring expenses
* less sales and customer support
* less corporate overhead
* **equals EBITDA**

Explain:

* fixed versus variable costs;
* installation-only versus recurring costs;
* the impact of technician and route frequency;
* consumables burden;
* route or installed-base density;
* operating leverage;
* capitalized operating costs; and
* recurring adjustments excluded from reported EBITDA.

Show reported and normalized margins separately. Identify the EBITDA definition used, reconcile material adjustments, and avoid mixing accounting standards or segment scopes without explanation. Analyze gross versus net presentation of commodity or recovered-material revenue, fuel and disposal pass-throughs, route density, internal versus third-party treatment, facility throughput, permit-constrained capacity, installation mix, and customer-site equipment economics.

3. Maintenance capex as a percentage of revenue

Define maintenance capex as physical investment required to preserve existing revenue and operating capacity.

Potential components include:

* replacement equipment;
* major refurbishment;
* replacement fleet;
* service vehicles and route trucks;
* mandatory technology upgrades;
* meter replacements, where applicable;
* processing equipment;
* facility maintenance;
* tanks, pumps, and customer-site equipment;
* equipment removal and reinstallation; and
* regulatory, environmental, and safety capital.

Separate maintenance capex from:

* new growth units;
* expansion facilities;
* acquisition capex;
* development projects; and
* discretionary growth investments.

Use at least two of the following estimation approaches wherever the available evidence permits, and explain why each approach is or is not available:

1. Direct disclosure
2. Replacement-cycle analysis
3. Total capex less identifiable growth capex
4. Asset-life and depreciation cross-check

For mobile-fleet businesses, distinguish gross replacement spending from net spending after normal disposal proceeds. For onsite equipment, include necessary replacement, refurbishment, removal, reinstallation, and service-vehicle support. For treatment or processing operations, include sustaining facility, environmental, permit-compliance, and safety capital. Do not label route expansion, new customer installations, acquisition-related fleet additions, or new processing capacity as maintenance capex.

4. Post-maintenance-capex conversion

Calculate:

Post-Maintenance-Capex Conversion = (EBITDA - Maintenance Capex) / EBITDA

State clearly that this is not full free-cash-flow conversion because it excludes interest, taxes, working capital, lease principal, and growth capex.

Show low, central, and high conversion consistent with the EBITDA-margin and maintenance-capex assumptions. Where capitalized customer acquisition, installation, equipment placement, route startup, technology, or other spending is necessary merely to preserve current revenue, calculate and discuss a separate economic sustaining-spend adjustment. Explain any mismatch between accounting maintenance capex and economic sustaining spend.

5. Initial contract tenor and effective customer tenure

Analyze separately:

1. Initial legal term
2. Renewal term
3. Effective customer tenure
4. Expected customer life

Consider:

* automatic renewal;
* cancellation rights;
* termination penalties;
* purchase options;
* minimum-volume commitments;
* price escalators;
* transferability;
* rebid frequency; and
* actual churn and retention.

Explain why the legal term may understate or overstate actual revenue durability. Distinguish installed-equipment life, route-service contract term, treatment or disposal agreement, site life, and effective customer relationship. If product supply, equipment, collection, and processing revenue have different contractual terms, show them separately before selecting a normalized company range.

6. Utilization exposure

Analyze three distinct concepts:

* **Asset utilization:** whether an owned asset is under a paying contract or actively deployed.
* **Customer usage:** how intensively the customer uses the asset or generates service volume.
* **Service-capacity utilization:** how productively routes, technicians, vehicles, treatment equipment, or facilities are employed.

Address:

* whether idle assets earn revenue;
* asset returns and remarketing;
* customer resizing;
* exposure to restaurant activity, site count, oil consumption, waste volume, service frequency, facility throughput, or other relevant demand;
* redeployment time and expense;
* residual-value protection; and
* the most relevant operating KPI.

For the installed onsite-system model, distinguish deployed equipment from customer oil throughput and route productivity. For the route, collection, treatment, and processing model, distinguish truck or route utilization from collected volume and processing-facility utilization. Classify overall utilization exposure as **Low**, **Moderate**, or **High** and explain the classification. Where feasible, provide low, central, and high values for the most relevant quantitative KPI as well.

7. New-unit or new-customer payback

Define fully burdened initial investment to include:

* equipment;
* installation;
* site work;

* commissioning;
* delivery;
* sales commission;
* customer-acquisition cost; and
* other startup investment.

Calculate annual unit-level cash contribution as:

* **Annual revenue**
* less direct product cost
* less service labor
* less transportation and route expense
* less consumables
* less direct repairs
* less maintenance-capex reserve
* **equals annual unit-level cash contribution**

Then calculate:

Payback Period = Fully Burdened Initial Investment / Annual Unit-Level Cash Contribution

Provide low, central, and high estimates. Show the numerator, denominator, units, and assumptions; do not calculate payback from revenue. Address whether the initial contract covers payback, how customer life compares with payback, and how residual or redeployment value affects downside protection.

For the installed onsite-system model, analyze customer-site payback using fully burdened installed equipment and customer acquisition against contribution from the complete relationship, while separately showing equipment-only and customer-acquisition payback where feasible. For the route, collection, treatment, and processing model, distinguish incremental route-customer payback from truck, route, or treatment-capacity investment; do not allocate an entire shared facility to one customer without an economically justified capacity method.

8. B2B versus B2C customer profile

Identify:

* commercial and industrial exposure;
* municipal and utility exposure;
* residential exposure;
* institutional exposure;
* national-account concentration; and
* customer concentration.

Classify each company as one of:

* predominantly B2B;
* predominantly B2C;
* mixed B2B/B2C; or
* B2B contract with residential end users.

Explain implications for:

* bargaining power;
* procurement sophistication;
* sales cycles;
* bid frequency;
* concentration;
* pricing;
* customer-acquisition costs;
* churn; and
* regulation.

Identify the legal buyer, invoice recipient, decision-maker, site operator, and end user. Separate local accounts from national accounts and explain how concentration affects density, pricing, retention, and bargaining power.

Required output for every company

For each company, first provide the full first-principles profile and metric analysis above. Then include one populated summary table with exactly these columns and metric rows:

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Normalized revenue growth						
Normalized EBITDA margin						
Maintenance capex % of revenue						
Post-maintenance-capex conversion						
Initial contract tenor						
Effective customer tenure						
Utilization exposure						
New-unit or customer payback						
B2B/B2C profile						

Complete every required cell in the delivered report with a company-specific conclusion; the blank cells above define the output schema and are not permission to omit a conclusion. Use percentages for growth, margin, maintenance capex, and conversion; years or months for tenor, tenure, and payback; and explicit qualitative classifications, supplemented by quantitative KPIs where available, for utilization and customer profile. If a value truly cannot be estimated, state **Unavailable**, explain why, and do not fabricate a range.

Evidence Status must be one of:

* Disclosed
* Calculated
* Inferred
* Triangulated
* Unavailable

If evidence status differs across the low, central, and high estimate, use the status that best describes the central conclusion and explain the difference in the metric narrative. **Confidence** must be High, Medium, or Low.

For each metric, provide a separate explanation covering all six items:

1. Conclusion
2. Observed facts
3. Root business mechanism
4. Calculation or driver bridge
5. Sensitivities
6. Confidence

Conclude each company profile with:

* a plain-English business description;
* principal business strengths;
* principal risks;
* key unavailable information; and
* the most important diligence questions.

Required subvertical synthesis

Begin synthesis only after all four individual profiles are complete.

1. Identify operating archetypes

At minimum, present and analyze the two mandatory archetypes exactly as named:

1. **Installed onsite-system model** - Restaurant Technologies
2. **Route, collection, treatment, and processing model** - Liquid Environmental Solutions, Waste Resources Management, and Wind River Environmental

Determine whether further sub-archetypes are necessary, particularly where treatment-asset ownership, disposal access, customer-site equipment, service mix, or recovered-material economics differ materially. Do not collapse structurally distinct models into one blended benchmark.

2. Classify every company

Classify each designated company as exactly one of:

* Core representative
* Secondary reference
* Mixed-business company
* Outlier
* Insufficient disclosure

Explain the classification. Base "Core representative" status on business-model relevance and evidence quality, not company size or familiarity. A company may be core to its own archetype even if it is not representative of the other archetype.

3. Develop the benchmark

Provide a separate populated benchmark table for each mandatory archetype using exactly these columns and rows:

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
Revenue growth						
EBITDA margin						
Maintenance capex % revenue						
Post-maintenance conversion						
Initial contract tenor						
Effective customer tenure						
Utilization exposure						
New-unit/customer payback						
B2B/B2C profile						

Complete every required cell in the delivered report with a supported conclusion. For an archetype represented by one designated company, make the table an evidence-based archetype range derived from that company's normalized operating mechanics and clearly label the limitation; do not create artificial cross-company precision.

Only add a subvertical-wide benchmark table if the economics support meaningful comparison across both archetypes. If added, use the same format, label it clearly as either a true benchmark or a broader reference range, and explain the basis. Do not force the two mandatory archetypes into one average.

For every archetype or subvertical benchmark metric, explain:

* why the central range is appropriate;

* which designated companies support it;
* what operating mechanism drives it;
* why certain designated companies fall above or below it;
* whether the differences are structural or temporary; and
* what evidence remains uncertain.

Do not use a simple average unless equal weighting is demonstrably appropriate. If weighting is used, explain the basis and show why it reflects the intended benchmark. Prefer mechanism-based ranges and representative-company evidence over mathematically precise but economically incoherent aggregation.

4. Final subvertical conclusion

Conclude with all of the following, distinguishing the two mandatory archetypes wherever one answer would be misleading:

1. Typical customer problem
2. Typical revenue model
3. Primary growth drivers
4. Primary margin drivers
5. Principal maintenance-capex requirements
6. Typical contract and customer duration
7. Typical utilization dynamics
8. Typical new-unit payback
9. Typical B2B/B2C profile
10. Largest source of variation among companies
11. Lowest-confidence conclusion
12. A **25-30-word** slide-ready business-model description

The slide-ready description must accurately encompass the subvertical without erasing the two archetypes. Count its words and ensure it is between 25 and 30 words inclusive.

Source and evidence standards

Prioritize sources in this order:

1. Audited filings and statutory accounts
2. Official earnings releases and investor materials
3. Official company operating information
4. Debt, securitization, and ratings materials
5. Official transaction and sponsor materials
6. Government and regulatory filings
7. Reputable financial and industry publications

Use company-specific evidence wherever possible. Cite every load-bearing factual assertion close to the claim it supports, using direct links or precise footnotes that identify the source, date, and relevant page or section when available. Cite the original source rather than search-result snippets, aggregators, or articles that merely repeat it. Do not cite multiple sources that only repeat the same originating disclosure.

Use sources to establish facts, not to replace reasoning. Clearly label disclosed facts, calculations, inferences, triangulations, and unavailable information. State when information is unavailable. Do not convert weak evidence into precise figures; avoid false precision. A smaller number of authoritative sources is preferable to a large number of weak or repetitive sources.

When a source covers a parent, mixed segment, different geography, or different period, disclose that limitation before using it. Keep currencies, percentages, periods, numerator definitions, and denominator definitions explicit. Do not silently combine inconsistent EBITDA, capex, contract, customer, commodity, volume, or facility definitions.

Required report structure

Use this order:

1. Executive summary and concise description of the subvertical
2. Scope, analysis date, definitions, and methodology
3. Individual company profile: Restaurant Technologies
4. Individual company profile: Liquid Environmental Solutions
5. Individual company profile: Waste Resources Management
6. Individual company profile: Wind River Environmental
7. Mandatory operating-archetype analysis and company classification
8. Separate benchmark table and metric-by-metric synthesis for the installed onsite-system model
9. Separate benchmark table and metric-by-metric synthesis for the route, collection, treatment, and processing model
10. Optional subvertical reference range only if economically supported
11. Final subvertical conclusion, including the 25-30-word slide-ready description
12. Key evidence gaps and consolidated diligence priorities
13. Source list

The executive summary may preview conclusions, but it must not replace or pre-empt the independent company work. The archetype and final benchmark conclusions must be derived only after the four profiles have been completed.

Completion requirements and final self-audit

Before ending the response, verify and correct the report against every requirement below:

1. Every designated company-Restaurant Technologies, Liquid Environmental Solutions, Waste Resources Management, and Wind River Environmental-has a complete individual profile.
2. No non-designated company has been added as a research-universe profile.

```
3. All common research instructions in this prompt have been followed.
4. Every liquid-environmental-services-specific instruction has been addressed explicitly, including both
  mandatory named archetypes and every listed archetype factor.
5. The report is fully self-contained and does not depend on unavailable files or other conversations.
6. No unexplained placeholder, "TBD," empty required table cell, or promised later work remains.
7. All individual company profiles are complete before the archetype synthesis and benchmark.
8. Every metric includes root-cause analysis, not just a sourced figure.
9. Every estimable metric has low, central, and high estimates.
10. Facts, calculations, inferences, triangulations, and unavailable information are distinguished through the
  required Evidence Status and narrative.
11. Organic growth is separated from acquisitions, commodity-price effects, and other perimeter effects.
12. Maintenance capex is separated from growth capex.
13. Initial legal contract tenor is separated from effective customer tenure and expected customer life.
14. Utilization is defined according to the actual asset, customer-usage, route, service-capacity, and
  processing model.
15. Payback is calculated using fully burdened investment and unit-level cash contribution, not revenue.
16. At least the installed onsite-system model and the route, collection, treatment, and processing model are
  identified and separately benchmarked.
17. No unjustified simple average is used, and the two mandatory archetypes are not forced into one average.
18. No comparison with CRA or Compactor Rentals of America appears.
19. Load-bearing facts are cited and the source hierarchy and evidence discipline are followed.
20. The entire designated subvertical report is completed in this single response and conversation.

Do not end with an offer to continue later. Deliver the complete, citation-supported report now.
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05 Transportation Solutions Prompt

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SHA-256: 0e65730902237042de2ada0425a7ba5e61d5666c2af52b1b7b214c39ec11732c

You are ChatGPT 5.6 Pro acting as an expert infrastructure, business-services, asset-leasing, and M&A research analyst. Prepare an independent, first-principles operating benchmark study of the Transportation Solutions subvertical and every company listed in the research universe below.

Operating Benchmark - Transportation Solutions

Mandate and analysis date

Treat this as a fresh analysis as of **July 18, 2026**. No prior conversation, predetermined company, assumed benchmark, or preferred conclusion should influence the work. Build each company analysis from company-specific evidence and the underlying economics, complete every individual company profile first, and only then develop the subvertical benchmark.

Do **not** compare the findings with CRA or Compactor Rentals of America. Do not use either as a comparator, benchmark, analogy, cross-check, or source of an assumed result.

Analyze every designated company. Do not omit a company because disclosure is limited; instead, provide the fullest defensible profile, mark unavailable information clearly, and lower confidence where appropriate. Proceed without requesting clarification. Document all reasonable assumptions, identity issues, conflicts in evidence, unresolved uncertainties, and their effects on the conclusions.

Complete the entire Transportation Solutions report in this conversation. Do not stop after a plan, source list, identity check, or partial set of profiles, and do not defer any company or the synthesis to another conversation.

Research universe and required operating archetypes

Analyze only the following designated companies, grouped into three separate operating archetypes:

1. Railcar and locomotive leasing

- VTG
- Akiem

2. Airport ground-support-equipment leasing

- Air Rail
- TCR

3. Outsourced yard logistics

- Lazer Logistics

Every company must receive its own full profile. Analyze and benchmark all three archetypes separately before considering whether any limited combined Transportation Solutions conclusions are economically meaningful. Do not force businesses with materially different customer, asset, labor, contract, utilization, or capital models into one blended average.

Mandatory Air Rail identity-resolution gate

The name **Air Rail** is potentially ambiguous. Before conducting any operating or metric analysis of Air Rail:

1. Resolve the exact legal entity and operating business represented by Air Rail using authoritative company, ownership, transaction, registry, and other primary-source context.
2. State the exact legal name, trading or operating name, jurisdiction, current ownership, former names if any, relevant transaction history, and the evidence linking that entity to airport ground-support-equipment leasing.
3. Distinguish it from similarly named, unrelated, or adjacent businesses.
4. Explain the resolution logic and assign High, Medium, or Low identity confidence.
5. If the public evidence does not conclusively resolve the name, identify the plausible candidates, explain which candidate is most defensible and why, state the remaining uncertainty, and continue using only that explicitly identified entity. Do not silently select an entity or invent certainty.

This identity resolution must be completed before Air Rail's company profile or any benchmark calculation begins.

Research objective

Develop a fundamental understanding of the following nine metrics for each company, for each of the three archetypes, and for the subvertical where a combined conclusion is defensible:

1. Revenue growth
2. EBITDA margin
3. Maintenance capex as a percentage of revenue
4. Post-maintenance-capex conversion
5. Initial contract tenor
6. Effective customer tenure
7. Utilization exposure
8. New-unit or new-customer payback
9. B2B versus B2C customer profile

For every metric, explain both:

```
- the most defensible normalized low, central, and high assumption under normal operating conditions; and
- why the underlying customer, asset, contract, operating-cost, or capital model naturally produces that result.

Do not rely on reported historical figures alone. The purpose is to understand the root business mechanisms and
  establish defensible normalized operating benchmarks.

## Governing analytical method

Use the following sequence for **every company and every metric**:

1. **Observed facts** - Identify the most relevant company-specific disclosures and their dates.
2. **Root business mechanism** - Explain the actual customer, asset, contract, operating-cost, or capital
  mechanism that drives the metric.
3. **Normalized assumption** - Develop low, central, and high estimates under normal operating conditions. State
   units and the normalization period.
4. **Sensitivities** - Identify the principal factors that could move the result above or below the central
  estimate.
5. **Confidence** - Assign High, Medium, or Low confidence and explain why.

Sources establish facts; they do not replace the analysis. Where direct disclosure is unavailable, derive the
  estimate transparently from operating drivers, comparable periods within the same company, asset lives, unit
  economics, contract evidence, and other company-specific evidence. Clearly label calculations, inferences, and
   triangulations. Avoid false precision.

Do not:

- simply average disconnected source estimates;
- repeat management language without analyzing it;
- assume recurring revenue automatically means low churn;
- assume high EBITDA margins automatically imply strong cash conversion;
- treat acquisition-driven growth as organic growth;
- treat contract tenor as customer tenure;
- treat total capex as maintenance capex;
- calculate payback using revenue rather than unit-level cash contribution;
- force companies or archetypes with materially different operating models into one average.

## Required first-principles company profile

Complete Sections A through H below for each company before estimating its metrics.

### A. Resolve the company and segment

Confirm and cite:

- exact legal and operating identity;
- current and former names;
- current ownership;
- relevant transaction history;
- geographic footprint;
- relevant operating segment; and
- latest reliable standalone reporting period.

For an acquired company, use the last reliable standalone information and clearly disclose the data period.
  Supplement it with later official operating facts only when those facts are clearly attributable to the same
  business. For a diversified company, isolate the relevant rail-leasing, locomotive-leasing,
  airport-equipment-leasing, or yard-logistics segment wherever possible; do not apply consolidated economics to
   the designated segment without a reasoned bridge.

### B. Define the economic unit

Identify the operating unit that most directly creates revenue and use it consistently in growth, utilization,
  capex, and payback analysis. Depending on the company, the appropriate unit may be one railcar, one
  locomotive, one airport ground-support-equipment unit, one full-service equipment placement, one airport or
  airline contract, one dedicated yard site, one yard tractor, or one service hour. If more than one unit is
  needed to describe the economics, define the hierarchy and explain which unit governs each metric.

### C. Explain the customer problem

Explain:

- what the customer needs;
- how essential or discretionary the need is;
- what happens if it is not addressed;
- whether the need is permanent, recurring, temporary, or project based; and
- whether demand is driven by cost savings, convenience, regulation, safety, outsourcing, or capacity.

### D. Explain customer alternatives

Determine whether the customer can:

- purchase the asset;
- use another rental or service provider;
- perform the activity internally;
- use a manual substitute;
- defer the decision;
- eliminate the need; or
- switch providers easily.
```

Explain why the customer selects and retains the company's solution and what makes switching economically, operationally, or contractually difficult or easy.

E. Map the revenue model

Identify whether revenue comes from:

- recurring equipment rental;
- fixed subscription;
- asset leasing;
- service fees;
- usage or volume;
- product and consumable sales;
- installation;
- delivery;
- monitoring or data;
- processing or disposal; or
- ancillary fees.

Separate recurring and transactional revenue. For mixed contracts, identify fixed, variable, reimbursable, pass-through, mobilization, maintenance, service-hour, and ancillary components wherever possible.

F. Map the operating model

Explain what the company must continue doing after acquiring the customer. Consider:

- technician visits;
- preventive maintenance;
- emergency repairs;
- route service;
- deliveries;
- consumables;
- inspections;
- monitoring;
- equipment removal;
- refurbishment;
- facility operations;
- processing;
- customer support; and
- billing and data services.

Identify which work is performed by employees, subcontractors, customers, manufacturers, or airport/rail counterparties, and distinguish costs incurred once at deployment from costs incurred throughout the revenue life.

G. Explain customer retention and churn

Identify why customers remain, including as relevant:

- physical installation;
- operational integration;
- switching disruption;
- cost savings;
- essentiality;
- service reliability;
- regulation;
- proprietary systems;
- contract restrictions; and
- limited alternatives.

Identify why customers leave, including as relevant:

- competitive switching;
- site closure;
- bankruptcy;
- insourcing;
- equipment purchase;
- poor service;
- repricing;
- technology replacement; and
- loss of concessions or contracts.

Distinguish customer retention from retention of an individual asset placement or site contract.

H. Map the capital model

Explain:

- upfront asset investment;
- installation or mobilization cost;
- customer-acquisition cost;
- asset ownership;
- useful life;
- replacement cycle;
- overhaul and refurbishment requirements;
- redeployment potential;
- residual value;

- whether the initial contract recovers the investment; and
- whether renewals or successive placements are required to generate the expected return.

The metric assumptions must follow from this operating analysis.

Transportation-archetype requirements

Railcar and locomotive leasing: VTG and Akiem

For each company, address at minimum:

- fleet size and the proportion of the relevant fleet on lease;
- initial lease terms, renewal structure, and effective customer relationships;
- off-lease periods, return rates, remarketing time, storage, and repositioning expense;
- asset specialization and the breadth or narrowness of alternative users;
- residual values and disposal proceeds;
- scheduled and unscheduled maintenance, heavy overhauls, regulatory inspections, and refurbishment;
- gross fleet replacement, net replacement spending after ordinary disposal proceeds, and growth-fleet additions;
- freight, passenger, industrial-production, commodity, and individual-customer demand as applicable; and
- the distinction between reported fleet utilization, revenue-earning utilization, time utilization, and dollar utilization.

Do not assume railcars and locomotives have identical useful lives, lease structures, maintenance cycles, redeployment risks, or residual-value behavior. Isolate each company's relevant leasing operation from unrelated logistics, workshop, forwarding, maintenance-for-third-parties, or other activities wherever possible.

Airport ground-support-equipment leasing: Air Rail and TCR

For each company, address at minimum:

- the scope of full-service leasing and which maintenance, repair, replacement, compliance, and fleet-management obligations are included;
- airline, ground-handler, and airport contracts, including who is the contractual customer and who uses the equipment;
- exposure to flight activity, passenger volumes, airline capacity, airport concessions, and ground-handler contract turnover;
- physical and revenue fleet utilization, spare-fleet requirements, and whether idle equipment earns revenue;
- maintenance intensity by equipment type, age, duty cycle, safety requirements, and service-level obligation;
- electrification, charging infrastructure, battery replacement, emissions rules, and technology-obsolescence risk;
- airport-specific approvals, deployment, airside access, mobilization, demobilization, and redeployment constraints; and
- residual value and the feasibility, time, and cost of moving equipment between airports, airlines, or countries.

Distinguish a long-lived customer relationship from the duration of an individual airline, ground-handler, airport, concession, or equipment-placement contract.

Outsourced yard logistics: Lazer Logistics

Address at minimum:

- labor intensity, wage rates, overtime, recruiting, training, turnover, and supervisory requirements;
- service-hour, shift, site, fixed-fee, variable-fee, cost-plus, and performance-based revenue economics;
- initial terms, renewals, rebids, termination rights, and effective tenure of customer contracts;
- ownership or leasing of yard tractors, trailers, hostlers, technology, and other fleet or equipment;
- site dedication, mobilization and demobilization, equipment redeployment, and the economics of a lost site;
- employee and site productivity, hours per move, moves per labor hour, staffing utilization, and route or network effects if relevant;
- the role of technology, scheduling, spotting, shuttle, trailer management, and other outsourced operating services; and
- whether the business is primarily asset leasing or outsourced operations, with an evidence-based conclusion and implications for every metric.

Do not allow Lazer Logistics to determine the asset-leasing benchmark without explicit, metric-by-metric justification. Its results should remain in a separate outsourced-operations archetype unless the evidence establishes genuine comparability for a particular metric.

Required treatment of the core metrics

1. Revenue growth

Decompose growth into all relevant drivers:

- price;
- new customers;
- new installed or deployed units;
- net customer retention;
- service frequency;
- customer usage or volumes;
- fleet additions;
- utilization;
- cross-selling;
- acquisitions; and
- foreign exchange.

Show latest reported growth and historical growth where available. Separate organic growth, acquisition contribution, foreign exchange, fleet investment, utilization recovery or decline, price, and mix. Develop normalized low, central, and high assumptions and explain structural tailwinds and constraints. Explain from first principles why growth should be structurally faster or slower than that of the other designated operating models rather than relying only on reported history.

2. EBITDA margin

Construct a driver-based margin bridge for the relevant company or segment:

Revenue
less product and consumable costs
less service and field labor
less delivery, route, and fuel costs
less repairs and maintenance
less facilities or processing costs
less technology and monitoring expenses
less sales and customer support
less corporate overhead
equals **EBITDA**

Use zero or not applicable only with an explanation. Explain fixed versus variable costs, installation-only versus recurring costs, technician and route frequency, consumables burden, fleet or installed-base density, operating leverage, capitalized operating costs, and recurring adjustments excluded from reported EBITDA. For asset owners, address depreciation only as a cross-check and do not confuse EBITDA with asset-level economic profitability. For Lazer Logistics, explicitly bridge labor and subcontracting costs. Show reported and normalized margins separately and reconcile adjusted, reported, and segment measures.

3. Maintenance capex

Define maintenance capex as the physical investment required to preserve existing revenue and operating capacity. Potential components include:

- replacement equipment;
- major refurbishment and overhauls;
- replacement fleet;
- service vehicles;
- mandatory technology upgrades;
- meter replacements where metering is embedded in relevant equipment;
- processing or workshop equipment;
- facility maintenance;
- equipment removal and reinstallation; and
- regulatory and safety capital.

Separate maintenance capex from new growth units, expansion facilities, acquisition capex, development projects, and discretionary growth investments. Use at least two estimation approaches wherever possible:

1. Direct disclosure
2. Replacement-cycle analysis
3. Total capex less identifiable growth capex
4. Asset-life and depreciation cross-check

For every mobile-fleet business, distinguish gross replacement spending from net replacement spending after ordinary disposal proceeds. Explain whether reported net capex masks the cash needed to replace assets, and show maintenance capex as a percentage of revenue on a consistent basis. Treat asset disposals separately and transparently.

4. Post-maintenance-capex conversion

Calculate:

Post-Maintenance-Capex Conversion = (EBITDA - Maintenance Capex) / EBITDA

State explicitly that this is not full free-cash-flow conversion because it excludes interest, taxes, working capital, lease principal, and growth capex. Where capitalized customer acquisition, installation, mobilization, contract setup, or technology spending is necessary merely to preserve current revenue, calculate and show a separate economic sustaining-spend adjustment. Do not net disposal proceeds into the primary gross-maintenance view without also showing the gross amount and a reconciled net view.

5. Initial contract tenor and effective customer tenure

Analyze separately:

1. Initial legal term
2. Renewal term
3. Effective customer tenure
4. Expected customer life

Consider automatic renewal, cancellation rights, termination penalties, purchase options, minimum-volume or minimum-use commitments, price escalators, transferability, rebid frequency, and actual churn or retention. Explain why the legal term may understate or overstate revenue durability. Distinguish an individual asset lease, equipment placement, airport or airline contract, and yard-site contract from a broader customer relationship.

6. Utilization exposure

Analyze separately:

- **Asset utilization** - whether an owned asset is under a paying contract;
- **Customer usage** - how intensively the customer uses the asset; and
- **Service-capacity utilization** - how productively routes, technicians, employees, vehicles, workshops, or facilities are employed.

Address whether idle assets earn revenue; asset returns and remarketing; customer resizing; exposure to freight, passenger, flight, industrial, warehouse, retail, manufacturing, or other activity; redeployment time and expense; and residual-value protection. Identify the most relevant operating KPI for each company. Classify overall utilization exposure as Low, Moderate, or High and explain the classification. Do not mistake high reported utilization for low earnings sensitivity without analyzing rate, downtime, service-capacity, and customer-volume effects.

7. New-unit or new-customer payback

Include in fully burdened initial investment:

- equipment;
- installation;
- site work;
- commissioning;
- delivery and mobilization;
- sales commission;
- customer-acquisition cost; and
- other startup investment.

Calculate annual unit-level cash contribution as:

Annual revenue
less direct product cost
less service labor
less transportation and route expense
less consumables
less direct repairs
less maintenance-capex reserve
equals **Annual Unit-Level Cash Contribution**

Then calculate:

Payback Period = Fully Burdened Initial Investment / Annual Unit-Level Cash Contribution

Provide low, central, and high estimates, using cash contribution rather than revenue. Test whether the initial contract covers payback, compare expected customer or placement life with payback, and account separately for residual and redeployment value. For labor-led outsourced operations, define a customer- or site-acquisition payback rather than forcing an asset-unit payback, and identify working-capital or mobilization investment if material.

8. B2B versus B2C profile

Identify:

- commercial and industrial exposure;
- municipal and utility exposure;
- residential exposure;
- institutional exposure;
- national-account concentration; and
- customer concentration.

Classify each company as one of:

- predominantly B2B;
- predominantly B2C;
- mixed B2B/B2C; or
- B2B contract with residential end users.

Explain implications for bargaining power, procurement sophistication, sales cycles, bid frequency, concentration, pricing, customer-acquisition costs, churn, and regulation. Identify both the contractual customer and ultimate end user where they differ.

Required company output

Present a separate, complete profile for each company. Use the same section order and units across companies where their economics are comparable. Each company profile must include this table:

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Normalized revenue growth						
Normalized EBITDA margin						
Maintenance capex % of revenue						
Post-maintenance-capex conversion						
Initial contract tenor						
Effective customer tenure						
Utilization exposure						
New-unit or customer payback						
B2B/B2C profile						

For numerical metrics, state percentages, years, months, or other units. For categorical metrics, use meaningful low/central/high scenarios or classifications rather than fabricated numbers. If a range cannot be estimated responsibly, write **Unavailable** and explain what evidence would be needed.

Evidence Status must be exactly one of:

- Disclosed
- Calculated
- Inferred
- Triangulated
- Unavailable

Confidence must be High, Medium, or Low. Evidence Status describes how the conclusion was obtained; Confidence describes its reliability. Do not combine the two.

For **each metric after the table**, provide a separate explanation with these six labeled elements:

1. Conclusion
2. Observed facts
3. Root business mechanism
4. Calculation or driver bridge
5. Sensitivities
6. Confidence

Conclude every company profile with:

- a plain-English business description;
- principal business strengths;
- principal risks;
- key unavailable information; and
- the most important diligence questions.

Required Transportation Solutions synthesis

Begin synthesis only after all five individual company profiles are complete.

1. Identify operating archetypes

At minimum, preserve and analyze these three archetypes independently:

1. Railcar and locomotive leasing
2. Airport ground-support-equipment leasing
3. Outsourced yard logistics

Determine whether further distinctions are required, such as railcars versus locomotives, full-service versus finance-like leases, or equipment leasing versus labor-led managed services. Do not force a blended benchmark where distinct archetypes should be presented separately. Any combined subvertical range must be accompanied by an explanation of what it does and does not represent.

2. Classify every company

Classify each company as exactly one of:

- Core representative
- Secondary reference
- Mixed-business company
- Outlier
- Insufficient disclosure

Explain the classification and identify the archetype for which the company is informative. A company may not be omitted merely because it is not a core representative.

3. Develop the benchmark

Produce a separate benchmark table for each archetype and, only if defensible, a clearly labeled combined Transportation Solutions table. Use this format:

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
Revenue growth						
EBITDA margin						
Maintenance capex % revenue						
Post-maintenance conversion						
Initial contract tenor						
Effective customer tenure						
Utilization exposure						
New-unit/customer payback						
B2B/B2C profile						

For every benchmark metric, explain:

- why the central range is appropriate;
- which companies support it;
- what operating mechanism drives it;
- why certain companies fall above or below it;
- whether differences are structural or temporary; and
- what evidence remains uncertain.

Do not use a simple average unless equal weighting is demonstrably appropriate. If weighting is used, explain the economic basis. Do not allow Lazer Logistics to determine any rail or airport asset-leasing benchmark without explicit, metric-specific evidence and justification.

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### 4. Final subvertical conclusion

Conclude with all of the following:

1. Typical customer problem
2. Typical revenue model
3. Primary growth drivers
4. Primary margin drivers
5. Principal maintenance-capex requirements
6. Typical contract and customer duration
7. Typical utilization dynamics
8. Typical new-unit payback
9. Typical B2B/B2C profile
10. Largest source of variation among companies
11. Lowest-confidence conclusion
12. A **25-30-word** slide-ready business-model description

Where a single "typical" conclusion would conceal the three archetypes, give archetype-specific answers and then
   state the narrow conclusion, if any, that applies across the subvertical.

## Source and evidence standards

Prioritize sources in this order:

1. Audited filings and statutory accounts
2. Official earnings releases and investor materials
3. Official company operating information
4. Debt, securitization, and ratings materials
5. Official transaction and sponsor materials
6. Government and regulatory filings
7. Reputable financial and industry publications

Use company-specific evidence wherever possible. Cite every load-bearing factual assertion close to the claim it
   supports, with enough source identification and date context for verification. Do not use search-result
  snippets as evidence. Do not cite multiple sources that merely repeat the same original information. Prefer a
  smaller number of authoritative sources to a large number of weak or repetitive sources.

Use sources to establish facts, not to replace reasoning. Distinguish disclosed facts from calculations,
  inferences, and triangulations. State when information is unavailable, disclose contradictory evidence, and
  avoid false precision. For acquired or privately held companies, clearly state the reporting period to which
  each financial fact applies. Do not present parent-company or post-acquisition figures as standalone company
  figures unless attribution is supportable.

## Required report order and completion standard

Deliver the report in this order:

1. Scope, analysis date, and concise methodology
2. Air Rail identity-resolution finding
3. Company profiles for VTG and Akiem
4. Company profiles for Air Rail and TCR
5. Company profile for Lazer Logistics
6. Company classification and three-archetype synthesis
7. Archetype benchmark tables and any defensible combined table
8. Final Transportation Solutions conclusion
9. Concise source list organized by company

## Final completion checklist

Before submitting the report, silently verify and correct it against every requirement below:

1. Every designated company-VTG, Akiem, Air Rail, TCR, and Lazer Logistics-has a complete individual profile.
2. No company from another subvertical has been added to the research universe or benchmark population.
3. Every common role, research-objective, analytical-method, first-principles company-profile, core-metric,
  company-output, synthesis, source, and completion instruction in this prompt has been followed.
4. All Transportation Solutions special instructions have been followed, including the Air Rail
  identity-resolution gate, separate analysis of all three operating archetypes, and the Lazer Logistics
  benchmark restriction.
5. The report is fully self-contained and does not rely on another conversation or an unavailable file.
6. The report contains no unexplained placeholders; unavailable evidence is labeled and explained rather than
  left blank.
7. All five individual company profiles are complete before any subvertical synthesis or benchmark is developed.
8. Every metric includes root-cause analysis based on the actual customer, asset, contract, operating-cost, or
  capital mechanism.
9. Every metric contains normalized low, central, and high estimates under normal operating conditions, or an
  explicit Unavailable conclusion with the missing evidence identified.
10. Disclosed facts, calculations, inferences, triangulations, and unavailable information are clearly
  distinguished using the required Evidence Status labels.
11. Organic growth is separated from acquisitions and other inorganic growth contributions.
12. Maintenance capex is separated from growth capex, and gross mobile-fleet replacement spending is
  distinguished from net spending after ordinary disposal proceeds.
13. Initial contract tenor, renewal term, effective customer tenure, and expected customer life are analyzed
  separately.
14. Asset utilization, customer usage, and service-capacity utilization are defined according to each company's
  actual operating model.
15. Payback uses fully burdened initial investment divided by annual unit-level cash contribution, not revenue.
16. The railcar and locomotive leasing, airport ground-support-equipment leasing, and outsourced yard-logistics
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  archetypes are identified and analyzed separately, with further distinctions where required.
17. No simple average is used unless equal weighting is demonstrably appropriate, and no materially different
  operating model is forced into a blended benchmark.
18. The findings are not compared with CRA or Compactor Rentals of America in any form.
19. Every load-bearing factual assertion is cited under the required source hierarchy and evidence discipline.
20. The complete Transportation Solutions report-including every company profile, all three archetype syntheses
  and benchmark tables, and the final conclusion-is finished in this single conversation.

The report is complete only when every item above is satisfied. Complete the full report now without requesting
  clarification.
```

06 Specialty Rental Prompt

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SHA-256: ad02a1ac42080c6a2dfca70889309d7a35f0b3610807fd67027543b7d3697f89

You are ChatGPT 5.6 Pro acting as an expert infrastructure, business-services, asset-leasing, and M&A research analyst. Prepare an independent, first-principles operating benchmark study of the Specialty Rental subvertical and every company listed in the research universe below.

Operating Benchmark - Specialty Rental

Mandate and analysis date

Treat this as a fresh analysis as of **July 18, 2026**. No prior conversation, predetermined company, assumed benchmark, or preferred conclusion should influence the work. Build each company analysis from company-specific evidence and the underlying economics, complete every individual company profile first, and only then develop the subvertical benchmark.

Do **not** compare the findings with CRA or Compactor Rentals of America. Do not use either as a comparator, benchmark, analogy, cross-check, or source of an assumed result.

Analyze every designated company. Do not omit a company because disclosure is limited; instead, provide the fullest defensible profile, mark unavailable information clearly, and lower confidence where appropriate. Proceed without requesting clarification. Document all reasonable assumptions, identity issues, conflicts in evidence, unresolved uncertainties, and their effects on the conclusions.

Complete the entire Specialty Rental report in this conversation. Do not stop after a plan, source list, identity check, or partial set of profiles, and do not defer any company or the synthesis to another conversation.

Research universe

Analyze only these designated companies:

- PODS
- Mobile Mini
- Trench Plate Rental
- General Finance Corporation
- Modulaire Group
- McGrath RentCorp
- Reliant Rental

Every company must receive its own full profile. Do not substitute a parent, successor, seller, buyer, sponsor, portfolio company, or similarly named business for a designated company without resolving and explaining the relationship.

Mandatory Reliant Rental identity-resolution gate

The name **Reliant Rental** is potentially ambiguous. Before conducting any operating or metric analysis of Reliant Rental:

1. Locate the relevant transaction context associated with the designated specialty-rental business, prioritizing authoritative transaction announcements, sponsor or buyer materials, company materials, statutory records, and other primary evidence.
2. Use that transaction context to resolve the exact legal entity and operating business represented by Reliant Rental; do **not** presume an answer from the name alone.
3. State the exact legal name, trading or operating name, jurisdiction, current and former ownership, former names if any, transaction parties, transaction date, business description, and evidence connecting the resolved entity to specialty rental.
4. Distinguish it from similarly named, unrelated, or adjacent rental businesses.
5. Explain the resolution logic and assign High, Medium, or Low identity confidence.
6. If the evidence does not conclusively resolve the identity, identify plausible candidates, explain which candidate is most defensible and why, disclose the remaining uncertainty, and continue using only that explicitly identified entity. Do not silently select an entity or invent certainty.

This identity resolution must be completed before Reliant Rental's company profile or any benchmark calculation begins.

Specialty-rental scope and model distinctions

Determine from evidence which activities and assets are relevant for each designated company. Explicitly distinguish, rather than blend without analysis:

- portable storage;
- mobile storage;
- modular space;
- temporary offices and classrooms;
- trench-safety equipment; and
- other specialty-rental assets.

Identify further distinctions where economically important, including on-site versus off-site storage, delivered versus branch-collected equipment, short-duration project rental versus multi-year placement, standardized versus highly specialized assets, and rental-plus-service versus rental-only economics.

Do not assume all specialty-rental companies have the same utilization, contract duration, margin, or capital requirements. Preserve separate archetypes whenever physical assets, customers, deployment models, branch/service requirements, or rental durations produce structurally different economics.

For **every acquired company**, use the last reliable standalone reporting period as the primary financial basis and clearly disclose its exact period end or date. Do not substitute consolidated buyer data for standalone company economics. Later official facts may be used only when clearly attributable to the acquired business, and any mixed-period inference must be explained.

Research objective

Develop a fundamental understanding of the following nine metrics for each company, for each identified specialty-rental archetype, and for the subvertical where a combined conclusion is defensible:

1. Revenue growth
2. EBITDA margin
3. Maintenance capex as a percentage of revenue
4. Post-maintenance-capex conversion
5. Initial contract tenor
6. Effective customer tenure
7. Utilization exposure
8. New-unit or new-customer payback
9. B2B versus B2C customer profile

For every metric, explain both:

- the most defensible normalized low, central, and high assumption under normal operating conditions; and
- why the underlying customer, asset, contract, operating-cost, or capital model naturally produces that result.

Do not rely on reported historical figures alone. The purpose is to understand the root business mechanisms and establish defensible normalized operating benchmarks.

Governing analytical method

Use the following sequence for **every company and every metric**:

1. **Observed facts** - Identify the most relevant company-specific disclosures and their dates.
2. **Root business mechanism** - Explain the actual customer, asset, contract, operating-cost, or capital mechanism that drives the metric.
3. **Normalized assumption** - Develop low, central, and high estimates under normal operating conditions. State units and the normalization period.
4. **Sensitivities** - Identify the principal factors that could move the result above or below the central estimate.
5. **Confidence** - Assign High, Medium, or Low confidence and explain why.

Sources establish facts; they do not replace the analysis. Where direct disclosure is unavailable, derive the estimate transparently from operating drivers, comparable periods within the same company, asset lives, unit economics, contract evidence, and other company-specific evidence. Clearly label calculations, inferences, and triangulations. Avoid false precision.

Do not:

- simply average disconnected source estimates;
- repeat management language without analyzing it;
- assume recurring revenue automatically means low churn;
- assume high EBITDA margins automatically imply strong cash conversion;
- treat acquisition-driven growth as organic growth;
- treat contract tenor as customer tenure;
- treat total capex as maintenance capex;
- calculate payback using revenue rather than unit-level cash contribution;
- force companies or archetypes with materially different operating models into one average.

Required first-principles company profile

Complete Sections A through H below for each company before estimating its metrics.

A. Resolve the company and segment

Confirm and cite:

- exact legal and operating identity;
- current and former names;
- current ownership;
- relevant transaction history;
- geographic footprint;
- relevant operating segment; and
- latest reliable standalone reporting period.

For every acquired company, use the last reliable standalone information and clearly disclose the exact data period. Supplement it with later official operating facts only when those facts are clearly attributable to the same business. For a diversified company, isolate the relevant portable-storage, mobile-storage, modular-space, temporary-office/classroom, trench-safety, or other specialty-rental segment wherever possible; do not apply consolidated economics without a reasoned bridge.

When a company has been merged, renamed, broken into segments, or acquired by another designated company, keep its historical standalone profile distinct. Explain overlaps without double counting the same period or business when developing the benchmark.

B. Define the economic unit

Identify the operating unit that most directly creates revenue and use it consistently in growth, utilization, capex, and payback analysis. Depending on the company and segment, the appropriate unit may be one portable storage container, mobile storage unit, modular building, temporary office, classroom, trench plate, trench shield, shoring unit, branch-deployed equipment unit, customer location, or rental contract. If more than one unit is required, define the hierarchy and explain which unit governs each metric.

C. Explain the customer problem

Explain:

- what the customer needs;
- how essential or discretionary the need is;
- what happens if it is not addressed;
- whether the need is permanent, recurring, temporary, or project based; and
- whether demand is driven by cost savings, convenience, regulation, safety, outsourcing, or capacity.

Distinguish planned capacity needs, emergency needs, construction and infrastructure projects, school or institutional capacity, business relocations, and consumer storage events where relevant.

D. Explain customer alternatives

Determine whether the customer can:

- purchase the asset;
- use another rental or service provider;
- perform the activity internally;
- use a manual substitute;
- defer the decision;
- eliminate the need; or
- switch providers easily.

Explain why the customer selects and retains the company's solution. Address local availability, delivery speed, branch density, product breadth, safety/compliance support, permitting, customization, switching disruption, transportation cost, and total cost of ownership as relevant.

E. Map the revenue model

Identify whether revenue comes from:

- recurring equipment rental;
- fixed subscription;
- asset leasing;
- service fees;
- usage or volume;
- product and consumable sales;
- installation;
- delivery;
- monitoring or data;
- processing or disposal; or
- ancillary fees.

Separate recurring and transactional revenue. Identify base rental, delivery, pickup, installation, site work, damage waiver, accessories, maintenance, sale of new or used units, and ancillary services. Distinguish original-equipment sales and fleet disposals from rental revenue.

F. Map the operating model

Explain what the company must continue doing after acquiring the customer. Consider:

- technician visits;
- preventive maintenance;
- emergency repairs;
- route service;
- deliveries;
- consumables;
- inspections;
- monitoring;
- equipment removal;
- refurbishment;
- facility operations;
- processing;
- customer support; and
- billing and data services.

Address branch operations, dispatch, trucking, installation, pickup, cleaning, repair, refurbishment, inventory control, fleet balancing, sales coverage, permitting assistance, and safety inspections. Distinguish costs incurred once at deployment or return from recurring costs during the rental.

G. Explain customer retention and churn

Identify why customers remain, including as relevant:

- physical installation;
- operational integration;
- switching disruption;
- cost savings;
- essentiality;
- service reliability;
- regulation;

- proprietary systems;
- contract restrictions; and
- limited alternatives.

Identify why customers leave, including as relevant:

- competitive switching;
- site closure;
- bankruptcy;
- insourcing;
- equipment purchase;
- poor service;
- repricing;
- technology replacement; and
- loss of concessions or contracts.

Also distinguish expected project completion or household move-out from avoidable customer churn. Separate the duration of an individual asset rental from the tenure of a repeat commercial or institutional relationship.

H. Map the capital model

Explain:

- upfront asset investment;
- installation and site-work cost;
- customer-acquisition cost;
- asset ownership;
- useful life;
- replacement cycle;
- repair and refurbishment requirements;
- redeployment potential;
- residual value;
- whether the initial contract recovers the investment; and
- whether renewals or successive rentals are required to generate the expected return.

The metric assumptions must follow from this operating analysis.

Specialty-rental operating requirements

For each company and relevant asset class, explicitly analyze:

- **Physical utilization:** the proportion of rentable units on rent, including the treatment of units in transit, under repair, held for sale, seasonally idle, or unavailable.
- **Dollar utilization:** rental revenue relative to original equipment cost, average fleet cost, or the company's disclosed denominator; reconcile definitions before comparing companies.
- **Rental rates:** price per unit and period, ancillary charges, mix, inflation, yield management, and rate changes on new versus existing contracts.
- **Rental duration:** initial legal or order term, minimum term, actual time on rent, extensions, renewals, and repeat-customer tenure.
- **Delivery and branch costs:** outbound and return transport, set-up, pickup, branch labor, dispatch, mileage, fuel, cranes or specialized handling, and fleet-balancing expense.
- **Repair and refurbishment:** routine repairs, between-rental work, damage, major refurbishment, useful-life extension, cleaning, code or safety upgrades, and capitalization policy.
- **Gross and net replacement capex:** gross cash purchases required to replace existing capacity, ordinary disposal proceeds, gains on sale, and the distinct net cash burden after disposals.
- **Residual value:** ordinary sale proceeds by age and asset type, liquidity of the used-equipment market, transport cost, and downside under distressed or technologically obsolete conditions.
- **Project and construction cyclicality:** exposure to project starts, completions, construction activity, infrastructure spending, nonresidential demand, education budgets, disaster response, and household moves as applicable.
- **Customer concentration:** national accounts, local accounts, end markets, geographies, individual projects, institutions, consumers, and top-customer exposure.
- **Asset redeployment:** standardization, alternate uses, return logistics, branch network, repair lead time, geographic mobility, site work, permitting, and the time and cost required to place a returned unit with another customer.
- **New-fleet payback:** fully burdened delivered fleet cost divided by cash contribution after direct service expense and a maintenance-capex reserve, with residual value shown separately rather than used to obscure the initial payback.

Reconcile company-specific definitions before comparing physical utilization, dollar utilization, fleet cost, rental rate, rental duration, or capex. Do not treat different denominators or accounting policies as comparable merely because they have similar labels.

Required treatment of the core metrics

1. Revenue growth

Decompose growth into all relevant drivers:

- price;
- new customers;
- new installed or deployed units;
- net customer retention;
- service frequency;
- customer usage or volumes;
- fleet additions;
- utilization;
- cross-selling;

- acquisitions; and
- foreign exchange.

Show latest reported growth and historical growth where available. Separate organic growth, acquisition contribution, foreign exchange, price, physical utilization, rental rates, rental duration, fleet investment, product mix, fleet sales, and ancillary services. Develop normalized low, central, and high assumptions and explain structural tailwinds and constraints. Explain from first principles why growth should be structurally faster or slower than that of the other designated operating models rather than relying only on reported history.

2. EBITDA margin

Construct a driver-based margin bridge for the relevant company or segment:

Revenue
less product and consumable costs
less service and field labor
less delivery, route, and fuel costs
less repairs and maintenance
less facilities or processing costs
less technology and monitoring expenses
less sales and customer support
less corporate overhead
equals **EBITDA**

Use zero or not applicable only with an explanation. Explain fixed versus variable costs, installation-only versus recurring costs, technician and route frequency, consumables burden, branch or installed-fleet density, operating leverage, capitalized operating costs, and recurring adjustments excluded from reported EBITDA. Identify gains on fleet disposals, acquisition adjustments, stock compensation, restructuring, and other recurring "one-time" adjustments where material. Show reported and normalized margins separately, reconcile adjusted and reported measures, and avoid applying consolidated margins to a segment without a bridge.

3. Maintenance capex

Define maintenance capex as the physical investment required to preserve existing revenue and operating capacity. Potential components include:

- replacement equipment;
- major refurbishment;
- replacement fleet;
- service vehicles;
- mandatory technology upgrades;
- meter replacements where metering is embedded in relevant equipment;
- processing equipment where relevant;
- facility and branch maintenance;
- equipment removal and reinstallation; and
- regulatory and safety capital.

Separate maintenance capex from new growth units, expansion branches or facilities, acquisition capex, development projects, and discretionary growth investments. Use at least two estimation approaches wherever possible:

1. Direct disclosure
2. Replacement-cycle analysis
3. Total capex less identifiable growth capex
4. Asset-life and depreciation cross-check

For these mobile-fleet businesses, distinguish gross replacement spending from net replacement spending after ordinary disposal proceeds. Show gross purchases, disposal proceeds, gains or losses, and resulting net cash spending separately wherever evidence permits. Do not assume used-equipment proceeds eliminate the physical replacement requirement.

4. Post-maintenance-capex conversion

Calculate:

Post-Maintenance-Capex Conversion = (EBITDA - Maintenance Capex) / EBITDA

State explicitly that this is not full free-cash-flow conversion because it excludes interest, taxes, working capital, lease principal, and growth capex. Where capitalized customer acquisition, installation, site preparation, delivery fleet, branch setup, or technology spending is necessary merely to preserve current revenue, calculate and show a separate economic sustaining-spend adjustment. Present gross-maintenance and disposal-adjusted views separately rather than allowing asset-sale proceeds to obscure the sustaining capital burden.

5. Initial contract tenor and effective customer tenure

Analyze separately:

1. Initial legal term
2. Renewal term
3. Effective customer tenure
4. Expected customer life

Consider automatic renewal, cancellation rights, termination penalties, purchase options, minimum-volume or minimum-rental commitments, price escalators, transferability, rebid frequency, and actual churn or retention. Explain why the legal term may understate or overstate revenue durability. Distinguish rental duration for a particular unit, project, site, classroom, office, or storage need from the customer's total repeat relationship.

6. Utilization exposure

Analyze separately:

- **Asset utilization** - whether an owned asset is under a paying contract;
- **Customer usage** - how intensively the customer uses the asset; and
- **Service-capacity utilization** - how productively branches, routes, technicians, drivers, vehicles, installation crews, or facilities are employed.

Address whether idle assets earn revenue; asset returns and remarketing; customer resizing; exposure to construction, project completion, office or school occupancy, household moves, industrial activity, infrastructure spending, weather, or other relevant activity; redeployment time and expense; and residual-value protection. Identify the most relevant operating KPI for each company. Report both physical and dollar utilization when available and explain their different signals. Classify overall utilization exposure as Low, Moderate, or High and explain the classification.

7. New-unit or new-customer payback

Include in fully burdened initial investment:

- equipment;
- installation;
- site work;
- commissioning;
- delivery and mobilization;
- sales commission;
- customer-acquisition cost; and
- other startup investment.

Calculate annual unit-level cash contribution as:

Annual revenue
less direct product cost
less service labor
less transportation and route expense
less consumables
less direct repairs
less maintenance-capex reserve
equals **Annual Unit-Level Cash Contribution**

Then calculate:

Payback Period = Fully Burdened Initial Investment / Annual Unit-Level Cash Contribution

Provide low, central, and high estimates, using cash contribution rather than revenue. Test whether the initial rental or customer relationship covers payback, compare expected unit placement and customer life with payback, and show residual or redeployment value separately. For fleet bought speculatively rather than for a signed customer, analyze new-fleet payback at normalized utilization and rate, including expected idle time and initial deployment costs.

8. B2B versus B2C profile

Identify:

- commercial and industrial exposure;
- municipal and utility exposure;
- residential exposure;
- institutional exposure;
- national-account concentration; and
- customer concentration.

Classify each company as one of:

- predominantly B2B;
- predominantly B2C;
- mixed B2B/B2C; or
- B2B contract with residential end users.

Explain implications for bargaining power, procurement sophistication, sales cycles, bid frequency, concentration, pricing, customer-acquisition costs, churn, and regulation. Identify both the contractual customer and ultimate end user where they differ.

Required company output

Present a separate, complete profile for each company. Use the same section order and units across companies where their economics are comparable. Each company profile must include this table:

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Normalized revenue growth						
Normalized EBITDA margin						
Maintenance capex % of revenue						
Post-maintenance-capex conversion						
Initial contract tenor						
Effective customer tenure						
Utilization exposure						

| New-unit or customer payback | | | | | | |
| B2B/B2C profile | | | | | | |

For numerical metrics, state percentages, years, months, or other units. For categorical metrics, use meaningful low/central/high scenarios or classifications rather than fabricated numbers. If a range cannot be estimated responsibly, write **Unavailable** and explain what evidence would be needed.

Evidence Status must be exactly one of:

- Disclosed
- Calculated
- Inferred
- Triangulated
- Unavailable

Confidence must be High, Medium, or Low. Evidence Status describes how the conclusion was obtained; Confidence describes its reliability. Do not combine the two.

For **each metric after the table**, provide a separate explanation with these six labeled elements:

1. Conclusion
2. Observed facts
3. Root business mechanism
4. Calculation or driver bridge
5. Sensitivities
6. Confidence

Conclude every company profile with:

- a plain-English business description;
- principal business strengths;
- principal risks;
- key unavailable information; and
- the most important diligence questions.

Required Specialty Rental synthesis

Begin synthesis only after all seven individual company profiles are complete.

1. Identify operating archetypes

Determine which companies represent portable storage, mobile storage, modular space, temporary offices and classrooms, trench-safety equipment, or other specialty-rental models. Create separate or overlapping archetypes only when supported by the actual business mix, and explain any overlap. Determine whether additional distinctions are necessary based on customer type, branch model, rental duration, asset standardization, installation intensity, or delivery economics.

Do not force a blended benchmark where distinct archetypes should be presented separately. Do not assume all specialty-rental companies have the same utilization, contract duration, margin, or capital requirements. Any combined subvertical range must explain what it represents and which structurally different economics it conceals.

2. Classify every company

Classify each company as exactly one of:

- Core representative
- Secondary reference
- Mixed-business company
- Outlier
- Insufficient disclosure

Explain the classification and identify the archetype for which the company is informative. A company may not be omitted merely because it is acquired, historical, mixed, or not a core representative.

3. Develop the benchmark

Produce a separate benchmark table for every material archetype and, only if defensible, a clearly labeled combined Specialty Rental table. Use this format:

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
Revenue growth						
EBITDA margin						
Maintenance capex % revenue						
Post-maintenance conversion						
Initial contract tenor						
Effective customer tenure						
Utilization exposure						
New-unit/customer payback						
B2B/B2C profile						

For every benchmark metric, explain:

- why the central range is appropriate;
- which companies support it;
- what operating mechanism drives it;
- why certain companies fall above or below it;

- whether differences are structural or temporary; and
- what evidence remains uncertain.

Do not use a simple average unless equal weighting is demonstrably appropriate. If weighting is used, explain the economic basis. Reconcile differences in utilization definitions, fleet-cost denominators, disposal treatment, accounting policies, and financial periods before making comparisons.

4. Final subvertical conclusion

Conclude with all of the following:

1. Typical customer problem
2. Typical revenue model
3. Primary growth drivers
4. Primary margin drivers
5. Principal maintenance-capex requirements
6. Typical contract and customer duration
7. Typical utilization dynamics
8. Typical new-unit payback
9. Typical B2B/B2C profile
10. Largest source of variation among companies
11. Lowest-confidence conclusion
12. A **25-30-word** slide-ready business-model description

Where a single "typical" conclusion would conceal distinct archetypes, give archetype-specific answers and then state the narrow conclusion, if any, that applies across the subvertical.

Source and evidence standards

Prioritize sources in this order:

1. Audited filings and statutory accounts
2. Official earnings releases and investor materials
3. Official company operating information
4. Debt, securitization, and ratings materials
5. Official transaction and sponsor materials
6. Government and regulatory filings
7. Reputable financial and industry publications

Use company-specific evidence wherever possible. Cite every load-bearing factual assertion close to the claim it supports, with enough source identification and date context for verification. Do not use search-result snippets as evidence. Do not cite multiple sources that merely repeat the same original information. Prefer a smaller number of authoritative sources to a large number of weak or repetitive sources.

Use sources to establish facts, not to replace reasoning. Distinguish disclosed facts from calculations, inferences, and triangulations. State when information is unavailable, disclose contradictory evidence, and avoid false precision. For every acquired company, clearly identify the last reliable standalone reporting period and the reporting date to which each financial fact applies. Do not present parent-company or post-acquisition figures as standalone company figures unless attribution is supportable.

Required report order and completion standard

Deliver the report in this order:

1. Scope, analysis date, and concise methodology
2. Reliant Rental identity-resolution finding based on transaction context
3. Seven individual company profiles, one each for PODS, Mobile Mini, Trench Plate Rental, General Finance Corporation, Modulaire Group, McGrath RentCorp, and Reliant Rental
4. Company classification and archetype synthesis
5. Archetype benchmark tables and any defensible combined table
6. Final Specialty Rental conclusion
7. Concise source list organized by company

Final completion checklist

Before submitting the report, silently verify and correct it against every requirement below:

1. Every designated company-PODS, Mobile Mini, Trench Plate Rental, General Finance Corporation, Modulaire Group, McGrath RentCorp, and Reliant Rental-has a complete individual profile.
2. No company from another subvertical has been added to the research universe or benchmark population.
3. Every common role, research-objective, analytical-method, first-principles company-profile, core-metric, company-output, synthesis, source, and completion instruction in this prompt has been followed.
4. All Specialty Rental special instructions have been followed, including the Reliant Rental transaction-context identity gate, the required asset and model distinctions, and the acquired-company standalone-period requirement.
5. The report is fully self-contained and does not rely on another conversation or an unavailable file.
6. The report contains no unexplained placeholders; unavailable evidence is labeled and explained rather than left blank.
7. All seven individual company profiles are complete before any subvertical synthesis or benchmark is developed.
8. Every metric includes root-cause analysis based on the actual customer, asset, contract, operating-cost, or capital mechanism.
9. Every metric contains normalized low, central, and high estimates under normal operating conditions, or an explicit Unavailable conclusion with the missing evidence identified.
10. Disclosed facts, calculations, inferences, triangulations, and unavailable information are clearly distinguished using the required Evidence Status labels.
11. Organic growth is separated from acquisitions and other inorganic growth contributions.
12. Maintenance capex is separated from growth capex, and gross mobile-fleet replacement spending is

distinguished from net spending after ordinary disposal proceeds.
13. Initial contract tenor, renewal term, effective customer tenure, and expected customer life are analyzed separately.
14. Asset utilization, customer usage, and service-capacity utilization are defined according to each company's actual operating model, including physical and dollar utilization where available.
15. Payback uses fully burdened initial investment divided by annual unit-level cash contribution, not revenue.
16. Portable storage, mobile storage, modular space, temporary offices and classrooms, trench-safety equipment, and other specialty-rental archetypes are identified and analyzed separately where their economics differ.
17. No simple average is used unless equal weighting is demonstrably appropriate, and no materially different operating model is forced into a blended benchmark.
18. The findings are not compared with CRA or Compactor Rentals of America in any form.
19. Every load-bearing factual assertion is cited under the required source hierarchy and evidence discipline.
20. The complete Specialty Rental report-including every company profile, all material archetype syntheses and benchmark tables, and the final conclusion-is finished in this single conversation.

The report is complete only when every item above is satisfied. Complete the full report now without requesting clarification.

07 Quality Control Checklist

research_prompts/07_quality_control_checklist.md

SHA-256: ee31d929caaaf5eb419fb7d6e1b624bde0fa747f36f369d1671e00b638da661f

Research Prompt Quality-Control Checklist

Analysis date checked: **July 18, 2026**

This checklist is the final release gate for the six independent research prompts. A prompt may be marked **Ready** only after every applicable item below has been checked in the generated file itself. Any issue found during review must be corrected before release.

Prompt scope and identity matrix

Prompt	Required title	Required company universe	Required exclusions / identity treatment	Review status
`01_home_comfort_prompt.md`	Operating Benchmark - Home Comfort	Reliance Home Comfort; Enercare	Vista excluded	Ready / Passed
`02_point_of_use_water_prompt.md`	Operating Benchmark - Point-of-Use Water	AquaVenture Holdings / Quench; Primo Water	Waterlogic excluded; Quench isolated from AquaVenture's unrelated operations	Ready / Passed
`03_submetering_prompt.md`	Operating Benchmark - Submetering	Energy Assets; ista; Techem; Smart Metering Systems; Calisen; CARMA	Meter economics isolated from unrelated energy-transition activities	Ready / Passed
`04_liquid_environmental_services_prompt.md`	Operating Benchmark - Liquid Environmental Services	Restaurant Technologies; Liquid Environmental Solutions; Waste Resources Management; Wind River Environmental	Installed onsite-system and route/collection/treatment/processing archetypes kept distinct	Ready / Passed
`05_transportation_solutions_prompt.md`	Operating Benchmark - Transportation Solutions	VTG; Akiem; Air Rail; TCR; Lazer Logistics	Exact Air Rail legal entity must be resolved before analysis; three archetypes required	Ready / Passed
`06_specialty_rental_prompt.md`	Operating Benchmark - Specialty Rental	PODS; Mobile Mini; Trench Plate Rental; General Finance Corporation; Modulaire Group; McGrath RentCorp; Reliant Rental	Exact Reliant Rental entity must be resolved from relevant transaction context; acquired-company data periods disclosed	Ready / Passed

Per-prompt release checks

Apply every row to each of the six prompt files.

#	Release check	Home Comfort	Point-of-Use Water	Submetering	Liquid Environmental Services	Transportation Solutions	Specialty Rental
1	Correct file name	?	?	?	?	?	?
2	Correct new-chat title	?	?	?	?	?	?
3	Correct and complete company list	?	?	?	?	?	?
4	Correct subvertical-specific instructions	?	?	?	?	?	?
5	Vista is excluded from every research universe, profile, and benchmark	?	?	?	?	?	?
6	Waterlogic is excluded from every research universe, profile, and benchmark	?	?	?	?	?	?
7	Comparison with CRA or Compactor Rentals of America is expressly prohibited	?	?	?	?	?	?
8	No unresolved drafting placeholders	?	?	?	?	?	?
9	No dependency on external local files or prior conversations	?	?	?	?	?	?
10	Complete definitions and treatment for all nine core metrics	?	?	?	?	?	?
11	Complete first-principles company-profile requirements	?	?	?	?	?	?
12	Complete subvertical-synthesis and benchmark requirements	?	?	?	?	?	?
13	Explicit observed-facts -> mechanism -> normalized-assumption -> sensitivities -> confidence method	?	?	?	?	?	?
14	Explicit citation hierarchy and evidence-status rules	?	?	?	?	?	?
15	Explicit instruction to profile every listed company before synthesis	?	?	?	?	?	?
16	Explicit instruction to complete the full report in the current chat without clarification	?	?	?	?	?	?
17	Low, central, and high estimates required	?	?	?	?	?	?
18	Organic growth separated from acquisitions	?	?	?	?	?	?
19	Maintenance capex separated from growth capex	?	?	?	?	?	?
20	Contract tenor separated from effective customer tenure and expected customer life	?	?	?	?	?	?
21	Asset, customer-usage, and service-capacity utilization defined for the actual model	?	?	?	?	?	?
22	Payback uses fully burdened investment and unit-level cash contribution, not revenue	?	?	?	?	?	?
23	Distinct archetypes required where economics differ	?	?	?	?	?	?
24	Unjustified simple averages prohibited	?	?	?	?	?	?
25	Facts, calculations, inferences, triangulations, and unavailable information distinguished	?	?	?	?	?	?
26	Current analysis date is stated as July 18, 2026	?	?	?	?	?	?

Content checks for every company profile

- [x] Resolve legal and operating identity, names, ownership, transactions, geography, relevant segment, and latest reliable standalone reporting period.
- [x] Define and consistently use the economic unit.
- [x] Explain the customer problem, essentiality, demand type, and demand drivers.
- [x] Explain purchase, rental, insourcing, manual, deferral, elimination, and switching alternatives as applicable.
- [x] Map recurring and transactional revenue separately.
- [x] Map ongoing service and operating obligations.

- [x] Explain retention and churn mechanisms.
- [x] Map upfront investment, ownership, useful life, replacement, refurbishment, redeployment, residual value, and renewal dependence.
- [x] Include the required nine-row company metric table with Low, Central, High, Evidence Status, Confidence, and Root Business Driver columns.
- [x] Follow each metric with Conclusion, Observed facts, Root business mechanism, Calculation or driver bridge, Sensitivities, and Confidence.
- [x] End with a plain-English description, strengths, risks, unavailable information, and diligence questions.

Synthesis checks for every prompt

- [x] Identify operating archetypes before creating a benchmark.
- [x] Classify every company as Core representative, Secondary reference, Mixed-business company, Outlier, or Insufficient disclosure.
- [x] Include the required nine-row subvertical benchmark table.
- [x] Explain support, mechanisms, outliers, structural versus temporary differences, and uncertainty for every benchmark metric.
- [x] Include all 12 required final conclusions, including the 25-30-word slide-ready description.
- [x] Use no simple average unless equal weighting is demonstrably appropriate.

Identity-resolution gate

- [x] `05_transportation_solutions_prompt.md` requires the analyst to resolve the exact legal and operating entity represented by **Air Rail before any company analysis**, document the evidence, and avoid silently substituting a similarly named company.
- [x] `06_specialty_rental_prompt.md` requires the analyst to resolve the exact legal and operating entity represented by **Reliant Rental using the relevant transaction context before any company analysis**, document the evidence, and preserve uncertainty if public evidence does not support a unique conclusion.

Final disposition

Ready / Passed. All six prompt files were read end to end and independently cross-reviewed. An automated scope and shared-requirement scan also passed. One literal common-list omission found during cross-review-"meter replacements, where applicable" in the Liquid Environmental Services maintenance-capex examples-was corrected before release.

Air Rail and Reliant Rental remain intentionally unresolved at prompt-preparation stage because no underlying company research was performed. Their prompts convert those ambiguities into mandatory, evidence-based identity-resolution gates that must be completed before company analysis begins.

ChatGPT

Operating Benchmark Study for Home Comfort

Scope, analysis date, and concise methodology

This report covers only the designated Home Comfort research universe: **Reliance Home Comfort** and **Enercare**. The analysis date is **July 18, 2026**. The objective is to isolate the recurring installed-equipment and home-services economics inside each company, separate them from one-time equipment sales where possible, and then form a defensible operating benchmark for the subvertical without using any outside benchmark population. For acquired or diversified businesses, I use the **latest reliable standalone period** for segment-level economics and explicitly label where current public disclosure becomes thin. [1]

The analytical method is first-principles rather than management-guided. For each company and metric, I start with observed company-specific facts, identify the operating mechanism that actually drives the result, normalize into low/central/high ranges, note the principal sensitivities, and assign a confidence level. Where direct disclosure is unavailable, I use clearly labeled triangulation from contract terms, portfolio continuity, equipment life, current buyout schedules, price cards, and ownership materials. Estimates are therefore **analytical outputs**, not management guidance. [2]

Two guardrails matter throughout. First, **contract tenor is not the same as customer tenure**. In Ontario home comfort, observed longevity often comes from residential inertia, home-sale transfer, and replacement convenience—not just legal enforcement. Second, **maintenance capex is not total capex**: in this model it is the sustaining physical spend required to preserve the installed revenue base, including replacement equipment and associated removal/reinstallation, while growth-originations and tuck-in acquisitions are excluded unless they are economically necessary to offset attrition. [3]

Reliance Home Comfort individual company profile

Resolve the company and segment

The current operating business is **Reliance Home Comfort**, and the legal operating entity identified in official company reporting is **Reliance Comfort Limited Partnership**. Owner reporting describes it as a Canadian home-services business offering sale and rental of water heaters, HVAC equipment, water purification, plumbing, electrical, smart-home solutions, and comfort protection plans. CK Infrastructure disclosed that it acquired Reliance in 2017 together with CK Asset at an acquisition enterprise value of about **C$4.6 billion**, and CKI's annual reports continue to disclose a **25% ownership interest**. Public materials reviewed here do **not** fully disclose the current cap table beyond CKI's stake, so the precise present-day ownership split outside CKI remains an unresolved public-data gap. [4]

The relevant reporting boundary is the **Home Comfort / residential home-services operation**, not all broader affiliated infrastructure activities. CKI's business profiles classify Reliance within its "contracted infrastructure" bucket and describe the business as serving **over 2 million customers**, primarily in Ontario, while 2024 project materials show operations in Ontario, Manitoba, Saskatchewan, Alberta, British

Columbia, Georgia, and Florida. CKI’s 2025 results materials further state that Reliance completed **two U.S. home-services acquisitions** in 2025. Because no full standalone Reliance financial statements were located in the public sources reviewed, the **latest reliable standalone operating snapshot** is a combination of current company contract/pricing pages and 2024–2026 owner presentations, not a complete standalone income statement. That lowers confidence for margin and capex precision. [5]

Relevant transaction history matters because it explains today's installed base and retention posture. Reliance was already a major Ontario water-heater rental operator before the 2017 acquisition. In 2014, the Canadian Competition Bureau cleared Reliance’s acquisition of National Energy Corporation but only after obtaining commitments and a consent agreement aimed at stopping anti-competitive return practices and reducing switching friction in Ontario. That is operationally important: part of historical tenure in this market was created by process friction, which means raw longevity statistics should not be naively interpreted as pure contractual protection. [6]

Define the economic unit

For recurring rental economics, the best economic unit is **one installed asset at one residential service address**, usually a water heater or HVAC system connected to an active billing account. That unit best captures acquisition spending, installation cost, monthly recurring revenue, service obligations, replacement requirements, and churn behavior. [7]

A second, distinct economic unit exists for **protection-plan-only customers**, because those contracts produce recurring revenue without provider-owned heavy equipment. A third unit exists for **one-time replacement or service jobs** such as purchased HVAC systems, plumbing, and electrical work. I do **not** blend those units without explanation. The recurring-rental benchmark is anchored on the installed-asset unit; protection plans are an adjacent recurring service layer; and one-time equipment sales are transactional revenue that should not drive the recurring benchmark. [8]

Explain the customer problem

The customer problem is simple and highly non-discretionary: the household needs functioning **hot water, heating, cooling, and related home systems**, and failures are urgent rather than optional. The customer can tolerate deferring a cosmetic renovation, but not a winter furnace failure or a dead water heater. Reliance’s current marketing positions the product explicitly around predictable monthly cost, same-day service, included repairs, and full replacement if the unit cannot be repaired. That indicates the problem being solved is not just equipment purchase; it is **outsourced reliability plus budget smoothing**. [9]

Demand is therefore driven by a combination of essentiality and convenience. Water heater rental pages frame the offer as “service, not a product,” with standard installation, repairs, labor, and replacement included, while current educational materials emphasize that Reliance’s minimum rental term is **7 to 10 years** rather than a long amortizing consumer loan and stress that the homeowner avoids surprise repair bills. The need is permanent at the household level, but service events are irregular, which is exactly what makes the rental-and-service bundle attractive: the provider absorbs lumpy maintenance risk while the customer pays a smooth monthly fee. [10]

Explain customer alternatives

Customers can buy equipment outright, finance it, use another rental provider, or rely on independent local contractors for repair and installation. Reliance itself openly presents “finance, rent or buy” as alternatives, which is useful evidence that the primary competition is not only other rental companies but also the purchase pathway. [11]

The reason customers initially choose Reliance is usually one of three things: no upfront cash outlay, outsourced repair-and-replacement risk, or the convenience of a single provider that installs, services, and bills the system over time. The reason they remain is more nuanced. Some retention comes from legal terms and buyout obligations, but a meaningful portion comes from physical installation, hassle of switching, uncertainty over future repair costs, and the fact that home systems are low-attention products most of the time. That distinction matters because the Competition Bureau’s 2014 action showed that this market can generate apparent stickiness through switching friction that is not equivalent to durable lawful contractual protection. [12]

Map the revenue model

Reliance’s current revenue model has three layers. The first is **recurring equipment rental**, led by water heaters and extended into HVAC, water treatment, and some adjacent equipment. The second is **recurring protection-plan revenue**. The third is **transactional revenue** from equipment sales, installations, plumbing, electrical, and other service work. CKI’s current business descriptions and Reliance’s website both show this broader mix clearly. [13]

Within rental economics, water heaters remain the most observable base. Reliance’s 2026 Ontario rental page shows monthly water-heater rental prices ranging roughly from **about C$20 per month for smaller electric tanks to just over C$100 per month for high-efficiency or tankless gas models**, with standard installation included. That directly supports two important analytical conclusions: first, water-heater rentals are recurring monthly monetization of an installed asset; second, unit economics vary materially by equipment type, so a blended company-level average must account for mix. [14]

Transaction revenue should be explicitly separated from recurring revenue. Reliance’s current portfolio now includes plumbing, electrical, smart-home offerings, and outright equipment sales, especially as the company has broadened beyond its historical water-heater core. Those categories can lift reported growth while diluting recurring-margin purity. For benchmark purposes, the recurring Home Comfort core is therefore rental plus protection plans, while one-time sales and one-off work are treated as adjacent but economically different. [15]

Map the operating model

After customer acquisition, Reliance must keep doing four things: bill accurately, maintain a dense field-service network, respond to failures, and replace equipment when repair is uneconomic. The current website explicitly promises 24/7/365 support, same-day availability, repairs, and full replacement if needed. That means the operating model is not “install and forget”; it is an installed-base service platform. [9]

The recurring service burden differs by category. Water heaters are relatively low-touch: they do not require the same routine maintenance cadence as HVAC systems. HVAC rentals, by contrast, pull more field labor because preventive maintenance and seasonal service are more common. Protection plans are labor-light until a service event occurs, but they still require call-center, dispatch, and technician capacity. Installation-only activities include initial setup and replacement installs. Predictable recurring work includes billing and some scheduled maintenance. Event-driven work includes emergency repairs and end-of-life replacements. The core economic advantage only appears when those obligations are spread over a very large installed base and a dense technician footprint. [16]

Explain customer retention and churn

Retention at Reliance is produced by a mix of **installation-based friction, convenience, service dependence, and contract structure**. The current water-heater offer emphasizes that after the initial term the agreement continues **month to month**, that rental includes installation, parts, labor, and replacement, and that the customer is paying for dependable hot water rather than buying a tank. Those design choices reduce voluntary churn because leaving the platform means the household must source equipment, installation, repair responsibility, and disposal on its own. [17]

But it would be analytically wrong to treat all observed retention as enforceable contractual protection. The Competition Bureau found that Reliance had used return policies and procedures that impeded switching and required changes to those practices. Current Reliance materials still require review of agreement-specific termination provisions and, in some cases, the use of prescribed return processes and agency documentation. So the right conclusion is that **revenue durability is real, but its drivers are mixed**: residential inertia and installed-equipment convenience are major drivers, while legal restrictions matter at the margin and vary by contract cohort. [18]

The principal churn vectors are household moves, buyouts, provider switching, dissatisfaction with service, repricing, outright equipment purchase, and technology migration. Yet the installed revenue stream can outlast any one homeowner because agreements can transfer with the home, and because an end-of-life replacement often resets the provider relationship even though the physical unit has changed. That is why customer tenure and equipment life must be modeled separately. [19]

Map the capital model

Reliance owns the rented equipment and pays for installation upfront. Current contract and product pages show standard installation included, provider ownership retained, and full replacement provided if the unit cannot be repaired. Water heaters are explicitly described as lasting **14 to 16 years**, and current educational materials say minimum rental terms are **7 to 10 years**. Put together, those facts imply a capital model in which the provider funds the asset, installation, and service wrapper and expects to earn back the investment over a multiyear billing stream rather than at the moment of installation. [20]

Redeployment value exists in theory because the provider owns the asset, but practical redeployment is low. Once a heater or HVAC unit is customized to a home, removed, and aged, most of the economic value is no longer in resale; it is in continued billing while the system remains installed and functioning. The real residual value is the **customer relationship at the address**, not the used hardware. That means the return on the initial investment often relies on either surviving beyond the first contract term, transferring with the home, or replacing the equipment and continuing the relationship. [21]

Operating benchmark table

The following table is derived from the company-specific operating facts and the first-principles analysis that follows.

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Normalized revenue growth	3%	5%	8%	Inferred	Medium	Mature installed base; price/mix and cross-sell drive more growth than net new households
Normalized EBITDA margin	43%	49%	55%	Triangulated	Low	High recurring gross margin offset by field labor and broader transactional mix
Maintenance capex % of revenue	9%	12%	15%	Triangulated	Low	Provider-owned equipment replacement plus removal/reinstallation to preserve installed revenue
Post-maintenance-capex conversion	68%	76%	84%	Calculated	Low	Strong recurring EBITDA partly offset by sustaining physical replacement burden

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Initial contract tenor	At-will legacy / non-term	7–10 years	10 years then month-to-month	Triangulated	Medium	Mix of legacy non-term water-heater forms and newer minimum-term rentals
Effective customer tenure	10 years	16 years	25 years	Inferred	Low	Residential inertia, home-sale transfer, and replacement convenience exceed stated term
Utilization exposure	Low	Low-Moderate	Moderate	Inferred	Medium	Fixed monthly billing on installed base; biggest utilization risk sits in technician capacity rather than customer usage
New-unit or customer payback	6 years	8 years	11 years	Triangulated	Low	Water-heater units are slower; higher-rent HVAC and premium equipment are faster

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
B2B/B2C profile	Predominantly B2C	Predominantly B2C	Mixed B2B/B2C	Disclosed	High	Revenue is primarily homeowner-facing, with some property-management and commercial adjacency

Metric explanations

Revenue growth

Conclusion. Reliance's normalized revenue growth is best framed at **3% to 8%**, with **5%** as the central assumption under normal operating conditions.

Observed facts. CKI's 2024 project update showed the customer base growing from **1.8 million households at acquisition to 2.1 million customers** by end-2023 and described product expansion from mainly water heaters into HVAC, water purification, plumbing, EV chargers, and more. CKI's 2025 results slides also said Reliance completed **two U.S. home-services acquisitions** in 2025. [22]

Root business mechanism. In a mature installed-base model, growth comes less from explosive customer additions and more from rental-rate increases, mix shift toward higher-revenue HVAC and premium water products, cross-sell into adjacent services, and tuck-in acquisitions. Gross originations can be large without high net growth because much activity merely replaces churned or retired units.

Calculation or driver bridge. The roughly 300,000-customer increase from acquisition to end-2023 implies modest installed-base expansion over six years, or only low-single-digit annual customer growth before any 2025 acquisitions. That supports a central case where **organic** growth is mostly price plus mix, with reported growth occasionally stepping higher when acquisitions are included. A 5% central assumption is therefore more defensible than extrapolating any single reported year.

Sensitivities. Faster heat-pump adoption, successful U.S. tuck-ins, and wider HVAC rental penetration would push growth higher. Housing softness, weaker discretionary replacement demand, or a higher mix of one-time sales rather than rentals would push sustained recurring growth lower.

Confidence. Medium. Direction is clear, but no current standalone revenue bridge or organic/acquired split was publicly available for Reliance. [23]

EBITDA margin

Conclusion. A normalized EBITDA margin of **43% to 55%**, with **49%** central, is most defensible for Reliance's relevant Home Comfort operation.

Observed facts. Public owner materials describe Reliance as a contracted-infrastructure-style recurring home-services business with over 2 million customers, but they do not publish a standalone EBITDA margin. Current marketing makes clear that rental pricing includes standard installation, repairs, labor, and replacement, and that the business now includes broader transactional categories such as plumbing and electrical work. [24]

Root business mechanism. Recurring rental revenue should carry very high gross margins because the equipment is already installed and water-heater service frequency is relatively low. Margin is then reduced by technician labor, dispatch, call-center support, bad debt, advertising, and the drag from one-time sales and service lines that are more labor- and parts-intensive.

Calculation or driver bridge. The central estimate assumes a business mix still anchored in high-margin monthly rentals and plans, but somewhat diluted versus a purer legacy water-heater portfolio because Reliance has broadened into plumbing, electrical, water purification, smart home, green products, and acquired U.S. home-service operations. That mix expansion likely lowers blended margin relative to the pure recurring core even if it lifts revenue growth.

Sensitivities. Margin rises with more rental mix, denser route utilization, and more premium HVAC rentals. Margin falls with a greater share of outright equipment sales, emergency labor intensity, wage inflation, or underutilized technicians in shoulder seasons.

Confidence. Low. The operating logic is strong, but there is no currently disclosed standalone profit bridge for Reliance in the public materials reviewed. [25]

Maintenance capex

Conclusion. Maintenance capex is best normalized at **9% to 15% of revenue**, with **12%** central.

Observed facts. Reliance's water-heater page states that standard installation is included, the provider owns the equipment, the customer receives full replacement if the unit cannot be repaired, and water heaters generally last **14 to 16 years**. Current educational materials indicate minimum rental terms of **7 to 10 years**, and the current product page states **10 years, then month-to-month** for water heaters. Contract forms also show buyout values tied to depreciated fair value and provider ownership of the unit. [26]

Root business mechanism. To preserve existing recurring revenue, Reliance must replace failed or obsolete owned equipment and fund the associated removal and reinstallation. In this model, replacement equipment is sustaining capital, not growth capex, whenever it preserves an existing billable address.

Calculation or driver bridge. A replacement-cycle approach is the only practical one with public data. Current water-heater installed cost ranges on Reliance's site run roughly from **C$1,000 to C$5,500 installed**, depending on product type. A mature portfolio with provider-funded replacement, limited salvage value, and a meaningful HVAC mix logically produces sustaining capital in the low-teens as a percent of total relevant revenue; a central **12%** of revenue is more defensible than a lower single-digit assumption. [14]

Sensitivities. Higher HVAC/heat-pump mix increases maintenance capex intensity because ticket sizes are larger. More protection-plan-only revenue lowers capex intensity. Faster unit turnover, harder water conditions, or code-driven replacement requirements would also increase sustaining needs.

Confidence. Low. There is no direct standalone maintenance-capex disclosure, so this range is triangulated from economics rather than reported accounting. [27]

Post-maintenance-capex conversion

Conclusion. Reported-accounting post-maintenance-capex conversion is best normalized at **68% to 84%**, with **76%** central. Economically, the true sustaining conversion could be lower if portion of originations merely replace attrition.

Observed facts. Reliance's model bundles installation, repairs, and replacement into a recurring monthly payment, which supports strong EBITDA but also clearly embeds a physical replacement obligation. Current contract and marketing materials show that the provider owns the unit and either replaces or buys out equipment based on age and contract terms. [28]

Root business mechanism. Cash conversion looks strong because recurring billing is front-loaded relative to intermittent service events. But it is not perfect because the provider eventually must replace owned hardware to keep the revenue stream alive. In other words, high EBITDA does not eliminate sustaining equipment spend.

Calculation or driver bridge. Using the central EBITDA margin of **49%** and central maintenance capex of **12% of revenue**, post-maintenance-capex conversion is about **76%**. The low and high cases follow directly from the scenario bounds. This is **not** full free-cash-flow conversion because it excludes interest, taxes, working capital, lease principal, and growth capex.

Sensitivities. Higher equipment replacement requirements, more HVAC mix, or more economically sustaining installation spending would push conversion down. More protection-plan revenue and stronger pricing would push it up.

Confidence. Low. The math is straightforward, but the underlying maintenance-capex input is triangulated rather than disclosed. [29]

Contract tenor and customer tenure

Conclusion. Reliance's legal contract tenor is mixed, but the best central assumption for **initial contract tenor** is **7 to 10 years**, while **effective customer tenure** is better thought of as **around 16 years** centrally and can run longer.

Observed facts. Reliance's educational page says its minimum rental term is **7 to 10 years**, while its current water-heater rental page states **10 years, then month-to-month**. At the same time, Reliance hosts a current Ontario **non-term** water-heater agreement and separately a current **84-month term** agreement, proving that multiple contract cohorts exist in the field. [30]

Root business mechanism. Legal terms protect early-period recovery, but longer-lived revenue usually comes from customer inertia, home-sale transfer, and the convenience of keeping the incumbent provider after the first unit reaches end of life. That is why effective tenure often exceeds both the minimum term and the life of a specific heater.

Calculation or driver bridge. A central legal-term assumption of 7–10 years fits current marketing and term documents. A central effective-tenure assumption of 16 years matches the fact that the typical heater lasts 14–16 years and that current materials explicitly move accounts month-to-month after the initial term. The high case assumes the provider maintains the relationship through replacement into a second equipment cycle.

Sensitivities. Tenure rises when service quality is high, transfer processes are smooth, and replacement decisions are easy. It falls if customers buy out older units, if competitive switching increases, or if regulators reduce friction further.

Confidence. Medium for legal term and **low** for effective tenure. The legal structure is observable; the true economic life of the address-level relationship is not directly disclosed. [31]

Utilization exposure

Conclusion. Reliance's utilization exposure is **Low to Low-Moderate**, with **Low-Moderate** central.

Observed facts. Reliance generates fixed monthly revenue from installed equipment and emphasizes that the customer receives support, repairs, and replacement regardless of how intensively the household uses the equipment. Current pages focus more on asset ownership and service response than on metered consumption. [32]

Root business mechanism. Asset utilization risk is mostly binary: the unit is either installed on a paying account or it is not. Household usage intensity does not normally change revenue because rentals are not volume-billed. The more relevant utilization issue is **service-capacity utilization**—technicians, dispatch, and installation crews are busier in heating or cooling peaks than in shoulder periods.

Calculation or driver bridge. The most relevant KPI in this model is net installed units or billable accounts, not gallons heated or compressor runtime. That makes utilization exposure much lower than in a traditional short-term rental fleet, though not zero because churn, buyouts, and seasonal field-labor swings still matter.

Sensitivities. Exposure rises if customer switching accelerates or if labor capacity is poorly matched to seasonal demand. It falls with steady installed-base growth and dense local route coverage.

Confidence. Medium. The qualitative classification follows directly from the contract and service model. [33]

New-unit or customer payback

Conclusion. Reliance's fully burdened new-unit payback is best normalized at **6 to 11 years**, with **8 years** central.

Observed facts. Reliance's water-heater site shows installed purchase costs of roughly **C$1,000 to C$5,500** and monthly rental prices of roughly **C$20 to C$100+**, with standard installation included. Current marketing also indicates that repairs, labor, and replacement are wrapped into the monthly fee. [34]

Root business mechanism. Payback is not driven by revenue alone. The installed unit must cover equipment cost, installation, sales effort, dispatch overhead, direct repair costs, and a reserve for future replacement. Water heaters have lower revenue but also lower recurring service burden. HVAC rentals have much higher monthly revenue and often faster payback despite higher installed cost.

Calculation or driver bridge. A low-end standard water heater can be slow to repay if monthly rent is near the lower end of the range, while premium water heaters, tankless units, and HVAC rentals should repay faster because revenue per unit is much higher. A blended central estimate of about 8 years is consistent with minimum terms near a decade and with the need for the first contract to recover most of the initial investment before month-to-month continuation becomes strong value creation.

Sensitivities. Payback improves with higher-rent equipment, lower acquisition cost, and dense installation routing. It worsens with builder concessions, higher code-compliance installation cost, or more service-intensive product categories.

Confidence. Low. No direct unit-contribution disclosure was located. This is a first-principles triangulation from current pricing, included services, and equipment cost. [34]

B2B versus B2C profile

Conclusion. Reliance is **predominantly B2C**.

Observed facts. CKI describes Reliance as providing home and commercial services, but current business descriptions still say it serves homeowners primarily in Ontario, and current consumer materials clearly target households. Reliance also maintains separate pages for commercial, builder, and property-management channels, confirming adjacency but not a fundamentally B2B revenue base. [35]

Root business mechanism. The installed rental base is mostly household and small-property oriented. Builders and property managers matter as originination channels, but the economic engine is monthly monetization of residential equipment at the dwelling.

Calculation or driver bridge. The central classification is therefore predominantly B2C, with the high-case scenario acknowledging some mixed exposure where the contract counterparty may be a builder or property manager even though the end use is residential.

Sensitivities. A more aggressive commercial expansion would move the business toward mixed B2B/B2C, but present evidence does not justify that as the central classification.

Confidence. High. The customer-facing positioning is explicit and consistent across sources. [35]

Profile conclusion

Plain-English business description. Reliance is a large installed-base home-comfort company that monetizes household hot-water and HVAC needs through monthly equipment rentals, protection plans, and adjacent service and replacement work. [36]

Principal business strengths. The biggest strengths are a very large installed customer base, recurring monthly billing, strong route-density potential, broad product adjacency, and a service model that solves urgent household failures without upfront customer cash. Those features support durable revenue and high underlying gross margins. [37]

Principal risks. The main risks are regulatory scrutiny over switching friction, limited public disclosure, integration risk from broader acquisitions, labor intensity in field service, and the possibility that outright purchase or financing options reduce rental penetration over time. [38]

Key unavailable information. No current standalone revenue, EBITDA, maintenance capex, customer-churn, or organic-versus-acquired-growth disclosure was found for Reliance. The exact current ownership split beyond CKI's 25% stake was also not fully disclosed in the public materials reviewed. [39]

Most important diligence questions. The most important diligence items are the current revenue split between rentals, protection plans, and transactional sales; water-heater versus HVAC mix; annual gross originations versus attrition; replacement exchange volume; direct service cost by product type; and the true legal-term mix between legacy non-term water-heater contracts and newer minimum-term rental forms.

Enercare individual company profile

Resolve the company and segment

The current corporate entity is **Enercare Inc.**, headquartered in Markham, Ontario. Its current official reporting under Canadian supply-chain legislation lists subsidiaries including **SE Canada Inc., Enercare Home and Commercial Services Inc., Enercare Home and Commercial Services Limited Partnership, Enercare HS Portfolio Limited Partnership, and HydroSolution, L.P.** Current company pages say Enercare serves **1.1 million customers throughout Ontario**, while the 2023 statutory report says Enercare provides heating, cooling, plumbing, electrical, and water-related products and services across Canada, including Ontario, Quebec, Alberta, Manitoba, Saskatchewan, and British Columbia. [40]

For transaction history, Brookfield Infrastructure announced in August 2018 that it would acquire Enercare in a transaction valued at **C$4.3 billion including debt**, and an official October 2018 release states the acquisition **closed on October 16, 2018**, Enercare shares were to be delisted, and **Enercare Solutions Inc.** would remain a reporting issuer because of its outstanding public debt. That is the key ownership event. Current operating pages do not separately disclose a later completed ownership change in the sources reviewed. [41]

The analytically relevant business for this benchmark is **Enercare Home Services**, not the whole historical public company. Historically, the public Enercare group also included Service Experts and sub-metering. The latest reliable standalone segment disclosure for Home Comfort recurring economics is therefore the **Enercare Home Services segment** in the 2017 AIF and the **nine months ended September 30, 2018** MD&A, which separately disclosed segment revenue, gross margin, EBITDA, and unit continuity. I exclude Service Experts and sub-metering from the benchmark logic except where needed to explain corporate boundaries. [42]

Define the economic unit

For recurring rental economics, the proper economic unit is **one installed rental unit at one customer location**. For Enercare, that unit was overwhelmingly a residential water heater: the 2017 AIF says about **95% of the Rental Portfolio** consisted of residential water heaters. [43]

A second economic unit is **one protection-plan contract**, because Enercare separately disclosed about **552,000** residential and commercial protection-plan customers and showed materially different continuity behavior for that portfolio. A third economic unit is the **one-time HVAC sale/install job**. Enercare Home Services disclosed all three streams separately, which is why it can be benchmarked more cleanly than Reliance. [44]

Explain the customer problem

The customer problem is the same core Household Infrastructure problem as Reliance's, but Enercare's legacy disclosure shows the product architecture very clearly. The Home Services business provided rentals of water heaters, HVAC equipment, and water-treatment solutions; it sold protection plans covering furnaces, air conditioners, plumbing, electrical systems, and appliances; and it also sold HVAC systems outright and performed other one-time services. [45]

What the customer is really buying is outsourced continuity of essential home systems plus a predictable payment structure. Current company pages still emphasize that Enercare sells, rents, repairs, and maintains heating, cooling, plumbing, and water systems, and that it offers payment, rental, or financing options. That combination of essentiality and budget smoothing is why this market can generate long-lived recurring revenue despite technically simple physical assets. [46]

Explain customer alternatives

Homeowners can buy equipment outright, rent from a different provider, purchase a protection plan only, or self-manage repair and replacement. Enercare's own current positioning makes those alternatives explicit by offering purchase, rental, financing, and plan products side by side. [47]

Customers initially select Enercare when they want low upfront cash cost, service certainty, or a bundled repair-and-replacement offer. They subsequently stay for three main reasons. First, the equipment is already installed and billing is already set up. Second, there is genuine residential inertia—people do not often revisit water-heater decisions. Third, the company can convert product failures and maintenance relationships into new rentals, sales, or plan renewals. Importantly, historical Enercare disclosure also shows that this retention was not purely contractual: in 2017, **53% of rental water-heater customers were**

under a non-buy-out contract that allowed termination and return, while **47%** were under a buy-out form of contract. [48]

Map the revenue model

Enercare's Home Services revenue model is unusually well disclosed. The 2017 AIF states that Home Services revenue consisted of **Rental Portfolio**, **Protection Plan Portfolio**, **HVAC Sales**, and **Other Services**. In 2017, those lines generated **C$333.3 million**, **C$92.8 million**, **C$22.7 million**, and **C$9.2 million**, respectively. That is the cleanest direct evidence in this report of how the subvertical monetizes the customer relationship. Rentals were about **73%** of Home Services revenue in 2017, protection plans about **20%**, and the rest largely transactional. [49]

The 2018 MD&A strengthens that picture. For the first nine months of 2018, Enercare Home Services generated **C$331.963 million of contracted revenue** and **C$26.591 million of sales and other services revenue**. The MD&A states that contracted revenue represented rentals plus protection plans, while sales and other services mainly related to residential furnaces, boilers, air conditioners, plumbing, and duct cleaning. In other words, the recurring core was still dominant, and one-time work was a much smaller but strategically useful attachment category. [50]

This matters for benchmarking because it shows how to isolate recurring and transactional economics. The recurring core is rental plus protection plans. HVAC sales and "other services" produce useful lead generation and attachment opportunities, but they should not be mistaken for recurring rental revenue. [51]

Map the operating model

Enercare's legacy disclosure is detailed enough to map the operating model directly. As of the 2017 AIF, Enercare Home Services had about **462 service and maintenance technicians**, about **194 independent contractors**, and about **223 installers and helpers** available to it. Call-center and customer-management systems dispatched work based on proximity and availability. [52]

The service burden varies sharply by product. The AIF says the Home Services business **did not conduct periodic scheduled maintenance of installed rental water heaters** in large part because of their reliable performance; by contrast, it **did conduct periodic scheduled maintenance of installed rental HVAC equipment** and certain protection plans. That asymmetry is one of the biggest reasons the water-heater-heavy installed base can sustain such high segment EBITDA margins: water heaters produce recurring revenue without a correspondingly high recurring labor burden. [52]

Installation crews remove, replace, and install water heaters and HVAC equipment, but the company said they generally do **not** install equipment in newly constructed housing, because that is usually done by the builder's HVAC contractor. So Enercare's field model is primarily retrofits, replacements, service calls, and plan-related maintenance spread across a dense Ontario installed base. [52]

Explain customer retention and churn

Retention at Enercare historically came from the same blend seen across this subvertical: installed equipment, emergency-repair dependence, convenience, renewal pathways, and some contractual buyout economics. The 2017 AIF is unusually candid on contract structure: water-heater customers were split between **non-buy-out** contracts, under which customers could terminate and return the unit, and **buy-out** contracts, under which early termination required purchase at a discounted buyout price based on equipment age. All rental HVAC equipment also used the buy-out form. [48]

Observed attrition was very low for rentals and much higher for protection plans. In the first nine months of 2018, rental attrition was **22.0 thousand** units on a starting base of **1,143.4 thousand**, or **1.9%** for nine months, while protection-plan attrition was **51.6 thousand** on a starting base of **552.0 thousand**, or **9.3%** for nine months. That difference is exactly what first principles would predict: installed rented equipment is sticky, while annual service-plan contracts churn materially more. [53]

At the same time, Enercare's own history is a warning against over-legalizing tenure. The Competition Bureau's 2014 fact sheet stated that Enercare, after acquiring Direct Energy's water-heater rental business, committed not to continue Direct Energy's anti-competitive return practices and disclosed that Enercare and Enercare Solutions owned roughly **1.1 million water heaters and other assets** rented primarily to residential customers in Ontario. So observed tenure in this market has always reflected both lawful inertia and market-structure frictions; using only contract language would misread the business. [54]

Map the capital model

Enercare's capital model is very clear. The Rental Portfolio consisted of provider-owned water heaters, HVAC equipment, water-treatment solutions, and related assets, with about **95%** of units being residential water heaters. The company purchased equipment from multiple suppliers, used installers and contractors to place units, and retained the replacement obligation when repair was uneconomic. [55]

Several disclosed facts are especially important for sustaining-capex analysis. First, Enercare said that in almost all cases it was **not economically viable to refurbish and re-install used water heaters**. Second, if it was not economical to repair the installed heater or HVAC unit, a **replacement unit under a new rental contract would typically be offered**. Third, current HVAC rental terms explicitly tie buyout values to the original **retail/total installed cost**, and Enercare's 2026 Ontario water-heater buyout table embeds original installed economics in the buyout schedule. This confirms that the business must repeatedly reinvest in equipment to preserve recurring revenue; it cannot rely on meaningful refurbish-and-redeploy economics. [56]

Current HVAC terms also show that the agreement runs until the **useful life** of the equipment unless the customer buys out earlier, and that the useful life ends when repair is no longer commercially reasonable. That is a very revealing contractual form: the legal term is linked to the economics of repair versus replace, which makes sustaining capital inseparable from revenue durability. [57]

Operating benchmark table

The following table is derived from the company-specific operating facts and the first-principles analysis that follows.

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Normalized revenue growth	3%	5%	7%	Calculated	Medium	Mature installed base; growth comes mainly from rental price, mix, and cross-sell rather than large net unit growth
Normalized EBITDA margin	50%	55%	58%	Disclosed	High	Water-heater-heavy recurring revenue with low scheduled service burden
Maintenance capex % of revenue	9%	12%	15%	Triangulated	Medium	Owned rental equipment must be replaced, and used water heaters are not economically redeployed
Post-maintenance-capex conversion	73%	79%	84%	Calculated	Medium	Very high recurring EBITDA offset by sustaining exchange/replacement spend
Initial contract tenor	At-will legacy water heater form	Mixed portfolio centered on ~10 years	Useful-life / 15-year-style buyout economics	Triangulated	Medium	Water-heater cohort mix ranges from legacy permissive contracts to useful-life buyout structures

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Effective customer tenure	12 years	18 years	25 years	Inferred	Medium	Installed-base inertia and transfer/ replacement pathways outlast stated initial terms
Utilization exposure	Low	Low-Moderate	Moderate	Inferred	High	Fixed rental revenue depends more on active accounts and technician density than customer usage intensity
New-unit or customer payback	6 years	8 years	10 years	Triangulated	Medium	Water-heater units are slower; HVAC rentals with 3–5x revenue per unit are faster
B2B/B2C profile	Predominantly B2C	Predominantly B2C	B2B contract with residential end users	Disclosed	High	Installed base is overwhelmingly residential even when builders or property managers originate the account

Metric explanations

Revenue growth

Conclusion. Enercare Home Services supports a normalized revenue growth range of **3% to 7%**, with **5%** central.

Observed facts. Home Services revenue rose from about **C$426.5 million in 2015 to C$457.9 million in 2017**, a roughly **3.6% CAGR**, while rental revenue rose from **C$298.1 million to C$333.3 million**, about **5.7% CAGR**. In the first nine months of 2018, Home Services revenue increased **6% year over year** to **C$359.8**

million. The same MD&A states that the increase was driven by higher contracted revenue and higher sales-and-other-services revenue. [51]

Root business mechanism. Enercare's growth is primarily a price-and-mix story living on top of a mature installed base. The installed base does grow, but not dramatically. Water-heater and protection-plan portfolios are already large and mature, so the biggest durable drivers are rental-rate increases, higher-value HVAC rental mix, and cross-selling from service calls into plans or replacements.

Calculation or driver bridge. The proof is in the 2018 continuity data: rental units rose only from **1,143.4 thousand to 1,150.7 thousand** in nine months, just **0.6%**, while the average monthly rental rate increased from **C$24.35 in 2017 to C$25.50 year-to-date 2018**. The 2018 revenue increase therefore came mostly from price and mix, not explosive unit growth. That is why a 5% central assumption is stronger than assuming unit growth will carry the model. [58]

Sensitivities. Faster HVAC-rental penetration, price increases, and plan cross-sell can push growth toward the high end. Slower housing turnover, lower sales conversion, or a return to weaker rental-rate growth would push it toward the low end.

Confidence. Medium. The historical segment evidence is strong, but current post-privatization segment growth is not publicly disclosed with the same clarity. [59]

EBITDA margin

Conclusion. Enercare Home Services supports a normalized EBITDA margin of **50% to 58%**, with **55%** central.

Observed facts. For the first nine months of 2018, Enercare Home Services reported revenue of **C$359.8 million**, gross margin of **C$284.0 million**, and EBITDA excluding intercompany income of **C$201.4 million**, implying an EBITDA margin of about **56.0%**. Contracted revenue was **C$332.0 million** of the total, or roughly **92%** of Home Services revenue. [50]

Root business mechanism. This business earns very high margin because most revenue comes from already-installed recurring contracts, and because the water-heater-heavy portfolio requires little scheduled maintenance. That produces a very favorable bridge: extremely high gross margin, then moderate SG&A and field-service overhead, then EBITDA.

Calculation or driver bridge. In 9M 2018, gross margin was about **78.9%** of revenue and SG&A about **22.3%** of revenue, leaving the disclosed EBITDA margin around **56%**. I set the central assumption slightly below the point estimate at 55% to reflect the possibility that a broader current product set and more labor-intensive categories dilute the pure 2018 segment economics modestly. [50]

Sensitivities. Margin rises with more rental and plan mix, denser technician routing, and stronger pricing. It falls with more HVAC sales, plumbing, or service-heavy jobs; more warranty/service frequency; or field-labor inflation.

Confidence. High. This is directly supported by granular segment disclosure. [60]

Maintenance capex

Conclusion. Maintenance capex is best normalized at **9% to 15% of revenue**, with **12%** central.

Observed facts. Enercare disclosed **33.6 thousand** rental asset exchanges—units retired and replaced—in the first nine months of 2018, equal to **2.9%** of opening rental units. It also disclosed that used water heaters are almost never economical to refurbish and re-install, and that if repair is uneconomic, the company typically offers a replacement unit under a new rental contract. Current Ontario buyout schedules show original installed values that range from roughly **C$1,800 to over C$4,000** for typical water-heater types, with higher values for premium categories, and HVAC buyout schedules are based on original retail/ total installed cost. [61]

Root business mechanism. This business is capital-light relative to EBITDA, but it is not no-capex. Because the provider owns the equipment, sustaining spend arrives through unit exchanges and end-of-life replacements needed to preserve the revenue stream.

Calculation or driver bridge. Annualizing the 2018 asset-exchange rate implies roughly **4%** of opening rental units were exchanged per year. Applying current installed-value evidence to that exchange cadence produces a sustaining-capex number in the low teens as a percent of Home Services revenue, especially once removal/reinstallation and vehicle/tool support are included. This is also lower than total D&A, which is appropriate because reported amortization contains non-cash and non-sustaining items that should not be treated as maintenance capex. [62]

Sensitivities. More HVAC mix, harder water conditions, stricter codes, or faster exchange frequency increase maintenance capex. More protection-plan revenue or longer-lived premium equipment can reduce it as a percentage of revenue.

Confidence. Medium. Direct maintenance-capex disclosure was unavailable, but the replacement-cycle evidence is unusually strong. [63]

Post-maintenance-capex conversion

Conclusion. Reported-accounting post-maintenance-capex conversion is best normalized at **73% to 84%**, with **79%** central. Economically adjusted sustaining conversion is lower than the accounting view because gross originations partially offset attrition.

Observed facts. Enercare Home Services generated a disclosed EBITDA margin of about **56%** in 9M 2018. During the same period, rental additions were **28.8 thousand** and rental attrition was **22.0 thousand**. That means much of gross installation activity merely preserved the installed base. Asset exchanges added another **33.6 thousand** units retired and replaced. [64]

Root business mechanism. EBITDA is high because recurring billing is strong and many operating costs are spread across the portfolio. But true sustaining cash generation is lower because the business must fund enough new installations to replace attrition and enough exchanges to replace old hardware, even when net growth is modest.

Calculation or driver bridge. Using a 55% central EBITDA margin and a 12% central maintenance-capex estimate gives conversion of about **79%**. The economically adjusted version would be lower because the business must install a large number of "new" units just to stay flat; with 9M 2018 data, attrition already consumed most of gross additions before any real growth occurred. That is why high EBITDA here should not be mistaken for frictionless free-cash-flow conversion. [64]

Sensitivities. Rising exchange frequency or higher acquisition/install spending to replace churn reduces conversion. Better pricing, denser routes, or more plan revenue increases it.

Confidence. Medium. The EBITDA input is disclosed and the math is sound; the maintenance-capex input remains triangulated. [64]

Contract tenor and customer tenure

Conclusion. Enercare's initial legal terms are mixed, but a central economic reading is **around 10 years or useful life for current-style rentals**, with **effective customer tenure around 18 years** centrally and longer in some cohorts.

Observed facts. In 2017, **53% of rental water-heater customers** were under the legacy non-buy-out form, while **47%** were under the buy-out form; all rental HVAC used the buy-out form. Current HVAC rental terms run until the equipment's **useful life**, and the current buyout schedule extends out to **15+ years**, with a residual 5% buyout floor at that age. [65]

Root business mechanism. The legal term varies by cohort, but the economic relationship often survives beyond any one form of contract because households move, equipment is replaced, and the address-level provider relationship persists. Observed low rental attrition in 2018 confirms that reality, but that durability cannot be credited solely to enforceable contracts because a large legacy cohort was cancellable.

Calculation or driver bridge. The low case reflects the legacy non-buy-out water-heater cohort. The central case reflects how the economically relevant mix had already shifted toward buy-out or useful-life-style obligations, especially for HVAC. The central customer-tenure assumption of 18 years is intentionally far below the naive inverse of rental attrition, because raw attrition overstates same-customer life when contracts transfer or reset on replacement.

Sensitivities. Tenure rises with more owner-occupier inertia, smooth transfer processes, and proactive replacement offerings. It falls with more competitive switching, more buyouts, or regulatory changes that reduce exit friction.

Confidence. Medium. The legal structure is well supported, but actual same-household tenure still requires inference. [66]

Utilization exposure

Conclusion. Enercare's utilization exposure is **Low to Low-Moderate**, with **Low-Moderate** central.

Observed facts. Enercare's recurring revenue is overwhelmingly contracted rather than usage-based, and the business does not meter customer consumption for billing within Home Services. The 2018 MD&A and

2017 AIF both focus on units/contracts outstanding, additions, attrition, and exchange volumes—not consumption intensity. [67]

Root business mechanism. Asset utilization is about whether the unit remains installed on a paying account. Customer usage does not materially change revenue. The more relevant utilization variable is service-capacity utilization—how efficiently installers and technicians are deployed through seasonal peaks and troughs.

Calculation or driver bridge. The correct KPI for this model is contract/unit continuity, not equipment usage hours. That makes utilization risk comparatively low, though not zero: poor labor planning or rising attrition can still hurt margin.

Sensitivities. Seasonal HVAC spikes, technician shortages, or weaker route density increase effective utilization risk. A more stable water-heater-heavy mix reduces it.

Confidence. High. The business model and operating KPIs make the classification clear. [68]

New-unit or customer payback

Conclusion. Enercare's new-unit payback is best normalized at **6 to 10 years**, with **8 years** central on a blended basis.

Observed facts. Historical Enercare disclosure said HVAC rental units generated **three to five times** the rental revenue of a traditional water heater. Average monthly rental rates in Enercare Home Services were **C$24.35** in 2017 and **C$25.50** year-to-date 2018. Current Ontario buyout schedules show a new conventional-vent CV50 water heater at about **C$1,904** of buyout value in year one, with higher values for power-vent, direct-vent, tankless, and combi units; current HVAC buyout economics are tied to original retail/total installed cost. [69]

Root business mechanism. Water-heater rentals are relatively cheap to install but also generate lower monthly revenue, so they repay more slowly. HVAC rentals require more capital but can repay faster because monthly revenue is much higher and contracts/buyout structures are more protective. The company-level blended payback lands in the middle because the portfolio is water-heater-heavy but mix is migrating upward.

Calculation or driver bridge. A blended central payback around 8 years is consistent with average water-heater rents in the mid-C$20s per month, with higher-rent HVAC units offsetting slower low-end tank economics. It is also consistent with the idea that the first contract or first equipment life should recover most of the provider's all-in investment before any residual value is realized through continuation or replacement.

Sensitivities. Payback shortens with more HVAC mix, higher monthly rents, and lower acquisition cost. It lengthens with builder concessions, high installation complexity, or rising service-event costs.

Confidence. Medium. The revenue and installed-value evidence is strong, but direct unit-contribution disclosure is unavailable. [69]

B2B versus B2C profile

Conclusion. Enercare is **predominantly B2C**.

Observed facts. The 2017 AIF states that about **95%** of the Rental Portfolio consisted of residential water heaters, and current company pages say Enercare serves **1.1 million customers throughout Ontario** with products and services centered on household heating, cooling, water, plumbing, and plans. Commercial, builder, and property-management offerings exist, but they do not appear to define the installed recurring core. [70]

Root business mechanism. The business is sold into homes, even when builders or property managers originate some accounts. That creates consumer-like churn dynamics, large account counts, low single-account concentration, and substantial dependence on service quality and brand trust rather than procurement bids from large enterprises.

Calculation or driver bridge. The correct classification is predominantly B2C, with the upper case recognizing that some contracts may be signed by builders or multifamily owners serving residential end users.

Sensitivities. Greater commercial emphasis or builder-channel concentration would move the business toward mixed B2B/B2C, but current evidence does not support that as the central case.

Confidence. High. This is directly supported by product mix and customer descriptions. [71]

Profile conclusion

Plain-English business description. Enercare is a highly recurring Ontario-centric home-comfort platform whose core economics come from installed rental water heaters and HVAC, supplemented by protection plans and replacement/service work. [51]

Principal business strengths. Its main strengths are a very large installed base, heavy residential water-heater concentration, widely distributed service infrastructure, strong contracted-revenue mix, and unusually transparent historical segment disclosure that confirms strong EBITDA economics. [72]

Principal risks. The main risks are regulatory scrutiny around the Ontario rental market, the difference between apparent longevity and true legal protection, labor and parts inflation, and the need to keep replacing provider-owned assets to preserve recurring revenue. [73]

Key unavailable information. The biggest public-data gap is current post-privatization segment disclosure comparable to the 2018 MD&A. Current customer counts, revenue mix, and product footprint are available, but full current segment financials are not. [74]

Most important diligence questions. The most important diligence items are current Ontario rental-unit mix by water-heater type and HVAC, current protection-plan continuity, gross additions needed to offset churn, exchange volume by product type, direct service cost per rental cohort, and whether the legacy non-buy-out cohort has materially rolled off.

Operating-archetype assessment and company classifications

The two companies do **not** require fully separate operating archetypes. Both businesses fundamentally monetize the same installed-residential-home-comfort model: provider-owned or provider-supported water-heating and HVAC equipment at household addresses, layered with protection/maintenance plans and attached replacement/service revenue. In both cases, the key economic engine is a dense installed base that turns urgent household infrastructure into monthly recurring payments plus service attachment opportunities. [75]

That said, there is an important **mix axis** inside the shared archetype. Enercare's historical Home Services disclosure reflects a more **water-heater-heavy, Ontario recurring-core archetype** with cleaner segmentation and higher visibility into recurring gross margins. Reliance's current public description reflects a **broader mixed home-services model** that still has the same recurring installed-base core, but with more adjacent one-time services, broader product breadth, and U.S. tuck-ins that can blur pure recurring economics. Those differences widen the benchmark range, but they do not justify forcing two separate benchmark tables from a universe of only two companies. [76]

Enercare is best classified as **Core representative**. The reason is not size; it is disclosure quality and operating purity. The 2017 AIF and 2018 MD&A provide direct evidence on rentals, plans, transactional sales, EBITDA, rates, continuity, and service structure within the relevant Home Services segment. That makes Enercare the best anchor for the recurring installed-equipment/home-services archetype. [77]

Reliance Home Comfort is best classified as **Mixed-business company**. Reliance clearly belongs in the subvertical and is highly relevant, but its currently disclosed business perimeter is broader—water heaters, HVAC, water purification, plumbing, electrical, smart home, green products, and U.S. acquisitions—and current public materials do not provide the same clean recurring-versus-transactional segment bridge. I therefore use Reliance to widen the observed operating range and to test benchmark reasonableness, but I do **not** let it mechanically set the central benchmark where disclosure is thinner. [78]

Home Comfort benchmark table

Because both companies fall within one installed home-comfort archetype, the benchmark below is a **single-archetype benchmark**, but its weighting is not a simple average. The **Core Central** ranges are anchored more heavily to Enercare's clearer recurring Home Services disclosure and then widened where Reliance's broader current operating mix shows structurally different outcomes.

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
Revenue growth	3%	5%	7%	3%–8%	Medium	Price, mix shift to HVAC/ premium units, cross-sell, modest net installed growth, and occasional M&A
EBITDA margin	48%	53%	56%	43%–56%	Medium	Very high recurring gross margin from installed rentals, offset by field labor and transactional mix
Maintenance capex % revenue	10%	12%	14%	9%–15%	Medium	Owned-equipment replacement, removal/ reinstallation, and low redeployment value
Post-maintenance conversion	73%	78%	82%	68%–84%	Medium	Strong EBITDA partly offset by sustaining hardware exchanges and in-practice replacement originations

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
Initial contract tenor	At-will / legacy cohorts	7–10 years	Useful-life / 15-year-style economics	Mixed legacy at-will to useful-life forms	Medium	Cohort mix between legacy water-heater forms and newer buyout / useful-life structures
Effective customer tenure	12 years	18 years	25 years	10–25 years	Medium	Residential inertia, home-sale transfer, and replacement convenience exceed legal term
Utilization exposure	Low	Low-Moderate	Moderate	Low to Moderate	High	Revenue rests on installed accounts; biggest utilization risk is technician capacity, not customer usage
New-unit/ customer payback	6.5 years	8 years	10 years	6–11 years	Medium	Lower-end water heaters repay slowly; HVAC and premium units repay faster
B2B/B2C profile	Predominantly B2C	Predominantly B2C	B2B contract with residential end users	Predominantly B2C to mixed edge cases	High	Households dominate, though builders and property managers can originate contracts

Metric-by-metric benchmark explanations

Revenue growth

The **5% central range** is appropriate because both companies are mature installed-base businesses where net unit growth is modest and revenue growth is primarily driven by pricing and mix rather than explosive customer adds. Enercare directly shows this: revenue rose 6% in 9M 2018 even though rental units barely grew, because average monthly rent and product mix moved up. Reliance's customer-base growth since acquisition was meaningful but still low-single-digit annually before recent U.S. tuck-ins, which points to the same structural conclusion. Reliance supports the high side of the observed range because its broader product footprint and acquisitions can produce higher reported growth, but that should not be mistaken for faster pure organic recurring growth. [79]

EBITDA margin

The **53% central EBITDA margin** is anchored mainly by Enercare's disclosed Home Services margin of roughly 56% in 9M 2018 and then tempered modestly for mix. The economic driver is straightforward: fixed monthly billing on installed assets, especially water heaters, creates very high gross margin because ongoing scheduled service frequency is low. Reliance widens the lower end because its current publicly described business perimeter includes more adjacent one-time categories and broader home services that are likely more labor- and parts-intensive than a pure water-heater-heavy portfolio. Those differences look structural, not just temporary. [80]

Maintenance capex as a percentage of revenue

The **12% central maintenance-capex ratio** reflects the fact that these are recurring businesses with owned installed assets, but they are not asset-light in the sense of requiring little sustaining physical reinvestment. Enercare most clearly supports this through disclosed exchange volumes and its statement that used water heaters are not generally economical to refurbish and reuse. Reliance current contract and pricing materials support the same conclusion through included replacement, installed-cost evidence, and equipment lives of roughly 14–16 years. Companies fall above or below the central range mainly because of mix: more water-heater and protection-plan revenue pushes capex intensity down; more HVAC and premium-product rental pushes it up. [81]

Post-maintenance-capex conversion

The **78% central conversion** is appropriate because EBITDA is genuinely strong, but some of that apparent cash richness must be given back through sustaining replacements. Enercare supports the central range best: its disclosed EBITDA is very high, but its disclosed attrition, additions, and exchanges show that a meaningful share of field investment merely preserves the installed base. Reliance supports a similar conclusion through contract terms that include full replacement and continued provider ownership. Because those mechanics are structural, not cyclical, this metric should remain meaningfully below simple EBITDA-to-cash intuition. The biggest uncertainty is how much "growth" installation spend is economically sustaining in any given year. [82]

Initial contract tenor

The benchmark **central range of 7–10 years** is appropriate for new-style rental contracts, but the broader observed range is explicitly mixed because legacy Ontario water-heater cohorts can be more permissive. Reliance's current public education and water-heater pages support the 7–10-year central case, while Enercare's historical and current documents show that some cohorts were cancellable legacy water-heater forms and others ran through buyout or useful-life economics. The variation is partly structural—different contract vintages and product categories—and not just temporary. [83]

Effective customer tenure

The **18-year central effective tenure** is intentionally longer than the initial legal term and longer than the life of many individual units. That is because the durable economic relationship sits at the address and provider level, not just in the first signed contract or the first installed heater. Enercare's very low rental attrition and Reliance's month-to-month continuation after the initial term both support that direction. But the central estimate is kept below a naive inverse-churn calculation because home sales, customer moves, contract transfers, and replacement resets mean observed revenue continuity is not identical to single-customer tenure. This is one of the report's lower-confidence areas, but the direction of travel is clear. [84]

Utilization exposure

The benchmark classification of **Low-Moderate** utilization exposure is appropriate because neither company is meaningfully exposed to customer usage volume in the way a metered or transactional service would be. Revenue depends far more on whether the equipment remains installed on an active account. The main utilization risk is therefore not the asset's operating hours; it is technician, installer, and dispatch capacity utilization, especially around seasonal spikes in heating and cooling demand. This is structurally different from short-term rental fleets and is one of the clearest conclusions in the report. [85]

New-unit or new-customer payback

The **8-year central payback** is the most defensible blended figure across the subvertical. Enercare provides the better support because it discloses average water-heater rental rates, states that HVAC rentals produce three to five times traditional water-heater revenue, and provides current installed-value and buyout schedules. Reliance's current installed-price and monthly-rent ranges support similar economics. Companies fall above or below the central range mainly because of product mix: low-end tank water heaters are slower, while HVAC and premium equipment are faster. This difference is structural. [86]

B2B versus B2C profile

The benchmark is firmly **predominantly B2C**. Enercare supports this most strongly through the disclosure that 95% of its rental units were residential water heaters, and Reliance supports it through repeated references to homeowners as the primary customer base. Some contracts may be originated via builders, home sellers, property managers, or small commercial channels, which is why the broader range can shade toward "B2B contract with residential end users." But the economic heart of the subvertical remains household-facing. [87]

Final Home Comfort conclusion

Typical customer problem. The typical customer needs uninterrupted hot water, heating, cooling, or related home-system functionality without large upfront cash outlay or surprise repair expense, and wants a provider to solve outages quickly. [88]

Typical revenue model. The subvertical's core revenue model is monthly recurring equipment rental and protection-plan revenue, layered with one-time HVAC replacement sales and ancillary service work. Water heaters still anchor the recurring base; HVAC broadens revenue per account. [89]

Primary growth drivers. The main growth drivers are rate increases, mix shift toward higher-value HVAC and premium equipment, cross-selling from service to plans or replacement, and occasional tuck-in acquisitions. Net installed-base growth is usually modest. [90]

Primary margin drivers. The biggest margin drivers are high recurring gross margins on already-installed assets, especially water heaters, offset by field labor, dispatch, customer support, and lower-margin transactional sales/service categories. [91]

Principal maintenance-capex requirements. Maintenance capex is driven mainly by replacement equipment, removal/reinstallation, and related sustaining field assets required to keep existing addresses billable. It is not mostly discretionary growth spending. [92]

Typical contract and customer duration. New-style legal terms center on roughly 7–10 years or useful-life-style buyout economics, but effective revenue durability usually extends well beyond that because of residential inertia, home-sale transfer, and replacement resets. [83]

Typical utilization dynamics. Asset utilization is mostly binary—installed on a paying account or not. Customer usage intensity has little direct revenue effect. The more important utilization issue is seasonal technician and installer productivity. [85]

Typical new-unit payback. Blended payback is typically around 8 years, slower for basic water heaters and faster for higher-value HVAC rental units. [93]

Typical B2B/B2C profile. The subvertical is predominantly B2C, though builders, property managers, and small commercial channels can originate or intermediate contracts. [87]

Largest source of variation among companies. The largest source of variation is not the basic archetype; it is **revenue mix**—how much of the business is pure recurring water-heater/HVAC rental and protection plans versus broader one-time home services, outright equipment sales, and acquisition-driven expansion. [94]

Lowest-confidence conclusion. The lowest-confidence conclusion is the precise level of maintenance capex and post-maintenance conversion at Reliance, because current standalone public financial disclosure is limited and requires more triangulation than Enercare. [95]

Slide-ready business-model description. Installed residential water-heating and HVAC providers monetize long-lived household relationships through monthly rental and protection-plan fees, plus replacement sales and service, supported by dense technician networks and sustaining capital.

Consolidated source list

Reliance Home Comfort

- CK Infrastructure 2017 annual report and chairman materials for transaction history, acquisition value, early customer count, geographic footprint, and original operating description. 96
- Reliance ESG report for current legal operating identity as Reliance Comfort Limited Partnership and customer/team scale. 97
- CKI 2024 and 2025 annual reports and 2024/2026 investor materials for current customer count, business scope, product expansion, geographic footprint, and 2025 U.S. acquisitions. 98
- Competition Bureau fact sheet and National acquisition release for Ontario switching-friction history and regulatory context. 99
- Reliance current consumer pages and contract forms for pricing, installation inclusion, equipment life, minimum term, month-to-month continuation, non-term and term water-heater agreements, and protection-plan scope. 100

Enercare

- Brookfield official announcement and completion release for 2018 take-private transaction, delisting, and continuing-reporting-issuer status of Enercare Solutions. 41
- Current Enercare website and 2023 statutory report for current legal scope, customer count, subsidiaries, and geographic footprint. 101
- Enercare Solutions 2017 AIF for Home Services business structure, 95% residential water-heater mix, protection-plan count, historical revenue mix, technician/installer model, contract-form mix, and service/maintenance characteristics. 102
- Enercare Solutions 9M 2018 MD&A for segmented Home Services revenue, gross margin, EBITDA, average rental rates, continuity, additions, attrition, and exchange volumes. 103
- Current Enercare HVAC terms and 2026 water-heater buyout table for useful-life structure, buyout economics, and installed-value cross-checks. 104
- Competition Bureau fact sheet for Enercare's commitments following the Direct Energy water-heater acquisition. 54

Synthesis topics

- Cross-company benchmark synthesis draws principally on Enercare's historical Home Services disclosures for segment economics and on Reliance's current contract, pricing, and owner materials for current market structure and mix breadth. 105
- Contract-versus-tenure and lawful-retention analysis is supported by the Competition Bureau's Ontario water-heater materials, Enercare's legacy contract-cohort disclosures, and current Reliance/Enercare contract pages. 106

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1 4 97 https://reliancehomecomfort.com/wp-content/uploads/2023/12/ESG-Annual-Impact-Report-2022-November-21-2022-FINAL-WEB.pdf
https://reliancehomecomfort.com/wp-content/uploads/2023/12/ESG-Annual-Impact-Report-2022-November-21-2022-FINAL-WEB.pdf

2 50 53 58 60 61 62 64 67 68 69 79 80 81 82 84 85 86 90 91 92 93 103 https://filecache.investorroom.com/mr5ircnw_enercare_ir/429/Enercare%20Solutions%20Inc.%20-%20MD%26A%20-%20Q3%202018%20-%20FINAL.PDF
https://filecache.investorroom.com/mr5ircnw_enercare_ir/429/Enercare%20Solutions%20Inc.%20-%20MD%26A%20-%20Q3%202018%20-%20FINAL.PDF

3 12 18 38 54 73 99 106 https://competition-bureau.canada.ca/en/how-we-foster-competition/education-and-outreach/news-releases/competition-bureau-strengthens-competition-ontarios-water-heater-industry-fact-sheet
https://competition-bureau.canada.ca/en/how-we-foster-competition/education-and-outreach/news-releases/competition-bureau-strengthens-competition-ontarios-water-heater-industry-fact-sheet

5 24 35 39 98 https://www.cki.com.hk/english/PDF_file/annualReport/2024/cki_ar24_full.pdf
https://www.cki.com.hk/english/PDF_file/annualReport/2024/cki_ar24_full.pdf

6 https://competition-bureau.canada.ca/en/how-we-foster-competition/education-and-outreach/news-releases/competition-bureau-clears-reliances-acquisition-national
https://competition-bureau.canada.ca/en/how-we-foster-competition/education-and-outreach/news-releases/competition-bureau-clears-reliances-acquisition-national

7 9 10 14 17 20 26 27 28 29 31 32 33 34 https://reliancehomecomfort.com/water-heaters/water-heater-rentals/
https://reliancehomecomfort.com/water-heaters/water-heater-rentals/

8 13 15 25 36 37 78 95 https://www.cki.com.hk/english/PDF_file/annualReport/2025/cki_ar25_full.pdf
https://www.cki.com.hk/english/PDF_file/annualReport/2025/cki_ar25_full.pdf

11 30 83 100 https://reliancehomecomfort.com/learning-centre/be-in-the-know/
https://reliancehomecomfort.com/learning-centre/be-in-the-know/

16 https://reliancehomecomfort.com/wp-content/uploads/2024/10/ON-MB-TCs.pdf
https://reliancehomecomfort.com/wp-content/uploads/2024/10/ON-MB-TCs.pdf

19 21 https://reliancehomecomfort.com/wp-content/uploads/2023/10/Water-Heater-Non-Term-Rental-Agreement-_-ON-_-for-web.pdf
https://reliancehomecomfort.com/wp-content/uploads/2023/10/Water-Heater-Non-Term-Rental-Agreement-_-ON-_-for-web.pdf

22 23 https://www.cki.com.hk/english/PDF_file/ir_presentation/20240320_1.pdf
https://www.cki.com.hk/english/PDF_file/ir_presentation/20240320_1.pdf

40 https://www.enercare.ca/wp-content/uploads/2024/06/Enercare_Annual-Report_2023-Financial-Year.pdf
https://www.enercare.ca/wp-content/uploads/2024/06/Enercare_Annual-Report_2023-Financial-Year.pdf

41 https://bip.brookfield.com/press-releases/bip/enercare-inc-be-acquired-brookfield-infrastructure-c43-billion-transaction
https://bip.brookfield.com/press-releases/bip/enercare-inc-be-acquired-brookfield-infrastructure-c43-billion-transaction

42 43 44 45 48 49 51 52 55 56 63 65 66 70 71 72 75 76 77 87 89 94 102 105 https://filecache.investorroom.com/mr5ircnw_enercare_ir/397/Enercare%20Solutions%20AIF%20-%20%202017%20-%20FINAL.PDF

https://filecache.investorroom.com/mr5ircnw_enercare_ir/397/Enercare%20Solutions%20AIF%20-%20%202017%20-%20FINAL.PDF

46 47 59 74 88 101 https://www.enercare.ca/about

https://www.enercare.ca/about

57 104 https://www.enercare.ca/wp-content/uploads/2023/10/RHVAC-Agreement-SAMPLE-TCs.pdf

https://www.enercare.ca/wp-content/uploads/2023/10/RHVAC-Agreement-SAMPLE-TCs.pdf

96 https://www.cki.com.hk/english/PDF_file/annualReport/2017/cki_ar17_full.pdf

https://www.cki.com.hk/english/PDF_file/annualReport/2017/cki_ar17_full.pdf

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Operating Benchmark Study of Point-of-Use Water

Scope, analysis date, and concise methodology

This report analyzes the Point-of-Use Water subvertical **as of July 18, 2026** and covers only the designated research universe: **AquaVenture Holdings / Quench** and **Primo Water**. For AquaVenture, the relevant operation is **Quench** and this report excludes desalination and unrelated infrastructure. For Primo Water, the report isolates **point-of-use filtration** wherever the evidence permits and explicitly separates that activity from bottled-water delivery, refill, exchange, dispenser sales, and other unrelated revenue streams. Where segment disclosure is incomplete, I state that limitation and widen the normalized ranges rather than forcing false precision. [1]

The method is first-principles rather than precedent-driven. For each company and each metric, I start with observed facts, then identify the operating mechanism, then derive low/central/high normalized assumptions, then test sensitivities, and finally assign confidence. In practice, that means this report does **not** treat acquisition-driven growth as organic growth, does **not** treat contract tenor as customer tenure, and does **not** treat total capex as maintenance capex. Reported figures are used to anchor the analysis, but the normalized conclusions come from the business model: installed assets, service cadence, field labor, route density, filter replacement, sanitation, customer mix, and utilization exposure. [2]

Two methodological limits matter. First, **Quench's last reliable standalone financial period is FY2019**, because AquaVenture was acquired by Culligan in 2020; current official Quench materials are useful for operating facts but not for full standalone financial statements. Second, **Primo Water's point-of-use filtration economics are not separately disclosed after the 2024 transaction that created Primo Brands**, and even before that the company grouped "Water Refill/Water Filtration" in one reported channel. Accordingly, Quench can support a much tighter benchmark range than Primo. [3]

AquaVenture Holdings and Quench

Resolve the company and segment. AquaVenture Holdings Limited was a public company until it agreed to be acquired by Culligan in December 2019; the merger announcement described AquaVenture as having two platforms, Quench and Seven Seas Water, and said Quench had more than **155,000 installed units** across the U.S. and Canada and served more than **55,000 institutional and commercial customers**. AquaVenture's March 2020 earnings release then described Quench as having approximately **160,000 rental units** in its installed asset base. The relevant profile here is therefore **Quench**, not Seven Seas Water, and the latest reliable standalone reporting period for Quench is **FY2019**. Today, Quench's operating identity is **Quench USA, Inc.**, which its official site describes as a **B2B subsidiary of Culligan International Group** focused on workplace drinking-water solutions. [4]

Define the economic unit. The most useful economic unit for Quench is the **installed point-of-use system under paying contract**, because revenue, service labor, filtration economics, maintenance, replacement cadence, and equipment payback all attach most directly to the installed machine. For customer-acquisition payback and tenure, however, the better unit is the **customer location / customer relationship**, because Quench had roughly **160,000 units** and more than **55,000 customers** in 2019, implying about **2.9 units per customer**. That means unit economics and customer economics should not be blurred together: a single customer relationship can support multiple machines, while installation and selling costs are partly shared at the location level. [5]

Explain the customer problem. Quench solves a recurring workplace need: reliable access to better-tasting water without bottled inventory, bottle handling, storage, or delivery scheduling. The current Quench site frames the value proposition as cleaner, safer, better-tasting water for workplaces, and notes that systems connect to the existing water supply, filter at the point of use, and are installed and maintained by Quench's service team. This is usually not a project-based need. It is a standing workplace amenity and, in many settings, a health-and-hospitality expectation. The need is most essential in workplaces with employees, visitors, or patients on-site every day; it is less essential in low-occupancy or highly distributed work environments. [6]

Explain customer alternatives. Customers can buy a system outright, use bottled-water delivery, rely on tap water without treatment, or switch to another point-of-use provider. Quench's own materials emphasize the elimination of bottling, packaging, labeling, transporting, bottle storage, and manual bottle handling; they also emphasize fixed monthly pricing, included maintenance, and included filter service. Those points are economically important. Bottled-water delivery remains a real substitute, but it is operationally heavier: more storage space, more manual handling, and more variable delivered-product cost. Buying the asset is also a substitute, but it transfers maintenance, sanitation, repairs, and replacement risk to the customer. Quench is therefore selected initially for lower hassle and outsourced service, and retained later because removal, replacement, re-plumbing, and retraining create mild but real switching friction. [7]

Map the revenue model. Quench's official operating model is recurring rental/service revenue. The company's AI-grounding facts page states that the primary sales model centers on **renting equipment on a month-to-month basis**, with installation, maintenance, and filter changes included in the monthly fee. The current website separately says the monthly price is fixed for service, maintenance, and filters, while installation fees are quoted separately by site. Historically, Quench also had non-rental revenue: AquaVenture disclosed both rental revenue and product sales margins in 2019, and company commentary attributed growth partly to dealer activities and direct rental. That means the correct point-of-use view is: **core recurring revenue from equipment rental/service**, plus some ancillary product and dealer-type revenue that should not be treated as identical to machine-lease economics. [8]

Map the operating model. After acquisition, Quench must continue to install, sanitize, maintain, repair, and periodically re-filter the installed base. The current site says Quench has **500+ service technicians**, that installation usually takes **an hour or less**, and that technicians handle everything from install to filter replacement. The official Quench facts page adds that preventive maintenance includes a routine **11-point inspection and maintenance service**. This matters because the revenue model is subscription-like, but the cost model is unmistakably physical and field-service-heavy: labor, dispatch, vans, route planning, parts, filters, sanitation, and occasional removals or re-installs. Route density therefore matters. A dense metro installed base lowers travel time, labor utilization drag, and service cost per machine; a sparse footprint does the opposite. [9]

Explain customer retention and churn. Legal contract term and real customer life diverge at Quench. The company's current materials emphasize month-to-month and short-term flexibility, so the legal term can be short; yet installed equipment, trusted service, and multi-unit customer relationships create longer economic lives. Retention mechanisms include physical installation, hassle avoidance, included maintenance, predictable billing, and the fact that the machine already works. Churn mechanisms include office closures, relocations, bankruptcies, headcount reductions, hybrid-work downsizing, competitive rebids, service failures, and deliberate migration back to bottled or owned systems. Because Quench is explicitly a workplace business, office-occupancy shifts matter. Kastle's July 2026 occupancy data show that average office attendance remained well below pre-pandemic baselines and was still strongly midweek-skewed, which means some sites will keep the machine under contract even with lighter draw, but others will resize or cancel when occupancy weakens enough to make the service feel discretionary. [10]

Map the capital model. Quench owns the equipment and rents it to customers; the official Quench facts page explicitly states that the business centers on rented equipment, and the benefits-of-renting page emphasizes service and maintenance within the monthly price. The capital model therefore starts with company-funded equipment, installation labor, fittings/hookup, and customer-acquisition spending. The key distinction is this: **equipment payback** is earned by the installed machine's recurring contribution; **customer-acquisition payback** is earned by the relationship, often across more than one unit. Renewals beyond the initial legal term are therefore economically important even if contracts are short, because the company funds the upfront asset and then monetizes it over time through recurring fees and service density. Redeployment value exists, but it is not frictionless, because removal, refurbishment, and reinstallation consume labor and can require localized inventory matching. [11]

AquaVenture Holdings and Quench metric table

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Normalized revenue growth	4%	8%	12%	Triangulated	Medium	Underpenetrated workplace placements, price, and tuck-in acquisitions offset by office downsizing
Normalized EBITDA margin	24%	28%	31%	Calculated	Medium	Fixed monthly rental/service revenue on a denser installed base, partly offset by field labor and filtration service

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Maintenance capex % of revenue	10%	13%	17%	Inferred	Low	Owned fleet of installed systems, refurbishment/ redeployment, and service vehicles
Post-maintenance-capex conversion	30%	54%	68%	Calculated	Low	Strong EBITDA offset by meaningful sustaining equipment replacement needs
Initial contract tenor	1 month	1–12 months	36 months	Inferred	Medium	Low-friction subscription selling and customer preference for flexibility
Effective customer tenure	3 years	6 years	9 years	Inferred	Medium	Installation stickiness, service continuity, and multi-unit relationships
Utilization exposure	Low customer-usage / Moderate asset	Moderate overall	Moderate-high in weak office markets	Triangulated	Medium	Revenue is mostly fixed-fee, but returned units and poor route density hurt economics
New-unit or customer payback	Equip. 1.8 yrs / CAC 0.6 yr	Equip. 2.7 yrs / CAC 1.0 yr	Equip. 3.8 yrs / CAC 1.5 yrs	Inferred	Low	Moderate installed cost, recurring monthly fees, and direct service contribution

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
B2B/B2C profile	Predominantly B2B	Predominantly B2B	Almost entirely workplace B2B	Disclosed	High	Contracting customer is the business or institution, not the end user

Table note: The ranges above are anchored to Quench's last standalone segment reporting in FY2019, its disclosed installed-unit base and customer base, and its current official operating model of rented point-of-use systems with fixed monthly fees, included filters, and included service. [12]

AquaVenture Holdings and Quench metric explanations

Revenue growth. Conclusion: normalized Quench revenue growth is best framed as **mid-single-digit to low-double-digit**, with **8%** as the most defensible central case. **Observed facts:** Quench revenue grew **45.4%** in 2019, but management said that a large percentage of year-over-year growth came from acquisitions; it also said Quench delivered **13.6% organic growth** in 2019 and completed six acquisitions in 2019 plus two more early in 2020. **Root business mechanism:** growth comes from new customer wins, additional units at existing locations, price, and M&A; unlike project businesses, it does not depend on water throughput. **Calculation or driver bridge:** the central case discounts the unusually acquisition-heavy 2019 reported rate and assumes organic placements plus price can still support high-single-digit growth in normal conditions, while acquisitions can move the reported number higher. **Sensitivities:** office-market health, branch density, sales productivity, small-tuck-in M&A, and hybrid-work downsizing. **Confidence: Medium**, because the 2019 organic disclosure is strong, but current standalone financials are not public. [13]

EBITDA margin. Conclusion: a **28%** normalized EBITDA margin is the best central estimate, with a defensible range of **24% to 31%**. **Observed facts:** Quench reported **$32.4 million** of adjusted EBITDA on **$115.2 million** of revenue in 2019, about **28.1%**. **Root business mechanism:** recurring fixed fees support good contribution margins once units are installed, but margin is pulled down by field labor, routing, sanitation, filter changes, repairs, and overhead. **Calculation or driver bridge:** the reported 2019 margin is the anchor; normalization widens around it based on route density and service intensity. **Sensitivities:** technician productivity, mix between simple coolers and more complex systems, facility costs, and the ratio of recurring rental revenue to ancillary product sales. **Confidence: Medium**, because the reported anchor is sound but today's private-company cost structure is undisclosed. [14]

Maintenance capex. Conclusion: Quench's sustaining physical capital burden is likely **meaningful rather than light**, with **13% of revenue** as the central estimate. **Observed facts:** current Quench materials confirm an owned rental model with included maintenance and filters; 2019 Quench commentary said rental margin pressure partly reflected higher depreciation from additional units placed on lease, which is exactly what one expects in an owned-fleet model. AquaVenture's 2019 consolidated capex rose to **$36.6 million**, but management said the increase was primarily driven by the unrelated AUC desalination operation, so the group figure is not a clean Quench proxy. **Root business mechanism:** Quench must replace aging machines, refurbish and redeploy returned units, refresh service vehicles/tools, and occasionally reinstall at moved sites to preserve existing earning capacity. **Calculation or driver bridge:**

because the legal rental model capitalizes the asset and then monetizes it over multiple years, sustaining capex should track the replacement cycle of the installed base rather than total company capex. **Sensitivities:** real useful life, redeployment success, failure rates, technician treatment of assets, and how much returned equipment is salvageable. **Confidence: Low**, because Quench does not disclose segment maintenance capex. [15]

Post-maintenance-capex conversion. Conclusion: Quench's economically adjusted conversion is likely **moderate, not elite**, with **54%** as the central estimate under the required formula. **Observed facts:** the reported adjusted EBITDA margin anchor is about **28%**, while the sustaining-capex burden appears meaningful in an owned-fleet, field-service model. **Root business mechanism:** recurring billing makes EBITDA look attractive, but sustaining the earning base requires asset replacement and vehicle/service investment. **Calculation or driver bridge:** using the central assumptions,

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produces roughly **54%**. **Sensitivities:** replacement-cycle assumptions dominate the result; if sustaining capex is closer to 10% of revenue, conversion is much higher, and if it is closer to 17%, conversion drops sharply. **Confidence: Low**, because the sustaining-capex estimate is inferred. [16]

Contract tenor and customer tenure. Conclusion: Quench's initial legal term is often **short and flexible**, but effective customer tenure is much longer; the best central economic life assumption is about **6 years**. **Observed facts:** Quench's official facts page says the primary sales model centers on renting equipment on a **month-to-month basis**; current marketing also stresses short-term contracts and contract lengths that suit the customer. **Root business mechanism:** a short legal term reduces sales friction, while installation and service continuity extend the real duration. **Calculation or driver bridge:** the central case assumes many signups begin effectively rolling-monthly or short-term, yet remain in place for multiple years unless occupancy, service quality, or provider choice changes materially. **Sensitivities:** office closures, relocations, competitive rebids, and how mission-critical the amenity feels at a given site. **Confidence: Medium**, because the legal-term signal is clear, but actual churn data are not disclosed. [11]

Utilization exposure. Conclusion: Quench has **moderate overall utilization exposure**. **Observed facts:** Quench's pricing is fixed monthly and includes service, maintenance, and filters; systems remain productive only while installed under contract, and the company runs a large field-service network of **500+ technicians**. **Root business mechanism:** end-user water draw is less important than contract status and route density. A lightly used but still-contracted machine continues to earn revenue; a returned machine does not. **Calculation or driver bridge:** asset utilization is tied to contract retention and redeployment speed; customer usage exposure is low because the fee is not primarily volumetric; service-capacity utilization is meaningful because technicians and routes must stay dense. **Sensitivities:** office occupancy, site closures, regional density, and the cost to recover and redeploy returned units. **Confidence: Medium**, because the contract mechanics are clear even without granular churn disclosure. [17]

New-unit or new-customer payback. Conclusion: Quench's **equipment payback** is likely around **2.7 years** centrally, while **customer-acquisition payback** is shorter, around **1 year**, because one customer often supports more than one machine. **Observed facts:** Quench had ~**160,000 rental units** and more than **55,000 customers** by late 2019; it charges fixed monthly fees with separate installation quotes and includes service, maintenance, and filters in the recurring fee. **Root business mechanism:** the machine generates recurring cash contribution after direct service costs, while the customer relationship can amortize sales and setup cost across multiple units and many years. **Calculation or driver bridge:** the central estimate uses FY2019 revenue per installed unit as an imperfect ceiling, adjusts for ancillary non-rental revenue, and then reserves direct service cost and sustaining-capex allowance before calculating payback. **Sensitivities:** actual installed cost, local labor for hookup, unit mix, service frequency, and true rental revenue per installed machine. **Confidence: Low**, because Quench does not publish installed cost or segment-level direct contribution per unit. [18]

B2B versus B2C profile. Conclusion: Quench is **predominantly B2B**, and in practice almost entirely so for benchmark purposes. **Observed facts:** Quench's official facts page describes it as a **B2B subsidiary** serving commercial industries; the live site says its products are for **workplaces** and directs residential customers to Culligan's separate residential offerings. **Root business mechanism:** the contractual customer is the workplace or institution that buys/leases the service, not the individual who drinks the water. **Calculation or driver bridge:** this means procurement sophistication is higher than in household channels, sales cycles can be longer, and national-account opportunities can create both scale and concentration benefits. **Sensitivities:** public-sector accounts, dealer-channel mix, and any expansion into adjacent channels. **Confidence: High**, because the company states it explicitly. [19]

AquaVenture Holdings and Quench profile conclusion

Plain-English business description. Quench is a workplace-focused point-of-use water service business that installs company-owned filtration dispensers at customer sites and earns recurring fees for the equipment, filters, sanitation, and upkeep. [20]

Principal business strengths. The strongest features are recurring revenue, included maintenance, route-density benefits, and a selling proposition that replaces bottle storage and delivery with fixed-fee on-site filtration. The multi-unit customer pattern also improves customer-acquisition payback relative to single-unit thinking. [21]

Principal risks. The largest risks are office closures or downsizing, hybrid-work effects on workplace amenity spending, route inefficiency in thinner territories, and the fact that recurring EBITDA still sits on top of an owned and serviced asset fleet that must be sustained. [22]

Key unavailable information. Missing items include current standalone revenue, gross margin, EBITDA, installed-unit growth, churn, net revenue retention, refurbishment rates, replacement-cycle data, and segment-specific maintenance capex. [8]

Most important diligence questions. The key diligence asks are: what percentage of the installed base is on truly month-to-month contracts; what is annual gross and net churn by site and by machine; what is the average installed cost by product family; how much of service is preventive versus reactive; and what percentage of returned units are redeployed versus scrapped. These answers would tighten the maintenance-capex and payback estimates materially. [11]

Primo Water

Resolve the company and segment. The designated profile is **Primo Water's point-of-use filtration operation**, but the current public parent is **Primo Brands Corporation**, which the 2024 Annual Report says was formerly **Triton US HoldCo, Inc.** and became the successor issuer to **Primo Water** on **November 8, 2024**. The company's 2026 full-year results release states that 2024 comparable GAAP information reflects a transaction period rather than a clean standalone period, while the 2024 Annual Report states that the company now operates as a **single segment**. The latest reliable full-year standalone period for pre-merger channel detail is therefore **FY2023** under **Primo Water Corporation**. [23]

Define the economic unit. For point-of-use filtration, the best economic unit is the **installed filtration system under service at a home or business site**. That is the unit that drives monthly plan revenue, installation cost, ongoing maintenance, and equipment replacement. For customer-acquisition analysis, the better unit is the **customer account**, because a household or workplace may buy filtration along with other offerings. The major complication is that public filings do **not** isolate filtration as a standalone reporting line; instead, FY2023 revenue grouped **"Water Refill/Water Filtration"** into one combined channel worth **$226.9 million**, up from **$192.0 million** in 2022. That means refill kiosks and point-of-use filtration cannot be cleanly separated from company filings alone. [24]

Explain the customer problem. Primo's filtration pages position the service around better-tasting tap water with low upfront commitment, consultation, installation, and ongoing maintenance included. The filtration pages say plans start at **$27.99 per month**, cover home or office use, and are offered in activated-carbon and reverse-osmosis formats, with floor, countertop, and under-sink models. The problem being solved is therefore similar to Quench's in fundamentals—better water, less hassle, no bottle logistics—but Primo serves both **home and business** consumers, which makes the use case less purely workplace-driven and less purely B2B. [25]

Explain customer alternatives. Customers can buy a dispenser or filtration device, rent from another provider, use bottled delivery, refill/exchange water, or simply consume municipal tap water. Primo itself explicitly presents both **rental** and **purchase** as options for dispensers and highlights "no contracts" as a benefit of purchase. Its services pages also place filtration alongside delivery, exchange, and refill, which means Primo's own channel architecture exposes filtration customers to many adjacent substitutes inside and outside the company. Retention is driven by convenience, installation, and maintenance inclusion; switching risk is higher than at Quench because the company's own ecosystem offers multiple alternative hydration formats and because the service agreement is very flexible. [26]

Map the revenue model. The company's website states that Primo offers direct tap-water filtration systems with monthly plans starting at **$27.99** and that the team handles installation and maintenance. The service-agreement page says bottles, dispensers, brewers, filters, and other equipment remain **Primo's property** and are leased rather than sold. The accounting footnote in the 2023 Annual Report says the company recognizes rental income on filtration, brewer, and dispensing equipment at customer locations. The reporting problem, however, is material: FY2023 public filings disclose a combined **Water Refill/Water Filtration** channel, not a filtration-only channel. So the correct analytical view is that Primo's POU economics are recurring-rental/service economics, but the public filing perimeter materially overstates apples-to-apples comparability because it includes refill. [27]

Map the operating model. Primo's filtration pages describe a process of quote, installation, and then ongoing maintenance/support; they also say select models have **IoT-enabled smart monitoring** that can alert Primo to output, purity, and leak issues and trigger service contact. The commercial-services page shows how the broader direct-service platform supports schools, healthcare, hospitality, and manufacturing, although that page also covers bottled products. The recurring operating activities for filtration specifically are technician installation, preventive maintenance, filter changes, sanitation, repairs, customer support, and billing. Unlike the refill channel, filtration economics depend on installed-base service density rather than kiosk throughput. [28]

Explain customer retention and churn. Primo's service agreement, updated February 6, 2024, says the agreement continues until the customer cancels or Primo terminates it, that the customer may cancel at any time without penalty after the three-day rescission window, and that cancellation becomes effective at the end of the next full billing period. That means the legal term is short and flexible. Effective tenure is therefore driven much more by convenience than by contractual lock-in. Installation and service continuity help, but the absence of punitive term protections means churn should be more sensitive to price, service quality, moving events, and substitute formats than in a heavy-equipment lease with long legal tenure. Compared with Quench, office-occupancy exposure is lower at the enterprise level because Primo's filtration offer explicitly serves homes as well as businesses, but workplace accounts still face the same hybrid-work risk logic as the broader category. [29]

Map the capital model. Primo owns leased customer equipment. Its 2023 Annual Report says customer equipment includes coolers, refill equipment, brewers, refrigerators, water-purification devices, and storage racks at customer locations, and it shows **customer equipment** with useful lives of **2 to 15 years** and net book value of **$235.0 million** at year-end 2023. That is not a filtration-only asset figure, but it confirms an asset-heavy service model with meaningful company-funded in-field equipment. The economic split between equipment payback and customer-acquisition payback is important here too. Equipment payback depends on the installed device and direct service burden; customer-acquisition payback depends on local marketing/sales cost and any cross-sell potential into delivery or other hydration products. [30]

Primo Water metric table

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Normalized revenue growth	1%	5%	8%	Triangulated	Low	Monthly-plan filtration growth, partly offset by easy cancellation and mixed-channel substitution

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Normalized EBITDA margin	12%	18%	24%	Inferred	Low	Recurring rental/ service economics moderated by mixed home/ business service burden and weaker disclosure
Maintenance capex % of revenue	5%	8%	12%	Inferred	Low	Owned customer equipment with 2–15 year lives plus field-service support assets
Post-maintenance-capex conversion	35%	56%	71%	Calculated	Low	EBITDA is supported by recurring revenue, but customer equipment still requires sustaining replacement
Initial contract tenor	1 month	1 month rolling	12 months	Disclosed	Medium	Service agreement is cancellable at any time after the rescission period

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Effective customer tenure	2 years	4 years	6 years	Inferred	Low	Convenience and installation support retention, but contractual lock-in is light
Utilization exposure	Low customer-usage / Moderate asset	Moderate overall	Moderate-to-high in weak service geographies	Triangulated	Low	Revenue is recurring while active, but cancellation flexibility and mixed channel reduce durability
New-unit or customer payback	Equip. 1.5 yrs / CAC 0.8 yr	Equip. 2.5 yrs / CAC 1.3 yrs	Equip. 3.5 yrs / CAC 2.0 yrs	Inferred	Low	Lower monthly-ticket systems, but maintenance and acquisition cost matter materially
B2B/B2C profile	Mixed, skew residential/ small business	Mixed B2B/B2C	Mixed with commercial overlays	Disclosed	Medium	Contracting customer can be a household, small office, or business account

Table note: These ranges are intentionally broad because Primo’s public filings combine **Water Refill/Water Filtration**, while current company materials describe filtration qualitatively and commercially but do not disclose filtration-only revenue, EBITDA, churn, or capex. [31]

Primo Water metric explanations

Revenue growth. Conclusion: the best normalized point-of-use filtration growth assumption for Primo is about **5%**, with low confidence. **Observed facts:** FY2023 disclosed **Water Refill/Water Filtration** revenue of **$226.9 million**, up from **$192.0 million** in 2022, but the company did not disclose filtration-only growth. **Root business mechanism:** filtration grows through monthly-plan adds, home/business consumer awareness, and selective commercial wins, but growth is tempered by easy cancellation and substitution across delivery, refill, exchange, and purchased equipment. **Calculation or driver bridge:** the 18.2% blended channel growth is not a clean filtration proxy because refill is included; the central case therefore discounts the disclosed blended growth materially. **Sensitivities:** share of refill in the channel, digital customer acquisition, competitive pricing, and cross-sell from broader water services. **Confidence: Low**, because the reporting boundary is mixed. [32]

EBITDA margin. Conclusion: a normalized filtration EBITDA margin of **18%** is the most defensible central case, but the range must be broad. **Observed facts:** Primo Water's 2023 filings disclose company-level operating income and channel revenue, but not filtration-only EBITDA; the company also states that it recognizes rental income on filtration equipment at customer locations. **Root business mechanism:** recurring monthly revenue should support decent economics, but field service, maintenance, filter replacement, and mixed home/business support reduce margin relative to software-like subscriptions. **Calculation or driver bridge:** because there is no filtration-only EBITDA disclosure, the central estimate is triangulated from the recurring-rental model, the lease/service agreement, and the mixed-channel reality that likely makes filtration less margin-rich than a pure, dense, commercial-only installed base. **Sensitivities:** actual mix of refill versus filtration, residential service density, and service-call frequency. **Confidence: Low**. [33]

Maintenance capex. Conclusion: the sustaining capital burden for Primo filtration is likely around **8% of revenue** centrally. **Observed facts:** customer equipment had useful lives of **2–15 years** and net book value of **$235.0 million** at the end of 2023, but those assets include coolers, refill equipment, brewers, refrigerators, water-purification devices, and racks. **Root business mechanism:** preserving current filtration revenue still requires replacement of aging devices, selected refurbishment, and service-support assets. **Calculation or driver bridge:** the wide useful-life range implies some light, shorter-life assets and some longer-lived equipment; because refill equipment contaminates the customer-equipment pool, the estimate must remain broad. **Sensitivities:** how much of customer equipment belongs to refill versus filtration, failure rates, and how much growth capex is embedded in the fleet build. **Confidence: Low**, because the segmentation is mixed. [30]

Post-maintenance-capex conversion. Conclusion: Primo filtration likely converts about **56%** of EBITDA after sustaining capex on the central case. **Observed facts:** the company runs an owned-equipment rental model for customer equipment and recognizes rental income on filtration systems at customer locations. **Root business mechanism:** recurring revenue supports EBITDA, but the asset base must still be refreshed over time. **Calculation or driver bridge:** using the required formula and the central assumptions,

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gives about **56%**. **Sensitivities:** the correct filtration-only capex burden is the key swing factor. **Confidence: Low**, because both numerator components are inferred rather than disclosed for filtration alone. [34]

Contract tenor and customer tenure. Conclusion: Primo's legal term is effectively **rolling month-to-month**, while effective customer tenure is probably several years but materially shorter and more fragile than at Quench. **Observed facts:** the service agreement says customers can cancel at any time without penalty after the initial rescission period, with cancellation effective at the end of the next full billing period. **Root business mechanism:** the low-friction contract helps conversion, but it shortens the contractual moat. **Calculation or driver bridge:** the central case therefore uses a one-month legal term and about **4 years** of effective customer life, reflecting convenience retention rather than contract lock-in. **Sensitivities:** moving events, service quality, pricing, and cross-sell stickiness into other products. **Confidence: Low-to-medium** on the legal term, **Low** on actual tenure. [35]

Utilization exposure. Conclusion: Primo filtration has **moderate** overall utilization exposure. **Observed facts:** Primo owns the equipment and rents it; the website markets filtration to both home and business consumers, and the service agreement makes service cancellable with limited friction. **Root business mechanism:** as with Quench, low water draw does not necessarily kill revenue while a machine remains under contract, but cancellation or nonrenewal does. **Calculation or driver bridge:** customer-usage exposure is low, because pricing is not fundamentally volume-driven; asset-utilization exposure is moderate, because returning or losing accounts strands equipment until it is recovered and redeployed; service-capacity utilization matters because mixed residential/business operations can be harder to densify. **Sensitivities:** local route density, residential mix, and competitive substitution from purchase, refill, or delivery. **Confidence: Low**, because no filtration-only churn or route metrics are disclosed. [36]

New-unit or new-customer payback. Conclusion: the central estimate is about **2.5 years** for equipment payback and **1.3 years** for customer-acquisition payback, but confidence is low. **Observed facts:** Primo's filtration plans start at **$27.99/month**, installation and maintenance are included, and the company keeps ownership of the equipment. **Root business mechanism:** lower-ticket equipment can repay fairly quickly if service intensity is light, but easy cancellation pushes the model to recover economics early. **Calculation or driver bridge:** the central range assumes a modest installed cost, recurring monthly contribution after direct service costs, and customer-acquisition spending that is meaningful but not as heavy as enterprise field sales. **Sensitivities:** actual installed-system cost, marketing cost per acquired account, and the share of customers taking filtration only versus broader services. **Confidence: Low**. [37]

B2B versus B2C profile. Conclusion: Primo filtration is **mixed B2B/B2C**. **Observed facts:** the company's filtration pages repeatedly market to **home or office** users, and the broader services pages recommend solutions for small workplaces, local shops, home offices, schools, hospitals, hotels, factories, and job sites. **Root business mechanism:** the contractual customer can be either a household or a workplace, which widens the funnel but changes procurement behavior and churn dynamics. **Calculation or driver bridge:** the mixed profile implies shorter sales cycles for many residential/small-business accounts than Quench, but also lower switching barriers and more fragmented retention drivers. **Sensitivities:** channel mix shifts, price, and whether commercial accounts become a larger share of filtration. **Confidence: Medium** on the mix, **Low** on exact weighting. [38]

Primo Water profile conclusion

Plain-English business description. Primo Water's filtration offering is a monthly subscription service that installs company-owned filtration equipment at home and business sites and bundles maintenance into the recurring fee. [37]

Principal business strengths. The strengths are low-friction monthly pricing, broad consumer awareness from a large hydration platform, home-and-business addressability, and maintenance inclusion that simplifies adoption. [36]

Principal risks. The biggest risks are weak public segmentation, easy cancellation, mixed-channel substitution inside the same company, and difficulty separating filtration economics from refill and other water formats. [39]

Key unavailable information. Missing items include filtration-only revenue, installed units, customer counts, churn, net revenue retention, segment EBITDA, direct service cost, and maintenance capex. [40]

Most important diligence questions. The key asks are: what share of the combined refill/filtration channel is true installed filtration; what is the residential versus commercial split; what is average paid tenure; what is the gross margin after service labor and filters; and what percentage of customer-equipment capex belongs to filtration. Without those answers, Primo can inform the boundary of the benchmark but not determine its core center. [41]

Archetypes and company classifications

The two companies do **not** sit in the subvertical with equal comparability. Quench is a **core workplace point-of-use rental-and-service model**: company-owned assets, fixed monthly fees, included filters and maintenance, field technicians, and clearly workplace-oriented B2B demand. Primo's filtration operation is economically related, but the public disclosure perimeter is fundamentally mixed, because reported revenue combines filtration with refill and the commercial front-end coexists with a significant home-consumer channel. That difference is structural, not cosmetic. Quench is therefore the only **core representative** in this small research universe, while Primo is a useful but bounded **mixed-business reference**. [42]

Quench — Core representative. Quench is the purest expression of the designated subvertical in this universe: workplace point-of-use systems, recurring rental/service economics, included filter replacement, meaningful field service, and explicit B2B positioning. It therefore gets the highest benchmark weight. [20]

Primo Water — Mixed-business company. Primo belongs in the research universe because it clearly offers water filtration units for home and business customers, recognizes rental income on filtration equipment, and has customer equipment deployed at customer sites. But it should not determine the benchmark center because the relevant POU economics are not separately disclosed and the reported channel includes refill. It is most useful as a boundary case that widens the observed range and highlights what happens when point-of-use filtration sits inside a broader hydration platform. [43]

Benchmark tables

Because this research universe contains only **one core representative** and one **mixed-business reference**, the defensible benchmark is a **core point-of-use workplace archetype benchmark** anchored on Quench, with the **Broader Observed Range** column widened to show where Primo's mixed-business evidence sits. This is deliberately **not** a simple average. [44]

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
Revenue growth	4%	8%	12%	1%–18%	Medium	New placements, pricing, net retention, office/site health, acquisition contribution
EBITDA margin	24%	28%	31%	12%–31%	Medium	Recurring fees, field service intensity, route density, product mix
Maintenance capex % revenue	10%	13%	17%	5%–17%	Low	Replacement systems, refurbishment, service vehicles, redeployment losses
Post-maintenance conversion	30%	54%	68%	35%–71%	Low	EBITDA quality versus sustaining asset burden
Initial contract tenor	1 month	1–12 months	36 months	1 month rolling to multiyear	Medium	Sales-friction reduction versus enterprise contracting needs
Effective customer tenure	3 years	6 years	9 years	2–9 years	Medium	Installed-base stickiness, service continuity, switching effort, office closures

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
Utilization exposure	Low customer-usage / Moderate asset	Moderate overall	Moderate-high	Low-to-moderate customer usage; moderate asset/service capacity	Medium	Fixed-fee contracts, risk of returns, route density, office occupancy
New-unit/ customer payback	Equip. 1.8 yrs / CAC 0.6 yr	Equip. 2.7 yrs / CAC 1.0 yr	Equip. 3.8 yrs / CAC 1.5 yrs	Equip. 1.5–3.8 yrs / CAC 0.8–2.0 yrs	Low	Installed cost, recurring contribution, CAC, filter/ service burden
B2B/B2C profile	Predominantly B2B	Predominantly B2B	Almost entirely workplace B2B	Predominantly B2B to mixed B2B/B2C	High	Contractual customer identity and route/service design

Benchmark note: The core benchmark is anchored on Quench, not on a weighted average of Quench and Primo, because Primo's public disclosures do not isolate point-of-use filtration from refill. The broader observed range shows where the mixed-business reference sits, but it is not the benchmark center. [42]

Metric-by-metric benchmark explanations

Revenue growth. The central **8%** core range is appropriate because Quench showed that pure-play workplace POU can produce solid organic growth when placements outpace attrition, but historical reported growth can be inflated by acquisitions. Primo supports the view that a broader hydration platform can post faster channel growth, but its **Water Refill/Water Filtration** disclosure is not filtration-pure, so it belongs in the broader observed range rather than the center. The difference is structural: pure B2B POU growth depends on placements, density, and customer retention; mixed hydration growth can be influenced by refill, retail behavior, and cross-channel effects. The weakest evidence here is current standalone Quench growth, which private ownership obscures. [45]

EBITDA margin. The benchmark center of **28%** is supported by Quench's reported FY2019 adjusted EBITDA margin and by the natural economics of recurring machine rental with dense field service. Primo widens the lower end because a mixed home/business platform with noisier disclosure and broader channel overlap can plausibly earn less on filtration than a dense workplace specialist. Differences above or below the center are mostly structural, not temporary: service intensity, density, and commercial focus matter more than one quarter's pricing. The remaining uncertainty is how much current Quench or Primo cost structure has shifted since the disclosed periods. [46]

Maintenance capex. The benchmark central estimate of **13% of revenue** is intentionally above what a naïve recurring-revenue investor might assume. Owned installed systems, removal/reinstallation, refurbishment, and service vehicles make this a real sustaining-capex model. Primo supports a lower broader bound because lighter-ticket home/business filtration may need less capital than a denser workplace fleet, but that conclusion is low confidence because refill equipment contaminates the public balance-sheet asset pool. Differences here are partly structural and partly data-driven. The lowest-confidence conclusion in the entire study is maintenance capex, especially for Primo. [47]

Post-maintenance conversion. The core central range of about **54%** flows directly from the prior two conclusions: good EBITDA, but meaningful sustaining capital. Quench supports that center. Primo widens the broader range because its likely filtration-only maintenance burden could be lower than Quench's, but the evidence is weak. The critical analytical point is that POU water should not be treated as a capital-light SaaS model just because invoicing is recurring. Where a provider owns and services the installed equipment, sustaining cash conversion is materially lower than EBITDA alone suggests. [48]

Initial contract tenor. The benchmark center is short because the best direct evidence points to **month-to-month or otherwise flexible contracts**, not long hard leases, especially for Quench and Primo's current service agreement. That is a structural feature of the category: low-friction adoption matters. Some enterprise or custom contracts can be longer, which is why the high end extends into multiyear territory. The category's real durability therefore comes less from legal term than from operating stickiness. [49]

Effective customer tenure. The core center of about **6 years** is supported by Quench's installed-service model: plumbing, service continuity, trust, and multi-unit locations create real stickiness beyond the legal term. Primo broadens the lower bound because easy cancellation and mixed B2B/B2C exposure weaken durability. Differences here are partly structural and partly customer-mix driven. Office closures and hybrid-work downsizing are the main reasons the effective-life estimate should not be set too high. [50]

Utilization exposure. The benchmark center is **moderate overall** because three utilization layers must be kept separate. Customer-usage exposure is low: a contracted machine still earns even if cups-per-day fall. Asset utilization is moderate: returned or idle machines stop earning until redeployed. Service-capacity utilization is also moderate because technician productivity and route density determine margin. Quench's workplace concentration makes office occupancy a more direct sensitivity; Primo's mixed home/business exposure softens that risk somewhat, but does not remove it. [51]

New-unit or new-customer payback. The core center of **~2.7 years equipment payback** and **~1 year customer-acquisition payback** is appropriate because the installed machine has meaningful physical cost, while customer-level selling cost is spread across a recurring relationship that can support multiple units. Quench supports the center; Primo broadens the lower and upper bounds because lighter-ticket systems may repay faster, but easier cancellation can demand earlier recovery. The differences are structural: product family, install complexity, service burden, and account size matter more than corporate scale alone. Evidence remains weakest because neither company discloses installed cost or direct unit contribution by product family. [52]

B2B/B2C profile. The benchmark center is **predominantly B2B** because the clearest pure-play representative, Quench, is explicitly a workplace B2B model. The broader observed range extends to **mixed B2B/B2C** because Primo's filtration business serves both homes and offices. That distinction has operating implications: B2B improves density and often lowers small-account churn but can lengthen sales cycles and increase procurement sophistication; mixed B2B/B2C shortens some sales cycles but increases fragmentation and can reduce tenure. This difference is structural, not temporary. [53]

Final point-of-use water conclusion

Typical customer problem. The category solves a simple but recurring problem: a site wants potable, better-tasting drinking water without bottle storage, delivery complexity, or self-managed maintenance. In workplaces, that need is an amenity and productivity issue; in homes and small offices, it is mostly convenience and taste. [54]

Typical revenue model. The core model is company-owned equipment placed at the customer site for a fixed recurring fee that usually includes filters, sanitation, maintenance, and service support. Installation may be billed separately. [55]

Primary growth drivers. Growth comes from new placements, additional units at existing customers, price, and selective acquisitions. In this subvertical, the main constraint is not water consumption but placement velocity, retention, and addressable-site growth. [56]

Primary margin drivers. Margins are driven by route density, technician productivity, service frequency, product mix, and how much ancillary non-rental revenue sits alongside the installed base. A denser workplace specialist generally outperforms a mixed, less isolated channel. [57]

Principal maintenance-capex requirements. The real sustaining burden is replacement of aging systems, refurbishment and redeployment of returned systems, service vehicles/tools, and sometimes removal/reinstallation work. Filters themselves are operating cost, not maintenance capex. [47]

Typical contract and customer duration. Contract terms can be short, but customer lives are usually much longer. The category's durability comes less from legal lock-in and more from installed convenience, service continuity, and customer inertia. [49]

Typical utilization dynamics. End-user draw is not the main utilization variable. The key variables are whether the machine remains under paying contract, how fast returned equipment can be recovered and redeployed, and whether technician routes remain dense enough to absorb periodic service efficiently. [9]

Typical new-unit payback. For the core archetype, equipment payback is usually best thought of in **roughly two-to-four years**, while customer-acquisition payback is shorter because one account can support several machines. [5]

Typical B2B/B2C profile. The benchmark center is **predominantly B2B workplace**, but the broader observed perimeter includes mixed B2B/B2C operators where filtration is bundled inside a larger consumer hydration platform. [58]

Largest source of variation among companies. The biggest source of variation is **operating perimeter**: a pure, workplace POU rental/service company behaves differently from a mixed water platform that reports filtration together with refill, delivery, or retail-adjacent channels. [59]

Lowest-confidence conclusion. The lowest-confidence conclusion is **maintenance capex**, especially for Primo, because neither company discloses point-of-use-specific sustaining capital and Primo's public asset and revenue pools are materially mixed. [60]

Slide-ready business-model description. Point-of-use water providers install and service filtration-equipped dispensers at customer sites, earning recurring fees for equipment, filters, sanitation, and support while trading route density and retention against office exposure. [55]

Consolidated source list

AquaVenture Holdings / Quench. Primary sources used were AquaVenture's December 23, 2019 merger-announcement press release with unit and customer counts; AquaVenture's March 5, 2020 FY2019 earnings release with Quench segment growth, EBITDA, acquisition activity, and installed-unit commentary; Quench's current official site pages covering the homepage, About Us, benefits of renting, FAQs, and company facts pages describing B2B orientation, fixed monthly fees, service inclusion, month-to-month rental orientation, 500+ technicians, and preventive maintenance. [61]

Primo Water / Primo Brands. Primary sources used were Primo Water's FY2023 Annual Report and accompanying notes for channel disclosure, customer-equipment accounting, rental-income policy, useful lives, and customer-equipment composition; Primo Brands' FY2024 Annual Report for the November 8, 2024 transaction and successor-issuer status; Primo Brands' February 26, 2026 full-year results release for current corporate perimeter and 2025 figures; and the current Water.com services, filtration, commercial-services, rental-versus-purchase, dispenser, and service-agreement pages for monthly plan structure, installation/maintenance inclusion, cancellability, and equipment ownership. [62]

Synthesis topics. For office-occupancy exposure, I used Kastle Systems' July 2026 occupancy barometer materials and related methodology pages, which showed ongoing hybrid-work patterns and sub-pre-pandemic attendance levels in covered office markets. Those data were used only to inform utilization exposure and office-risk discussion, not to replace company-specific analysis. [63]

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[1] [3] [4] [5] [12] [18] [50] [52] [61] https://www.sec.gov/Archives/edgar/data/1422841/000110465919075681/tm1927296d1_ex99-2.htm

https://www.sec.gov/Archives/edgar/data/1422841/000110465919075681/tm1927296d1_ex99-2.htm

[2] [13] [45] [56] https://aquaventure2019ir.q4web.com/investors/press-releases/press-release-details/2020/AquaVenture-Holdings-Limited-Announces-Fourth-Quarter-and-Full-Year-2019-Earnings-Results/default.aspx

https://aquaventure2019ir.q4web.com/investors/press-releases/press-release-details/2020/AquaVenture-Holdings-Limited-Announces-Fourth-Quarter-and-Full-Year-2019-Earnings-Results/default.aspx

[6] [9] [15] [17] [47] [51] [54] [57] https://quench.culligan.com/

https://quench.culligan.com/

[7] https://quench.culligan.com/water-dispenser-rental-benefits/

https://quench.culligan.com/water-dispenser-rental-benefits/

[8] [10] [11] [19] [20] [21] [42] [44] [49] [53] [55] [58] https://quench.culligan.com/facts-culligan-quench/

https://quench.culligan.com/facts-culligan-quench/

14 16 46 48 https://aquaventure2019ir.q4web.com/investors/press-releases/press-release-details/2020/AquaVenture-Shareholders-Approve-Planned-Acquisition/default.aspx
https://aquaventure2019ir.q4web.com/investors/press-releases/press-release-details/2020/AquaVenture-Shareholders-Approve-Planned-Acquisition/default.aspx

22 63 https://www.kastle.com/safety-wellness/getting-america-back-to-work/
https://www.kastle.com/safety-wellness/getting-america-back-to-work/

23 https://www.sec.gov/Archives/edgar/data/2042694/000114036125009678/ny20042792x4_ars.pdf
https://www.sec.gov/Archives/edgar/data/2042694/000114036125009678/ny20042792x4_ars.pdf

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https://www.sec.gov/Archives/edgar/data/884713/000114036124015948/ny20019840x5_ars.pdf

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https://www.water.com/services/water-filtration/

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https://www.water.com/modal/dispenser-rental-vs-purchase/

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Operating Benchmark Study of the Submetering Subvertical

Executive summary and subvertical overview

This report evaluates six designated companies as of **July 18, 2026**: Energy Assets, ista, Techem, Smart Metering Systems, Calisen, and CARMA. The evidence shows that the research universe does **not** represent one single economic model. Instead, it separates into two principal archetypes: **building submetering and billing platforms** that monetize long-lived installed devices plus recurring billing/data workflows, and **meter-asset-owner platforms** that finance and own larger, more expensive metering infrastructure under utility-facing contracts. Energy Assets sits between these groups as a broader metering/data/network platform rather than a pure apartment submetering company. [1]

Across the six companies, the most defensible central conclusion is that the **core submetering benchmark** is better represented by **ista and Techem**, with **CARMA** as a useful but lower-disclosure secondary reference. In that archetype, recurring revenue is created by a combination of installed measurement hardware, legally or operationally necessary consumption allocation, annual or periodic billing, and property-level workflow integration. That mix naturally produces mid-single-digit normalized growth, EBITDA margins around the mid-40s to low-50s, modest-to-moderate maintenance capex relative to revenue, long effective customer lives, and utilization exposure that is lower than typical route-based field services because the device continues earning while fixed in place. [2]

By contrast, **Calisen and Smart Metering Systems** are economically closer to **capital-heavy meter-asset providers** than to classic apartment heat-cost-allocation platforms. Their recurring rental economics are durable, but their capital model is much heavier because each incremental meter carries substantially more upfront funded asset cost. In those models, normalized EBITDA margins can still be very high because depreciation sits below EBITDA and field service is partly installation-led, yet post-maintenance-capex conversion is structurally lower than reported EBITDA would suggest. In other words, these businesses can look extremely profitable on EBITDA while still consuming large amounts of sustaining and replacement capital over the life of the estate. [3]

The single largest source of variation across the universe is therefore **asset intensity**. Where the company owns relatively inexpensive apartment-level submeters and monetizes billing/process services, payback is usually shorter and maintenance capex lower. Where the company owns more expensive smart utility meters and effectively finances the asset for the supplier market, payback is longer and sustaining capex consumes a much larger share of EBITDA over time. That is why this report provides archetype-level benchmarks rather than a single blended “submetering average.” [4]

This annual-report page illustrates how **Calisen’s** 2023 growth was driven by a larger revenue-generating meter portfolio and how that also carried higher meter depreciation. [5]

This **Techem** page shows the report's own segmentation between submetering and contracting, which is critical to isolating the relevant economics. [6]

Scope, definitions, and methodology

The analysis date is **July 18, 2026**. For each company, I first resolved the relevant legal and operating identity, then defined the relevant economic unit, then separated submetering or metering economics from unrelated activities such as EV charging, battery storage, utility network construction, or energy contracting where the evidence permitted. For acquired or formerly public businesses, I used the **last reliable standalone financial disclosure** and then updated operating facts with later official company or transaction materials without silently blending incompatible scopes. [7]

The nine core metrics are interpreted as follows. **Revenue growth** means normalized organic growth in the relevant operating unit, excluding acquisition, perimeter, and FX where determinable. **EBITDA margin** is normalized operating EBITDA rather than headline sponsor-adjusted cash earnings without qualification. **Maintenance capex** is the spend needed to preserve existing revenue and capacity, not rollout or new-customer capex. **Post-maintenance-capex conversion** is defined strictly as $(\mathrm{EBITDA} - \mathrm{Maintenance\ Capex})/\mathrm{EBITDA}$, and it is explicitly **not** full free-cash-flow conversion because it excludes interest, tax, working capital, lease principal, and growth capex. **Initial contract tenor** is legal or formal starting term; **effective customer tenure** is the expected revenue life after auto-renewal, inertia, transfer, and replacement effects; **utilization exposure** is the combination of asset utilization, customer usage dependence, and service-capacity utilization; **payback** is fully burdened initial investment divided by unit-level annual cash contribution; and **B2B/B2C profile** is based on the legal buyer and invoiced counterparty, not the resident end user. [8]

I used five evidence labels. **Disclosed** means the number was directly reported by the company or an official source. **Calculated** means it is an arithmetic derivation from disclosed figures. **Inferred** means the estimate follows directly from the business model with limited hard-company data. **Triangulated** means multiple facts from different reliable sources were connected to estimate the metric. **Unavailable** means the evidence was too weak or contaminated to support even a defensible range. Confidence reflects source quality, segment purity, recency, consistency, and inference burden. [9]

The most important methodological choice is the separation of **reported EBITDA** from **normalized economics**. For example, Calisen reports very high EBITDA margins because it owns the meter estate and books depreciation below EBITDA; that does not mean its maintenance capex is low. Likewise, Techem and ista benefit from regulation-supported recurring billing, but that does not mean contract tenor and customer life are identical or that all capex is sustaining. The causal bridge matters more than the reported ratio. [10]

Company profiles covering Energy Assets, ista, and Techem

Energy Assets

Company and segment resolution

The relevant legal entities in accessible public sources are **Energy Assets Group Holdings Limited** and **Energy Assets Group Limited**. Companies House shows Energy Assets Group Limited as an active private company incorporated in 2012, formerly named **Energy Assets Group plc**, with last accounts made up to **31 March 2025**; Energy Assets Group Holdings Limited filed group accounts to the same date in July 2025. The company website states that the group is now branded **Energy Assets Group**, with a registered office in Darwen and a broad utility-services footprint. Publicly accessible sources do **not** provide enough detail to isolate the full 31 March 2025 financial statements via this tooling, so the operating analysis relies primarily on official operating disclosures and Companies House metadata, with reduced confidence on financial ratios. [11]

The company is **not** a pure apartment submetering platform. Its own site describes a combined offering across metering and data, network construction, network ownership, water, fibre, and low-carbon utility services; it also says the group is the only company of scale in the UK with combined metering/data services, local network ownership, and network construction. The relevant segment for this benchmark is therefore the **metering and data** business, not the full multi-utility construction-and-network platform. That contamination is material and lowers confidence. [12]

Economic unit, customer problem, alternatives, revenue, operating model, retention, and capital model

The most relevant economic unit is **one installed metering asset under management**, usually paired with a related data/reading/service relationship. The company says it owns, operates, and manages **over 1.5 million assets across Great Britain**, serves major energy suppliers, TPIs, contractors, local authorities, and public/private organizations, and offers gas, electricity, water, and data services as well as installation, exchanges, maintenance, and support. In other words, revenue is created partly by the installed meter asset and partly by the associated data and field-service workflow. [12]

The customer problem is broader than apartment heat-cost allocation. For commercial users, suppliers, and developers, the need is accurate metering, compliant data handling, and reliable field service; for some customers the need is essential because metering is required for billing and network operation, while for developers it can be project-linked and installation-timed. Alternatives exist—other metering providers, internal utility relationships, or customer-owned assets—but Energy Assets argues its value proposition is the integrated offering across metering, data, network ownership, and construction. That creates practical retention through fewer handoffs, incumbent knowledge of asset history, and single-point-of-contact workflow benefits, but weaker lock-in than the classic apartment submetering model. [12]

Revenue appears to come from a mixture of recurring meter services, AMR/smart metering, data services, installation and exchanges, maintenance/support, and broader utility services. Because install-led and network-related activities remain in the wider group, EBITDA and capex for the pure metering subsegment are not cleanly disclosed. The capital model therefore looks more hybrid than ista/Techem/CARMA: a portion of revenue is likely recurring and asset-backed, but a portion remains activity-based and field-labor-intensive. That hybrid model is why I classify Energy Assets as a **mixed-business company**, not a core representative of apartment submetering. [13]

Metric analysis

Revenue growth. Conclusion: 3% / 5% / 8% normalized. **Observed facts:** the company presents itself as a scaled integrated metering/data/platform business with more than 1.5 million managed assets and a 31 March 2025 reporting date, but accessible public sources do not provide a clean metering segment revenue bridge. **Root business mechanism:** growth comes from meter estate additions, production of new developer/supplier relationships, and data-service attach, but is damped by exposure to project-led installation activity and the mixed composition of the wider group. **Bridge:** because recurring metering/data should grow more steadily than network construction, I normalize the metering subset at mid-single digits rather than extrapolating any whole-group growth. **Sensitivities:** UK metering regulation, supplier procurement cycles, new-build activity, and the split between recurring data revenue and one-off installation work. **Confidence: Low**, because the segment is not publicly isolated. [14]

EBITDA margin. Conclusion: 18% / 23% / 28% normalized for the relevant metering/data operation. **Observed facts:** Energy Assets explicitly combines metering/data with network construction and network ownership, so no clean reported margin is available for the relevant unit. **Root business mechanism:** compared with apartment submetering, Energy Assets likely carries more field labor, installation, and customer-specific project work; compared with network construction, the metering/data subset should benefit from recurring service revenue and installed-base density. **Bridge:** the central estimate assumes a hybrid mix where recurring meter/data revenue lifts margin above pure contracting, but recurring field work and subscale data products keep it below ista/Techem. **Sensitivities:** the share of revenue from recurring asset/data services, technician intensity, procurement pricing, and subcontracting. **Confidence: Low**. [12]

Maintenance capex as a percentage of revenue. Conclusion: 6% / 9% / 13%. **Observed facts:** no direct capex disclosure is available for the relevant metering subsegment; the group also owns and operates 1.5 million assets, which implies at least some replacement and refresh burden. **Root business mechanism:** maintenance capex is driven by periodic meter replacements, communications upgrades, handheld/IT/data infrastructure refresh, and service vehicles, but the capital load should be below UK smart MAP models because the asset base appears more mixed and less dominated by utility-owned smart meter financing. **Bridge:** the range triangulates from a replacement-cycle view of smaller metering assets plus supporting service and communications equipment. **Sensitivities:** ownership mix, who funds replacements, and whether AMR/comms upgrades are expensed, capitalized, or customer-funded. **Confidence: Low**. [12]

Post-maintenance-capex conversion. Conclusion: 35% / 61% / 79%. **Observed facts:** this is a calculation from the inferred margin and maintenance-capex ranges. **Root business mechanism:** conversion is limited by sustaining asset replacement and field infrastructure refresh, but supported by recurring service revenue once the device is installed. **Bridge:** at the central point, $(23\% - 9\%)/23\% \approx 61\%$. **Sensitivities:** any shift toward installation-heavy work or customer-funded capex would move the figure materially. **Confidence: Low**. [12]

Initial contract tenor. Conclusion: 3 / 5 / 8 years. **Observed facts:** the company serves energy suppliers, developers, TPIs, and public/private organizations, but does not publicly disclose standard metering/data contract terms in accessible sources. **Root business mechanism:** framework contracts in utility and commercial metering are usually longer than ad hoc field-service jobs because the meter/data setup and service history matter after installation. **Bridge:** I assume medium-duration framework or service contracts rather than month-to-month economic relationships. **Sensitivities:** customer class and whether the revenue stream is meter rental, data service, or installation support. **Confidence: Low**. [12]

Effective customer tenure. Conclusion: 5 / 8 / 12 years. **Observed facts:** integrated service breadth and installed assets support retention, but the business does not show the same regulatory billing lock-in profile as ista or Techem. **Root business mechanism:** installed assets, switching complexity, and data continuity extend practical life beyond contractual term, but procurement rebids and broader utility project cycles make tenure shorter than in classic residential submetering. **Bridge:** I therefore place effective tenure above initial legal term but well below the longest continental European submetering relationships. **Sensitivities:** ownership changes at sites, procurement centralization, and whether the meter remains owned by Energy Assets after customer handoff. **Confidence: Low**. [12]

Utilization exposure. Conclusion: Moderate / Moderate / High. **Observed facts:** the company both manages installed assets and performs field and construction-linked work. **Root business mechanism:** installed meters reduce asset-utilization volatility, but service-capacity utilization depends on installation and maintenance demand, making the model more exposed than pure billing-led platforms. **Bridge:** the most relevant KPI is likely the number of installed/managed assets per field or data-service head, but no public KPI is disclosed. **Sensitivities:** new-build cycles, supplier work volumes, and route density. **Confidence: Low**. [12]

New-unit or new-customer payback. Conclusion: 2.0 / 3.5 / 5.0 years. **Observed facts:** no unit economics are disclosed, but the group's mix includes asset installation and recurring data services. **Root business mechanism:** payback should be faster than UK smart MAP models because unit capex per metering point is likely lower and revenue includes data/service fees, but slower than pure software because field installation and support are required. **Bridge:** the central estimate assumes moderate equipment and installation cost recovered by recurring service income over several years. **Sensitivities:** customer-funded installation, meter ownership, and the share of first-year revenue that is one-off versus recurring. **Confidence: Low**. [12]

B2B versus B2C profile. Conclusion: Predominantly B2B / Predominantly B2B / Predominantly B2B. **Observed facts:** named customer groups include energy suppliers, TPIs, contractors, local authorities, developers, and public/private institutions. **Root business mechanism:** the legal buyer is typically an organization, not the end resident. **Bridge:** even where the metered end user is residential, the economic counterparty is business or institutional. **Sensitivities:** none material. **Confidence: Medium**, because customer categories are directly disclosed. [12]

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Normalized revenue growth	3%	5%	8%	Inferred	Low	Installed-asset additions plus data attach, offset by mixed project/install exposure
Normalized EBITDA margin	18%	23%	28%	Inferred	Low	Hybrid recurring meter/data revenue but meaningful field-service and broader mixed-business costs
Maintenance capex % of revenue	6%	9%	13%	Inferred	Low	Meter replacement, comms refresh, vehicles, and data infrastructure sustainment
Post-maintenance-capex conversion	35%	61%	79%	Calculated	Low	Moderate recurring cash flow offset by sustaining refresh spend
Initial contract tenor	3 yrs	5 yrs	8 yrs	Inferred	Low	Framework utility/commercial service contracts rather than one-off jobs

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Effective customer tenure	5 yrs	8 yrs	12 yrs	Inferred	Low	Installed-asset inertia and single-provider workflow benefits
Utilization exposure	Moderate	Moderate	High	Inferred	Low	Fixed installed assets reduce volatility, but field/service capacity remains cycle-sensitive
New-unit or customer payback	2.0 yrs	3.5 yrs	5.0 yrs	Inferred	Low	Moderate device and install cost recovered through recurring service/data income
B2B/B2C profile	Predominantly B2B	Predominantly B2B	Predominantly B2B	Disclosed	Medium	Suppliers, developers, TPIs, authorities, and organizations are the contracting buyers

Plain-English business description: Energy Assets is a UK metering-and-data provider embedded in a broader multi-utility platform, serving suppliers, developers, and institutions with combined meter, data, service, and network capabilities. [15]

Principal strengths: integrated offering, large installed asset base, national coverage, and multiple customer touchpoints. **Principal risks:** segment contamination, weaker pure-submetering comparability, procurement exposure, and limited accessible disclosure. **Key unavailable information:** clean segment revenue, EBITDA, capex, churn, and contract tenor. **Most important diligence questions:** what share of

EBITDA comes from recurring metering/data; who owns the devices; what is the replacement cycle by asset class; and how much revenue is installation/project-led versus recurring? [13]

ista

Company and segment resolution

ista SE describes itself as one of the leading energy service providers for buildings, with **over 45 million wireless devices** in operation, products and services used in **over 14 million homes and commercial properties**, more than **460,000 customers**, operations across **20 countries**, and **€1,220 million** of 2024 sales on the corporate page. Its audited 2025 key-figures page reports **€1,279.0 million** of 2025 sales, **€260 million** of total investment (capex), and notes that the report covers ista SE, parent **Trionista SE**, and the wider group with an editorial deadline of **31 May 2026**. [16]

The most recent accessible ownership evidence indicates that CVC agreed the sale of ista to the CKP/CKI buyer group in 2017, and S&P's 2026 rating commentary is issued on **Trionista SE**, which I use as the current financing parent in available sources. Because a current public owner page was not available in accessible text, I anchor ownership at the last clearly evidenced corporate parent structure and note moderate uncertainty on the ultimate holding chain. [17]

The relevant segment here is the **core building submetering and related billing/data platform**, not every adjacent digital building service. ista's own product descriptions emphasize digital consumption recording, radio-read heat and hot-water devices, and sustainable building solutions. The central economic unit is therefore best defined as **one managed utilization unit or one connected device footprint attached to a dwelling/commercial property**, with billing/data workflows layered on top. [18]

Economic unit, customer problem, alternatives, revenue, operating model, retention, and capital model

The customer problem is continuous and regulation-shaped: landlords and property managers need legally compliant, auditable, and practical allocation of heating and water costs and increasingly want digital consumption visibility and process automation. ista says it makes buildings more climate-friendly, safe, and comfortable; it also says it invests over **€100 million per year** in digital devices and infrastructure and already has more than **25 million connected devices** in use for radio-reading of heat and hot-water consumption. That shows a model in which the physical devices matter, but the monetization does not stop at meter ownership: recurring billing, data capture, analytics, and property workflow integration are essential pieces of the value chain. [18]

Customer alternatives include self-provision, another submetering service provider, or manual reading substitutes. But the root retention mechanism is stronger than "recurring revenue" alone. Once a provider's devices are installed across a building, switching requires site access, replacement or interoperability work, resident communication, data migration, and rebilling setup. Because the service also connects to regulated or quasi-mandatory cost allocation, the decision is sticky in practice even where the formal contract is not perpetual. That naturally extends effective tenure well beyond any first legal term. [19]

The capital model is meaningful but not as heavy as UK smart meter asset ownership. ista's audited 2025 key figures show total investment of **€260 million** on **€1,279 million** of sales, while S&P's 2026 rating snippet says capex amounted to **12%-15% of revenue during 2023-2025**. Because ista's groups of devices are smaller, more numerous, and typically tied to billing/process revenue rather than pure asset rent, economic maintenance capex should sit below total capex and below UK MAP models, but well above software-only service businesses. [20]

Metric analysis

Revenue growth. Conclusion: 4% / 6% / 8% normalized. **Observed facts:** sales rose from **€1,216.1 million in 2024** to **€1,279.0 million in 2025**, or about **5.2%**, on the audited key-figures page. **Root business mechanism:** mature installed-base revenue grows through indexation, device upgrades, digital-service attachment, and measured penetration of additional solutions into existing buildings, not through high-churn logo expansion. **Bridge:** the central estimate rounds slightly above the most recent reported rate because the model benefits from regulation-linked demand and digital upsell, but not from the very high rollout-driven growth seen in utility smart meter asset platforms. **Sensitivities:** acquisition contribution, smart-building service attach, and energy-policy-driven adoption of digital devices. **Confidence: Medium**. [21]

EBITDA margin. Conclusion: 42% / 45% / 48% normalized. **Observed facts:** Fitch reported that ista generated an EBITDA margin of **over 40%** in 2016, while S&P's April 2026 commentary says the company still generates a **high EBITDA margin** despite heavy capex. **Root business mechanism:** margin is supported by recurring billing/data revenue, dense installed bases, limited consumables, and relatively low variable cost per additional managed unit once devices are installed; it is restrained by field service, device procurement, and ongoing digital infrastructure spend. **Bridge:** the central estimate assumes today's margin remains above 40% but below the very highest rental-heavy UK MAP margin profiles because ista's model still includes a larger service and process component. **Sensitivities:** labor inflation, digital device mix, procurement costs, and the split between recurring billing services and installation-heavy work. **Confidence: Medium**. [22]

Maintenance capex as a percentage of revenue. Conclusion: 8% / 10% / 13%. **Observed facts:** audited 2025 total investment was **€260 million**, or about **20.3% of sales**, while S&P says capex ran at **12%-15% of revenue during 2023-2025**. ista also says it invests more than **€100 million** per year in digital devices and infrastructure. **Root business mechanism:** only a fraction of total capex is required to preserve the existing installed base; the remainder supports digital rollout, new devices, and broader platform expansion. Apartment-level submeter devices are less capital-intensive than UK supplier smart meters, and recurring billing/data revenue is partly preserved by replacing batteries, communication modules, and devices on scheduled cycles. **Bridge:** I treat the central estimate as the sustaining portion of recent capex intensity, not the whole amount. **Sensitivities:** how aggressively ista is digitizing legacy estates and the proportion of capex aimed at new connected-device deployment. **Confidence: Medium**. [23]

Post-maintenance-capex conversion. Conclusion: 69% / 78% / 83%. **Observed facts:** this is calculated from the normalized margin and maintenance-capex assumptions. **Root business mechanism:** the model converts well after sustaining capex because billing/data economics are sticky and labor-light at scale, but conversion is not extreme because device refresh is real and cannot be ignored. **Bridge:** at the central point, $(45\% - 10\%)/45\% \approx 78\%$. **Sensitivities:** whether additional digital rollout is economically sustaining rather than genuinely growth-oriented. **Confidence: Medium**. [24]

Initial contract tenor. Conclusion: 5 / 8 / 10 years. **Observed facts:** accessible public sources do not give a standard contract term, but the business serves large property portfolios and relies on installed device estates plus recurring billing/data services. **Root business mechanism:** property managers typically choose a provider for a building or portfolio, installing device infrastructure that is not frictionless to replace. **Bridge:** I therefore assume formal initial terms are multi-year, not month-to-month, and usually shorter than the effective revenue life. **Sensitivities:** country-specific legal regimes and whether the deal is device rental, billing-only, or a fuller service bundle. **Confidence: Medium-Low**. [19]

Effective customer tenure. Conclusion: 10 / 14 / 18 years. **Observed facts:** ista manages more than 14 million homes and commercial properties with more than 45 million wireless devices and over 460,000 customers, implying a highly embedded installed estate. **Root business mechanism:** renewal inertia is created by device installation, access logistics, billing history, software/workflow integration, and the low salience of the service versus the hassle of switching. **Bridge:** effective tenure therefore materially exceeds initial legal term. **Sensitivities:** portfolio sales, regulatory standardization that reduces switching friction, and device interoperability. **Confidence: Medium**. [18]

Utilization exposure. Conclusion: Low / Low / Moderate. **Observed facts:** once installed, the device continues to participate in billing regardless of daily energy use volatility, and ista's model depends on an installed estate rather than route density alone. **Root business mechanism:** asset utilization is primarily a function of whether the dwelling/building remains in service and whether the provider remains appointed; customer usage affects bill volume more than the existence of recurring service revenue. **Bridge:** the model has lower utilization exposure than mobile fleets or volume-linked waste services, though service-capacity utilization still matters. **Sensitivities:** vacancy, refurbishments, and country-level rollout cycles for digital replacements. **Confidence: Medium**. [18]

New-unit or new-customer payback. Conclusion: 3.0 / 4.0 / 5.5 years. **Observed facts:** no public unit-payback disclosure is accessible, but ista's annual investment level and connected-device investment program show a meaningful, recurring upfront spend. **Root business mechanism:** apartment-level devices are cheaper than utility smart meters, while recurring gross contribution includes device economics plus billing/data service. That combination tends to shorten payback versus UK MAP models. **Bridge:** the central estimate assumes device-plus-install cost is earned back across several annual billing cycles, with attractive lifetime economics if the site remains retained. **Sensitivities:** install complexity, labor, and the share of economics captured in service fees versus hardware rent. **Confidence: Medium-Low**. [25]

B2B versus B2C profile. Conclusion: B2B contract with residential end users / B2B contract with residential end users / B2B contract with residential end users. **Observed facts:** ista explicitly serves buildings, homes, and commercial properties through institutional customers and reports over 460,000 customers against more than 14 million utilization units. **Root business mechanism:** the resident consumes the measured utility, but the property owner or manager is typically the contractual buyer. **Bridge:** the right classification is therefore B2B contract with residential end users. **Sensitivities:** commercial-property mix by country. **Confidence: High**. [26]

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Normalized revenue growth	4%	6%	8%	Calculated	Medium	Indexed installed-base growth plus digital-service upsell
Normalized EBITDA margin	42%	45%	48%	Triangulated	Medium	Recurring billing/data revenue on dense installed device estates
Maintenance capex % of revenue	8%	10%	13%	Triangulated	Medium	Device refresh, communications upgrades, and sustaining digital infrastructure
Post-maintenance-capex conversion	69%	78%	83%	Calculated	Medium	Strong recurring contribution after moderate sustaining refresh spend
Initial contract tenor	5 yrs	8 yrs	10 yrs	Inferred	Medium-Low	Multi-year building/ portfolio service contracts
Effective customer tenure	10 yrs	14 yrs	18 yrs	Inferred	Medium	Installed-device inertia, workflow integration, and low switching salience
Utilization exposure	Low	Low	Moderate	Inferred	Medium	Fixed installed assets with limited daily usage dependence

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
New-unit or customer payback	3.0 yrs	4.0 yrs	5.5 yrs	Inferred	Medium-Low	Moderate device/install cost recovered through recurring billing and service income
B2B/B2C profile	B2B contract with residential end users	B2B contract with residential end users	B2B contract with residential end users	Disclosed	High	Landlords/ property managers buy; residents consume

Plain-English business description: ista is a large European building-services platform that installs and connects apartment-level meters and allocators, then monetizes those devices through recurring billing, data, and sustainability workflows. [18]

Principal strengths: scale, international density, large connected-device base, and strong recurring economics. **Principal risks:** country-level regulation changes, procurement and labor inflation, and the need to keep digital replacement capex disciplined. **Key unavailable information:** public current contract-term disclosure and a clean current EBITDA bridge by subsegment. **Most important diligence questions:** what proportion of capex is growth versus sustaining; what is actual customer retention by cohort; and how much margin comes from billing/data versus device ownership? [27]

Techem

Company and segment resolution

Techem describes itself as a leading provider for smart and sustainable buildings, with operations in **18 countries**, more than **4,300 employees**, and services to over **13.5 million apartments**. The July 2025 owner announcement says Techem had over **440,000 customers**, serviced more than **13 million dwellings**, and had about **62 million installed devices** at that time. The accessible 2025 sustainability report then updates the installed device base to around **70 million devices** and explicitly separates the business into **Energy Services**—the submetering business—and **Energy Efficiency Solutions**—contracting and consulting. [28]

That segmentation is crucial. For benchmark purposes, the relevant unit is **Techem's ESG/ESI submetering segment**, not the EES contracting business. Techem's report says the business activity generating the most revenue is **submetering**, meaning consumption-based billing of heating costs, and provides segment revenue on page 50: **€1,041.5 million** for ESG/ESI submetering and **€131.9 million** for EES contracting in fiscal 2025, against **€1,173.4 million** total group revenue. [29]

On ownership, the latest official transaction evidence available in public text is the **July 14, 2025** announcement that Partners Group's infrastructure business would take a controlling stake, with GIC, TPG

Rise Climate, and Mubadala as minority investors, following a €6.7 billion valuation. I therefore use that consortium as the ownership basis for this report. [30]

Economic unit, customer problem, alternatives, revenue, operating model, retention, and capital model

The economic unit is best defined as **one serviced residential unit or one installed set of apartment-level measuring devices attached to a building billing relationship**. The customer problem is compliance and cost allocation: property owners/managers of centrally heated multi-unit buildings need measured and allocated heating and water costs, increasingly in digital form, and Techem's installed digital radio infrastructure enables both billing and broader process services. The report describes Techem's devices and sensors as an interoperable digital radio infrastructure built over decades, which then supports optimization, automation, and additional applications. [31]

Alternatives exist: other submetering providers, manual reading, or insourcing. But the installed base, data continuity, and workflow integration make switching disruptive. Techem's submetering service therefore combines legal or quasi-mandatory demand, installed physical infrastructure, and recurring billing/data service. That is why the model tends to have long effective customer tenures and relatively low day-to-day utilization exposure once the site is won. [32]

The capital model is moderate rather than light. The company's submetering estate is large, and the rating/public materials show meaningful capex. Fitch said it expected about **€190 million** of annual capex in 2023-2025, while S&P reported **€557 million** of adjusted EBITDA on **€1.17 billion** of fiscal 2025 revenue. This means Techem is not "software-like"; the devices and digital infrastructure are capitalized and matter. But relative to UK smart meter owners, Techem's unit devices are less expensive and the service stack captures more recurring contribution beyond the hardware itself. [33]

Metric analysis

Revenue growth. Conclusion: 4% / 6% / 8% normalized for core submetering. **Observed facts:** group revenue increased from **€1,058.2 million in 2024** to **€1,173.4 million in 2025**, while the report separately shows submetering revenue rising from **€933.5 million** to **€1,041.5 million**. S&P also referenced close to **11%** revenue growth in fiscal 2025. **Root business mechanism:** normalized growth is driven by device digitalization, additional metering points, price/indexation, and adjacent digital services, but not all of 2025's growth should be annualized because some expansion reflects broader platform activity and favorable recent tailwinds. **Bridge:** I therefore normalize to mid-single digits for steady-state submetering, above GDP-like growth but below rollout-heavy UK MAP economics. **Sensitivities:** regulatory expansion, digitization rates, pricing, and acquisitions. **Confidence: High** on direction, **Medium** on the normalized range. [34]

EBITDA margin. Conclusion: 45% / 49% / 53% normalized for submetering. **Observed facts:** S&P says Techem generated **€557 million** of adjusted EBITDA on **€1.17 billion** of 2025 revenue, implying a total-group margin near **47.5%**. Because EES contracting is the less relevant and likely lower-margin energy-contracting business, the core submetering segment should sit modestly above that total-group figure. **Root business mechanism:** recurring billing, radio-read devices, and dense installed property portfolios support high margins; field service and device procurement prevent margins from drifting toward software levels. **Bridge:** the central estimate therefore places submetering margin just above total-group adjusted margin. **Sensitivities:** direct procurement costs, labor, and the share of contracting revenue in the mix. **Confidence: Medium-High**. [35]

Maintenance capex as a percentage of revenue. Conclusion: 7% / 9% / 12%. **Observed facts:** Fitch expected capex of about **€190 million** per year in 2023-2025, which is about **16%** of 2025 total revenue. The sustainability report also makes clear that part of the group sits in contracting, where capex serves heat-generating plants and other assets outside pure submetering. **Root business mechanism:** sustaining submetering capex is driven by device replacement cycles, battery life, communications upgrades, and installation labor when legacy equipment is replaced, but not all group capex is required just to preserve existing submetering revenue. **Bridge:** I treat the central estimate as roughly a little over half of recent total capex intensity, recognizing both true sustaining device refresh and ongoing digitalization. **Sensitivities:** regulatory device-refresh requirements and whether smart-metering additions are growth or preservation of the existing installed base. **Confidence: Medium**. [36]

Post-maintenance-capex conversion. Conclusion: 73% / 82% / 87%. **Observed facts:** calculated from the normalized margin and maintenance-capex assumptions. **Root business mechanism:** the combination of high-margin recurring billing/data revenue and moderate sustaining capex produces strong post-maintenance conversion. **Bridge:** at the central point, $(49\% - 9\%)/49\% \approx 82\%$. **Sensitivities:** if more digital rollout is economically sustaining than assumed, true conversion would be lower. **Confidence: Medium**. [37]

Initial contract tenor. Conclusion: 5 / 8 / 10 years. **Observed facts:** accessible sources do not disclose a standard term, but Techem operates at building scale for property owners/managers and installs long-lived device estates. **Root business mechanism:** building-level service awards typically require multi-year commitment because replacing an incumbent submetering setup is cumbersome. **Bridge:** the central estimate assumes medium-duration initial terms with renewals, not perpetual economic commitments. **Sensitivities:** country regime and whether the contract includes only billing or bundled metering infrastructure. **Confidence: Medium-Low**. [32]

Effective customer tenure. Conclusion: 10 / 15 / 20 years. **Observed facts:** Techem services over 13 million dwellings and 62-70 million installed devices, all within an interoperable radio infrastructure. **Root business mechanism:** switching requires replacing or reconfiguring devices, data flows, and billing processes across entire buildings or portfolios; the service is also comparatively low cost relative to rent and utilities, reducing the incentive to rebid aggressively. **Bridge:** effective tenure therefore materially exceeds the initial legal term. **Sensitivities:** antitrust or interoperability changes and portfolio ownership turnover. **Confidence: Medium**. [32]

Utilization exposure. Conclusion: Low / Low / Moderate. **Observed facts:** Techem's main revenue source is submetering and billing, not high-throughput transactional services. **Root business mechanism:** once the installed device set is active in a building, the asset remains productive regardless of short-term usage swings; customer usage changes the bill amount more than the existence of the service. **Bridge:** field-service capacity still has some seasonality, so I do not classify the model as zero-exposure. **Sensitivities:** vacancy, heating demand regulation, and fixed-network rollout pace. **Confidence: Medium**. [38]

New-unit or new-customer payback. Conclusion: 2.5 / 3.5 / 5.0 years. **Observed facts:** no unit-payback disclosure is public, but the device base is huge, capex meaningful, and EBITDA high. **Root business mechanism:** apartment-level device and install costs are moderate, while recurring contribution includes both device economics and ongoing billing/data income. **Bridge:** that combination supports a shorter payback than UK MAPs and a longer payback than billing-only software. **Sensitivities:** install access cost, portfolio density, and new-service attach. **Confidence: Medium-Low**. [39]

B2B versus B2C profile. Conclusion: B2B contract with residential end users / B2B contract with residential end users / B2B contract with residential end users. **Observed facts:** Techem explicitly serves property managers and owners of multi-tenant residential buildings and commercial properties. **Root business mechanism:** the legal customer is the owner/manager; residents are the measured end users. **Bridge:** the correct classification is B2B contract with residential end users. **Sensitivities:** commercial-property share. **Confidence: High**. [40]

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Normalized revenue growth	4%	6%	8%	Calculated	Medium	Digitalized installed-base growth plus pricing and service expansion
Normalized EBITDA margin	45%	49%	53%	Triangulated	Medium-High	High-value recurring billing/data on dense submeter estates
Maintenance capex % of revenue	7%	9%	12%	Triangulated	Medium	Device replacement, batteries, comms upgrades, and sustaining install work
Post-maintenance-capex conversion	73%	82%	87%	Calculated	Medium	High recurring contribution with moderate sustaining capital needs

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Initial contract tenor	5 yrs	8 yrs	10 yrs	Inferred	Medium-Low	Building-level multi-year service awards
Effective customer tenure	10 yrs	15 yrs	20 yrs	Inferred	Medium	Device installation, billing-data migration cost, and low switching salience
Utilization exposure	Low	Low	Moderate	Inferred	Medium	Installed assets stay productive with limited daily usage dependence
New-unit or customer payback	2.5 yrs	3.5 yrs	5.0 yrs	Inferred	Medium-Low	Moderate device/install cost recovered through recurring billing and service margin
B2B/B2C profile	B2B contract with residential end users	B2B contract with residential end users	B2B contract with residential end users	Disclosed	High	Owners/ managers contract; tenants consume

Plain-English business description: Techem is a large European submetering and building-services provider that installs apartment-level measuring devices and turns them into recurring billing, data, and building-efficiency revenue. [41]

Principal strengths: leading market position, large installed base, explicit segment disclosure separating submetering from contracting, and regulation-supported demand. **Principal risks:** capital discipline during digitization, procurement/labor inflation, antitrust scrutiny, and contamination from contracting if investors use total-group metrics mechanically. **Key unavailable information:** publicly accessible standard contract terms and explicit submetering EBITDA by segment. **Most important diligence questions:** what share of capex is true maintenance, what are renewal rates by country, and how much margin dilution comes from non-submetering activities? [42]

Company profiles covering Smart Metering Systems, Calisen, and CARMA

Smart Metering Systems

Company and segment resolution

Smart Metering Systems is now **Smart Metering Systems Limited**, with the historical public-company disclosure housed on the corporate reports page. That page shows **2023** and **2024** annual reports remain available, along with a **2025 interim report**. The company announced in February 2025 that it completed the combination with **Horizon Energy Infrastructure** and **Smart Meter Assets**, creating a larger group managing more than **6 million meters** across the UK and expanding into water metering and German smart meter financing. However, the last fully text-accessible detailed public standalone financial disclosure available through the tooling is the **FY2022 full-year results** published on March 14, 2023. I therefore use FY2022 as the last clean standalone public base and then update operating facts with 2025 owner-level releases. [43]

The prompt requires me to isolate the relevant operation. SMS is **not** a pure apartment submetering company. The relevant segment is its **smart meter funding, installation, and management platform**, especially the recurring meter rental and data economics represented by **ILARR**. In FY2022, ILARR was **£97.1 million**, pre-exceptional EBITDA was **£63.8 million**, group revenue was **£135.5 million**, and the smart meter portfolio stood at about **2.1 million** meters. [44]

Economic unit, customer problem, alternatives, revenue, operating model, retention, and capital model

The best economic unit is **one owned smart meter under contract**, with secondary economics from meter data and associated management services. The customer problem is utility-facing: suppliers need financed smart meters, field installation, meter health, and ongoing portfolio performance. SMS's own 2025 "Smart Meter Health" materials emphasize that smart meters only deliver value if they remain connected and functional, which highlights the separation between installation revenue and ongoing service/data economics. [45]

Customer alternatives include supplier self-funding, competing meter asset providers, or alternative service contractors. But retention comes from funded asset ownership, meter portfolio scale, integration into supplier operational workflows, and the practical difficulty of replacing a working estate before the expected life has been harvested. That said, this is still more exposed to rollout pace, supplier procurement, and capital markets than classic apartment submetering. [46]

The capital model is the central economic fact. Each new meter requires financed equipment and installation before the company earns recurring rental/data revenue. In 2022 SMS installed **480,000** smart meters, had a contracted pipeline of about **2.17 million** additional meters, and said that those meters would further increase ILARR by about **£50 million**. That implied around **£23** of incremental annual recurring revenue per pipeline meter. Given UK smart meter funding requirements, that recurring yield strongly suggests a multi-year payback and heavy dependence on long-lived asset utilization rather than on light-service economics. [47]

Metric analysis

Revenue growth. Conclusion: 5% / 8% / 11% normalized for the relevant smart-metering platform. **Observed facts:** FY2022 group revenue increased **25%** to **£135.5 million**, ILARR increased **13%** to **£97.1 million**, and in 2025 the group said it continued to grow organically by increasing owned meters and capex deployed, while also winning a **1 million meter** water contract with Southern Water. **Root business mechanism:** recurring growth is driven by new meter funding and installation flowing into the ILARR base, plus meter-health/data services. **Bridge:** I normalize well below the FY2022 headline growth because rollout phases and acquisitions inflate reported growth. **Sensitivities:** supplier rollout schedules, financed-install pace, and contract mix. **Confidence: Medium**. [48]

EBITDA margin. Conclusion: 42% / 47% / 52% normalized. **Observed facts:** FY2022 pre-exceptional EBITDA was **£63.8 million** on **£135.5 million** revenue, or about **47.1%**. **Root business mechanism:** margin is strong because recurring meter-rental and data revenue is layered onto financed assets, and installation costs are partly capitalized into the estate rather than fully burdening EBITDA immediately. **Bridge:** I use the FY2022 pre-exceptional margin as the central estimate for the relevant platform, recognizing that batteries and other CaRe assets contaminate the broader group. **Sensitivities:** installation mix, meter-health service intensity, financing/accounting treatment, and adjacent-business dilution. **Confidence: Medium**. [49]

Maintenance capex as a percentage of revenue. Conclusion: 20% / 25% / 32%. **Observed facts:** FY2022 public materials do not split sustaining from growth capex, but the business model is explicitly asset-funded and installation-led, and later group materials say organic growth continued by increasing the number of meters owned and capex deployed. **Root business mechanism:** even after rollout growth is stripped out, an owner of millions of smart meters must fund meter replacements, communications refresh, removal/reinstallation, and associated field labor over the life of the estate. **Bridge:** because smart utility meters are much more capital-intensive than apartment heat allocators, maintenance capex as a share of revenue is structurally much higher than for Techem/ista. **Sensitivities:** actual useful life, supplier-funded removals, and how much of current capex is still growth rollout rather than steady-state replacement. **Confidence: Medium-Low**. [48]

Post-maintenance-capex conversion. Conclusion: 24% / 47% / 62%. **Observed facts:** calculated from the normalized margin and capex assumptions. **Root business mechanism:** recurring rental revenue is durable, but a large share of EBITDA is economically reserved for replacement of the financed meter estate. **Bridge:** at the central point, $(47\% - 25\%)/47\% \approx 47\%$. **Sensitivities:** the true steady-state replacement cycle. **Confidence: Medium-Low**. [46]

Initial contract tenor. Conclusion: 3 / 5 / 10 years. **Observed facts:** public accessible sources do not disclose a standard starting term, but SMS's economics are built around long-lived recurring meter-rental and data contracts, not one-time jobs. **Root business mechanism:** suppliers need meter-estate funding and service continuity, which normally requires multi-year relationship economics. **Bridge:** I assume medium-term legal frameworks with longer economic lives. **Sensitivities:** contract form, supplier cancellation rights, and compensation terms. **Confidence: Low-Medium**. [45]

Effective customer tenure. Conclusion: 8 / 12 / 16 years. **Observed facts:** the company's recurring revenue base is explicitly represented by ILARR, and the model is built on ownership and management of installed smart meters. **Root business mechanism:** once the meter is funded and installed, the customer relationship normally persists until replacement, supplier change with take-over mechanics, or compensation-driven early removal. **Bridge:** economic tenure therefore substantially exceeds the initial formal term. **Sensitivities:** supplier insolvency, policy shifts, and early removal compensation. **Confidence: Medium-Low**. 50

Utilization exposure. Conclusion: Low / Moderate / High. **Observed facts:** the asset keeps earning once installed, but the business also depends on installation run-rate and supplier-side rollout decisions. **Root business mechanism:** asset utilization on the installed base is relatively defensible, but service-capacity utilization and growth depend on rollout volume. **Bridge:** that makes the overall exposure higher than apartment submetering but lower than pure mobile-asset rental. **Sensitivities:** installation pace, meter failure, and supplier portfolio churn. **Confidence: Medium**. 45

New-unit or new-customer payback. Conclusion: 4.0 / 5.5 / 7.0 years. **Observed facts:** SMS said its pipeline of about **2.17 million** meters would add roughly **£50 million** of ILARR, implying about **£23** annual recurring revenue per meter. **Root business mechanism:** with funded hardware plus field installation, the numerator is meaningful; with annual recurring rental/data revenue per meter in the low-twenties, payback cannot be short unless the capital cost is far lower than peers, which is unlikely. **Bridge:** the central estimate therefore falls in the mid-single-digit years. **Sensitivities:** per-meter installed capital cost, retention length, and meter-health service contribution. **Confidence: Medium-Low**. 47

B2B versus B2C profile. Conclusion: Predominantly B2B / Predominantly B2B / Predominantly B2B. **Observed facts:** SMS contracts with energy suppliers and infrastructure counterparties; residents use the metered service but are not the contract buyer in the core model. **Root business mechanism:** procurement sophistication and concentration sit at the supplier level. **Bridge:** the model is clearly B2B. **Sensitivities:** none material. **Confidence: High**. 46

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Normalized revenue growth	5%	8%	11%	Triangulated	Medium	New financed meters convert into ILARR and service revenue
Normalized EBITDA margin	42%	47%	52%	Calculated	Medium	Asset-rental and data economics lift EBITDA before replacement capex

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Maintenance capex % of revenue	20%	25%	32%	Inferred	Medium-Low	Large funded smart-meter estate requires real replacement and upgrade spend
Post-maintenance-capex conversion	24%	47%	62%	Calculated	Medium-Low	High EBITDA but capital-heavy sustaining asset model
Initial contract tenor	3 yrs	5 yrs	10 yrs	Inferred	Low-Medium	Multi-year supplier frameworks around funded meter estates
Effective customer tenure	8 yrs	12 yrs	16 yrs	Inferred	Medium-Low	Installed-asset persistence and removal friction
Utilization exposure	Low	Moderate	High	Inferred	Medium	Installed assets are sticky, but rollout/service capacity remains volume-sensitive
New-unit or customer payback	4.0 yrs	5.5 yrs	7.0 yrs	Triangulated	Medium-Low	Hardware-plus-install cost recovered through low annual rental per meter

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
B2B/B2C profile	Predominantly B2B	Predominantly B2B	Predominantly B2B	Disclosed	High	Energy suppliers and infrastructure counterparties buy the service

Plain-English business description: SMS is a UK smart-meter infrastructure owner and service platform that funds, installs, and manages utility-facing smart meters, earning recurring rental and data income over long asset lives. [50]

Principal strengths: visible recurring revenue base, large pipeline, and scale in funded smart metering. **Principal risks:** capital intensity, dependence on rollout pace, and contamination from non-core CaRe assets if not separated. **Key unavailable information:** current clean standalone segment margin and sustaining-versus-growth capex split. **Most important diligence questions:** what is steady-state replacement capex per meter, what are actual supplier contract termination mechanics, and how much of 2025+ growth is acquisition/perimeter rather than organic? [48]

Calisen

Company and segment resolution

Calisen's official reports page shows annual reports through **2023**, with the 2023 annual report titled **The Smarter Energy Report 2023**. It says Calisen operates across four business units—**MAP Services**, **Installation Services**, **Data Services**, and **Plug Me In**—and that MAP Services owns and manages metering assets. For this benchmark, the relevant segment is **MAP Services plus closely linked data/ maintenance services**, while **Plug Me In**, EV charging, heat pumps, and other adjacent transition activities are excluded from the core economics. [51]

In 2023 Calisen reported **15.2 million revenue-generating meters**, including **12.0 million** owned and managed smart meters, and total revenue of **£358.2 million**. MAP Services revenue rose to **£293.9 million**, Data Services revenue to **£23.2 million**, and Installation Services revenue to **£34.4 million**. Underlying EBITDA was **£265.4 million**, or a **74.1%** margin, and average capital expenditure per smart meter was **£219**. The company also discloses that, for accounting purposes, retailer arrangements are treated as **month-to-month contracts**, even though meter income continues when the end consumer switches supplier and customer-arrangement economics are assessed over up to **15 years** or, after review, **10-20 years** depending on the contract. [52]

The latest accessible ownership update is the December 2024 announcement that **EQT and GIC** would acquire a majority stake in Calisen; EQT's current-portfolio page says Calisen owns and manages **over 16 million meters** under long-term contracts with major energy suppliers. I therefore use the 2024-2026 ownership basis as EQT/GIC-led control with Equitix still associated historically, while the last clean detailed operating disclosure remains the 2023 report. [53]

Economic unit, customer problem, alternatives, revenue, operating model, retention, and capital model

The best economic unit is **one owned revenue-generating meter**. Calisen's own accounting notes make the underlying model clear: it owns the meters, charges meter rental and related services, and continues to collect the meter point charge when the consumer switches supplier unless the physical meter is removed. That is not classic apartment submetering; it is a funded smart-meter-asset model sitting next to installation and data services. [54]

The customer problem is supplier-facing and policy-backed. Calisen says it works on behalf of major energy retailers and that those retailers have been set **mandatory targets by Ofgem** to equip residential customers with smart meters. The company procures devices, installs them, and supports them through service life. That creates demand that is partly regulatory and partly outsourcing/funding driven. [55]

The revenue model must be separated carefully. MAP Services provides the recurring meter-asset economics; Installation Services is more transactional; Data Services provides recurring support and reading/data workflows. The company's own disclosures show why reported EBITDA is not enough: 2023 total capex was **£355.5 million**, average capex per smart meter was **£219**, and depreciation on metering equipment alone was **£144.8 million**. In a model like this, a very high EBITDA margin is consistent with very meaningful ongoing sustaining economics because the meter estate itself is the business. [56]

Metric analysis

Revenue growth. Conclusion: 4% / 7% / 10% normalized for the relevant MAP-led platform. **Observed facts:** total revenue increased from **£285.8 million** to **£358.2 million** in 2023, while MAP Services revenue increased from **£236.2 million** to **£293.9 million**. The smart meter portfolio grew from **8.2 million** to **12.0 million** smart meters, partly due to the MapleCo acquisition and partly due to ongoing installations. **Root business mechanism:** growth is driven by adding revenue-generating meters and, secondarily, data and maintenance services; acquisition and rollout can materially overstate organic base growth. **Bridge:** I normalize to high-single digits only in the optimistic case and strip out acquisition/perimeter effects for the central case. **Sensitivities:** supplier rollout pace, capital availability, ARPM mix, and M&A. **Confidence: Medium**. [57]

EBITDA margin. Conclusion: 68% / 72% / 76% normalized. **Observed facts:** underlying EBITDA in 2023 was **£265.4 million** on **£358.2 million** revenue, a **74.1%** margin. **Root business mechanism:** the meter estate earns recurring revenue with modest direct opex relative to revenue, while depreciation and replacement capital sit below EBITDA. The mix of installation services can dilute or inflate the group figure depending on activity, but MAP dominates. **Bridge:** I use a central estimate slightly below reported 2023 to reflect normalization and possible mix changes, while preserving the core truth that this is a very high-EBITDA, asset-owner model. **Sensitivities:** installation mix, compensation income, and ARPM by contract type. **Confidence: High** on the general level, **Medium** on the normalized range. [58]

Maintenance capex as a percentage of revenue. Conclusion: 28% / 35% / 45%. **Observed facts:** total 2023 capex was **£355.5 million**, or almost the full revenue base, and average smart-meter capex was **£219**; smart meter useful life was reviewed to **10-20 years**, and meter depreciation alone was **£144.8 million** in 2023. **Root business mechanism:** a large share of total capex is growth rollout, but an owner of millions of smart meters must ultimately replace and upgrade those meters to preserve current rental revenue.

Because asset cost per meter is large relative to annual revenue per meter, economic maintenance capex is necessarily material. **Bridge:** the central estimate treats sustaining needs as meaningfully below reported total capex, but still large compared with service-led submetering platforms. **Sensitivities:** actual replacement cycles, regulatory obsolescence, communications-module refresh, and customer-funded removals or compensation. **Confidence: Medium**. [59]

Post-maintenance-capex conversion. Conclusion: 34% / 51% / 63%. **Observed facts:** calculated from the normalized margin and maintenance-capex range. **Root business mechanism:** Calisen converts a large share of revenue into EBITDA but must reserve a large share of that EBITDA for the replacement economics of the estate. **Bridge:** at the central point, $(72\% - 35\%)/72\% \approx 51\%$. **Sensitivities:** if more of the current meter estate has to be refreshed on the short end of the 10-20 year range, true conversion is lower. **Confidence: Medium**. [60]

Initial contract tenor. Conclusion: 0.1 / 0.1 / 1.0 years. **Observed facts:** Calisen explicitly says that retailer energy orders are treated as **month-to-month contracts** for accounting purposes because the energy retailer can terminate subject to appropriate consideration. **Root business mechanism:** the legal term is short in accounting form, but that does not describe economic duration. **Bridge:** I therefore report the initial legal term as month-to-month rather than pretending the legal term equals asset life. **Sensitivities:** contract documentation and compensation terms. **Confidence: High**. [54]

Effective customer tenure. Conclusion: 10 / 15 / 20 years. **Observed facts:** Calisen says it continues to collect MPC when the underlying consumer moves to a new energy retailer unless the meter is removed, and its customer arrangements are modeled over around **15 years**, with smart meter UEL now **10-20 years** depending on contract. **Root business mechanism:** the meter stays in the property and typically continues earning across end-customer switching; revenue durability is anchored more by physical meter persistence and compensation mechanics than by the initial legal order term. **Bridge:** the central estimate aligns with the company's own average economic-life language. **Sensitivities:** premature meter removals, contract changes, and technology obsolescence. **Confidence: High**. [61]

Utilization exposure. Conclusion: Low / Moderate / Moderate. **Observed facts:** once a meter is installed it generally earns ongoing revenue unless removed, but portfolio growth depends on installation volumes and supplier relationships. **Root business mechanism:** asset utilization of the installed base is relatively sticky; service-capacity utilization in installation teams is more variable; customer usage itself does not meaningfully determine core rental revenue. **Bridge:** I classify the overall model as moderate because the installed base is stable but installation capacity and rollout pace still matter. **Sensitivities:** supplier rollout pauses, meter removals, and portfolio mix. **Confidence: Medium**. [62]

New-unit or new-customer payback. Conclusion: 5.0 / 6.5 / 8.5 years. **Observed facts:** average 2023 capex per smart meter was **£219**; ARPM was **£23.8**, with smart ARPM **£25.3**; longer-duration contracts carry lower annual rental but more guaranteed years. **Root business mechanism:** high upfront meter-plus-install spend is recovered through annual rental/data income over many years, not quickly. **Bridge:** even before deducting direct service cost and a maintenance-capex reserve, revenue/capex math already implies a long payback; the unit-level cash contribution shortens it somewhat only if operating cost per installed meter is very low. **Sensitivities:** actual gross contribution per meter, compensation on early removal, and contract mix. **Confidence: Medium**. [63]

B2B versus B2C profile. Conclusion: Predominantly B2B / Predominantly B2B / Predominantly B2B. **Observed facts:** Calisen works on behalf of energy retailers, which are the legal counterparties; households are end users, not the core customer. **Root business mechanism:** concentration and pricing power sit at supplier level. **Bridge:** the correct classification is predominantly B2B. **Sensitivities:** none material. **Confidence: High**. [64]

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Normalized revenue growth	4%	7%	10%	Triangulated	Medium	Revenue-generating meter additions plus data services, moderated for M&A/ rollout effects
Normalized EBITDA margin	68%	72%	76%	Calculated	Medium	Asset-rental economics dominate, with depreciation below EBITDA
Maintenance capex % of revenue	28%	35%	45%	Triangulated	Medium	Expensive funded meter estate requires real replacement and technology-refresh capital
Post-maintenance-capex conversion	34%	51%	63%	Calculated	Medium	Very high EBITDA partly offset by heavy sustaining asset economics

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Initial contract tenor	0.1 yrs	0.1 yrs	1.0 yrs	Disclosed	High	Accounting contract form is month-to-month
Effective customer tenure	10 yrs	15 yrs	20 yrs	Triangulated	High	Meter remains revenue-generating across supplier switching until removal
Utilization exposure	Low	Moderate	Moderate	Triangulated	Medium	Installed assets are sticky, but rollout and installation capacity matter
New-unit or customer payback	5.0 yrs	6.5 yrs	8.5 yrs	Triangulated	Medium	Large upfront meter spend recovered through modest annual rental/data contribution
B2B/B2C profile	Predominantly B2B	Predominantly B2B	Predominantly B2B	Disclosed	High	Energy retailers contract; homes consume

Plain-English business description: Calisen is a UK smart-meter asset owner whose core economics come from funding and owning millions of meters, then earning recurring rental and data income over long asset lives. [65]

Principal strengths: huge portfolio, strong EBITDA, and durable economics anchored in installed meters and supplier outsourcing. **Principal risks:** heavy capital intensity, rollout dependence, contract-form misreadings, and contamination from installation and non-core energy-transition activities. **Key unavailable information:** current clean post-2023 segment bridge and explicit sustaining capex split. **Most important diligence questions:** what share of capex is true estate maintenance, what are actual removal/compensation rates, and how much value sits in MAP versus installation/data? [66]

CARMA

Company and segment resolution

CARMA Corp is an Ontario-based Canadian submetering and billing platform. Its official website says it provides trusted submetering solutions, end-to-end service and support, and real-time energy insights from **over 11 million daily meter reads**. The contact page describes meter data management and customer care for submetering systems. LinkedIn says CARMA was founded in **1977**, was the first Measurement Canada-approved supplier of electronic submetering systems, has installed **over 350,000 meters**, and has **over 150,000 suites under contract**. The September 2025 CVC DIF acquisition announcement is even more useful: it says CARMA is one of Canada's largest submetering providers, metering **over 135,000 units** across more than **1,000** multi-residential and commercial buildings in Ontario, Alberta, British Columbia, and Nova Scotia, and that its submetering portfolio is supported by **long-term contracts with inflation protection**. [67]

Ownership history is clearer than financial disclosure. MNP states that in **August 2019**, TerraNova Partners LP and Michael Platt acquired majority control of Carma Industries Inc. and Carma Billing Services Inc.; the 2025 CVC announcement then says CVC DIF signed a definitive agreement to acquire CARMA from TerraNova. I therefore use the 2025 pending/acquired-by-CVC DIF basis as the ownership frame, but financial disclosure remains private and limited. [68]

The relevant segment is the company's **core Canadian submetering and billing operation** across residential, commercial, and institutional buildings. The economic unit is **one metered suite or unit under long-term contract**, with device ownership, data collection, and billing services all contributing to revenue. [69]

Economic unit, customer problem, alternatives, revenue, operating model, retention, and capital model

The customer problem is straightforward and highly recurring: condo boards, apartment owners, developers, and institutions want utility costs fairly allocated, consumption visible, and buildings run more efficiently. CARMA's own market pages emphasize lower maintenance fees, stable budgets, tenant satisfaction, and lower operating costs; the CVC release emphasizes lower utility costs for residents and sustainability benefits. This demand is less directly statutory than in some continental European heat-cost allocation markets, but it is strongly supported by owner economics and long-term building operations. [69]

Alternatives include master metering, manual allocation, or competing submetering vendors. But retention is supported by installed hardware, billing platform integration, resident onboarding, moving-in/moving-out workflows, and building-level contract administration. CARMA's billing-services pages and enrollment forms make clear that the company sits in the ongoing resident-account workflow, not just the initial hardware sale. That is why its economic profile is closer to ista/Techem than to installation-only contractors. [70]

The capital model is moderate. Compared with UK smart-meter MAPs, the devices are cheaper and building-level billing creates a larger service contribution relative to hardware cost. Compared with pure software, however, CARMA clearly has physical device ownership/management economics and customer-care workflows. Because no public financials are available, range estimation is inference-heavy and confidence is lower than for ista/Techem. [71]

Metric analysis

Revenue growth. Conclusion: 6% / 9% / 12% normalized. **Observed facts:** CARMA's public sources show ongoing geographic reach, over 135,000 units, more than 1,000 buildings, over 11 million daily reads, and a 2025 sponsor acquisition framing the company as a growth platform. **Root business mechanism:** growth comes from new building wins, retrofit conversions, additions of new suites under contract, and some cross-selling into essential building services. **Bridge:** I set the central estimate above mature European incumbents because CARMA is smaller and operating in a consolidation-friendly North American market, but below rollout-style hypergrowth because each win still requires installation and onboarding. **Sensitivities:** condo/apartment construction, retrofit penetration, and M&A. **Confidence: Medium-Low**. [72]

EBITDA margin. Conclusion: 28% / 33% / 38% normalized. **Observed facts:** no public EBITDA is disclosed. **Root business mechanism:** CARMA combines recurring billing and device economics with customer care and field/service functions. That should produce margins above installation-heavy contractors, but below the largest European incumbents because of smaller scale and likely higher customer-support intensity. **Bridge:** the central range is therefore below ista/Techem but solidly above commoditized field-service businesses. **Sensitivities:** cost to serve residents, bad debt handling, labor, and the mix between manufacturing/equipment versus billing. **Confidence: Low-Medium**. [73]

Maintenance capex as a percentage of revenue. Conclusion: 7% / 10% / 14%. **Observed facts:** no direct capex disclosure is public, but the company has large installed meter counts, daily read infrastructure, and long-term contracts. **Root business mechanism:** sustaining capex comes from meter replacement, communications nodes, server/platform refresh, and occasional removal/reinstallation in retrofit settings. **Bridge:** because unit hardware is materially cheaper than UK smart meter MAP assets and billing services are an important part of the revenue stack, maintenance capex should be closer to European building submetering than to Calisen/SMS. **Sensitivities:** ownership mix, retrofit rate, and communications technology changes. **Confidence: Low-Medium**. [74]

Post-maintenance-capex conversion. Conclusion: 50% / 70% / 82%. **Observed facts:** calculated from inferred margin and maintenance-capex assumptions. **Root business mechanism:** good recurring billing economics support conversion, but physical-device upkeep and account-service costs prevent the model from matching software-like cash conversion. **Bridge:** at the central point, $(33\% - 10\%)/33\% \approx 70\%$. **Sensitivities:** true customer-support intensity and sustaining IT spend. **Confidence: Low-Medium**. [69]

Initial contract tenor. Conclusion: 5 / 7 / 10 years. **Observed facts:** CVC explicitly says CARMA's submetering portfolio is supported by **long-term contracts with inflation protection**. **Root business mechanism:** building-level submetering agreements need enough duration to amortize equipment, billing setup, and resident administration. **Bridge:** I therefore anchor the central estimate in the upper-middle single-digit years. **Sensitivities:** new construction versus retrofit and province-specific utility practices. **Confidence: Medium**. [75]

Effective customer tenure. Conclusion: 8 / 12 / 16 years. **Observed facts:** the company has long-term contracts, over 1,000 buildings, and a resident-account workflow. **Root business mechanism:** once devices, billing, and move-in/move-out processes are embedded, customers typically stay longer than the original term because switching is disruptive for both property management and residents. **Bridge:** effective tenure therefore runs materially beyond legal tenor. **Sensitivities:** property sales, board changes, and customer-service quality. **Confidence: Medium-Low**. [76]

Utilization exposure. Conclusion: Low / Low / Moderate. **Observed facts:** CARMA emphasizes long-term contracted units and high-frequency daily reads. **Root business mechanism:** once a suite is metered and billed, revenue depends more on contract persistence than on daily consumption variation. **Bridge:** the model is low-exposure on asset utilization, though building occupancy and meter functionality still matter. **Sensitivities:** vacancy, resident payment friction, and communication failures. **Confidence: Medium-Low**. [77]

New-unit or new-customer payback. Conclusion: 3.0 / 4.5 / 6.0 years. **Observed facts:** no payback disclosure is public. **Root business mechanism:** suite-level meters and onboarding are cheaper than funded utility smart meters, while recurring billing and data services enhance annual gross contribution. **Bridge:** the central estimate is longer than pure billing software but shorter than UK MAPs. **Sensitivities:** installer access, retrofit complexity, and resident-account support cost. **Confidence: Low-Medium**. [78]

B2B versus B2C profile. Conclusion: B2B contract with residential end users / B2B contract with residential end users / B2B contract with residential end users. **Observed facts:** CARMA markets to condominiums, apartments, offices, retail, and institutions, while also handling resident billing relationships. **Root business mechanism:** the building owner/manager or developer is the true customer, while residents are end users and often invoice recipients. **Bridge:** I classify the model as B2B contract with residential end users. **Sensitivities:** commercial/institutional mix. **Confidence: High**. [79]

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Normalized revenue growth	6%	9%	12%	Inferred	Medium-Low	New building wins, retrofit conversions, and suite additions
Normalized EBITDA margin	28%	33%	38%	Inferred	Low-Medium	Recurring billing/device economics offset by customer-care and field-service costs

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Maintenance capex % of revenue	7%	10%	14%	Inferred	Low-Medium	Suite-meter replacement, communications refresh, and sustaining IT/ platform spend
Post-maintenance-capex conversion	50%	70%	82%	Calculated	Low-Medium	Good recurring billing economics with real but moderate sustaining capex
Initial contract tenor	5 yrs	7 yrs	10 yrs	Triangulated	Medium	Long-term inflation-linked building contracts
Effective customer tenure	8 yrs	12 yrs	16 yrs	Inferred	Medium-Low	Installed devices plus billing/ workflow integration extend revenue life
Utilization exposure	Low	Low	Moderate	Inferred	Medium-Low	Contracted suites continue earning with limited daily consumption dependence
New-unit or customer payback	3.0 yrs	4.5 yrs	6.0 yrs	Inferred	Low-Medium	Moderate suite-level hardware/ install cost recovered through recurring billing margin

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
B2B/B2C profile	B2B contract with residential end users	B2B contract with residential end users	B2B contract with residential end users	Disclosed	High	Property owners/ managers buy; residents are metered and billed

Plain-English business description: CARMA is a Canadian suite-level submetering and billing platform that installs meters in multifamily and commercial buildings, then monetizes them through long-term contracted billing and data workflows. [69]

Principal strengths: long-term inflation-protected contracts, broad Canadian footprint, and end-to-end billing workflow integration. **Principal risks:** limited public disclosure, customer-service execution, and regulatory variability by province. **Key unavailable information:** revenue, EBITDA, capex, and churn by cohort. **Most important diligence questions:** what is actual per-suite contribution, what loss rates occur at contract renewal, and what percentage of capex is device replacement versus growth? [80]

Archetypes, classification, and benchmark synthesis

Operating archetypes and company classification

The evidence supports **three economically distinct categories**.

The first and most representative archetype is **building submetering and billing platforms**. These companies install relatively low-cost devices at suite or apartment level, then earn recurring revenue through billing, allocation, data collection, and related workflow integration. The physical meter matters, but the billing/data layer is a major share of the customer value proposition. **ista** and **Techem** are the clearest examples, while **CARMA** is a smaller North American version with less public disclosure. [81]

The second archetype is **meter-asset owners and utility-facing funded smart meter platforms**. These companies own larger, more expensive metering assets under supplier-facing contracts and therefore show very high EBITDA margins but much heavier sustaining and replacement economics. **Calisen** and **Smart Metering Systems** fit here. [82]

The third category is **mixed metering/data/network service platforms**, where the relevant metering business exists but sits inside a larger services portfolio that includes network construction or ownership. **Energy Assets** fits this category. [13]

I classify the six companies as follows: **ista — Core representative; Techem — Core representative; CARMA — Secondary reference; Energy Assets — Mixed-business company; Smart Metering Systems — Mixed-business company; Calisen — Mixed-business company.** None of the six is treated as an "outlier" in the sense of being irrelevant to metering economics, but SMS and Calisen are economically different enough from apartment submetering that they should not anchor the core benchmark. [83]

Benchmark tables

Building submetering and billing platform benchmark

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
Revenue growth	4%	6%	8%	4%–12%	Medium	Indexation, device refresh, digital upsell, retrofit wins
EBITDA margin	44%	47%	51%	28%–53%	Medium	Recurring billing/data revenue on installed device estates
Maintenance capex % revenue	7%	9%	12%	7%–14%	Medium	Device replacement cycles, batteries, comms upgrades, sustaining IT
Post-maintenance conversion	74%	81%	86%	50%–87%	Medium	High recurring contribution with moderate sustaining hardware spend
Initial contract tenor	5 yrs	8 yrs	10 yrs	5–10 yrs	Medium-Low	Building-level multi-year contracts
Effective customer tenure	10 yrs	14 yrs	18 yrs	8–20 yrs	Medium	Installed-device and billing-workflow stickiness

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
Utilization exposure	Low	Low	Moderate	Low–Moderate	Medium	Assets earn while installed; daily usage matters less than appointment persistence
New-unit/ customer payback	2.5 yrs	4.0 yrs	5.5 yrs	2.5–6.0 yrs	Medium-Low	Moderate suite-level capex recovered through recurring billing and service margin
B2B/B2C profile	B2B contract with residential end users	B2B contract with residential end users	B2B contract with residential end users	Same for most peers	High	Owners/ managers buy, residents consume

The **core central range** above is supported primarily by **ista** and **Techem**, with **CARMA** widening the broader observed range on growth, margin, and disclosure. Revenue growth centers around **6%** because the installed base is mature and sticky, but still benefits from indexation, digital replacement, and upsell. EBITDA centers around the high-40s because these businesses monetize both hardware presence and recurring billing/data workflow, yet still carry field service and procurement costs. Maintenance capex centers around **9%** because the devices are real and must be refreshed, but they are cheaper and monetized more richly than utility-owned smart meters. Effective tenure centers around the mid-teens because switching is physically and operationally disruptive. [84]

Meter-asset owner and installer benchmark

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
Revenue growth	4%	7%	10%	5%–11%	Medium	Meter portfolio additions, rollout pace, supplier wins, data attach

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
EBITDA margin	45%	60%	74%	42%–76%	Medium	Asset-rental economics dominate EBITDA while depreciation sits below it
Maintenance capex % revenue	22%	30%	40%	20%–45%	Medium-Low	Expensive funded meter estate, replacement cycles, communications refresh
Post-maintenance conversion	33%	50%	63%	24%–63%	Medium-Low	Strong EBITDA but heavy economic sustaining spend
Initial contract tenor	0.1 yrs	3 yrs	8 yrs	month-to-month to 10 yrs	Low-Medium	Legal form differs from economic life; supplier frameworks vary
Effective customer tenure	8 yrs	13 yrs	18 yrs	8–20 yrs	Medium	Installed asset persistence and compensation/ removal mechanics
Utilization exposure	Low	Moderate	Moderate	Low–High	Medium	Installed meters are sticky but rollout and service capacity remain volume-sensitive

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
New-unit/ customer payback	4.0 yrs	6.0 yrs	8.0 yrs	4.0–8.5 yrs	Medium-Low	High upfront hardware-plus-install cost versus modest annual rental per meter
B2B/B2C profile	Predominantly B2B	Predominantly B2B	Predominantly B2B	Same for both peers	High	Energy suppliers and infrastructure counterparties buy the service

This second benchmark is supported by **Calisen** and **SMS**. The central difference versus the core submetering archetype is not revenue durability but **capital intensity**. EBITDA appears very strong because the asset is funded upfront and depreciation is below EBITDA. But maintenance capex is much higher because preserving revenue requires periodic replacement of a large, expensive meter estate. This is why post-maintenance conversion converges toward roughly **50%** despite apparently impressive EBITDA margins. [85]

Subvertical reference range

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
Revenue growth	4%	6%	9%	3%–12%	Medium	Installed-base additions, pricing, digitalization, selective M&A
EBITDA margin	33%	47%	72%	18%–76%	Medium	Service-led platforms sit below funded asset-owner models

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
Maintenance capex % revenue	8%	12%	35%	6%–45%	Medium-Low	Physical-device replacement burden is the main differentiator
Post-maintenance conversion	61%	78%	82%	24%–87%	Medium-Low	Billing/data-rich models convert better than asset-heavy MAP models
Initial contract tenor	5 yrs	7 yrs	8 yrs	month-to-month to 10 yrs	Low-Medium	Legal form varies materially by model
Effective customer tenure	8 yrs	14 yrs	18 yrs	5–20 yrs	Medium	Installed assets, data migration, billing integration, switching disruption
Utilization exposure	Low	Low-Moderate	Moderate	Low–High	Medium	Most assets keep earning while installed, but service-capacity sensitivity varies
New-unit/ customer payback	3.0 yrs	4.5 yrs	6.5 yrs	2.0–8.5 yrs	Medium-Low	Higher where meter ownership absorbs more upfront capital

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
B2B/B2C profile	B2B contract with residential end users	B2B contract with residential end users	Predominantly B2B	Predominantly B2B or B2B-with-residential-end-users	High	Legal customers are owners, managers, suppliers, or institutions

Metric-by-metric synthesis

Revenue growth is structurally higher in the capital-backed UK meter-asset models when rollout is active, because growth there is strongly a function of funded meter additions. In core apartment submetering, growth is steadier and more organic because it comes from portfolio penetration, indexation, digital upgrades, and adjacent billing/data services rather than pure funded asset rollout. That is why I center core growth at **6%**, not at the 10%+ reported rates seen in very strong recent years. [86]

EBITDA margin is the most misleading metric if used without first-principles analysis. The highest reported margins appear in Calisen, but that is exactly where sustaining asset economics are heaviest. Core submetering platforms like Techem and ista, by contrast, look slightly lower on EBITDA but often better after maintenance capex because their device estates are less capital-intensive relative to annual recurring revenue. [87]

Maintenance capex is the key differentiator. For ista/Techem/CARMA, sustaining capex mainly reflects scheduled device refresh, battery life, communications upgrades, and supporting software/data systems. For Calisen/SMS, sustaining capex must also absorb the economics of replacing expensive smart utility meters and comms hardware whose annual rental yield is relatively modest compared with installed cost. That alone explains why a blended benchmark would be economically incoherent. [88]

Contract tenor and effective tenure diverge materially. Calisen is the cleanest case: legal orders are month-to-month for accounting, but economic customer life is closer to asset life because revenue continues across supplier switching until removal. The apartment submetering platforms more likely start with explicit multi-year building agreements, but the broad conclusion is the same: **effective customer life is much longer than the stated initial term**. [89]

Utilization exposure is generally low to moderate across the subvertical because the asset earns while installed. The more important exposure is not daily consumption but **appointment persistence** and **service-capacity utilization**. The meter-asset-owner archetype is slightly more exposed because installation and rollout teams must stay productive, while the classic submetering archetype is steadier because recurring billing continues even if day-to-day usage fluctuates. [90]

Payback maps directly to asset intensity. Suite-level submetering and billing platforms typically recover a device and onboarding cost in about **3-5 years**. Utility smart-meter asset owners often need **5-8 years** because annual rent per meter is lower relative to funded capital per installed unit. [91]

Final conclusions, evidence gaps, and source list

The **typical customer problem** in core submetering is accurate and operationally manageable allocation of heating, water, or other utility costs inside multi-unit buildings, increasingly with digital transparency and energy-saving behavior support. The **typical revenue model** combines installed measurement hardware with recurring billing, monitoring, data, and compliance workflows. The **primary growth drivers** are digital replacement, portfolio penetration, indexation, adjacent data services, and selective new-building or retrofit wins. The **primary margin drivers** are installed-base density, low customer-switching frequency, and recurring billing/process income. The **principal maintenance-capex requirements** are device replacements, batteries, communications/network upgrades, and sustaining software/data infrastructure. **Typical contract and customer duration** diverge, but initial contracts are commonly multi-year while effective customer life is often a decade or more. **Typical utilization dynamics** are favorable because installed devices keep earning while in place. **Typical new-unit payback** is around **3-5 years** in core building submetering and **5-8 years** in funded smart-meter asset ownership. The **typical customer profile** is best described as **B2B contract with residential end users**. The **largest source of variation** among companies is asset intensity. The **lowest-confidence conclusion** in this report is the precise normalization of Energy Assets and CARMA EBITDA/capex because public segment disclosure is limited. [92]

The archetype-level takeaway is therefore simple. If the business is primarily a **building submetering and billing platform**, it behaves like a sticky, recurring, moderately capital-intensive service-and-device model with strong EBITDA and good post-maintenance cash conversion. If the business is primarily a **meter-asset owner**, it behaves more like a financed infrastructure asset with long duration and strong EBITDA but much heavier economic reinvestment needs. The same “metering” label can hide very different cash economics. [93]

Slide-ready business-model description:
Submetering providers install measurement devices in buildings and turn them into recurring billing, data, and compliance revenue; returns depend on contract stickiness, replacement capital, and workflow integration.
Word count: 26. [94]

Consolidated diligence priorities: first, obtain clean segment financials for Energy Assets and CARMA; second, separate growth capex from true sustaining replacement capex at SMS and Calisen; third, confirm actual contract duration and churn cohort data for ista and Techem; fourth, test whether digitalization capex currently booked as growth is economically required to preserve today’s revenue base. These four diligence points would change valuation more than any small adjustment to reported growth or EBITDA. [95]

Source list: Energy Assets official corporate pages and Companies House filings; ista corporate site, audited key figures, and progress-report materials; S&P and Fitch rating snippets on ista and Techem; Techem corporate site, sustainability report, and owner press release; SMS historical results and reports pages, FY2022 results, FY2025 interim update, and 2025 combination announcement; Calisen reports page, 2023 annual report, current site, and 2024 ownership announcement; CARMA official site, billing and contact pages, CVC DIF acquisition announcement, MNP transaction note, and LinkedIn company profile. [96]

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1 12 13 14 15 96 https://www.energyassets.co.uk/our-utility-services/our-business-services/energy-assets/
https://www.energyassets.co.uk/our-utility-services/our-business-services/energy-assets/

2 16 18 19 26 81 83 92 94 https://www.ista.com/corporate/
https://www.ista.com/corporate/

3 5 10 52 55 56 57 58 59 60 62 63 64 65 66 82 85 86 87 91 https://www.calisen.com/media/0jzkligx/calisen-smarter-energy-report-2023.pdf
https://www.calisen.com/media/0jzkligx/calisen-smarter-energy-report-2023.pdf

4 6 29 31 34 38 39 90 https://www.techem.com/content/dam/techem-no/documents/om-oss/techem_b%C3%A6rekraftsrapport_2025_EN.pdf.coredownload.inline.pdf
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7 11 95 https://find-and-update.company-information.service.gov.uk/company/12390729/filing-history
https://find-and-update.company-information.service.gov.uk/company/12390729/filing-history

8 54 61 89 https://www.calisen.com/media/ca3hxwpp/2022-calisen_annual_report_2022_04-04-23.pdf
https://www.calisen.com/media/ca3hxwpp/2022-calisen_annual_report_2022_04-04-23.pdf

9 20 21 23 24 25 27 84 88 https://www.ista.com/corporate/sustainability/key-figures/
https://www.ista.com/corporate/sustainability/key-figures/

17 https://www.cvc.com/media/news/2017/2017-07-26-cvc-capital-partners-fund-v-agrees-the-sale-of-its-investment-in-ista/
https://www.cvc.com/media/news/2017/2017-07-26-cvc-capital-partners-fund-v-agrees-the-sale-of-its-investment-in-ista/

22 https://www.fitchratings.com/research/corporate-finance/no-immediate-impact-on-cki-ratings-from-proposed-german-ista-buy-31-07-2017
https://www.fitchratings.com/research/corporate-finance/no-immediate-impact-on-cki-ratings-from-proposed-german-ista-buy-31-07-2017

28 41 https://www.techem.com/corp/en
https://www.techem.com/corp/en

30 32 40 42 https://www.techem.com/corp/en/news-and-media/press-releases/next-growth-phase-at-Techem
https://www.techem.com/corp/en/news-and-media/press-releases/next-growth-phase-at-Techem

33 36 https://www.fitchratings.com/research/corporate-finance/fitch-affirms-techem-at-b-outlook-stable-09-03-2023
https://www.fitchratings.com/research/corporate-finance/fitch-affirms-techem-at-b-outlook-stable-09-03-2023

35 37 93 https://www.spglobal.com/ratings/en/regulatory/article/-/view/type/HTML/id/3568419
https://www.spglobal.com/ratings/en/regulatory/article/-/view/type/HTML/id/3568419

43 https://www.smsenergy.com/results-reports-presentations/
https://www.smsenergy.com/results-reports-presentations/

44 45 46 47 48 49 50 https://d2ygkcw3bu7ib2.cloudfront.net/app/uploads/2023/03/SMS-Full-year-results-2022-RNS-FINAL.pdf
https://d2ygkcw3bu7ib2.cloudfront.net/app/uploads/2023/03/SMS-Full-year-results-2022-RNS-FINAL.pdf

[51] https://www.calisen.com/knowledge-hub/reports
https://www.calisen.com/knowledge-hub/reports

[53] https://www.calisen.com/knowledge-hub/news/eqt-and-gic-acquire-majority-stake-in-calisen
https://www.calisen.com/knowledge-hub/news/eqt-and-gic-acquire-majority-stake-in-calisen

[67] [69] [72] [73] [78] [79] https://carmacorp.com/
https://carmacorp.com/

[68] https://www.mnp.ca/en/services/corporate-finance/transactions/carma-corp-priority-submetering-solutions
https://www.mnp.ca/en/services/corporate-finance/transactions/carma-corp-priority-submetering-solutions

[70] https://carmacorp.com/carma-billing-services/
https://carmacorp.com/carma-billing-services/

[71] [74] [75] [76] [77] [80] https://www.cvc.com/media/news/2025/cvc-dif-to-acquire-carma-corp-a-leading-canadian-submetering-and-essential-building-services-platform/
https://www.cvc.com/media/news/2025/cvc-dif-to-acquire-carma-corp-a-leading-canadian-submetering-and-essential-building-services-platform/

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Operating Benchmark of Liquid Environmental Services

Executive summary and subvertical overview

Liquid environmental services is not one business model. The designated universe clearly separates into two structurally different archetypes: an **installed onsite-system model** centered on automated fresh-oil delivery, onsite storage, filtration, monitoring, and used-oil recovery at restaurant locations; and a **route, collection, treatment, and processing model** centered on pumping, hauling, treatment, recycling, disposal access, and route density across grease traps, septic, grit, wastewater, and related non-hazardous liquid waste streams. Restaurant Technologies is the clearest example of the first model, while Liquid Environmental Solutions, Waste Resources Management, and Wind River Environmental are variations of the second. The economics differ enough that a single blended benchmark would be misleading. [1]

Across the four companies, the most durable customer problem is not “waste” in the abstract. It is **keeping kitchens, food-service sites, wastewater systems, and related facilities operating without spills, sewer blockages, compliance failures, labor inefficiency, safety incidents, or unplanned downtime**. That makes recurring service demand real, but it does not make the economics identical. In the installed onsite-system model, revenue durability comes from embedded workflow integration and customer-site equipment. In the route-processing model, durability comes from recurring compliance needs, route density, treatment and disposal access, local permits, and the operational friction of switching vendors. [2]

The strongest disclosed economics in the universe come from Liquid Environmental Solutions, whose official operating materials show national scale of **1,020 employees, 670 field vehicles, 95 facility and partner locations, and 129,500 customer locations served**, while S&P described the post-transaction credit as supporting **ratings-adjusted EBITDA margins above 30%**. By contrast, Restaurant Technologies offers national scale and unusually sticky installed relationships—**45,000+ commercial kitchens, 41 depots, and 1,200+ employees**—but its model carries more customer-site equipment deployment and product-cost complexity, making reported growth and margin more sensitive to installation timing, oil price pass-through, and funding structure. WRM and Wind River both show meaningful vertical integration and M&A-driven platform building, but with less pure disclosure and, in Wind River’s case especially, a broader mix of septic, plumbing, drain, and municipal-industrial activities that contaminates archetype purity. [3]

Using July 18, 2026 as the analysis date, the most defensible normalized conclusions are these. The installed onsite-system archetype appears capable of **mid-single-digit to low-double-digit normalized growth**, **high-teens to mid-20s EBITDA margins**, **low maintenance capex as a percent of revenue but higher economic sustaining spend than accounting maintenance alone**, **roughly two-to-three-year customer payback**, and **effective customer lives well beyond initial legal terms**. The route-processing archetype appears capable of **high-single-digit to low-teens normalized growth**, **low-20s to low-30s EBITDA margins depending route density and treatment ownership**, **moderate maintenance capex tied to fleet and treatment infrastructure**, **strong post-maintenance conversion when density is established**, and **short legal terms but durable practical tenure in dense local markets**. These conclusions are driven less by reported history alone than by who owns the treatment bottleneck, who controls the route, what must be serviced on a recurring basis, and whether recovered-value streams augment margin or simply offset disposal cost. [4]

This report treats hard facts, calculations, inferences, and unavailable items separately. Because all four companies are private and public disclosure is uneven, several metrics—especially legal contract tenor, maintenance capex, and clean standalone EBITDA for mixed-service platforms—necessarily rely on triangulation and first-principles normalization rather than direct disclosure. Confidence is therefore highest for business-model mechanics, medium for broad margin and growth normalization, and lowest for legal contract term and strict maintenance-capex precision at the private-company level. [5]

Scope, definitions, and methodology

The scope is limited to four designated companies and no others as research-universe profiles: Restaurant Technologies, Liquid Environmental Solutions, Waste Resources Management, and Wind River Environmental. The analysis date is **July 18, 2026**, and all cited facts are taken from information publicly available on or before that date. When a disclosed metric comes from a prior period, that period is stated explicitly in the related narrative. [6]

For **revenue growth**, I separate structural growth drivers from perimeter effects. Acquisitions at LES, WRM, and Wind River are treated as perimeter growth unless there is evidence that the acquired capacity merely formalized preexisting internal throughput. At Restaurant Technologies, reported revenue growth is separated conceptually into customer count, installed units, service/delivery activity, fresh-oil pricing, and recovered-oil economics; commodity pass-through is not treated as the same thing as organic volume growth. [7]

For **EBITDA margin**, I reconstruct the bridge from revenue less product/consumables, field labor, transportation/route costs, facilities/treatment/disposal, repairs, customer support, and overhead. Reported private-credit margin disclosures are used where available, especially for LES, but normalized margins adjust for temporary integration loads, route buildout, and mix distortion from product pass-through or newly added facilities. [8]

For **maintenance capex**, I use four filters. First, I prefer direct disclosure where available, though it is sparse here. Second, I use replacement-cycle analysis for trucks, vac units, and treatment equipment. Third, I subtract obvious growth capex conceptually from total economic asset deployment when the business is clearly adding branches, customer-site systems, or new processing sites. Fourth, I cross-check against the asset intensity implied by the disclosed fleet and facility base. Because the companies are private, central estimates here are substantially inferential and confidence is lower than for growth and customer profile. [9]

For **post-maintenance-capex conversion**, I use the prompt's definition: **(EBITDA − Maintenance Capex) / EBITDA**. I state it explicitly as **not full free-cash-flow conversion**, because it excludes interest, taxes, working capital, lease principal, and growth capex. This distinction matters especially for Restaurant Technologies, where S&P's discussion of cash burn under high interest rates suggests that even a decent EBITDA profile can coexist with weak equity cash flow after financing and growth spending. [10]

For **initial contract tenor, effective tenure, utilization exposure, and payback**, the evidence is thinner and the analysis therefore relies on operating mechanics. Where no hard legal term is disclosed, I state that clearly and normalize around what the model economically requires: the minimum period needed to recover install or route-acquisition spending, the practical friction of switching, the recurrence of service demand, and the redeployability of assets. This is a first-principles benchmark study, not a comp-sheet exercise, so the ranges below prioritize mechanism over false precision. [11]

Individual company profile: Restaurant Technologies

First-principles company profile

Restaurant Technologies, Inc., commonly branded as **Restaurant Technologies** or **RTI**, is a private company headquartered in Mendota Heights, Minnesota. Public operating materials describe it as the leading provider of automated cooking-oil management for **over 45,000 commercial kitchens nationwide**, served through **41 depots** and **more than 1,200 employees**. S&P's 2022 credit commentary referred to the financing entity as **Double Eagle Buyer (d/b/a Restaurant Technologies)**, and Energy Capital Partners' portfolio materials indicate ECP owned the company as of July 2026. A 2022 deal report stated ECP acquired Restaurant Technologies from the infrastructure investing business of Goldman Sachs Asset Management. S&P also noted in 2022 that about **60% of revenue** came from quick-service restaurants. The latest reliable operating facts are therefore a mix of 2024-2026 company website disclosures and 2022-2024 credit commentary. [12]

The economic unit is best defined as the **customer location with installed Total Oil Management equipment**, because that unit ties together the up-front asset, recurring fresh-oil delivery, filtration and monitoring, used-oil pickup, and customer workflow integration. That is a better unit than a gallon of oil alone, because the customer relationship value depends on the installed storage, pump, filtration, and safe-disposal system—not just on commodity supply. Alternative units matter too: the depot route is the service-capacity unit, and gallons or pounds of oil are the usage unit. But for growth, retention, and payback, the installed location is the right core lens. [13]

The customer problem is persistent and operationally painful. Fryer-oil handling is labor intensive, messy, and hazardous if done manually; restaurants also need consistent oil quality, safe disposal, and dependable recycling. RTI's marketing proposition is not simply "we sell oil." It is "we automate your oil," offering bulk oil delivery, automated filtration and monitoring, hands-free used-oil disposal, and recycling. The economic buyer is usually a restaurant operator, chain procurement function, franchisee, or foodservice manager; the site operator is the kitchen staff; the end user benefit is safer, more consistent kitchen operations. Failure to solve the problem creates spill, burn, slip, labor, and food-quality risk. [14]

Customer alternatives exist, but they are weaker once RTI is installed. A restaurant can buy oil in jugs or choose a manual fresh-oil and rendering setup, and it can theoretically switch to another oil service provider. Yet RTI's installed equipment, automated level monitoring, workflow redesign, and closed-loop service materially raise switching disruption. The company explicitly emphasizes the elimination of jugs and rendering tanks and a full-service automatic system that delivers fresh oil and removes used oil in the same service loop. That physical embedding is why legal term and economic tenure diverge. A contract may not be exceptionally long, but practical tenure can be. [15]

The revenue model combines several streams: fresh cooking-oil supply, service/delivery, installed-system economics, used-oil pickup and recycling, and ancillary value from automated filtration/monitoring. The company's public materials do not provide a clean revenue split, but the disclosed offer makes clear that fresh-oil delivery and used-oil recovery are both part of the same integrated solution. That means reported revenue can be distorted by oil commodity price swings, by gross-versus-net presentation of recovered-oil value, and by installation timing. It also means customer-level contribution must be evaluated on the **whole relationship**, not on an equipment-rental line alone. [16]

The operating model is recurring and route-based after installation. RTI must monitor levels, schedule deliveries, service the installed system, collect used oil, and recycle it. The depot network is therefore economically important: the more dense the installed base around a depot, the better labor and truck absorption becomes. At the same time, the model is different from a pure pumping-and-treatment route business because RTI's route activity is tied to customer-site installed hardware and fresh-oil replenishment rather than only to pumping frequency. The onsite system also means customer throughput matters: a slower restaurant consumes less fresh oil and generates less used oil even if the installed asset remains under contract. [17]

Retention is driven by physical installation, workflow integration, safety value, chain consistency, and the operational nuisance of reverting to manual handling. Churn can still occur if a site closes, a franchise changes operators, a competitor underprices the service, or a location's fryer volume drops below the economics of an installed system. But the combination of customer-site hardware and labor savings makes the effective relationship much stickier than a generic delivered commodity. RTI's own survey-based marketing claim of average customer savings—**$250 per week**, **9 labor hours per week**, and **102 pounds of oil per week**—is directionally important because it shows why customers renew even where the legal term is not fully disclosed. [18]

The capital model includes customer-site tanks, pumps, filtration and monitoring components, installation labor, and supporting route and depot infrastructure. The key economic question is whether the company recovers this install investment during the initial term or only over longer customer lives. The answer is likely "usually over the full customer relationship, not purely the first invoice cycle," because RTI is an embedded-service platform, not an invoice-light brokerage. That helps explain why S&P highlighted cash-flow deficits and liquidity pressure in late 2024: the model can generate durable revenue visibility while still consuming cash if installation growth, equipment deployment, truck support, and financing costs outrun near-term reported EBITDA. [19]

Core metric analysis

Revenue growth. Conclusion: **Low 5%, central 8%, high 12% normalized revenue growth.** Observed facts: S&P expected low-teens growth in 2018; the business served 45,000+ kitchens by 2026; and S&P's 2022 commentary showed heavy exposure to QSR customers, which supports scale growth but not hyperscaling. Root mechanism: growth comes from new installed customer locations, same-store oil throughput, modest pricing, and recovered-oil economics, while commodity pass-through can inflate reported revenue without corresponding volume growth. Calculation bridge: I normalize below historical credit-case growth because mature national footprint and QSR concentration imply slower underlying unit growth than early expansion years, and because oil price pass-through should not be treated as organic volume. Sensitivities: restaurant traffic, chain rollout wins, depot infill, input oil pricing, and used-oil commodity values. Confidence: **Medium**, because scale is disclosed but organic-versus-commodity decomposition is not. [20]

EBITDA margin. Conclusion: **Low 18%, central 21%, high 25% normalized EBITDA margin.** Observed facts: RTI is a closed-loop provider that both delivers fresh oil and picks up used oil; S&P flagged ongoing cash-flow deficits in 2024, which implies that any healthy EBITDA must be considered separately from financing and growth burden. Root mechanism: margin is helped by route density, service stickiness, and recovered-oil value, but burdened by product cost, field service, depot/truck support, and the complexity of managing installed hardware. Calculation bridge: this margin range is lower than LES's disclosed >30% because RTI carries a much larger fresh-oil product component and customer-site equipment/service bundle. Sensitivities: commodity spreads, gross-vs-net recovered-oil presentation, depot density, and installation mix. Confidence: **Medium-Low**, because no clean current standalone EBITDA margin is publicly disclosed. [21]

Maintenance capex as a percentage of revenue. Conclusion: **Low 2.5%, central 3.5%, high 4.5%.** Observed facts: RTI operates 41 depots and serves 45,000+ kitchens, implying a meaningful installed-asset fleet plus route support assets. Root mechanism: maintenance capex is driven by replacement and refurbishment of customer-site systems and service vehicles, but growth installations must be excluded. Calculation bridge: a low-to-mid single-digit revenue burden fits a model with distributed equipment but relatively long-lived customer-site assets; however, economic sustaining spend may be higher than accounting maintenance because removal/reinstallation and non-capitalized service effort are part of preserving the installed base. Sensitivities: asset life, failure rates, truck age, and whether refurbishments are expensed or capitalized. Confidence: **Low**, because there is no direct disclosure. [22]

Post-maintenance-capex conversion. Conclusion: **Low 75%, central 83%, high 90%.** Observed facts: this is a calculation from the normalized EBITDA and maintenance-capex assumptions above, not full free cash flow; S&P's 2024 cash-burn commentary reinforces that accounting post-maintenance conversion can still overstate equity cash generation. Root mechanism: once a location is installed, the model is service-recurring and should convert reasonably well before financing and growth capex, but installations and fleet support mean it will not convert like a pure software-like subscription. Calculation bridge: central case uses roughly 21% EBITDA less 3.5% maintenance capex, divided by EBITDA. Sensitivities: true sustaining spend on installed hardware, route-density losses, and commodity-driven margin compression. Confidence: **Low-Medium**, because the formula is straightforward but the maintenance input is inferential. [23]

Initial contract tenor. Conclusion: **Low 3 years, central 4 years, high 5 years.** Observed facts: RTI discloses no public standard contract term, and the FAQ emphasizes flexibility and no minimum order requirement rather than legal duration. Root mechanism: because RTI funds or supports meaningful install economics and changes kitchen workflow, it likely seeks a multi-year initial term long enough to underwrite installation and customer-acquisition cost. Calculation bridge: I center around a mid-single-digit-year commercial-service term typical for embedded onsite equipment, while explicitly marking low confidence. Sensitivities: chain bargaining power, franchise structure, and whether equipment is customer-funded. Confidence: **Low**, because legal term is not disclosed. [24]

Effective customer tenure. Conclusion: **Low 6 years, central 9 years, high 12 years.** Observed facts: RTI has an installed, automated, hands-free system embedded in 45,000+ kitchens and markets measurable labor, safety, and waste savings. Root mechanism: effective tenure is longer than legal term because equipment is installed onsite, workflows are redesigned, and reverting to jugs/manual disposal is operationally unattractive. Calculation bridge: I extend customer life well beyond initial tenure because the system behaves more like embedded facility infrastructure than like a low-friction spot service. Sensitivities: site closures, franchisor changes, competitive repricing, and restaurant traffic declines. Confidence: **Medium**, because the mechanism is strong even if exact churn is undisclosed. [25]

Utilization exposure. Conclusion: **Moderate overall.** Observed facts: RTI's economic model requires customer-site deployment plus recurring oil throughput and route service. Root mechanism: asset utilization risk is lower than in a pure rental fleet because installed systems are embedded, but customer-usage risk remains meaningful—if a restaurant uses less fryer oil, revenue from supply and recycling activity can decline even while the asset stays installed. Calculation bridge: the most relevant KPI is probably gallons or pounds per installed kitchen and route density per depot, not simply "assets on rent." Sensitivities: restaurant traffic, menu mix, fryer intensity, and redeployment time when sites close. Confidence: **Medium** on classification, **Low** on exact KPI range because the company does not publish utilization data. [26]

New-unit or new-customer payback. Conclusion: **Low 1.5 years, central 2.5 years, high 3.5 years.** Observed facts: RTI advertises average customer savings of $250 per week and 9 labor hours per week; the model requires customer-site equipment and installation, with national chain deployment and depot-backed recurring service. Root mechanism: payback depends on the full relationship contribution—fresh-oil margin, service, and used-oil recovery—not on revenue alone, while the installed system and route support make the up-front investment real. Calculation bridge: if customer-site hardware, installation, and acquisition cost are in the low-to-mid five figures and annual unit cash contribution is several thousand dollars, a two-to-three-year payback is economically plausible. Sensitivities: install cost variability by kitchen layout, customer throughput, and used-oil value. Confidence: **Low-Medium**, because the savings evidence is real but RTI does not disclose unit-level contribution. [24]

B2B versus B2C profile. Conclusion: **Predominantly B2B.** Observed facts: RTI serves commercial kitchens, including quick-service and full-service restaurant chains, grocery delis, hotels, casinos, universities, and hospitals; S&P said about 60% of revenue came from QSRs in 2022. Root mechanism: the legal buyer is a business or institutional operator, and national-account concentration likely matters materially in pricing and rollout dynamics. Calculation bridge: residential exposure appears negligible. Sensitivities: none that would change the classification, though chain concentration affects bargaining power. Confidence: **High**. [27]

The table below summarizes my normalized conclusions for Restaurant Technologies. It is driven by the cited installed-system facts, national depot footprint, QSR concentration, service design, and the distinction between accounting conversion and full cash generation. [28]

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Normalized revenue growth	5%	8%	12%	Triangulated	Medium	New installed kitchens, same-site oil throughput, chain wins, modest pricing, commodity-pass-through normalization
Normalized EBITDA margin	18%	21%	25%	Inferred	Medium-Low	Integrated service value offset by oil product cost, route labor, depot/truck support, and install/service complexity
Maintenance capex % of revenue	2.5%	3.5%	4.5%	Inferred	Low	Replacement/ refurbishment of installed systems and service vehicles
Post-maintenance-capex conversion	75%	83%	90%	Calculated	Low-Medium	Decent recurring EBITDA before financing, moderated by distributed equipment upkeep
Initial contract tenor	3 yrs	4 yrs	5 yrs	Inferred	Low	Need to underwrite installation and customer-acquisition spend

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Effective customer tenure	6 yrs	9 yrs	12 yrs	Triangulated	Medium	Physical installation, workflow integration, safety/labor benefits, switching disruption
Utilization exposure	Moderate	Moderate	Moderate	Inferred	Medium	Revenue depends on installed sites staying active and consuming oil, not just equipment deployment
New-unit or customer payback	1.5 yrs	2.5 yrs	3.5 yrs	Triangulated	Low-Medium	Installed-equipment investment recovered from total relationship contribution, not oil revenue alone
B2B/B2C profile	Predominantly B2B	Predominantly B2B	Predominantly B2B	Disclosed	High	Commercial kitchen customers, especially chains and institutions

Plain-English business description: Restaurant Technologies installs and supports automated cooking-oil systems in commercial kitchens, then monetizes the recurring relationship through fresh-oil supply, filtration, service, and used-oil collection/recycling. [29]

Principal strengths: customer-site integration, strong practical retention, national depot network, safety/labor value proposition, and a closed-loop system that can benefit from both service revenue and recovered-oil value. Principal risks: commodity and gross/net presentation noise, financing burden, installation cash intensity, and concentrated exposure to restaurant traffic and chain procurement. Key unavailable information: current standalone revenue, adjusted EBITDA, maintenance capex, churn, and contract terms. Most important diligence questions: who funds installation, how recovered-oil value is booked, what percent of revenue is commodity pass-through, what true maintenance capex is by depot and truck class, and how much of customer life is contractually versus behaviorally protected. [30]

Individual company profile: Liquid Environmental Solutions

First-principles company profile

Liquid Environmental Solutions, usually branded **LES**, is a private, national non-hazardous liquid-waste company founded in 2002. The company's 2026 website describes it as "the nation's leader in non-hazardous liquid waste management," while the home page reports **1,020 employees, 670 field vehicles, 95 LES facility and partner locations, and 129,500 customer locations served**. The current operating scope includes drain line cleaning, grease trap and interceptor cleaning, oil-water-separator and grit-trap cleaning, used cooking oil collection, and wastewater disposal. The latest reliable operating period is therefore the 2026 company website, supplemented by earlier official acquisition releases and 2024-2025 ratings commentary. [31]

Ownership changed recently. Houlihan Lokey's June 2025 environmental-services market update said LES, then a portfolio company of **Audax Private Equity**, had **agreed to be acquired by Infrastructure at Goldman Sachs Alternatives**. S&P Market Intelligence separately reported in July 2025 that Audax agreed to sell Liquid Environmental Solutions to Goldman Sachs Asset Management. Because I do not have an official closing press release in the retrieved record, I treat **Goldman Sachs Alternatives / Goldman Sachs Asset Management** as the likely current owner as of July 2026 with **Medium confidence**, while clearly noting that the precise closing evidence in hand is indirect. [32]

The economic unit is best analyzed in three layers: the **route customer** for incremental payback, the **truck/route** for service-capacity utilization, and the **permitted treatment facility** for margin and strategic advantage. That is the correct structure because LES is not merely a brokered hauler. Its own operating materials emphasize collection, transportation, treatment, recycling, reclamation, and disposal, and historical releases quantified owned networks of branches and recycling facilities. The route customer matters for sales economics, but the facility matters for structural margin. [33]

The customer problem is recurring and compliance oriented. Restaurants, grocery and convenience stores, industrial and manufacturing facilities, car washes, and related sites generate grease-trap waste, grit, oily wastewater, used cooking oil, and other non-hazardous liquid streams that must be removed, documented, and processed properly. If the service is not performed, the customer faces sewer backups, foul odors, downtime, drain failure, regulatory problems, and environmental liability. LES's offer is therefore part sanitation, part compliance, part outsourced logistics, and part circular-economy processing. [34]

Customer alternatives do exist. A site can use another hauler, sometimes self-handle certain waste streams, or split providers across trap cleaning, used-oil collection, and industrial wastewater disposal. But LES's model gains force when it combines route service with treatment and beneficial recovery. Historical acquisitions make this explicit. In 2019, LES said the Carolinas Resource Recovery acquisition added two wastewater treatment facilities specializing in the recovery of brown grease as a green-energy feedstock, and the A-1 Restaurant Services acquisition added a treatment facility dedicated to grease-trap waste and used-cooking-oil processing. That reduces third-party dependence and increases capacity control. [35]

The revenue model is diversified within the route-processing archetype. Official materials show revenue from grease trap and interceptor cleaning, oil-water-separator and grit-trap service, used cooking oil collection, wastewater disposal, and drain cleaning. Historically, LES also highlighted beneficial recovery of materials from grease traps, used cooking oil, waste oil, industrial wastewater, and related streams. That means reported revenue can be a blend of service fees, hauling, treatment/disposal economics, and recovered-material value. The key distinction for normalization is that **volume growth, price growth, acquisition growth, and recovered-value movements are not the same thing**. [36]

The operating model is exactly what the second mandatory archetype requires: scheduled route density, service frequency, waste volumes, truck productivity, permitted treatment infrastructure, and recovery pathways. LES has historically disclosed both branch networks and treatment-facility networks, and the current website's "facility and partner locations" language implies a hybrid model in which some disposal or treatment capacity is partnered rather than owned. That matters economically. Owned, permitted plants increase margin and defensibility; partner locations increase flexibility but reduce the depth of structural moat. [37]

Retention is driven by recurrence and density. Grease traps do not clean themselves, used oil accumulates, oil-water separators and grit traps require service, and industrial wastewater cannot simply be ignored. Customers stay when the provider is reliable, compliant, and operationally easy to use. They leave when service quality slips, pricing gets aggressive, or a local operator with denser routes and better disposal economics undercuts the incumbent. In practice, LES's combination of national-account reach and physical processing infrastructure supports lower churn than a pure hauling broker would likely achieve. [34]

The capital model is meaningful but attractive once density is established. LES's disclosed 2026 asset base—670 field vehicles and 95 facility/partner locations—along with earlier disclosures of 24 wastewater treatment plants and more than 300 vacuum trucks, show that this is a capital-using service platform rather than an asset-light coordinator. But the capital is network shared, not dedicated to a single customer. That creates favorable incremental economics: once trucks and a plant are in place, adding a dense new customer on an existing route can pay back quickly. [38]

Core metric analysis

Revenue growth. Conclusion: **Low 7%, central 10%, high 14% normalized revenue growth.** Observed facts: LES has expanded from historical footprints of 49 service branches and 20 liquid-waste recycling facilities, then 54 branches and 24 treatment plants in 2019, to 95 facility and partner locations and 129,500 customer locations served by 2026. Root mechanism: growth comes from new route customers, increased route density, cross-sell across waste streams, facility throughput, and acquisitions, while recovered-material value and perimeter expansion can distort reported growth. Calculation bridge: I normalize below headline footprint growth because much of the branch/facility build has been acquisition-led. Sensitivities: M&A pace, national-account wins, local route density, plant capacity, and recovered-value pricing. Confidence: **Medium-High**. [38]

EBITDA margin. Conclusion: **Low 28%, central 31%, high 34% normalized EBITDA margin.** Observed facts: S&P said in July 2025 that the post-acquisition credit for "VRS Buyer Inc. (dba Liquid Tech ...)" supported **ratings-adjusted EBITDA margins above 30%**. LES's operating structure includes treatment, recycling, reclamation, and disposal, not just hauling. Root mechanism: margin is elevated when route density is high and owned or controlled treatment capacity captures the spread between collection price and disposal cost. Calculation bridge: the central estimate anchors directly to S&P’s >30% statement, with the range reflecting the mix between owned facilities and partner disposal plus integration effects. Sensitivities: plant utilization, third-party disposal dependence, route density, labor/fuel, and recovered-material value. Confidence: **High** for broad range, **Medium** for exact central point. [39]

Maintenance capex as a percentage of revenue. Conclusion: **Low 3.0%, central 4.0%, high 5.0%.** Observed facts: LES disclosed 670 field vehicles and a national footprint of facility and partner locations; earlier releases identified 24 wastewater treatment plants and 300+ vacuum trucks. Root mechanism: maintenance capex is driven by vac-truck replacement, pumps, tanks, pretreatment infrastructure, and environmental/safety upkeep at processing sites. Calculation bridge: a mid-single-digit revenue burden is consistent with a scaled mobile fleet plus treatment assets, but well below total growth capex in acquisitive or expanding periods. Sensitivities: truck age profile, owned-vs-partner plant mix, and regulatory capital requirements. Confidence: **Medium-Low**, because there is no direct maintenance-capex disclosure. [38]

Post-maintenance-capex conversion. Conclusion: **Low 82%, central 87%, high 91%.** Observed facts: this is calculated from the normalized margin and maintenance ranges, not full free cash flow. Root mechanism: the model should convert well after maintenance because routes recur and the network is shared across many customers, but treatment assets prevent it from converting like a pure brokered service. Calculation bridge: central case approximates 31% EBITDA and 4% maintenance capex. Sensitivities: true sustaining capex for treatment, environmental compliance spending, and route inefficiency in new markets. Confidence: **Medium-Low**. [40]

Initial contract tenor. Conclusion: **Low 1 year, central 2 years, high 3 years.** Observed facts: LES does not publicly disclose standard legal terms. Root mechanism: grease trap, used-oil, and related service contracts are usually operational-service relationships with recurring schedules rather than long finance-like leases; the company’s economics rely more on route density and service continuity than on very long initial legal terms. Calculation bridge: I center around short-to-medium commercial-service agreements with likely auto-renewal features, while recognizing that national accounts may negotiate different terms. Sensitivities: customer type, national-account procurement, and local competitive intensity. Confidence: **Low**. [31]

Effective customer tenure. Conclusion: **Low 5 years, central 7 years, high 9 years.** Observed facts: LES serves 129,500 customer locations and markets itself as the partner for life on recurring non-hazardous liquid-waste needs. Root mechanism: effective tenure exceeds initial term because the service need is permanent and recurring, but retention is weaker than RTI’s installed-equipment model because customers can more readily switch route providers if service disappoints. Calculation bridge: central tenure reflects recurring compliance demand and moderate switching friction without assuming extreme lock-in. Sensitivities: service reliability, local route competition, and disposal access in each market. Confidence: **Medium**. [34]

Utilization exposure. Conclusion: **Moderate to High, central Moderate-High.** Observed facts: LES relies on 670 field vehicles and a network of facilities and partners. Root mechanism: idle trucks and underfilled routes directly impair economics; underutilized facilities dilute fixed-cost absorption. On the other hand, recurring service frequency and broad end-market diversity reduce cycle risk relative to project-based hauling. Calculation bridge: the most relevant KPIs are route density, gallons per stop, and facility throughput. Sensitivities: customer-site volume changes, route balance, labor availability, and plant capacity. Confidence: **Medium** on classification, **Low** on exact KPI range. [31]

New-unit or new-customer payback. Conclusion: **Low 0.5 years, central 1.0 year, high 2.0 years** for an incremental route customer added to an existing dense market; longer for new-market branch buildout. Observed facts: the network is national and asset shared, with substantial existing fleet and facility infrastructure. Root mechanism: most incremental customer acquisitions in an established route require modest incremental servicing cost compared with a full new-truck or new-plant investment. Calculation bridge: I explicitly do **not** allocate an entire facility to one new customer; the relevant payback is the incremental route-account contribution after labor, transportation, and a maintenance reserve. Sensitivities: whether the customer sits on an existing dense route, service frequency, and disposal complexity. Confidence: **Medium-Low**. [37]

B2B versus B2C profile. Conclusion: **Predominantly B2B.** Observed facts: LES targets restaurants, grocery and convenience stores, industrial and manufacturing facilities, automotive maintenance, power plants, petrochemical sites, shipbuilding/repair, and even environmental-services counterparties. Root mechanism: the legal buyer is overwhelmingly commercial or industrial, not residential. Calculation bridge: while a few waste streams may originate from broad municipal or utility ecosystems, the company's invoiced relationships are business-to-business. Sensitivities: none that would change the classification. Confidence: **High**. [41]

The table below summarizes my normalized conclusions for LES. It reflects the best disclosed margin evidence in the universe, together with LES's explicitly national route-and-processing infrastructure and the distinction between acquisition-led footprint expansion and true organic route density growth. [42]

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Normalized revenue growth	7%	10%	14%	Triangulated	Medium-High	Route density, cross-sell, facility throughput, national-account wins, excluding acquisition/perimeter effects
Normalized EBITDA margin	28%	31%	34%	Triangulated	High	High route density plus treatment/recovery ownership and recurring compliance service

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Maintenance capex % of revenue	3.0%	4.0%	5.0%	Inferred	Medium-Low	Fleet replacement, treatment infrastructure upkeep, environmental/ safety capital
Post-maintenance-capex conversion	82%	87%	91%	Calculated	Medium-Low	Shared-network economics and strong EBITDA offset by fleet/ plant upkeep
Initial contract tenor	1 yr	2 yrs	3 yrs	Inferred	Low	Recurring service contracts, usually shorter legally than their practical duration
Effective customer tenure	5 yrs	7 yrs	9 yrs	Triangulated	Medium	Recurring compliance need, route reliability, vendor switching friction
Utilization exposure	Moderate	Moderate-High	High	Inferred	Medium	Route fill, gallons per stop, and plant throughput drive profitability
New-unit or customer payback	0.5 yrs	1.0 yr	2.0 yrs	Triangulated	Medium-Low	Incremental route accounts are added to shared fleet and plant infrastructure

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
B2B/B2C profile	Predominantly B2B	Predominantly B2B	Predominantly B2B	Disclosed	High	Commercial and industrial counterparties dominate

Plain-English business description: Liquid Environmental Solutions is a scaled national route-and-processing company that collects non-hazardous liquid waste, treats or channels it through controlled infrastructure, and monetizes recurring compliance and recovery needs. [43]

Principal strengths: dense recurring route economics, strong disclosure versus private peers, national account reach, and meaningful treatment/recovery infrastructure. Principal risks: integration from acquisitions, third-party partner dependence in some markets, exposure to fuel/labor/throughput utilization, and incomplete public evidence on current legal terms and sustaining capex. Key unavailable information: standalone revenue, exact organic growth, current owned-vs-partner facility mix, and churn by end market. Most important diligence questions: what percentage of volume is processed internally, what fraction of EBITDA comes from recovered-material spreads, how much of "95 facility and partner locations" is actually owned, and how quickly new markets reach route-density breakeven. [44]

Individual company profile: Waste Resources Management

First-principles company profile

Waste Resources Management, LLC, or **WRM**, is a private, vertically integrated environmental-services platform focused on non-hazardous liquid waste collection, processing, recycling, and transport. Its official website says WRM owns and operates a portfolio of companies focused on liquid waste, with **grease trap waste as the nexus waste stream**, and serves **over 10,000 clients** through SouthWaste Disposal, Silver City Processing, and McDonald Farms Enterprises. The company's service area page identifies operations in **Florida, Nevada, Texas, and Colorado**. A January 2024 WRM announcement said the company joined **Ridgewood Infrastructure**, and WRM's partner page confirms Ridgewood as a key partner. A May 2026 WRM press release said the company employed **over 300 people** and had agreed to acquire the majority of Darling Ingredients' trap-grease business, expanding its footprint in **13 U.S. metro markets**. [45]

The legal and operating-identity challenge with WRM is segment purity. WRM itself is the platform holding company, but one of its operating brands—McDonald Farms—also offers bulk water, roll-off containers, rail services, and broader environmental services. That means the relevant liquid-environmental-service segment is not perfectly isolated. The best way to handle that contamination is not to pretend it is absent, but to reduce confidence where mix could distort margin and capital-intensity conclusions. The latest reliable operating facts are from 2024-2026 official WRM materials. [46]

The economic unit, for benchmarking purposes, is the **incremental route customer connected to WRM's collection and processing network**, with a secondary lens on the **processing or transfer node**. That is because WRM's infrastructure is clearly shared across customers and because the company explicitly describes itself as vertically integrated. A single customer does not "own" a whole facility economically;

rather, customer value depends on fitting into an existing network that can collect, treat, dewater, recycle, or transport the waste stream efficiently. [47]

The customer problem is similar to LES's, but WRM appears more deliberately focused on **grease-trap-centered liquid-waste ecosystems**. WRM says grease-trap waste is the nexus stream, while other premium liquid waste streams are managed market by market to maximize efficiency and profitability. That indicates a strategy of building local density around grease, then layering in adjacent services such as hydro jetting, trap repair and replacement, plumbing, septic, sludge, and vacuum-truck support. The buyer is typically a business or institutional site operator—restaurants, grocery, hotels/casinos, prisons, stadiums, hospitals, school districts, and commercial car washes are specifically named. [48]

Customer alternatives are straightforward in theory but narrower in practice when vertical integration is real. Customers can hire another grease or liquid-waste hauler, self-perform some maintenance, or unbundle services. Yet WRM's portfolio logic is that the best economics come when the same platform can collect, transport, process, and market the recovered outputs. The company states that it recycles certain streams into renewable-energy feedstocks, compost and soil amendments, animal-feed and animal-health compounds, and dust-control products. That increases the chance that recovered value is an economic contributor, not merely a disposal cost offset. [49]

The revenue model is therefore mixed across route service, processing, recovery, and ancillary field work. SouthWaste focuses on grease-trap maintenance and liquid-waste collection; Silver City adds grease and grit trap cleaning, septic maintenance, and plumbing; McDonald Farms includes wastewater treatment and remediation, vacuum trucks, grease and grit trap services, groundwater disposal, hauling, and other support services. That breadth is a strength for customer wallet share, but it also means reported margin can vary meaningfully by branch mix and stream mix. [50]

The operating model is dense-market, shared-asset, and explicitly vertical. WRM is not presented as a simple contractor using only third-party disposal. Instead, the company describes itself as owning and operating a portfolio focused on processing, recycling, and transport. The Darling transaction is especially important here. Management said the deal would expand WRM's footprint in 13 metro markets and drive "further vertical integration across the WRM platform," which strongly suggests that treatment capacity and collection density are core to economic returns. [51]

Retention should be solid but not as strong as an installed-equipment lease-like model. Customers remain where outgoing service is reliable, emergency response is strong, and the platform can solve multiple related problems. They leave where service misses, line cleaning/plumbing competitors take share, or waste-stream economics shift locally. Because WRM's official positioning stresses "around the clock response" and portfolio breadth, I infer that customer stickiness is driven by integrated service bundles and local route reliability rather than by long legal lockups. [48]

The capital model includes vac trucks, specialty service units, processing assets, and possibly transfer or terminal infrastructure depending on brand. It is more capital-intensive than a broker, less customer-site-dedicated than RTI, and likely slightly less standardized than LES because of the broader McDonald Farms asset footprint. The good news is that much of that capital is shared across customers, making incremental route-customer payback relatively quick in dense markets. The bad news is that mix complexity lowers precision in estimating true maintenance capex. [46]

Core metric analysis

Revenue growth. Conclusion: **Low 7%, central 10%, high 13% normalized revenue growth.** Observed facts: WRM has over 10,000 clients, operates in four states, joined Ridgewood in 2024, and in 2026 announced a transaction that would expand its footprint in 13 metro markets. Root mechanism: growth comes from local route density, cross-selling around grease-centered customer relationships, and acquisitions that expand market coverage. Calculation bridge: the normalized range strips out the 2026 Darling perimeter step-up and focuses on what a vertically integrated mid-market platform can grow organically in dense areas. Sensitivities: M&A timing, branch density, and recovered-value markets. Confidence: **Medium**. 52

EBITDA margin. Conclusion: **Low 22%, central 26%, high 30% normalized EBITDA margin.** Observed facts: WRM positions itself as vertically integrated, with grease-trap waste as the nexus stream and recovered-value outputs across energy, compost, animal-feed, and dust-control applications. Root mechanism: margin should benefit when WRM controls collection and processing economics, but the platform's broader service mix and McDonald Farms contamination likely keep it below LES's clean >30% range on a consolidated basis. Calculation bridge: I anchor the range between a lower-margin field-services operator and a higher-margin treatment-enabled platform. Sensitivities: segment mix, disposal control, labor/fuel, and the percentage of revenue from ancillary plumbing or bulky support services. Confidence: **Medium-Low**. 53

Maintenance capex as a percentage of revenue. Conclusion: **Low 3.5%, central 4.5%, high 6.0%.** Observed facts: WRM owns operating companies involved in transport, processing, recycling, vacuum trucks, wastewater remediation, and related infrastructure. Root mechanism: maintaining route capacity and processing capability requires periodic truck replacement, pump/tank upkeep, and sustaining plant or terminal capital. Calculation bridge: the broad asset mix likely pushes maintenance capex slightly above LES if lower-scale local operations dominate, though this is a low-confidence estimate because the company discloses no capex. Sensitivities: asset age, McDonald Farms infrastructure requirements, and whether certain remediation or rail-linked assets sit inside the consolidated perimeter. Confidence: **Low**. 46

Post-maintenance-capex conversion. Conclusion: **Low 73%, central 83%, high 88%.** Observed facts: this is calculated from the margin and maintenance assumptions above. Root mechanism: shared infrastructure helps conversion, but mix complexity and heavier sustaining needs than a pure collection-only model restrain it. Calculation bridge: central case approximates 26% EBITDA and 4.5% maintenance capex. Sensitivities: true treatment-asset sustaining burden and branch mix. Confidence: **Low-Medium**. 54

Initial contract tenor. Conclusion: **Low 1 year, central 2 years, high 3 years.** Observed facts: WRM does not disclose contract terms publicly. Root mechanism: the service mix suggests ordinary recurring service agreements rather than long lease-style arrangements, though bundled grease/plumbing/emergency relationships can carry auto-renewing commercial terms. Calculation bridge: I use short-to-medium commercial-service agreements as the most defensible structure. Sensitivities: end market, emergency-service content, and local competition. Confidence: **Low**. 48

Effective customer tenure. Conclusion: **Low 4 years, central 6 years, high 8 years.** Observed facts: WRM serves more than 10,000 clients across essential recurring waste streams and ancillary services. Root mechanism: practical tenure should exceed legal term because the service need recurs and multi-service vendors become embedded in customer operations, but switching remains easier than in an onsite-installed model. Calculation bridge: I place WRM slightly below LES on tenure because of somewhat less clearly disclosed national-account scale and more mixed-service local operating-company structure. Sensitivities: service quality, density advantages, and share of wallet from ancillary plumbing/jetting. Confidence: **Medium-Low**. [49]

Utilization exposure. Conclusion: **Moderate to High, central Moderate-High.** Observed facts: WRM's model depends on collection, transportation, processing, and recycling assets, and the Darling transaction was justified partly by vertical integration and market expansion. Root mechanism: underfilled routes, underutilized processing, and weak local density all hurt returns sharply; conversely, tight market density creates strong operating leverage. Calculation bridge: the relevant KPIs are stops per route, gallons per haul, and processing-node throughput. Sensitivities: local route balance, metro density, and premium-stream mix. Confidence: **Medium** on classification. [55]

New-unit or new-customer payback. Conclusion: **Low 0.75 years, central 1.25 years, high 2.25 years** for an incremental customer on existing infrastructure. Observed facts: WRM already operates scaled brands and a shared network, and management emphasized that the Darling transaction would create new opportunities for organic growth across the platform. Root mechanism: incremental customers on an existing route should be attractive because they leverage shared truck and processing capacity; new-market entries remain far slower. Calculation bridge: I reserve a maintenance-capex burden in the denominator and exclude allocating a whole facility to one account. Sensitivities: route adjacency, waste type complexity, and emergency-service requirements. Confidence: **Low-Medium**. [56]

B2B versus B2C profile. Conclusion: **Predominantly B2B, with some mixed exposure.** Observed facts: WRM explicitly lists restaurants, grocery stores, hotels/casinos, prisons, stadiums, hospitals, school districts, and commercial car washes. Some operating brands also provide residential septic services, but the platform's named client universe is overwhelmingly commercial or institutional. Root mechanism: the invoice recipient and decision-maker are usually business or institutional operators. Calculation bridge: I classify WRM as predominantly B2B but note contamination from mixed-service subsidiaries. Sensitivities: residential share by subsidiary. Confidence: **Medium-High**. [50]

The table below summarizes my normalized conclusions for WRM. Because disclosure is thinner and the portfolio includes some non-core or adjacent services, several rows are more inferential than for LES and should be read with correspondingly lower precision. [57]

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Normalized revenue growth	7%	10%	13%	Triangulated	Medium	Route density, cross-sell around grease nexus, excluding acquisitions

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Normalized EBITDA margin	22%	26%	30%	Inferred	Medium-Low	Vertical integration and recovered-value upside offset by mixed-service platform contamination
Maintenance capex % of revenue	3.5%	4.5%	6.0%	Inferred	Low	Vac-truck and processing-asset replacement plus broader operating-asset upkeep
Post-maintenance-capex conversion	73%	83%	88%	Calculated	Low-Medium	Shared network supports conversion, but asset mix is heavier and less pure
Initial contract tenor	1 yr	2 yrs	3 yrs	Inferred	Low	Recurring commercial service relationships, likely auto-renewing and localized
Effective customer tenure	4 yrs	6 yrs	8 yrs	Triangulated	Medium-Low	Integrated service bundle and recurring need support retention, but switching remains feasible

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Utilization exposure	Moderate	Moderate-High	High	Inferred	Medium	Route fill and processing-node throughput are core profit drivers
New-unit or customer payback	0.75 yrs	1.25 yrs	2.25 yrs	Triangulated	Low-Medium	Incremental dense-market customers leverage shared route and processing infrastructure
B2B/B2C profile	Predominantly B2B	Predominantly B2B	Mixed B2B/B2C	Triangulated	Medium-High	Commercial/institutional clients dominate, with some subsidiary residential activity

Plain-English business description: WRM is a grease-centered, vertically integrated liquid-waste platform that collects, processes, transports, and recycles non-hazardous liquid waste while selling adjacent services around the same customer relationship. [55]

Principal strengths: explicit vertical integration, clear focus on grease-trap-centered waste streams, shared network economics, and sponsor backing from Ridgewood Infrastructure. Principal risks: mixed-business contamination, lower disclosure depth, integration risk from acquisitions, and uncertainty around exact treatment ownership and sustaining capex. Key unavailable information: revenue, EBITDA, margin by brand, organic growth, churn, and contract terms. Most important diligence questions: what share of EBITDA comes from processing versus service, how much of McDonald Farms is in-scope liquid environmental services, what owned-permit base exists by market, and how fast the Darling asset acquisition can be densified economically. [58]

Individual company profile: Wind River Environmental

First-principles company profile

Wind River Environmental is a private environmental-services platform based in Marlborough, Massachusetts. Gryphon Investors announced a majority investment in the company in April 2017, and Gryphon's current portfolio page still listed Wind River Environmental as a portfolio company as of July

2026. Wind River's own materials describe it as the leading provider on the East Coast, with **over 1,100 experts, 1,000 vehicles, and 100 local teams**, operations across **16 states**, **more than 90 acquisitions**, and ownership of **16 treatment facilities** processing a combined **more than 200 million gallons of waste annually**. The company also describes itself as having a **75+ year legacy of service**. [59]

Segment purity is the central issue with Wind River. This is not a pure grease or non-hazardous liquid-waste hauler. The company serves residential, business/commercial, and municipal/industrial customers and offers septic pumping, grease-trap cleaning, drain cleaning, plumbing, catch-basin service, sludge removal, hydroexcavation, trenchless pipelining, pump sales, portable-toilet rentals, septic-tank sales, and treatment-facility management. That breadth makes Wind River highly relevant to the route-processing archetype, but also a **mixed-business company** whose consolidated economics are broader than grease-trap or commercial liquid-waste service alone. [60]

The economic unit therefore differs by service line. For benchmarking the designated archetype, the best unit is the **recurring route or service customer tied to Wind River's collection and treatment network**, not the whole company. For residential septic, the "customer relationship" may recur on multi-year pumping cycles; for commercial grease and municipal-industrial work, the unit looks more like a recurring service account on a dense local route. At the network level, the treatment facility is again a structural margin unit. [61]

The customer problem is broad but always essential. Homes and businesses with septic systems require pumping, maintenance, and repair. Restaurants and chains require grease-trap service. Municipal and industrial customers require sludge removal, lift-station work, catch-basin cleaning, wastewater solutions, and sometimes treatment-facility management. If these services fail, the consequences include backups, flooding, odors, non-compliance, and operational interruption. Wind River's own local-market pages emphasize essential wastewater services and the ability to handle emergencies. [62]

Customer alternatives vary by service. Residential septic customers have local alternatives and can defer pumping for some period, though not indefinitely. Commercial grease and municipal-industrial customers can switch vendors more often, but route reliability, emergency capability, permit access, disposal access, and paperwork quality all matter. Wind River's marketplace proposition is that the platform's scale, treatment ownership, and family-of-brands structure provide "local team, strong backing" and "complete coverage." [63]

The revenue model is highly diversified. Residential service includes septic pumping, inspections, additives, repair, installation, and drain-field work. Business/commercial service includes grease traps, preventative maintenance, sludge removal, waste disposal, hydroexcavation, and septic for franchises and chains. Municipal/industrial includes lift stations, catch basins, sludge removal, waste disposal, treatment-facility management, vacuum-truck services, and disaster recovery. Because that mix includes both recurring route work and more project-like specialty services, margin and growth are structurally noisier than at a cleaner pure-play like LES. [64]

The operating model is route-and-processing heavy, but with substantial adjacency. The East Coast scale is unusually large for this subvertical: more than 90 acquisitions, 100 local teams, 1,000 vehicles, and 16 treatment facilities. The treatment-facility page also highlights a food-and-beverage wastewater solution in which waste is hauled to anaerobic digesters for sustainable energy creation, and the company's Earth

Farms composting history shows that Wind River also monetizes organic-waste recovery and compost pathways where permitted and economical. [65]

Retention is strong in practice but mixed by customer type. Residential septic relationships can last for years because households prefer trusted local vendors and pumping cycles repeat. Commercial and municipal relationships are also recurring, but route-service quality matters a great deal and switching costs are lower than in the installed onsite-system model. Wind River's acquisition page helps here: it explicitly seeks businesses with both commercial and residential customers and core services including septic pumping, grease traps, and wastewater treatment, typically with **$2 million to $30 million** of annual revenue. That acquisition target tells us the platform is stitching together sticky local operators rather than building only from greenfield branches. [66]

The capital model is heavier than it first appears. Wind River owns a large vehicle fleet and treatment-facility portfolio and runs emergency and specialty services that likely require more varied equipment than a simpler grease-route operator. At the same time, most of that capital is shared and locally densified through acquisitions. The operating economics therefore favor incumbent density and treatment ownership, but maintenance capex and working complexity are almost certainly higher than at a narrower pure-play route processor. [67]

Core metric analysis

Revenue growth. Conclusion: **Low 6%, central 9%, high 12% normalized revenue growth.** Observed facts: Wind River has completed more than 90 acquisitions and operates across 16 states; its 2023 and 2024 acquisition releases continued to describe the company as a consolidator with a vertical-integration model. Root mechanism: reported growth is likely heavily acquisition influenced, while underlying organic growth comes from density, cross-sell, and recurring wastewater-service demand. Calculation bridge: I normalize organic growth below the company's likely reported growth because acquisition activity is a major perimeter driver. Sensitivities: acquisition pace, local-route share gains, weather/emergency work, and residential septic cycling. Confidence: **Medium**. [68]

EBITDA margin. Conclusion: **Low 18%, central 22%, high 26% normalized EBITDA margin.** Observed facts: Wind River owns and operates 16 treatment facilities and a large vehicle fleet, but also sells broader services such as plumbing, trenchless, and residential work. Root mechanism: treatment ownership and route density support attractive margins, but the mixed service mix and specialty labor content should lower consolidated EBITDA versus a cleaner high-density processor like LES. Calculation bridge: I center the range in the low 20s because the company has genuine infrastructure advantages, yet wider service breadth dilutes pure route-processing economics. Sensitivities: mix of residential/commercial/municipal, emergency work, and facility utilization. Confidence: **Medium-Low**. [69]

Maintenance capex as a percentage of revenue. Conclusion: **Low 4.0%, central 5.5%, high 7.0%.** Observed facts: Wind River disclosed 1,000 vehicles and 16 treatment facilities. Root mechanism: the fleet is large, specialized, and maintenance-hungry, while treatment facilities also require environmental, safety, and equipment upkeep. Calculation bridge: a higher maintenance-capex burden than LES is justified by fleet diversity and mixed-service complexity. Sensitivities: truck age, specialty-equipment content, and regulatory capex. Confidence: **Low-Medium**. [70]

Post-maintenance-capex conversion. Conclusion: **Low 68%, central 75%, high 85%.** Observed facts: this is calculated from the margin and maintenance assumptions above and is explicitly not full free cash flow. Root mechanism: recurring route economics help conversion, but heavier fleet/facility upkeep and broader service complexity depress it relative to cleaner national processors. Calculation bridge: central case approximates 22% EBITDA and 5.5% maintenance capex. Sensitivities: true sustaining spend and mix of specialty/project services. Confidence: **Low-Medium**. [71]

Initial contract tenor. Conclusion: **Low 0.5 years, central 1 year, high 2 years.** Observed facts: Wind River does not publicly disclose standard legal terms, and much of its service mix appears operational rather than lease-like. Root mechanism: many services are likely governed by short service agreements, preventative-maintenance schedules, or recurring callout relationships rather than long locked terms. Calculation bridge: I therefore assume short initial legal duration, especially for residential and local commercial services. Sensitivities: municipal contracts, franchise chain agreements, and treatment-management relationships. Confidence: **Low**. [62]

Effective customer tenure. Conclusion: **Low 5 years, central 8 years, high 10 years.** Observed facts: Wind River emphasizes local continuity, family-of-brands identity, and 75+ years of service, while its acquisition strategy seeks strong local brands with existing customer bases. Root mechanism: practical tenure is longer than legal term because septic, grease, and wastewater needs recur and trusted local operators are sticky, especially once folded into a bigger network. Calculation bridge: I place effective tenure above WRM and roughly comparable to LES for commercial service, with residential relationships extending longer even if service intervals are less frequent. Sensitivities: quality of local integration, customer-service consistency after acquisitions, and local competition. Confidence: **Medium**. [72]

Utilization exposure. Conclusion: **Moderate to High, central Moderate-High.** Observed facts: Wind River has over 1,000 vehicles and 16 treatment facilities processing more than 200 million gallons annually. Root mechanism: route fill, technician productivity, and facility throughput are all central to profitability; idle equipment earns nothing. However, recurring demand and broad end-market mix reduce pure project cyclicality. Calculation bridge: the most relevant quantitative KPI would be gallons per truck day and treatment-plant throughput versus permitted capacity, but those figures are not publicly disclosed. Sensitivities: weather, emergency spikes, labor, route density, and residential pumping cycles. Confidence: **Medium**. [71]

New-unit or new-customer payback. Conclusion: **Low 0.75 years, central 1.5 years, high 3.0 years** for incremental dense-market route customers; much longer for de novo branches or acquisitions. Observed facts: Wind River's growth model is acquisition heavy, and the platform has shared route and treatment infrastructure. Root mechanism: an incremental customer in a dense local market should pay back quickly, but a boutique local acquisition or forced branch build includes substantial hidden startup cost. Calculation bridge: I reserve direct service labor, transportation, and a maintenance-capex reserve in the denominator and explicitly distinguish incremental route-account economics from full-platform expansion economics. Sensitivities: route adjacency, service frequency, and mix of residential versus commercial accounts. Confidence: **Low-Medium**. [73]

B2B versus B2C profile. Conclusion: **Mixed B2B/B2C.** Observed facts: Wind River explicitly serves homes, businesses, municipalities, and industrial settings. Root mechanism: the company invoices both residential households and commercial/institutional counterparties, which affects sales cycles, pricing, churn, and route design. Calculation bridge: because residential is real rather than incidental, a "predominantly B2B" label would overstate segment purity for the consolidated company. Sensitivities: the exact revenue split is undisclosed. Confidence: **High** on the mixed classification. [74]

The table below summarizes my normalized conclusions for Wind River Environmental. Because the company is a broad mixed-service consolidator rather than a clean segment pure-play, the central values should be treated as route-processing-relevant economics embedded inside a more diversified operating perimeter. [75]

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Normalized revenue growth	6%	9%	12%	Triangulated	Medium	Recurring service demand and route density, after stripping acquisition effects
Normalized EBITDA margin	18%	22%	26%	Inferred	Medium-Low	Treatment ownership and density help, but mixed services dilute pure route-processing margins
Maintenance capex % of revenue	4.0%	5.5%	7.0%	Inferred	Low-Medium	Large vehicle fleet plus treatment-facility and specialty-equipment upkeep
Post-maintenance-capex conversion	68%	75%	85%	Calculated	Low-Medium	Recurring routes convert reasonably, but sustaining capex is heavier than at cleaner peers
Initial contract tenor	0.5 yrs	1 yr	2 yrs	Inferred	Low	Many services are short-form recurring operational agreements

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Effective customer tenure	5 yrs	8 yrs	10 yrs	Triangulated	Medium	Essential recurring need plus local-brand trust and network effects
Utilization exposure	Moderate	Moderate-High	High	Inferred	Medium	Vehicle productivity, route fill, and treatment throughput are central
New-unit or customer payback	0.75 yrs	1.5 yrs	3.0 yrs	Triangulated	Low-Medium	Incremental route customers leverage shared branch, fleet, and treatment assets
B2B/B2C profile	Mixed B2B/B2C	Mixed B2B/B2C	Mixed B2B/B2C	Disclosed	High	Residential, commercial, municipal, and industrial customers all matter

Plain-English business description: Wind River Environmental is an East Coast wastewater-services consolidator that combines septic, grease, plumbing, municipal-industrial service, and treatment-facility ownership across a dense family-of-brands network. [76]

Principal strengths: strong East Coast density, owned treatment capacity, broad service capability, emergency responsiveness, and a proven acquisition roll-up engine. Principal risks: mixed-business contamination, integration risk, less clean comparability to pure grease-route processors, and potentially heavier sustaining capex. Key unavailable information: segment revenue split, EBITDA by service line, organic growth, contract structure, and end-market mix. Most important diligence questions: what percentage of EBITDA comes from residential septic versus commercial grease and municipal-industrial work, what treatment facilities are truly core to the benchmark segment, how much specialty work is project-based, and whether acquired local brands retain original customer-tenure characteristics after integration. [77]

Archetypes, benchmarks, final conclusions, evidence gaps, and source list

Operating archetypes and company classification

The designated universe supports the two mandatory archetypes exactly as instructed.

Installed onsite-system model — Restaurant Technologies. Restaurant Technologies is the **core representative** for this archetype. The company installs customer-site oil-management systems and monetizes an integrated relationship spanning fresh-oil delivery, filtration, monitoring, and used-oil collection/recycling at the customer site. The core economic distinction is that the customer relationship is organized around **embedded onsite equipment** first and recurring route service second. Evidence quality is good on operating scope but weaker on margin and contract detail, so this is an evidence-based archetype range rather than a multi-company statistical peer set. [78]

Route, collection, treatment, and processing model — Liquid Environmental Solutions, Waste Resources Management, and Wind River Environmental. LES is the **core representative** because it is the cleanest disclosed national route-and-treatment operator in the set and has the best direct margin evidence. WRM is a **secondary reference** because its operating mechanics are highly relevant but disclosure is thinner and there is some adjacent-business contamination. Wind River Environmental is a **mixed-business company** because it is clearly relevant to the archetype but broad residential septic, plumbing, trenchless, and municipal-industrial activities make a single consolidated benchmark less pure. None of the three route-processing companies should be treated as an outlier or as insufficient disclosure overall, though WRM carries lower-confidence metric precision. [79]

Benchmark synthesis for the installed onsite-system model

Restaurant Technologies alone represents this mandatory archetype, so the table below is an **evidence-based archetype range derived from Restaurant Technologies' normalized operating mechanics** rather than a cross-company average. That limitation matters. A single-company archetype range can still be useful when the business model is structurally distinct and the mechanisms are clear. [80]

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
Revenue growth	5%	8%	12%	5%–12%	Medium	New installed kitchens, chain wins, same-site throughput, net of commodity pass-through

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
EBITDA margin	18%	21%	25%	18%–25%	Medium-Low	Service value and sticky workflows offset by oil product costs and install/service burden
Maintenance capex % revenue	2.5%	3.5%	4.5%	2.5%–4.5%	Low	Installed-system and service-vehicle replacement/refurbishment
Post-maintenance conversion	75%	83%	90%	75%–90%	Low-Medium	Recurring service base with moderate sustaining physical capital
Initial contract tenor	3 yrs	4 yrs	5 yrs	3–5 yrs	Low	Equipment and installation economics require multi-year underwriting
Effective customer tenure	6 yrs	9 yrs	12 yrs	6–12 yrs	Medium	Physical embedding and workflow switching friction
Utilization exposure	Moderate	Moderate	Moderate	Moderate	Medium	Revenue depends on both deployed assets and fryer-oil throughput

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
New-unit/ customer payback	1.5 yrs	2.5 yrs	3.5 yrs	1.5–3.5 yrs	Low-Medium	Fully burdened install recovered from whole-relationship contribution
B2B/B2C profile	Predominantly B2B	Predominantly B2B	Predominantly B2B	Predominantly B2B	High	Commercial kitchen buyer base

The central range is appropriate because the installed onsite-system model has two unusual traits. First, it is stickier than a normal service route because equipment is embedded inside the customer workflow. Second, it is less margin-rich than a pure route-processor because fresh-oil product costs and customer-site hardware complicate the unit economics. That combination explains why Restaurant Technologies can have strong practical retention and still experience cash stress when financing costs are high and install growth is aggressive. The biggest structural difference versus the route-processing model is that the customer-site asset is part of the moat. [80]

Benchmark synthesis for the route, collection, treatment, and processing model

This archetype is supported by all three designated companies, but LES carries the most direct evidence, WRM reinforces the grease-centered vertical-integration logic, and Wind River broadens the range because of its mixed-service East Coast wastewater platform. I therefore use a **mechanism-based benchmark**, not an equal-weighted average. [81]

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
Revenue growth	6%	9%	13%	6%–14%	Medium	Route density, cross-sell, throughput, modest price, excluding acquisition/ perimeter growth

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
EBITDA margin	22%	28%	32%	18%–34%	Medium-High	Density plus treatment/ disposal control; lower where service mix is broader or third-party dependence is higher
Maintenance capex % revenue	3.5%	5.0%	6.5%	3.0%–7.0%	Medium-Low	Vac-truck replacement, treatment infrastructure, environmental/ safety capital
Post-maintenance conversion	77%	82%	89%	68%–91%	Medium-Low	High recurring contribution once density exists, moderated by fleet and plant upkeep
Initial contract tenor	1 yr	2 yrs	3 yrs	0.5–3 yrs	Low	Short recurring service agreements with renewal and habitual continuity
Effective customer tenure	5 yrs	7 yrs	10 yrs	4–10 yrs	Medium	Recurring compliance need, route reliability, switching friction, local brand trust

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
Utilization exposure	Moderate	Moderate-High	High	Moderate–High	Medium	Route fill, gallons per stop, technician productivity, treatment throughput
New-unit/ customer payback	0.5 yrs	1.25 yrs	2.25 yrs	0.5–3.0 yrs	Medium-Low	Incremental dense-market customers leverage shared network assets
B2B/B2C profile	Predominantly B2B	Predominantly B2B	Mixed B2B/ B2C	Predominantly B2B to Mixed	High	Commercial/ institutional share dominates pure-play platforms; residential matters in mixed operators

The central range is appropriate because the route-processing archetype earns money from **density and control**, not from any single invoice line. LES sits above the central margin range when owned or controlled treatment, route density, and recovery are strongest and the business mix is cleanest. WRM supports the same mechanics but with slightly lower confidence and somewhat more mix complexity. Wind River falls below the cleanest processors on margin and above them on service breadth because its mixed residential/ septic/plumbing/municipal model dilutes the purity of grease- and wastewater-route economics. Those differences are mostly **structural**, not temporary. [82]

I do **not** present a single subvertical-wide benchmark table blending both archetypes. The economics do not support that move. Restaurant Technologies' installed onsite-system model is built around customer-site equipment, fresh-oil supply, and embedded workflow integration, while the other three companies are built around route density, treatment/disposal control, and shared processing assets. A single averaged margin, payback, or tenor would conceal the true drivers. [83]

Final subvertical conclusion

Typical customer problem. Customers need recurring, compliant handling of cooking oil, grease, wastewater, septic, or related non-hazardous liquid streams to avoid downtime, labor inefficiency, safety incidents, sewer problems, and environmental liability. In the installed onsite-system model, the problem is kitchen oil handling. In the route-processing model, it is recurring waste removal and compliant downstream treatment/disposal. [84]

Typical revenue model. Installed onsite systems monetize an integrated relationship spanning hardware-enabled service, fresh-oil delivery, filtration, monitoring, and used-oil recovery. Route-processing platforms monetize recurring pumping/collection, hauling, treatment, disposal, and recovered-material economics, often with ancillary plumbing, hydro jetting, or emergency services layered in. [83]

Primary growth drivers. The biggest common drivers are customer additions, route density expansion, cross-sell, and selective M&A. The biggest archetype difference is that Restaurant Technologies also depends on new customer-site installs and oil-throughput trends, while the route-processing companies depend more on route fill, treatment capacity, and local disposal control. Acquisitions are especially important at LES, WRM, and Wind River and must be separated from organic growth. [85]

Primary margin drivers. In the installed onsite-system model, margin is driven by install density, field-service efficiency, fresh-oil product spread, and recovered-oil economics. In the route-processing model, margin is driven by route density, truck productivity, treatment/disposal control, permits, and whether recovered material is monetized or only offsets disposal cost. This is why LES can sit above 30% adjusted EBITDA margin while broader mixed-service platforms normalize lower. [86]

Principal maintenance-capex requirements. Restaurant Technologies' maintenance capex is centered on installed-system refurbishment and service vehicles. The route-processing archetype requires recurring vacuum-truck replacement plus treatment-facility, tank, pump, environmental, and safety capital. Wind River likely has the heaviest sustaining burden because of scale and service breadth. [87]

Typical contract and customer duration. Initial legal terms appear shorter in the route-processing model and longer where customer-site equipment is installed. Effective customer tenure is longer than legal term in both archetypes, but for different reasons: physical embedding at RTI and recurring compliance plus route habit at LES, WRM, and Wind River. The legal-term conclusion is the least directly disclosed part of the study. [88]

Typical utilization dynamics. Installed onsite systems have mixed exposure: assets can remain deployed while customer throughput softens. Route-processing platforms have more direct service-capacity exposure because underfilled routes and underutilized treatment assets hit margins quickly. The most important KPI in the route-processing model is not customer count by itself, but customer count **per route, per truck, and per plant capacity**. [84]

Typical new-unit payback. Restaurant Technologies' customer-site payback is materially longer because the install itself is part of the economic unit. Route-processing archetype payback on an incremental dense-market customer is shorter because trucks and treatment assets are shared across many accounts, though new-branch or acquisition payback is much longer than incremental customer payback. [89]

Typical B2B/B2C profile. The subvertical is mostly B2B, but not uniformly so. Restaurant Technologies and LES are predominantly B2B. WRM is predominantly B2B with some mixed exposure. Wind River is genuinely mixed B2B/B2C because residential septic is a real part of the business. [90]

Largest source of variation among companies. The single biggest source of variation is **who owns the bottleneck asset**. At RTI, the bottleneck is the installed customer-site system and ongoing oil workflow. In the route-processing model, the bottleneck is route density plus treatment/disposal access. That one difference cascades into margin, payback, tenure, and utilization exposure. [83]

Lowest-confidence conclusion. Initial legal contract tenor is the lowest-confidence metric across the universe, followed closely by strict maintenance capex, because the companies are private and do not disclose those items directly in the retrieved record. Growth, customer profile, and core operating mechanics are materially stronger conclusions than legal term or accounting maintenance precision. [91]

Slide-ready business-model description. Liquid environmental services combines installed kitchen oil-management networks and route-based collection-treatment platforms that handle regulated, recurring waste streams for dispersed commercial sites, monetizing service density, compliance, and recovered materials. **Word count: 29.**

Key evidence gaps and consolidated diligence priorities

The biggest evidence gaps are current standalone revenue and EBITDA for all four private companies; clean organic-versus-acquisition growth splits for LES, WRM, and Wind River; current contract forms and renewal mechanics across all four; true maintenance capex by asset class; and the exact accounting treatment of recovered materials and pass-through commodity exposures. These gaps matter most for valuation and cash-conversion work, but they do not overturn the first-principles operating conclusions above. [92]

The most important diligence priorities are therefore practical. For Restaurant Technologies: who funds installation, how much cash is tied up per new location, and how recovered-oil value is booked. For LES: what share of throughput is internally treated versus partner-routed, and how much EBITDA depends on recovered-value spreads. For WRM: what percentage of EBITDA comes from core liquid environmental services versus adjacent support services. For Wind River: what the revenue and margin split is among residential septic, commercial grease, municipal-industrial wastewater, and plumbing/drain/trenchless. Across all four: what true customer churn is by cohort, what route density KPIs management uses internally, and what sustaining capital is required to hold today's revenue flat. [84]

Source list

Restaurant Technologies sources used in this report include the company's official operating pages and contact page for current scale and service design, S&P credit snippets for historical growth expectations, QSR mix, and 2024 liquidity commentary, and Energy Capital Partners plus deal reporting for ownership history. [93]

Liquid Environmental Solutions sources used in this report include the official company home page, locations page, official acquisition releases for Carolinas Resource Recovery and A-1 Restaurant Services, the LES onboarding page for used-cooking-oil recovery language, Houlihan Lokey's 2025 market update and S&P/Market Intelligence materials for ownership and margin evidence. [94]

Waste Resources Management sources used in this report include WRM's official website, the official Ridgewood-related announcement and partners page, and WRM's 2026 Darling transaction announcement for current scale, ownership, metro expansion, and business-model description. [45]

Wind River Environmental sources used in this report include Wind River's official About, Acquisitions, Treatment Facilities, Locations, and local operating pages, along with Gryphon Investors' 2017 transaction announcement and current portfolio entry. [95]

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[1] [6] [12] [14] [16] [21] [25] [27] [28] [29] [78] [80] [90] [93] Restaurant Technologies | Cooking Oil Delivery & Recycling
https://www.rti-inc.com/?utm_source=chatgpt.com

[2] [11] [13] [15] [17] [18] [24] [26] [83] [84] [86] [88] [89] Commercial & Restaurant Cooking Oil Management Systems
https://www.rti-inc.com/oil-management/?utm_source=chatgpt.com

[3] [4] [9] [31] [34] [36] [37] [38] [41] [43] [44] [79] [81] [94] Home | Liquid Environmental Solutions
https://www.liquidenviro.com/

[5] [8] [39] [40] [42] [82] Research Update: VRS Buyer Inc. (dba Liquid Tech
https://www.spglobal.com/ratings/en/regulatory/article/-/view/sourceId/101636728?utm_source=chatgpt.com

[7] [35] [85] Liquid Environmental Solutions Acquires Carolinas Resource Recovery | Liquid Environmental Solutions
https://www.liquidenviro.com/blog-news/liquid-environmental-solutions-acquires-carolinas-resource-recovery

[10] [19] [23] [91] [92] Double Eagle Buyer Inc. Outlook Revised To Negati
https://www.spglobal.com/ratings/pt/regulatory/article/-/view/type/HTML/id/3292027?utm_source=chatgpt.com

[20] Restaurant Technologies Inc. Assigned 'B-' Issuer
https://www.spglobal.com/ratings/en/regulatory/article/-/view/type/HTML/id/2098325?utm_source=chatgpt.com

[22] [87] Contact RTI
https://www.rti-inc.com/contact-rti/?utm_source=chatgpt.com

[30] Double Eagle Buyer (d/b/a Restaurant Technologies
https://www.spglobal.com/ratings/en/regulatory/article/-/view/type/HTML/id/2806652?utm_source=chatgpt.com

[32] Environmental Services Market Update | Q2 2025
https://cdn.hl.com/pdf/2025/bus-environmental-services-q2-2025.pdf?utm_source=chatgpt.com

[33] Liquid Environmental Solutions
https://www.liquidenviro.com/locations

[45] [46] [47] [48] [49] [50] [52] [53] [54] [55] [57] Waste Resource Management | Liquid Waste Management
https://wrmco.com/

[51] [56] WRM Acquires a Majority of Darling Ingredients' Trap Grease Business
https://www.prnewswire.com/news-releases/wrm-acquires-a-majority-of-darling-ingredients-trap-grease-business-302771051.html

[58] WRM & Ridgewood Unite for Wastewater Solutions
https://wrmco.com/a-new-chapter-for-wrm-joining-forces-with-ridgewood-infrastructure-to-enhance-u-s-water-sector-leadership/?utm_source=chatgpt.com

[59] Private Equity Firm Gryphon Investors Announces Majority ...
https://www.gryphon-inv.com/news/private-equity-firm-gryphon-investors-announces-majority-investment-wind-river-environmental/?utm_source=chatgpt.com

60 61 62 64 Wastewater Treatment Facilities | Wind River Environmental
https://www.wrenvironmental.com/about-us/treatment-facilities/

63 72 74 Septic Services in Milford, PA | Wind River Environmental
https://www.wrenvironmental.com/pike-county/

65 67 68 69 70 71 73 75 76 77 95 About Us | Wind River Environmental
https://www.wrenvironmental.com/about-us/

66 Acquisitions | Wind River Environmental
https://www.wrenvironmental.com/about-us/acquisitions/

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Transportation Solutions Operating Benchmark

Executive summary

Transportation Solutions is **not** one uniform operating model. In this research universe, the economics split into four meaningfully different models: **railcar leasing** at VTG, **locomotive leasing** at Akiem, **airport ground-support-equipment full-service leasing** at Air Rail and TCR, and **labor-led outsourced yard logistics** at Lazer Logistics. Those models differ materially in asset life, contract structure, maintenance burden, utilization mechanics, customer switching friction, and payback shape, so a single blended benchmark would conceal more than it reveals. [1]

The most robust benchmark conclusions are archetype-specific. **Railcar leasing** appears to support lower normalized growth but high EBITDA margins and moderate sustaining capital intensity because wagons are long-lived, redeployable across many industrial flows, and often leased on medium-term contracts, though intermodal fleets remain more utilization-sensitive than specialized tank wagons. **Locomotive leasing** is even more capital-intensive and technically specialized, with longer contract tenors and more approval, overhaul, and interoperability complexity; that tends to support strong margins but also higher sustaining-upgrade risk and slower redeployment after contract loss. **Airport GSE leasing** is structurally more service-heavy than pure railcar leasing because maintenance response, airport approvals, spare fleets, electrification, and battery/charging planning are embedded in the value proposition. **Outsourced yard logistics** is primarily a people-and-process business with some equipment support, so margins are lower but maintenance capex is much lighter and payback is faster. [2]

The mandatory identity-resolution gate for **Air Rail** resolves to **AIR RAIL, S.L.**, a Madrid-based Spanish company registered under CIF **B-80330947**, operating the **air-rail.org** website. The same operating business is the one IFM Investors agreed to acquire in 2025 and lists in its infrastructure portfolio with a **75% fund ownership stake** and **initial fund investment in February 2026**, while founder **José Manuel García Prieto** retained **25%** at signing. The business is **mixed** across airport GSE and rail-machinery activities, but IFM's asset description and the company's own service pages clearly tie the entity to airport GSE leasing, maintenance, and distribution. Identity confidence is **High**. [3]

At the company level, my central operating conclusions are these. **VTG's railcar-leasing operation** normalizes to modest organic growth, high segment EBITDA margins, gross maintenance capex in the mid-to-high teens as a percent of leasing revenue, and a payback that usually requires more than one contract cycle but is supported by long asset life and residual value. **Akiem** normalizes to somewhat faster growth than VTG because locomotive demand is being pushed by fleet modernization, cross-border traffic, and electrification, but it carries more technical concentration risk and heavier overhaul/upgrade requirements. **TCR** sets the clearest global benchmark for airport GSE full-service rental: solid growth, mid-30s-or-lower EBITDA margins depending on service mix, moderate-to-high utilization exposure, and five-to-seven-year asset paybacks at the unit level. **Air Rail** fits the same archetype but with lower confidence because public disclosure is thinner and the company has a mixed airport/rail platform. **Lazer Logistics** should remain in a separate outsourced-operations bucket: it is not a clean asset lessor, and that distinction matters for every metric except the broad B2B customer profile. [4]

The lowest-confidence area in the full study is **maintenance capex for private companies**, especially where public reporting does not separate gross replacement spending, refurbishment, battery replacement, and growth fleet additions. The strongest conclusions are around **business model structure, contract durability, utilization mechanics, and payback logic**, because those are better disclosed in official operating materials and transaction sources than full private-company segment cash flow statements. [5]

Scope, analysis date, and concise methodology

This is a fresh analysis **as of July 18, 2026**, covering only the designated companies: **VTG, Akiem, Air Rail, TCR, and Lazer Logistics**. I did **not** compare any result with CRA or Compactor Rentals of America, and I did **not** allow Lazer Logistics to determine the asset-leasing benchmark except where a metric-specific mechanism supports cross-archetype comparison. The report completes all five company profiles **before** synthesis.

Methodologically, each company and each metric follows the same sequence: **observed facts**, **root business mechanism**, **normalized low/central/high assumption**, **sensitivities**, and **confidence**. Priority was given to official or primary sources: audited or statutory accounts where available, official company and ownership materials, sponsor and transaction materials, ratings-style financing materials where official company disclosure referenced them, and only then selective industry publications for context. Where direct disclosure was unavailable, I used transparent first-principles triangulation from the company's asset life, maintenance obligations, service content, contract structure, fleet scale, ownership documents, and official operating statements. Numeric precision is intentionally bounded: low/central/high ranges are meant to represent **normalized operating economics**, not point forecasts. [1]

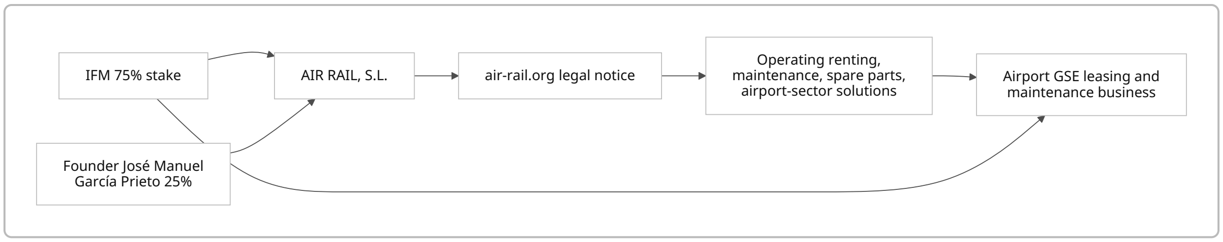
A final caution applies across the report: several companies here are private or no longer public. That means the **quality of evidence differs by metric and by company**. VTG has the best current statutory financial visibility among the group. Akiem has strong current operating disclosure but much thinner current public financial detail. Air Rail and TCR have strong recent strategic and operational disclosure but much more limited public standalone group financial reporting. Lazer has excellent operating-model disclosure and sponsor-scale facts but limited public financial disclosure. I therefore use the required **Evidence Status** labels rigorously, and I lower confidence wherever current public evidence is limited. [6]

Air Rail identity-resolution finding

The most defensible resolved entity is **AIR RAIL, S.L.**, headquartered at **C/ Alsasua nº 16, 1ª izquierda, 28023 Madrid**, with Spanish tax/registration identifier **C.I.F. B-80330947**, as stated in the legal notice on the company website **air-rail.org**. That same website presents "Operating Renting," airport equipment distribution, maintenance contracts, technical service, overhaul, spare parts, and airport-sector machinery as core offerings. It also states that the company was **founded in 1992**, began in Spain, and operates across multiple countries. [7]

The most important ownership evidence comes from **IFM Investors**. IFM first announced in July 2025 that it had agreed to acquire **75% of Air Rail**, describing it as the **largest airport GSE owner and lessor in Spain and Portugal**, with founder **José Manuel García Prieto** retaining **25%**. IFM then listed **Air Rail** in its infrastructure portfolio with **initial fund investment in February 2026**, **75% fund ownership stake**, and a description of the company as a leader in airport GSE leasing, distribution, and maintenance with **over 4,700 GSE units across circa 60 airports in eight countries** as of **31 March 2026**. [8]

The evidence chain linking the legal entity to the airport GSE leasing business is therefore direct:



That linkage is stronger than any alternative reading of "Air Rail." The phrase can generically describe intermodal transport concepts, and the company itself also sells **railway shunting and maintenance equipment**, which could have created naming ambiguity. But the legal notice, the operating website, and the owner's portfolio page all point to the **same legal and operating business**. The only substantive uncertainty is **segment purity**: AIR RAIL, S.L. is **not** a pure-play airport GSE lessor, because its own site also markets rail-sector products and services. For the benchmark work below, I therefore isolate the **airport GSE leasing/maintenance/distribution activity** wherever possible and label confidence lower when mixed-company disclosure prevents a clean segment cut. [9]

Identity confidence: High. The exact legal entity is resolved. The remaining uncertainty is **segment isolation**, not entity identity. [10]

Company profiles for VTG and Akiem

VTG

First-principles profile

A. Resolve the company and segment. The relevant entity is **VTG GmbH**, headquartered in Hamburg, using **2024 consolidated accounts** as the latest reliable standalone reporting period in the sources reviewed. The 2022 ownership transaction announced the sale of a **72.5% majority stake** from Morgan Stanley Infrastructure Partners and Joachim Herz Stiftung to a consortium of **GIP and ADIA**, with **OMERS** retaining its stake. VTG's 2024 accounts still show GIP-, ADIA-, and OMERS-related holding structures and board representation, indicating continued consortium ownership. The same 2024 accounts also emphasize a strategic **focus on the European core wagon business**, which is the relevant segment for this benchmark. [11]

B. Define the economic unit. The governing economic unit is **one revenue-earning rail freight wagon under lease**. For utilization, maintenance capex, and payback, the wagon is the right unit. For customer tenure, the more relevant analytical unit is often the **customer fleet program** or **lane-specific wagon pool**, because customers lease groups of wagons tied to industrial flows rather than making many isolated single-wagon decisions. VTG's own historical public reporting describes wagon hire as the core business and frames wagons as deeply embedded in customer production and transport flows. [12]

C. Explain the customer problem. Customers need rail freight capacity that is reliable, regulatorily compliant, and tailored to specific cargo flows without tying up their own capital or maintenance organization. In chemicals, petroleum, agriculture, paper, steel, autos, and intermodal transport, the wagon acts like a **mobile production asset** rather than a generic rental item. Where a customer's production system, freight schedule, and safety requirements depend on specialized wagons, leasing solves both capital intensity and fleet-flexibility problems. [12]

D. Explain customer alternatives. Customers can buy wagons outright, lease from another owner, or sometimes redesign transport flows around truck or other rail operators. But switching is not frictionless: fleets are cargo-specific, maintenance windows are regulated, and wagon availability matters to plant throughput. Specialized tank and industrial wagons are harder to substitute than standard intermodal wagons. That is one reason VTG's 2024 accounts report that **tank-wagon utilization remained high**, while **intermodal utilization softened**, showing that alternative-user breadth differs by subfleet. [13]

E. Map the revenue model. The relevant segment is primarily **recurring wagon lease revenue**, sometimes supported by fleet management, maintenance, and related service revenue. VTG's broader group also includes rail logistics, but its own accounts identify wagon-fleet investment and wagon leasing quality leadership as core strengths. For benchmark purposes, I exclude broader forwarding/logistics economics and focus on the asset-leasing component. [14]

F. Map the operating model. After winning the customer, VTG must keep the wagon legally operable and revenue-earning: inspections, wheelset and brake work, component replacement, workshop visits, field maintenance coordination, and remarketing/redeployment when off lease. The company's 2024 investment disclosure explicitly says capital spending went mainly into **new freight wagons** and **maintenance of the fleet in continental Europe**, with further investment into UK wagon-leasing activities. That means the post-sale operating model is not passive finance leasing; it is active fleet stewardship. [13]

G. Explain customer retention and churn. Customers stay because wagon supply is operationally important, specialization can be high, and changing fleets risks disruption. Customers leave when lanes disappear, industrial output falls, a fleet becomes mismatched to current needs, or a competitor offers better availability or price. In VTG's case, intermodal churn risk is structurally higher than tank-wagon churn risk because the 2024 accounts show intermodal utilization weakness while tank-wagon utilization held up better. Customer retention is therefore stronger than single-asset retention, but not uniform across fleet categories. [13]

H. Map the capital model. VTG is capital-heavy. In 2024 it generated **€405 million** of operating cash flow and spent **€536 million** on fixed-asset investment, mainly on new wagons and fleet maintenance. The group also maintained a **€400 million CAPEX line** and stated that 2024 investment outflows were mainly caused by investment into the wagon fleet. This is the classic pattern of a scaled railcar owner: returns depend on long asset lives, high utilization, disciplined maintenance, residual value, and frequent reuse of the same asset across successive contracts rather than full recovery in a single short lease. [15]

VTG metric table

The table below reflects the **railcar-leasing economics only**, not the full mixed VTG group.

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Normalized revenue growth	1%	4%	7%	Triangulated	Medium	Fleet additions, price, and utilization mix by wagon type
Normalized EBITDA margin	52%	58%	65%	Inferred	Medium	High asset gross margin offset by workshops, maintenance, and repositioning
Maintenance capex % of revenue	14%	18%	24%	Calculated	Medium	Long-lived wagons require recurring heavy maintenance and periodic replacement
Post-maintenance-capex conversion	54%	69%	78%	Calculated	Medium	High EBITDA partially consumed by sustaining fleet spend
Initial contract tenor	2 yrs	4 yrs	7 yrs	Inferred	Medium	Mix of standard/intermodal versus specialized wagon programs
Effective customer tenure	6 yrs	9 yrs	12 yrs	Inferred	Medium	Fleet integration and multi-cycle reuse within customer relationships

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Utilization exposure	Moderate	Moderate	High	Inferred	Medium	Off-lease risk is manageable in standard fleets but rises in cyclical intermodal pockets
New-unit or new-customer payback	7 yrs	9 yrs	12 yrs	Triangulated	Medium	Wagon cost, lease rate, and maintenance reserve govern cash recovery
B2B/B2C profile	Predominantly B2B	Predominantly B2B	Predominantly B2B with some customer concentration	Disclosed	High	Industrial shippers, rail operators, and logistics firms

The largest observed anchors are VTG's 2024 disclosures that investment was mainly for **new wagons and fleet maintenance**, that intermodal utilization remained under pressure while tank-wagon utilization stayed strong, and that the group remained strategically centered on wagon-leasing quality leadership in Europe. [13]

VTG metric analysis

Revenue growth

Conclusion. Normalized railcar-leasing revenue growth is **1% / 4% / 7%**.
Observed facts. VTG's 2024 accounts describe continued focus on the European core wagon business, significant wagon-fleet investment, stable tank-wagon utilization, and weaker intermodal utilization. [13]
Root business mechanism. Growth comes from lease-rate resets, new wagons placed on lease, utilization changes, and modest share gains in specialized equipment; it does **not** naturally compound like software or an outsourced labor network.
Calculation or driver bridge. A central case assumes roughly low-single-digit price/mix plus selective fleet additions and stable utilization in specialized wagons, partially offset by cyclical intermodal softness.
Sensitivities. European industrial production, commodity flows, intermodal demand, wagon availability, and regulatory rail competitiveness.
Confidence. Medium because the operating direction is clearly disclosed, but current public segment revenue detail is limited in the accessed sources.

EBITDA margin

Conclusion. Normalized segment EBITDA margin is **52% / 58% / 65%**.
Observed facts. VTG's wagon leasing is capital-heavy and service-light relative to airport GSE or yard logistics, but not maintenance-free; 2024 investment disclosure explicitly includes fleet maintenance, and the company emphasizes wagon-lease quality leadership rather than pure logistics services. [14]
Root business mechanism. Lease revenue has high gross margin once wagons are placed, but workshops, regulatory upkeep, storage, and repositioning prevent margins from reaching "pure finance lease" levels across the whole railcar portfolio.
Calculation or driver bridge. Central margin assumes strong asset-level economics in tank and specialized wagons, diluted by workshop/service costs and weaker intermodal dollar utilization during softer periods.
Sensitivities. Fleet mix, repair intensity, storage days, and whether off-lease wagons remain revenue-earning.
Confidence. Medium because current public segment EBITDA is not cleanly separated in the accessed materials.

Maintenance capex

Conclusion. Gross maintenance capex normalizes at **14% / 18% / 24% of relevant leasing revenue**.
Observed facts. VTG disclosed **€536 million** of 2024 fixed-asset investment and said the main uses were **new wagons** and **fleet maintenance** in continental Europe, with UK wagon-leasing investment also relevant. [13]
Root business mechanism. Wagons are long-lived, but they are not static. Preserving earning capacity requires inspections, brake/wheel work, component renewal, refurbishment, and some replacement CAPEX as older or unsuitable wagons are cycled out.
Calculation or driver bridge. Because public disclosure does not isolate growth from sustaining spend, I triangulate from the life-cycle reality of rail wagons and the fact that current heavy investment reflects both growth and preservation. Central assumption uses a gross view; a net-after-ordinary-disposals view would be lower.
Sensitivities. Average fleet age, specialization, regulatory standards, and whether heavy overhauls are expensed or capitalized.
Confidence. Medium.

Post-maintenance-capex conversion

Conclusion. Post-maintenance-capex conversion normalizes at **54% / 69% / 78%**.
Observed facts. VTG produced strong operating cash flow in 2024 but also required substantial fleet investment, with management explicitly linking investment outflows to wagon-fleet spending. [13]
Root business mechanism. Railcar leasing converts EBITDA to cash well **before** sustaining investment, but part of that EBITDA must be recycled back into the fleet to keep earning power intact.
Calculation or driver bridge. Using the central EBITDA margin and sustaining-capex assumptions above yields roughly 69% post-maintenance conversion. This is **not** free-cash-flow conversion because it excludes interest, tax, working capital, and growth CAPEX.
Sensitivities. Growth-vs-sustain split in capex, asset disposals, and maintenance deferral.
Confidence. Medium.

Initial contract tenor and effective customer tenure

Conclusion. Initial contract tenor is **2 / 4 / 7 years**, while effective customer tenure is **6 / 9 / 12 years**.
Observed facts. VTG's historical public materials describe wagon leasing as fundamental to customer production flows and note that customers tend to hire wagons over the **medium to long term**. VTG Rail UK's Mendip Rail example also shows a relationship tied to a **10-year agreement** and fleet replacement cycle. [16]
Root business mechanism. Individual leases can be shorter than the economic relationship because fleets are reused, resized, and refreshed over time.
Calculation or driver bridge. Standard wagons and intermodal pools pull the low end down; specialized industrial and long-program fleets pull the high end up.
Sensitivities. Wagon type, customer procurement structure, lane stickiness, and macro demand.
Confidence. Medium.

Utilization exposure

Conclusion. Utilization exposure is **Moderate**, with a high-case shift to **High** in more cyclical intermodal pockets.
Observed facts. VTG disclosed that **intermodal fleet utilization weakened in 2024**, while **tank-wagon utilization held at a high level** and rental-price development was positive. [13]
Root business mechanism. The key KPI is not only "fleet utilization" but **revenue-earning utilization by wagon type**. Standard fleets can usually be redeployed, but that still takes time and can involve storage and repositioning cost.
Calculation or driver bridge. I classify the portfolio as Moderate overall because specialized wagons are sticky and diversified, but utilization clearly moves with industrial cycles and fleet mix.
Sensitivities. Intermodal demand, customer insolvency, regional imbalances, and remarketing speed.
Confidence. Medium.

New-unit or new-customer payback

Conclusion. New-wagon payback normalizes at **7 / 9 / 12 years**.
Observed facts. VTG remains an active wagon investor and keeps a dedicated CAPEX line for fleet investment. The company's business model relies on repeated long-life monetization of wagons. [13]
Root business mechanism. B2B rail wagons are expensive, but their useful lives are long and residual value exists. What matters is annual unit-level cash contribution after a maintenance reserve, not headline lease revenue.
Calculation or driver bridge. Central case assumes lease-rate economics that do **not** fully repay a new wagon in one short contract but do support acceptable recovery over one initial term plus renewals or redeployment.
Sensitivities. Wagon specification, purchase cost, grant support, maintenance intensity, and residual value.
Confidence. Medium.

B2B versus B2C profile

Conclusion. VTG is **predominantly B2B**.
Observed facts. Historical public materials describe customers including shippers, railway companies, and rail logistics companies across chemicals, petroleum, agriculture, autos, steel, and paper. [17]
Root business mechanism. Procurement is professional, contracts are negotiated, and customer value

rests on uptime, safety, and lane economics rather than consumer demand.
Calculation or driver bridge. No B2C revenue stream is economically relevant to the segment.
Sensitivities. Customer concentration by sector and subfleet.
Confidence. High.

VTG conclusion

Plain-English business description. VTG's relevant business is owning and managing large fleets of freight wagons that industrial customers and rail operators lease to move goods across European rail networks.

Principal strengths. Scale, deep wagon specialization, long asset lives, fleet redeployability, and customer embedding in industrial transport flows. [14]

Principal risks. Intermodal cyclicality, sustaining-capex opacity, off-lease/storage drag, and the difficulty of separating pure wagon-leasing economics from broader group activities. [13]

Key unavailable information. Clean current public railcar-segment revenue, segment EBITDA, exact July 2026 wagon count, gross-versus-net maintenance capex split, and disclosed average contract tenor.

Most important diligence questions. What share of current capex is sustaining versus growth? What is revenue-earning utilization by wagon family? What are average off-lease days and repositioning costs? How much of lease pricing has reset upward since 2023? What percentage of the fleet is under multi-year master programs versus shorter opportunistic leases?

Akiem

First-principles profile

A. Resolve the company and segment. The relevant business is **Akiem**, headquartered in **Saint-Ouen, France**, founded in **2008** and operating across **22 European countries**. Official history on Akiem's site states that **La Caisse/CDPQ became the sole shareholder** after buying out SNCF and DWS, while a current governance page says CDPQ is the **majority shareholder**. I therefore treat current ownership as **CDPQ-controlled**, with some uncertainty around any minority holdings. The latest detailed public financial fact I found is from official 2021 transaction materials; current public operating facts extend through the **April 2026 refinancing** and 2025/2026 operating updates. [18]

B. Define the economic unit. The core economic unit is **one locomotive under lease**, optionally bundled with maintenance/full-service obligations. Passenger-train activity exists, but the designated Transportation Solutions archetype is the **locomotive leasing operation**, so that is the governing unit for growth, utilization, capex, and payback. [19]

C. Explain the customer problem. Rail operators need traction capacity that is available, interoperable, and reliable without committing huge capital to owned fleets. Akiem's own materials describe the company as a partner for fleet strategy that supplies locomotives and passenger trains, finances new and used rolling stock, and lets customers focus on their core operations. The need is mission-critical, recurring, and increasingly shaped by cross-border interoperability and decarbonization requirements. [19]

D. Explain customer alternatives. Customers can buy locomotives, finance them directly, use sale-and-leaseback, or lease from another lessor. Switching is materially harder than in many rental markets because locomotives require approvals, compatibility with national systems, maintenance support, and highly reliable uptime. Those frictions are part of why Akiem has built ECM-certified maintenance infrastructure and dedicated country footprints. [19]

E. Map the revenue model. Revenue comes from **leasing solutions**, **industrial and maintenance solutions**, **full service**, and now some passenger rolling-stock financing/management. The current company is therefore not a bare lessor; it is a leasing-plus-maintenance platform. That raises service content and customer stickiness but lowers comparability with a pure finance-style lessor. [20]

F. Map the operating model. Akiem maintains workshops, spare parts, industrial/logistical support, and ECM-certified maintenance systems. Its official materials emphasize operational maintenance, general overhauls, spare-parts management, and major upgrades such as **ERTMS baseline 3**, with roughly **250 locomotives** being equipped under that program. This is an asset-owner model that actively protects asset availability and future compliance. [19]

G. Explain customer retention and churn. Customers remain when Akiem is solving a long-term traction problem with equipment that is already approved, maintained, and integrated into service plans. Relationships often outlast any single legal contract because fleets are resized, refreshed, and reconfigured over time. Customers leave when concessions change, traffic plans change, operators fail, or locomotives become technically mismatched to network or emissions requirements. [21]

H. Map the capital model. Akiem is extremely capital-intensive. Official transaction materials described a 2021 fleet of **over 600 locomotives and 46 passenger trains**, with roughly **€220 million** of revenue and **€150 million** of EBITDA. Since then, Akiem has expanded orders materially, completed a **€1.52 billion green refinancing** in April 2026, and emphasized that the transaction supports future growth projects. This model depends on long contract lives, long asset lives, and meaningful residual values, but also on regular heavy maintenance, upgrades, and occasional mid-life programs. [22]

Akiem metric table

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Normalized revenue growth	3%	6%	10%	Triangulated	Medium	Fleet additions, modernization demand, and lease-rate mix

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Normalized EBITDA margin	55%	62%	70%	Triangulated	Medium	High locomotive lease margins tempered by maintenance-service content
Maintenance capex % of revenue	12%	18%	25%	Inferred	Medium	Heavy overhauls, compliance upgrades, and mid-life programs
Post-maintenance-capex conversion	55%	71%	83%	Calculated	Medium	Strong lease EBITDA offset by sustaining overhauls and upgrades
Initial contract tenor	5 yrs	7 yrs	10 yrs	Inferred	Medium	High-value locomotives need longer legal terms to support economics
Effective customer tenure	8 yrs	12 yrs	15 yrs	Inferred	Medium	Switching friction, approvals, and repeated fleet program extensions
Utilization exposure	Moderate	Moderate	High	Inferred	Medium	Long contracts reduce churn, but each off-lease unit is expensive and harder to redeploy

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
New-unit or new-customer payback	8 yrs	10 yrs	13 yrs	Triangulated	Medium	Large upfront unit cost with strong residual value and multi-cycle earnings
B2B/B2C profile	Predominantly B2B	Predominantly B2B	Predominantly B2B with public-authority exposure	Disclosed	High	Freight/ passenger operators and transport authorities

Akiem's strongest disclosed anchors are the 2021 revenue/EBITDA/fleet data in official transaction materials, the current official operating model on the company site, and the 2026 green refinancing used to support future growth. [23]

Akiem metric analysis

Revenue growth

Conclusion. Normalized revenue growth is **3% / 6% / 10%**.
Observed facts. Akiem's official site shows major recent orders from Siemens, Alstom, and Newag, while the 2026 refinancing was explicitly framed as supporting future growth projects. Official history also records expansion in maintenance capacity and passenger rolling stock. [21]
Root business mechanism. Growth is primarily driven by new locomotives placed on lease, higher-value interoperable fleet mix, and secular demand for outsourced traction capacity.
Calculation or driver bridge. Central case assumes moderate annual fleet growth plus pricing/mix from newer and greener assets, without assuming M&A-led growth every year.
Sensitivities. European freight volumes, public transport funding, OEM delivery schedules, and cross-border approval timing.
Confidence. Medium.

EBITDA margin

Conclusion. Normalized EBITDA margin is **55% / 62% / 70%**.
Observed facts. Official 2021 transaction materials showed roughly **€150 million EBITDA on €220 million revenue**, implying exceptionally strong leasing economics, while current company materials make clear that Akiem also performs substantial maintenance and industrial services. [24]
Root business mechanism. Locomotive lease economics are strong because each asset is scarce and mission-critical, but full-service maintenance, workshops, and upgrade programs add recurring operating cost.
Calculation or driver bridge. Central margin accepts that 2021 demonstrated very strong profitability, but normalizes lower to reflect broader service mix, passenger-train activity, and maintenance intensity.

Sensitivities. Share of full-service contracts, workshop absorption, and locomotive mix.
Confidence. Medium.

Maintenance capex

Conclusion. Maintenance capex normalizes at **12% / 18% / 25% of revenue**.
Observed facts. Akiem explicitly maintains locomotives, performs overhauls, manages spare parts, and is equipping around **250 locomotives** with ERTMS. Its April 2026 refinancing was designed not only to refinance debt but also to support growth projects. [21]
Root business mechanism. Locomotives have long lives, but they also require major scheduled and unscheduled work. Sustaining investment is more lumpy than for wagons because each unit is more complex and more valuable.
Calculation or driver bridge. Central case assumes that part of annual capex is genuine sustaining spend for overhauls and compliance upgrades; the high case reflects fleet age, electrification/ERTMS intensity, and passenger-train complexity.
Sensitivities. Asset age, maintenance packaging, grant support, and capitalized upgrade treatment.
Confidence. Medium.

Post-maintenance-capex conversion

Conclusion. Post-maintenance-capex conversion is **55% / 71% / 83%**.
Observed facts. Akiem's lease-heavy model produces strong EBITDA, but the company also runs certified maintenance and major technical programs. [25]
Root business mechanism. EBITDA is strong because the assets are scarce and contracts are long, but sustaining spend remains meaningful because availability and compliance are part of the commercial promise.
Calculation or driver bridge. Applying the central margin and maintenance-capex assumptions yields roughly 71% conversion before interest, taxes, working capital, and growth capex.
Sensitivities. Timing of overhauls and technical retrofits.
Confidence. Medium.

Initial contract tenor and effective customer tenure

Conclusion. Initial legal term is **5 / 7 / 10 years**; effective customer tenure is **8 / 12 / 15 years**.
Observed facts. Akiem's business is built around integrated support from "diagnosing your needs until you finally hand back the rolling stock," and its growth historically includes sale-and-leaseback and framework agreements for large numbers of locomotives. [19]
Root business mechanism. A high-value locomotive typically needs a longer first term than a wagon or trailer to justify purchase and mobilization. Effective tenure can be much longer because operators often continue leasing similar units from the same lessor.
Calculation or driver bridge. Central case assumes long but not ultra-long first contracts, with customer life extending through renewals, fleet refreshes, or successor assets.
Sensitivities. Passenger versus freight mix, public tenders, and residual-value confidence.
Confidence. Medium.

Utilization exposure

Conclusion. Utilization exposure is **Moderate overall**, but the downside skew is meaningful because each off-lease locomotive is expensive and remarketing is slower than for wagons.
Observed facts. Akiem stresses pan-European reach, interoperability, proximity to customers, and maintenance capacity; the ERTMS upgrade program also shows that technical compatibility and compliance matter to asset usability. [19]
Root business mechanism. Long contracts reduce near-term churn, but each asset is specialized, approvals are not trivial, and technical mismatch can delay redeployment.
Calculation or driver bridge. I classify the asset-utilization risk as Moderate rather than Low because contract structure is strong, but off-lease pain is large when it occurs.
Sensitivities. Border-crossing approvals, freight cycles, and operator credit.
Confidence. Medium.

New-unit or new-customer payback

Conclusion. New-locomotive payback is **8 / 10 / 13 years**.
Observed facts. Akiem continues to place large locomotive orders and refinance for growth, implying that individual asset economics are long-dated and capital-intensive. [21]
Root business mechanism. A locomotive's upfront cost is high enough that cash recovery usually depends on a long first term plus residual value or second placements.
Calculation or driver bridge. Central case assumes strong annual cash contribution but also a substantial maintenance-capex reserve; residual value shortens the effective economic recovery period relative to a "zero residual" model.
Sensitivities. Unit cost, maintenance responsibility, utilization, and resale values.
Confidence. Medium.

B2B versus B2C profile

Conclusion. Akiem is **predominantly B2B**, with some public-authority and regional-transport exposure at the contractual level.
Observed facts. Official materials say Akiem serves freight and passenger carriers, manufacturers, and local authorities. [19]
Root business mechanism. Even where passengers are the end users, the contractual customer is still a professional operator or transport authority.
Calculation or driver bridge. The business is entirely institutional from a sales and retention perspective.
Sensitivities. Passenger-train growth would increase public-authority exposure but not create true B2C economics.
Confidence. High.

Akiem conclusion

Plain-English business description. Akiem buys, leases, and maintains locomotives and some passenger rolling stock so rail operators can access traction capacity without owning and servicing the full fleet themselves.

Principal strengths. Large pan-European installed base, technical know-how, ECM-certified maintenance ecosystem, long contract tenors, and strong strategic position in fleet modernization. [21]

Principal risks. Higher single-unit concentration than wagon leasing, lumpy overhaul and compliance capex, and mixed-business disclosure that combines leasing with maintenance and passenger activity. [25]

Key unavailable information. Current public consolidated revenue, current EBITDA, contract-book duration, gross sustaining capex by fleet family, and precise current cap table under CDPQ control.

Most important diligence questions. What percentage of EBITDA comes from leasing versus maintenance services? How many years of weighted-average remaining lease term are on the locomotive fleet? What is the annual heavy-overhaul reserve per locomotive family? How much of ERTMS spend is customer-funded, grant-funded, or lessor-funded? What has actual redeployment time been for returned locomotives?

Company profiles for Air Rail and TCR

Air Rail

First-principles profile

A. Resolve the company and segment. The resolved legal entity is **AIR RAIL, S.L.** of Madrid. IFM's 2025 and 2026 materials tie that entity to airport GSE leasing and maintenance, with **75% IFM ownership** and founder **José Manuel García Prieto** retaining **25%**. The key caveat is that the company is **mixed**: its own website markets both airport GSE and rail-sector machinery/services. For this benchmark I isolate the **airport GSE leasing and maintenance segment**, using the latest reliable official operating facts from **31 March 2026** where available. [26]

B. Define the economic unit. The primary economic unit is **one GSE unit deployed under an operating rental/full-service arrangement**. But airport GSE economics are also shaped by the **airport-station cluster** because maintenance coverage, spare fleets, approvals, and redeployment are airport-specific. A single stair unit or pushback tractor matters less economically than a contracted cluster of units tied to one customer at one airport. [27]

C. Explain the customer problem. The customer needs mission-critical airside equipment without committing capital and without building a full maintenance and lifecycle-management organization. Air Rail's services solve availability, fleet flexibility, maintenance response, and decarbonization problems. If that need is not addressed, turnarounds slow, compliance risk rises, and handlers or airlines can lose operational performance. [27]

D. Explain customer alternatives. Customers can buy equipment, self-maintain, use other lessors, or rely on airport/handler arrangements. Manual substitution is very limited for core equipment categories. Switching is easier than in locomotives but harder than generic equipment rental because airports impose access, operating, and safety constraints, and because fleet maintenance responsiveness matters. [27]

E. Map the revenue model. Air Rail's site discloses revenue lines including **Operating Renting**, **Maintenance Contracts**, **Technical Service**, **Distribution of new equipment**, **Original Spare Parts**, **Second-hand purchase and sale**, and **Reconditioning & Overhaul**. That means the company combines recurring rental with service, distribution, parts, and resale/refurbishment revenue. For benchmarking I focus primarily on the recurring leased-fleet and maintenance component. [28]

F. Map the operating model. After deployment, Air Rail performs preventive, corrective, and full-service maintenance; technical service; parts support; telemetry-based fleet monitoring; and overhaul/reconditioning. The website explicitly states "preventive, corrective and full service maintenance contracts" and highlights telemetry for real-time control of machines. The company also runs a substantial parts inventory and used/refurbished activity. [28]

G. Explain customer retention and churn. Customers remain when equipment availability is high, response times are good, and Air Rail helps them defer capex while meeting electrification and operational goals. Churn risk comes from handler rebids, airline network changes, airport concession changes, insourcing, or loss of a key airport relationship. Importantly, an **airport-site contract** is not the same thing as a whole-customer relationship: the same ground handler might stay with Air Rail in one airport and leave in another. [29]

H. Map the capital model. Air Rail grew from **180 units in 2014 to more than 4,000 in 2024** on its own site, while IFM's latest portfolio page says **over 4,700 GSE units** at **circa 60 airports in eight countries** as of March 2026. The company also says it has met and exceeded its goal of having **60% of the fleet electric by 2025**, while IFM estimates **68% of motorized units electrified** as of March 2026. This is a capital-intensive, fleet-owning model with residual value and redeployment potential, but the economics depend on airport-specific placement, maintenance density, and lifecycle management. [27]

Air Rail metric table

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Normalized revenue growth	5%	8%	12%	Triangulated	Medium	Airport expansion, electrification replacement, and full-service penetration
Normalized EBITDA margin	20%	27%	33%	Inferred	Low	Recurring fleet rental plus maintenance/service/distro mix
Maintenance capex % of revenue	9%	13%	18%	Inferred	Low	Shorter-lived assets, batteries, refurbishment, and safety/compliance spend

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Post-maintenance-capex conversion	35%	52%	65%	Calculated	Low	Service-heavy EBITDA and meaningful sustaining fleet spend
Initial contract tenor	3 yrs	5 yrs	7 yrs	Inferred	Medium	Ground-handler/airline contracts and airport-specific deployments
Effective customer tenure	5 yrs	8 yrs	12 yrs	Inferred	Medium	Multi-airport relationships and repeated renewals/ outplacements
Utilization exposure	Moderate	Moderate-High	High	Inferred	Medium	Idle spare units often do not earn revenue and airport transfers are frictional
New-unit or new-customer payback	4 yrs	5.5 yrs	8 yrs	Triangulated	Low	Equipment cost, service content, utilization, and redeployment value
B2B/B2C profile	Predominantly B2B	Predominantly B2B	Predominantly B2B with handler concentration	Disclosed	High	Airlines, handlers, airports

This table is lower-confidence than TCR's because Air Rail is a **mixed airport/rail company** and public standalone segment reporting is thin. The official anchors that matter most are the legal-entity evidence, the current ownership and scale statements from IFM, and the company's own disclosed revenue/solution lines. [3]

Air Rail metric analysis

Revenue growth

Conclusion. Normalized growth is **5% / 8% / 12%**.
Observed facts. IFM described Air Rail as the largest GSE owner/lessor in Spain and Portugal in 2025, and its 2026 portfolio page showed over **4,700 units** across **circa 60 airports** in **eight countries**. [8]
Root business mechanism. Growth comes from new airport entries, more units per airport, electrification-driven replacement cycles, and cross-selling maintenance and lifecycle services.
Calculation or driver bridge. Central assumption reflects a still-growing but not hyper-growth airport infrastructure platform.
Sensitivities. Air traffic, handler market share, airport approvals, and pace of European expansion.
Confidence. Medium.

EBITDA margin

Conclusion. Normalized EBITDA margin is **20% / 27% / 33%**.
Observed facts. Air Rail combines operating rental with maintenance, parts, distribution, and overhaul. IFM describes a full-service-rental led model rather than simple equipment broking. [30]
Root business mechanism. Recurring lease revenue supports decent margins, but the maintenance and service obligations are heavier than for railcars, and commercial distribution activity can dilute margin.
Calculation or driver bridge. Central case sits below TCR's likely global-scale benchmark because Air Rail is smaller and structurally mixed.
Sensitivities. Share of revenue from pure rental versus distribution and repair, technician productivity, and fleet age.
Confidence. Low.

Maintenance capex

Conclusion. Maintenance capex is **9% / 13% / 18% of revenue**.
Observed facts. Air Rail discloses operating rental, maintenance contracts, overhaul/reconditioning, and a heavily electrified fleet. IFM's 2026 page highlights one of the youngest and most electrified fleets in the market. [27]
Root business mechanism. GSE assets have shorter lives than railcars, and sustaining economics include refurbishment, major repairs, and eventually battery and charging-related replacements.
Calculation or driver bridge. Central estimate assumes a meaningful sustaining burden but still below the heaviest rail-asset cases because unit costs and asset lives are shorter.
Sensitivities. Fleet age, battery replacement cadence, and how much charging infrastructure sits on the customer versus lessor side.
Confidence. Low.

Post-maintenance-capex conversion

Conclusion. Post-maintenance-capex conversion is **35% / 52% / 65%**.
Observed facts. The model is a blend of recurring rental and substantial service obligations, with a young/electrified fleet requiring ongoing maintenance planning. [27]
Root business mechanism. EBITDA is real, but preserving airport service levels and fleet readiness consumes a larger share of cash than in a pure long-life railcar lessor.

Calculation or driver bridge. Central case uses the margin and capex assumptions above and excludes interest, taxes, working capital, and growth capex.
Sensitivities. Refurbishment cadence, airport redeployment friction, and service-level commitments.
Confidence. Low.

Initial contract tenor and effective customer tenure

Conclusion. Initial term is **3 / 5 / 7 years**; effective customer tenure is **5 / 8 / 12 years**.
Observed facts. Air Rail's customers include handlers, airlines, and airports, and IFM describes the company as operating at many airports with long-term customer relationships. [31]
Root business mechanism. Legal life is often bounded by handler contracts, airport arrangements, or fleet plans, but customers can continue across multiple placements and airports.
Calculation or driver bridge. Central case assumes several-year initial contracts with repeated renewals or re-awards where service is good and airports are retained.
Sensitivities. Handler rebids, airport policies, and customer concentration.
Confidence. Medium.

Utilization exposure

Conclusion. Utilization exposure is **Moderate-to-High**.
Observed facts. Air Rail's business is tied to airports and mission-critical GSE; IFM describes airport-specific geographic deployment and emphasizes the fleet's electrified composition. [31]
Root business mechanism. Idle GSE generally does **not** earn revenue, yet some spare equipment must be held to guarantee service. Moving assets between airports or countries is feasible but not frictionless.
Calculation or driver bridge. Central classification is Moderate-High because contracts soften volatility, but sudden demand changes or customer losses can create idle fleets and transfer costs.
Sensitivities. Flight activity, customer wins/losses, airport approvals, and electrified-charging compatibility.
Confidence. Medium.

New-unit or new-customer payback

Conclusion. Payback normalizes at **4 / 5.5 / 8 years**.
Observed facts. Air Rail's offering includes everything from stairs and conveyor belts to maintenance and telemetry, and IFM characterizes the model as full-service rental. [27]
Root business mechanism. GSE unit economics recover faster than locomotives because units are cheaper, but the payback is not trivial because service, maintenance, and airport-specific mobilization matter.
Calculation or driver bridge. Central case assumes a blended fleet and a meaningful maintenance-capex reserve.
Sensitivities. Equipment type, utilization intensity, electrification, charger funding, and residual value.
Confidence. Low.

B2B versus B2C profile

Conclusion. Air Rail is **predominantly B2B**.
Observed facts. IFM and the company site both describe customers as airports, airlines, and ground handlers. [29]
Root business mechanism. The contractual customer is always institutional.
Calculation or driver bridge. End users are passengers and cargo shippers, but revenue is contractually

B2B.
Sensitivities. Customer concentration by major handlers or airport accounts.
Confidence. High.

Air Rail conclusion

Plain-English business description. Air Rail's relevant segment leases and maintains airside equipment for airports, airlines, and ground handlers, while also selling equipment, parts, and refurbishment services.

Principal strengths. Iberian market position, high fleet electrification, airport embeddedness, and full-service operating model. [32]

Principal risks. Mixed-company disclosure, ground-handler rebid exposure, airport redeployment friction, and battery/charging obsolescence risk. [33]

Key unavailable information. Public consolidated revenue, EBITDA, segment mix between airport and rail activities, contract duration statistics, and gross sustaining capex.

Most important diligence questions. What share of revenue is airport GSE rental versus equipment distribution? What is the weighted-average remaining contract life by airport? What portion of chargers and batteries is funded by Air Rail? What are actual idle-unit days by equipment family? How concentrated is EBITDA in top handlers and airports?

TCR

First-principles profile

A. Resolve the company and segment. The relevant business is **TCR Group**, headquartered in **Steenokkerzeel/Brussels Airport, Belgium**. TCR describes itself as the world's leading or largest independent GSE lessor and full-service provider. Ownership is in transition: GIP announced in March 2026 that it had entered a definitive agreement to acquire TCR from **3i Infrastructure**, and **3i** said the transaction was expected to complete in **Q3 2026**. As of the analysis date, the best reading is that TCR is **still in signed sale but pre-close status**. [34]

B. Define the economic unit. The primary unit is **one revenue-earning GSE asset under contract**, but the economically decisive unit is often the **station/airport fleet package** because TCR's economics depend on workshop density, spare pools, technician coverage, and airport embeddedness. [35]

C. Explain the customer problem. Customers need GSE that is available, compliant, maintained, and increasingly electrified without tying up capex or internal maintenance capacity. TCR frames its offering around reliability, safety, fleet scaling, and predictable cost, all of which are mission-critical to aircraft turnaround. [36]

D. Explain customer alternatives. Airlines, airports, and handlers can own fleets, sell and lease them back, self-maintain, or use other lessors. But TCR's value proposition is broader than financing: it includes maintenance, ramp assistance, telematics, fleet planning, refurbishment, conversion, and electrification support. That creates more switching friction than pure ownership-versus-rental economics alone would suggest. [36]

E. Map the revenue model. TCR explicitly breaks out **Full Service Rental**, **Maintenance & Ramp Assistance**, **Sale & Lease Back**, **Fleet Management**, and **Electrification & Decarbonisation**. Its GSE-services page also references pre-owned sales and consultancy/training. So recurring revenue is dominated by leased fleet contracts, but there is meaningful ancillary service and lifecycle revenue around the core. [36]

F. Map the operating model. TCR operates an extensive service platform with **100+ workshops**, **around 1,000 technicians**, large spare-parts capability, telematics/fleet-management tools, and ECO centres for refurbishment/reuse. This is a classic full-service rental model where maintenance intensity and station density are major margin drivers. [37]

G. Explain customer retention and churn. Customers remain because turnaround reliability is critical, downtime is costly, and TCR can manage the fleet lifecycle better than many customers can internally. Churn comes from opponent lessors, handler rebids, airline network changes, airport concession changes, insourcing, or poor service. TCR’s long-term relationships matter, but an individual station contract can still turn over. [38]

H. Map the capital model. TCR’s scale is exceptional: the current official count is **44,000+ assets**, **100+ workshops**, **1,800+ employees**, and operations across **24 countries**; sponsor materials still describe **more than 200 airports worldwide**, while earlier 3i materials referenced **more than 230 airports across more than 20 countries**. The model requires continuous fleet renewal, refurbishment, and increasingly electrification-related capex, but also benefits from refurbishment, resale, and fleet density. [39]

TCR metric table

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Normalized revenue growth	6%	9%	13%	Triangulated	Medium	Airport growth, outsourcing, electrification, and global expansion
Normalized EBITDA margin	26%	32%	38%	Inferred	Medium	Scaled full-service rental with dense maintenance network
Maintenance capex % of revenue	10%	14%	18%	Inferred	Medium	Refurbishment, replacement fleet, electric transition, and batteries

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Post-maintenance-capex conversion	42%	56%	69%	Calculated	Medium	Healthy rental EBITDA partly recycled into sustaining fleet spend
Initial contract tenor	3 yrs	5 yrs	7 yrs	Inferred	Medium	Handler/airline/airport contracts and sale-leaseback structures
Effective customer tenure	6 yrs	9 yrs	14 yrs	Inferred	Medium	Deep airport embeddedness and multiservice relationships
Utilization exposure	Moderate	Moderate-High	High	Inferred	Medium	Paying deployment matters and idle/reserve fleets can dilute returns
New-unit or new-customer payback	4 yrs	5.5 yrs	7.5 yrs	Triangulated	Medium	Unit cost, service labor, and resale/refurbishment value
B2B/B2C profile	Predominantly B2B	Predominantly B2B	Predominantly B2B with some customer concentration	Disclosed	High	Handlers, airlines, airports

The operational anchors are unusually strong for a private company: TCR explicitly discloses the structure of its service offering, network scale, workforce scale, electrification goals, and sponsor ownership transition. [40]

TCR metric analysis

Revenue growth

Conclusion. Normalized growth is **6% / 9% / 13%**.
Observed facts. TCR is the largest independent GSE lessor, serves **200+ airports**, and continues to invest in

electrification and fleet management capabilities; 3i and GIP both framed it as a scaled growth platform. [41]
Root business mechanism. Growth comes from new airport/station wins, wider outsourcing, larger per-airport fleets, and electrification projects that refresh installed bases.
Calculation or driver bridge. Central case assumes continued outsourcing and green-fleet adoption without requiring extraordinary M&A.
Sensitivities. Passenger activity, handler market structure, and airport capex policies.
Confidence. Medium.

EBITDA margin

Conclusion. Normalized EBITDA margin is **26% / 32% / 38%**.
Observed facts. TCR runs a scaled full-service network with 100+ workshops and ~1,000 technicians. A local TCR UK entity reported a **10.9% operating-profit margin** on **£67.9 million** of revenue in FY2024, but that local figure is not the right benchmark for the global full-service lessor. [42]
Root business mechanism. Scale and dense maintenance infrastructure support margins, but the business is still more labor- and maintenance-intensive than railcar leasing.
Calculation or driver bridge. Central case places TCR above smaller mixed airport platforms because global density helps purchasing, workshop utilization, and spares management.
Sensitivities. Mix of rental versus service-only work, technician productivity, and fleet age.
Confidence. Medium.

Maintenance capex

Conclusion. Maintenance capex is **10% / 14% / 18% of revenue**.
Observed facts. TCR's own solution pages emphasize lifecycle management, refurbishment, conversions, electrification, fleet planning, and ECO-centre refurbishment; it also runs one of the world's largest GSE fleets. [43]
Root business mechanism. Full-service GSE leasing requires recurring sustaining spend: unit replacement, refurbishment, major repairs, emissions/electrification retrofits, and eventually battery-related capital.
Calculation or driver bridge. Central assumption reflects the shorter life and higher wear of airport GSE relative to wagons, moderated by refurbishment and resale/reuse economics.
Sensitivities. Electric mix, battery treatment, airport ownership of charging infrastructure, and residual values by equipment family.
Confidence. Medium.

Post-maintenance-capex conversion

Conclusion. Post-maintenance-capex conversion is **42% / 56% / 69%**.
Observed facts. TCR's business deliberately absorbs maintenance and lifecycle obligations rather than outsourcing them back to the customer. [36]
Root business mechanism. EBITDA is solid, but sustaining fleet readiness, compliance, and electrification consume meaningful cash.
Calculation or driver bridge. Central case follows directly from the above margin and maintenance-capex assumptions.
Sensitivities. Fleet age, refurbishment recoveries, and spare-fleet intensity.
Confidence. Medium.

Initial contract tenor and effective customer tenure

Conclusion. Initial term is **3 / 5 / 7 years**; effective customer tenure is **6 / 9 / 14 years**.
Observed facts. GIP described TCR as having **long-term contractual relationships** and a diversified customer base. TCR's own offerings include full-service rental, sale-and-leaseback, and multi-year fleet planning. [38]
Root business mechanism. Contracts are long enough to support equipment deployment and station setup, but customer relationships often continue across contract renewals, additional stations, or refreshed fleets.
Calculation or driver bridge. Central case reflects the airport-GSE norm of multi-year terms with longer economic relationships where operational performance is strong.
Sensitivities. Handler rebid cycles, airport tendering, and customer credit.
Confidence. Medium.

Utilization exposure

Conclusion. Utilization exposure is **Moderate-to-High**.
Observed facts. TCR explicitly addresses additional GSE on demand, fleet right-sizing, idle-fleet waste, and telematics-based fleet planning. It also notes that up to **600 motorized GSEs** are available in the used/refurbished channel, supported by **44,000+ assets**. [36]
Root business mechanism. Idle GSE does not typically earn, but a service-quality promise requires some spare coverage. Utilization must be analyzed as paid deployment, actual operating intensity, and workshop/technician utilization.
Calculation or driver bridge. I classify central exposure as Moderate-High because TCR can redeploy and optimize at scale, but flight activity and contract turnover still move earnings.
Sensitivities. Airport traffic, customer resizing, and transfer friction between airports or countries.
Confidence. Medium.

New-unit or new-customer payback

Conclusion. Payback is **4 / 5.5 / 7.5 years**.
Observed facts. TCR's business includes equipment supply, maintenance, fleet management, and electrification support, plus resale/refurbishment through ECO centres. [43]
Root business mechanism. The capital burden is meaningful but unit costs are much lower than locomotives, and refurbishment/resale help recover value beyond the first customer placement.
Calculation or driver bridge. Central case assumes mid-single-digit-year cash recovery on a blended equipment basis after service costs and a maintenance reserve.
Sensitivities. Equipment family, battery aging, contract term, and secondary-market liquidity.
Confidence. Medium.

B2B versus B2C profile

Conclusion. TCR is **predominantly B2B**.
Observed facts. TCR's customers are "predominantly independent ground handling companies, airlines and airports." [44]
Root business mechanism. Sales cycles are institutional, tender-driven, and operationally sophisticated.
Calculation or driver bridge. Passenger and cargo users are indirect end users only.

Sensitivities. Concentration in a few large handlers at specific airports.
Confidence. High.

TCR conclusion

Plain-English business description. TCR owns, rents, maintains, refurbishes, and upgrades airport ground-support equipment so airlines, handlers, and airports can run safe, reliable, and increasingly electrified ground operations.

Principal strengths. Global scale, dense service footprint, explicit full-service lifecycle model, refurbishment ecosystem, and strong sponsor validation. [45]

Principal risks. Flight-activity sensitivity, handler contract turnover, electrification capex demands, and the possibility that idle reserve fleets dilute returns. [35]

Key unavailable information. Full current public consolidated revenue, EBITDA, maintenance-capex split, weighted-average contract term, and airport-level utilization statistics.

Most important diligence questions. What share of group EBITDA comes from full-service rental? What is the paid-utilization rate by major equipment category? What proportion of electrification capex is customer-recoverable? How much spare capacity is contractually required? What are gross and net refurbishment economics by unit family?

Company profile for Lazer Logistics

Lazer Logistics

First-principles profile

A. Resolve the company and segment. The business is **Lazer Logistics**, formerly **Lazer Spot**, headquartered in **Alpharetta/Atlanta, Georgia**. The company says it was rebranded from Lazer Spot to Lazer Logistics to reflect the broader value delivered beyond spotting, and that **EQT Partners and Lazer management** acquired it in **2023**. EQT describes it as North America's largest provider of outsourced yard management and trailer spotting services, covering **550+ customer sites**, with **5,000 employees**, **6,000 fleet assets**, and **9 million annual service hours** across **39 states and territories**. [46]

B. Define the economic unit. The governing economic unit is **one customer site contract**, not one tractor. A yard tractor matters for maintenance and mobilization, but the true economic unit for growth, utilization, and payback is the **site-level operating program** or, secondarily, the **service hour/shift** because labor, supervision, schedules, and customer integration drive value. [47]

C. Explain the customer problem. Customers need trailers spotted, gates managed, yard bottlenecks removed, and freight moved between dock, yard, and nearby nodes without building an internal hostler operation. Lazer pitches improved velocity, visibility, control, lower idle time, lower fuel cost, and fewer wasted moves. The need is recurring and operationally important: if the yard is inefficient, dock turns slow, throughput drops, and site labor is wasted. [48]

D. Explain customer alternatives. Customers can run hostlers internally, use local trucking/cartage, manage gates themselves, buy tractors, or accept manual/less optimized yard processes. Switching is operationally easier than in locomotives or airport GSE, but still disruptive because the provider becomes embedded in staffing, safety protocols, schedules, and site systems. That is why Lazer emphasizes technology, analytics, and “The Power Triangle” of technology, operations, and people. [48]

E. Map the revenue model. Lazer’s disclosed solutions include **yard spotting**, **shuttle services**, **EV yard trucks**, **trailer rental**, **yard tech/NexusYMS**, **drayage**, **gate management**, and **shunting services**. This implies a mixed revenue model of recurring site-service fees, labor/service-hour revenue, equipment rental, and increasing software/technology attachment, but the business is clearly led by outsourced operations rather than vehicle finance. [49]

F. Map the operating model. Lazer recruits and schedules drivers, manages safety, supervises site labor, runs or coordinates tractors and trailers, provides EV expertise, and layers in proprietary technology and analytics. EQT’s portfolio page quantifies that model at **9 million annual service hours**, which is the clearest signal that labor utilization is central to economics. [50]

G. Explain customer retention and churn. Customers stay because Lazer is operationally embedded and can lower idle time, increase throughput, and remove management burden. Lazer’s website cites customer results such as substantially lower idle time and major fuel savings, and customer testimonials reference lower equipment costs and faster issue resolution. Customers leave when sites close, volumes fall, rebids are lost, in-house teams are rebuilt, or labor/service quality slips. [48]

H. Map the capital model. Lazer is capital-light **relative to rail and airport lessors**, but not asset-free. EQT cites **6,000 fleet assets**, while the company’s site markets trailer rental and EV yard trucks. Sustaining spend therefore includes yard tractors, trailers, EV infrastructure interfaces, and site technology, but the dominant economic burden is labor, recruiting, scheduling, and site management rather than depreciating high-value mobile equipment. This is why Lazer should be treated as an **outsourced operations archetype**, not a core asset-lessor comparator. [50]

Lazer Logistics metric table

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Normalized revenue growth	4%	7%	11%	Triangulated	Medium	New site wins, cross-sell, EV adoption, and selective tuck-ins
Normalized EBITDA margin	10%	14%	18%	Inferred	Medium	Labor-heavy outsourcing economics with some tech and asset support

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Maintenance capex % of revenue	2%	4%	7%	Inferred	Medium	Tractors, trailers, and sustaining site tech, but far less than asset lessors
Post-maintenance-capex conversion	55%	71%	84%	Calculated	Medium	Lower capex burden allows better post-maintenance cash conversion
Initial contract tenor	1 yr	3 yrs	5 yrs	Inferred	Medium	Site-service agreements and operational outsourcing contracts
Effective customer tenure	4 yrs	7 yrs	10 yrs	Inferred	Medium	Embedded operations and switching disruption extend life beyond legal term
Utilization exposure	High	High	High	Inferred	High	Labor and site-capacity utilization are the core earnings lever
New-unit or new-customer payback	0.8 yrs	1.5 yrs	2.5 yrs	Triangulated	Medium	Mobilization, recruiting, and limited equipment spend recover quickly if site volume holds
B2B/B2C profile	Predominantly B2B	Predominantly B2B	Predominantly B2B with blue-chip concentration	Disclosed	High	Warehousing, retail, manufacturing, food, 3PLs

The decisive pieces of evidence are the official and sponsor descriptions of Lazer as an outsourced yard-management provider, the scale metrics from EQT, and the company's own presentation of technology-enabled, labor-led yard operations. [51]

Lazer Logistics metric analysis

Revenue growth

Conclusion. Normalized growth is **4% / 7% / 11%**.
Observed facts. Lazer has expanded through acquisitions, broadened offerings beyond spotting, and now covers **550+ sites** with **9 million service hours**. [52]
Root business mechanism. Growth comes from winning new sites, adding adjacent services at existing sites, technology attach, and occasional tuck-in acquisitions.
Calculation or driver bridge. Central case assumes steady new-site wins and cross-sell, but not constant acquisition-driven expansion.
Sensitivities. Warehouse/logistics volumes, labor availability, and customer site consolidation.
Confidence. Medium.

EBITDA margin

Conclusion. Normalized EBITDA margin is **10% / 14% / 18%**.
Observed facts. Lazer's business is built on people, operations, and technology, with 5,000 employees and 9 million service hours. The company markets operational savings, not finance-style asset spreads. [50]
Root business mechanism. Labor, supervision, recruiting, and schedule efficiency dominate cost. That structurally caps margins below asset-leasing models.
Calculation or driver bridge. Central case assumes decent scale and tech support, but a gross-revenue denominator that includes substantial labor content.
Sensitivities. Wage inflation, overtime, turnover, subcontracting, and contract pricing structure.
Confidence. Medium.

Maintenance capex

Conclusion. Maintenance capex is **2% / 4% / 7% of revenue**.
Observed facts. Lazer does own or control fleet assets and offers trailer rental and EV yard trucks, but the sponsor description makes clear that the core model is site-service and service hours, not high-value asset leasing. [50]
Root business mechanism. Sustaining spend exists, but it is secondary to payroll and operating expense.
Calculation or driver bridge. Central case includes tractor/trailer replacement, shop tools, and sustaining site technology while excluding pure growth fleet.
Sensitivities. EV mix, ownership versus aggregated equipment, and capitalized software.
Confidence. Medium.

Post-maintenance-capex conversion

Conclusion. Post-maintenance-capex conversion is **55% / 71% / 84%**.
Observed facts. The model is labor-heavy with relatively limited sustaining capital compared with rail or airport lessors. [50]
Root business mechanism. Lower capex means more EBITDA survives after maintenance spending,

although working capital and labor volatility still matter.
Calculation or driver bridge. Central case follows from the central margin and capex assumptions and remains well below true free-cash-flow conversion because it excludes working capital and lease principal.
Sensitivities. Site losses, uncompensated mobilization, and capex treatment of tractors or software.
Confidence. Medium.

Initial contract tenor and effective customer tenure

Conclusion. Initial term is **1 / 3 / 5 years**; effective customer tenure is **4 / 7 / 10 years**.
Observed facts. Lazer is a deeply embedded site operator with broad solution scope, and customer testimonials point to long-running operational relationships. [48]
Root business mechanism. Legal contracts need not be long because the provider can be replaced, but actual replacement is disruptive once staffing, safety, and yard systems are integrated.
Calculation or driver bridge. Central case assumes relatively short legal terms but materially longer effective tenure.
Sensitivities. Rebids, customer insourcing, and site closures.
Confidence. Medium.

Utilization exposure

Conclusion. Utilization exposure is **High**.
Observed facts. EQT states that Lazer runs **9 million annual service hours**, which makes service-capacity utilization more important than simple equipment deployment. [53]
Root business mechanism. Earnings are highly sensitive to labor utilization, staffing density, idle paid hours, overtime, and shift design.
Calculation or driver bridge. I classify exposure as High because the core bottleneck is not whether a tractor is leased, but whether labor and site capacity are productively used.
Sensitivities. Volume swings, absenteeism, hiring difficulty, and supervision quality.
Confidence. High.

New-unit or new-customer payback

Conclusion. New-site payback is **0.8 / 1.5 / 2.5 years**.
Observed facts. Lazer's economic expansion is site-based and labor-led; its offerings can be mobilized with a combination of people, tractors, trailers, and tech rather than by funding a single huge long-life asset. [47]
Root business mechanism. Initial investment is mainly recruiting, training, mobilization, and any tractors/tech needed to launch the site.
Calculation or driver bridge. Central case assumes one to two years to recover those up-front costs through site-level cash contribution.
Sensitivities. Ramp speed, customer launch issues, labor turnover, and whether the customer requires Lazer-owned equipment.
Confidence. Medium.

B2B versus B2C profile

Conclusion. Lazer is **predominantly B2B**.
Observed facts. Lazer markets to logistics/distribution, industrial/manufacturing, food and beverage, retail and grocery, and consumer goods customers. [48]

Root business mechanism. Contracts are signed by warehouses, manufacturers, retailers, and 3PLs, not households.
Calculation or driver bridge. There is no meaningful B2C revenue.
Sensitivities. Some blue-chip concentration is likely, but still within B2B.
Confidence. High.

Lazer Logistics conclusion

Plain-English business description. Lazer runs customer yards for warehouses, manufacturers, retailers, and 3PLs by supplying drivers, site processes, tractors, trailers, and technology to keep freight moving.

Principal strengths. Scale, embedded site relationships, strong EV positioning, operational technology, and broad North American footprint. [50]

Principal risks. Wage inflation, turnover, rebid risk, site volume volatility, and the need to maintain service levels in a labor-intensive model. [47]

Key unavailable information. Public EBITDA, contract-term statistics, customer concentration, wage pass-through mechanics, and site-level churn data.

Most important diligence questions. What share of revenue is fixed-fee versus variable labor/service-hour? What is annual site retention by revenue? How much EBITDA is generated by trailer rental and technology versus operations labor? What is driver turnover by region? What was the average payback on the last 50 new sites?

Archetypes, classification, and benchmark synthesis

Company classification

VTG — Core representative. VTG is the clearest **railcar leasing** representative in the universe, even though the legal group contains other logistics activities. Its current statutory accounts and strategic focus on European wagon leasing make it directly informative for the railcar sub-archetype. [54]

Akiem — Mixed-business company. Akiem is highly informative for **locomotive leasing**, but less clean than VTG because it combines locomotive leasing with maintenance services and some passenger-train activity. It is still essential to the rail-asset analysis, but it should not be forced into a simple wagon benchmark. [25]

Air Rail — Mixed-business company. Air Rail is clearly relevant to **airport GSE leasing**, but it also sells and services rail-sector equipment and lacks clean public segment financial reporting. It is informative, but not a pure-play global benchmark. [9]

TCR — Core representative. TCR is the strongest **airport GSE full-service leasing** benchmark in the set because its operating model, scale, and sponsor documentation are unusually explicit. [45]

Lazer Logistics — Core representative. Lazer is the clearest **outsourced yard logistics** representative here, and it should remain a separate archetype because its economics are labor-led rather than asset-led. [50]

Archetype findings

Further distinction **is required** inside the rail bucket. The prompt's minimum archetype of "Railcar and locomotive leasing" is too broad for a single central benchmark because wagons and locomotives have different unit costs, technical complexity, overhaul cycles, redeployment friction, and contract lives. I therefore present **separate railcar** and **locomotive** benchmarks, then a narrow comment on what they have in common. The same logic applies less strongly in airport GSE: Air Rail and TCR differ in scale and disclosure quality, but they are still close enough economically to support a common **airport GSE full-service leasing** benchmark. Lazer remains separate. [1]

Railcar leasing benchmark

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
Revenue growth	1%	4%	7%	1%–7%	Medium	Price resets, fleet additions, and wagon-type utilization
EBITDA margin	52%	58%	65%	52%–65%	Medium	Lease spread on long-lived wagons, reduced by upkeep and repositioning
Maintenance capex % revenue	14%	18%	24%	14%–24%	Medium	Recurring inspections, overhauls, and periodic replacement
Post-maintenance conversion	54%	69%	78%	54%–78%	Medium	High EBITDA with meaningful sustaining fleet spend

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
Initial contract tenor	2 yrs	4 yrs	7 yrs	2–7 yrs	Medium	Standard versus specialized wagon mix
Effective customer tenure	6 yrs	9 yrs	12 yrs	6–12 yrs	Medium	Fleet integration into industrial flows
Utilization exposure	Moderate	Moderate	High	Moderate–High	Medium	Off-lease risk concentrated in cyclical subfleets
New-unit/ customer payback	7 yrs	9 yrs	12 yrs	7–12 yrs	Medium	Long asset life plus residual value
B2B/B2C profile	Predominantly B2B	Predominantly B2B	Predominantly B2B	B2B	High	Industrial shippers and rail operators

The central railcar benchmark is supported mainly by **VTG**. The main structural driver is that wagons are durable, broadly reusable industrial assets whose value can be captured over multiple placements. What pushes results above or below the benchmark is **fleet specialization**: tank and other niche wagons tend to deliver better durability than cyclical intermodal wagons. [14]

Locomotive leasing benchmark

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
Revenue growth	3%	6%	10%	3%–10%	Medium	Modernization demand, operator outsourcing, and new deliveries

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
EBITDA margin	55%	62%	70%	55%–70%	Medium	Scarce high-value assets plus service content
Maintenance capex % revenue	12%	18%	25%	12%–25%	Medium	Heavy overhauls, ERTMS, and technical refresh
Post-maintenance conversion	55%	71%	83%	55%–83%	Medium	Strong lease economics partly offset by heavy sustain spend
Initial contract tenor	5 yrs	7 yrs	10 yrs	5–10 yrs	Medium	Unit value and approval complexity require longer terms
Effective customer tenure	8 yrs	12 yrs	15 yrs	8–15 yrs	Medium	High switching friction and repeated fleet programs
Utilization exposure	Moderate	Moderate	High	Moderate–High	Medium	Long contracts help, but each returned unit is costly to replace
New-unit/customer payback	8 yrs	10 yrs	13 yrs	8–13 yrs	Medium	Large upfront cost and residual value
B2B/B2C profile	Predominantly B2B	Predominantly B2B	Predominantly B2B with institutional/public exposure	B2B	High	Freight and passenger operators, transport authorities

The locomotive benchmark is supported mainly by **Akiem**. The central range is higher-growth and longer-tenor than railcars because each contract solves a more capital-intensive, technically complex customer problem. Differences versus railcars are **structural**, not temporary. [18]

Airport GSE full-service leasing benchmark

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
Revenue growth	5%	8%	12%	5%–13%	Medium	Outsourcing, airport growth, electrification, and fleet right-sizing
EBITDA margin	24%	30%	36%	20%–38%	Medium	Rental revenue plus service density minus technician/ fleet-management cost
Maintenance capex % revenue	10%	14%	18%	9%–18%	Medium	Refurbishment, replacement fleet, electric transition, and battery risk
Post-maintenance conversion	40%	55%	67%	35%–69%	Medium	Meaningful EBITDA but also meaningful sustaining fleet spend
Initial contract tenor	3 yrs	5 yrs	7 yrs	3–7 yrs	Medium	Airport/ handler contract structure and sale-leaseback tenor

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
Effective customer tenure	6 yrs	9 yrs	13 yrs	5–14 yrs	Medium	Embedded airport relationships and repeated fleet placements
Utilization exposure	Moderate	Moderate-High	High	Moderate–High	Medium	Paid deployment, spare-fleet requirement, and airport transfer friction
New-unit/customer payback	4 yrs	5.5 yrs	7.5 yrs	4–8 yrs	Medium	Moderate unit cost, service burden, and resale/refurbishment value
B2B/B2C profile	Predominantly B2B	Predominantly B2B	Predominantly B2B	B2B	High	Handlers, airlines, airports

The central airport GSE range is supported most strongly by **TCR**, with **Air Rail** widening the observed band on disclosure and mix. The structural driver is that airport GSE lessors are really **fleet-and-service operators**: growth is supported by outsourcing and electrification, but margins and cash conversion are held back by maintenance response, spare fleets, and airport-specific deployment friction. Differences between Air Rail and TCR are partly structural—especially scale and purity—and partly disclosure-related. [55]

Outsourced yard logistics benchmark

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
Revenue growth	4%	7%	11%	4%–11%	Medium	New site wins, cross-sell, and modest tuck-ins

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
EBITDA margin	10%	14%	18%	10%–18%	Medium	Labor-heavy service economics
Maintenance capex % revenue	2%	4%	7%	2%–7%	Medium	Tractors, trailers, and site tech only
Post-maintenance conversion	55%	71%	84%	55%–84%	Medium	Light sustaining capex compared with asset lessors
Initial contract tenor	1 yr	3 yrs	5 yrs	1–5 yrs	Medium	Service contracts rarely need long legal terms
Effective customer tenure	4 yrs	7 yrs	10 yrs	4–10 yrs	Medium	Operational embedding extends actual life
Utilization exposure	High	High	High	High	High	Labor and site-capacity productivity dominate earnings
New-unit/ customer payback	0.8 yrs	1.5 yrs	2.5 yrs	0.8–2.5 yrs	Medium	Recruiting and mobilization recover faster than heavy equipment spend
B2B/B2C profile	Predominantly B2B	Predominantly B2B	Predominantly B2B	B2B	High	Warehousing, retail, manufacturing, 3PLs

Lazer alone supports this archetype, but the logic is still robust because the business model is so clearly labor-led. This benchmark should **not** be blended into asset leasing on EBITDA or capex metrics. The only truly comparable overarching conclusion is that the customer base is overwhelmingly B2B and operationally mission-critical. [50]

Combined Transportation Solutions benchmark

A **single combined numeric Transportation Solutions benchmark is not economically defensible** for most metrics. The dispersion between wagons, locomotives, airport GSE, and outsourced yard labor is structural, not noise. A combined average would blur exactly the questions the benchmark is supposed to answer: how much of EBITDA must be reinvested, how long initial contracts need to be, how hard assets are to redeploy, and whether payback is driven by equipment value or by site mobilization. The only combined conclusions that survive aggregation are qualitative: the universe is overwhelmingly **B2B**, typically **mission-critical**, and often wins through a combination of capital relief plus operational complexity transfer. [1]

Final Transportation Solutions conclusion

Typical customer problem. There is no single universal customer problem. By archetype: railcar customers need freight-carrying assets embedded in industrial flows; locomotive customers need traction capacity and compliance; airport GSE customers need safe, reliable ramp equipment without owning and maintaining it; yard-logistics customers need bottlenecks removed and trailer flow managed. Across the group, the common thread is outsourcing a mission-critical operational bottleneck. [56]

Typical revenue model. Archetype-specific. Rail models are mainly recurring lease revenue with maintenance/services attached to varying degrees. Airport GSE is recurring full-service rental plus maintenance and fleet-management services. Yard logistics is a recurring outsourced-operations model with some asset and tech attachment. [57]

Primary growth drivers. Railcar growth is mostly fleet additions, price, and utilization. Locomotive growth is fueled by modernization and scarce interoperable traction. Airport GSE growth is driven by outsourcing, airport/handler fleet turnover, and electrification. Yard-logistics growth comes from new site wins and adjacent-service expansion. [58]

Primary margin drivers. Rail asset margins are driven by lease spread, fleet mix, and off-lease drag. Airport GSE margins are driven by station density, workshop productivity, and maintenance burden. Yard-logistics margins are driven primarily by labor utilization, wage pass-through, and supervisory efficiency. [1]

Principal maintenance-capex requirements. Wagons need inspections, overhauls, and periodic replacement. Locomotives add heavier overhaul and compliance-upgrade needs. Airport GSE needs refurbishment, replacement, and increasingly batteries/charging-related investment. Yard logistics has comparatively light sustaining capex, mainly tractors, trailers, and site tech. [1]

Typical contract and customer duration. Railcars sit in a medium-term legal / long-term economic relationship. Locomotives are longer and stickier. Airport GSE usually sits in mid-length site contracts but longer customer relationships. Yard logistics often has shorter legal terms but surprisingly durable site relationships when service is strong. [59]

Typical utilization dynamics. Railcars and locomotives are driven by asset deployment under paying contracts and redeployment speed after return. Airport GSE has three layers: paid deployment, actual ramp use, and technician/workshop productivity. Yard logistics is dominated by service-capacity utilization—hours, shifts, staffing density, and idle time. [1]

Typical new-unit payback. Railcars and locomotives generally need **multi-year recovery supported by residual value**. Airport GSE generally recovers in **mid-single-digit years** on a blended basis. Yard-logistics site mobilization typically recovers in **one to two years** if the site ramps properly. [60]

Typical B2B/B2C profile. The subvertical is overwhelmingly **predominantly B2B**. Even where passengers or household consumers are the ultimate end users, the contractual customer is still a professional operator, airport, handler, authority, manufacturer, warehouse, or logistics provider. [61]

Largest source of variation among companies. The largest source of variation is **whether the company is really an asset owner/lessor or a labor-led outsourced operator**. That single distinction cascades into margin structure, maintenance capex, utilization exposure, payback, and contract duration. [62]

Lowest-confidence conclusion. The lowest-confidence conclusion is the exact **maintenance-capex percentage of revenue** for the private companies, especially Air Rail and TCR, because public disclosure does not cleanly separate sustaining fleet replacement, refurbishment, electrification, and growth capex. [63]

Slide-ready business-model description. Transportation Solutions comprises mission-critical B2B rail, airport, and yard-flow platforms that monetize owned equipment or embedded operations by reducing customer capital needs, uptime risk, and process complexity.

Concise source list organized by company

VTG. Main evidence came from the **VTG GmbH 2024 consolidated accounts** for ownership structures, capex, cash flow, financing, and operating commentary; the **2022 official VTG ownership transaction release**; and older official VTG annual-report materials for the underlying wagon-leasing business model. [64]

Akiem. Main evidence came from Akiem's official **About** and **Sustainability** pages, the **2026 green refinancing press release**, and official **2022 sale materials** from Akiem/CDPQ/SNCF describing the 2021 financial and fleet base. [65]

Air Rail. Main evidence came from the **air-rail.org legal notice**, the company's official site pages describing operating rental and maintenance, and IFM's **2025 acquisition announcement** plus **2026 portfolio page**. [66]

TCR. Main evidence came from TCR's official **About** and **ESG** pages, its detailed **GSE Services** page, GIP's **March 2026 acquisition announcement**, and **3i Infrastructure's March 2026 sale announcement**. A local cross-check came from **TCR UK Limited FY2024 accounts**. [67]

Lazer Logistics. Main evidence came from Lazer's official **homepage** and **About** page, EQT's **portfolio page**, and the official **2023 acquisition announcement**. [68]

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[1] [2] [4] [5] [6] [13] [14] [15] [54] [57] [58] [60] [62] [64] https://www.lobbyregister.bundestag.de/media/f3/0a/614644/VTG_GmbH_KAP_31-12-2024_Final_TE_24-004268.pdf
https://www.lobbyregister.bundestag.de/media/f3/0a/614644/VTG_GmbH_KAP_31-12-2024_Final_TE_24-004268.pdf

[3] [7] [9] [10] [26] [66] https://air-rail.org/en/legal/
https://air-rail.org/en/legal/

[8] [32] https://www.ifminvestors.com/news-and-insights/media-centre/ifm-investors-announces-the-acquisition-of-air-rail-and-strengthens-its-position-in-the-spanish-infrastructure-market/
https://www.ifminvestors.com/news-and-insights/media-centre/ifm-investors-announces-the-acquisition-of-air-rail-and-strengthens-its-position-in-the-spanish-infrastructure-market/

[11] https://www.vtg.com/news-and-insights/press-releases/detail/morgan-stanley-infrastructure-partners-and-joachim-herz-stiftung-agree-to-sell-majority-stake-in-vtg-aktiengesellschaft-to-a-consortium-of-gip-and-adia
https://www.vtg.com/news-and-insights/press-releases/detail/morgan-stanley-infrastructure-partners-and-joachim-herz-stiftung-agree-to-sell-majority-stake-in-vtg-aktiengesellschaft-to-a-consortium-of-gip-and-adia

[12] [16] [17] [56] [59] https://www.vtg.com/fileadmin/VTG/Dokumente/Investor_Relations/Berichte/VTG_GB14-engl.pdf
https://www.vtg.com/fileadmin/VTG/Dokumente/Investor_Relations/Berichte/VTG_GB14-engl.pdf

[18] [19] [20] [21] [25] [61] [65] https://www.akiem.com/en/about-akiem/
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[22] [23] [24] https://www.akiem.com/en/sncf-and-dws-finalize-the-sale-of-akiem-the-leading-provider-of-locomotive-leasing-services-in-france-and-europe-to-cdpq/
https://www.akiem.com/en/sncf-and-dws-finalize-the-sale-of-akiem-the-leading-provider-of-locomotive-leasing-services-in-france-and-europe-to-cdpq/

[27] [28] [30] https://air-rail.org/en/
https://air-rail.org/en/

[29] [31] [33] [63] https://www.ifminvestors.com/capabilities/infrastructure/our-portfolio/air-rail/
https://www.ifminvestors.com/capabilities/infrastructure/our-portfolio/air-rail/

[34] [67] https://www.tcr-group.com/about-us
https://www.tcr-group.com/about-us

[35] [36] [37] [39] [40] [42] [43] [45] [55] https://www.tcr-group.com/solution/gse-services
https://www.tcr-group.com/solution/gse-services

[38] [41] https://www.global-infra.com/news/global-infrastructure-partners-agrees-to-acquire-tcr/
https://www.global-infra.com/news/global-infrastructure-partners-agrees-to-acquire-tcr/

[44] https://www.3i.com/infrastructure/our-portfolio/tcr/
https://www.3i.com/infrastructure/our-portfolio/tcr/

[46] [52] https://www.lazerlogistics.com/about
https://www.lazerlogistics.com/about

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Specialty Rental Operating Benchmark Study

Executive Summary

This report assesses the Specialty Rental subvertical as of **July 18, 2026**, using a first-principles approach rather than a screen of reported margins or a simple peer average. The designated research universe is: **PODS, Mobile Mini, Trench Plate Rental, General Finance Corporation, Modulaire Group, McGrath RentCorp, and Reliant Rental**. For acquired businesses, I anchor the analysis to the **last reliable standalone period** and then use later facts only where they are clearly attributable to the same business. I do **not** compare the findings to CRA or Compactor Rentals of America. 1

The central analytical conclusion is that **"specialty rental" is not one operating model**. The designated companies split into at least four distinct archetypes: **consumer-led delivered portable storage**, **B2B branch-based portable/mobile storage**, **modular space and temporary classrooms/offices**, and **trench-safety equipment rental**. Those archetypes differ materially in contract form, delivery intensity, utilization measurement, redeployment friction, maintenance burden, and payback logic. The widest structural gap is between **PODS-style consumer moving/storage**, where customer acquisition and logistics complexity are unusually high, and **modular space/classroom rental**, where delivered assets stay on rent much longer and relationship tenure is often measured in years rather than months. 2

Across the universe, the most robustly evidenced normalized economics sit with the formerly public businesses. **Mobile Mini Storage Solutions** and **McGrath Portable Storage** show the clearest portable-storage evidence: long-lived steel assets, high residual values, rental revenue that is primarily recurring, and EBITDA margins that can exceed many general-equipment rental categories because the assets have no engines, few moving parts, and limited technical obsolescence. **Modulaire** and **McGrath Mobile Modular** support a separate modular-space benchmark characterized by higher delivery and site-service complexity but much longer initial rental periods and stronger effective commercial tenure. **General Finance** confirms that portable services can recover equipment cost quickly in revenue terms, but it also demonstrates why revenue payback is not enough: true cash payback stretches longer once delivery, refurbishment, and sustaining capital are reserved. 3

At the subvertical level, the most defensible central normalized ranges by archetype are as follows. **Consumer-led portable storage**: revenue growth about **4%**, EBITDA margin about **33%**, maintenance capex about **7% of revenue**, post-maintenance conversion about **79%**, and payback around **3.5 years**. **B2B portable/mobile storage**: revenue growth about **5%**, EBITDA margin about **40%**, maintenance capex about **6%**, post-maintenance conversion about **85%**, and payback about **4.5 years**. **Modular space/classrooms**: revenue growth about **6%**, EBITDA margin about **34%**, maintenance capex about **8%**, post-maintenance conversion about **76%**, and payback about **6.0 years**. **Trench safety**: revenue growth about **6%**, EBITDA margin about **33%**, maintenance capex about **7%**, post-maintenance conversion about **79%**, and payback about **3.5 years**. These are normalized operating assumptions, not management guidance. 4

The lowest-confidence conclusion in the report is **Reliant Rental**. The best transaction-context evidence I found ties the name **"Reliant Rental"** to trademark filings made by **Vander Intermediate Holding III**

Corporation, a BlueLine Rental holding-company affiliate, not to a clearly resolved standalone specialty-rental operating company in portable storage, modular space, temporary classrooms/offices, trench safety, or an adjacent specialty-rental niche. Because the evidence does not establish a distinct legal operating business within the user's intended specialty-rental scope, I continue with the **most defensible identified candidate only**, mark confidence **Low**, and exclude it from any load-bearing benchmark weighting. [5]

Scope, methodology, and Reliant Rental identity resolution

This study uses a consistent sequence for each company and metric: **observed facts**, **root business mechanism**, **normalized low/central/high assumption**, **sensitivities**, and **confidence**. For acquired companies, I use the last reliable standalone reporting period: **Mobile Mini year ended December 31, 2019**; **General Finance year ended June 30, 2020**; **Trench Plate Rental immediately pre-merger in 2021 using transaction disclosures and operating evidence**; and **PODS using the best available standalone company and ratings evidence, but with materially lower confidence because current full financial disclosure is limited**. For current independent reporters, I use **McGrath year ended December 31, 2025** and **Modulaire Group Holdings Ltd. year ended December 31, 2024** as the latest formal period evidenced in the browsed record. [6]

The key methodological guardrail is that **maintenance capex is not total capex**. In these fleets, the sustaining burden is usually best approximated by replacement-cycle logic, depreciation and residual-value cross-checks, and management commentary on asset lives and maintenance intensity. A second guardrail is that **legal contract term is not customer tenure**. Portable storage, modular space, and trench-safety rental often renew informally, extend automatically, or repeat across projects, so the durability of the customer relationship typically exceeds the duration of any one unit placement. [7]

Reliant Rental identity resolution. The strongest primary trail starts with the **Canadian Trade-marks Journal**, which lists **RELIANT RENTAL** as a mark filed in January 2014 by **Vander Intermediate Holding III Corporation**. Separate trademark-search evidence also links the same name to Vander and shows the U.S. filing as **dead**, not an active trademark. Vander is important because official United Rentals acquisition materials identify **Vander Holding Corporation and its subsidiaries, including BlueLine Rental, LLC**, as the BlueLine business acquired in 2018. Put differently, the transaction-context evidence ties "Reliant Rental" to the **BlueLine/Vander general equipment-rental corporate family**, not to a clearly evidenced standalone specialty-rental operator in the asset classes the user designated. [5]

On the evidence available, the **most defensible resolution** is that "Reliant Rental" was **a trademarked but not clearly commercialized brand or proposed brand associated with the Vander/BlueLine family**, rather than a separate legal operating company with standalone financials in portable storage, modular space, temporary offices/classrooms, trench safety, or another clearly defined specialty-rental niche. I did **not** find authoritative transaction announcements, sponsor materials, statutory accounts, or company operating materials establishing a distinct **Reliant Rental** legal entity with standalone specialty-rental operations. I therefore treat the resolved entity as: **the "Reliant Rental" trademark/brand candidate associated with Vander Intermediate Holding III Corporation and the BlueLine corporate family; jurisdiction evidenced in the trademark record as a U.S. corporate applicant operating in North America; former/current ownership trail running through Vander/BlueLine to United Rentals' 2018 acquisition of BlueLine**. Identity confidence is **Low**. [5]

That choice has two consequences. First, the company profile is completed, but most operating metrics are marked **Unavailable** because the evidence does not support a clean specialty-rental operating profile. Second, **Reliant Rental is not used as a load-bearing input** in the benchmark ranges except as a disclosed identity-resolution exception. That is more defensible than silently substituting BlueLine's general-equipment-rental economics for a specialty-rental company that the evidence does not actually resolve. [8]

Company profiles

PODS

A. Resolve the company and segment. PODS was founded in 1998 and describes itself as the pioneer of portable storage. Its official site says it provides residential and commercial moving and storage in the U.S., Canada, and the U.K., and that the network has completed more than **2 million long-distance moves** and over **7 million initial deliveries**. Official ownership history in the browsed record shows Ontario Teachers' agreed to acquire PODS from Arcapita in late 2014 and completed the deal in early 2015. I did not find a later primary ownership-change announcement in the browsed record, so I treat PODS as still appearing to be Ontario Teachers-owned, but I mark that as an inference. Because detailed current standalone financial statements were not publicly available in the browsed record, evidence quality is materially weaker than for the former public comparators. [9]

B. Define the economic unit. The core unit is **one portable storage container placement**, but economics are actually two-layered: the **container-month on rent** and the **container move event**. That distinction matters because PODS earns monthly storage revenue, but also incurs meaningful truck, driver, dispatch, and linehaul costs tied to delivery, pickup, redelivery, and long-distance repositioning. [10]

C. Explain the customer problem. PODS solves a combination of **temporary storage**, **staged moving**, **renovation overflow**, and **transition-timing mismatch**. The need is often temporary rather than permanent, but it is highly time-sensitive and operationally important because the alternative is self-moving, renting a truck, making multiple storage trips, or coordinating tightly timed traditional movers. That is why the business can monetize convenience and schedule flexibility, not just square footage. [11]

D. Explain customer alternatives. Customers can use self-storage plus their own transportation, truck rental, full-service movers, or competing portable-storage providers. PODS wins when the combined value of **delivered storage**, **time flexibility**, **temporary on-site use**, and **integrated long-distance handling** is higher than the cost premium versus those alternatives. Switching is easy before a container is delivered, but harder once the container is loaded and in the system. [12]

E. Map the revenue model. Revenue is a mix of **recurring monthly storage rent**, **transport/delivery fees**, **long-distance move charges**, and ancillary services such as labor referrals, packing supplies, and car shipping connections. That means revenue is less "pure rental" than in B2B container leasing: part of the customer value comes from logistics orchestration. [13]

F. Map the operating model. After winning the order, PODS must schedule drop-off, maintain container inventory, manage local yards/storage centers, reposition containers, coordinate linehaul for long-distance moves, and handle pickup/redelivery. The need to physically move containers and maintain storage infrastructure creates a more logistics-heavy cost model than branch-based B2B storage rental. [10]

G. Explain retention and churn. Household customers churn naturally when a move ends or renovation finishes. That is not "bad churn"; it is the intended end of the use case. Commercial users can repeat over multiple projects, but the consumer-heavy demand mix means effective customer tenure is structurally lower than in modular classrooms or contractor-led trench rental. [13]

H. Map the capital model. The capital base includes containers, specialized delivery equipment, yards/ storage centers, and logistics systems. Container life is long, but economics depend on getting enough loaded months and enough move-event gross profit to recover both container investment and delivery-system burden. The 2024 S&P update reported weaker-than-expected earnings, with **sales down 7.5%** and **EBITDA down 24%** in the third quarter, which is consistent with a model where fixed logistics and yard costs can pressure margins quickly when consumer demand softens. [14]

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Normalized revenue growth	1%	4%	7%	Triangulated	Low	Household-move volume, price, and commercial repeat demand
Normalized EBITDA margin	28%	33%	38%	Inferred	Low	Container rent plus logistics gross profit minus trucking/ yard overhead
Maintenance capex % of revenue	5%	7%	9%	Inferred	Low	Container refresh, delivery fleet/yard upkeep, sustaining systems
Post-maintenance-capex conversion	68%	79%	87%	Calculated	Low	Margin less sustaining container/ logistics capital
Initial contract tenor	1 mo	1 mo	3 mo	Inferred	Medium	Consumer storage is usually billed monthly with flexible duration

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Effective customer tenure	3 mos	8 mos	18 mos	Inferred	Low	Move/ renovation duration plus repeat commercial uses
Utilization exposure	Moderate	Moderate-High	High	Inferred	Low	Consumer move volumes and loaded-container/yard utilization
New-unit or customer payback	2.5 yrs	3.5 yrs	5.0 yrs	Triangulated	Low	Container-month contribution plus move-event gross profit
B2B/B2C profile	Mixed	Mixed leaning B2C	Mixed	Inferred	Medium	Large residential mix, with meaningful but secondary commercial use

Primary evidence for PODS profile: official PODS operating description and service model; PODS ownership event; S&P's 2024 earnings deterioration note. [9]

Revenue growth. Conclusion: normalized growth is best framed at **1% / 4% / 7%**. **Observed facts:** PODS operates in residential and commercial moving/storage, has completed millions of deliveries and long-distance moves, and S&P said third-quarter 2024 sales were down **7.5%** year over year. **Root business mechanism:** growth is driven by move volumes, pricing, container availability, and cross-sell of storage-plus-move services rather than simply by same-store rental rate. **Calculation or driver bridge:** a central case assumes low-single-digit price plus modest unit growth offset by cyclical consumer softness. **Sensitivities:** housing turnover, migration, renovation demand, and linehaul capacity. **Confidence: Low**, because full standalone revenue history was not publicly available in the browsed record. [15]

EBITDA margin. Conclusion: 28% / 33% / 38%. **Observed facts:** S&P cited a sharper EBITDA decline than sales decline in 2024, implying meaningful operating leverage. **Root business mechanism:** PODS monetizes convenience, but the model carries trucking, labor, yard, and dispatch costs that are more substantial than for "leave-it-on-site" B2B container rental. **Calculation or driver bridge:** margins should sit below the best container-only B2B storage lessors but above low-value trucking businesses. **Sensitivities:** route density, fuel, driver labor, and long-distance move mix. **Confidence: Low**. [14]

Maintenance capex. Conclusion: 5% / 7% / 9% of revenue. **Observed facts:** PODS operates a delivered-container model with storage centers and specialized delivery operations. **Root business mechanism:** sustaining spend is not just container replacement; it also includes yard/handling equipment, delivery-fleet upkeep, and systems needed to preserve service quality. **Calculation or driver bridge:** container lives are long, but logistics-support assets shorten the true sustaining cycle. **Sensitivities:** fleet age, damage rates, and network intensity. **Confidence: Low**. [16]

Post-maintenance-capex conversion. Conclusion: 68% / 79% / 87%. **Observed facts:** the business has meaningful fixed logistics overhead, and recent EBITDA volatility shows that fixed costs matter. **Root business mechanism:** once sustaining container and logistics capital are reserved, conversion remains healthy but not as strong as the cleanest branch-based B2B storage models. **Calculation or driver bridge:** calculated as $(EBITDA - maintenance\ capex)/EBITDA$. **Sensitivities:** move-event mix and sustaining fleet assumptions. **Confidence: Low**. [14]

Initial contract tenor. Conclusion: central legal tenor is effectively **monthly**. **Observed facts:** PODS markets flexible timing and "unlimited time, control, and flexibility." **Root business mechanism:** the customer need is temporary and uncertain in length, so short legal billing cycles maximize conversion. **Calculation or driver bridge:** one month is the practical central initial tenor, although long-distance packages can embed longer effective use. **Sensitivities:** residential versus commercial mix. **Confidence: Medium**. [17]

Effective customer tenure. Conclusion: 3 months / 8 months / 18 months. **Observed facts:** the core use cases are local moves, long-distance moves with storage, renovations, and temporary overflow. **Root business mechanism:** tenure is bounded by life events more than by switching friction. **Calculation or driver bridge:** central tenure assumes a mix of short residential engagements and somewhat longer storage extensions. **Sensitivities:** renovation activity and commercial share. **Confidence: Low**. [18]

Utilization exposure. Conclusion: Moderate to high. **Observed facts:** the business depends on delivered containers and storage-center throughput; S&P flagged earnings weakness when sales softened. **Root business mechanism:** idle containers do not monetize, and underutilized routes/yards depress margin quickly. **Calculation or driver bridge:** exposure is higher than for long-term classroom rental, but lower than the shortest project equipment rental because containers can be stored awaiting redeployment. **Sensitivities:** seasonality, housing turnover, storage-center occupancy, and linehaul demand. **Confidence: Low**. [14]

New-unit or customer payback. Conclusion: 2.5 / 3.5 / 5.0 years. **Observed facts:** PODS combines monthly rent with logistics revenue and has long-lived containers. **Root business mechanism:** payback depends on both storage months and profitable move events. **Calculation or driver bridge:** central payback assumes a multiyear container life, moderate loaded utilization, and a direct reserve for move-related service cost. **Sensitivities:** local route density and damage/repair. **Confidence: Low**. [16]

B2B versus B2C. Conclusion: mixed B2B/B2C, leaning B2C. **Observed facts:** PODS explicitly serves residential and commercial moving/storage. **Root business mechanism:** retail residential demand is a major driver, but commercial jobs expand use cases and off-season utilization. **Calculation or driver bridge:** the residential event-driven mix is too large to classify as predominantly B2B. **Sensitivities:** commercial project demand and housing-market softness. **Confidence: Medium**. [19]

Plain-English business description: PODS rents portable containers and sells a bundled moving-and-storage service in which the unit, not just the labor, becomes the customer's temporary storage and relocation platform. [19]

Principal strengths: brand leadership in portable storage, strong convenience value proposition, long-lived assets, and the ability to monetize both storage months and move events. **Principal risks:** housing and migration cyclicality, trucking/labor cost inflation, and lower legal contract tenure than institutional or B2B rental models. **Key unavailable information:** current standalone revenue, EBITDA, fleet counts, and capex by type. **Most important diligence questions:** What is current residential versus commercial mix, what share of revenue is move-event versus monthly storage, and what is the true sustaining capex burden including delivery equipment and yards? [15]

Mobile Mini

A. Resolve the company and segment. Mobile Mini, Inc. was a Delaware public company with principal offices in Phoenix and filed its last full standalone annual report for the year ended **December 31, 2019**. It described itself as the world's leading provider of portable storage solutions and also a provider of tank and pump solutions. In March 2020, WillScot and Mobile Mini announced a merger; the July 2020 closing created the combined WillScot Mobile Mini business. For specialty-rental benchmarking, the relevant segment is **Storage Solutions**, not Tank & Pump. [20]

B. Define the economic unit. The governing unit is the **storage-solution unit on rent**: steel storage container or ground-level office. That unit governs utilization, rate, payback, and sustaining capital. [21]

C. Explain the customer problem. Customers need secure temporary on-site storage or portable ground-level office capacity without buying, transporting, maintaining, or yarding the box themselves. The problem is recurring in construction, retail, industrial, commercial, and government use, but usually temporary at the site level. [22]

D. Explain customer alternatives. Customers can buy containers, rent from local providers, use warehouse/self-storage, or postpone deployment. Mobile Mini's differentiators were fleet breadth, security features, availability, geographic network, and service responsiveness. [23]

E. Map the revenue model. In 2019, rental revenues were about **95% of total revenue**. Revenue came primarily from rental, with smaller sales of new/used units and ancillary services. Storage Solutions contributed **$490.7 million** of 2019 revenue, about **80%** of total company revenue. [24]

F. Map the operating model. Mobile Mini rented rather than sold, maintained the fleet, delivered and picked up units, and optimized location-level availability. Because containers are simple steel assets, recurring service intensity is lower than for powered equipment; the important operating tasks are branch stocking, transport, refurbishment, and yard utilization. [25]

G. Explain retention and churn. Unit-level churn is driven by project completion and site needs ending. Customer-level tenure is longer because contractors, retailers, and industrial accounts often repeat across jobs and geographies. The largest Storage Solutions customer represented only **about 6%** of rental revenues in 2019; the top 20 were about **14%**, which supports a diversified repeat-account structure. [26]

H. Map the capital model. Storage containers and steel ground-level offices had estimated useful lives of **30 years** and residual values of **55%** of original cost. Ordinary repair and maintenance were expensed as incurred; fleet cost included acquisition and first-in-service freight/customization. Storage Solutions had annual average utilization of **79.1% by units** and **77.9% by original equipment cost** in 2019, demonstrating strong fleet loading. [27]

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Normalized revenue growth	2%	5%	8%	Triangulated	High	Rate, units on rent, and branch-market density
Normalized EBITDA margin	36%	41%	44%	Disclosed	High	High rental mix, simple assets, and strong residual economics
Maintenance capex % of revenue	4%	6%	8%	Inferred	Medium	Long asset lives, refurbish-and-redeploy model
Post-maintenance-capex conversion	78%	85%	91%	Calculated	Medium	High margin offset by modest sustaining fleet burden
Initial contract tenor	1 mo	3 mos	12 mos	Inferred	Medium	Month-based rentals with some longer commercial placements
Effective customer tenure	2 yrs	4 yrs	7 yrs	Inferred	Medium	Repeat contractor/ industrial relationships across jobs
Utilization exposure	Moderate	Moderate	High	Disclosed	High	Revenue depends on units on rent and local fleet balance
New-unit or customer payback	3.5 yrs	4.5 yrs	6.0 yrs	Triangulated	Medium	30-year lives and high residual values, but service reserve matters

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
B2B/B2C profile	Pred. B2B	Pred. B2B	Pred. B2B	Disclosed	High	Contractors, retail, industrial, commercial, government

Primary evidence for Mobile Mini profile: 2019 10-K and 2020 merger announcement. [28]

Revenue growth. Conclusion: 2% / 5% / 8%. **Observed facts:** Storage Solutions rental revenue grew **4.0%** in 2019; North America Storage Solutions rental revenue grew **6.3%**; year-over-year yield rose **3.5%** on a 28-day equivalent basis. **Root business mechanism:** growth comes from rate, units on rent, and network-driven availability rather than high-frequency usage. **Calculation or driver bridge:** a central case reflects modest volume growth plus price increases roughly in line with 2019 evidence. **Sensitivities:** construction activity, local competition, and managed-rental mix. **Confidence: High**. [29]

EBITDA margin. Conclusion: 36% / 41% / 44%. **Observed facts:** Storage Solutions adjusted EBITDA margin was **41.0%** in 2019. Rental revenues were about **95%** of company revenue. **Root business mechanism:** margins benefit from recurring rental, long-lived units, limited moving parts, and strong residual values. **Calculation or driver bridge:** the central case anchors to disclosed 2019 Storage Solutions EBITDA margin. **Sensitivities:** transport cost, maintenance inflation, and local price competition. **Confidence: High**. [30]

Maintenance capex. Conclusion: 4% / 6% / 8% of revenue. **Observed facts:** containers and ground-level offices have **30-year** lives and **55%** residual values; ordinary repairs are expensed. 2020 planned net fleet capex was said to be growth-focused. **Root business mechanism:** the sustaining burden is modest because older containers still rent and resale values remain meaningful. **Calculation or driver bridge:** central maintenance capex is well below depreciation because a large share of gross fleet purchases is growth rather than replacement. **Sensitivities:** corrosion, branch mix, and freight/customization needs. **Confidence: Medium**. [31]

Post-maintenance-capex conversion. Conclusion: 78% / 85% / 91%. **Observed facts:** disclosed EBITDA margin is strong and sustaining capex appears modest relative to revenue. **Root business mechanism:** the business converts EBITDA well because it does not need engine-heavy replacement cycles. **Calculation or driver bridge:** computed from the EBITDA and maintenance-capex ranges. **Sensitivities:** true replacement need if steel prices or damage rates rise. **Confidence: Medium**. [32]

Initial contract tenor. Conclusion: 1 month / 3 months / 12 months. **Observed facts:** the company rented ground-level offices and storage containers on a month-based basis; specific long fixed terms were not emphasized in the 10-K. **Root business mechanism:** customers want flexibility because storage needs often follow project timing. **Calculation or driver bridge:** central initial tenor is short-to-intermediate, even though realized duration can be longer. **Sensitivities:** national accounts and office-unit mix. **Confidence: Medium**. [23]

Effective customer tenure. Conclusion: 2 / 4 / 7 years. **Observed facts:** approximately **74,000 customers** were served in 2019, with diversified concentration; service and availability were emphasized as causes of loyalty. **Root business mechanism:** job-by-job container duration is short, but customer relationships recur.

Calculation or driver bridge: effective customer life is therefore measured in years, not months. **Sensitivities:** local service failures or aggressive competitor pricing. **Confidence: Medium**. [33]

Utilization exposure. Conclusion: Moderate, with a high case in downturns. **Observed facts:** annual average Storage Solutions utilization by units was **79.1%** and by original equipment cost **77.9%** in 2019. **Root business mechanism:** revenue is directly linked to fleet on rent; idle assets earn nothing, but assets are readily redeployable and durable. **Calculation or driver bridge:** that makes exposure meaningful but not catastrophic at normal utilization levels. **Sensitivities:** local oversupply and macro construction slowdown. **Confidence: High**. [34]

New-unit or customer payback. Conclusion: 3.5 / 4.5 / 6.0 years. **Observed facts:** industry yield rose **3.5%**, utilization was near **79%**, and assets have **30-year** lives with **55%** residuals. **Root business mechanism:** gross revenue payback is quick enough to be attractive, but cash payback must reserve for delivery, service, and maintenance. **Calculation or driver bridge:** central payback assumes annual revenue at roughly high-20s percent of cost at normalized utilization, with substantial direct margin. **Sensitivities:** monthly rates and freight/customization cost. **Confidence: Medium**. [35]

B2B versus B2C. Conclusion: predominantly B2B. **Observed facts:** key end markets were construction, retail, consumer services, industrial, commercial, and governmental, but the operating framing and account structure were commercial. **Root business mechanism:** the company sold largely to businesses that needed temporary secure storage and jobsite offices. **Calculation or driver bridge:** B2C does not appear load-bearing in the standalone profile. **Sensitivities:** none material. **Confidence: High**. [22]

Plain-English business description: Mobile Mini's Storage Solutions business was a national B2B network that owned long-lived steel containers and ground-level offices, monetizing availability, security, delivery, and redeployment. [20]

Principal strengths: scale, residual-value-rich fleet, high rental mix, diversified customers, and strong disclosed EBITDA. **Principal risks:** price competition, local utilization swings, freight cost, and cyclicality in project demand. **Key unavailable information:** contract-tenor distribution by product line. **Most important diligence questions:** What share of gross capex was true replacement versus growth, and how variable are direct delivery and refurbishment costs across markets? [36]

Trench Plate Rental

A. Resolve the company and segment. The company referred to in primary transaction materials is **Trench Plate Rental Company (TPRC)**, founded in **1979** and based in **Artesia, California**. In March 2021, National Trench Safety and TPRC announced a definitive agreement to merge, with the combined company majority-owned by Tailwind Capital. The relevant standalone period is therefore the pre-merger operating business described in 2021 transaction materials and prior operating references. [37]

B. Define the economic unit. The most useful unit is not one customer contract but the **equipment-set on rent**: trench plates, trench shields/boxes, shoring equipment, and traffic-safety accessories deployed to a jobsite. Revenue often travels with the set, not just a single item. [38]

C. Explain the customer problem. The customer is excavating or crossing an open trench and must solve **worker safety**, **regulatory compliance**, **site access**, and in many cases **traffic continuity**. Unlike discretionary equipment rental, the need is partly safety- and code-driven. If it is not addressed, the project cannot proceed safely or legally. [38]

D. Explain customer alternatives. Customers can buy equipment, rent from local trench-safety specialists, or attempt less sophisticated approaches. Switching exists, but it is constrained by engineering support, inventory availability, branch coverage, and the risk of job shutdown or safety incidents. That gives trench specialists more solution value than a plain steel-rental comparison suggests. [38]

E. Map the revenue model. Revenue is mainly **equipment rental**, supplemented by solution/engineering support and associated traffic-safety products. The combined NTS/TPRC release explicitly described both companies as “pure-play providers of trench safety solutions.” [39]

F. Map the operating model. The company needs branches, dispatch, specialized inventory, engineering support, and field service around job setup and returns. Steel assets are durable, but the business is operationally branch-intensive because retrieval, cleaning, inspection, and redeployment all matter. [40]

G. Explain retention and churn. Asset rental duration ends when the excavation or crossing phase ends. Customer relationships can last much longer because utilities, civil contractors, and underground contractors repeat across projects and geographies. That is why project duration should not be confused with customer tenure. [38]

H. Map the capital model. TPRC had grown to **37 branches** and served more than **5,000 customers** by the time of the 2021 merger announcement. The assets are heavy steel and engineered systems with long lives and scrap/resale value, but they require transport and branch handling. That combination usually supports strong returns when branch density and project utilization are good, despite more service intensity than plain storage containers. [41]

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Normalized revenue growth	3%	6%	10%	Triangulated	Medium	Underground utility/ infrastructure activity and branch expansion
Normalized EBITDA margin	28%	33%	38%	Inferred	Medium	Safety-value pricing offset by branch/transport/ engineering cost
Maintenance capex % of revenue	5%	7%	10%	Inferred	Medium	Fleet replacement, repair, inspection, specialty transport equipment

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Post-maintenance-capex conversion	64%	79%	87%	Calculated	Medium	Good branch density plus durable steel fleet
Initial contract tenor	1 wk	2 mos	6 mos	Inferred	Medium	Job-phase rental rather than long legal lease
Effective customer tenure	3 yrs	5 yrs	8 yrs	Inferred	Medium	Repeat contractor and utility relationships
Utilization exposure	Moderate-High	High	High	Inferred	Medium	Project starts/ completions directly control on-rent levels
New-unit or customer payback	2.5 yrs	3.5 yrs	5.0 yrs	Triangulated	Medium	High safety value and steel residual/ scrap support
B2B/B2C profile	Pred. B2B	Pred. B2B	Pred. B2B	Disclosed	High	Utilities, underground contractors, infrastructure-related customers

Primary evidence for Trench Plate Rental profile: the 2021 NTS/TPRC merger announcement and contemporary operating references. [42]

Revenue growth. Conclusion: 3% / 6% / 10%. **Observed facts:** TPRC had scaled to **37 branches** and more than **5,000 customers** by 2021. **Root business mechanism:** growth is driven by branch density, share gains in specialized excavation safety, and underlying underground-infrastructure activity. **Calculation or driver bridge:** the central case assumes modest same-branch growth plus selective network expansion. **Sensitivities:** utility work, muni budgets, and private nonresidential civil work. **Confidence: Medium**. [41]

EBITDA margin. Conclusion: 28% / 33% / 38%. **Observed facts:** the company was large enough to be described as one of the country's largest trench-safety providers, and the combined company was attractive enough for sponsor ownership and merger formation. **Root business mechanism:** pricing is supported by safety-critical demand and engineering content, but margins are held back by branch and transport intensity. **Calculation or driver bridge:** this should sit below best-in-class portable storage and near good specialty-rental field-service businesses. **Sensitivities:** local competition and transport intensity. **Confidence: Medium**. [37]

Maintenance capex. Conclusion: 5% / 7% / 10%. **Observed facts:** equipment includes trench plates, shoring, shields, and related traffic products. **Root business mechanism:** steel assets last a long time, but sustaining spend is more than corrosion repainting; it includes repairs, certifications, lifting/handling gear, and branch support. **Calculation or driver bridge:** central sustaining burden is moderate, not trivial. **Sensitivities:** damage rates and engineering-system mix. **Confidence: Medium**. 40

Post-maintenance-capex conversion. Conclusion: 64% / 79% / 87%. **Observed facts:** revenues are branch-driven and project-linked. **Root business mechanism:** strong gross economics of steel fleets are partly diluted by sustaining support equipment and branch infrastructure. **Calculation or driver bridge:** computed from the EBITDA and maintenance assumptions. **Sensitivities:** utilization and equipment damage. **Confidence: Medium**. 38

Initial contract tenor. Conclusion: one week / two months / six months. **Observed facts:** the business serves excavation and trenching jobs whose duration maps to project phases. **Root business mechanism:** the legal rental period is usually short and extendable rather than multi-year fixed. **Calculation or driver bridge:** central initial tenor therefore fits project-phase timing. **Sensitivities:** utility outage work versus large civil projects. **Confidence: Medium**. 38

Effective customer tenure. Conclusion: 3 / 5 / 8 years. **Observed facts:** TPRC served thousands of customers and operated nationally. **Root business mechanism:** civil contractors and utilities repeatedly need trench-safety solutions even if individual jobs roll off. **Calculation or driver bridge:** customer life is long relative to unit rental duration. **Sensitivities:** customer concentration and regional exposure. **Confidence: Medium**. 43

Utilization exposure. Conclusion: High. **Observed facts:** the business is tied directly to underground construction and infrastructure-related industries. **Root business mechanism:** when project starts slow or jobs finish, heavy steel inventory falls off rent immediately; branch labor and yards remain. **Calculation or driver bridge:** redeployability helps, but exposure stays high because demand is project-specific. **Sensitivities:** public-works funding and contractor backlog. **Confidence: Medium**. 37

New-unit or customer payback. Conclusion: 2.5 / 3.5 / 5.0 years. **Observed facts:** the assets are durable, value-bearing steel systems rented into safety-critical work. **Root business mechanism:** good day-rates and long physical lives support attractive returns if branch density is adequate. **Calculation or driver bridge:** central payback is shorter than modular space because the ticket is smaller and the realization per unit of cost can be high. **Sensitivities:** transport distance and mix of basic plates versus engineered systems. **Confidence: Medium**. 40

B2B versus B2C. Conclusion: predominantly B2B. **Observed facts:** the customer base is underground construction and infrastructure related. **Root business mechanism:** contracts are to contractors, utilities, and civil project owners rather than consumers. **Calculation or driver bridge:** the company is a straightforward B2B specialist. **Sensitivities:** none material. **Confidence: High**. 37

Plain-English business description: Trench Plate Rental rented heavy steel and engineered trench-safety systems to contractors and utilities that could not safely or legally perform excavation work without them. 38

Principal strengths: safety-critical demand, branch-based solution selling, long-lived steel assets, repeat contractor relationships. **Principal risks:** project cyclicality, branch fixed-cost leverage, and transport/logistics intensity. **Key unavailable information:** standalone revenue, EBITDA, fleet cost, and disposal proceeds. **Most important diligence questions:** What is gross versus net replacement capex, what percentage of fleet is engineered/high-yield versus commoditized plates, and how concentrated is demand by project type and region? [37]

General Finance Corporation

A. Resolve the company and segment. General Finance Corporation's last reliable full standalone annual period is the fiscal year ended **June 30, 2020**, filed before its 2021 sale to United Rentals for about **$996 million enterprise value**, including assumed net debt. Its portable-services industry exposure included **portable storage, modular space, and liquid containment**, operated through **Pac-Van**, **Lone Star**, and Asia-Pacific leasing businesses. For this study, the relevant specialty-rental segments are portable/mobile storage, mobile offices, and modular units; frac tanks are discussed only where necessary to bridge company evidence. [44]

B. Define the economic unit. The governing unit is the **lease fleet unit on rent**, with separate economics by asset type: storage containers, office containers, mobile offices, modular units, and frac tanks. That distinction matters because monthly lease rates and utilization differ sharply by asset class. [45]

C. Explain the customer problem. Customers need temporary storage, office, or modular space near the site, or in the Asia-Pacific operations, containerized space for moving/removals, storage, and related uses. The need is practical and often recurring, but normally temporary at the unit level. [46]

D. Explain customer alternatives. Customers can buy boxes or portable buildings, use local lessors, or rely on fixed facilities. General Finance won by supplying long-lived specialty units through branded leasing businesses with established regional networks. [47]

E. Map the revenue model. In FY2020 consolidated revenue, North America leasing contributed about **$164.7 million**, Asia-Pacific leasing about **$63.1 million**, and non-lease sales were material as well. For specialty-rental analysis, the recurring leasing line is the anchor, while sale of fleet/manufactured units is treated as ancillary and disposal-related. [48]

F. Map the operating model. The company had to acquire, customize, deliver, maintain, and re-lease units. Direct maintenance was expensed as incurred. In North America, the fleet spanned storage containers, office containers, mobile offices, modular units, and frac tanks, each with different yields and utilization patterns. [49]

G. Explain retention and churn. General Finance said its typical lease term had averaged **over twelve months**, and historical average monthly fleet utilization had averaged between **70% and 85%**. That points to a model with meaningfully longer site duration than PODS or B2B storage containers alone, especially in mobile offices/modular, but still with project roll-off risk. [50]

H. Map the capital model. The company stated that its portable storage, modular space, and liquid containment units had long useful lives of **20-30 years**, very low maintenance requirements, and modest rate sensitivity to age; it also said the business could **return original equipment cost through revenue**

within four years on average. But revenue payback is not cash payback. Depreciation policies showed lease fleet assets depreciated over **5-20 years** to residual values of up to **70%**. That indicates high residual support and modest economic obsolescence. [51]

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Normalized revenue growth	2%	5%	8%	Triangulated	High	Lease rates, utilization, and mix by unit type
Normalized EBITDA margin	24%	29%	34%	Inferred	Medium	Attractive fleet economics diluted by mixed assets and manufacturing/ sales mix
Maintenance capex % of revenue	6%	8%	11%	Triangulated	Medium	Long lives and high residuals, but broad fleet variety raises sustaining need
Post-maintenance-capex conversion	54%	72%	82%	Calculated	Medium	Mid-range margins with moderate sustaining burden
Initial contract tenor	12 mos	18 mos	36 mos	Disclosed	High	Company-stated typical lease term over twelve months
Effective customer tenure	3 yrs	5 yrs	7 yrs	Inferred	Medium	Repeat commercial users, even though unit durations end
Utilization exposure	Moderate	Moderate	High	Disclosed	High	Historically 70%–85% lease-fleet utilization
New-unit or customer payback	4.5 yrs	5.5 yrs	7.0 yrs	Triangulated	High	Revenue can recover cost in four years, but cash payback takes longer

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
B2B/B2C profile	Mixed B2B/B2C	Pred. B2B with some B2C	Pred. B2B	Inferred	Medium	NA business commercial; APAC includes removals/storage retail exposure

Primary evidence for General Finance profile: FY2020 10-K and 2021 United Rentals acquisition announcement. [52]

Revenue growth. Conclusion: 2% / 5% / 8%. **Observed facts:** FY2020 North America Leasing revenue was **$164.7 million** and Asia-Pacific Leasing revenue **$63.1 million**; utilization and monthly rates by asset type were disclosed. **Root business mechanism:** growth comes from rate, utilization, and unit mix—especially storage containers, office containers, and modular units. **Calculation or driver bridge:** central growth assumes modest rate growth plus normalized utilization recovery from FY2020 softness. **Sensitivities:** oil-and-gas exposure through Lone Star and cyclical project demand. **Confidence: High**. [53]

EBITDA margin. Conclusion: 24% / 29% / 34%. **Observed facts:** the assets had long lives and low maintenance, but the company also had manufacturing and sales activity, and FY2020 included Lone Star goodwill impairment. **Root business mechanism:** lease fleet economics are good, but consolidated margins are mixed by asset type and non-lease activities. **Calculation or driver bridge:** central margin is below pure container lessors and below best portable-modular specialists because the business mix was broader and less clean. **Sensitivities:** frac-tank weakness and sales mix. **Confidence: Medium**. [54]

Maintenance capex. Conclusion: 6% / 8% / 11%. **Observed facts:** fleet lives are **20-30 years** for portable storage/modular/liquid containment, with residuals up to **70%** and repairs generally expensed. **Root business mechanism:** sustaining capital is moderate because old units still rent and resale values remain good, but variant-heavy fleets require ongoing replacement. **Calculation or driver bridge:** central maintenance capex is above simple container businesses because the fleet mix is more diverse. **Sensitivities:** unit-type mix and refurbishment standards. **Confidence: Medium**. [55]

Post-maintenance-capex conversion. Conclusion: 54% / 72% / 82%. **Observed facts:** margins are good but not elite, while maintenance capex is not minimal. **Root business mechanism:** conversion is healthy because these are long-life assets, but diversified fleet mix lowers the result relative to a pure storage-container lessor. **Calculation or driver bridge:** formulaically derived from margin and sustaining-capex assumptions. **Sensitivities:** deterioration in utilization. **Confidence: Medium**. [56]

Initial contract tenor. Conclusion: 12 / 18 / 36 months. **Observed facts:** management said the typical lease term had averaged **over twelve months**. **Root business mechanism:** portable services are often tied to project or site duration and can extend. **Calculation or driver bridge:** central tenor therefore sits comfortably above one year. **Sensitivities:** storage-container share versus modular/mobile office share. **Confidence: High**. [50]

Effective customer tenure. Conclusion: 3 / 5 / 7 years. **Observed facts:** historical utilization was **70% to 85%**, and the company operated branded leasing networks. **Root business mechanism:** customers repeat even as assets cycle off rent. **Calculation or driver bridge:** central effective tenure exceeds initial contract term materially. **Sensitivities:** competitive pricing and project concentration. **Confidence: Medium**. [50]

Utilization exposure. Conclusion: Moderate. **Observed facts:** the company explicitly reported historical lease-fleet utilization between **70% and 85%** and disclosed by-asset utilization rates in FY2020. **Root business mechanism:** assets earn rent only on lease, but the fleet is durable and redeployable. **Calculation or driver bridge:** exposure is lower than trench safety and higher than very long-term institutional classroom rental. **Sensitivities:** macro/geographic mix. **Confidence: High**. [57]

New-unit or customer payback. Conclusion: 4.5 / 5.5 / 7.0 years. **Observed facts:** management said original equipment cost could be returned through revenue within **four years on average**. **Root business mechanism:** that is gross revenue payback, not cash payback. **Calculation or driver bridge:** after delivery, direct service cost, and a maintenance reserve, cash payback should be longer; I center it near **5.5 years**. **Sensitivities:** mix of modular units versus containers and regional delivery cost. **Confidence: High**. [57]

B2B versus B2C. Conclusion: predominantly B2B, but not pure B2B. **Observed facts:** North America leasing is clearly commercial, while Asia-Pacific disclosure referenced consumer/removals and storage sectors. **Root business mechanism:** the company sold into both businesses and, in some geographies, consumer-linked storage/moving uses. **Calculation or driver bridge:** central classification is mostly B2B with some B2C exposure. **Sensitivities:** geography. **Confidence: Medium**. [58]

Plain-English business description: General Finance was a diversified portable-services lessor whose relevant specialty-rental businesses rented long-lived containers, mobile offices, and modular units through regional operating brands. [44]

Principal strengths: long asset lives, strong residual values, utilization discipline, and average lease terms longer than a year. **Principal risks:** mixed fleet quality, exposure to weaker liquid-containment demand, and the difficulty of separating growth fleet capex from replacement fleet capex. **Key unavailable information:** clean segment-only EBITDA and sustaining capex for portable storage/modular excluding liquid containment. **Most important diligence questions:** What is the portable-storage/modular-only margin bridge, how much of gross fleet spend is replacement, and what is customer concentration by end market and region? [59]

Modulaire Group

A. Resolve the company and segment. The current legal filing anchor in the browsed record is **MODULAIRE GROUP HOLDINGS LIMITED** (Companies House company number **13449748**), with full accounts filed for **December 31, 2024** and prior years. The official Modulaire site describes the group as a leader in **European modular services and infrastructure**, operating in **23 countries**, with over **~330,000 modular space and portable storage units** and **5,000 remote accommodation rooms**. Historical primary evidence from a 2021 SEC-filed interim report stated that the ultimate parent was Modulaire Holding S.à r.l., principally owned by **TDR Capital** at that time. I do not assert a later beneficial-owner change without a primary source in the browsed record. [60]

B. Define the economic unit. The economic unit is primarily the **modular unit on rent**, sometimes bundled into a larger project or room package. For remote accommodation, the unit can scale from one room to a camp configuration, but utilization and pricing are still governed by rentable modular capacity. [61]

C. Explain the customer problem. Customers need temporary or semi-permanent space to **work, learn, live**, or house workers remotely. Needs are often institutional and B2B: education capacity, public administration, defense, energy, infrastructure, and business services. That raises essentiality and often lengthens project or relationship duration. [62]

D. Explain customer alternatives. Alternatives include permanent construction, buying modular units, or using other rental providers. Modulaire wins through fleet availability, geographic footprint, customization, and solution capability across public-sector and project end markets. Switching is costlier than for containers because transport, installation, and specification matter more. [62]

E. Map the revenue model. Revenue includes leasing and servicing of modular assets, and in 1Q22 the group posted **€370 million** of total revenue and **€116 million** of EBITDA including acquisitions. Revenue per unit grew **5%**, average utilization was **86%**, and units on rent increased meaningfully. That points to a model driven by price, utilization, and fleet growth, not by high-frequency consumables. [63]

F. Map the operating model. This is a full-service modular-space platform: manufacture/specification sourcing, transport, installation, project preparation, field servicing, and redeployment. Operational excellence and transformation initiatives remain important because modular work creates meaningful yard, transport, and installation complexity. [61]

G. Explain retention and churn. Unit placements can run from one project cycle to several years; customer relationships can last even longer where public-sector, infrastructure, education, and defense users repeatedly need space. That is why modular has materially longer effective tenure than PODS-like consumer storage. [62]

H. Map the capital model. Modulaire is fleet-intensive but benefits from high utilization, long asset lives, and redeployment. In 1Q22, average utilization was **86%**, and 2025 commentary said H1 margin improved **60 bps** year over year. Those facts support a good modular-space model, but one with more delivery and service intensity than pure containers. The evidence is strongest on utilization and reported EBITDA, weaker on gross-versus-net replacement capex because investor financial reports were password-protected. [64]

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Normalized revenue growth	3%	6%	9%	Triangulated	High	Price, units on rent, and public-sector/ project pipeline

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Normalized EBITDA margin	27%	31%	35%	Disclosed	High	Strong utilization with meaningful install/service cost
Maintenance capex % of revenue	6%	8%	11%	Inferred	Medium	Refurbishment, replacement modules, yard/ fleet support
Post-maintenance-capex conversion	59%	74%	83%	Calculated	Medium	Good margins but real sustaining fleet requirements
Initial contract tenor	12 mos	18 mos	36 mos	Inferred	Medium	Modular projects and classrooms/ offices usually stay longer
Effective customer tenure	4 yrs	7 yrs	10 yrs	Inferred	Medium	Public-sector and repeat enterprise relationships
Utilization exposure	Moderate	Moderate	Moderate-High	Disclosed	High	High fleet loading, but space demand tied to project pipeline
New-unit or customer payback	4.5 yrs	6.0 yrs	8.0 yrs	Triangulated	Medium	Longer asset lives and good utilization offset higher delivered cost
B2B/B2C profile	Pred. B2B	Pred. B2B	Pred. B2B	Disclosed	High	Public administration, construction, energy, industry, infrastructure

Primary evidence for Modulaire profile: official company news, Companies House filings, and the 2021 SEC-filed interim report identifying historical parentage. [65]

Revenue growth. Conclusion: 3% / 6% / 9%. **Observed facts:** 1Q22 revenue grew to **€370 million**, units on rent increased, and revenue per unit rose **5%**; 2025 commentary still cited good pipelines in defense, energy, infrastructure, and public sector. **Root business mechanism:** growth is driven by the installed unit base, price, and public/project demand. **Calculation or driver bridge:** central growth assumes mid-single-digit price-and-volume expansion in normal markets. **Sensitivities:** Germany recovery, public budgets, and M&A. **Confidence: High**. [64]

EBITDA margin. Conclusion: 27% / 31% / 35%. **Observed facts:** 1Q22 EBITDA was **€116 million** on **€370 million** of revenue, implying just over **31%** reported EBITDA. **Root business mechanism:** modular rental has strong recurring revenue once installed, but installation, project preparation, transport, and service are more cost-intensive than container-only rental. **Calculation or driver bridge:** the central case anchors to reported 1Q22 performance. **Sensitivities:** mix of leasing versus sales/service and large-contract timing. **Confidence: High**. [64]

Maintenance capex. Conclusion: 6% / 8% / 11%. **Observed facts:** utilization is very high and the group emphasizes refurbishment and circularity in its ESG framing. **Root business mechanism:** units are redeployable and long-lived, but preserving rentability requires refurbishment and occasional significant reset work. **Calculation or driver bridge:** central sustaining burden sits above plain storage containers but below engine-heavy equipment fleets. **Sensitivities:** classroom/building spec complexity and geographic handling cost. **Confidence: Medium**. [66]

Post-maintenance-capex conversion. Conclusion: 59% / 74% / 83%. **Observed facts:** EBITDA is solid, but modular fleets require real refurbishment capital. **Root business mechanism:** conversion remains good because the assets stay on rent for relatively long periods, yet a meaningful refurb reserve is essential. **Calculation or driver bridge:** computed from margin less maintenance reserve. **Sensitivities:** age distribution and install/removal intensity. **Confidence: Medium**. [64]

Initial contract tenor. Conclusion: 12 / 18 / 36 months. **Observed facts:** modular demand comes from institutional and project users, not quick-turn consumer events. **Root business mechanism:** once a modular building is installed, the customer typically keeps it through a school year, project phase, or longer. **Calculation or driver bridge:** central initial tenor is therefore longer than portable storage. **Sensitivities:** event space versus classrooms versus site accommodation. **Confidence: Medium**. [62]

Effective customer tenure. Conclusion: 4 / 7 / 10 years. **Observed facts:** Modulaire serves more than **52,000** customers globally with emphasis on public administration and infrastructure-related sectors. **Root business mechanism:** end-user relationships recur across projects and public-sector cycles. **Calculation or driver bridge:** central tenure meaningfully exceeds initial contract term. **Sensitivities:** public spending and customer concentration by market. **Confidence: Medium**. [67]

Utilization exposure. Conclusion: Moderate. **Observed facts:** average utilization was **86%** in 1Q22. **Root business mechanism:** utilization matters, but high installed durations and public-sector exposure reduce volatility versus shorter-cycle project rental. **Calculation or driver bridge:** central exposure is moderate, rising to moderate-high in a construction downturn. **Sensitivities:** redeployment lag and macro slowdown. **Confidence: High**. [63]

New-unit or customer payback. Conclusion: 4.5 / 6.0 / 8.0 years. **Observed facts:** the group runs at high utilization and positive price/mix, but modular units are more expensive delivered assets than containers. **Root business mechanism:** payback requires several years of rental cash contribution, though then benefits from long life and reuse. **Calculation or driver bridge:** central payback assumes good occupancy and a meaningful but not excessive service burden. **Sensitivities:** installation intensity and asset-spec complexity. **Confidence: Medium**. [64]

B2B versus B2C. Conclusion: predominantly B2B. **Observed facts:** Modulaire explicitly serves public administration, construction, energy, infrastructure, industry, and business services. **Root business mechanism:** contracting counterparties are institutions and enterprises, not households. **Calculation or driver bridge:** clean B2B classification. **Sensitivities:** remote accommodation mix, but not enough to change classification. **Confidence: High**. [68]

Plain-English business description: Modulaire is a large modular-space platform that rents, installs, services, and redeploys temporary and semi-permanent space for public-sector, infrastructure, industrial, and enterprise customers. [69]

Principal strengths: scale, high utilization, public-sector/infrastructure exposure, and long effective customer life. **Principal risks:** transport/install complexity, project timing, and limited public detail on sustaining capex. **Key unavailable information:** full 2024 income statement and cash-flow detail in line-readable form from the browsed record. **Most important diligence questions:** What is gross replacement capex versus growth capex, what share of EBITDA comes from leasing versus sales/service, and how concentrated is exposure to weaker Germany/nonresidential markets? [70]

McGrath RentCorp

A. Resolve the company and segment. McGrath RentCorp filed a 10-K for the year ended **December 31, 2025** and remained a standalone public company in that filing. The in-scope specialty-rental segments are **Mobile Modular** and **Portable Storage**; **TRS-RenTelco** is outside the target subvertical and is excluded from archetype benchmarking except where company-level disclosures cannot be cleanly unbundle corporate overhead. In 2025, Mobile Modular had rental equipment cost of **$1.49 billion** with average utilization of **73.0%**; Portable Storage had rental equipment cost of **$245.1 million** with average utilization of **60.8%**. [71]

B. Define the economic unit. Mobile Modular is best analyzed as the **modular unit on rent**; Portable Storage as the **container on rent**. Those are the correct units for rate, utilization, and payback. [72]

C. Explain the customer problem. Mobile Modular solves temporary classroom, education, commercial, and jobsite space needs; Portable Storage solves secure jobsite storage and office-container needs. These are practical capacity and convenience problems rather than discretionary usage. [72]

D. Explain customer alternatives. Customers can build, buy, or rent from competitors. McGrath wins through standardized fleet quality, responsive delivery/service, and in modular, the ability to customize and install quickly. In Portable Storage, competition turns more on local availability, delivery, and service. [72]

E. Map the revenue model. For 2024, Mobile Modular had **$318.1 million** of rental revenue and **$127.6 million** of rental-related services revenue; Portable Storage had **$70.0 million** of rental revenue and **$17.7**

million of rental-related services revenue. That mix matters because modular and storage have both recurring rent and transaction/transport service revenue. [73]

F. Map the operating model. Mobile Modular requires procurement to specification, refurbishment, delivery, installation, and in-field maintenance. Portable Storage requires delivery/pickup and lower recurring service while on rent. McGrath separately discloses direct depreciation and service costs by segment, showing the economics are meaningfully different. [74]

G. Explain retention and churn. Mobile Modular says typical initial rental terms are **12 to 24 months**; Portable Storage rentals are typically **1 to 12 months**. Effective customer life is longer because contractors, schools, and commercial accounts return repeatedly. Management explicitly links service quality to repeat business in modular. [75]

H. Map the capital model. Mobile Modular depreciates buildings over an **18-year** life to a **50% residual value**; Portable Storage depreciates containers over **25 years** to **62.5% residual value**. Monthly rental rates are typically **2% to 4% of original acquisition cost** for Portable Storage. These are unusually favorable residual assumptions that point to robust unit economics. They also explain why maintenance capex should be materially below depreciation under normal conditions. [76]

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Normalized revenue growth	3%	6%	9%	Triangulated	High	Modular price/units plus portable storage rate and utilization
Normalized EBITDA margin	35%	39%	43%	Triangulated	High	Attractive modular and storage margins, long lives, strong residuals
Maintenance capex % of revenue	6%	8%	11%	Triangulated	High	Refurbish-and-redeploy modular fleet plus container replacement reserve
Post-maintenance-capex conversion	69%	79%	86%	Calculated	High	High recurring margin offset by real refurb/replacement need

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Initial contract tenor	6 mos	18 mos	24 mos	Triangulated	High	Blend of 1–12 month storage and 12–24 month modular
Effective customer tenure	3 yrs	6 yrs	9 yrs	Inferred	Medium	Repeat institutional and commercial relationships
Utilization exposure	Moderate	Moderate	Moderate-High	Disclosed	High	Segment average utilization and redeployment capacity
New-unit or customer payback	4.0 yrs	5.5 yrs	7.0 yrs	Triangulated	High	Strong residual-heavy economics but lower modular revenue yield on cost
B2B/B2C profile	Pred. B2B	Pred. B2B	Pred. B2B	Disclosed	High	Education, contractors, commercial and industrial accounts

Primary evidence for McGrath profile: 2025 10-K segment disclosures. [77]

Revenue growth. Conclusion: 3% / 6% / 9%. **Observed facts:** in 2025, Portable Storage rental revenues decreased **3%** on lower average equipment on rent and slightly lower rates; in 2024 Mobile Modular posted materially larger revenue and utilization. **Root business mechanism:** company-level specialty-rental growth is a blend of modular growth with higher average duration and storage growth with shorter-cycle utilization/rate movement. **Calculation or driver bridge:** central growth assumes modular mid-single-digit expansion plus modest storage recovery. **Sensitivities:** education budgets, construction starts, and storage fleet loading. **Confidence: High**. [74]

EBITDA margin. Conclusion: 35% / 39% / 43%. **Observed facts:** in 2024 Mobile Modular operating income was **$173.1 million** on **$635.4 million** revenue, and Portable Storage operating income was **$36.0 million** on **$94.5 million** revenue; both segments also bear depreciation and direct service costs. **Root business mechanism:** margins are helped by long asset lives and residual values, but modular incurs more service/install complexity than storage. **Calculation or driver bridge:** central margin blends modular and storage while excluding TRS from the benchmark logic. **Sensitivities:** corporate allocation and mix shift between modular and storage. **Confidence: High**. 78

Maintenance capex. Conclusion: 6% / 8% / 11%. **Observed facts:** modular buildings have **18-year** lives to **50%** residual; containers **25-year** lives to **62.5%** residual. **Root business mechanism:** sustaining replacement is meaningfully below depreciation, but modular refurbishment is real and cannot be ignored. **Calculation or driver bridge:** a central blended specialty-rental maintenance capex near **8%** is consistent with the fleet profile. **Sensitivities:** fleet age and classroom-spec refurbishment standards. **Confidence: High**. 72

Post-maintenance-capex conversion. Conclusion: 69% / 79% / 86%. **Observed facts:** high segment margins and long asset lives support solid cash conversion. **Root business mechanism:** modular and storage both benefit from low technical obsolescence; capital is the main friction. **Calculation or driver bridge:** computed from the EBITDA and maintenance assumptions. **Sensitivities:** if modular refurbishment is heavier than assumed, this comes down. **Confidence: High**. 79

Initial contract tenor. Conclusion: 6 / 18 / 24 months. **Observed facts:** Mobile Modular's typical initial term is **12–24 months**; Portable Storage is typically **1–12 months**. **Root business mechanism:** company-level tenor is a weighted blend of these two structurally different businesses. **Calculation or driver bridge:** central value lands around **18 months**, with storage at the low end and modular at the high end. **Sensitivities:** revenue mix by segment. **Confidence: High**. 80

Effective customer tenure. Conclusion: 3 / 6 / 9 years. **Observed facts:** modular management emphasizes customer loyalty and repeat business; storage markets diversified commercial users. **Root business mechanism:** companies and schools repeat while units move on and off rent. **Calculation or driver bridge:** central customer life is measured in years, not contract months. **Sensitivities:** service reliability and competitive repricing. **Confidence: Medium**. 72

Utilization exposure. Conclusion: Moderate. **Observed facts:** average utilization in 2025 was **73.0%** for Mobile Modular and **60.8%** for Portable Storage. **Root business mechanism:** revenue depends on on-rent fleet cost, but long lives and strong redeployability soften downside. **Calculation or driver bridge:** exposure is higher in storage than in modular, but central company classification is moderate. **Sensitivities:** project roll-offs and new placement timing. **Confidence: High**. 73

New-unit or customer payback. Conclusion: 4.0 / 5.5 / 7.0 years. **Observed facts:** Portable Storage rates are **2%–4% of original cost per month**, while modular terms are longer but revenue yield on cost is lower. **Root business mechanism:** storage boxes pay back faster; modular pays back over more years but with longer continuation and better relationship durability. **Calculation or driver bridge:** central specialty-rental payback blends those two. **Sensitivities:** segment mix and ancillary service burden. **Confidence: High**. 81

B2B versus B2C. Conclusion: predominantly B2B. **Observed facts:** Mobile Modular's largest markets include education and commercial uses; Portable Storage markets construction, retail, commercial, industrial, education, and healthcare. **Root business mechanism:** contracting counterparties are overwhelmingly businesses and institutions. **Calculation or driver bridge:** clean B2B classification. **Sensitivities:** negligible. **Confidence: High**. 72

Plain-English business description: McGrath's in-scope specialty-rental business rents and services modular buildings and storage containers through a high-residual, redeployable fleet model aimed at business and institutional customers. [82]

Principal strengths: unusually favorable fleet lives/residuals, segment disclosure quality, strong modular economics, and diversified B2B end markets. **Principal risks:** lower storage utilization, mix shifts, and the need not to blend modular economics with out-of-scope test equipment. **Key unavailable information:** a fully standalone specialty-rental-only cash-flow statement excluding TRS. **Most important diligence questions:** What portion of modular fleet purchases is genuine replacement, how much of segment service revenue is low-margin transport, and how much cross-selling exists between modular and storage? [83]

Reliant Rental

A. Resolve the company and segment. As discussed in the identity gate, the most defensible candidate is **the "Reliant Rental" trademark/brand associated with Vander Intermediate Holding III Corporation**, which sits in the BlueLine corporate family later acquired by United Rentals in 2018. I did **not** find primary evidence of a standalone legal operating company named Reliant Rental, nor did I find a transaction that clearly identified a specialty-rental operating business in portable storage, modular space, temporary classrooms/offices, trench safety, or another in-scope specialty niche. [5]

B. Define the economic unit. Unavailable. Without a resolved operating business, the economic unit cannot be specified responsibly. [8]

C. Explain the customer problem. Unavailable. The name alone does not establish the actual product, asset class, or end-user need. [84]

D. Explain customer alternatives. Unavailable. Same reason. [84]

E. Map the revenue model. Unavailable. I found no primary financial or operating disclosure for a resolved specialty-rental business under this name. [8]

F. Map the operating model. Unavailable. [8]

G. Explain retention and churn. Unavailable. [8]

H. Map the capital model. Unavailable. [8]

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Normalized revenue growth	Unavailable	Unavailable	Unavailable	Unavailable	Low	Identity unresolved

Metric	Low	Central	High	Evidence Status	Confidence	Root Business Driver
Normalized EBITDA margin	Unavailable	Unavailable	Unavailable	Unavailable	Low	Identity unresolved
Maintenance capex % of revenue	Unavailable	Unavailable	Unavailable	Unavailable	Low	Identity unresolved
Post-maintenance-capex conversion	Unavailable	Unavailable	Unavailable	Unavailable	Low	Identity unresolved
Initial contract tenor	Unavailable	Unavailable	Unavailable	Unavailable	Low	Identity unresolved
Effective customer tenure	Unavailable	Unavailable	Unavailable	Unavailable	Low	Identity unresolved
Utilization exposure	Unavailable	Unavailable	Unavailable	Unavailable	Low	Identity unresolved
New-unit or customer payback	Unavailable	Unavailable	Unavailable	Unavailable	Low	Identity unresolved
B2B/B2C profile	Unavailable	Unavailable	Unavailable	Unavailable	Low	Identity unresolved

Primary evidence for Reliant Rental profile: trademark records and BlueLine/Vander transaction materials. [5]

Revenue growth. Conclusion: Unavailable. Observed facts: only trademark and BlueLine-family evidence were located. **Root business mechanism:** unresolved. **Calculation or driver bridge:** not possible responsibly. **Sensitivities:** identity itself. **Confidence: Low**. [8]

EBITDA margin. Conclusion: Unavailable. Observed facts: no standalone operating disclosure. **Root business mechanism:** unresolved. **Calculation or driver bridge:** none. **Sensitivities:** identity itself. **Confidence: Low**. [8]

Maintenance capex. Conclusion: Unavailable. Observed facts: no resolved asset class. **Root business mechanism:** unresolved. **Calculation or driver bridge:** none. **Sensitivities:** identity itself. **Confidence: Low**. [84]

Post-maintenance-capex conversion. Conclusion: Unavailable. Observed facts: no EBITDA or capital basis. **Root business mechanism:** unresolved. **Calculation or driver bridge:** none. **Sensitivities:** identity itself. **Confidence: Low**. [84]

Initial contract tenor. Conclusion: Unavailable. Observed facts: no contracts or operating description. **Root business mechanism:** unresolved. **Calculation or driver bridge:** none. **Sensitivities:** identity itself. **Confidence: Low**. [84]

Effective customer tenure. Conclusion: Unavailable. Observed facts: no customer evidence. **Root business mechanism:** unresolved. **Calculation or driver bridge:** none. **Sensitivities:** identity itself. **Confidence: Low**. [84]

Utilization exposure. Conclusion: Unavailable. Observed facts: no asset evidence. **Root business mechanism:** unresolved. **Calculation or driver bridge:** none. **Sensitivities:** identity itself. **Confidence: Low**. [8]

New-unit or customer payback. Conclusion: Unavailable. Observed facts: no unit economics. **Root business mechanism:** unresolved. **Calculation or driver bridge:** none. **Sensitivities:** identity itself. **Confidence: Low**. [8]

B2B versus B2C. Conclusion: Unavailable. Observed facts: no resolved operating business. **Root business mechanism:** unresolved. **Calculation or driver bridge:** none. **Sensitivities:** identity itself. **Confidence: Low**. [84]

Plain-English business description: On the current evidence, Reliant Rental is best treated as an unresolved brand/trademark candidate linked to the Vander/BlueLine family, not a clearly evidenced standalone specialty-rental operator. **Principal strengths / risks / key unavailable information / diligence questions:** the core diligence question is the identity itself—what exact legal entity, if any, was intended, and what in-scope specialty-rental assets did it operate? [8]

Archetype synthesis and benchmark tables

The designated companies support **four material archetypes**. The first is **consumer-led delivered portable storage**, represented by PODS. The second is **B2B portable/mobile storage**, represented by Mobile Mini Storage Solutions, McGrath Portable Storage, and the storage portions of General Finance. The third is **modular space and temporary offices/classrooms**, represented by Modulaire, McGrath Mobile Modular, and the modular/mobile-office portion of General Finance. The fourth is **trench-safety equipment rental**, represented by Trench Plate Rental. These should not be blended without explanation because portable storage legal terms can be monthly while modular often starts closer to one to two years, and trench safety is short-contract but safety-critical. [85]

Company classification is as follows. **PODS:** *Core representative* for consumer-led portable storage. **Mobile Mini:** *Core representative* for B2B portable/mobile storage, with Tank & Pump excluded from the specialty-rental benchmark. **Trench Plate Rental:** *Core representative* for trench safety. **General Finance:** *Mixed-business company* and *secondary reference* for portable storage/modular because liquid containment and manufacturing/sales complicate the mix. **Modulaire Group:** *Core representative* for modular space/

classrooms and *secondary portable-storage reference*. **McGrath RentCorp:** *Mixed-business company* but also a *core representative* for modular space and a *secondary reference* for portable storage, with TRS excluded. **Reliant Rental:** *Insufficient disclosure*. [86]

Archetype benchmark tables

Consumer-led delivered portable storage

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
Revenue growth	1%	4%	7%	Demand can swing negative in weak housing periods	Low	Residential move volumes, pricing, move-event mix
EBITDA margin	28%	33%	38%	High operating leverage to route/yard utilization	Low	Container rent plus logistics margin minus delivery/yard overhead
Maintenance capex % revenue	5%	7%	9%	Higher than pure containers because logistics support matters	Low	Container refresh, delivery fleet, yard support
Post-maintenance conversion	68%	79%	87%	Often healthy, but not as clean as B2B container lessors	Low	Margin strength offset by logistics sustaining burden
Initial contract tenor	1 mo	1 mo	3 mos	Typically short and flexible	Medium	Temporary consumer need
Effective customer tenure	3 mos	8 mos	18 mos	Mostly event-driven	Low	Move/renovation duration and storage extensions

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
Utilization exposure	Moderate	Moderate-High	High	Higher in housing downturns	Low	Loaded-container, route, and yard utilization
New-unit/ customer payback	2.5 yrs	3.5 yrs	5.0 yrs	Faster if move fees are strong, slower if local utilization weakens	Low	Container-month plus move-event contribution
B2B/B2C profile	Mixed	Mixed leaning B2C	Mixed	B2C exception within sector	Medium	Residential moving/ storage core

B2B portable/mobile storage

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
Revenue growth	2%	5%	8%	Mobile Mini and McGrath support mid-single-digit central growth	High	Rate, units on rent, and local availability
EBITDA margin	33%	40%	44%	Mobile Mini and best storage segments sit near or above 40%	High	Simple steel assets, long lives, strong residuals
Maintenance capex % revenue	4%	6%	8%	Lower than modular and trench safety	Medium	Container replacement/ refurbishment
Post-maintenance conversion	76%	85%	91%	Strongest conversion archetype here	Medium	High margin and modest sustaining capex
Initial contract tenor	1 mo	3 mos	12 mos	Usually short legal term, extendable in practice	Medium	Project-centered jobsite storage use

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
Effective customer tenure	2 yrs	4 yrs	7 yrs	Customer life much longer than contract life	Medium	Contractor and commercial repeat demand
Utilization exposure	Moderate	Moderate	High	Supported by redeployability	High	Units on rent versus total fleet
New-unit/ customer payback	3.5 yrs	4.5 yrs	6.0 yrs	Faster than modular, slower than simple revenue/yield heuristics imply	Medium	2%–4% monthly yield on cost at normalized occupancy
B2B/B2C profile	Pred. B2B	Pred. B2B	Pred. B2B	Cleanest B2B storage archetype	High	Contractors, commercial, industrial, institutional users

Modular space and temporary classrooms/offices

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
Revenue growth	3%	6%	9%	Modulaire and McGrath support mid-single-digit to high-single-digit ranges	High	Price, units on rent, project/ public-sector pipeline
EBITDA margin	28%	34%	38%	Modulaire 1Q22 implies ~31%+; McGrath modular can run higher	High	Strong recurring rent after install, offset by delivery/ service cost

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
Maintenance capex % revenue	6%	8%	11%	Above containers because refurbishment is more material	Medium	Refurbishment, resets, yard support
Post-maintenance conversion	61%	76%	84%	Good but not as strong as container lessors	Medium	Solid rent economics, real sustaining fleet spend
Initial contract tenor	12 mos	18 mos	36 mos	Longest archetype here	High	Installed customer space tends to remain
Effective customer tenure	4 yrs	7 yrs	10 yrs	Public/ institutional relationships extend tenure	Medium	Repeat school, government, infrastructure demand
Utilization exposure	Moderate	Moderate	Moderate-High	Higher if new projects slow sharply	High	Fleet loading and redeployment speed
New-unit/ customer payback	4.5 yrs	6.0 yrs	8.0 yrs	Long lives and reuse compensate for bigger upfront ticket	Medium	Delivered module cost versus multiyear rent stream
B2B/B2C profile	Pred. B2B	Pred. B2B	Pred. B2B	Education/ public sector important	High	Institutional and enterprise customers

Trench-safety equipment

Metric	Core Low	Core Central	Core High	Broader Observed Range	Confidence	Root Drivers
Revenue growth	3%	6%	10%	Branch growth and infrastructure work can lift top end	Medium	Underground utility/ infrastructure activity
EBITDA margin	28%	33%	38%	Better than general rental where solution value is strong	Medium	Safety-critical pricing offset by branch/ transport cost
Maintenance capex % revenue	5%	7%	10%	Durable steel fleet but nontrivial sustaining burden	Medium	Repair, inspection, replacement, handling gear
Post-maintenance conversion	64%	79%	87%	Healthy if branch density stays high	Medium	Solid gross yield on long-lived steel assets
Initial contract tenor	1 wk	2 mos	6 mos	Shortest contract archetype, but not shortest customer life	Medium	Job-phase duration
Effective customer tenure	3 yrs	5 yrs	8 yrs	Repeat contractor and utility accounts	Medium	Multi-project customer relationships
Utilization exposure	Moderate-High	High	High	Most directly tied to project starts and completions	Medium	On-rent project fleet loading
New-unit/ customer payback	2.5 yrs	3.5 yrs	5.0 yrs	Good because steel yields well when on rent	Medium	Safety pricing plus residual/ scrap support
B2B/B2C profile	Pred. B2B	Pred. B2B	Pred. B2B	No meaningful B2C role	High	Contractors and utilities

A narrow combined all-archetype numerical table is only **partially defensible**, because it conceals major differences in contract duration and cost model. Still, as a broad envelope for the designated specialty-rental universe **excluding unresolved Reliant weighting**, the central tendency is roughly **5% revenue growth**, **34% EBITDA margin**, **7% maintenance capex as a percentage of revenue**, **79% post-maintenance conversion**, **an initial tenor around 12 months only because modular heavily skews the**

average, **effective customer tenure around 5 years**, **moderate utilization exposure**, and **cash payback around 4.8 years**. That envelope is directionally useful but analytically inferior to the archetype tables above. [87]

Final Specialty Rental conclusion

The **typical customer problem** is temporary space, storage, or excavation safety that is operationally necessary but not worth solving through outright ownership for every site or event. Depending on the archetype, the need is driven by convenience, capital avoidance, regulation/safety, or temporary capacity. [88]

The **typical revenue model** is recurring rental revenue wrapped with varying degrees of delivery, installation, pickup, refurbishment, or related service fees. The cleanest recurring-rental model sits in B2B storage containers; the heaviest service wrapper sits in consumer portable moving/storage and modular space installation. [89]

The **primary growth drivers** are rental rate, units on rent, fleet additions, geographic density, end-market demand, and selective acquisition. The specific mix varies sharply: PODS leans on residential mobility, modular leans on public/infrastructure demand, and trench safety leans on underground utility and civil work. [90]

The **primary margin drivers** are asset simplicity versus service intensity. Margins are highest where assets are long-lived, standardized, and lightly serviced while on rent; they compress where delivery, installation, route density, branch engineering, or linehaul matter more. That is why B2B storage containers screen best, modular sits next, and PODS/trench safety need more model-specific analysis. [91]

The **principal maintenance-capex requirements** are replacement and refurbishment of rental fleet, plus support assets needed to preserve existing capacity. Container-heavy businesses have the lightest sustaining burden relative to revenue because of long lives and strong residuals; modular requires more refurbishment; trench safety requires heavier handling/inspection support; PODS also must sustain logistics-support assets. [92]

The **typical contract and customer duration** bifurcate. Legal contract tenor is short in portable storage and trench safety, longer in modular space, but customer tenure is almost always longer than unit tenure because repeat commercial or institutional accounts reuse the service. General Finance and McGrath make this distinction especially clear. [93]

The **typical utilization dynamic** is that idle owned assets do not earn revenue, but the degree of pain differs with redeployment friction. Standard containers redeploy readily; modular units redeploy more slowly because installation matters; trench safety is very sensitive to project starts; PODS adds route and yard utilization on top of container loading. [94]

The **typical new-unit payback** is not captured by simplistic revenue yield. A better central view is roughly **3.5 years** for consumer portable storage, **4.5 years** for B2B storage containers, **6.0 years** for modular space, and **3.5 years** for trench safety. The biggest single mistake would be to assume management's "years to cover cost through revenue" equals cash payback. [95]

The **typical B2B/B2C profile** is overwhelmingly **B2B**, with **PODS as the major exception** because it materially serves households as well as businesses. That difference matters because B2C drives shorter tenure, higher acquisition-cost content, and more event-driven volume. [96]

The **largest source of variation** among companies is the combination of **asset standardization plus service intensity**. Standard containers left on a jobsite have one economic profile; modular classrooms that must be installed and serviced have another; trench safety adds engineering and extreme project sensitivity; PODS overlays consumer logistics. [97]

The **lowest-confidence conclusion** is Reliant Rental. The evidence supports only a low-confidence link to a BlueLine/Vander trademark/brand trail, not a clean standalone specialty-rental operating company. [98]

Slide-ready business-model description: Specialty rental leases long-lived, redeployable storage, space, or safety assets and wraps them with delivery, installation, and service, converting temporary customer problems into recurring, asset-backed revenue. [88]

Concise source list by company

PODS: official PODS “About Us” and storage/moving pages; Ontario Teachers acquisition announcement and completion coverage; S&P ratings update on 2024 earnings. [99]

Mobile Mini: 2019 Form 10-K; WillScot/Mobile Mini merger announcement. [100]

Trench Plate Rental: National Trench Safety / TPRC merger announcement; official NTS operating description; contemporary operating references on branch footprint. [42]

General Finance Corporation: FY2020 Form 10-K; United Rentals acquisition announcement. [101]

Modulaire Group: Companies House filing history for Modulaire Group Holdings Ltd.; official Modulaire news and 1Q22 report highlights; SEC-filed 2021 interim parentage disclosure. [102]

McGrath RentCorp: 2025 Form 10-K segment disclosures. [103]

Reliant Rental: Canadian trademark journal and related trademark-search evidence for Reliant Rental; United Rentals/BlueLine acquisition materials establishing the Vander/BlueLine transaction context. [5]

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[1] [2] [4] [9] [10] [11] [12] [13] [15] [16] [17] [18] [19] [85] [86] [87] [88] [96] [99] https://www.pods.com/aboutus
https://www.pods.com/aboutus

[3] [6] [7] [20] [21] [22] [23] [24] [25] [26] [27] [28] [29] [30] [31] [32] [33] [34] [35] [36] [89] [91] [92] [94] [100] https://www.annualreports.com/Click/21340
https://www.annualreports.com/Click/21340

[5] [8] [84] [98] https://ic.gc.ca/opic-cipo/tmj/eng/view.html?edition=06-03&file=Journal_en.html&year=2015
https://ic.gc.ca/opic-cipo/tmj/eng/view.html?edition=06-03&file=Journal_en.html&year=2015

14 90 https://www.spglobal.com/ratings/en/regulatory/article/-/view/type/HTML/id/3293453
https://www.spglobal.com/ratings/en/regulatory/article/-/view/type/HTML/id/3293453

37 38 39 41 42 43 https://www.ntsafety.com/national-trench-safety-to-merge-with-trench-plate-rental-company/
https://www.ntsafety.com/national-trench-safety-to-merge-with-trench-plate-rental-company/

40 https://www.ntsafety.com/
https://www.ntsafety.com/

44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 93 95 101 https://www.sec.gov/Archives/edgar/data/1342287/000119312520242828/d939869d10k.htm
https://www.sec.gov/Archives/edgar/data/1342287/000119312520242828/d939869d10k.htm

60 65 70 102 https://find-and-update.company-information.service.gov.uk/company/13449748/filing-history
https://find-and-update.company-information.service.gov.uk/company/13449748/filing-history

61 62 69 https://www.modulairegroup.com/news/modulaire-group-h1-2025-update
https://www.modulairegroup.com/news/modulaire-group-h1-2025-update

63 64 66 https://www.modulairegroup.com/news/modulaire-group-releases-first-quarter-2022-financial-report
https://www.modulairegroup.com/news/modulaire-group-releases-first-quarter-2022-financial-report

67 68 https://www.modulairegroup.com/news/modulaire-group-pricing-announcement
https://www.modulairegroup.com/news/modulaire-group-pricing-announcement

71 72 73 74 75 76 77 78 79 80 81 82 83 97 103 https://investors.mgrc.com/sec-filings/sec-filings/content/0001193125-26-071463/0001193125-26-071463.pdf
https://investors.mgrc.com/sec-filings/sec-filings/content/0001193125-26-071463/0001193125-26-071463.pdf

Home Comfort - Draft JSON

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      and HVAC equipment, plus protection plans and one-time service. A dense installed base converts essential
      household needs into recurring monthly billing. Pricing, premium-equipment mix, and cross-sell drive
      growth; low-touch water heaters support high margins. Equipment exchanges, removal, and reinstallation
      still consume sustaining capital. Enercare anchors the benchmark with cleaner segment disclosure; Reliance
       widens the range through its broader, less separable mix.",
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          modest net units, and occasional M&A; acquisitions are not organic growth.",
        "evidence": "Enercare revenue rose 6% in 9M 2018 while units rose only 0.6% and monthly rent increased
          from C$24.35 to C$25.50. Reliance customer growth was low-single-digit annually before U.S.
          tuck-ins.",
        "confidence": "Medium - strong Enercare history, limited current standalone bridges."
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        "metric": "EBITDA margin",
        "range": "Core low / central / high: 48% / 53% / 56%; broader observed range: 43%-56%.",
        "rationale": "Monthly revenue on installed, low-touch water heaters creates high recurring gross margin;
           technicians, dispatch, support, and transactional services dilute it.",
        "evidence": "Enercare disclosed C$201.4 million EBITDA on C$359.8 million 9M 2018 revenue, about 56%,
          with contracted revenue near 92%. Reliance has no standalone margin and a broader mix.",
        "confidence": "Medium - Enercare is disclosed; Reliance's range is low-confidence and triangulated."
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        "metric": "Maintenance capex / revenue",
        "range": "Core low / central / high: 10% / 12% / 14%; broader observed range: 9%-15%.",
        "rationale": "Providers replace owned end-of-life units, including removal and reinstallation, to
          preserve billable addresses. Water heaters and plans lower intensity; HVAC raises it.",
        "evidence": "Enercare exchanged 33.6 thousand assets in 9M 2018, 2.9% of opening units, and found used
          heaters uneconomic to redeploy. Reliance cites 14-16-year lives and included replacement.",
        "confidence": "Medium - replacement-cycle derived; Reliance-specific capex is low-confidence."
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        "metric": "Post-maintenance-capex conversion",
        "range": "Core low / central / high: 73% / 78% / 82%; broader observed range: 68%-84%.",
        "rationale": "Recurring billing creates strong EBITDA, but exchanges prevent software-like conversion.
          This measure excludes interest, taxes, working capital, lease principal, and growth capex.",
        "evidence": "The 78% center follows 53% EBITDA margin and 12% maintenance capex. Enercare additions,
          attrition, and exchanges show that some new installation spend merely preserves revenue.",
        "confidence": "Medium - direct arithmetic, derived capex; Reliance is low-confidence."
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          15-year-style economics. Broader range spans mixed legacy at-will to useful-life forms.",
        "rationale": "Newer rentals use minimum-term, buyout, or useful-life structures to recover capital;
          legacy Ontario water-heater cohorts can be cancellable or non-term.",
        "evidence": "Reliance cites 7-10-year minimums, 10 years then monthly, and separate non-term and
          84-month forms. Enercare historically split heaters between non-buyout and buyout contracts.",
        "confidence": "Medium - forms are observable; current cohort mix is not."
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        "rationale": "Address-level continuity, residential inertia, home-sale transfer, and convenient
          replacement can carry revenue beyond both legal term and the first asset.",
        "evidence": "Enercare rental attrition was 1.9% over 9M 2018; Reliance continues monthly after initial
          terms. Short-period attrition is not mechanically converted into customer life.",
        "confidence": "Medium - same-household tenure is inferred and differs from equipment life."
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        "rationale": "Asset utilization is largely binary: installed on a paying account or not. Household usage
           intensity does not normally change fixed rent. The more important exposure is seasonal technician,
          installer, and dispatch capacity.",
        "evidence": "Company reporting emphasizes units, contracts, attrition, exchanges, service response, and
          installed accounts rather than consumption or runtime.",
        "confidence": "High - contract mechanics and operating KPIs make the qualitative classification clear."
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        "rationale": "Fully burdened payback includes equipment, installation, acquisition, service, repair, and
           replacement reserve. Basic heaters repay slower; HVAC and premium units repay faster.",
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        "evidence": "Reliance showed C$1,000-C$5,500 installed cost and C$20-C$100+ monthly rent. Enercare said
          HVAC rentals earned three to five times traditional heater revenue.",
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        "metric": "B2B vs B2C",
        "range": "Low / central: Predominantly B2C; high: B2B contract with residential end users. Broader
          range: predominantly B2C to mixed edge cases.",
        "rationale": "Households are the economic center, although builders, property managers, and small
          commercial channels may originate or intermediate accounts.",
        "evidence": "Enercare disclosed that about 95% of its rental portfolio consisted of residential water
          heaters. Reliance's owner and consumer materials identify homeowners as the primary customer base.",
        "confidence": "High - product mix and customer positioning are explicit."
      }
    ]
  },
  "page2": {
    "overview": "Both companies monetize essential household infrastructure through installed assets and
      service. Rentals and protection plans are recurring; outright HVAC sales and one-time work are distinct.
      Retention reflects convenience and address continuity, but also historical switching friction, so customer
       life is not enforceable term. Returns depend on dense field service, recovery of installation cost, and
      continued billing or replacement after the first unit ages out.",
    "businesses": [
      {
        "name": "Reliance Home Comfort",
        "what_it_does": "Reliance Comfort Limited Partnership serves more than 2 million customers, mainly in
          Ontario, with reported western Canadian, Georgia, and Florida operations. It rents and sells water
          heaters and HVAC and offers water purification, plumbing, electrical, smart-home, and protection
          plans. It solves urgent hot-water, heating, and cooling needs without upfront purchase or surprise
          repair risk. The benchmark isolates rentals and plans from one-time work.",
        "economic_model": "The core unit is an installed asset at a residential address; plans and one-time jobs
           are separate units. Reliance funds equipment and installation, retains ownership, bills monthly,
          provides repair and replacement, and operates technicians, dispatch, and call centers. Essentiality,
          installation, budget smoothing, transfers, and replacement convenience support retention; moves,
          buyouts, switching, poor service, repricing, purchase, and technology migration drive churn. Used
          water heaters have limited redeployment value, so the address relationship is the key residual
          asset.",
        "supporting_facts": "Acquired in 2017 for about C$4.6 billion, Reliance grew from 1.8 million customers
          to 2.1 million by end-2023 before two 2025 U.S. acquisitions. Its mix extends beyond rentals into
          HVAC, water purification, plumbing, electrical, smart-home and one-time services. Ontario materials
          show 7-10-year minimums; one offer continues month-to-month after year 10, with 14-16-year heater
          lives. Current offers show C$1,000-C$5,500 installed purchase costs and C$20-C$100+ monthly rent, with
           installation, repair and replacement included. Homeowners are the core customer; direct unit
          contribution and sustaining capex are undisclosed."
      },
      {
        "name": "Enercare",
        "what_it_does": "Enercare is an Ontario-centric platform serving about 1.1 million customers. The
          relevant business is Home Services, excluding former Service Experts and submetering. It rents water
          heaters, HVAC, and water-treatment equipment, sells plans, and performs equipment sales and service,
          offering predictable cost and outsourced continuity for essential systems. Brookfield acquired it in
          2018; the clean segment base is 2017 and 9M 2018.",
        "economic_model": "The core unit is an installed rental unit, about 95% residential water heaters
          historically; plans and one-time HVAC jobs are separate. Rentals were 73% of 2017 revenue and plans
          20%. Heaters require little scheduled maintenance; HVAC and plans use more field labor. Ownership,
          installation, response, and buyout or useful-life terms support retention; moves, buyouts, switching,
          repricing, and poor service cause churn. Used water heaters are replaced rather than redeployed,
          making exchanges sustaining capital and allowing relationships to continue through replacement.",
        "supporting_facts": "In 9M 2018, Home Services revenue grew 6% to C$359.8 million; contracted revenue
          was C$332.0 million, EBITDA was C$201.4 million (56%), units grew 0.6%, average monthly rent rose from
           C$24.35 to C$25.50 and 33.6 thousand assets were exchanged. Rental attrition was 1.9% versus 9.3% for
           plans. About 95% of rentals were residential water heaters, which were almost never economical to
          refurbish and reinstall. HVAC rentals generated three to five times heater revenue; sustaining capex
          and current segment data are unavailable."
      }
    ],
    "watch_items": [
      "Separate recurring rentals and plans from one-time sales, service, and acquisition-led growth.",
      "Obtain current product mix, gross additions, attrition, exchange volume, and direct service cost by
        cohort.",
      "Test how much installation spending labeled as growth merely replaces churn and is economically
        sustaining.",
      "Distinguish legal protection from residential inertia and historical switching friction.",
      "Keep address-level customer tenure separate from the life of any individual heater or HVAC unit."
    ]
  }
}
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Point Of Use Water - Draft JSON

tmp/two_pagers/drafts/point-of-use-water.json

SHA-256: 9b181321cff8471c76e9c496716d862227041865eb62c4bd7e3fd2a478f9ad5b


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{
  "title": "Point-of-Use Water - Two-Page Operating Benchmark",
  "subtitle": "AquaVenture Holdings / Quench and Primo Water | Source report dated July 18, 2026",
  "page1": {
    "thesis": "The core archetype is Quench's workplace point-of-use model: company-owned filtration dispensers
      installed for fixed recurring fees including filters, sanitation, maintenance, and support. Primo is a
      mixed reference because filings combine filtration with refill and the offer spans homes and businesses.
      The benchmark therefore anchors on Quench, not a simple average. Recurring billing and route density
      support margins, but technicians and owned equipment add physical intensity. Legal terms can be short;
      durability comes from installation, service continuity, and switching effort.",
    "metrics": [
      {
        "metric": "Revenue growth",
        "range": "Core low / central / high: 4% / 8% / 12%; broader observed range: 1%-18%.",
        "rationale": "New placements, added units, price, retention, and acquisitions drive growth, not water
          throughput. Office health constrains Quench; mixed channels add refill effects.",
        "evidence": "Quench grew 45.4% in 2019, with 13.6% organic growth. Primo's combined Refill/Filtration
          channel grew 18.2% in 2023, but filtration-only growth was undisclosed.",
        "confidence": "Medium - Quench's anchor is strong but dated; Primo is mixed."
      },
      {
        "metric": "EBITDA margin",
        "range": "Core low / central / high: 24% / 28% / 31%; broader observed range: 12%-31%.",
        "rationale": "Monthly fees create recurring contribution; technicians, travel, filters, sanitation,
          repairs, and mix consume margin. Dense workplace routes should outperform fragmented home/business
          service.",
        "evidence": "Quench reported $32.4 million EBITDA on $115.2 million 2019 revenue, a 28.1% margin.
          Primo's 12%-24% filtration range is inferred.",
        "confidence": "Medium - Quench is reported; Primo is low-confidence."
      },
      {
        "metric": "Maintenance capex / revenue",
        "range": "Core low / central / high: 10% / 13% / 17%; broader observed range: 5%-17%.",
        "rationale": "Sustaining capital includes replacement systems, returned-unit refurbishment, service
          vehicles, removal, and reinstallation. Filters are operating cost.",
        "evidence": "Quench owns about 160,000 rental units. Primo disclosed $235.0 million net customer
          equipment with 2-15-year lives, but it includes refill and other assets.",
        "confidence": "Low - filtration-specific sustaining capex is undisclosed."
      },
      {
        "metric": "Post-maintenance-capex conversion",
        "range": "Core low / central / high: 30% / 54% / 68%; broader observed range: 35%-71%.",
        "rationale": "Recurring fees support EBITDA, but replacement and service assets absorb cash. This metric
           is not full free cash flow.",
        "evidence": "The 54% center uses Quench's 28% margin and 13% capex estimate. Primo's 56% case uses
          inferred 18% margin and 8% capex.",
        "confidence": "Low - capex and Primo EBITDA are derived."
      },
      {
        "metric": "Initial contract tenor",
        "range": "Core low / central / high: 1 month / 1-12 months / 36 months; broader range: one-month rolling
           to multiyear.",
        "rationale": "Flexible terms reduce sales friction; some enterprise accounts run longer. Durability
          relies more on operating stickiness than legal lock-in.",
        "evidence": "Quench describes primarily month-to-month rental. Primo permits cancellation without
          penalty after rescission, effective after the next full billing period.",
        "confidence": "Medium - terms are direct; account mix is not."
      },
      {
        "metric": "Effective customer tenure",
        "range": "Core low / central / high: 3 / 6 / 9 years; broader observed range: 2-9 years.",
        "rationale": "Plumbing, service, billing, and multi-unit relationships extend life; office closure,
          downsizing, hybrid work, easy cancellation, and substitutes shorten it.",
        "evidence": "Quench averaged about 2.9 units per customer. Primo's rolling cancellation supports a lower
           4-year company center.",
        "confidence": "Medium core; churn is undisclosed and Primo is low-confidence."
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        "metric": "Utilization exposure",
        "range": "Low: low customer-usage / moderate asset; central: moderate overall; high: moderate-high.
          Broader range: low-to-moderate customer usage and moderate asset/service capacity.",
        "rationale": "A lightly used contracted machine still earns; a returned unit does not until redeployed.
          Route density and technician productivity govern service economics.",
        "evidence": "Quench uses fixed pricing and 500+ technicians, with office-occupancy exposure. Primo
          diversifies office risk but can be harder to densify.",
        "confidence": "Medium - churn, idle fleet, and route KPIs are unavailable."
      },
      {
        "metric": "New-unit or customer payback",
        "range": "Core equipment low / central / high: 1.8 / 2.7 / 3.8 years; core CAC: 0.6 / 1.0 / 1.5 years.
          Broader equipment range: 1.5-3.8 years; broader CAC range: 0.8-2.0 years.",
        "rationale": "Equipment payback uses installed cost and contribution after service and sustaining
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          reserve. CAC payback is shorter because one account can support multiple machines.",
        "evidence": "Quench had 160,000 units and 55,000-plus customers, about 2.9 units per account. Primo
          plans start at $27.99 monthly; installed cost is undisclosed.",
        "confidence": "Low - both paybacks are derived."
      },
      {
        "metric": "B2B vs B2C",
        "range": "Core low / central: Predominantly B2B; high: almost entirely workplace B2B. Broader range:
          predominantly B2B to mixed B2B/B2C.",
        "rationale": "Workplace customers support density but add procurement complexity. Mixed home/business
          models broaden the funnel while increasing fragmentation and weakening contractual durability.",
        "evidence": "Quench is explicitly B2B and directs residential users elsewhere. Primo markets to homes,
          offices, schools, healthcare, hospitality, and manufacturing.",
        "confidence": "High core; Primo's exact mix is undisclosed."
      }
    ]
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    "overview": "Point-of-use water replaces bottle logistics with filtration connected to a site's supply. The
      core unit is a contracted machine; CAC and tenure sit at the account because one customer may support
      several systems. Providers fund equipment and installation, then filter, sanitize, repair, recover,
      refurbish, and redeploy. Draw matters less than contract status. The strategic divide is dense workplace
      B2B versus mixed platforms whose filtration economics are obscured by refill, delivery, retail, and
      household channels.",
    "businesses": [
      {
        "name": "AquaVenture Holdings / Quench",
        "what_it_does": "The relevant AquaVenture operation is Quench, excluding Seven Seas Water and
          desalination. Quench USA, Inc., now a Culligan subsidiary, installs company-owned filtration
          dispensers for institutional and commercial workplaces in the U.S. and Canada. Fixed-fee plumbed
          filtration replaces bottle storage, handling, and delivery.",
        "economic_model": "The core unit is a machine; CAC and tenure sit at the customer location. Monthly rent
           includes maintenance and filters; installation may be separate. Multiple machines spread account
          setup cost. More than 500 technicians install, inspect, sanitize, change filters, and repair, making
          route density central. Working equipment, included service, predictable bills, and re-plumbing support
           retention; office closure, relocation, downsizing, hybrid work, rebids, and poor service cause churn.
           Returns require recovery, refurbishment, and reinstallation before earning again.",
        "supporting_facts": "FY2019 Quench had about 160,000 company-owned rental units and 55,000-plus
          customers (2.9 units per account), with $115.2 million revenue, $32.4 million adjusted EBITDA and
          13.6% organic growth. Rentals are primarily month-to-month with fixed fees including maintenance and
          filters, supported by 500-plus technicians. Official materials identify Quench as workplace B2B and
          direct residential users to Culligan; churn, replacement cycles, installed cost, unit contribution and
           capex are unavailable."
      },
      {
        "name": "Primo Water",
        "what_it_does": "Primo Water's installed filtration offer now sits inside Primo Brands after the
          November 2024 transaction. It places company-owned carbon or reverse-osmosis systems in homes and
          businesses, with floor, countertop, and under-sink formats from $27.99 monthly. It solves taste and
          convenience needs without bottle logistics or self-managed maintenance. The benchmark excludes
          delivery, refill, exchange, dispenser sales, and unrelated products where possible.",
        "economic_model": "The core unit is a filtration system; the account governs CAC and cross-sell. Primo
          owns, installs, leases, maintains, filters, sanitizes, repairs, bills, and sometimes monitors devices.
           Rolling cancellation lowers sales friction but makes retention depend on convenience, service, price,
           moves, and cross-sell. Home/business reach reduces office concentration but fragments routes and
          creates internal purchase, refill, exchange, and delivery substitutes. Cancelled equipment must be
          recovered and redeployed.",
        "supporting_facts": "FY2023 combined Refill/Filtration revenue was $226.9 million versus $192.0 million,
           so filtration growth is not separable. Customer equipment had $235.0 million net book value and
          2-15-year lives but included non-filtration assets. Plans start at $27.99 monthly; cancellation
          without penalty is permitted after the rescission period and takes effect after the next full billing
          period. Current materials market to homes and businesses, but the mix and filtration-only EBITDA,
          capex and payback inputs are unavailable."
      }
    ],
    "watch_items": [
      "Obtain filtration-only revenue, installed units, churn, net retention, margin, service cost, and
        sustaining capex.",
      "Separate organic placements and price from tuck-in acquisitions at Quench and refill-driven growth at
        Primo.",
      "Measure returned-unit recovery, refurbishment, redeployment, and scrap rates by product family.",
      "Track office occupancy, customer downsizing, route density, and technician productivity by geography.",
      "Keep equipment payback distinct from account-level CAC payback and include direct service plus
        maintenance reserves."
    ]
  }
}
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Submetering - Draft JSON

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{
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  "subtitle": "Two-page benchmark brief | Analysis date: July 18, 2026 | Energy Assets, ista, Techem, Smart
    Metering Systems, Calisen, and CARMA",
  "page1": {
    "thesis": "Submetering contains two different core cash models. Building submetering and billing platforms
      such as ista and Techem install relatively low-cost devices and monetize recurring billing, data, and
      compliance workflows; CARMA is a lower-disclosure reference. Meter-asset owners such as Calisen and Smart
      Metering Systems finance larger utility meters and earn rental and data income over long asset lives.
      Energy Assets is a mixed metering, data, network, and installation platform. The first model has
      mid-single-digit growth, high-40s EBITDA margins, moderate reinvestment, and strong conversion. The second
       can report higher EBITDA, but replacement capital and longer payback materially reduce its cash
      economics.",
    "metrics": [
      {
        "metric": "Revenue growth",
        "range": "Subvertical reference: 4% / 6% / 9% low-central-high; broader observed range 3%-12%.",
        "rationale": "Building platforms grow through indexation, digital refresh, penetration, and data upsell.
           Meter-asset growth can accelerate with funded rollout, but acquisitions and rollout effects are not
          normalized organic growth.",
        "evidence": "ista grew sales about 5.2% in 2025; Techem submetering rose from EUR933.5 million to
          EUR1,041.5 million. Calisen growth included installation and acquisition; SMS FY2022 ILARR grew 13%.",
        "confidence": "Medium"
      },
      {
        "metric": "EBITDA margin",
        "range": "Subvertical reference: 33% / 47% / 72%; broader observed range 18%-76%.",
        "rationale": "Dense billing/data estates have low marginal cost. Asset owners look richer because
          depreciation and replacement capital sit below EBITDA; service-led and mixed platforms sit lower.",
        "evidence": "Techem reported EUR557 million EBITDA on EUR1.17 billion 2025 revenue; ista evidence
          supports 40%+. Calisen reported 74.1% in 2023; SMS reported about 47.1% in FY2022.",
        "confidence": "Medium"
      },
      {
        "metric": "Maintenance capex / revenue",
        "range": "Subvertical reference: 8% / 12% / 35%; broader observed range 6%-45%.",
        "rationale": "Building platforms replace cheaper devices, batteries, communications, and IT.
          Utility-facing owners must replace expensive funded meter estates, making asset intensity the key
          structural divide.",
        "evidence": "ista reported EUR260 million total 2025 investment. Calisen disclosed GBP355.5 million 2023
           capex, GBP219 per smart meter, and GBP144.8 million meter depreciation; clean sustaining splits are
          unavailable.",
        "confidence": "Medium-Low"
      },
      {
        "metric": "Post-maintenance-capex conversion",
        "range": "Subvertical reference: 61% / 78% / 82%; broader observed range 24%-87%.",
        "rationale": "Billing/data platforms convert well after device refresh. Asset owners reserve much more
          EBITDA for replacement, so high reported margins can yield only roughly 50% conversion.",
        "evidence": "Derived as (EBITDA - maintenance capex) / EBITDA. Central company cases: ista 78%, Techem
          82%, SMS 47%, Calisen 51%. This is not full free-cash-flow conversion.",
        "confidence": "Medium-Low because sustaining capex is derived"
      },
      {
        "metric": "Initial contract tenor",
        "range": "Subvertical reference: 5 / 7 / 8 years; broader observed range month-to-month to 10 years.",
        "rationale": "Building awards need term to amortize installation. Utility-meter economics may rely on
          supplier frameworks and removal compensation, so legal form can be far shorter than asset life.",
        "evidence": "CARMA has long-term, inflation-protected contracts; ista and Techem publish no standard
          term. Calisen treats retailer orders as month-to-month although income can survive supplier
          switching.",
        "confidence": "Low-Medium"
      },
      {
        "metric": "Effective customer tenure",
        "range": "Subvertical reference: 8 / 14 / 18 years; broader observed range 5-20 years.",
        "rationale": "Installation, billing history, data migration, and low switching salience extend building
          tenure. Utility meters can keep earning across supplier changes until physical removal.",
        "evidence": "ista manages 45 million-plus devices; Techem has roughly 62-70 million. Calisen models
          arrangements around 15 years and reviewed smart-meter lives at 10-20 years.",
        "confidence": "Medium"
      },
      {
        "metric": "Utilization exposure",
        "range": "Subvertical reference: Low / Low-Moderate / Moderate; broader observed range Low-High.",
        "rationale": "Installed meters normally earn regardless of daily consumption. Exposure rises when
          installation teams, field service, and funded rollout pipelines must remain productive or meters are
          removed.",
        "evidence": "ista, Techem, and CARMA monetize installed units and billing. SMS and Calisen also depend
          on installation pace and supplier programs, although deployed estates remain sticky.",
        "confidence": "Medium"
      },
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      {
        "metric": "New-unit / customer payback",
        "range": "Subvertical reference: 3.0 / 4.5 / 6.5 years; broader observed range 2.0-8.5 years.",
        "rationale": "Cheaper suite devices plus billing/data yield roughly three-to-five-year payback. Funded
          utility meters pair higher installed cost with modest annual rent, extending payback.",
        "evidence": "Calisen disclosed GBP219 capex per meter and GBP23.8 ARPM. SMS expected 2.17 million
          pipeline meters to add about GBP50 million ILARR, or roughly GBP23 annually each.",
        "confidence": "Medium-Low"
      },
      {
        "metric": "B2B vs B2C",
        "range": "Subvertical reference: B2B contract with residential end users / B2B contract with residential
           end users / predominantly B2B.",
        "rationale": "Owners and managers buy building submetering while residents consume and may receive
          bills. Suppliers procure funded utility meters; households are users, not core customers.",
        "evidence": "ista has 460,000-plus customers against 14 million-plus units; Techem and CARMA market to
          buildings. SMS and Calisen contract with energy suppliers and retailers.",
        "confidence": "High"
      }
    ]
  },
  "page2": {
    "overview": "The six companies span three categories: core building submetering and billing, capital-heavy
      meter-asset ownership, and a mixed metering/data/network platform. The profiles below use the unit that
      drives revenue and highlight why similar recurring-revenue labels conceal different field-service,
      replacement-capital, and contract economics.",
    "businesses": [
      {
        "name": "Energy Assets",
        "what_it_does": "UK metering-and-data provider inside a wider network construction, ownership, water,
          fibre, and utility-services platform; customers need compliant metering, data, and field service.",
        "economic_model": "Unit: installed asset with attached data/service. The offering spans recurring
          metering/data, installation, exchanges, and maintenance; public sources do not quantify revenue mix.
          Integrated workflows likely aid retention, while project exposure and segment contamination weaken
          comparability. Capital supports meters, communications, IT, and vehicles.",
        "supporting_facts": "Energy Assets manages more than 1.5 million assets. Clean segment revenue, EBITDA,
          sustaining capex, churn and contract-term disclosures are unavailable, so its company data do not
          anchor the numerical estimates."
      },
      {
        "name": "ista",
        "what_it_does": "European platform installing apartment meters and allocators, then providing recurring
          consumption allocation, billing, data, and sustainability workflows to owners and managers.",
        "economic_model": "Unit: managed dwelling/device footprint. Regulation-shaped demand and dense estates
          support margin; access, hardware, resident communication, and data migration deter switching. Capital
          refreshes devices, batteries, communications, and digital infrastructure.",
        "supporting_facts": "2025 sales were EUR1,279.0 million versus EUR1,216.1 million, up 5.2%; EBITDA
          margin evidence was above 40%. Total investment was EUR260 million, or 20.3% of sales, and S&P cited
          total capex at 12%-15% of revenue; neither measure is sustaining capex."
      },
      {
        "name": "Techem",
        "what_it_does": "European provider installing apartment devices and monetizing recurring heating/water
          allocation, billing, data, and digital building services for owners and managers.",
        "economic_model": "Unit: serviced dwelling/device set. Regulation-shaped demand, radio infrastructure,
          and embedded billing deter switching; field service and procurement remain costs. The benchmark
          excludes energy contracting.",
        "supporting_facts": "2025 submetering revenue was EUR1,041.5 million versus EUR933.5 million, up 11.6%;
          adjusted EBITDA was EUR557 million on EUR1,173.4 million group revenue, or 47.5%. Fitch expected about
           EUR190 million of annual capex in 2023-2025, roughly 16% of 2025 group revenue."
      },
      {
        "name": "Smart Metering Systems",
        "what_it_does": "UK platform funding, installing, and managing supplier-facing smart meters for
          recurring rent and data income; batteries and EV-charging are excluded.",
        "economic_model": "Unit: owned contracted meter. Ownership and workflow integration support tenure;
          rollout and supplier procurement drive growth and service utilization. Hardware-plus-installation
          makes replacement capital heavy.",
        "supporting_facts": "FY2022 ILARR rose 13% to GBP97.1 million; pre-exceptional EBITDA was GBP63.8
          million on GBP135.5 million revenue, or 47.1%. SMS installed 480,000 meters; its 2.17 million-meter
          pipeline implied about GBP50 million ILARR, or GBP23 per meter annually. Sustaining capex is
          undisclosed."
      },
      {
        "name": "Calisen",
        "what_it_does": "UK asset owner funding meters for energy retailers; the benchmark centers on MAP
          Services and excludes EV charging, heat pumps, and other transition activities.",
        "economic_model": "Unit: owned revenue-generating meter. Policy drives installs; charges can survive
          supplier switching. Month-to-month legal orders contrast with 10-20-year economics. High EBITDA
          requires heavy replacement capital.",
        "supporting_facts": "2023 MAP revenue: GBP293.9m vs GBP236.2m; smart meters: 12.0m vs 8.2m. Group
          EBITDA/revenue: GBP265.4m/GBP358.2m (74.1%). Total capex: GBP355.5m; meter depreciation: GBP144.8m;
          capex/rent per meter: GBP219/GBP23.8 annually. Retailer orders: month-to-month; arrangements: 15
          years; meter lives: 10-20 years."
      },
      {
        "name": "CARMA",
        "what_it_does": "Canadian suite-level metering and billing platform helping owners and managers allocate
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          utilities, provide consumption visibility, and administer resident accounts.",
        "economic_model": "Unit: contracted metered suite. Devices, data, billing, and move workflows likely
          deter switching; property decisions or service failures are plausible churn vectors. Capital funds
          meters, communications, IT, and reinstallation.",
        "supporting_facts": "CARMA meters 135,000-plus units in 1,000-plus buildings under long-term
          inflation-protected contracts and processes 11 million-plus daily reads; revenue, EBITDA, capex and
          churn are undisclosed."
      }
    ],
    "watch_items": [
      "Obtain clean metering-segment financials for Energy Assets and CARMA; both materially widen the observed
        range but have the weakest disclosure.",
      "Separate rollout and acquisition capex from true estate replacement at Smart Metering Systems and
        Calisen, including communications refresh and removal or reinstallation costs.",
      "Confirm standard contract terms, renewal cohorts, and churn for ista and Techem; legal terms are inferred
         even though the retention mechanism is well supported.",
      "Test whether digitalization capex currently described as growth is economically required to preserve
        existing billing and data revenue."
    ]
  }
}
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Liquid Environmental Services - Draft JSON

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  "subtitle": "Two-page benchmark brief | Analysis date: July 18, 2026 | Restaurant Technologies, Liquid
    Environmental Solutions, Waste Resources Management, and Wind River Environmental",
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    "thesis": "The universe contains two structurally different models that should not be averaged. Restaurant
      Technologies represents an installed onsite-system model: customer-site oil equipment anchors a recurring
      fresh-oil, service, monitoring, and used-oil recovery relationship. LES, WRM, and Wind River represent
      route, collection, treatment, and processing: shared trucks, dense routes, permits, disposal access, and
      treatment throughput drive returns. Onsite embedding supports longer legal terms and customer payback but
      brings product and installation complexity. Route-processing offers quicker incremental-customer payback
      and higher margins when density and owned treatment are strong, but it has greater route, vehicle, and
      plant utilization exposure.",
    "metrics": [
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        "metric": "Revenue growth",
        "range": "Onsite system: 5% / 8% / 12% low-central-high. Route-processing: 6% / 9% / 13%.",
        "rationale": "Onsite growth comes from installed kitchens, chain wins, throughput, and price after
          removing commodity pass-through. Route growth reflects density, cross-sell, treatment throughput, and
          new accounts after excluding acquisitions.",
        "evidence": "RTI serves 45,000-plus kitchens through 41 depots but has no organic/commodity bridge. LES
          reaches 129,500 locations and 95 facility or partner sites; WRM and Wind River are active acquirers.",
        "confidence": "Onsite: Medium; route-processing: Medium"
      },
      {
        "metric": "EBITDA margin",
        "range": "Onsite system: 18% / 21% / 25%. Route-processing: 22% / 28% / 32%, with an observed 18%-34%
          company range.",
        "rationale": "RTI benefits from sticky service and recovered-oil value but bears fresh-oil,
          installation, depot, and field costs. Route margin rises when dense collection feeds controlled
          treatment; mixed services or third-party disposal dilute it.",
        "evidence": "Historical S&P commentary cited ratings-adjusted EBITDA margins above 30% for LES, the
          cleanest disclosed route processor. RTI has no current standalone margin; Wind River's plumbing,
          residential, and specialty-service mix supports a lower 18%-26% company range.",
        "confidence": "Onsite: Medium-Low; route-processing: Medium-High"
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      {
        "metric": "Maintenance capex / revenue",
        "range": "Onsite system: 2.5% / 3.5% / 4.5%. Route-processing: 3.5% / 5.0% / 6.5%.",
        "rationale": "RTI sustains installed tanks, pumps, filtration, monitoring, and vehicles; new installs
          are growth capex. Route processors replace vacuum trucks and maintain treatment, environmental, and
          safety systems.",
        "evidence": "RTI has 41 depots and 45,000-plus sites; LES reports 670 field vehicles; Wind River reports
           1,000 vehicles and 16 plants. None discloses maintenance capex, so ranges use replacement cycles and
          asset intensity.",
        "confidence": "Onsite: Low; route-processing: Medium-Low"
      },
      {
        "metric": "Post-maintenance-capex conversion",
        "range": "Onsite system: 75% / 83% / 90%. Route-processing: 77% / 82% / 89%.",
        "rationale": "Both models convert recurring EBITDA well before financing, working capital, and growth
          capex. RTI maintains distributed equipment; route operators reserve cash for fleet and plants, with
          mixed platforms weakest.",
        "evidence": "The ranges are derived as (EBITDA - maintenance capex) / EBITDA. Company central cases are
          RTI 83%, LES 87%, WRM 83%, and Wind River 75%. S&P's RTI cash-burn commentary shows why this metric is
           not full equity cash flow.",
        "confidence": "Low-Medium for both archetypes because maintenance capex is inferred"
      },
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        "metric": "Initial contract tenor",
        "range": "Onsite system: 3 / 4 / 5 years. Route-processing: 1 / 2 / 3 years, with an observed 0.5-3-year
           company range.",
        "rationale": "RTI likely needs a multi-year term to underwrite onsite equipment. Route service uses
          shorter commercial agreements, schedules, or callout relationships; renewal behavior matters more than
           lease-like lockup.",
        "evidence": "No company publishes a standard term. RTI emphasizes flexibility, not duration; LES and WRM
           describe recurring services, while Wind River's residential/local mix supports shorter
          arrangements.",
        "confidence": "Low for both archetypes; this is the report's lowest-confidence metric"
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        "metric": "Effective customer tenure",
        "range": "Onsite system: 6 / 9 / 12 years. Route-processing: 5 / 7 / 10 years.",
        "rationale": "Equipment, workflow, safety value, and the nuisance of manual oil handling make RTI
          sticky. Route customers stay because service recurs and reliability, compliance, and local disposal
          access matter, though switching is easier.",
        "evidence": "RTI has equipment embedded in more than 45,000 commercial kitchens. LES serves 129,500
          locations; WRM bundles grease and adjacent services; Wind River retains trusted local brands across a
          75-plus-year operating legacy. Cohort churn is not disclosed.",
        "confidence": "Medium for both archetypes"
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        "metric": "Utilization exposure",
        "range": "Onsite system: Moderate / Moderate / Moderate. Route-processing: Moderate / Moderate-High /
          High.",
        "rationale": "RTI equipment can remain deployed while oil throughput softens, so gallons per kitchen and
           depot density matter. Routes are more exposed to stops, gallons per haul, truck/technician
          productivity, and plant throughput.",
        "evidence": "RTI combines 45,000-plus installed kitchens with 41 depots. LES has 670 field vehicles;
          Wind River has 1,000 vehicles and plants processing more than 200 million gallons annually. Public
          route-fill and permitted-capacity utilization KPIs are unavailable.",
        "confidence": "Medium on classification; Low on exact KPI levels"
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        "metric": "New-unit / customer payback",
        "range": "Onsite system: 1.5 / 2.5 / 3.5 years. Route-processing: 0.5 / 1.25 / 2.25 years for an
          incremental customer in an existing dense market.",
        "rationale": "RTI recovers hardware, installation, acquisition, and route support from
          whole-relationship contribution. Nearby route customers leverage shared trucks and plants; de novo
          branches and acquisitions take longer.",
        "evidence": "RTI cites $250 and nine labor hours of weekly customer savings, not unit contribution. LES,
           WRM, and Wind River share fleets and plants; the benchmark does not allocate a whole asset to one
          account.",
        "confidence": "Onsite: Low-Medium; route-processing: Medium-Low"
      },
      {
        "metric": "B2B vs B2C",
        "range": "Onsite system: predominantly B2B across low-central-high. Route-processing: predominantly B2B
          / predominantly B2B / mixed B2B-B2C.",
        "rationale": "RTI sells to commercial kitchens and institutions. LES and most WRM customers are
          commercial or industrial. Wind River materially serves households alongside business, municipal, and
          industrial accounts, widening the route archetype's customer mix.",
        "evidence": "About 60% of RTI revenue was QSR-related in 2022. LES targets commercial/industrial
          generators; WRM names commercial/institutional users; Wind River serves homes, businesses,
          municipalities, and industry.",
        "confidence": "High"
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    "overview": "The companies solve recurring, essential oil and wastewater problems, but the bottleneck asset
      differs. At RTI it is the installed customer-site system and oil workflow. In route-processing it is local
       density plus permitted treatment or disposal access. That difference drives margin, payback, legal term,
      and utilization.",
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        "name": "Restaurant Technologies",
        "what_it_does": "Installs automated cooking-oil systems and provides fresh-oil delivery, filtration,
          monitoring, used-oil removal, and recycling, solving kitchen labor, safety, quality, and disposal
          problems.",
        "economic_model": "Unit: installed location; gallons measure usage and depot routes service capacity.
          Revenue mixes product, service, equipment, and recovery value. Equipment/workflow embedding retains
          customers; closures, operator changes, low throughput, service failures, or repricing cause churn.
          Capital funds onsite systems, vehicles, and depots.",
        "supporting_facts": "RTI serves 45,000-plus kitchens through 41 depots; about 60% of 2022 revenue was
          QSR-related, and S&P expected low-teens growth in 2018. RTI advertises average customer savings of
          $250 and 9 labor hours per week; current margin, maintenance capex, contract terms and provider
          contribution are undisclosed."
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        "name": "Liquid Environmental Solutions",
        "what_it_does": "Collects, transports, treats, recycles, and disposes of grease, used oil, wastewater,
          grit, and related streams for commercial/industrial customers needing recurring sanitation and
          compliance.",
        "economic_model": "Units: route customer for payback, truck/route for utilization, treatment plant for
          margin. Revenue mixes service, hauling, treatment/disposal, and recovery value. Plants and permits
          capture spread; recurring need, reliability, and processing access retain accounts. Poor service,
          pricing, dense rivals, or partner dependence create churn/margin risk.",
        "supporting_facts": "LES expanded from 49 branches/20 recycling facilities historically and 54
          branches/24 treatment plants in 2019 to 95 facility or partner sites and 129,500 customer locations by
           2026. Historical S&P commentary cited ratings-adjusted EBITDA margins above 30%; current scale
          includes 670 field vehicles, versus 300-plus vacuum trucks in earlier releases."
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        "name": "Waste Resources Management",
        "what_it_does": "Grease-centered platform collecting, processing, transporting, and recycling liquid
          waste while cross-selling jetting, trap repair, plumbing, septic, sludge, and remediation to
          commercial/institutional sites.",
        "economic_model": "Unit: route customer linked to shared processing. Revenue mixes collection,
          treatment, recovery, transport, and field work. Vertical integration and bundled reliability retain
          accounts; switching remains easier than onsite equipment. Capital includes vacuum trucks, specialty
          units, pumps, tanks, and processing assets.",
        "supporting_facts": "WRM serves 10,000-plus clients in four states; a 2026 transaction adds 13 metros,
          and the company has 300-plus employees. Revenue, EBITDA, maintenance capex and contract terms are
          undisclosed; the McDonald Farms combination broadens the service mix beyond pure grease routes."
      },
      {
        "name": "Wind River Environmental",
        "what_it_does": "East Coast consolidator spanning residential septic, commercial grease,
          plumbing/drains, municipal-industrial wastewater, specialty services, and treatment management,
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          solving backup, compliance, and downtime risks.",
        "economic_model": "Unit: route/service customer; plants drive network margin. Revenue mixes routes,
          septic cycles, emergencies, and projects. Local trust, recurring need, disposal access, and response
          retain customers; service inconsistency, competition, and integration cause churn. Its varied
          fleet/plants create density but heavy maintenance and mixed-business exposure.",
        "supporting_facts": "Wind River has 1,000 vehicles and 16 plants processing 200 million-plus gallons
          annually across 16 states, plus 90-plus acquisitions and a 75-plus-year operating legacy. It
          explicitly serves residential and commercial customers; revenue, EBITDA, maintenance capex and
          contract terms are undisclosed."
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    "watch_items": [
      "Obtain current standalone revenue and EBITDA for all four private companies and separate organic growth
        from acquisitions at LES, WRM, and Wind River.",
      "Confirm contract forms, renewal mechanics, and cohort churn. Initial legal tenor is the lowest-confidence
         conclusion in the report.",
      "Build maintenance capex by asset class: RTI onsite systems and service vehicles; route-company vacuum
        trucks, tanks, pumps, plants, and environmental and safety capital.",
      "Clarify recovered-material accounting and commodity pass-through: fresh and used oil at RTI; internal
        treatment, partner routing, and recovery spreads at LES; recovered outputs at WRM and Wind River.",
      "Preserve the payback distinction between an installed RTI location, an incremental dense-route customer,
        and a full new branch, plant, or acquisition."
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Transportation Solutions - Draft JSON

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  "subtitle": "Two-page source brief | Four structurally distinct archetypes | Analysis date: July 18, 2026",
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    "thesis": "Transportation Solutions is not a single benchmark. The report separates railcar leasing,
      locomotive leasing, airport ground-support-equipment (GSE) full-service leasing, and outsourced yard
      logistics because asset life, service intensity, contract structure, utilization and capital recovery
      differ structurally. Rail lessors earn high margins on scarce, long-lived assets but reinvest materially
      and recover capital over many years. Airport GSE combines rental with maintenance, spare fleets and
      airport-specific deployment, lowering margins and conversion. Lazer is labor-led outsourced operations,
      not an asset lessor: margins are lower, sustaining capex and payback are much lighter, and
      service-capacity utilization dominates. Only the mission-critical, predominantly B2B customer profile
      carries cleanly across all four.",
    "metrics": [
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        "metric": "Revenue growth",
        "range": "Railcar 1% / 4% / 7%; Loco 3% / 6% / 10%; GSE 5% / 8% / 12%; Yard 4% / 7% / 11%.",
        "rationale": "Railcar growth is price, fleet additions and utilization; locomotives add modernization
          demand. GSE gains from outsourcing and electrification. Yard growth is new sites and cross-sell.",
        "evidence": "VTG reported strong tank-wagon but weaker intermodal utilization; Akiem has placed major
          orders and completed a EUR1.52 billion green refinancing; Air Rail and TCR disclosed expanding fleets;
           Lazer serves 550+ sites.",
        "confidence": "Medium; normalized and triangulated, with private-company segment disclosure limiting
          precision."
      },
      {
        "metric": "EBITDA margin",
        "range": "Railcar 52% / 58% / 65%; Loco 55% / 62% / 70%; GSE 24% / 30% / 36% (observed 20%-38%); Yard
          10% / 14% / 18%.",
        "rationale": "Rail earns high lease spreads, tempered by maintenance and off-lease drag. GSE bears
          technician and workshop cost. Yard payroll and shift efficiency cap margins.",
        "evidence": "Akiem's official 2021 transaction data showed roughly EUR150 million EBITDA on EUR220
          million revenue. TCR runs 100+ workshops and about 1,000 technicians. Lazer reports 5,000 employees
          and 9 million annual service hours.",
        "confidence": "Medium; Air Rail's 20% / 27% / 33% range is Low due to mixed, private reporting."
      },
      {
        "metric": "Maintenance capex / revenue",
        "range": "Railcar 14% / 18% / 24%; Loco 12% / 18% / 25%; GSE 10% / 14% / 18%; Yard 2% / 4% / 7%.",
        "rationale": "Wagons need overhaul and replacement; locomotives add ERTMS and compliance work. GSE needs
           refurbishment and batteries. Yard sustains tractors and tech only.",
        "evidence": "VTG spent EUR536 million on fixed assets in 2024, mainly new wagons and fleet maintenance.
          Akiem is upgrading about 250 locomotives for ERTMS. Air Rail and TCR explicitly provide full lifecycle
           maintenance and refurbishment.",
        "confidence": "Medium overall; exact private-company gross sustaining spend is the report's
          lowest-confidence area."
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      {
        "metric": "Post-maintenance-capex conversion",
        "range": "Railcar 54% / 69% / 78%; Loco 55% / 71% / 83%; GSE 40% / 55% / 67% (observed 35%-69%); Yard
          55% / 71% / 84%.",
        "rationale": "The measure is (EBITDA minus maintenance capex) / EBITDA, not full free cash flow. Rail
          and GSE margins are high, but sustaining fleet spend absorbs cash; Lazer's lower margin is offset by a
           much lighter physical-capital burden.",
        "evidence": "Ranges are calculated directly from each archetype's normalized EBITDA and
          maintenance-capex assumptions and exclude interest, taxes, working capital, lease principal and growth
           capex.",
        "confidence": "Medium; derived, with sensitivity to the growth-versus-sustaining capex split."
      },
      {
        "metric": "Initial contract tenor",
        "range": "Railcar 2 / 4 / 7 years; Loco 5 / 7 / 10; GSE 3 / 5 / 7; Yard 1 / 3 / 5.",
        "rationale": "High unit value and approvals require longer locomotive terms. Wagon and GSE packages are
          medium term. Yard mobilization needs less legal duration than large assets.",
        "evidence": "VTG materials describe medium-to-long-term wagon hire; Akiem supports rolling stock through
           handback; TCR is described as having long-term contractual relationships. Lazer is embedded
          operationally but replaceable at rebid.",
        "confidence": "Medium; largely inferred because weighted-average contract terms are not publicly
          disclosed."
      },
      {
        "metric": "Effective customer tenure",
        "range": "Railcar 6 / 9 / 12 years; Loco 8 / 12 / 15; GSE 6 / 9 / 13 (observed 5-14); Yard 4 / 7 / 10.",
        "rationale": "Customer life outlasts one asset or site contract. Industrial wagon pools, approved
          locomotive fleets, multi-airport placements and deeply integrated yard staffing are resized, renewed
          or refreshed without changing provider.",
        "evidence": "The report cites fleet integration in industrial flows, Akiem's framework and lifecycle
          support, TCR's long-term relationships, and Lazer customer testimony about embedded operating
          performance.",
        "confidence": "Medium; tenure is inferred from integration, not disclosed churn cohorts."
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        "metric": "Utilization exposure",
        "range": "Railcar Moderate / Moderate / High; Loco Moderate / Moderate / High; GSE Moderate /
          Moderate-to-High / High; Yard High / High / High.",
        "rationale": "Rail depends on paid deployment and redeployment; locomotive downtime is concentrated. GSE
           adds ramp use and spare fleets. Yard depends on productive labor hours.",
        "evidence": "VTG's intermodal utilization weakened while tank wagons stayed high. TCR discusses
          idle-fleet waste and telematics right-sizing. Lazer's 9 million annual service hours make
          service-capacity utilization the decisive KPI.",
        "confidence": "Medium for asset models; High for Lazer's classification."
      },
      {
        "metric": "New-unit / customer payback",
        "range": "Railcar 7 / 9 / 12 years; Loco 8 / 10 / 13; GSE 4 / 5.5 / 7.5; Yard 0.8 / 1.5 / 2.5.",
        "rationale": "Rail needs multiple placements and residual value. GSE is cheaper but service-heavy. Yard
          launch spend is mainly recruiting, training, limited equipment and tech.",
        "evidence": "All ranges are triangulated from fully burdened investment and unit- or site-level cash
          contribution after direct cost and a maintenance reserve; none uses revenue payback alone.",
        "confidence": "Medium; Air Rail's company-specific payback is Low confidence because segment financials
          are thin."
      },
      {
        "metric": "B2B versus B2C",
        "range": "Railcar, Loco, GSE and Yard: predominantly B2B; Loco includes public-authority exposure.",
        "rationale": "Customers buy capital relief, uptime, compliance and outsourcing. Contracts sit with
          operators, authorities, airports, handlers, industrial firms and 3PLs.",
        "evidence": "Each company's official customer description is institutional; the report found no
          economically relevant direct B2C revenue stream in the benchmarked segments.",
        "confidence": "High."
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    "overview": "Five businesses map to four models: VTG is the railcar reference; Akiem is a mixed
      leasing-plus-maintenance locomotive platform; TCR is the core airport-GSE reference and Air Rail a smaller
       mixed airport/rail reference; Lazer is a separate outsourced-operations archetype. The economic unit
      shifts from one earning wagon or locomotive, to a GSE station package, to a yard-site contract and service
       hour.",
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        "name": "VTG",
        "what_it_does": "Leases and manages European freight wagons for industrial shippers and rail
          operators.",
        "economic_model": "Unit: earning wagon; revenue: recurring lease plus services. Customers avoid capital
          and secure compliant, cargo-specific capacity. VTG maintains and redeploys fleets; integration retains
           customers, while lane loss and intermodal weakness create churn. Long life and residuals offset heavy
           sustaining work.",
        "supporting_facts": "2024 operating cash flow was EUR405m and fixed-asset investment EUR536m, mainly new
           wagons and maintenance; tank-wagon utilization stayed high, intermodal utilization softened and
          rental pricing was positive. Customers hire wagons medium-to-long term, including a cited 10-year
          agreement, and are industrial shippers, rail operators and logistics firms. Standalone leasing EBITDA,
           sustaining capex and unit payback are undisclosed."
      },
      {
        "name": "Akiem",
        "what_it_does": "Leases and maintains locomotives and some passenger rolling stock across Europe.",
        "economic_model": "Unit: locomotive under lease/full service. Operators outsource scarce, compliant
          traction. Akiem runs workshops, overhauls and upgrades; approvals and uptime support retention, while
          concession loss or technical mismatch cause churn. Long contracts, residuals and repeat placements
          fund heavy capital needs.",
        "supporting_facts": "2021: 600-plus locomotives, 46 passenger trains, EUR220m revenue and EUR150m
          EBITDA, or roughly 68%. A EUR1.52bn 2026 refinancing, major OEM orders and ERTMS upgrades on about 250
           locomotives support expansion. Customers include carriers, manufacturers and local authorities;
          weighted tenor, utilization, sustaining capex and payback are undisclosed."
      },
      {
        "name": "Air Rail",
        "what_it_does": "AIR RAIL, S.L. leases, maintains and refurbishes airport GSE; it also has rail
          activities.",
        "economic_model": "Unit: deployed GSE/station package. Rental, maintenance, parts and resale solve
          airline/handler capex and uptime needs. Airport service embeds relationships; rebids and transfer
          friction cause churn. Fleets, batteries and charging drive capital needs.",
        "supporting_facts": "Air Rail expanded from 180 units in 2014 to 4,700-plus by March 2026 across about
          60 airports and eight countries; 68% of motorized units were electric. It serves airports, airlines
          and handlers through rental, maintenance, parts, resale and reconditioning. Segment revenue, EBITDA,
          sustaining capex, contract tenure, utilization and payback are undisclosed."
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        "name": "TCR",
        "what_it_does": "Rents, maintains, refurbishes and electrifies GSE for airlines, handlers and
          airports.",
        "economic_model": "Unit: GSE/airport fleet package. Full-service rent and lifecycle services solve
          turnaround uptime and capital needs. Workshops, technicians and spares embed TCR; handler rebids drive
           churn. Scale aids utilization but requires renewal and reserve fleet.",
        "supporting_facts": "TCR has 44,000-plus assets, 100-plus workshops, about 1,000 technicians and
          200-plus airports. GIP describes long-term customer relationships; TCR provides multi-year fleet
          planning, refurbishment, electrification, fleet right-sizing and used-equipment resale for handlers,
          airlines and airports. EBITDA, sustaining capex, weighted tenor, paid utilization and unit payback are
           undisclosed."
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      },
      {
        "name": "Lazer Logistics",
        "what_it_does": "Runs warehouse and industrial yards using drivers, equipment, processes and
          technology.",
        "economic_model": "Unit: site contract/service hour. Recurring site and labor revenue solves
          trailer-flow bottlenecks. Lazer staffs and supervises yards; integration supports retention, while
          rebids, closures and volume loss cause churn. Labor utilization dominates limited equipment capital.",
        "supporting_facts": "EQT reports 550-plus sites, 5,000 employees, 6,000 fleet assets and 9m annual
          service hours. Lazer expanded through acquisitions and serves logistics, manufacturing, food, retail
          and consumer-goods companies. Its labor-led site programs are embedded but rebiddable; organic growth,
           EBITDA, contract tenure, labor utilization, sustaining capex and site-launch payback are
          undisclosed."
      }
    ],
    "watch_items": [
      "Do not blend the four archetypes: the asset-owner versus labor-operator distinction drives margins,
        capex, utilization and payback.",
      "Private-company maintenance capex is the weakest metric; obtain gross replacement, refurbishment,
        battery/electrification and growth-fleet splits.",
      "Measure paid utilization by asset family, off-lease days and redeployment cost rather than relying on
        headline fleet counts.",
      "Separate legal term from customer life and isolate airport-, lane- or site-level churn from whole-account
         retention.",
      "Air Rail's legal identity is resolved, but its airport GSE segment remains mixed with rail activities;
        TCR's ownership was signed but not closed at the analysis date."
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Specialty Rental - Draft JSON

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    "thesis": "Specialty rental splits into four economically distinct models: consumer-led delivered portable
      storage, B2B portable/mobile storage, modular space and temporary classrooms/offices, and trench-safety
      equipment. A narrow combined central envelope - about 5% growth, 34% EBITDA margin, 7% maintenance capex,
      79% post-maintenance conversion and 4.8-year payback - is directionally useful but inferior to the
      archetype ranges. Standard B2B containers have the cleanest recurring rent, longest asset lives and
      strongest cash conversion. Modular units stay installed longer but cost more to deliver and refurbish.
      Trench safety is short-duration and project-sensitive but safety-critical. PODS overlays consumer
      acquisition, moving logistics, routes and yards. Reliant Rental remains unresolved and is excluded from
      benchmark weighting.",
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        "metric": "Revenue growth",
        "range": "Consumer storage 1% / 4% / 7%; B2B storage 2% / 5% / 8%; Modular 3% / 6% / 9%; Trench 3% / 6%
          / 10%.",
        "rationale": "Growth is rate, units on rent and fleet additions: home moves for PODS, local availability
          for containers, project pipelines for modular and utility work for trench.",
        "evidence": "Mobile Mini's 2019 Storage Solutions rental revenue grew 4.0% and yield 3.5%; Modulaire
          reported more units on rent and 5% revenue-per-unit growth in 1Q22; TPRC had reached 37 branches and
          5,000+ customers by 2021.",
        "confidence": "High for B2B storage and modular; Medium for trench safety; Low for consumer storage."
      },
      {
        "metric": "EBITDA margin",
        "range": "Consumer storage 28% / 33% / 38%; B2B storage 33% / 40% / 44%; Modular 28% / 34% / 38%; Trench
          28% / 33% / 38%.",
        "rationale": "Simple steel containers maximize margin. Modular adds installation; trench adds branch,
          engineering and hauling cost; PODS bears delivery, yards and linehaul.",
        "evidence": "Mobile Mini disclosed a 41.0% Storage Solutions adjusted EBITDA margin in 2019. Modulaire
          reported EUR116 million EBITDA on EUR370 million revenue in 1Q22, just over 31%. PODS showed operating
          leverage when Q3 2024 sales fell 7.5% and EBITDA fell 24%.",
        "confidence": "High for B2B storage and modular; Medium for trench; Low for PODS."
      },
      {
        "metric": "Maintenance capex / revenue",
        "range": "Consumer storage 5% / 7% / 9%; B2B storage 4% / 6% / 8%; Modular 6% / 8% / 11%; Trench 5% / 7%
          / 10%.",
        "rationale": "Containers have long lives and high residuals; PODS also sustains delivery equipment,
          yards and systems. Modular requires refurbishment and resets. Trench fleets need inspection, repair,
          replacement and handling gear despite durable steel assets.",
        "evidence": "Mobile Mini assets had 30-year lives and 55% residuals. McGrath uses 18 years/50% for
          modular and 25 years/62.5% for storage. Ranges are inferred from replacement cycles.",
        "confidence": "Medium except Low for PODS; gross-versus-net replacement remains an important diligence
          gap."
      },
      {
        "metric": "Post-maintenance-capex conversion",
        "range": "Consumer storage 68% / 79% / 87%; B2B storage 76% / 85% / 91%; Modular 61% / 76% / 84%; Trench
          64% / 79% / 87%.",
        "rationale": "The calculation is (EBITDA minus maintenance capex) / EBITDA, before interest, tax,
          working capital and growth capex. B2B containers convert best because margins are high and sustaining
          fleet needs modest; modular gives up more cash to refurbishment.",
        "evidence": "Every range is calculated from the report's normalized margin and maintenance-capex
          assumptions rather than reported free cash flow.",
        "confidence": "Medium; Low for PODS because both input assumptions are lower-confidence."
      },
      {
        "metric": "Initial contract tenor",
        "range": "Consumer storage 1 / 1 / 3 months; B2B storage 1 / 3 / 12 months; Modular 12 / 18 / 36 months;
          Trench 1 week / 2 months / 6 months.",
        "rationale": "Consumer moves and trench phases need flexibility; commercial containers are month-based
          but extend. Installed modular space stays through school years or project phases, creating the longest
          initial terms.",
        "evidence": "PODS markets flexible monthly use; Mobile Mini used month-based rentals; McGrath states
          1-12 months for storage and 12-24 months for modular; General Finance reported a typical lease
          averaging more than 12 months.",
        "confidence": "High for modular; Medium for the other archetypes."
      },
      {
        "metric": "Effective customer tenure",
        "range": "Consumer storage 3 / 8 / 18 months; B2B storage 2 / 4 / 7 years; Modular 4 / 7 / 10 years;
          Trench 3 / 5 / 8 years.",
        "rationale": "PODS tenure ends with the move or renovation. Commercial contractors, schools,
          institutions and utilities return across projects, so customer life materially exceeds any one unit's
          rental duration.",
        "evidence": "Mobile Mini served about 74,000 customers with low concentration; Modulaire serves 52,000+
          customers; TPRC served 5,000+ customers. The report infers relationship tenure from repeat-account
          structures rather than disclosed churn cohorts.",
        "confidence": "Medium except Low for PODS."
```

```
      },
      {
        "metric": "Utilization exposure",
        "range": "Consumer storage Moderate / Moderate-to-High / High; B2B storage Moderate / Moderate / High;
          Modular Moderate / Moderate / Moderate-to-High; Trench Moderate-to-High / High / High.",
        "rationale": "Idle fleet earns no rent. Containers redeploy readily; modular units face installation and
           removal friction; trench inventory rolls off with project completion. PODS adds route and yard
          utilization to loaded-container exposure.",
        "evidence": "Mobile Mini's 2019 utilization was 79.1% by units and 77.9% by original cost; Modulaire
          reported 86% in 1Q22; McGrath's 2025 utilization was 73.0% for Mobile Modular and 60.8% for Portable
          Storage.",
        "confidence": "High for B2B storage and modular; Medium for trench; Low for PODS."
      },
      {
        "metric": "New-unit / customer payback",
        "range": "Consumer storage 2.5 / 3.5 / 5.0 years; B2B storage 3.5 / 4.5 / 6.0; Modular 4.5 / 6.0 / 8.0;
          Trench 2.5 / 3.5 / 5.0.",
        "rationale": "Cash payback reserves delivery, service and maintenance. Containers and trench steel have
          residual value; modular costs more; PODS needs rent plus move-event profit.",
        "evidence": "General Finance said revenue could recover original equipment cost within four years, but
          the report extends cash payback after direct costs. McGrath storage rents at 2%-4% of original cost
          monthly; long lives and residuals support downside value.",
        "confidence": "Medium; Low for PODS. All ranges are triangulated rather than direct cash-payback
          disclosures."
      },
      {
        "metric": "B2B versus B2C",
        "range": "Consumer storage Mixed / Mixed leaning B2C / Mixed; B2B storage, Modular and Trench:
          predominantly B2B.",
        "rationale": "PODS materially serves households whose event-driven demand creates shorter tenure and
          more acquisition cost. Other archetypes contract mainly with contractors, commercial and industrial
          users, schools, government, institutions and utilities.",
        "evidence": "Company disclosures identify PODS as residential and commercial; Mobile Mini's account
          structure was commercial; Modulaire and McGrath emphasize institutional/enterprise markets; TPRC
          served contractors and utilities.",
        "confidence": "High for the three B2B archetypes; Medium for PODS's exact mix."
      }
    ]
  },
  "page2": {
    "overview": "Seven designated names yield four usable archetypes. Mobile Mini is the clearest B2B storage
      reference; Modulaire and McGrath support modular; TPRC anchors trench safety; PODS is the consumer-led
      delivered-storage exception. General Finance bridges storage and modular but has mixed assets and sales.
      Reliant Rental is an identity exception, not a benchmark input. Acquired companies use their last reliable
       standalone periods.",
    "businesses": [
      {
        "name": "PODS",
        "what_it_does": "Delivers containers for residential and commercial moving/storage.",
        "economic_model": "Unit: container-month plus move event; revenue is rent and logistics fees. Customers
          solve moves or renovations. PODS runs drivers, yards and linehaul; life-event completion causes churn.
           Container and delivery assets require capital.",
        "supporting_facts": "PODS reports 2 million-plus long-distance moves and 7 million-plus deliveries;
          rentals renew monthly and serve residential and commercial moving/storage. Q3 2024 sales fell 7.5% and
           adjusted EBITDA fell 24%. Normalized margin, fleet count, sustaining capex, utilization, effective
          tenure and unit payback are not publicly disclosed."
      },
      {
        "name": "Mobile Mini",
        "what_it_does": "Rented storage containers and ground-level offices to commercial users.",
        "economic_model": "Unit: box on rent; revenue was mostly rental. Branches deliver and refurbish simple
          assets solving temporary jobsite needs. Projects end placements, but accounts repeat. Long lives and
          residuals limit capital needs.",
        "supporting_facts": "In 2019, Storage rental revenue grew 4.0%, North America grew 6.3%, yield rose
          3.5%, EBITDA margin was 41.0% and unit utilization was 79.1%. Month-based rentals served about 74,000
          predominantly commercial customers. Assets had 30-year lives and 55% residual values; direct
          sustaining capex and cash payback were undisclosed."
      },
      {
        "name": "Trench Plate Rental",
        "what_it_does": "Rented trench plates, shoring and safety systems to contractors and utilities.",
        "economic_model": "Unit: jobsite equipment set; revenue is rental/support. Branches solve mandatory
          excavation safety, then inspect and redeploy steel. Projects end rentals, but contractors repeat.
          Residual value helps capital recovery; project timing drives utilization.",
        "supporting_facts": "Before its 2021 merger, Trench Plate Rental had 37 branches and 5,000-plus
          customers. Rentals to contractors and utilities were short and extendable; durable steel plates,
          shoring and safety systems were inspected and redeployed between projects. Revenue growth, EBITDA,
          sustaining capex, utilization, churn and payback were not disclosed."
      },
      {
        "name": "General Finance Corporation",
        "what_it_does": "Rented containers, mobile offices and modular units through regional brands.",
        "economic_model": "Unit: fleet asset; revenue is lease plus sales. Customers need temporary space. GFC
          delivered, maintained and re-leased units; projects cause churn, while accounts repeat. Long
          lives/residuals support capital recovery; mixed assets dilute margins.",
        "supporting_facts": "Typical leases averaged over 12 months and historical fleet utilization was
          70%-85%. Management said revenue recovered original equipment cost within four years on average.
          Assets had 20-30-year lives and residuals up to 70%. North American leasing was commercial; segment
```

```
        EBITDA, sustaining capex and churn cohorts were undisclosed."
      },
      {
        "name": "Modulaire Group",
        "what_it_does": "Rents and installs modular space for public, infrastructure and enterprise users.",
        "economic_model": "Unit: module on rent; revenue is lease/service. Customers need temporary space.
          Modulaire transports, installs and refurbishes units; installation and repeat projects support
          retention. Resets and yards raise capital needs versus containers.",
        "supporting_facts": "In 1Q22, revenue was EUR370m and EBITDA EUR116m; revenue per unit rose 5%, units on
           rent increased and utilization was 86%. Modulaire serves 52,000-plus public, infrastructure and
          enterprise customers and emphasizes fleet refurbishment. Weighted contract terms, sustaining capex,
          customer tenure and direct payback are undisclosed."
      },
      {
        "name": "McGrath RentCorp",
        "what_it_does": "Rents modular buildings and storage containers to businesses and institutions.",
        "economic_model": "Unit: module/container; revenue is rent plus delivery/service. Customers need
          classrooms, jobsite space or storage. McGrath installs and refurbishes; repeat accounts outlast
          placements. Strong residuals limit replacement, though modular resets consume capital.",
        "supporting_facts": "In 2025, Portable Storage rental revenue fell 3%; utilization was 73.0% for modular
           and 60.8% for storage. Initial terms are 12-24 months and 1-12 months; storage rents at 2%-4% of
          original cost monthly. Modular and storage assets use 18-year/50% and 25-year/62.5% life/residual
          assumptions and serve business and institutional customers."
      },
      {
        "name": "Reliant Rental",
        "what_it_does": "No standalone specialty-rental operator is resolved; evidence points only to a
          Vander/BlueLine brand candidate.",
        "economic_model": "Unavailable: no supported asset, unit, revenue, customer, operating, capital or churn
           model. BlueLine economics cannot be substituted.",
        "supporting_facts": "A 2014 trademark links Reliant to Vander; BlueLine sold in 2018. All metrics are
          Unavailable, Low confidence and unweighted."
      }
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    "watch_items": [
      "Keep the four archetypes separate: asset standardization and service intensity are the largest sources of
         economic variation.",
      "Use gross replacement and refurbishment requirements, not depreciation or net capex alone, to assess
        sustaining capital.",
      "Track physical and dollar utilization with consistent denominators; delivery, yard and branch utilization
         can matter as much as units on rent.",
      "Separate unit rental duration from repeat-account life; short monthly or job-phase contracts can coexist
        with multi-year customer tenure.",
      "Resolve Reliant Rental's exact legal and operating identity before assigning any metric or including it
        in a benchmark."
    ]
  }
}
```

Evidence Audit Matrix

tmp/two_pagers/compendium_parts/evidence-audit-matrix.txt

SHA-256: de40426a28a4a6193466c60d17c5f57c0dfc4abcef470226f1cbaca8ad1a5dea

```
EVIDENCE AUDIT MATRIX

Every retained estimate has at least a disclosed anchor or bounded operating-mechanism support in the
  corresponding Deep Research report. Whenever a Page 1 estimate uses comparable-company data, the complete
  supporting datapoints are repeated in that company's Page 2 Reference data section. Any component explicitly
  rated Low is shown as Not estimated with a first-principles directional framework in the final brief. The
  concise briefs omit inline source callouts; this evidence matrix and the embedded research reports preserve
  the source mapping and underlying links. Subsector categories are an analytical taxonomy used to organize the
  business set.

HOME COMFORT OPERATING BENCHMARK
================================

PAGE 1 - ESTIMATES AND ASSERTIONS
Metric: Revenue growth
Final display: 5%
Support: Medium
Decision: Retain
Source: Deep Research pp. 7, 18, 24, 26
Audit rationale: Enercare's disclosed 6% revenue growth with only 0.6% unit growth and Reliance's
  low-single-digit pre-acquisition customer growth bound a mature 5% normalized rate.

Metric: EBITDA margin
Final display: 53%
Support: Medium
Decision: Retain
Source: Deep Research pp. 18-19, 24, 26
Audit rationale: Enercare's disclosed roughly 56% margin anchors the estimate; Reliance's broader HVAC and
  field-service mix supports a downward normalization to 53%.

Metric: Maintenance capex / revenue
Final display: 12%
Support: Medium
Decision: Retain as modeled
Source: Deep Research pp. 10, 19, 24, 26
Audit rationale: Asset exchanges, limited used-water-heater redeployment and 14-16-year lives support low-teens
  sustaining intensity, but neither company discloses a clean sustaining split.

Metric: Customer tenor
Final display: Initial / effective: 7-10 / 18 yrs
Support: Medium
Decision: Retain as modeled
Source: Deep Research pp. 8, 15, 25, 27
Audit rationale: Certain Ontario forms support 7-10-year initial terms; monthly continuation and low
  short-period attrition support an 18-year normalized effective-tenure assumption, not an observed cohort life.

Metric: Utilization exposure
Final display: Low-Moderate
Support: High
Decision: Retain
Source: Deep Research pp. 25, 27
Audit rationale: Installed paying accounts earn fixed rent independent of runtime; account vacancy, removal and
  service capacity are the relevant exposures.

Metric: New-unit / customer payback
Final display: 8 yrs
Support: Medium
Decision: Retain as triangulated
Source: Deep Research pp. 8, 11, 21, 25, 27
Audit rationale: Disclosed installed costs, monthly rents and HVAC revenue multiples bound the eight-year
  estimate, while undisclosed direct contribution prevents High confidence.

Metric: B2B vs B2C
Final display: Predominantly B2C
Support: High
Decision: Retain
Source: Deep Research pp. 13, 15, 25, 27
Audit rationale: Enercare disclosed an approximately 95% residential water-heater portfolio and Reliance is
  primarily homeowner-facing.

PAGE 2 - BUSINESS FACTS
Business: Reliance Home Comfort
Support: Medium-High
Decision: Retain after scope wording correction
Source: Deep Research pp. 1-11
Audit rationale: Operating scale and services are directly sourced. Contract terms are limited to certain
  Ontario forms, pricing is illustrative, and redeployment is framed as an analytical mechanism.

Business: Enercare
Support: High
Decision: Retain after asset wording correction
Source: Deep Research pp. 12-23
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Audit rationale: 2017 and 9M 2018 filings directly support portfolio mix, revenue, EBITDA, attrition and
  exchanges; the redeployment conclusion is narrowed to used water heaters.


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POINT-OF-USE WATER OPERATING BENCHMARK
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PAGE 1 - ESTIMATES AND ASSERTIONS
Metric: Revenue growth
Final display: 8%
Support: Medium
Decision: Retain
Source: Deep Research pp. 5, 15-16
Audit rationale: Quench disclosed 13.6% organic growth in 2019; 8% is a dated-result normalization that allows
  for current office exposure.

Metric: EBITDA margin
Final display: 28%
Support: Medium
Decision: Retain
Source: Deep Research pp. 5, 15-16
Audit rationale: The 28% estimate is directly anchored to Quench's $32.4 million EBITDA on $115.2 million of
  2019 revenue.

Metric: Maintenance capex / revenue
Final display: Not estimated
Support: Low
Decision: Omit numeric estimate
Source: Deep Research pp. 1-2, 15, 17
Audit rationale: A 160,000-unit fleet proves physical capital, but filtration-only sustaining capex and
  replacement-cycle economics are not disclosed.

Metric: Customer tenor
Final display: Initial / effective: 1-12 mos / 6 yrs
Support: Medium
Decision: Retain as modeled
Source: Deep Research pp. 2, 6-7, 15, 17
Audit rationale: Month-to-month or flexible terms are disclosed; six years is a modeled effective-tenure
  assumption based on plumbing, recurring service and multi-unit relationships.

Metric: Utilization exposure
Final display: Moderate overall
Support: Medium
Decision: Retain
Source: Deep Research pp. 2, 6, 16-17
Audit rationale: Fixed billing reduces water-usage exposure, while returned equipment, office occupancy, route
  density and technician productivity create moderate asset/service exposure.

Metric: New-unit / customer payback
Final display: Not estimated
Support: Low
Decision: Omit numeric estimate
Source: Deep Research pp. 2, 13, 16-17
Audit rationale: Units per account and monthly pricing cannot establish payback without installed cost, direct
  unit contribution and customer-acquisition cost.

Metric: B2B vs B2C
Final display: Predominantly B2B
Support: High
Decision: Retain
Source: Deep Research pp. 1-2, 16-17
Audit rationale: Quench is explicitly workplace B2B and directs residential users elsewhere; Primo is a
  disclosed mixed boundary case.

PAGE 2 - BUSINESS FACTS
Business: AquaVenture Holdings / Quench
Support: Medium-High
Decision: Retain with FY2019 date
Source: Deep Research pp. 1-7
Audit rationale: FY2019 filings support units, customers, revenue, EBITDA and growth. Current churn, replacement
   cycles and unit economics remain expressly unavailable.

Business: Primo Water
Support: Medium
Decision: Retain with mixed-business caveat
Source: Deep Research pp. 8-15
Audit rationale: Annual reports and current service terms support the offering, asset base and cancellation
  mechanics; filtration-only economics remain unavailable.


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SUBMETERING OPERATING BENCHMARK
===============================

PAGE 1 - ESTIMATES AND ASSERTIONS
Metric: Revenue growth
Final display: Building: 6%; Meter owner: 7%
```

```
Support: Medium
Decision: Retain
Source: Deep Research pp. 9, 13, 18, 21, 31-32, 36
Audit rationale: Building estimates normalize ista and Techem results; the meter-owner estimate is deliberately
  below rollout- and acquisition-inflated SMS growth.

Metric: EBITDA margin
Final display: Building: 47%; Meter owner: 60%
Support: Medium
Decision: Retain with perimeter caveat
Source: Deep Research pp. 13, 18-22, 31, 33-34, 36
Audit rationale: Building economics are bracketed by ista and Techem's group proxy; meter-owner economics are
  bracketed by SMS and Calisen, with replacement capital below EBITDA.

Metric: Maintenance capex / revenue
Final display: Building: 9%; Meter owner: 30%
Support: Medium-Low
Decision: Retain as modeled
Source: Deep Research pp. 11, 22-23, 31, 33, 36
Audit rationale: Device refresh supports the building estimate and owned-meter estates support heavier capital,
  but rollout and replacement spending are not cleanly separated.

Metric: Customer tenor
Final display: Building: 8 / 14 yrs; Meter owner: 3 / 13 yrs
Support: Medium-Low
Decision: Retain as modeled
Source: Deep Research pp. 31, 33, 35-36
Audit rationale: Initial contract disclosures are thin; installed-device persistence, billing integration and
  supplier-switch compensation support longer modeled effective tenure.

Metric: Utilization exposure
Final display: Building: Low; Meter owner: Moderate
Support: Medium
Decision: Retain
Source: Deep Research pp. 32-33, 35-36
Audit rationale: Installed assets earn despite consumption variation, while meter-owner rollout and field-team
  productivity add operating exposure.

Metric: New-unit / customer payback
Final display: Building: 4.0 yrs; Meter owner: 6.0 yrs
Support: Medium-Low
Decision: Retain as modeled
Source: Deep Research pp. 22-23, 32, 34-36
Audit rationale: Building ranges use lower-cost device and onboarding economics; disclosed meter capex and
  annual rent support a bounded meter-owner payback.

Metric: B2B vs B2C
Final display: Building: B2B; residential end users; Meter owner: Predominantly B2B
Support: High
Decision: Retain
Source: Deep Research pp. 32, 34, 36
Audit rationale: Owners and managers contract for building services; suppliers and retailers contract with meter
   owners.

PAGE 2 - BUSINESS FACTS
Business: ista
Support: Medium-High
Decision: Retain with capex label
Source: Deep Research pp. 8-11
Audit rationale: Corporate disclosures support device, unit, customer and sales scale; EUR260 million is labeled
   total investment, not maintenance capex.

Business: Techem
Support: Medium-High
Decision: Retain with perimeter correction
Source: Deep Research pp. 12-16
Audit rationale: Submetering revenue and group EBITDA are stated on separate, correctly labeled perimeters
  rather than combined into a false segment margin.

Business: CARMA
Support: Medium-High
Decision: Retain with analytical churn wording
Source: Deep Research pp. 26-29
Audit rationale: Official sources support units, buildings, contracts and daily reads; churn mechanisms are
  identified as analytical rather than disclosed.

Business: Smart Metering Systems
Support: Medium-High
Decision: Retain with group-perimeter correction
Source: Deep Research pp. 17-20
Audit rationale: The 6 million-plus meter scale is attributed to the enlarged SMS / HEI / SMA group; the clean
  FY2022 base is separately labeled.

Business: Calisen
Support: High
Decision: Retain with capex label
Source: Deep Research pp. 21-25
Audit rationale: The annual report supports meters, revenue, EBITDA and rent; GBP355.5 million is labeled total
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  capex, not sustaining capex.

Business: Energy Assets
Support: Medium
Decision: Retain with revenue-mix caveat
Source: Deep Research pp. 3-7
Audit rationale: Managed-asset scale and service breadth are sourced; the card no longer implies a quantified
  revenue mix and states the segment-data gap.


-
LIQUID ENVIRONMENTAL SERVICES OPERATING BENCHMARK
=================================================

PAGE 1 - ESTIMATES AND ASSERTIONS
Metric: Revenue growth
Final display: Onsite: 8%; Route: 9%
Support: Medium
Decision: Retain
Source: Deep Research pp. 4-5, 9-11, 25, 27
Audit rationale: Onsite growth is tied to installed-site wins, pricing and throughput; route growth is tied to
  density, cross-sell and treatment throughput after excluding acquisitions conceptually.

Metric: EBITDA margin
Final display: Onsite: 21%; Route: 28%
Support: Medium-Low
Decision: Retain
Source: Deep Research pp. 5, 10-11, 26, 28
Audit rationale: Historical LES credit commentary supports route margins; onsite RTI margin remains
  mechanism-based because no standalone margin is disclosed.

Metric: Maintenance capex / revenue
Final display: Onsite: Not estimated; Route: 5.0%
Support: Medium-Low
Decision: Partially omit
Source: Deep Research pp. 4-5, 9-11, 20-22, 26, 28
Audit rationale: Onsite sustaining capex has Low support and is omitted. Observable route fleets and treatment
  plants support the retained 5% route estimate, though no clean sustaining split is disclosed.

Metric: Customer tenor
Final display: Onsite: Not est. / 9 yrs; Route: Not est. / 7 yrs
Support: Medium
Decision: Partially omit
Source: Deep Research pp. 5, 26, 28
Audit rationale: Both undisclosed initial legal terms are omitted. Nine-year onsite and seven-year route values
  remain as modeled effective tenure based on embedding and recurring service needs.

Metric: Utilization exposure
Final display: Onsite: Moderate; Route: Moderate-High
Support: Medium
Decision: Retain
Source: Deep Research pp. 4-5, 9-11, 20-22, 26, 29
Audit rationale: Installed sites and depots support onsite exposure; route stops, gallons, vehicles and
  treatment throughput support the higher route classification.

Metric: New-unit / customer payback
Final display: Onsite: 2.5 yrs; Route: 1.25 yrs
Support: Medium-Low
Decision: Retain with scope caveat
Source: Deep Research pp. 4, 6, 27, 29
Audit rationale: Onsite is triangulated from relationship economics and route applies only to an incremental
  account within an existing dense market; neither is a disclosed cohort payback.

Metric: B2B vs B2C
Final display: Onsite: Predominantly B2B; Route: Predominantly B2B
Support: High
Decision: Retain
Source: Deep Research pp. 27, 29-30
Audit rationale: RTI and the clean route processors serve commercial or institutional customers; Wind River is
  the explicit residential-commercial boundary case.

PAGE 2 - BUSINESS FACTS
Business: Restaurant Technologies
Support: Medium-High
Decision: Retain after unsupported phrase removal
Source: Deep Research pp. 3-8
Audit rationale: Operating scale and customer mix are sourced; the unsupported 'despite durable revenue' phrase
  is removed and data gaps remain explicit.

Business: Liquid Environmental Solutions
Support: Medium-High
Decision: Retain with dated-margin caveat
Source: Deep Research pp. 9-13
Audit rationale: Operating scale is sourced and the above-30% margin is explicitly labeled historical S&P credit
   commentary, not a current disclosure.

Business: Waste Resources Management
Support: Medium-High
```

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Decision: Retain with comparability wording
Source: Deep Research pp. 14-19
Audit rationale: Scale and acquisition facts are sourced; the McDonald Farms combination is framed as a
  broader-service comparability issue rather than an unsupported purity conclusion.

Business: Wind River Environmental
Support: High
Decision: Retain with direct customer-mix wording
Source: Deep Research pp. 20-24
Audit rationale: Official pages support scale, plants and residential-commercial service lines; the card avoids
  an unsupported absolute mix claim.


-
TRANSPORTATION SOLUTIONS OPERATING BENCHMARK
============================================

PAGE 1 - ESTIMATES AND ASSERTIONS
Metric: Revenue growth
Final display: Railcar: 4%; Loco: 6%; GSE: 8%; Yard: 7%
Support: Medium
Decision: Retain
Source: Deep Research pp. 31-35
Audit rationale: Archetype estimates are tied to rail pricing/utilization, locomotive modernization, GSE
  outsourcing/electrification and yard-site wins.

Metric: EBITDA margin
Final display: Railcar: 58%; Loco: 62%; GSE: 30%; Yard: 14%
Support: Medium
Decision: Retain
Source: Deep Research pp. 10, 12, 21, 23, 25-29, 31, 33-36
Audit rationale: Akiem anchors locomotive economics; disclosed asset scale and service intensity support the
  relative ordering across GSE and labor-led yard operations.

Metric: Maintenance capex / revenue
Final display: Railcar: 18%; Loco: 18%; GSE: 14%; Yard: 4%
Support: Medium
Decision: Retain as modeled
Source: Deep Research pp. 4, 7, 10, 13-15, 31, 33-36
Audit rationale: VTG investment, locomotive upgrades, GSE lifecycle service and yard's lighter asset base
  support relative ordering; private-company sustaining splits remain unavailable.

Metric: Customer tenor
Final display: Railcar: 4 / 9 yrs; Loco: 7 / 12 yrs; GSE: 5 / 9 yrs; Yard: 3 / 7 yrs
Support: Medium
Decision: Retain as modeled
Source: Deep Research pp. 32-36
Audit rationale: Asset value, approval burden and mobilization complexity support the terms, but the values are
  archetype assumptions rather than disclosed weighted-average cohorts.

Metric: Utilization exposure
Final display: Railcar: Moderate; Loco: Moderate; GSE: Moderate-High; Yard: High
Support: Medium
Decision: Retain
Source: Deep Research pp. 32-36
Audit rationale: Rail deployment, GSE spare-fleet mechanics and Lazer labor-hour productivity support the
  classifications; the scalar confidence is Medium.

Metric: New-unit / customer payback
Final display: Railcar: 9 yrs; Loco: 10 yrs; GSE: 5.5 yrs; Yard: 1.5 yrs
Support: Medium
Decision: Retain as modeled
Source: Deep Research pp. 32-36
Audit rationale: Asset cost, service intensity, residual value and site-launch investment support bounded
  archetype paybacks; Air Rail's weaker company evidence is not used alone.

Metric: B2B vs B2C
Final display: All models: Predominantly B2B
Support: High
Decision: Retain
Source: Deep Research pp. 32-36
Audit rationale: Institutional counterparties are explicit across rail, airport GSE and yard logistics.

PAGE 2 - BUSINESS FACTS
Business: VTG
Support: Medium-High
Decision: Retain with investment label
Source: Deep Research pp. 3-9
Audit rationale: 2024 accounts support operating cash flow and total fixed-asset investment; maintenance is not
  presented as separately quantified.

Business: Akiem
Support: Medium-High
Decision: Retain with 2021 date
Source: Deep Research pp. 9-15
Audit rationale: Fleet and financial facts retain their 2021 date; current upgrade and refinancing facts are
  separately dated.
```

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Business: Lazer Logistics
Support: Medium-High
Decision: Retain without revenue-mix claim
Source: Deep Research pp. 25-30
Audit rationale: Official and sponsor sources support sites, employees, fleet and service hours; the card
  describes a labor-led operating unit without implying a disclosed revenue split.

Business: Air Rail
Support: Medium
Decision: Retain with mixed-platform caveat
Source: Deep Research pp. 15-20
Audit rationale: Ownership and current GSE scale are sourced; the card retains the airport/rail perimeter caveat
   and does not present unsupported segment financials.

Business: TCR
Support: Medium-High
Decision: Retain
Source: Deep Research pp. 20-25
Audit rationale: Official company and sponsor sources support asset, workshop, employee, country and airport
  scale; operating economics are marked analyst interpretation.


-
SPECIALTY RENTAL OPERATING BENCHMARK
====================================

PAGE 1 - ESTIMATES AND ASSERTIONS
Metric: Revenue growth
Final display: Consumer: Not estimated; B2B storage: 5%; Modular: 6%; Trench: 6%
Support: Medium
Decision: Partially omit
Source: Deep Research pp. 9, 28-30, 32
Audit rationale: PODS lacks a clean organic bridge, so consumer is omitted. Mobile Mini, modular operators and
  trench-safety evidence support the retained B2B, modular and trench values.

Metric: EBITDA margin
Final display: Consumer: Not estimated; B2B storage: 40%; Modular: 34%; Trench: 33%
Support: Medium
Decision: Partially omit
Source: Deep Research pp. 9, 28-32
Audit rationale: PODS lacks a normalized standalone margin, so consumer is omitted. Mobile Mini directly anchors
   B2B storage; modular is disclosed and trench remains mechanism-based.

Metric: Maintenance capex / revenue
Final display: Consumer: Not estimated; B2B storage: 6%; Modular: 8%; Trench: 7%
Support: Medium
Decision: Partially omit
Source: Deep Research pp. 8-10, 22, 24, 28-32
Audit rationale: PODS's container, delivery and yard sustaining burden is undisclosed, so consumer is omitted.
  Long lives and residuals support the retained archetypes.

Metric: Customer tenor
Final display: Consumer: 1 mo / Not est.; B2B storage: 3 mos / 4 yrs; Modular: 18 mos / 7 yrs; Trench: 2 mos / 5
   yrs
Support: Medium
Decision: Partially omit
Source: Deep Research pp. 28-32
Audit rationale: PODS supports a one-month initial term but not an eight-month effective life; that component is
   omitted. Other archetypes have term evidence and modeled repeat tenure.

Metric: Utilization exposure
Final display: Consumer: Not estimated; B2B storage: Moderate; Modular: Moderate; Trench: High
Support: Medium
Decision: Partially omit
Source: Deep Research pp. 8, 10, 18, 20-24, 29-32
Audit rationale: PODS lacks comparable fleet, yard and route KPIs, so consumer is omitted. Mobile Mini,
  Modulaire and McGrath disclose utilization; trench is qualitatively project-sensitive.

Metric: New-unit / customer payback
Final display: Consumer: Not estimated; B2B storage: 4.5 yrs; Modular: 6.0 yrs; Trench: 3.5 yrs
Support: Medium
Decision: Partially omit
Source: Deep Research pp. 15, 17, 29-32
Audit rationale: PODS acquisition and logistics contribution is undisclosed, so consumer is omitted. General
  Finance and McGrath evidence supports the retained durable-asset archetypes.

Metric: B2B vs B2C
Final display: Consumer: Mixed; B2C lean; B2B storage: Predominantly B2B; Modular: Predominantly B2B; Trench:
  Predominantly B2B
Support: Medium
Decision: Retain
Source: Deep Research pp. 29-32
Audit rationale: PODS is explicitly residential-commercial; the remaining archetypes serve commercial,
  institutional, government or utility customers.

PAGE 2 - BUSINESS FACTS
Business: PODS
Support: Medium
```

```
Decision: Retain after factual correction
Source: Deep Research pp. 3-7
Audit rationale: Operating scale is sourced. The card now says sales fell 7.5% and adjusted EBITDA fell 24%, and
   it explicitly notes missing standalone fleet and capex data.

Business: Mobile Mini
Support: High
Decision: Retain
Source: Deep Research pp. 7-10
Audit rationale: The 2019 annual report directly supports revenue, margin, utilization, useful lives and
  residual values; sustaining-capital conclusions remain analytical.

Business: General Finance Corporation
Support: High
Decision: Retain with utilization-period label
Source: Deep Research pp. 14-17
Audit rationale: The FY2020 filing supports revenue, useful lives and residuals; 70%-85% is labeled historical
  average fleet utilization.

Business: Modulaire Group
Support: High
Decision: Retain with 1Q22 date
Source: Deep Research pp. 17-21
Audit rationale: The card attaches the 1Q22 date to revenue, EBITDA and utilization, while current scale is
  separately stated.

Business: McGrath RentCorp
Support: High
Decision: Retain
Source: Deep Research pp. 21-25
Audit rationale: The 2025 filing directly supports segment utilization, lives and residual assumptions.

Business: Trench Plate Rental
Support: Medium-High
Decision: Retain
Source: Deep Research pp. 10-14
Audit rationale: The merger announcement supports branch and customer scale; standalone economics are not stated
   as disclosed facts.


-
OMISSION LOG
============

- Point-of-Use Water: maintenance capex / revenue 13%.
- Point-of-Use Water: equipment payback 2.7 years.
- Point-of-Use Water: customer-acquisition-cost payback 1.0 year.
- Liquid Environmental Services: onsite maintenance capex / revenue 3.5%.
- Liquid Environmental Services: onsite initial contract term 4 years.
- Liquid Environmental Services: route initial contract term 2 years.
- Specialty Rental consumer/PODS: revenue growth 4%.
- Specialty Rental consumer/PODS: EBITDA margin 33%.
- Specialty Rental consumer/PODS: maintenance capex / revenue 7%.
- Specialty Rental consumer/PODS: effective customer tenure 8 months.
- Specialty Rental consumer/PODS: utilization exposure Moderate-High.
- Specialty Rental consumer/PODS: payback 3.5 years.
```

Source URL Index

tmp/two_pagers/compendium_parts/source-url-index.txt

SHA-256: 9b0f86feff3c75ad3b0686333ff5024190a8c89ba99b8ffef5104a083204e14f

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SOURCE URL INDEX

URLs below are extracted from the hyperlink annotations embedded in each Deep Research PDF.
The index preserves the report page on which each source link appears.

01-home-comfort-gpt-5.6-sol-deep-research.pdf
=============================================
Pages 12, 29, 30: https://bip.brookfield.com/press-releases/bip/enercare-inc-be-acquired-brookfield-infrastructu
  re-c43-billion-transaction
Pages 2, 30: https://competition-bureau.canada.ca/en/how-we-foster-competition/education-and-outreach/news-relea
  ses/competition-bureau-clears-reliances-acquisition-national
Pages 1, 3, 4, 12, 15, 22, 29, 30: https://competition-bureau.canada.ca/en/how-we-foster-competition/education-a
  nd-outreach/news-releases/competition-bureau-strengthens-competition-ontarios-water-heater-industry-fact-sheet
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  7/Enercare%20Solutions%20AIF%20-%20%202017%20-%20FINAL.PDF
Pages 1, 14, 15, 18, 19, 20, 21, 26, 27, 28, 29, 30: https://filecache.investorroom.com/mr5ircnw_enercare_ir/429
  /Enercare%20Solutions%20Inc.%20-%20MD%26A%20-%20Q3%202018%20-%20FINAL.PDF
Pages 3, 10, 27, 28, 29, 30: https://reliancehomecomfort.com/learning-centre/be-in-the-know/
Pages 2, 3, 4, 8, 9, 10, 11, 30: https://reliancehomecomfort.com/water-heaters/water-heater-rentals/
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  -ON-_-for-web.pdf
Pages 1, 29, 30: https://reliancehomecomfort.com/wp-content/uploads/2023/12/ESG-Annual-Impact-Report-2022-Novemb
  er-21-2022-FINAL-WEB.pdf
Pages 4, 30: https://reliancehomecomfort.com/wp-content/uploads/2024/10/ON-MB-TCs.pdf
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Pages 13, 18, 22, 28, 29, 31: https://www.enercare.ca/about
Pages 15, 29, 31: https://www.enercare.ca/wp-content/uploads/2023/10/RHVAC-Agreement-SAMPLE-TCs.pdf
Pages 12, 30: https://www.enercare.ca/wp-content/uploads/2024/06/Enercare_Annual-Report_2023-Financial-Year.pdf

02-point-of-use-water-gpt-5.6-sol-deep-research.pdf
===================================================
Pages 1, 5, 16, 18, 19: https://aquaventure2019ir.q4web.com/investors/press-releases/press-release-details/2020/
  AquaVenture-Holdings-Limited-Announces-Fourth-Quarter-and-Full-Year-2019-Earnings-Results/default.aspx
Pages 5, 6, 16, 17, 20: https://aquaventure2019ir.q4web.com/investors/press-releases/press-release-details/2020/
  AquaVenture-Shareholders-Approve-Planned-Acquisition/default.aspx
Pages 2, 6, 17, 18, 19: https://quench.culligan.com/
Pages 2, 3, 6, 7, 14, 16, 17, 18, 19: https://quench.culligan.com/facts-culligan-quench/
Pages 2, 19: https://quench.culligan.com/water-dispenser-rental-benefits/
Pages 7, 19, 20: https://www.kastle.com/safety-wellness/getting-america-back-to-work/
Pages 1, 2, 5, 7, 17, 18, 19:
  https://www.sec.gov/Archives/edgar/data/1422841/000110465919075681/tm1927296d1_ex99-2.htm
Pages 8, 20: https://www.sec.gov/Archives/edgar/data/2042694/000114036125009678/ny20042792x4_ars.pdf
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  https://www.sec.gov/Archives/edgar/data/884713/000114036124015948/ny20019840x5_ars.pdf
Pages 8, 20: https://www.water.com/modal/dispenser-rental-vs-purchase/
Pages 8, 9, 13, 14, 20: https://www.water.com/services/water-filtration/
Pages 9, 13, 20: https://www.water.com/support/service-agreement/

03-submetering-gpt-5.6-sol-deep-research.pdf
============================================
Pages 26, 27, 28, 30, 39: https://carmacorp.com/
Pages 27, 39: https://carmacorp.com/carma-billing-services/
Pages 17, 18, 19, 21, 38:
  https://d2ygkcw3bu7ib2.cloudfront.net/app/uploads/2023/03/SMS-Full-year-results-2022-RNS-FINAL.pdf
Pages 2, 3, 37, 38: https://find-and-update.company-information.service.gov.uk/company/12390729/filing-history
Pages 21, 39: https://www.calisen.com/knowledge-hub/news/eqt-and-gic-acquire-majority-stake-in-calisen
Pages 21, 39: https://www.calisen.com/knowledge-hub/reports
Pages 1, 2, 21, 22, 23, 24, 25, 26, 30, 34, 36, 38:
  https://www.calisen.com/media/0jzkligx/calisen-smarter-energy-report-2023.pdf
Pages 2, 22, 23, 36, 38: https://www.calisen.com/media/ca3hxwpp/2022-calisen_annual_report_2022_04-04-23.pdf
Pages 8, 38: https://www.cvc.com/media/news/2017/2017-07-26-cvc-capital-partners-fund-v-agrees-the-sale-of-its-i
  nvestment-in-ista/
Pages 27, 28, 30, 39: https://www.cvc.com/media/news/2025/cvc-dif-to-acquire-carma-corp-a-leading-canadian-subme
  tering-and-essential-building-services-platform/
Pages 1, 3, 4, 5, 7, 8, 30, 37, 38:
  https://www.energyassets.co.uk/our-utility-services/our-business-services/energy-assets/
Pages 13, 14, 38:
  https://www.fitchratings.com/research/corporate-finance/fitch-affirms-techem-at-b-outlook-stable-09-03-2023
Pages 9, 38: https://www.fitchratings.com/research/corporate-finance/no-immediate-impact-on-cki-ratings-from-pro
  posed-german-ista-buy-31-07-2017
Pages 1, 8, 10, 12, 30, 37, 38: https://www.ista.com/corporate/
Pages 2, 9, 10, 12, 32, 36, 38: https://www.ista.com/corporate/sustainability/key-figures/
Pages 26, 39:
  https://www.mnp.ca/en/services/corporate-finance/transactions/carma-corp-priority-submetering-solutions
Pages 17, 38: https://www.smsenergy.com/results-reports-presentations/
Pages 14, 37, 38: https://www.spglobal.com/ratings/en/regulatory/article/-/view/type/HTML/id/3568419
Pages 1, 2, 12, 13, 14, 15, 36, 38: https://www.techem.com/content/dam/techem-no/documents/om-oss/techem_b%C3%A6
  rekraftsrapport_2025_EN.pdf.coredownload.inline.pdf
Pages 12, 16, 38: https://www.techem.com/corp/en
```

```
Pages 13, 14, 15, 17, 38:
  https://www.techem.com/corp/en/news-and-media/press-releases/next-growth-phase-at-Techem

04-liquid-environmental-services-gpt-5.6-sol-deep-research.pdf
==============================================================
Pages 9, 32: https://cdn.hl.com/pdf/2025/bus-environmental-services-q2-2025.pdf?utm_source=chatgpt.com
Pages 14, 15, 16, 17, 19, 31, 32: https://wrmco.com/
Pages 19, 32: https://wrmco.com/a-new-chapter-for-wrm-joining-forces-with-ridgewood-infrastructure-to-enhance-u-
  s-water-sector-leadership/?utm_source=chatgpt.com
Pages 20, 32: https://www.gryphon-inv.com/news/private-equity-firm-gryphon-investors-announces-majority-investme
  nt-wind-river-environmental/?utm_source=chatgpt.com
Pages 1, 2, 9, 10, 11, 12, 14, 25, 27, 31, 32: https://www.liquidenviro.com/
Pages 2, 9, 30, 32:
  https://www.liquidenviro.com/blog-news/liquid-environmental-solutions-acquires-carolinas-resource-recovery
Pages 9, 32: https://www.liquidenviro.com/locations
Pages 15, 17, 32: https://www.prnewswire.com/news-releases/wrm-acquires-a-majority-of-darling-ingredients-trap-g
  rease-business-302771051.html
Pages 1, 2, 3, 4, 5, 6, 8, 25, 27, 30, 31, 32: https://www.rti-inc.com/?utm_source=chatgpt.com
Pages 5, 30, 32: https://www.rti-inc.com/contact-rti/?utm_source=chatgpt.com
Pages 1, 3, 4, 5, 6, 29, 30, 31, 32: https://www.rti-inc.com/oil-management/?utm_source=chatgpt.com
Pages 2, 11, 12, 29, 32:
  https://www.spglobal.com/ratings/en/regulatory/article/-/view/sourceId/101636728?utm_source=chatgpt.com
Pages 5, 32:
  https://www.spglobal.com/ratings/en/regulatory/article/-/view/type/HTML/id/2098325?utm_source=chatgpt.com
Pages 9, 32:
  https://www.spglobal.com/ratings/en/regulatory/article/-/view/type/HTML/id/2806652?utm_source=chatgpt.com
Pages 2, 4, 5, 31, 32:
  https://www.spglobal.com/ratings/pt/regulatory/article/-/view/type/HTML/id/3292027?utm_source=chatgpt.com
Pages 21, 22, 23, 24, 32, 33: https://www.wrenvironmental.com/about-us/
Pages 21, 33: https://www.wrenvironmental.com/about-us/acquisitions/
Pages 20, 22, 33: https://www.wrenvironmental.com/about-us/treatment-facilities/
Pages 20, 22, 23, 33: https://www.wrenvironmental.com/pike-county/

05-transportation-solutions-gpt-5.6-sol-deep-research.pdf
=========================================================
Pages 15, 16, 18, 19, 39: https://air-rail.org/en/
Pages 1, 2, 3, 15, 17, 30, 38, 39: https://air-rail.org/en/legal/
Pages 26, 28, 29, 30, 31, 37, 40: https://eqtgroup.com/en/about/current-portfolio/lazer-logistics
Pages 24, 39: https://www.3i.com/infrastructure/our-portfolio/tcr/
Pages 9, 10, 12, 13, 14, 15, 30, 34, 38, 39: https://www.akiem.com/en/about-akiem/
Pages 10, 12, 39: https://www.akiem.com/en/sncf-and-dws-finalize-the-sale-of-akiem-the-leading-provider-of-locom
  otive-leasing-services-in-france-and-europe-to-cdpq/
Pages 21, 23, 24, 39: https://www.global-infra.com/news/global-infrastructure-partners-agrees-to-acquire-tcr/
Pages 16, 19, 20, 38, 39: https://www.ifminvestors.com/capabilities/infrastructure/our-portfolio/air-rail/
Pages 3, 18, 20, 39: https://www.ifminvestors.com/news-and-insights/media-centre/ifm-investors-announces-the-acq
  uisition-of-air-rail-and-strengthens-its-position-in-the-spanish-infrastructure-market/
Pages 25, 26, 29, 30, 39, 40: https://www.lazerlogistics.com/
Pages 25, 28, 39: https://www.lazerlogistics.com/about
Pages 1, 2, 4, 6, 7, 8, 9, 30, 31, 32, 37, 38, 39:
  https://www.lobbyregister.bundestag.de/media/f3/0a/614644/VTG_GmbH_KAP_31-12-2024_Final_TE_24-004268.pdf
Pages 20, 38, 39: https://www.tcr-group.com/about-us
Pages 20, 21, 22, 23, 24, 25, 30, 35, 39: https://www.tcr-group.com/solution/gse-services
Pages 3, 4, 8, 37, 39: https://www.vtg.com/fileadmin/VTG/Dokumente/Investor_Relations/Berichte/VTG_GB14-engl.pdf
Pages 3, 39: https://www.vtg.com/news-and-insights/press-releases/detail/morgan-stanley-infrastructure-partners-
  and-joachim-herz-stiftung-agree-to-sell-majority-stake-in-vtg-aktiengesellschaft-to-a-consortium-of-gip-and-ad
  ia

06-specialty-rental-gpt-5.6-sol-deep-research.pdf
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Pages 17, 20, 21, 34, 35:
  https://find-and-update.company-information.service.gov.uk/company/13449748/filing-history
Pages 2, 3, 25, 26, 27, 34:
  https://ic.gc.ca/opic-cipo/tmj/eng/view.html?edition=06-03&file=Journal_en.html&year=2015
Pages 21, 22, 23, 24, 25, 34, 35:
  https://investors.mgrc.com/sec-filings/sec-filings/content/0001193125-26-071463/0001193125-26-071463.pdf
Pages 1, 2, 7, 8, 9, 10, 33, 34: https://www.annualreports.com/Click/21340
Pages 18, 20, 21, 35: https://www.modulairegroup.com/news/modulaire-group-h1-2025-update
Pages 20, 21, 35: https://www.modulairegroup.com/news/modulaire-group-pricing-announcement
Pages 18, 20, 21, 35:
  https://www.modulairegroup.com/news/modulaire-group-releases-first-quarter-2022-financial-report
Pages 11, 13, 35: https://www.ntsafety.com/
Pages 10, 11, 12, 13, 14, 34, 35:
  https://www.ntsafety.com/national-trench-safety-to-merge-with-trench-plate-rental-company/
Pages 1, 3, 4, 5, 6, 7, 27, 28, 33, 34: https://www.pods.com/aboutus
Pages 14, 15, 16, 17, 33, 34, 35:
  https://www.sec.gov/Archives/edgar/data/1342287/000119312520242828/d939869d10k.htm
Pages 4, 5, 6, 33, 35: https://www.spglobal.com/ratings/en/regulatory/article/-/view/type/HTML/id/3293453
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extract_source_text.py

tmp/two_pagers/extract_source_text.py

SHA-256: 7f27d969154cd778b60f552ee8df15f9306df9d6b3745de4aeefe0ec3f12f5d3

```
from pathlib import Path

from pypdf import PdfReader


ROOT = Path(__file__).resolve().parents[2]
SOURCE_DIR = ROOT / "output" / "pdf"
TEXT_DIR = ROOT / "tmp" / "two_pagers" / "source_text"
TEXT_DIR.mkdir(parents=True, exist_ok=True)

for source in sorted(SOURCE_DIR.glob("0*-gpt-5.6-sol-deep-research.pdf")):
    reader = PdfReader(str(source))
    text = "\n\n".join(page.extract_text() or "" for page in reader.pages)
    destination = TEXT_DIR / f"{source.stem}.txt"
    destination.write_text(text, encoding="utf-8")
    print(f"{destination.name}: {len(text):,} characters")
```

build_evidence_audit.py

tmp/two_pagers/build_evidence_audit.py

SHA-256: 052e6464496bab6dfb0cf80b3253af3c6fdbc0aecee3fab5480d3c2cc514dae8

```
from __future__ import annotations

import json
from pathlib import Path

import generate_two_pagers as briefs


OUTPUT = Path(__file__).resolve().parent / "evidence_audit.json"


METRIC_AUDIT: dict[str, dict[str, tuple[str, str, str]]] = {
    "home-comfort": {
        "Revenue growth": (
            "Medium",
            "Retain",
            "Enercare's disclosed 6% revenue growth with only 0.6% unit growth and Reliance's low-single-digit
              pre-acquisition customer growth bound a mature 5% normalized rate.",
        ),
        "EBITDA margin": (
            "Medium",
            "Retain",
            "Enercare's disclosed roughly 56% margin anchors the estimate; Reliance's broader HVAC and
              field-service mix supports a downward normalization to 53%.",
        ),
        "Maintenance capex / revenue": (
            "Medium",
            "Retain as modeled",
            "Asset exchanges, limited used-water-heater redeployment and 14-16-year lives support low-teens
              sustaining intensity, but neither company discloses a clean sustaining split.",
        ),
        "Customer tenor": (
            "Medium",
            "Retain as modeled",
            "Certain Ontario forms support 7-10-year initial terms; monthly continuation and low short-period
              attrition support an 18-year normalized effective-tenure assumption, not an observed cohort
              life.",
        ),
        "Utilization exposure": (
            "High",
            "Retain",
            "Installed paying accounts earn fixed rent independent of runtime; account vacancy, removal and
              service capacity are the relevant exposures.",
        ),
        "New-unit / customer payback": (
            "Medium",
            "Retain as triangulated",
            "Disclosed installed costs, monthly rents and HVAC revenue multiples bound the eight-year estimate,
              while undisclosed direct contribution prevents High confidence.",
        ),
        "B2B vs B2C": (
            "High",
            "Retain",
            "Enercare disclosed an approximately 95% residential water-heater portfolio and Reliance is
              primarily homeowner-facing.",
        ),
    },
    "point-of-use-water": {
        "Revenue growth": (
            "Medium",
            "Retain",
            "Quench disclosed 13.6% organic growth in 2019; 8% is a dated-result normalization that allows for
              current office exposure.",
        ),
        "EBITDA margin": (
            "Medium",
            "Retain",
            "The 28% estimate is directly anchored to Quench's $32.4 million EBITDA on $115.2 million of 2019
              revenue.",
        ),
        "Maintenance capex / revenue": (
            "Low",
            "Omit numeric estimate",
            "A 160,000-unit fleet proves physical capital, but filtration-only sustaining capex and
              replacement-cycle economics are not disclosed.",
        ),
        "Customer tenor": (
            "Medium",
            "Retain as modeled",
            "Month-to-month or flexible terms are disclosed; six years is a modeled effective-tenure assumption
              based on plumbing, recurring service and multi-unit relationships.",
        ),
```

```
        "Utilization exposure": (
            "Medium",
            "Retain",
            "Fixed billing reduces water-usage exposure, while returned equipment, office occupancy, route
              density and technician productivity create moderate asset/service exposure.",
        ),
        "New-unit / customer payback": (
            "Low",
            "Omit numeric estimate",
            "Units per account and monthly pricing cannot establish payback without installed cost, direct unit
              contribution and customer-acquisition cost.",
        ),
        "B2B vs B2C": (
            "High",
            "Retain",
            "Quench is explicitly workplace B2B and directs residential users elsewhere; Primo is a disclosed
              mixed boundary case.",
        ),
    },
    "submetering": {
        "Revenue growth": (
            "Medium",
            "Retain",
            "Building estimates normalize ista and Techem results; the meter-owner estimate is deliberately
              below rollout- and acquisition-inflated SMS growth.",
        ),
        "EBITDA margin": (
            "Medium",
            "Retain with perimeter caveat",
            "Building economics are bracketed by ista and Techem's group proxy; meter-owner economics are
              bracketed by SMS and Calisen, with replacement capital below EBITDA.",
        ),
        "Maintenance capex / revenue": (
            "Medium-Low",
            "Retain as modeled",
            "Device refresh supports the building estimate and owned-meter estates support heavier capital, but
              rollout and replacement spending are not cleanly separated.",
        ),
        "Customer tenor": (
            "Medium-Low",
            "Retain as modeled",
            "Initial contract disclosures are thin; installed-device persistence, billing integration and
              supplier-switch compensation support longer modeled effective tenure.",
        ),
        "Utilization exposure": (
            "Medium",
            "Retain",
            "Installed assets earn despite consumption variation, while meter-owner rollout and field-team
              productivity add operating exposure.",
        ),
        "New-unit / customer payback": (
            "Medium-Low",
            "Retain as modeled",
            "Building ranges use lower-cost device and onboarding economics; disclosed meter capex and annual
              rent support a bounded meter-owner payback.",
        ),
        "B2B vs B2C": (
            "High",
            "Retain",
            "Owners and managers contract for building services; suppliers and retailers contract with meter
              owners.",
        ),
    },
    "liquid-environmental-services": {
        "Revenue growth": (
            "Medium",
            "Retain",
            "Onsite growth is tied to installed-site wins, pricing and throughput; route growth is tied to
              density, cross-sell and treatment throughput after excluding acquisitions conceptually.",
        ),
        "EBITDA margin": (
            "Medium-Low",
            "Retain",
            "Historical LES credit commentary supports route margins; onsite RTI margin remains mechanism-based
              because no standalone margin is disclosed.",
        ),
        "Maintenance capex / revenue": (
            "Medium-Low",
            "Partially omit",
            "Onsite sustaining capex has Low support and is omitted. Observable route fleets and treatment
              plants support the retained 5% route estimate, though no clean sustaining split is disclosed.",
        ),
        "Customer tenor": (
            "Medium",
            "Partially omit",
            "Both undisclosed initial legal terms are omitted. Nine-year onsite and seven-year route values
              remain as modeled effective tenure based on embedding and recurring service needs.",
        ),
```

```
        "Utilization exposure": (
            "Medium",
            "Retain",
            "Installed sites and depots support onsite exposure; route stops, gallons, vehicles and treatment
              throughput support the higher route classification.",
        ),
        "New-unit / customer payback": (
            "Medium-Low",
            "Retain with scope caveat",
            "Onsite is triangulated from relationship economics and route applies only to an incremental account
               within an existing dense market; neither is a disclosed cohort payback.",
        ),
        "B2B vs B2C": (
            "High",
            "Retain",
            "RTI and the clean route processors serve commercial or institutional customers; Wind River is the
              explicit residential-commercial boundary case.",
        ),
    },
    "transportation-solutions": {
        "Revenue growth": (
            "Medium",
            "Retain",
            "Archetype estimates are tied to rail pricing/utilization, locomotive modernization, GSE
              outsourcing/electrification and yard-site wins.",
        ),
        "EBITDA margin": (
            "Medium",
            "Retain",
            "Akiem anchors locomotive economics; disclosed asset scale and service intensity support the
              relative ordering across GSE and labor-led yard operations.",
        ),
        "Maintenance capex / revenue": (
            "Medium",
            "Retain as modeled",
            "VTG investment, locomotive upgrades, GSE lifecycle service and yard's lighter asset base support
              relative ordering; private-company sustaining splits remain unavailable.",
        ),
        "Customer tenor": (
            "Medium",
            "Retain as modeled",
            "Asset value, approval burden and mobilization complexity support the terms, but the values are
              archetype assumptions rather than disclosed weighted-average cohorts.",
        ),
        "Utilization exposure": (
            "Medium",
            "Retain",
            "Rail deployment, GSE spare-fleet mechanics and Lazer labor-hour productivity support the
              classifications; the scalar confidence is Medium.",
        ),
        "New-unit / customer payback": (
            "Medium",
            "Retain as modeled",
            "Asset cost, service intensity, residual value and site-launch investment support bounded archetype
              paybacks; Air Rail's weaker company evidence is not used alone.",
        ),
        "B2B vs B2C": (
            "High",
            "Retain",
            "Institutional counterparties are explicit across rail, airport GSE and yard logistics.",
        ),
    },
    "specialty-rental": {
        "Revenue growth": (
            "Medium",
            "Partially omit",
            "PODS lacks a clean organic bridge, so consumer is omitted. Mobile Mini, modular operators and
              trench-safety evidence support the retained B2B, modular and trench values.",
        ),
        "EBITDA margin": (
            "Medium",
            "Partially omit",
            "PODS lacks a normalized standalone margin, so consumer is omitted. Mobile Mini directly anchors B2B
               storage; modular is disclosed and trench remains mechanism-based.",
        ),
        "Maintenance capex / revenue": (
            "Medium",
            "Partially omit",
            "PODS's container, delivery and yard sustaining burden is undisclosed, so consumer is omitted. Long
              lives and residuals support the retained archetypes.",
        ),
        "Customer tenor": (
            "Medium",
            "Partially omit",
            "PODS supports a one-month initial term but not an eight-month effective life; that component is
              omitted. Other archetypes have term evidence and modeled repeat tenure.",
        ),
        "Utilization exposure": (
```

```
            "Medium",
            "Partially omit",
            "PODS lacks comparable fleet, yard and route KPIs, so consumer is omitted. Mobile Mini, Modulaire
              and McGrath disclose utilization; trench is qualitatively project-sensitive.",
        ),
        "New-unit / customer payback": (
            "Medium",
            "Partially omit",
            "PODS acquisition and logistics contribution is undisclosed, so consumer is omitted. General Finance
               and McGrath evidence supports the retained durable-asset archetypes.",
        ),
        "B2B vs B2C": (
            "Medium",
            "Retain",
            "PODS is explicitly residential-commercial; the remaining archetypes serve commercial,
              institutional, government or utility customers.",
        ),
    },
}


BUSINESS_AUDIT: dict[str, dict[str, tuple[str, str, str]]] = {
    "home-comfort": {
        "Reliance Home Comfort": (
            "Medium-High",
            "Retain after scope wording correction",
            "Operating scale and services are directly sourced. Contract terms are limited to certain Ontario
              forms, pricing is illustrative, and redeployment is framed as an analytical mechanism.",
        ),
        "Enercare": (
            "High",
            "Retain after asset wording correction",
            "2017 and 9M 2018 filings directly support portfolio mix, revenue, EBITDA, attrition and exchanges;
              the redeployment conclusion is narrowed to used water heaters.",
        ),
    },
    "point-of-use-water": {
        "AquaVenture Holdings / Quench": (
            "Medium-High",
            "Retain with FY2019 date",
            "FY2019 filings support units, customers, revenue, EBITDA and growth. Current churn, replacement
              cycles and unit economics remain expressly unavailable.",
        ),
        "Primo Water": (
            "Medium",
            "Retain with mixed-business caveat",
            "Annual reports and current service terms support the offering, asset base and cancellation
              mechanics; filtration-only economics remain unavailable.",
        ),
    },
    "submetering": {
        "Energy Assets": (
            "Medium",
            "Retain with revenue-mix caveat",
            "Managed-asset scale and service breadth are sourced; the card no longer implies a quantified
              revenue mix and states the segment-data gap.",
        ),
        "ista": (
            "Medium-High",
            "Retain with capex label",
            "Corporate disclosures support device, unit, customer and sales scale; EUR260 million is labeled
              total investment, not maintenance capex.",
        ),
        "Techem": (
            "Medium-High",
            "Retain with perimeter correction",
            "Submetering revenue and group EBITDA are stated on separate, correctly labeled perimeters rather
              than combined into a false segment margin.",
        ),
        "Smart Metering Systems": (
            "Medium-High",
            "Retain with group-perimeter correction",
            "The 6 million-plus meter scale is attributed to the enlarged SMS / HEI / SMA group; the clean
              FY2022 base is separately labeled.",
        ),
        "Calisen": (
            "High",
            "Retain with capex label",
            "The annual report supports meters, revenue, EBITDA and rent; GBP355.5 million is labeled total
              capex, not sustaining capex.",
        ),
        "CARMA": (
            "Medium-High",
            "Retain with analytical churn wording",
            "Official sources support units, buildings, contracts and daily reads; churn mechanisms are
              identified as analytical rather than disclosed.",
        ),
    },
```

```
    "liquid-environmental-services": {
        "Restaurant Technologies": (
            "Medium-High",
            "Retain after unsupported phrase removal",
            "Operating scale and customer mix are sourced; the unsupported 'despite durable revenue' phrase is
              removed and data gaps remain explicit.",
        ),
        "Liquid Environmental Solutions": (
            "Medium-High",
            "Retain with dated-margin caveat",
            "Operating scale is sourced and the above-30% margin is explicitly labeled historical S&P credit
              commentary, not a current disclosure.",
        ),
        "Waste Resources Management": (
            "Medium-High",
            "Retain with comparability wording",
            "Scale and acquisition facts are sourced; the McDonald Farms combination is framed as a
              broader-service comparability issue rather than an unsupported purity conclusion.",
        ),
        "Wind River Environmental": (
            "High",
            "Retain with direct customer-mix wording",
            "Official pages support scale, plants and residential-commercial service lines; the card avoids an
              unsupported absolute mix claim.",
        ),
    },
    "transportation-solutions": {
        "VTG": (
            "Medium-High",
            "Retain with investment label",
            "2024 accounts support operating cash flow and total fixed-asset investment; maintenance is not
              presented as separately quantified.",
        ),
        "Akiem": (
            "Medium-High",
            "Retain with 2021 date",
            "Fleet and financial facts retain their 2021 date; current upgrade and refinancing facts are
              separately dated.",
        ),
        "Air Rail": (
            "Medium",
            "Retain with mixed-platform caveat",
            "Ownership and current GSE scale are sourced; the card retains the airport/rail perimeter caveat and
               does not present unsupported segment financials.",
        ),
        "TCR": (
            "Medium-High",
            "Retain",
            "Official company and sponsor sources support asset, workshop, employee, country and airport scale;
              operating economics are marked analyst interpretation.",
        ),
        "Lazer Logistics": (
            "Medium-High",
            "Retain without revenue-mix claim",
            "Official and sponsor sources support sites, employees, fleet and service hours; the card describes
              a labor-led operating unit without implying a disclosed revenue split.",
        ),
    },
    "specialty-rental": {
        "PODS": (
            "Medium",
            "Retain after factual correction",
            "Operating scale is sourced. The card now says sales fell 7.5% and adjusted EBITDA fell 24%, and it
              explicitly notes missing standalone fleet and capex data.",
        ),
        "Mobile Mini": (
            "High",
            "Retain",
            "The 2019 annual report directly supports revenue, margin, utilization, useful lives and residual
              values; sustaining-capital conclusions remain analytical.",
        ),
        "Trench Plate Rental": (
            "Medium-High",
            "Retain",
            "The merger announcement supports branch and customer scale; standalone economics are not stated as
              disclosed facts.",
        ),
        "General Finance Corporation": (
            "High",
            "Retain with utilization-period label",
            "The FY2020 filing supports revenue, useful lives and residuals; 70%-85% is labeled historical
              average fleet utilization.",
        ),
        "Modulaire Group": (
            "High",
            "Retain with 1Q22 date",
            "The card attaches the 1Q22 date to revenue, EBITDA and utilization, while current scale is
              separately stated.",
```

```
        ),
        "McGrath RentCorp": (
            "High",
            "Retain",
            "The 2025 filing directly supports segment utilization, lives and residual assumptions.",
        ),
    },
}


OMISSIONS = [
    "Point-of-Use Water: maintenance capex / revenue 13%.",
    "Point-of-Use Water: equipment payback 2.7 years.",
    "Point-of-Use Water: customer-acquisition-cost payback 1.0 year.",
    "Liquid Environmental Services: onsite maintenance capex / revenue 3.5%.",
    "Liquid Environmental Services: onsite initial contract term 4 years.",
    "Liquid Environmental Services: route initial contract term 2 years.",
    "Specialty Rental consumer/PODS: revenue growth 4%.",
    "Specialty Rental consumer/PODS: EBITDA margin 33%.",
    "Specialty Rental consumer/PODS: maintenance capex / revenue 7%.",
    "Specialty Rental consumer/PODS: effective customer tenure 8 months.",
    "Specialty Rental consumer/PODS: utilization exposure Moderate-High.",
    "Specialty Rental consumer/PODS: payback 3.5 years.",
]


def final_display(slug: str, metric: str) -> str:
    items = briefs.ESTIMATES[slug][metric]
    return "; ".join(f"{label}: {value}" if label else value for label, value in items)


def build() -> dict:
    reports = []
    for slug in briefs.ORDER:
        metrics = []
        for metric in briefs.REVISED_METRICS:
            support, decision, rationale = METRIC_AUDIT[slug][metric]
            metrics.append(
                {
                    "metric": metric,
                    "final_display": final_display(slug, metric),
                    "support": support,
                    "decision": decision,
                    "source": f"Deep Research {briefs.METRIC_SOURCES[slug][metric]}",
                    "audit_rationale": rationale,
                }
            )
        businesses = []
        for business in briefs.BUSINESS_ORDER[slug]:
            support, decision, rationale = BUSINESS_AUDIT[slug][business]
            businesses.append(
                {
                    "business": business,
                    "support": support,
                    "decision": decision,
                    "source": f"Deep Research {briefs.BUSINESS_SOURCES[slug][business]}",
                    "audit_rationale": rationale,
                }
            )
        reports.append(
            {
                "slug": slug,
                "title": briefs.DISPLAY_TITLES[slug],
                "lens_source": f"Deep Research {briefs.LENS_SOURCES[slug]}",
                "metrics": metrics,
                "businesses": businesses,
            }
        )
    return {
        "title": "Operating Benchmark Evidence Audit",
        "analysis_date": "2026-07-18",
        "policy": (
            "Every retained estimate has at least a disclosed anchor or bounded operating-mechanism support in
              the corresponding Deep Research report. "
            "Whenever a Page 1 estimate uses comparable-company data, the complete supporting datapoints are
              repeated in that company's Page 2 Reference data section. "
            "Any component explicitly rated Low is shown as Not estimated with a first-principles directional
              framework in the final brief. "
            "The concise briefs omit inline source callouts; this evidence matrix and the embedded research
              reports preserve the source mapping and underlying links. "
            "Subsector categories are an analytical taxonomy used to organize the business set."
        ),
        "reports": reports,
        "omissions": OMISSIONS,
    }


def validate(audit: dict) -> None:
```

```
    assert len(audit["reports"]) == 6
    assert len(audit["omissions"]) == 12
    for report in audit["reports"]:
        assert len(report["metrics"]) == 7, report["slug"]
        assert len(report["businesses"]) == len(briefs.BUSINESS_ORDER[report["slug"]])
        for metric in report["metrics"]:
            if metric["support"] == "Low":
                assert "Not estimated" in metric["final_display"], metric
            if metric["decision"] == "Partially omit":
                assert "Not est" in metric["final_display"], metric
            assert "Deep Research pp." in metric["source"], metric
        for business in report["businesses"]:
            assert "Deep Research pp." in business["source"], business


def main() -> None:
    audit = build()
    validate(audit)
    OUTPUT.write_text(json.dumps(audit, indent=2), encoding="utf-8")
    print(OUTPUT)


if __name__ == "__main__":
    main()
```

generate_two_pagers.py

tmp/two_pagers/generate_two_pagers.py

SHA-256: 785e4fba4d7b98d8f6f71f9418379e2bedc65dedbf52627e6b69f47ad897b453

```
from __future__ import annotations

import html
import json
import math
import re
from pathlib import Path

from pypdf import PdfReader, PdfWriter
from reportlab.lib import colors
from reportlab.lib.enums import TA_CENTER, TA_LEFT, TA_RIGHT
from reportlab.lib.pagesizes import letter, landscape
from reportlab.lib.styles import ParagraphStyle
from reportlab.pdfbase import pdfmetrics
from reportlab.pdfbase.ttfonts import TTFont
from reportlab.pdfgen import canvas
from reportlab.platypus import Paragraph, Table, TableStyle


ROOT = Path(__file__).resolve().parents[2]
DRAFT_DIR = ROOT / "tmp" / "two_pagers" / "drafts"
OUTPUT_DIR = ROOT / "output" / "pdf" / "two-pagers"
OUTPUT_DIR.mkdir(parents=True, exist_ok=True)

ORDER = [
    "home-comfort",
    "point-of-use-water",
    "submetering",
    "liquid-environmental-services",
    "transportation-solutions",
    "specialty-rental",
]

FONT_DIR = Path("/System/Library/Fonts/Supplemental")
pdfmetrics.registerFont(TTFont("Arial", str(FONT_DIR / "Arial.ttf")))
pdfmetrics.registerFont(TTFont("Arial-Bold", str(FONT_DIR / "Arial Bold.ttf")))
pdfmetrics.registerFont(TTFont("Arial-Italic", str(FONT_DIR / "Arial Italic.ttf")))
pdfmetrics.registerFont(
    TTFont("Arial-BoldItalic", str(FONT_DIR / "Arial Bold Italic.ttf"))
)
pdfmetrics.registerFontFamily(
    "Arial",
    normal="Arial",
    bold="Arial-Bold",
    italic="Arial-Italic",
    boldItalic="Arial-BoldItalic",
)

PAGE_W, PAGE_H = landscape(letter)
MARGIN_X = 42
FOOTER_Y = 17

# Exact palette from the weekly email briefing.
PLUM = colors.HexColor("#442142")
GOLD = colors.HexColor("#B4A87D")
MASTHEAD = colors.HexColor("#F9F8FA")
CARD = colors.HexColor("#FDFCFB")
SECTION = colors.HexColor("#F8F9FA")
BORDER = colors.HexColor("#E5E7EB")
EDGE = colors.HexColor("#D1D5DB")
BODY = colors.HexColor("#3F3F46")
META = colors.HexColor("#52525B")
MICRO = colors.HexColor("#4B5563")
SOURCE = colors.HexColor("#5B5563")
QUIET = colors.HexColor("#F4F5F7")
WHITE = colors.white


# One simple central estimate for each metric. Heterogeneous subverticals retain
# separate archetype estimates instead of presenting a misleading blend.
REVISED_METRICS = [
    "Revenue growth",
    "EBITDA margin",
    "Maintenance capex / revenue",
    "Customer tenor",
    "Utilization exposure",
    "New-unit / customer payback",
    "B2B vs B2C",
]

DISPLAY_TITLES = {
    "home-comfort": "Home Comfort Operating Benchmark",
```

```
    "point-of-use-water": "Point-of-Use Water Operating Benchmark",
    "submetering": "Submetering Operating Benchmark",
    "liquid-environmental-services": "Liquid Environmental Services Operating Benchmark",
    "transportation-solutions": "Transportation Solutions Operating Benchmark",
    "specialty-rental": "Specialty Rental Operating Benchmark",
}

ESTIMATES: dict[str, dict[str, list[tuple[str, str]]]] = {
    "home-comfort": {
        "Revenue growth": [("", "5%")],
        "EBITDA margin": [("", "53%")],
        "Maintenance capex / revenue": [("", "12%")],
        "Customer tenor": [("Initial / effective", "7-10 / 18 yrs")],
        "Utilization exposure": [("", "Low-Moderate")],
        "New-unit / customer payback": [("", "8 yrs")],
        "B2B vs B2C": [("", "Predominantly B2C")],
    },
    "point-of-use-water": {
        "Revenue growth": [("", "8%")],
        "EBITDA margin": [("", "28%")],
        "Maintenance capex / revenue": [("", "Not estimated")],
        "Customer tenor": [("Initial / effective", "1-12 mos / 6 yrs")],
        "Utilization exposure": [("", "Moderate overall")],
        "New-unit / customer payback": [("", "Not estimated")],
        "B2B vs B2C": [("", "Predominantly B2B")],
    },
    "submetering": {
        "Revenue growth": [("Building", "6%"), ("Meter owner", "7%")],
        "EBITDA margin": [("Building", "47%"), ("Meter owner", "60%")],
        "Maintenance capex / revenue": [("Building", "9%"), ("Meter owner", "30%")],
        "Customer tenor": [("Building", "8 / 14 yrs"), ("Meter owner", "3 / 13 yrs")],
        "Utilization exposure": [("Building", "Low"), ("Meter owner", "Moderate")],
        "New-unit / customer payback": [("Building", "4.0 yrs"), ("Meter owner", "6.0 yrs")],
        "B2B vs B2C": [
            ("Building", "B2B; residential end users"),
            ("Meter owner", "Predominantly B2B"),
        ],
    },
    "liquid-environmental-services": {
        "Revenue growth": [("Onsite", "8%"), ("Route", "9%")],
        "EBITDA margin": [("Onsite", "21%"), ("Route", "28%")],
        "Maintenance capex / revenue": [("Onsite", "Not estimated"), ("Route", "5.0%")],
        "Customer tenor": [("Onsite", "Not est. / 9 yrs"), ("Route", "Not est. / 7 yrs")],
        "Utilization exposure": [("Onsite", "Moderate"), ("Route", "Moderate-High")],
        "New-unit / customer payback": [("Onsite", "2.5 yrs"), ("Route", "1.25 yrs")],
        "B2B vs B2C": [("Onsite", "Predominantly B2B"), ("Route", "Predominantly B2B")],
    },
    "transportation-solutions": {
        "Revenue growth": [("Railcar", "4%"), ("Loco", "6%"), ("GSE", "8%"), ("Yard", "7%")],
        "EBITDA margin": [("Railcar", "58%"), ("Loco", "62%"), ("GSE", "30%"), ("Yard", "14%")],
        "Maintenance capex / revenue": [("Railcar", "18%"), ("Loco", "18%"), ("GSE", "14%"), ("Yard", "4%")],
        "Customer tenor": [("Railcar", "4 / 9 yrs"), ("Loco", "7 / 12 yrs"), ("GSE", "5 / 9 yrs"), ("Yard", "3 /
           7 yrs")],
        "Utilization exposure": [("Railcar", "Moderate"), ("Loco", "Moderate"), ("GSE", "Moderate-High"),
          ("Yard", "High")],
        "New-unit / customer payback": [("Railcar", "9 yrs"), ("Loco", "10 yrs"), ("GSE", "5.5 yrs"), ("Yard",
          "1.5 yrs")],
        "B2B vs B2C": [("All models", "Predominantly B2B")],
    },
    "specialty-rental": {
        "Revenue growth": [("Consumer", "Not estimated"), ("B2B storage", "5%"), ("Modular", "6%"), ("Trench",
          "6%")],
        "EBITDA margin": [("Consumer", "Not estimated"), ("B2B storage", "40%"), ("Modular", "34%"), ("Trench",
          "33%")],
        "Maintenance capex / revenue": [("Consumer", "Not estimated"), ("B2B storage", "6%"), ("Modular", "8%"),
           ("Trench", "7%")],
        "Customer tenor": [("Consumer", "1 mo / Not est."), ("B2B storage", "3 mos / 4 yrs"), ("Modular", "18
          mos / 7 yrs"), ("Trench", "2 mos / 5 yrs")],
        "Utilization exposure": [("Consumer", "Not estimated"), ("B2B storage", "Moderate"), ("Modular",
          "Moderate"), ("Trench", "High")],
        "New-unit / customer payback": [("Consumer", "Not estimated"), ("B2B storage", "4.5 yrs"), ("Modular",
          "6.0 yrs"), ("Trench", "3.5 yrs")],
        "B2B vs B2C": [
            ("Consumer", "Mixed; B2C lean"),
            ("B2B storage", "Predominantly B2B"),
            ("Modular", "Predominantly B2B"),
            ("Trench", "Predominantly B2B"),
        ],
    },
}


THESIS_SUMMARIES = {
    "home-comfort": "Installed residential equipment and protection plans convert essential household needs into
       recurring monthly billing. Pricing, premium-equipment mix and cross-sell support growth, while equipment
      exchange and reinstallation remain the central sustaining-capital burden.",
    "point-of-use-water": "The core archetype is Quench's workplace point-of-use model: installed,
      provider-owned filtration equipment for a recurring fee. Route density and recurring billing support
```

```
      margins, while equipment ownership, service intensity and customer-site turnover constrain conversion and
      tenure.",
    "submetering": "Building billing platforms and meter-asset owners are economically distinct. Billing
      platforms earn recurring data and compliance fees with moderate reinvestment; meter owners report higher
      EBITDA but require materially heavier replacement capital and longer payback.",
    "liquid-environmental-services": "Two models should remain separate: installed onsite oil systems and
      route-based collection, treatment and processing. Onsite embedding supports tenure; route density and
      owned treatment support margin, but increase fleet and plant utilization exposure.",
    "transportation-solutions": "Four models require separate estimates: railcar leasing, locomotive leasing,
      airport GSE leasing and outsourced yard logistics. Asset lessors earn higher margins with heavier
      reinvestment; yard logistics is labor-led, lower-margin and much lighter-capital.",
    "specialty-rental": "Consumer storage, B2B storage, modular space and trench safety differ in rental
      duration, service intensity and capital recovery. Simple standardized assets generally support stronger
      margins and conversion, while modular and project-driven fleets carry greater reset or utilization risk.",
}


UNIVERSE_SUMMARIES = {
    "home-comfort": "Both businesses monetize installed residential equipment and essential service continuity.
      The evidence bridge focuses on recurring rental and protection-plan economics, not one-time HVAC sales or
      service work.",
    "point-of-use-water": "Point-of-use providers replace bottle logistics with installed filtration, recurring
      service and route density. Quench is the clean workplace B2B anchor; Primo remains a mixed reference
      because filtration is combined with other water channels.",
    "submetering": "The six companies span building billing, meter-asset ownership and a mixed metering
      platform. Their profiles show why similar recurring-revenue labels conceal different field-service,
      replacement-capital and contract economics.",
    "liquid-environmental-services": "All four businesses solve recurring oil or wastewater needs, but the
      bottleneck differs: installed customer-site systems versus dense routes and permitted treatment capacity.
      That distinction drives the estimates.",
    "transportation-solutions": "Five businesses map to four models: railcar, locomotive, airport GSE and yard
      logistics. The economic unit shifts from an earning asset, to a station fleet package, to a site contract
      and productive service hour.",
    "specialty-rental": "Six resolved operators map to consumer storage, B2B storage, modular space and trench
      safety. Reliant Rental remains unresolved and is explicitly excluded from the estimates.",
}


SUBSECTOR_CATEGORIES: dict[str, list[tuple[str, list[str]]]] = {
    "home-comfort": [
        ("Diversified home comfort", ["Reliance Home Comfort"]),
        ("Rental-led equipment & plans", ["Enercare"]),
    ],
    "point-of-use-water": [
        ("Workplace POU filtration", ["AquaVenture Holdings / Quench"]),
        ("Diversified water platform", ["Primo Water"]),
    ],
    "submetering": [
        ("Building submetering & billing", ["ista", "Techem", "CARMA"]),
        ("Meter asset ownership", ["Smart Metering Systems", "Calisen"]),
        ("Mixed metering & utility services", ["Energy Assets"]),
    ],
    "liquid-environmental-services": [
        ("Installed onsite oil systems", ["Restaurant Technologies"]),
        ("Commercial liquid-waste routes", ["Liquid Environmental Solutions", "Waste Resources Management"]),
        ("Diversified wastewater services", ["Wind River Environmental"]),
    ],
    "transportation-solutions": [
        ("Railcar leasing", ["VTG"]),
        ("Locomotive leasing", ["Akiem"]),
        ("Airport GSE leasing", ["Air Rail", "TCR"]),
        ("Yard logistics", ["Lazer Logistics"]),
    ],
    "specialty-rental": [
        ("Consumer moving & storage", ["PODS"]),
        ("B2B portable storage", ["Mobile Mini", "General Finance Corporation"]),
        ("Modular space", ["Modulaire Group", "McGrath RentCorp"]),
        ("Trench safety rental", ["Trench Plate Rental"]),
    ],
}


BUSINESS_ORDER = {
    "home-comfort": ["Reliance Home Comfort", "Enercare"],
    "point-of-use-water": ["AquaVenture Holdings / Quench", "Primo Water"],
    "submetering": ["ista", "Techem", "CARMA", "Smart Metering Systems", "Calisen", "Energy Assets"],
    "liquid-environmental-services": ["Restaurant Technologies", "Liquid Environmental Solutions", "Waste
      Resources Management", "Wind River Environmental"],
    "transportation-solutions": ["VTG", "Akiem", "Lazer Logistics", "Air Rail", "TCR"],
    "specialty-rental": ["PODS", "Mobile Mini", "General Finance Corporation", "Modulaire Group", "McGrath
      RentCorp", "Trench Plate Rental"],
}


BUSINESS_CATEGORY = {
    slug: {
        company: category
        for category, companies in category_groups
```

```
        for company in companies
    }
    for slug, category_groups in SUBSECTOR_CATEGORIES.items()
}


ASSERTIONS: dict[str, dict[str, tuple[str, str]]] = {
    "home-comfort": {
        "Revenue growth": (
            "A 5% estimate sits below Enercare's 6% revenue growth in 9M 2018, when units grew only 0.6% and
              pricing and mix did most of the work. Reliance's low-single-digit customer growth before
              acquisitions also supports a mature organic rate below the disclosed Enercare result.",
            "Medium",
        ),
        "EBITDA margin": (
            "A 53% estimate sits just below Enercare's disclosed 56% margin. Recurring water-heater rentals
              carry strong contribution, while Reliance's broader HVAC, technician, dispatch and
              transactional-service mix should dilute the pure rental result.",
            "Medium",
        ),
        "Maintenance capex / revenue": (
            "The 12% estimate reflects recurring equipment replacement, removal and reinstallation: Enercare
              exchanged 33,600 assets in 9M 2018 and found used water heaters uneconomic to redeploy, while
              Reliance cites 14-16-year equipment lives. Neither company provides a clean sustaining-capex
              split.",
            "Medium",
        ),
        "Customer tenor": (
            "Certain Reliance Ontario forms support a 7-10-year initial term. The 18-year effective tenure is a
              normalized assumption because billing can continue monthly, equipment can be replaced at the same
              address and Enercare reported 1.9% rental attrition over nine months; it is not an observed cohort
               life.",
            "Medium",
        ),
        "Utilization exposure": (
            "Exposure is Low-Moderate because an installed unit earns fixed monthly rent regardless of household
               usage. The residual risk is account vacancy, equipment removal and seasonal technician or
              installer capacity rather than consumption volume.",
            "High",
        ),
        "New-unit / customer payback": (
            "An eight-year estimate triangulates Reliance's C$1,000-C$5,500 installed cost against
              C$20-C$100-plus monthly rent, then allows for installation, service and replacement reserves.
              Direct unit contribution is undisclosed.",
            "Medium",
        ),
        "B2B vs B2C": (
            "The model is predominantly B2C because roughly 95% of Enercare's rental portfolio was residential
              water heaters and Reliance primarily markets to homeowners. Builders and property managers may
              originate accounts, but households remain the economic center.",
            "High",
        ),
    },
    "point-of-use-water": {
        "Revenue growth": (
            "The 8% estimate sits below Quench's 13.6% organic growth in 2019 because that result is dated and
              current workplace demand is more exposed to office occupancy. Primo's Refill/Filtration growth is
              not a clean filtration-only measure.",
            "Medium",
        ),
        "EBITDA margin": (
            "The 28% estimate is anchored directly to Quench's 2019 result of $32.4 million of EBITDA on $115.2
              million of revenue. Recurring fees support the margin, while technicians, travel, filters,
              sanitation and repairs constrain it.",
            "Medium",
        ),
        "Maintenance capex / revenue": (
            "Sustaining capital should rise with provider-owned unit age, filter and sanitation frequency,
              service-call intensity and a dispersed route footprint. It should fall with standardized long-life
               equipment, dense routes, high reuse after redeployment and customer-funded site work.",
            "Low",
        ),
        "Customer tenor": (
            "The short initial term reflects Quench's primarily month-to-month rentals and Primo's flexible
              cancellation language. Six years is a modeled effective-tenure assumption based on plumbing,
              recurring service and multi-unit relationships; customer churn cohorts are unavailable.",
            "Medium",
        ),
        "Utilization exposure": (
            "Exposure is Moderate because fixed fees decouple revenue from water consumption, but a returned
              machine earns nothing until redeployed. Office occupancy, route density and productivity across
              Quench's 500-plus technicians remain meaningful variables.",
            "Medium",
        ),
        "New-unit / customer payback": (
            "Payback should shorten when recurring fees and units per customer are high relative to equipment,
              installation and acquisition cost, especially within dense routes and long-lived accounts. It
              should lengthen with plumbing complexity, frequent service, customer churn, idle redeployment time
```

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                "and low technician density.",
            "Low",
        ),
        "B2B vs B2C": (
            "The core model is predominantly B2B because Quench explicitly serves workplaces and directs
              residential users elsewhere. Primo broadens the universe into homes and other channels, but its
              exact filtration customer mix is undisclosed.",
            "High",
        ),
    },
    "submetering": {
        "Revenue growth": (
            "The 6% building-platform estimate is anchored near ista's 5.2% growth and normalizes Techem's
              stronger result. The 7% meter-owner estimate is deliberately below SMS's 13% ILARR growth because
              funded rollout and acquisitions inflated reported rates.",
            "Medium",
        ),
        "EBITDA margin": (
            "The 47% building estimate uses Techem's roughly 48% group-level margin and ista's 40%-plus
              evidence; Techem's submetering revenue is disclosed separately, so this is a normalized proxy
              rather than a segment margin. The 60% meter-owner estimate sits between SMS and Calisen.",
            "Medium",
        ),
        "Maintenance capex / revenue": (
            "The 9% building-platform estimate reflects device, battery, communications and IT refresh. The 30%
              meter-owner estimate reflects the heavier replacement capital required by owned estates; rollout
              and growth capital are not cleanly separated from sustaining spend.",
            "Medium-Low",
        ),
        "Customer tenor": (
            "These are modeled terms, not disclosed weighted-average cohorts. Building installation and data
              migration support an 8-year initial and 14-year effective relationship; meter-owner legal forms
              can be short, while compensation can survive supplier switching for roughly 13 years.",
            "Medium-Low",
        ),
        "Utilization exposure": (
            "Building-platform exposure is Low because installed meters continue earning regardless of daily
              consumption. Meter-owner exposure is Moderate because installation throughput, supplier rollout
              decisions, meter removals and field-team productivity still matter.",
            "Medium",
        ),
        "New-unit / customer payback": (
            "The modeled four-year building estimate centers the lower-cost device, billing and data cases. The
              modeled six-year meter-owner estimate is bounded by disclosed meter capex and annual rent evidence
               across SMS and Calisen.",
            "Medium-Low",
        ),
        "B2B vs B2C": (
            "Building owners and managers contract for submetering even when residents consume the utility and
              receive bills. Meter owners contract with energy suppliers or retailers, making both archetypes
              economically B2B.",
            "High",
        ),
    },
    "liquid-environmental-services": {
        "Revenue growth": (
            "The 8% onsite estimate reflects installed-kitchen wins, pricing and throughput across RTI's
              45,000-plus sites. The 9% route estimate reflects modestly greater upside from density, cross-sell
               and treatment throughput.",
            "Medium",
        ),
        "EBITDA margin": (
            "The 21% onsite estimate is mechanism-based because RTI has no standalone margin disclosure. The 28%
               route estimate is set below historical S&P commentary citing LES margins above 30%, allowing for
              dated evidence, mixed services and third-party disposal.",
            "Medium-Low",
        ),
        "Maintenance capex / revenue": (
            "Onsite sustaining capital should rise with provider ownership of tanks, pumps, filters and
              monitoring hardware, shorter replacement cycles and a dispersed installed base. It should fall
              with long-lived standardized systems, customer-funded installation and dense service routes; route
               processing remains more exposed to truck and treatment-plant renewal, supporting the retained 5%
              estimate.",
            "Medium-Low",
        ),
        "Customer tenor": (
            "Onsite contracts should be longer when the provider funds embedded equipment and bundles
              monitoring, compliance and maintenance, and shorter when customers own equipment or can switch
              without removal cost. Route contracts should be longer with dedicated capacity and bundled
              treatment, but shorter where work is bid per pickup; effective tenure tends to exceed the legal
              term when reliability and compliance make switching disruptive.",
            "Medium",
        ),
        "Utilization exposure": (
            "Onsite exposure is Moderate because deployed equipment remains in place, but gallons per kitchen
              and depot density affect economics. Route exposure is Moderate-High because stops per route, truck
               productivity and plant throughput determine contribution.",
            "Medium",
```

```
        ),
        "New-unit / customer payback": (
            "The 2.5-year onsite estimate allows recovery of hardware, installation and route support from the
              full customer relationship. The 1.25-year route estimate applies only to an incremental customer
              added within an existing dense market.",
            "Medium-Low",
        ),
        "B2B vs B2C": (
            "Both archetypes are predominantly B2B because RTI serves commercial kitchens and institutions,
              while LES and WRM focus on commercial and industrial generators. Wind River's residential exposure
               does not change the route-processing center.",
            "High",
        ),
    },
    "transportation-solutions": {
        "Revenue growth": (
            "Railcar growth is held to 4% because mature fleets depend mainly on rate, utilization and measured
              additions; locomotives receive 6% for modernization demand. GSE and yard services receive 8% and
              7% for outsourcing, electrification and new-site wins.",
            "Medium",
        ),
        "EBITDA margin": (
            "Railcar and locomotive estimates remain high because lease revenue sits on long-lived assets; the
              62% locomotive estimate is normalized below Akiem's roughly 68% result. GSE carries workshop
              costs, while yard services are labor-led.",
            "Medium",
        ),
        "Maintenance capex / revenue": (
            "Railcar and locomotive estimates reflect overhaul, replacement, compliance and ERTMS work; GSE's
              14% reflects refurbishment, batteries and workshop support. Yard services require only tractors
              and technology at 4%.",
            "Medium",
        ),
        "Customer tenor": (
            "Initial terms lengthen with asset value, approval burden and mobilization complexity, making
              locomotives longest and yard contracts shortest. Effective tenure is longer because fleets,
              airport packages and embedded yard operations are commonly renewed or refreshed.",
            "Medium",
        ),
        "Utilization exposure": (
            "Rail exposure is Moderate because leased deployment matters but assets can be redeployed across
              long economic lives. GSE adds station activity and spare-fleet risk, while yard services are High
              because contribution depends on productive labor hours.",
            "Medium",
        ),
        "New-unit / customer payback": (
            "Rail assets receive nine-to-ten-year paybacks because high upfront cost must be recovered over
              multiple placements and residual value matters. GSE is cheaper but service-heavy at 5.5 years,
              while yard launch spend can recover in about 1.5 years.",
            "Medium",
        ),
        "B2B vs B2C": (
            "All four models are predominantly B2B because contractual buyers are rail operators, public
              authorities, airports, ground handlers, industrial firms and logistics providers. The benchmarked
              operations have no meaningful direct consumer revenue.",
            "High",
        ),
    },
    "specialty-rental": {
        "Revenue growth": (
            "Consumer storage growth should rise with housing turnover, renovation, household dislocation,
              pricing and local container capacity; slower moves, higher acquisition cost and delivery
              constraints reduce it. Retained B2B, modular and trench estimates reflect rate, fleet additions
              and project demand.",
            "Medium",
        ),
        "EBITDA margin": (
            "Consumer margins should rise with container occupancy, rental extensions, route density and pricing
               that covers delivery; paid acquisition, empty miles, repeated handling, yard expense and damage
              reduce them. Retained B2B margin is anchored to Mobile Mini, with modular and trench lower because
               installation, engineering and hauling add cost.",
            "Medium",
        ),
        "Maintenance capex / revenue": (
            "Consumer sustaining capital should rise with asset age, damage, corrosion, road miles and
              repositioning; durable standardized containers, disciplined refurbishment and high recovery values
               reduce it. Retained B2B, modular and trench values reflect long asset lives, residuals and
              differing reset needs.",
            "Medium",
        ),
        "Customer tenor": (
            "PODS supports a one-month initial term. Effective consumer tenure should lengthen with uncertain
              move dates, extended renovations and passive monthly renewal, and shorten with defined moves,
              price pressure or easy substitution. B2B storage, modular and trench relationships persist through
               projects and repeat use.",
            "Medium",
        ),
        "Utilization exposure": (
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            "Consumer utilization exposure should rise with fixed local fleets, seasonal moving demand, depot
              imbalance and limited redeployment, and fall with broad coverage, flexible transfers and balanced
              demand. B2B storage and modular are Moderate; trench is High because inventory rolls off with
              projects.",
            "Medium",
        ),
        "New-unit / customer payback": (
            "Consumer payback should shorten with high occupancy, longer rentals, strong delivery economics and
              repeated reuse, and lengthen with acquisition cost, empty transport, yard expense, damage and idle
               time. Retained B2B, modular and trench values use General Finance's four-year revenue recovery
              evidence and allow for cost, maintenance and residual value.",
            "Medium",
        ),
        "B2B vs B2C": (
            "Consumer storage is mixed with a B2C lean because household moves are a material demand source.
              Storage, modular and trench customers are principally contractors, companies, schools,
              institutions, government bodies and utilities.",
            "Medium",
        ),
    },
}


# Page maps feed the evidence audit and source compendium. The concise briefs
# intentionally omit inline references while the embedded research PDFs retain
# the underlying primary-source hyperlinks.
LENS_SOURCES = {
    "home-comfort": "pp. 23-28",
    "point-of-use-water": "pp. 15-19",
    "submetering": "pp. 31-37",
    "liquid-environmental-services": "pp. 25-32",
    "transportation-solutions": "pp. 31-39",
    "specialty-rental": "pp. 28-34",
}

METRIC_SOURCES: dict[str, dict[str, str]] = {
    "home-comfort": {
        "Revenue growth": "pp. 7, 18, 24, 26",
        "EBITDA margin": "pp. 18-19, 24, 26",
        "Maintenance capex / revenue": "pp. 10, 19, 24, 26",
        "Customer tenor": "pp. 8, 15, 25, 27",
        "Utilization exposure": "pp. 25, 27",
        "New-unit / customer payback": "pp. 8, 11, 21, 25, 27",
        "B2B vs B2C": "pp. 13, 15, 25, 27",
    },
    "point-of-use-water": {
        "Revenue growth": "pp. 5, 15-16",
        "EBITDA margin": "pp. 5, 15-16",
        "Maintenance capex / revenue": "pp. 1-2, 15, 17",
        "Customer tenor": "pp. 2, 6-7, 15, 17",
        "Utilization exposure": "pp. 2, 6, 16-17",
        "New-unit / customer payback": "pp. 2, 13, 16-17",
        "B2B vs B2C": "pp. 1-2, 16-17",
    },
    "submetering": {
        "Revenue growth": "pp. 9, 13, 18, 21, 31-32, 36",
        "EBITDA margin": "pp. 13, 18-22, 31, 33-34, 36",
        "Maintenance capex / revenue": "pp. 11, 22-23, 31, 33, 36",
        "Customer tenor": "pp. 31, 33, 35-36",
        "Utilization exposure": "pp. 32-33, 35-36",
        "New-unit / customer payback": "pp. 22-23, 32, 34-36",
        "B2B vs B2C": "pp. 32, 34, 36",
    },
    "liquid-environmental-services": {
        "Revenue growth": "pp. 4-5, 9-11, 25, 27",
        "EBITDA margin": "pp. 5, 10-11, 26, 28",
        "Maintenance capex / revenue": "pp. 4-5, 9-11, 20-22, 26, 28",
        "Customer tenor": "pp. 5, 26, 28",
        "Utilization exposure": "pp. 4-5, 9-11, 20-22, 26, 29",
        "New-unit / customer payback": "pp. 4, 6, 27, 29",
        "B2B vs B2C": "pp. 27, 29-30",
    },
    "transportation-solutions": {
        "Revenue growth": "pp. 31-35",
        "EBITDA margin": "pp. 10, 12, 21, 23, 25-29, 31, 33-36",
        "Maintenance capex / revenue": "pp. 4, 7, 10, 13-15, 31, 33-36",
        "Customer tenor": "pp. 32-36",
        "Utilization exposure": "pp. 32-36",
        "New-unit / customer payback": "pp. 32-36",
        "B2B vs B2C": "pp. 32-36",
    },
    "specialty-rental": {
        "Revenue growth": "pp. 9, 28-30, 32",
        "EBITDA margin": "pp. 9, 28-32",
        "Maintenance capex / revenue": "pp. 8-10, 22, 24, 28-32",
        "Customer tenor": "pp. 28-32",
        "Utilization exposure": "pp. 8, 10, 18, 20-24, 29-32",
        "New-unit / customer payback": "pp. 15, 17, 29-32",
```

```
        "B2B vs B2C": "pp. 29-32",
    },
}

BUSINESS_SOURCES: dict[str, dict[str, str]] = {
    "home-comfort": {
        "Reliance Home Comfort": "pp. 1-11",
        "Enercare": "pp. 12-23",
    },
    "point-of-use-water": {
        "AquaVenture Holdings / Quench": "pp. 1-7",
        "Primo Water": "pp. 8-15",
    },
    "submetering": {
        "Energy Assets": "pp. 3-7",
        "ista": "pp. 8-11",
        "Techem": "pp. 12-16",
        "Smart Metering Systems": "pp. 17-20",
        "Calisen": "pp. 21-25",
        "CARMA": "pp. 26-29",
    },
    "liquid-environmental-services": {
        "Restaurant Technologies": "pp. 3-8",
        "Liquid Environmental Solutions": "pp. 9-13",
        "Waste Resources Management": "pp. 14-19",
        "Wind River Environmental": "pp. 20-24",
    },
    "transportation-solutions": {
        "VTG": "pp. 3-9",
        "Akiem": "pp. 9-15",
        "Air Rail": "pp. 15-20",
        "TCR": "pp. 20-25",
        "Lazer Logistics": "pp. 25-30",
    },
    "specialty-rental": {
        "PODS": "pp. 3-7",
        "Mobile Mini": "pp. 7-10",
        "Trench Plate Rental": "pp. 10-14",
        "General Finance Corporation": "pp. 14-17",
        "Modulaire Group": "pp. 17-21",
        "McGrath RentCorp": "pp. 21-25",
    },
}


def clean(value: object) -> str:
    text = "" if value is None else str(value)
    replacements = {
        "\u2010": "-",
        "\u2011": "-",
        "\u2012": "-",
        "\u2013": "-",
        "\u2014": "-",
        "\u2212": "-",
        "\u00a0": " ",
        "\u2022": ";",
        "\u2192": "to",
        "\u2026": "...",
    }
    for source, target in replacements.items():
        text = text.replace(source, target)
    return re.sub(r"\s+", " ", text).strip()


def h(value: object) -> str:
    return html.escape(clean(value), quote=False)


def compact(value: object, max_chars: int) -> str:
    text = clean(value)
    if len(text) <= max_chars:
        return text
    sentences = re.split(r"(?<=[.!?])\s+", text)
    selected: list[str] = []
    for sentence in sentences:
        candidate = " ".join([*selected, sentence]).strip()
        if len(candidate) <= max_chars:
            selected.append(sentence)
        else:
            break
    if selected:
        return " ".join(selected)
    clipped = text[: max_chars - 3].rsplit(" ", 1)[0]
    return clipped.rstrip(" ,;:") + "..."


def paragraph_style(
    name: str,
```

```
    font_size: float,
    leading: float,
    *,
    color: colors.Color = BODY,
    font: str = "Arial",
    alignment: int = TA_LEFT,
) -> ParagraphStyle:
    return ParagraphStyle(
        name,
        fontName=font,
        fontSize=font_size,
        leading=leading,
        textColor=color,
        alignment=alignment,
        allowWidows=0,
        allowOrphans=0,
    )


def draw_tracked_text(
    c: canvas.Canvas,
    x: float,
    y: float,
    text: str,
    *,
    font: str,
    size: float,
    color: colors.Color,
    tracking: float,
) -> None:
    text_object = c.beginText(x, y)
    text_object.setFont(font, size)
    text_object.setFillColor(color)
    text_object.setCharSpace(tracking)
    text_object.textOut(clean(text))
    # Character spacing is part of the PDF text state. Reset it explicitly so
    # later Paragraph and Table text uses the widths ReportLab calculated.
    text_object.setCharSpace(0)
    c.drawText(text_object)


def draw_footer(c: canvas.Canvas, page_number: int, title: str) -> None:
    c.setStrokeColor(BORDER)
    c.setLineWidth(0.6)
    c.line(MARGIN_X, 28, PAGE_W - MARGIN_X, 28)
    c.setFillColor(SOURCE)
    c.setFont("Arial", 6.25)
    descriptor = (
        "Directional framework shown where a point estimate is unsupported."
        if page_number == 1
        else "Business profiles grouped by subsector category."
    )
    c.drawString(
        MARGIN_X,
        FOOTER_Y,
        descriptor,
    )
    c.drawRightString(
        PAGE_W - MARGIN_X,
        FOOTER_Y,
        f"Page {page_number} of 2",
    )


def draw_header(c: canvas.Canvas, title: str, page_label: str) -> None:
    c.setFillColor(WHITE)
    c.rect(0, 0, PAGE_W, PAGE_H, fill=1, stroke=0)
    c.setFillColor(PLUM)
    c.rect(0, PAGE_H - 69, PAGE_W, 69, fill=1, stroke=0)
    c.setFillColor(GOLD)
    c.rect(0, PAGE_H - 69, 7, 69, fill=1, stroke=0)
    c.setFillColor(WHITE)
    title_size = 19.0
    while pdfmetrics.stringWidth(clean(title), "Arial-Bold", title_size) > PAGE_W - 2 * MARGIN_X - 150:
        title_size -= 0.25
    c.setFont("Arial-Bold", title_size)
    c.drawString(MARGIN_X, PAGE_H - 41, clean(title))

    c.setFillColor(WHITE)
    c.setFont("Arial-Bold", 7.0)
    c.drawRightString(PAGE_W - MARGIN_X, PAGE_H - 39, clean(page_label).upper())
    c.setFillColor(GOLD)
    c.rect(MARGIN_X, PAGE_H - 57, 64, 2, fill=1, stroke=0)


def draw_lens_box(c: canvas.Canvas, label: str, text: str, y_top: float) -> float:
    box_h = 52
    x = MARGIN_X
```

```
    width = PAGE_W - 2 * MARGIN_X
    c.setFillColor(EDGE)
    c.roundRect(x + 1.5, y_top - box_h - 1.5, width, box_h, 5, fill=1, stroke=0)
    c.setFillColor(MASTHEAD)
    c.setStrokeColor(BORDER)
    c.setLineWidth(0.65)
    c.roundRect(x, y_top - box_h, width, box_h, 5, fill=1, stroke=1)
    c.setFillColor(GOLD)
    c.rect(x, y_top - box_h, 4, box_h, fill=1, stroke=0)
    label_style = paragraph_style(
        "lens_label", 6.7, 8.1, color=GOLD, font="Arial-Bold", alignment=TA_CENTER
    )
    body_style = paragraph_style("lens", 8.15, 9.5, color=MICRO)
    label_copy = h(label).replace(" ", "<br/>", 1)
    table = Table(
        [[Paragraph(label_copy, label_style), Paragraph(h(text), body_style)]],
        colWidths=[132, width - 136],
        rowHeights=[box_h],
        hAlign="LEFT",
    )
    table.setStyle(
        TableStyle(
            [
                ("VALIGN", (0, 0), (-1, -1), "MIDDLE"),
                ("LEFTPADDING", (0, 0), (0, 0), 14),
                ("RIGHTPADDING", (0, 0), (0, 0), 14),
                ("LEFTPADDING", (1, 0), (1, 0), 14),
                ("RIGHTPADDING", (1, 0), (1, 0), 14),
                ("TOPPADDING", (0, 0), (-1, -1), 5),
                ("BOTTOMPADDING", (0, 0), (-1, -1), 5),
                ("LINEBEFORE", (1, 0), (1, 0), 0.45, BORDER),
            ]
        )
    )
    _, table_h = table.wrap(width - 4, box_h)
    if table_h > box_h:
        raise ValueError(f"Lens copy overflow for {label}: {table_h:.1f}pt")
    table.drawOn(c, x + 4, y_top - box_h)
    return y_top - box_h - 10


def draw_category_box(c: canvas.Canvas, slug: str, y_top: float) -> float:
    box_h = 60
    x = MARGIN_X
    width = PAGE_W - 2 * MARGIN_X
    c.setFillColor(EDGE)
    c.roundRect(x + 1.5, y_top - box_h - 1.5, width, box_h, 5, fill=1, stroke=0)
    c.setFillColor(MASTHEAD)
    c.setStrokeColor(BORDER)
    c.setLineWidth(0.65)
    c.roundRect(x, y_top - box_h, width, box_h, 5, fill=1, stroke=1)
    c.setFillColor(GOLD)
    c.rect(x, y_top - box_h, 4, box_h, fill=1, stroke=0)

    label_style = paragraph_style(
        "category_box_label", 6.6, 8.0, color=GOLD, font="Arial-Bold"
    )
    label = Paragraph("SUBSECTOR<br/>CATEGORIES", label_style)
    _, label_h = label.wrap(126, box_h - 14)
    label.drawOn(c, x + 14, y_top - (box_h + label_h) / 2)

    groups = SUBSECTOR_CATEGORIES[slug]
    category_x = x + 146
    category_width = width - 158
    category_style = paragraph_style(
        "category_box_item", 6.65, 8.1, color=META
    )
    cells = []
    for category, companies in groups:
        members = " / ".join(companies)
        cells.append(
            Paragraph(
                f"<font color='#442142'><b>{h(category)}</b></font><br/>"
                f"<font size='6.1' color='#52525B'>{h(members)}</font>",
                category_style,
            )
        )
    col_width = category_width / len(cells)
    table = Table([cells], colWidths=[col_width] * len(cells), hAlign="LEFT")
    table.setStyle(
        TableStyle(
            [
                ("VALIGN", (0, 0), (-1, -1), "MIDDLE"),
                ("LEFTPADDING", (0, 0), (-1, -1), 9),
                ("RIGHTPADDING", (0, 0), (-1, -1), 9),
                ("TOPPADDING", (0, 0), (-1, -1), 5),
                ("BOTTOMPADDING", (0, 0), (-1, -1), 5),
                ("LINEBEFORE", (1, 0), (-1, 0), 0.45, BORDER),
```

```
            ]
        )
    )
    _, table_h = table.wrap(category_width, box_h - 10)
    if table_h > box_h - 10:
        raise ValueError(f"{slug}: subsector category box overflow")
    table.drawOn(c, category_x, y_top - (box_h + table_h) / 2)
    return y_top - box_h - 10


def estimate_markup(items: list[tuple[str, str]]) -> str:
    if len(items) == 1 and not items[0][0]:
        return f"<font color='#442142'><b>{h(items[0][1])}</b></font>"
    if len(items) <= 2:
        return "<br/>".join(
            f"<font color='#52525B'><b>{h(label)}</b></font>  "
            f"<font color='#442142'><b>{h(value)}</b></font>"
            for label, value in items
        )
    long_items = any(len(clean(label)) + len(clean(value)) > 20 for label, value in items)
    group_size = 1 if long_items else 2
    lines: list[str] = []
    for index in range(0, len(items), group_size):
        cells = []
        for label, value in items[index : index + group_size]:
            cells.append(
                f"<font color='#52525B'><b>{h(label)}</b></font> "
                f"<font color='#442142'><b>{h(value)}</b></font>"
            )
        lines.append("   |   ".join(cells))
    return "<br/>".join(lines)


def estimate_flowable(
    items: list[tuple[str, str]], inner_width: float, font_size: float
) -> Paragraph | Table:
    estimate_style = paragraph_style(
        "estimate_value", font_size, font_size + 1.5, color=PLUM
    )

    def cell(label: str, value: str) -> Paragraph:
        if label:
            separator = " " if len(clean(label)) + len(clean(value)) <= 20 else "<br/>"
            markup = (
                f"<font size='6.1' color='#52525B'><b>{h(label)}</b></font>{separator}"
                f"<font color='#442142'><b>{h(value)}</b></font>"
            )
        else:
            markup = f"<font color='#442142'><b>{h(value)}</b></font>"
        return Paragraph(markup, estimate_style)

    if len(items) <= 2:
        return Paragraph(
            "<br/>".join(
                (
                    f"<font size='6.1' color='#52525B'><b>{h(label)}</b></font> "
                    f"<font color='#442142'><b>{h(value)}</b></font>"
                )
                if label
                else f"<font color='#442142'><b>{h(value)}</b></font>"
                for label, value in items
            ),
            estimate_style,
        )

    rows = []
    for index in range(0, len(items), 2):
        row = [cell(label, value) for label, value in items[index : index + 2]]
        if len(row) == 1:
            row.append(Paragraph("", estimate_style))
        rows.append(row)
    nested = Table(rows, colWidths=[inner_width / 2, inner_width / 2], hAlign="LEFT")
    nested.setStyle(
        TableStyle(
            [
                ("VALIGN", (0, 0), (-1, -1), "TOP"),
                ("LEFTPADDING", (0, 0), (-1, -1), 0),
                ("RIGHTPADDING", (0, 0), (-1, -1), 5),
                ("TOPPADDING", (0, 0), (-1, -1), 1),
                ("BOTTOMPADDING", (0, 0), (-1, -1), 2),
                ("LINEBEFORE", (1, 0), (1, -1), 0.35, BORDER),
                ("LEFTPADDING", (1, 0), (1, -1), 7),
                ("LINEABOVE", (0, 1), (-1, 1), 0.35, BORDER),
                ("TOPPADDING", (0, 1), (-1, 1), 4),
            ]
        )
    )
    return nested
```

```
def metric_table(slug: str, max_height: float) -> Table:
    col_widths = [106, 190, 326, 86]
    header_style = paragraph_style(
        "metric_header", 7.4, 8.5, color=WHITE, font="Arial-Bold"
    )
    confidence_style = paragraph_style(
        "metric_confidence", 7.05, 8.2, color=PLUM, font="Arial-Bold", alignment=TA_CENTER
    )
    body_size = 8.05

    def build(size: float) -> Table:
        metric_style = paragraph_style(
            "metric_name", size + 0.45, size + 1.65, color=PLUM, font="Arial-Bold"
        )
        estimate_style = paragraph_style(
            "metric_estimate", size + 0.8, size + 2.05, color=PLUM
        )
        assertion_style = paragraph_style(
            "metric_assertion", size, size + 1.35, color=BODY
        )
        data: list[list[Paragraph]] = [
            [
                Paragraph("METRIC", header_style),
                Paragraph("ESTIMATE", header_style),
                Paragraph("ASSERTION", header_style),
                Paragraph("CONFIDENCE", header_style),
            ]
        ]
        estimate_map = ESTIMATES[slug]
        assertion_map = ASSERTIONS[slug]
        for name in REVISED_METRICS:
            if name not in estimate_map:
                raise ValueError(f"{slug}: missing estimate for {name}")
            if name not in assertion_map:
                raise ValueError(f"{slug}: missing assertion for {name}")
            metric_label = h(name)
            if name == "Customer tenor":
                metric_label += "<br/><font size='6.3' color='#52525B'>INITIAL / EFFECTIVE</font>"
            assertion, confidence = assertion_map[name]
            data.append(
                [
                    Paragraph(metric_label, metric_style),
                    estimate_flowable(estimate_map[name], col_widths[1] - 16, size + 0.55),
                    Paragraph(h(assertion), assertion_style),
                    Paragraph(h(confidence).upper(), confidence_style),
                ]
            )
        table = Table(data, colWidths=col_widths, repeatRows=1, hAlign="LEFT")
        table.setStyle(
            TableStyle(
                [
                    ("BACKGROUND", (0, 0), (-1, 0), PLUM),
                    ("LINEBELOW", (0, 0), (-1, 0), 1.0, GOLD),
                    ("LINEBEFORE", (0, 0), (0, 0), 4.0, GOLD),
                    ("VALIGN", (0, 0), (-1, -1), "TOP"),
                    ("VALIGN", (3, 1), (3, -1), "MIDDLE"),
                    ("LEFTPADDING", (0, 0), (-1, -1), 8),
                    ("RIGHTPADDING", (0, 0), (-1, -1), 8),
                    ("TOPPADDING", (0, 0), (-1, 0), 7),
                    ("BOTTOMPADDING", (0, 0), (-1, 0), 7),
                    ("TOPPADDING", (0, 1), (-1, -1), 6.5),
                    ("BOTTOMPADDING", (0, 1), (-1, -1), 6.5),
                    ("LINEBELOW", (0, 1), (-1, -1), 0.45, BORDER),
                    ("LINEBEFORE", (1, 0), (1, -1), 0.35, BORDER),
                    ("LINEBEFORE", (2, 0), (2, -1), 0.35, BORDER),
                    ("LINEBEFORE", (3, 0), (3, -1), 0.35, BORDER),
                    ("ROWBACKGROUNDS", (0, 1), (-1, -1), [WHITE, CARD]),
                    ("BACKGROUND", (3, 1), (3, -1), SECTION),
                    ("BOX", (0, 0), (-1, -1), 0.65, BORDER),
                ]
            )
        )
        return table

    table = build(body_size)
    _, table_h = table.wrap(sum(col_widths), max_height)
    while table_h > max_height and body_size > 7.75:
        body_size -= 0.1
        table = build(body_size)
        _, table_h = table.wrap(sum(col_widths), max_height)
    if table_h > max_height:
        raise ValueError(f"{slug}: metric table is {table_h:.1f}pt, above {max_height:.1f}pt")
    return table


def draw_page_one(c: canvas.Canvas, document: dict, slug: str) -> None:
```

```
    title = DISPLAY_TITLES[slug]
    draw_header(c, title, "PAGE 1 | ESTIMATES")
    y_top = draw_lens_box(
        c,
        "SUBVERTICAL LENS",
        THESIS_SUMMARIES[slug],
        PAGE_H - 88,
    )
    max_h = y_top - 32
    table = metric_table(slug, max_h)
    _, table_h = table.wrap(PAGE_W - 2 * MARGIN_X, max_h)
    table.drawOn(c, MARGIN_X, y_top - table_h)
    draw_footer(c, 1, title)


def compact_words(value: object, max_words: int) -> str:
    text = clean(value)
    words = text.split()
    if len(words) <= max_words:
        return text
    clipped = " ".join(words[:max_words]).rstrip(" ,;:")
    return clipped + "..."


def complete_sentences(value: object, max_words: int, *, evidence: bool = False) -> str:
    text = clean(value)
    sentences = [sentence.strip() for sentence in re.split(r"(?<=[.!?])\s+", text) if sentence.strip()]
    if evidence:
        filtered = [
            sentence
            for sentence in sentences
            if not re.match(r"^(Central:|All metrics are|Financial ranges are)", sentence, re.IGNORECASE)
        ]
        if filtered:
            sentences = filtered
    selected: list[str] = []
    used_words = 0
    for sentence in sentences:
        sentence_words = len(sentence.split())
        if selected and used_words + sentence_words > max_words:
            break
        selected.append(sentence)
        used_words += sentence_words
        if used_words >= max_words:
            break
    return " ".join(selected) if selected else text


def business_sections(business: dict, word_limit: int) -> list[tuple[str, str]]:
    facts = business.get("supporting_facts", "")
    if isinstance(facts, list):
        facts = " ".join(clean(item) for item in facts)
    # Comparable-company datapoints are the audit trail for Page 1 estimates.
    # Never truncate this section; fit the descriptive copy around the complete
    # reference block instead.
    reference_copy = clean(facts)
    reference_words = len(reference_copy.split())
    descriptive_budget = max(24, word_limit - reference_words)
    business_words = max(10, round(descriptive_budget * 0.40))
    economics_words = max(14, descriptive_budget - business_words)
    what = complete_sentences(business.get("what_it_does", ""), business_words)
    model = complete_sentences(business.get("economic_model", ""), economics_words)
    return [
        ("Reference data", reference_copy),
        ("Business", what),
        ("Economics", model),
    ]


def business_markup(business: dict, word_limit: int) -> str:
    return "<br/><br/>".join(
        f"<font color='#442142'><b>{h(label)}.</b></font> {h(copy)}"
        for label, copy in business_sections(business, word_limit)
    )


def card_layout(count: int) -> tuple[int, int]:
    if count <= 2:
        return 2, 1
    if count <= 4:
        return 2, 2
    return 3, 2


def fit_card_paragraph(
    business: dict,
    width: float,
    height: float,
```

```
    word_limit: int,
    *,
    spacious: bool,
) -> tuple[list[Paragraph], list[float], int]:
    size = 8.35 if spacious else 7.45
    leading = 10.6 if spacious else 8.9
    style = paragraph_style("card_body", size, leading, color=BODY)
    for active_limit in range(word_limit, max(39, word_limit - 30), -4):
        if spacious:
            paragraphs = [
                Paragraph(
                    f"<font color='#442142'><b>{h(label)}.</b></font> {h(copy)}",
                    style,
                )
                for label, copy in business_sections(business, active_limit)
            ]
        else:
            paragraphs = [Paragraph(business_markup(business, active_limit), style)]
        heights = [paragraph.wrap(width, height)[1] for paragraph in paragraphs]
        minimum_spacing = 10 * (len(paragraphs) + 1) if spacious else 0
        if sum(heights) + minimum_spacing <= height:
            return paragraphs, heights, active_limit
    raise ValueError(
        f"Business card copy does not fit: {clean(business.get('name', ''))}"
    )


def draw_card(
    c: canvas.Canvas,
    business: dict,
    x: float,
    y: float,
    width: float,
    height: float,
    *,
    word_limit: int,
    category: str,
    spacious: bool,
) -> None:
    c.setFillColor(EDGE)
    c.roundRect(x + 2, y - 2, width, height, 6, fill=1, stroke=0)
    c.setFillColor(CARD)
    c.setStrokeColor(BORDER)
    c.setLineWidth(0.65)
    c.roundRect(x, y, width, height, 6, fill=1, stroke=1)

    header_h = 42
    c.setFillColor(PLUM)
    c.rect(x, y + height - header_h, width, header_h, fill=1, stroke=0)
    c.setFillColor(GOLD)
    c.rect(x, y + height - header_h, 4, header_h, fill=1, stroke=0)
    name_style = paragraph_style("card_name", 8.4, 9.2, color=WHITE, font="Arial-Bold")
    name_p = Paragraph(h(business.get("name", "")), name_style)
    _, name_h = name_p.wrap(width - 26, 18)
    if name_h > 18:
        raise ValueError(f"Card title overflow: {business.get('name', '')}")
    name_p.drawOn(c, x + 12, y + height - 9 - name_h)
    category_size = 5.7
    category_text = clean(category).upper()
    while pdfmetrics.stringWidth(category_text, "Arial-Bold", category_size) > width - 24:
        category_size -= 0.15
    c.setFillColor(GOLD)
    c.setFont("Arial-Bold", category_size)
    c.drawString(x + 12, y + height - header_h + 8, category_text)
    c.setStrokeColor(BORDER)
    c.line(x, y + height - header_h, x + width, y + height - header_h)

    body_top = y + height - header_h - 10
    body_floor = y + 10
    inner_w = width - 28
    inner_h = body_top - body_floor
    paragraphs, heights, _ = fit_card_paragraph(
        business,
        inner_w,
        inner_h,
        word_limit,
        spacious=spacious,
    )
    if spacious:
        gap = (inner_h - sum(heights)) / (len(paragraphs) + 1)
        cursor = body_top - gap
        for paragraph, paragraph_h in zip(paragraphs, heights):
            cursor -= paragraph_h
            paragraph.drawOn(c, x + 14, cursor)
            cursor -= gap
        if cursor < body_floor - 0.1:
            raise ValueError(f"Card body overflow: {business.get('name', '')}")
    else:
```

```
        body_bottom = body_top - heights[0]
        if body_bottom < body_floor:
            raise ValueError(f"Card body overflow: {business.get('name', '')}")
        paragraphs[0].drawOn(c, x + 14, body_bottom)


def draw_unresolved_callout(c: canvas.Canvas, business: dict, y: float) -> None:
    x = MARGIN_X
    width = PAGE_W - 2 * MARGIN_X
    height = 48
    c.setFillColor(CARD)
    c.setStrokeColor(GOLD)
    c.setLineWidth(0.8)
    c.roundRect(x, y, width, height, 5, fill=1, stroke=1)
    c.setFillColor(GOLD)
    c.rect(x, y, 4, height, fill=1, stroke=0)
    draw_tracked_text(
        c,
        x + 13,
        y + 32,
        "UNRESOLVED IDENTITY - EXCLUDED FROM ESTIMATES",
        font="Arial-Bold",
        size=6.3,
        color=GOLD,
        tracking=0.25,
    )
    c.setFillColor(PLUM)
    c.setFont("Arial-Bold", 8.5)
    c.drawString(x + 13, y + 15, clean(business.get("name", "")))
    body = complete_sentences(business.get("what_it_does", ""), 26) + " " + complete_sentences(
        business.get("supporting_facts", ""), 22, evidence=True
    )
    style = paragraph_style("unresolved", 7.25, 8.5, color=META)
    p = Paragraph(h(body), style)
    body_w = width - 248
    _, ph = p.wrap(body_w, height - 12)
    if ph > height - 12:
        raise ValueError("Unresolved identity callout overflow")
    p.drawOn(c, x + 232, y + (height - ph) / 2)


def draw_page_two(c: canvas.Canvas, document: dict, slug: str) -> None:
    title = DISPLAY_TITLES[slug]
    draw_header(c, title, "PAGE 2 | BUSINESS PROFILES")
    y_top = draw_category_box(c, slug, PAGE_H - 88)

    business_by_name = {
        clean(business.get("name", "")): business
        for business in document["page2"]["businesses"]
        if clean(business.get("name", "")) != "Reliant Rental"
    }
    businesses = [business_by_name[name] for name in BUSINESS_ORDER[slug]]
    if set(business_by_name) != set(BUSINESS_ORDER[slug]):
        raise ValueError(f"{slug}: business ordering does not match category assignments")

    cols, rows = card_layout(len(businesses))
    gap = 9
    grid_top = y_top
    grid_bottom = 38
    available_grid_h = grid_top - grid_bottom
    if rows == 1:
        card_h = available_grid_h
        grid_start_top = grid_top
    else:
        card_h = (available_grid_h - gap * (rows - 1)) / rows
        grid_start_top = grid_top
    grid_width = PAGE_W - 2 * MARGIN_X
    card_w = (grid_width - gap * (cols - 1)) / cols
    spacious = len(businesses) <= 2
    word_limit = 220 if spacious else 110 if len(businesses) <= 4 else 92

    for row in range(rows):
        row_businesses = businesses[row * cols : (row + 1) * cols]
        if not row_businesses:
            continue
        row_width = len(row_businesses) * card_w + (len(row_businesses) - 1) * gap
        row_x = MARGIN_X + (grid_width - row_width) / 2
        y = grid_start_top - (row + 1) * card_h - row * gap
        for col, business in enumerate(row_businesses):
            x = row_x + col * (card_w + gap)
            name = clean(business.get("name", ""))
            draw_card(
                c,
                business,
                x,
                y,
                card_w,
                card_h,
```

```
                word_limit=word_limit,
                category=BUSINESS_CATEGORY[slug][name],
                spacious=spacious,
            )

    draw_footer(c, 2, title)


def generate_one(path: Path) -> Path:
    document = json.loads(path.read_text(encoding="utf-8"))
    slug = path.stem
    sequence = ORDER.index(slug) + 1
    output = OUTPUT_DIR / f"{sequence:02d}-{slug}-two-page-brief.pdf"
    c = canvas.Canvas(str(output), pagesize=landscape(letter), pageCompression=1)
    c.setTitle(clean(document["title"]) + " - Two-Page Brief")
    c.setAuthor("Codex")
    draw_page_one(c, document, slug)
    c.showPage()
    draw_page_two(c, document, slug)
    c.save()
    pages = len(PdfReader(str(output)).pages)
    if pages != 2:
        raise ValueError(f"{output.name}: expected 2 pages, got {pages}")
    return output


def merge(outputs: list[Path]) -> Path:
    combined = OUTPUT_DIR / "all-subverticals-two-page-briefs.pdf"
    writer = PdfWriter()
    for output in outputs:
        reader = PdfReader(str(output))
        for page in reader.pages:
            writer.add_page(page)
    with combined.open("wb") as handle:
        writer.write(handle)
    return combined


def main() -> None:
    draft_paths = [DRAFT_DIR / f"{slug}.json" for slug in ORDER]
    missing = [path.name for path in draft_paths if not path.exists()]
    if missing:
        raise ValueError(f"Missing draft JSON files: {', '.join(missing)}")
    outputs = [generate_one(path) for path in draft_paths]
    combined = merge(outputs)
    for output in [*outputs, combined]:
        print(output)


if __name__ == "__main__":
    main()
```

make_qa_sheets.py

tmp/two_pagers/make_qa_sheets.py

SHA-256: 706e3cc70b128bddf5314b741737162849de653936b6f797ad357af7284bdf27

```
from pathlib import Path

from PIL import Image, ImageDraw, ImageFont


ROOT = Path(__file__).resolve().parent / "rendered"
FILES = sorted(
    path
    for path in ROOT.glob("0*-two-page-brief-*.png")
    if path.stem.endswith("-1") or path.stem.endswith("-2")
)
FONT = ImageFont.load_default()
THUMB_W = 500
THUMB_H = 386
LABEL_H = 22
GAP = 12


def build(files: list[Path], destination: Path) -> None:
    rows = 3
    canvas = Image.new(
        "RGB",
        (2 * THUMB_W + 3 * GAP, rows * (THUMB_H + LABEL_H) + (rows + 1) * GAP),
        "#D9D9DE",
    )
    draw = ImageDraw.Draw(canvas)
    for index, path in enumerate(files):
        row, column = divmod(index, 2)
        x = GAP + column * (THUMB_W + GAP)
        y = GAP + row * (THUMB_H + LABEL_H + GAP)
        with Image.open(path) as source:
            image = source.convert("RGB")
            image.thumbnail((THUMB_W, THUMB_H), Image.Resampling.LANCZOS)
        canvas.paste(image, (x, y + LABEL_H))
        draw.text((x, y + 3), path.stem, fill="#111111", font=FONT)
    canvas.save(destination, quality=92)


build(FILES[:6], ROOT / "qa-sheet-1.jpg")
build(FILES[6:12], ROOT / "qa-sheet-2.jpg")


COMPENDIUM_FILES = sorted(
    ROOT.glob("compendium-page-*.png"),
    key=lambda path: int(path.stem.rsplit("-", 1)[-1]),
)


def build_compendium(files: list[Path], destination: Path) -> None:
    columns = 4
    rows = 4
    thumb_w = 270
    thumb_h = 350
    label_h = 24
    gap = 10
    sheet = Image.new(
        "RGB",
        (
            columns * thumb_w + (columns + 1) * gap,
            rows * (thumb_h + label_h) + (rows + 1) * gap,
        ),
        "#D9D9DE",
    )
    draw = ImageDraw.Draw(sheet)
    for index, path in enumerate(files[: columns * rows]):
        row, column = divmod(index, columns)
        x = gap + column * (thumb_w + gap)
        y = gap + row * (thumb_h + label_h + gap)
        with Image.open(path) as source:
            image = source.convert("RGB")
            image.thumbnail((thumb_w, thumb_h), Image.Resampling.LANCZOS)
        sheet.paste(image, (x + (thumb_w - image.width) // 2, y + label_h))
        draw.text((x, y + 3), path.stem, fill="#111111", font=FONT)
    sheet.save(destination, quality=92)


if COMPENDIUM_FILES:
    build_compendium(COMPENDIUM_FILES, ROOT / "compendium-qa-sheet.jpg")
```

qa_two_pagers.py

tmp/two_pagers/qa_two_pagers.py

SHA-256: 5ecf57304e1ce42e696b2126fe8202b8327508205922b1db851200c9b4034e8b

```
from pathlib import Path
import json
import re

import pdfplumber
from pypdf import PdfReader


ROOT = Path(__file__).resolve().parents[2]
OUTPUT = ROOT / "output" / "pdf" / "two-pagers"
DRAFTS = ROOT / "tmp" / "two_pagers" / "drafts"

EXPECTED = {
    "01-home-comfort-two-page-brief.pdf": ["Reliance Home Comfort", "Enercare"],
    "02-point-of-use-water-two-page-brief.pdf": [
        "AquaVenture Holdings / Quench",
        "Primo Water",
    ],
    "03-submetering-two-page-brief.pdf": [
        "Energy Assets",
        "ista",
        "Techem",
        "Smart Metering Systems",
        "Calisen",
        "CARMA",
    ],
    "04-liquid-environmental-services-two-page-brief.pdf": [
        "Restaurant Technologies",
        "Liquid Environmental Solutions",
        "Waste Resources Management",
        "Wind River Environmental",
    ],
    "05-transportation-solutions-two-page-brief.pdf": [
        "VTG",
        "Akiem",
        "Air Rail",
        "TCR",
        "Lazer Logistics",
    ],
    "06-specialty-rental-two-page-brief.pdf": [
        "PODS",
        "Mobile Mini",
        "Trench Plate Rental",
        "General Finance Corporation",
        "Modulaire Group",
        "McGrath RentCorp",
    ],
}

CATEGORY_LABELS = {
    "01-home-comfort-two-page-brief.pdf": [
        "Diversified home comfort",
        "Rental-led equipment & plans",
    ],
    "02-point-of-use-water-two-page-brief.pdf": [
        "Workplace POU filtration",
        "Diversified water platform",
    ],
    "03-submetering-two-page-brief.pdf": [
        "Building submetering & billing",
        "Meter asset ownership",
        "Mixed metering & utility services",
    ],
    "04-liquid-environmental-services-two-page-brief.pdf": [
        "Installed onsite oil systems",
        "Commercial liquid-waste routes",
        "Diversified wastewater services",
    ],
    "05-transportation-solutions-two-page-brief.pdf": [
        "Railcar leasing",
        "Locomotive leasing",
        "Airport GSE leasing",
        "Yard logistics",
    ],
    "06-specialty-rental-two-page-brief.pdf": [
        "Consumer moving & storage",
        "B2B portable storage",
        "Modular space",
        "Trench safety rental",
    ],
}
```

```
METRICS = [
    "Revenue growth",
    "EBITDA margin",
    "Maintenance capex",
    "Customer tenor",
    "Utilization exposure",
    "payback",
    "B2B",
]

BANNED = [
    "\u2010",
    "\u2011",
    "\u2012",
    "\u2013",
    "\u2014",
    "\u2212",
    "[object Object]",
    "...",
    "GUGGENHEIM",
    "INFRASTRUCTURE COVERAGE & ADVISORY",
    "Two-page source brief",
    "MIXED BY ARCHETYPE",
]

LOW_SUPPORT_FORBIDDEN = {
    "02-point-of-use-water-two-page-brief.pdf": [
        "Maintenance capex / revenue 13%",
        "Equipment 2.7 yrs",
        "CAC 1.0 yr",
    ],
    "04-liquid-environmental-services-two-page-brief.pdf": [
        "Onsite 3.5%",
        "Onsite 4 / 9 yrs",
        "Route 2 / 7 yrs",
    ],
    "06-specialty-rental-two-page-brief.pdf": [
        "Consumer 4%",
        "Consumer 33%",
        "Consumer 7%",
        "Consumer 1 / 8 mos",
        "Consumer Moderate-High",
        "Consumer 3.5 yrs",
    ],
}

FIRST_PRINCIPLES_REQUIRED = {
    "02-point-of-use-water-two-page-brief.pdf": [
        "Sustaining capital should rise",
        "Payback should shorten",
    ],
    "04-liquid-environmental-services-two-page-brief.pdf": [
        "Onsite sustaining capital should rise",
        "Onsite contracts should be longer",
    ],
    "06-specialty-rental-two-page-brief.pdf": [
        "Consumer storage growth should rise",
        "Consumer margins should rise",
        "Consumer sustaining capital should rise",
        "Effective consumer tenure should lengthen",
        "Consumer utilization exposure should rise",
        "Consumer payback should shorten",
    ],
}


for filename, businesses in EXPECTED.items():
    path = OUTPUT / filename
    reader = PdfReader(str(path))
    assert len(reader.pages) == 2, (filename, len(reader.pages))
    page1 = reader.pages[0].extract_text() or ""
    page2 = reader.pages[1].extract_text() or ""
    page1_normalized = re.sub(r"\s+", " ", page1)
    page2_normalized = re.sub(r"\s+", " ", page2)
    slug = re.sub(r"^\d{2}-", "", filename).replace("-two-page-brief.pdf", "")
    draft = json.loads((DRAFTS / f"{slug}.json").read_text(encoding="utf-8"))
    assert len(page1) > 1_500, (filename, "page1", len(page1))
    assert len(page2) > 1_000, (filename, "page2", len(page2))
    assert "ESTIMATE" in page1, (filename, "missing ESTIMATE header")
    assert "BENCHMARK RANGE" not in page1, (filename, "legacy benchmark header")
    assert "ASSERTION" in page1, (filename, "missing assertion header")
    assert "CONFIDENCE" in page1, (filename, "missing confidence header")
    assert "SUBSECTOR" in page2 and "CATEGORIES" in page2, (
        filename,
        "missing subsector categories",
    )
    assert "BUSINESS UNIVERSE" not in page2, (filename, "legacy universe label")
    assert page2_normalized.count("Reference data.") == len(businesses), (
```

```
        filename,
        "each company must have one visible Reference data section",
        page2_normalized.count("Reference data."),
        len(businesses),
    )
    assert "Evidence." not in page2_normalized, (filename, "legacy Evidence label")
    assert "Post-maintenance" not in page1, (filename, "legacy conversion metric")
    assert "Initial contract tenor" not in page1, (filename, "uncombined tenor metric")
    assert "Effective customer tenure" not in page1, (filename, "uncombined tenure metric")
    assert page1_normalized.lower().count("customer tenor") == 1, (filename, "combined tenor row")
    for metric in METRICS:
        assert metric.lower() in page1_normalized.lower(), (filename, metric)
    for business in businesses:
        assert business.lower() in page2.lower(), (filename, business)
    # Page 1 estimates are supported by company-specific datapoints held in
    # supporting_facts. The complete block must survive rendering on Page 2;
    # a source-only fact that is clipped from the company card is a QA failure.
    draft_businesses = {
        item["name"]: item
        for item in draft["page2"]["businesses"]
        if item["name"] != "Reliant Rental"
    }
    assert set(draft_businesses) == set(businesses), (
        filename,
        "draft/PDF company set mismatch",
    )
    for business_name, business in draft_businesses.items():
        reference_data = business.get("supporting_facts", "")
        if isinstance(reference_data, list):
            reference_data = " ".join(str(item) for item in reference_data)
        reference_data = re.sub(r"\s+", " ", reference_data).strip()
        assert reference_data in page2_normalized, (
            filename,
            business_name,
            "full company reference data was clipped or omitted",
            reference_data,
        )
    assert "Directional framework shown where a point estimate is unsupported." in page1_normalized
    assert "Business profiles grouped by subsector category." in page2_normalized
    for removed_callout in ["Source: DR", "[DR", "DR =", "source links"]:
        assert removed_callout not in page1 + page2, (filename, removed_callout)
    assert "KEY DILIGENCE" not in page2, (filename, "key diligence remains")
    assert "Analyst interpretation." not in page2, (filename, "interpretation callout remains")
    assert not re.search(r"\b(?:we|our)\b", page1 + page2, re.IGNORECASE), (
        filename,
        "first-person language remains",
    )
    for category in CATEGORY_LABELS[filename]:
        assert category.lower() in page2.lower(), (filename, category)
    if filename == "06-specialty-rental-two-page-brief.pdf":
        assert "Reliant Rental" not in page2, (filename, "Reliant Rental was not removed")
        assert "sales fell 7.5% and adjusted EBITDA fell 24%" in page2_normalized, (
            filename,
            "PODS correction missing",
        )
        assert "EBITDA 24%" not in page2_normalized, (filename, "incorrect PODS wording")
    if filename == "01-home-comfort-two-page-brief.pdf":
        assert "Ontario materials show 7-10-year minimums" in page2_normalized, (
            filename,
            "Reliance form scope correction missing",
        )
        assert "almost never economical to refurbish and reinstall" in page2_normalized, (
            filename,
            "Enercare asset scope correction missing",
        )
    if filename == "03-submetering-two-page-brief.pdf":
        assert "EUR557 million on EUR1,173.4 million group revenue, or 47.5%" in page2_normalized, (
            filename,
            "Techem perimeter correction missing",
        )
        assert "GBP63.8 million on GBP135.5 million revenue, or 47.1%" in page2_normalized, (
            filename,
            "SMS perimeter correction missing",
        )
        assert "GBP265.4m/GBP358.2m (74.1%)" in page2_normalized, (
            filename,
            "Calisen group-margin denominator missing",
        )
    if filename == "04-liquid-environmental-services-two-page-brief.pdf":
        assert "Historical S&P commentary" in page2_normalized, (
            filename,
            "LES dated-margin caveat missing",
        )
        assert "despite durable revenue" not in page2_normalized, (
            filename,
            "unsupported RTI phrase remains",
        )
    for forbidden in LOW_SUPPORT_FORBIDDEN.get(filename, []):
```

```
        assert forbidden.lower() not in page1_normalized.lower(), (
            filename,
            "low-support estimate remains",
            forbidden,
        )
    for required in FIRST_PRINCIPLES_REQUIRED.get(filename, []):
        assert required.lower() in page1_normalized.lower(), (
            filename,
            "missing first-principles framework",
            required,
        )
    full_text = page1 + page2
    for token in BANNED:
        assert token not in full_text, (filename, token)
    media = reader.pages[0].mediabox
    assert round(float(media.width)) == 792, (filename, media.width)
    assert round(float(media.height)) == 612, (filename, media.height)
    with pdfplumber.open(path) as plumber_pdf:
        for page_number, page in enumerate(plumber_pdf.pages, start=1):
            for char in page.chars:
                if not str(char.get("text", "")).strip():
                    continue
                assert float(char["x0"]) >= 40.0, (
                    filename,
                    page_number,
                    "left overflow",
                    char,
                )
                assert float(char["x1"]) <= 752.0, (
                    filename,
                    page_number,
                    "right overflow",
                    char,
                )
    print(
        f"PASS {filename}: pages=2, text={len(full_text):,}, businesses={len(businesses)}"
    )

combined = PdfReader(str(OUTPUT / "all-subverticals-two-page-briefs.pdf"))
assert len(combined.pages) == 12, len(combined.pages)
print("PASS all-subverticals-two-page-briefs.pdf: pages=12")
```

qa_source_compendium.py

tmp/two_pagers/qa_source_compendium.py

SHA-256: b72af58a1da22e32bb57217730bac4ac927ec29b98094c14d7b8b698fc228a3a

```
from __future__ import annotations

import hashlib
import json
import re
from pathlib import Path

from pypdf import PdfReader


ROOT = Path(__file__).resolve().parents[2]
OUTPUT = ROOT / "output" / "pdf" / "operating-benchmark-source-compendium.pdf"
MANIFEST = ROOT / "tmp" / "two_pagers" / "source_compendium_manifest.json"
AUDIT = ROOT / "tmp" / "two_pagers" / "evidence_audit.json"

BANNED_DASHES = ["\u2010", "\u2011", "\u2012", "\u2013", "\u2014", "\u2212"]


def sha256(path: Path) -> str:
    digest = hashlib.sha256()
    with path.open("rb") as handle:
        for chunk in iter(lambda: handle.read(1024 * 1024), b""):
            digest.update(chunk)
    return digest.hexdigest()


def normalize(text: str) -> str:
    return re.sub(r"\s+", " ", text).strip()


def annotations(page) -> list[str]:
    values = []
    for reference in page.get("/Annots") or []:
        annotation = reference.get_object()
        action = annotation.get("/A")
        if action and action.get("/URI"):
            values.append(str(action.get("/URI")))
    return sorted(values)


manifest = json.loads(MANIFEST.read_text(encoding="utf-8"))
audit = json.loads(AUDIT.read_text(encoding="utf-8"))
reader = PdfReader(str(OUTPUT))

assert len(reader.pages) == manifest["total_pages"]
assert sha256(OUTPUT) == manifest["output_sha256"]
assert reader.outline, "Missing compendium bookmarks"
assert len(audit["reports"]) == 6
assert sum(len(report["metrics"]) for report in audit["reports"]) == 42
assert sum(len(report["businesses"]) for report in audit["reports"]) == 25
assert len(audit["omissions"]) == 12

cursor = manifest["front_matter_pages"] + 1
for part in manifest["parts"]:
    assert part["start_page"] == cursor, (part["title"], cursor, part["start_page"])
    source_path = Path(part["source"])
    if not source_path.is_absolute():
        source_path = ROOT / source_path
    assert source_path.exists(), source_path
    assert sha256(source_path) == part["sha256"], source_path
    start_index = part["start_page"] - 1
    end_index = start_index + part["pages"]
    segment = reader.pages[start_index:end_index]
    assert len(segment) == part["pages"]
    if part["kind"] == "pdf":
        source_reader = PdfReader(str(source_path))
        assert len(source_reader.pages) == len(segment)
        for offset, (source_page, merged_page) in enumerate(
            zip(source_reader.pages, segment), start=1
        ):
            assert normalize(source_page.extract_text() or "") == normalize(
                merged_page.extract_text() or ""
            ), (part["title"], offset, "text changed")
            assert annotations(source_page) == annotations(merged_page), (
                part["title"],
                offset,
                "annotations changed",
            )
            assert round(float(source_page.mediabox.width), 2) == round(
                float(merged_page.mediabox.width), 2
            )
            assert round(float(source_page.mediabox.height), 2) == round(
```

```
                float(merged_page.mediabox.height), 2
            )
    else:
        first_text = normalize(segment[0].extract_text() or "")
        assert part["title"] in first_text, (part["title"], "missing text title")
        assert part["sha256"] in first_text, (part["title"], "missing source hash")
        for page in segment:
            text = page.extract_text() or ""
            for character in BANNED_DASHES:
                assert character not in text, (part["title"], character)
    cursor += part["pages"]

assert cursor - 1 == len(reader.pages)

for page_number, page in enumerate(reader.pages, start=1):
    text = page.extract_text() or ""
    assert text.strip(), (page_number, "blank or non-extractable page")
    width = round(float(page.mediabox.width))
    height = round(float(page.mediabox.height))
    assert (width, height) in {(612, 792), (792, 612)}, (page_number, width, height)

front_text = "\n".join(
    reader.pages[index].extract_text() or ""
    for index in range(manifest["front_matter_pages"])
)
assert "comp inputs repeat on company pages" in front_text
assert "Documented Derivatives and Exact Duplicates" in front_text

final_part = manifest["parts"][-1]
assert final_part["title"] == "Combined Final Two-Page Briefs"
final_start = final_part["start_page"] - 1
final_pages = reader.pages[final_start : final_start + final_part["pages"]]
assert len(final_pages) == 12
for index, page in enumerate(final_pages):
    text = normalize(page.extract_text() or "")
    if index % 2 == 0:
        assert "Directional framework shown where a point estimate is unsupported." in text
    else:
        assert "Business profiles grouped by subsector category." in text
        assert text.count("Reference data.") >= 2
        assert "Evidence." not in text
        assert "KEY DILIGENCE" not in text

assert sum(
    normalize(page.extract_text() or "").count("Reference data.")
    for index, page in enumerate(final_pages)
    if index % 2 == 1
) == 25

final_text = " ".join(normalize(page.extract_text() or "") for page in final_pages)
for removed_callout in ["Source: DR", "[DR", "DR =", "Analyst interpretation."]:
    assert removed_callout not in final_text, removed_callout
assert not re.search(r"\b(?:we|our)\b", final_text, re.IGNORECASE)
for forbidden in [
    "Equipment 2.7 yrs",
    "CAC 1.0 yr",
    "Onsite 3.5%",
    "Onsite 4 / 9 yrs",
    "Route 2 / 7 yrs",
    "Consumer 1 / 8 mos",
    "Consumer Moderate-High",
]:
    assert forbidden not in final_text, forbidden

research_parts = [
    part for part in manifest["parts"] if part["section"] == "Deep Research Reports"
]
preserved_links = 0
for part in research_parts:
    start = part["start_page"] - 1
    for page in reader.pages[start : start + part["pages"]]:
        preserved_links += len(annotations(page))
assert preserved_links >= 1_200, preserved_links

print(
    "PASS operating-benchmark-source-compendium.pdf: "
    f"pages={len(reader.pages)}, parts={len(manifest['parts'])}, "
    "inline-source-callouts=0, key-diligence-sections=0, "
    f"preserved-links={preserved_links}"
)
```

generate_source_compendium.py

tmp/two_pagers/generate_source_compendium.py

SHA-256: 2c0900e79dbebe192778a58ed71fa5198042d50c9244c0bd337cad62c0c39971

```
from __future__ import annotations

import hashlib
import json
import math
import re
import textwrap
from dataclasses import dataclass
from datetime import date
from pathlib import Path
from typing import Iterable

from pypdf import PdfReader, PdfWriter
from reportlab.lib import colors
from reportlab.lib.enums import TA_LEFT
from reportlab.lib.pagesizes import landscape, letter
from reportlab.lib.styles import ParagraphStyle
from reportlab.lib.units import inch
from reportlab.pdfbase import pdfmetrics
from reportlab.pdfbase.ttfonts import TTFont
from reportlab.pdfgen import canvas
from reportlab.platypus import Paragraph, Table, TableStyle


ROOT = Path(__file__).resolve().parents[2]
ATTACHMENT = Path(
    "/Users/mikeberry6/.codex/attachments/"
    "3a6e9158-9f45-4701-a9f6-39d4ecedcb4e/pasted-text.txt"
)
TMP_DIR = ROOT / "tmp" / "two_pagers" / "compendium_parts"
OUTPUT = ROOT / "output" / "pdf" / "operating-benchmark-source-compendium.pdf"
MANIFEST_OUTPUT = ROOT / "tmp" / "two_pagers" / "source_compendium_manifest.json"
AUDIT_INPUT = ROOT / "tmp" / "two_pagers" / "evidence_audit.json"
TMP_DIR.mkdir(parents=True, exist_ok=True)
OUTPUT.parent.mkdir(parents=True, exist_ok=True)


FONT_DIR = Path("/System/Library/Fonts/Supplemental")
pdfmetrics.registerFont(TTFont("Arial", str(FONT_DIR / "Arial.ttf")))
pdfmetrics.registerFont(TTFont("Arial-Bold", str(FONT_DIR / "Arial Bold.ttf")))
pdfmetrics.registerFont(TTFont("Arial-Italic", str(FONT_DIR / "Arial Italic.ttf")))

PLUM = colors.HexColor("#442142")
GOLD = colors.HexColor("#B4A87D")
INK = colors.HexColor("#27272A")
BODY = colors.HexColor("#3F3F46")
META = colors.HexColor("#52525B")
SOURCE = colors.HexColor("#5B5563")
LIGHT = colors.HexColor("#F8F9FA")
BORDER = colors.HexColor("#E5E7EB")
WHITE = colors.white


PROMPTS = [
    ROOT / "research_prompts" / "00_launch_manifest.md",
    ROOT / "research_prompts" / "01_home_comfort_prompt.md",
    ROOT / "research_prompts" / "02_point_of_use_water_prompt.md",
    ROOT / "research_prompts" / "03_submetering_prompt.md",
    ROOT / "research_prompts" / "04_liquid_environmental_services_prompt.md",
    ROOT / "research_prompts" / "05_transportation_solutions_prompt.md",
    ROOT / "research_prompts" / "06_specialty_rental_prompt.md",
    ROOT / "research_prompts" / "07_quality_control_checklist.md",
]

RESEARCH_PDFS = [
    ROOT / "output" / "pdf" / "01-home-comfort-gpt-5.6-sol-deep-research.pdf",
    ROOT / "output" / "pdf" / "02-point-of-use-water-gpt-5.6-sol-deep-research.pdf",
    ROOT / "output" / "pdf" / "03-submetering-gpt-5.6-sol-deep-research.pdf",
    ROOT / "output" / "pdf" / "04-liquid-environmental-services-gpt-5.6-sol-deep-research.pdf",
    ROOT / "output" / "pdf" / "05-transportation-solutions-gpt-5.6-sol-deep-research.pdf",
    ROOT / "output" / "pdf" / "06-specialty-rental-gpt-5.6-sol-deep-research.pdf",
]

DRAFTS = [
    ROOT / "tmp" / "two_pagers" / "drafts" / "home-comfort.json",
    ROOT / "tmp" / "two_pagers" / "drafts" / "point-of-use-water.json",
    ROOT / "tmp" / "two_pagers" / "drafts" / "submetering.json",
    ROOT / "tmp" / "two_pagers" / "drafts" / "liquid-environmental-services.json",
    ROOT / "tmp" / "two_pagers" / "drafts" / "transportation-solutions.json",
    ROOT / "tmp" / "two_pagers" / "drafts" / "specialty-rental.json",
]
```

```
METHODS = [
    ROOT / "tmp" / "two_pagers" / "extract_source_text.py",
    ROOT / "tmp" / "two_pagers" / "build_evidence_audit.py",
    ROOT / "tmp" / "two_pagers" / "generate_two_pagers.py",
    ROOT / "tmp" / "two_pagers" / "make_qa_sheets.py",
    ROOT / "tmp" / "two_pagers" / "qa_two_pagers.py",
    ROOT / "tmp" / "two_pagers" / "qa_source_compendium.py",
    ROOT / "tmp" / "two_pagers" / "generate_source_compendium.py",
]

FINAL_PACK = (
    ROOT / "output" / "pdf" / "two-pagers" / "all-subverticals-two-page-briefs.pdf"
)

DERIVED_DUPLICATES = [
    *sorted((ROOT / "tmp" / "two_pagers" / "source_text").glob("*.txt")),
    *sorted((ROOT / "output" / "pdf" / "two-pagers").glob("0[1-6]-*.pdf")),
    *sorted(
        (ROOT / "public" / "one-off-requests" / "operating-benchmark-two-pagers").glob(
            "*.pdf"
        )
    ),
    *sorted((ROOT / "tmp" / "two_pagers" / "rendered").glob("*")),
]


@dataclass
class Part:
    title: str
    section: str
    source_path: Path
    pdf_path: Path
    pages: int
    kind: str
    sha256: str
    start_page: int = 0


def sha256(path: Path) -> str:
    digest = hashlib.sha256()
    with path.open("rb") as handle:
        for chunk in iter(lambda: handle.read(1024 * 1024), b""):
            digest.update(chunk)
    return digest.hexdigest()


def clean_text(value: str) -> str:
    replacements = {
        "\u2010": "-",
        "\u2011": "-",
        "\u2012": "-",
        "\u2013": "-",
        "\u2014": "-",
        "\u2212": "-",
        "\u2018": "'",
        "\u2019": "'",
        "\u201c": '"',
        "\u201d": '"',
        "\u2026": "...",
        "\u2022": "*",
        "\u2192": "->",
        "\u00a0": " ",
    }
    for old, new in replacements.items():
        value = value.replace(old, new)
    return value.encode("cp1252", errors="replace").decode("cp1252")


def display_path(path: Path) -> str:
    try:
        return str(path.relative_to(ROOT))
    except ValueError:
        return str(path)


def wrap_source_lines(text: str, max_chars: int = 112) -> list[str]:
    lines: list[str] = []
    for raw in clean_text(text).splitlines() or [""]:
        expanded = raw.expandtabs(4).rstrip()
        if not expanded:
            lines.append("")
            continue
        indent = re.match(r"^\s*", expanded).group(0)
        continuation = indent + "  "
        wrapped = textwrap.wrap(
            expanded,
            width=max_chars,
            initial_indent="",
```

```
            subsequent_indent=continuation,
            replace_whitespace=False,
            drop_whitespace=False,
            break_long_words=True,
            break_on_hyphens=False,
        )
        lines.extend(wrapped or [""])
    return lines


def render_text_artifact(path: Path, title: str, output: Path) -> int:
    page_w, page_h = letter
    left = 42
    right = page_w - 42
    top = page_h - 52
    bottom = 40
    leading = 7.55
    font_size = 6.15
    first_header = 76
    later_header = 35

    lines = wrap_source_lines(path.read_text(encoding="utf-8", errors="replace"))
    first_capacity = math.floor((top - first_header - bottom) / leading)
    later_capacity = math.floor((top - later_header - bottom) / leading)
    page_chunks: list[list[str]] = []
    page_chunks.append(lines[:first_capacity])
    remaining = lines[first_capacity:]
    while remaining:
        page_chunks.append(remaining[:later_capacity])
        remaining = remaining[later_capacity:]
    if not page_chunks:
        page_chunks = [[]]

    c = canvas.Canvas(str(output), pagesize=letter, pageCompression=1)
    c.setTitle(clean_text(title))
    c.setAuthor("Codex")
    digest = sha256(path)
    source_label = display_path(path)
    total = len(page_chunks)
    for index, chunk in enumerate(page_chunks, start=1):
        c.setFillColor(PLUM)
        c.rect(0, page_h - 22, page_w, 22, fill=1, stroke=0)
        c.setFillColor(GOLD)
        c.rect(0, page_h - 25, page_w, 3, fill=1, stroke=0)
        if index == 1:
            c.setFillColor(PLUM)
            c.setFont("Arial-Bold", 17)
            c.drawString(left, top, clean_text(title))
            c.setFillColor(META)
            c.setFont("Arial", 7.3)
            c.drawString(left, top - 17, clean_text(source_label))
            c.setFont("Arial", 6.2)
            c.drawString(left, top - 29, f"SHA-256: {digest}")
            y = top - 51
        else:
            c.setFillColor(PLUM)
            c.setFont("Arial-Bold", 8)
            c.drawString(left, top, clean_text(title))
            c.setFillColor(META)
            c.setFont("Arial", 6.4)
            c.drawRightString(right, top, clean_text(source_label))
            y = top - 18

        c.setFillColor(INK)
        c.setFont("Courier", font_size)
        for line in chunk:
            c.drawString(left, y, line)
            y -= leading

        c.setStrokeColor(BORDER)
        c.line(left, 29, right, 29)
        c.setFillColor(SOURCE)
        c.setFont("Arial", 6.5)
        c.drawString(left, 18, "Operating Benchmark Source Compendium")
        c.drawRightString(right, 18, f"Artifact page {index} of {total}")
        c.showPage()
    c.save()
    return total


def source_url_index() -> str:
    lines = [
        "SOURCE URL INDEX",
        "",
        "URLs below are extracted from the hyperlink annotations embedded in each Deep Research PDF.",
        "The index preserves the report page on which each source link appears.",
        "",
    ]
```

```
    for pdf_path in RESEARCH_PDFS:
        reader = PdfReader(str(pdf_path))
        by_url: dict[str, set[int]] = {}
        for page_number, page in enumerate(reader.pages, start=1):
            for reference in page.get("/Annots") or []:
                annotation = reference.get_object()
                action = annotation.get("/A")
                if not action or not action.get("/URI"):
                    continue
                url = str(action.get("/URI"))
                if "chatgpt.com/?utm_src=deep-research-pdf" in url:
                    continue
                by_url.setdefault(url, set()).add(page_number)
        lines.extend([pdf_path.name, "=" * min(90, len(pdf_path.name))])
        for url in sorted(by_url):
            pages = ", ".join(str(page) for page in sorted(by_url[url]))
            lines.append(f"Pages {pages}: {url}")
        lines.append("")
    return "\n".join(lines)


def write_generated_text(path: Path, text: str) -> None:
    path.write_text(clean_text(text), encoding="utf-8")


def audit_summary_text() -> str:
    if not AUDIT_INPUT.exists():
        raise FileNotFoundError(
            f"Missing {AUDIT_INPUT}; complete the evidence audit before building the compendium."
        )
    audit = json.loads(AUDIT_INPUT.read_text(encoding="utf-8"))
    lines = [
        "EVIDENCE AUDIT MATRIX",
        "",
        clean_text(audit.get("policy", "")),
        "",
    ]
    for report in audit.get("reports", []):
        lines.extend([report["title"].upper(), "=" * len(report["title"]), ""])
        lines.append("PAGE 1 - ESTIMATES AND ASSERTIONS")
        for row in report.get("metrics", []):
            lines.extend(
                [
                    f"Metric: {row['metric']}",
                    f"Final display: {row['final_display']}",
                    f"Support: {row['support']}",
                    f"Decision: {row['decision']}",
                    f"Source: {row['source']}",
                    f"Audit rationale: {row['audit_rationale']}",
                    "",
                ]
            )
        lines.append("PAGE 2 - BUSINESS FACTS")
        for business in report.get("businesses", []):
            lines.extend(
                [
                    f"Business: {business['business']}",
                    f"Support: {business['support']}",
                    f"Decision: {business['decision']}",
                    f"Source: {business['source']}",
                    f"Audit rationale: {business['audit_rationale']}",
                    "",
                ]
            )
        lines.extend(["", "-"])
    lines.extend(
        [
            "OMISSION LOG",
            "============",
            "",
            *[f"- {item}" for item in audit.get("omissions", [])],
        ]
    )
    return "\n".join(lines)


def build_parts() -> list[Part]:
    required = [ATTACHMENT, *PROMPTS, *RESEARCH_PDFS, *DRAFTS, *METHODS, FINAL_PACK]
    missing = [display_path(path) for path in required if not path.exists()]
    if missing:
        raise FileNotFoundError("Missing compendium inputs: " + ", ".join(missing))

    generated_audit = TMP_DIR / "evidence-audit-matrix.txt"
    generated_urls = TMP_DIR / "source-url-index.txt"
    write_generated_text(generated_audit, audit_summary_text())
    write_generated_text(generated_urls, source_url_index())

    specifications: list[tuple[str, str, Path, str]] = [
```

```
        ("Original Request and Scope", "Original request", ATTACHMENT, "text"),
    ]
    specifications.extend(
        ("Research Prompts and Controls", path.stem.replace("_", " ").title(), path, "text")
        for path in PROMPTS
    )
    specifications.extend(
        ("Deep Research Reports", PdfReader(str(path)).metadata.title or path.stem, path, "pdf")
        for path in RESEARCH_PDFS
    )
    specifications.extend(
        ("Draft Synthesis Records", path.stem.replace("-", " ").title() + " - Draft JSON", path, "text")
        for path in DRAFTS
    )
    specifications.extend(
        [
            ("Evidence and Sources", "Evidence Audit Matrix", generated_audit, "text"),
            ("Evidence and Sources", "Source URL Index", generated_urls, "text"),
        ]
    )
    specifications.extend(
        ("Methodology and Reproducibility", path.name, path, "text") for path in METHODS
    )
    specifications.append(
        ("Final Two-Page Briefs", "Combined Final Two-Page Briefs", FINAL_PACK, "pdf")
    )

    parts: list[Part] = []
    for index, (section, title, path, kind) in enumerate(specifications, start=1):
        if kind == "text":
            output = TMP_DIR / f"{index:02d}-{path.stem}.pdf"
            pages = render_text_artifact(path, title, output)
        else:
            output = path
            pages = len(PdfReader(str(path)).pages)
        parts.append(
            Part(
                title=clean_text(str(title)),
                section=section,
                source_path=path,
                pdf_path=output,
                pages=pages,
                kind=kind,
                sha256=sha256(path),
            )
        )
    return parts


def paragraph_style(name: str, size: float, leading: float, *, color=BODY, font="Arial"):
    return ParagraphStyle(
        name,
        fontName=font,
        fontSize=size,
        leading=leading,
        textColor=color,
        alignment=TA_LEFT,
        spaceAfter=0,
        spaceBefore=0,
    )


def draw_front_footer(c: canvas.Canvas, page_number: int) -> None:
    page_w, _ = letter
    c.setStrokeColor(BORDER)
    c.line(42, 30, page_w - 42, 30)
    c.setFillColor(SOURCE)
    c.setFont("Arial", 7)
    c.drawString(42, 18, "Operating Benchmark Source Compendium")
    c.drawRightString(page_w - 42, 18, str(page_number))


def draw_cover(c: canvas.Canvas) -> None:
    page_w, page_h = letter
    c.setFillColor(LIGHT)
    c.rect(0, 0, page_w, page_h, fill=1, stroke=0)
    c.setFillColor(PLUM)
    c.rect(0, page_h - 235, page_w, 235, fill=1, stroke=0)
    c.setFillColor(GOLD)
    c.rect(0, page_h - 242, page_w, 7, fill=1, stroke=0)
    c.setFillColor(WHITE)
    c.setFont("Arial-Bold", 26)
    c.drawString(48, page_h - 105, "Operating Benchmark")
    c.drawString(48, page_h - 139, "Source Compendium")
    c.setFillColor(colors.HexColor("#E9E3EA"))
    c.setFont("Arial", 11)
    c.drawString(48, page_h - 170, "Complete provenance, evidence audit, and final briefs")
```

```
    c.setFillColor(PLUM)
    c.setFont("Arial-Bold", 11)
    c.drawString(48, page_h - 300, "SIX SUBVERTICALS")
    c.setFillColor(BODY)
    c.setFont("Arial", 10)
    for index, label in enumerate(
        [
            "Home Comfort",
            "Point-of-Use Water",
            "Submetering",
            "Liquid Environmental Services",
            "Transportation Solutions",
            "Specialty Rental",
        ]
    ):
        c.drawString(62, page_h - 326 - index * 23, label)
        c.setFillColor(GOLD)
        c.circle(50, page_h - 322 - index * 23, 2.5, fill=1, stroke=0)
        c.setFillColor(BODY)

    c.setFillColor(META)
    c.setFont("Arial", 8.5)
    c.drawString(48, 86, f"Compiled {date.today().strftime('%B %d, %Y')}")
    c.drawString(48, 70, "Evidence policy: comp inputs repeat on company pages; unsupported estimates are
      omitted.")
    c.showPage()


def draw_control_page(c: canvas.Canvas, parts: list[Part], page_number: int) -> None:
    page_w, page_h = letter
    c.setFillColor(PLUM)
    c.setFont("Arial-Bold", 20)
    c.drawString(42, page_h - 58, "Document Control and Reading Guide")
    c.setFillColor(GOLD)
    c.rect(42, page_h - 70, 86, 3, fill=1, stroke=0)

    research_pages = sum(part.pages for part in parts if part.section == "Deep Research Reports")
    unique_artifacts = len(parts)
    values = [
        ("Canonical artifacts reproduced", str(unique_artifacts)),
        ("Deep Research pages", str(research_pages)),
        ("Final brief pages", str(len(PdfReader(str(FINAL_PACK)).pages))),
        ("Duplicate derivative files listed", str(len(DERIVED_DUPLICATES))),
    ]
    table = Table(
        [[Paragraph(f"<b>{label}</b>", paragraph_style("k", 8, 10, color=META)),
          Paragraph(value, paragraph_style("v", 16, 18, color=PLUM, font="Arial-Bold"))]
         for label, value in values],
        colWidths=[260, 110],
        rowHeights=[38] * len(values),
    )
    table.setStyle(
        TableStyle(
            [
                ("BACKGROUND", (0, 0), (-1, -1), LIGHT),
                ("BOX", (0, 0), (-1, -1), 0.6, BORDER),
                ("INNERGRID", (0, 0), (-1, -1), 0.4, BORDER),
                ("VALIGN", (0, 0), (-1, -1), "MIDDLE"),
                ("LEFTPADDING", (0, 0), (-1, -1), 12),
            ]
        )
    )
    table.wrapOn(c, 370, 170)
    table.drawOn(c, 42, page_h - 260)

    policy = [
        ("Evidence policy", "The concise briefs omit inline source callouts. Every comparable-company input used
           for a Page 1 estimate is repeated in that company's Page 2 Reference data section; the evidence
          matrix and embedded Deep Research reports preserve page mapping and source hyperlinks."),
        ("Estimate policy", "Numeric or qualitative estimates rated Low are removed from the final brief. Each
          affected row or archetype remains visible as 'Not estimated' with a first-principles directional
          framework."),
        ("Completeness policy", "Exact public deployment copies, machine-extracted report text, individual
          briefs duplicated by the combined pack, and rendered QA images are inventoried but not reproduced
          twice."),
        ("Normalization", "Unicode dash variants in rendered text appendices are normalized to ASCII hyphens;
          substantive source wording is otherwise preserved."),
    ]
    y = page_h - 305
    for label, body in policy:
        c.setFillColor(PLUM)
        c.setFont("Arial-Bold", 9)
        c.drawString(42, y, label)
        p = Paragraph(clean_text(body), paragraph_style(f"p{y}", 8.2, 11, color=BODY))
        _, h = p.wrap(page_w - 84, 80)
        p.drawOn(c, 42, y - 8 - h)
        y -= h + 30
    draw_front_footer(c, page_number)
```

```
    c.showPage()


def front_matter_page_count(parts: list[Part]) -> int:
    toc_rows = len(parts) + len({part.section for part in parts})
    toc_pages = max(1, math.ceil(toc_rows / 31))
    duplicate_pages = max(1, math.ceil(len(DERIVED_DUPLICATES) / 43))
    return 2 + toc_pages + duplicate_pages


def assign_start_pages(parts: list[Part], front_pages: int) -> None:
    cursor = front_pages + 1
    for part in parts:
        part.start_page = cursor
        cursor += part.pages


def draw_toc(c: canvas.Canvas, parts: list[Part], start_page_number: int) -> int:
    page_w, page_h = letter
    rows: list[tuple[str, str, int, bool]] = []
    last_section = None
    for part in parts:
        if part.section != last_section:
            rows.append((part.section.upper(), "", part.start_page, True))
            last_section = part.section
        rows.append((part.title, f"{part.pages} pp.", part.start_page, False))

    chunks = [rows[index : index + 31] for index in range(0, len(rows), 31)] or [[]]
    for offset, chunk in enumerate(chunks):
        page_number = start_page_number + offset
        c.setFillColor(PLUM)
        c.setFont("Arial-Bold", 20)
        c.drawString(42, page_h - 58, "Table of Contents")
        c.setFillColor(GOLD)
        c.rect(42, page_h - 70, 86, 3, fill=1, stroke=0)
        y = page_h - 98
        for title, page_count, start_page, is_section in chunk:
            if is_section:
                c.setFillColor(LIGHT)
                c.rect(42, y - 11, page_w - 84, 20, fill=1, stroke=0)
                c.setFillColor(PLUM)
                c.setFont("Arial-Bold", 7.2)
                c.drawString(50, y - 2, clean_text(title))
                c.drawRightString(page_w - 50, y - 2, str(start_page))
                y -= 27
            else:
                c.setFillColor(INK)
                c.setFont("Arial", 8.2)
                clipped = title if len(title) <= 74 else title[:71] + "..."
                c.drawString(54, y, clean_text(clipped))
                c.setFillColor(META)
                c.setFont("Arial", 7.3)
                c.drawRightString(page_w - 85, y, page_count)
                c.setFillColor(PLUM)
                c.setFont("Arial-Bold", 8.2)
                c.drawRightString(page_w - 50, y, str(start_page))
                y -= 18
        draw_front_footer(c, page_number)
        c.showPage()
    return len(chunks)


def draw_duplicate_inventory(
    c: canvas.Canvas, start_page_number: int, expected_pages: int
) -> int:
    page_w, page_h = letter
    entries = [
        f"{display_path(path)} | SHA-256 {sha256(path)}" for path in DERIVED_DUPLICATES if path.is_file()
    ]
    chunks = [entries[index : index + 43] for index in range(0, len(entries), 43)] or [[]]
    if len(chunks) != expected_pages:
        raise ValueError("Duplicate inventory page count changed unexpectedly")
    for offset, chunk in enumerate(chunks):
        page_number = start_page_number + offset
        c.setFillColor(PLUM)
        c.setFont("Arial-Bold", 20)
        c.drawString(42, page_h - 58, "Documented Derivatives and Exact Duplicates")
        c.setFillColor(GOLD)
        c.rect(42, page_h - 70, 86, 3, fill=1, stroke=0)
        intro = Paragraph(
            "These files were created or deployed during the exercise but are not reproduced in the body because
               their substantive content appears in a canonical artifact.",
            paragraph_style("dup_intro", 8.2, 11, color=BODY),
        )
        _, intro_h = intro.wrap(page_w - 84, 50)
        intro.drawOn(c, 42, page_h - 89 - intro_h)
        y = page_h - 122
        for entry in chunk:
```

```
            path_text, digest = entry.split(" | SHA-256 ")
            c.setFillColor(INK)
            c.setFont("Arial", 6.8)
            c.drawString(48, y, clean_text(path_text[:112]))
            c.setFillColor(SOURCE)
            c.setFont("Courier", 5.7)
            c.drawString(60, y - 9, digest)
            y -= 15
        draw_front_footer(c, page_number)
        c.showPage()
    return len(chunks)


def render_front_matter(parts: list[Part], output: Path) -> int:
    expected = front_matter_page_count(parts)
    assign_start_pages(parts, expected)
    c = canvas.Canvas(str(output), pagesize=letter, pageCompression=1)
    c.setTitle("Operating Benchmark Source Compendium - Front Matter")
    c.setAuthor("Codex")
    draw_cover(c)
    draw_control_page(c, parts, 2)
    toc_pages = draw_toc(c, parts, 3)
    duplicate_start = 3 + toc_pages
    duplicate_pages = expected - 2 - toc_pages
    draw_duplicate_inventory(c, duplicate_start, duplicate_pages)
    c.save()
    actual = len(PdfReader(str(output)).pages)
    if actual != expected:
        raise ValueError(f"Front matter expected {expected} pages, got {actual}")
    return actual


def merge_compendium(front_matter: Path, parts: list[Part]) -> int:
    writer = PdfWriter()
    writer.add_metadata(
        {
            "/Title": "Operating Benchmark Source Compendium",
            "/Author": "Codex",
            "/Subject": "Complete source material, evidence audit, and final operating benchmark briefs",
        }
    )
    front_reader = PdfReader(str(front_matter))
    for page in front_reader.pages:
        writer.add_page(page)
    writer.add_outline_item("Front Matter", 0)

    section_parents = {}
    for part in parts:
        reader = PdfReader(str(part.pdf_path))
        page_index = len(writer.pages)
        for page in reader.pages:
            writer.add_page(page)
        if part.section not in section_parents:
            section_parents[part.section] = writer.add_outline_item(part.section, page_index)
        writer.add_outline_item(part.title, page_index, parent=section_parents[part.section])
    with OUTPUT.open("wb") as handle:
        writer.write(handle)
    return len(writer.pages)


def write_manifest(parts: list[Part], front_pages: int, total_pages: int) -> None:
    data = {
        "title": "Operating Benchmark Source Compendium",
        "generated": date.today().isoformat(),
        "output": display_path(OUTPUT),
        "output_sha256": sha256(OUTPUT),
        "front_matter_pages": front_pages,
        "total_pages": total_pages,
        "parts": [
            {
                "section": part.section,
                "title": part.title,
                "source": display_path(part.source_path),
                "kind": part.kind,
                "sha256": part.sha256,
                "pages": part.pages,
                "start_page": part.start_page,
            }
            for part in parts
        ],
        "documented_but_not_reproduced": [
            {
                "path": display_path(path),
                "sha256": sha256(path),
                "reason": "Exact deployment copy, machine extraction, duplicate individual brief, or rendered QA
                   derivative.",
            }
            for path in DERIVED_DUPLICATES
```

```
            if path.is_file()
        ],
    }
    MANIFEST_OUTPUT.write_text(json.dumps(data, indent=2), encoding="utf-8")


def verify(parts: list[Part], front_pages: int, total_pages: int) -> None:
    reader = PdfReader(str(OUTPUT))
    if len(reader.pages) != total_pages:
        raise ValueError("Output page count changed after write")
    expected = front_pages + sum(part.pages for part in parts)
    if total_pages != expected:
        raise ValueError(f"Expected {expected} total pages, got {total_pages}")
    if not reader.outline:
        raise ValueError("Compendium has no bookmarks")
    for page_number, page in enumerate(reader.pages, start=1):
        box = page.mediabox
        if float(box.width) <= 0 or float(box.height) <= 0:
            raise ValueError(f"Invalid page box on page {page_number}")
        if not (page.extract_text() or "").strip():
            raise ValueError(f"Blank or non-extractable page {page_number}")


def main() -> None:
    parts = build_parts()
    front = TMP_DIR / "00-front-matter.pdf"
    front_pages = render_front_matter(parts, front)
    assign_start_pages(parts, front_pages)
    total_pages = merge_compendium(front, parts)
    write_manifest(parts, front_pages, total_pages)
    verify(parts, front_pages, total_pages)
    print(OUTPUT)
    print(f"pages={total_pages}")
    print(MANIFEST_OUTPUT)


if __name__ == "__main__":
    main()
```

Home Comfort Operating Benchmark

SUBVERTICAL LENS	
	Installed residential equipment and protection plans convert essential household needs into recurring monthly billing. Pricing, premium-equipment mix and cross-sell support growth, while equipment exchange and reinstallation remain the central sustaining-capital burden.

METRIC	ESTIMATE	ASSERTION	CONFIDENCE
Revenue growth	**5%**	A 5% estimate sits below Enercare's 6% revenue growth in 9M 2018, when units grew only 0.6% and pricing and mix did most of the work. Reliance's low-single-digit customer growth before acquisitions also supports a mature organic rate below the disclosed Enercare result.	MEDIUM
EBITDA margin	**53%**	A 53% estimate sits just below Enercare's disclosed 56% margin. Recurring water-heater rentals carry strong contribution, while Reliance's broader HVAC, technician, dispatch and transactional-service mix should dilute the pure rental result.	MEDIUM
Maintenance capex / revenue	**12%**	The 12% estimate reflects recurring equipment replacement, removal and reinstallation: Enercare exchanged 33,600 assets in 9M 2018 and found used water heaters uneconomic to redeploy, while Reliance cites 14-16-year equipment lives. Neither company provides a clean sustaining-capex split.	MEDIUM
Customer tenor INITIAL / EFFECTIVE	Initial / effective **7-10 / 18 yrs**	Certain Reliance Ontario forms support a 7-10-year initial term. The 18-year effective tenure is a normalized assumption because billing can continue monthly, equipment can be replaced at the same address and Enercare reported 1.9% rental attrition over nine months; it is not an observed cohort life.	MEDIUM
Utilization exposure	**Low-Moderate**	Exposure is Low-Moderate because an installed unit earns fixed monthly rent regardless of household usage. The residual risk is account vacancy, equipment removal and seasonal technician or installer capacity rather than consumption volume.	HIGH
New-unit / customer payback	**8 yrs**	An eight-year estimate triangulates Reliance's C$1,000-C$5,500 installed cost against C$20-C$100-plus monthly rent, then allows for installation, service and replacement reserves. Direct unit contribution is undisclosed.	MEDIUM
B2B vs B2C	**Predominantly B2C**	The model is predominantly B2C because roughly 95% of Enercare's rental portfolio was residential water heaters and Reliance primarily markets to homeowners. Builders and property managers may originate accounts, but households remain the economic center.	HIGH

Directional framework shown where a point estimate is unsupported.

Home Comfort Operating Benchmark

SUBSECTOR CATEGORIES

Diversified home comfort
Reliance Home Comfort

Rental-led equipment & plans
Enercare

Reliance Home Comfort

DIVERSIFIED HOME COMFORT

Reference data. Acquired in 2017 for about C$4.6 billion, Reliance grew from 1.8 million customers to 2.1 million by end-2023 before two 2025 U.S. acquisitions. Its mix extends beyond rentals into HVAC, water purification, plumbing, electrical, smart-home and one-time services. Ontario materials show 7-10-year minimums; one offer continues month-to-month after year 10, with 14-16-year heater lives. Current offers show C$1,000-C$5,500 installed purchase costs and C$20-C$100+ monthly rent, with installation, repair and replacement included. Homeowners are the core customer; direct unit contribution and sustaining capex are undisclosed.

Business. Reliance Comfort Limited Partnership serves more than 2 million customers, mainly in Ontario, with reported western Canadian, Georgia, and Florida operations. It rents and sells water heaters and HVAC and offers water purification, plumbing, electrical, smart-home, and protection plans. It solves urgent hot-water, heating, and cooling needs without upfront purchase or surprise repair risk.

Economics. The core unit is an installed asset at a residential address; plans and one-time jobs are separate units. Reliance funds equipment and installation, retains ownership, bills monthly, provides repair and replacement, and operates technicians, dispatch, and call centers. Essentiality, installation, budget smoothing, transfers, and replacement convenience support retention; moves, buyouts, switching, poor service, repricing, purchase, and technology migration drive churn. Used water heaters have limited redeployment value, so the address relationship is the key residual asset.

Enercare

RENTAL-LED EQUIPMENT & PLANS

Reference data. In 9M 2018, Home Services revenue grew 6% to C$359.8 million; contracted revenue was C$332.0 million, EBITDA was C$201.4 million (56%), units grew 0.6%, average monthly rent rose from C$24.35 to C$25.50 and 33.6 thousand assets were exchanged. Rental attrition was 1.9% versus 9.3% for plans. About 95% of rentals were residential water heaters, which were almost never economical to refurbish and reinstall. HVAC rentals generated three to five times heater revenue; sustaining capex and current segment data are unavailable.

Business. Enercare is an Ontario-centric platform serving about 1.1 million customers. The relevant business is Home Services, excluding former Service Experts and submetering. It rents water heaters, HVAC, and water-treatment equipment, sells plans, and performs equipment sales and service, offering predictable cost and outsourced continuity for essential systems.

Economics. The core unit is an installed rental unit, about 95% residential water heaters historically; plans and one-time HVAC jobs are separate. Rentals were 73% of 2017 revenue and plans 20%. Heaters require little scheduled maintenance; HVAC and plans use more field labor. Ownership, installation, response, and buyout or useful-life terms support retention; moves, buyouts, switching, repricing, and poor service cause churn. Used water heaters are replaced rather than redeployed, making exchanges sustaining capital and allowing relationships to continue through replacement.

Point-of-Use Water Operating Benchmark

SUBVERTICAL LENS	
	The core archetype is Quench's workplace point-of-use model: installed, provider-owned filtration equipment for a recurring fee. Route density and recurring billing support margins, while equipment ownership, service intensity and customer-site turnover constrain conversion and tenure.

METRIC	ESTIMATE	ASSERTION	CONFIDENCE
Revenue growth	**8%**	The 8% estimate sits below Quench's 13.6% organic growth in 2019 because that result is dated and current workplace demand is more exposed to office occupancy. Primo's Refill/Filtration growth is not a clean filtration-only measure.	MEDIUM
EBITDA margin	**28%**	The 28% estimate is anchored directly to Quench's 2019 result of $32.4 million of EBITDA on $115.2 million of revenue. Recurring fees support the margin, while technicians, travel, filters, sanitation and repairs constrain it.	MEDIUM
Maintenance capex / revenue	**Not estimated**	Sustaining capital should rise with provider-owned unit age, filter and sanitation frequency, service-call intensity and a dispersed route footprint. It should fall with standardized long-life equipment, dense routes, high reuse after redeployment and customer-funded site work.	LOW
Customer tenor INITIAL / EFFECTIVE	Initial / effective **1-12 mos / 6 yrs**	The short initial term reflects Quench's primarily month-to-month rentals and Primo's flexible cancellation language. Six years is a modeled effective-tenure assumption based on plumbing, recurring service and multi-unit relationships; customer churn cohorts are unavailable.	MEDIUM
Utilization exposure	**Moderate overall**	Exposure is Moderate because fixed fees decouple revenue from water consumption, but a returned machine earns nothing until redeployed. Office occupancy, route density and productivity across Quench's 500-plus technicians remain meaningful variables.	MEDIUM
New-unit / customer payback	**Not estimated**	Payback should shorten when recurring fees and units per customer are high relative to equipment, installation and acquisition cost, especially within dense routes and long-lived accounts. It should lengthen with plumbing complexity, frequent service, customer churn, idle redeployment time and low technician density.	LOW
B2B vs B2C	**Predominantly B2B**	The core model is predominantly B2B because Quench explicitly serves workplaces and directs residential users elsewhere. Primo broadens the universe into homes and other channels, but its exact filtration customer mix is undisclosed.	HIGH

Directional framework shown where a point estimate is unsupported.

Point-of-Use Water Operating Benchmark

SUBSECTOR CATEGORIES		
	Workplace POU filtration AquaVenture Holdings / Quench	**Diversified water platform** Primo Water

AquaVenture Holdings / Quench

WORKPLACE POU FILTRATION

Reference data. FY2019 Quench had about 160,000 company-owned rental units and 55,000-plus customers (2.9 units per account), with $115.2 million revenue, $32.4 million adjusted EBITDA and 13.6% organic growth. Rentals are primarily month-to-month with fixed fees including maintenance and filters, supported by 500-plus technicians. Official materials identify Quench as workplace B2B and direct residential users to Culligan; churn, replacement cycles, installed cost, unit contribution and capex are unavailable.

Business. The relevant AquaVenture operation is Quench, excluding Seven Seas Water and desalination. Quench USA, Inc., now a Culligan subsidiary, installs company-owned filtration dispensers for institutional and commercial workplaces in the U.S. and Canada. Fixed-fee plumbed filtration replaces bottle storage, handling, and delivery.

Economics. The core unit is a machine; CAC and tenure sit at the customer location. Monthly rent includes maintenance and filters; installation may be separate. Multiple machines spread account setup cost. More than 500 technicians install, inspect, sanitize, change filters, and repair, making route density central. Working equipment, included service, predictable bills, and re-plumbing support retention; office closure, relocation, downsizing, hybrid work, rebids, and poor service cause churn. Returns require recovery, refurbishment, and reinstallation before earning again.

Primo Water

DIVERSIFIED WATER PLATFORM

Reference data. FY2023 combined Refill/Filtration revenue was $226.9 million versus $192.0 million, so filtration growth is not separable. Customer equipment had $235.0 million net book value and 2-15-year lives but included non-filtration assets. Plans start at $27.99 monthly; cancellation without penalty is permitted after the rescission period and takes effect after the next full billing period. Current materials market to homes and businesses, but the mix and filtration-only EBITDA, capex and payback inputs are unavailable.

Business. Primo Water's installed filtration offer now sits inside Primo Brands after the November 2024 transaction. It places company-owned carbon or reverse-osmosis systems in homes and businesses, with floor, countertop, and under-sink formats from $27.99 monthly. It solves taste and convenience needs without bottle logistics or self-managed maintenance.

Economics. The core unit is a filtration system; the account governs CAC and cross-sell. Primo owns, installs, leases, maintains, filters, sanitizes, repairs, bills, and sometimes monitors devices. Rolling cancellation lowers sales friction but makes retention depend on convenience, service, price, moves, and cross-sell. Home/business reach reduces office concentration but fragments routes and creates internal purchase, refill, exchange, and delivery substitutes. Cancelled equipment must be recovered and redeployed.

Submetering Operating Benchmark

SUBVERTICAL LENS	
	Building billing platforms and meter-asset owners are economically distinct. Billing platforms earn recurring data and compliance fees with moderate reinvestment; meter owners report higher EBITDA but require materially heavier replacement capital and longer payback.

METRIC	ESTIMATE	ASSERTION	CONFIDENCE
Revenue growth	Building **6%** Meter owner **7%**	The 6% building-platform estimate is anchored near ista's 5.2% growth and normalizes Techem's stronger result. The 7% meter-owner estimate is deliberately below SMS's 13% ILARR growth because funded rollout and acquisitions inflated reported rates.	MEDIUM
EBITDA margin	Building **47%** Meter owner **60%**	The 47% building estimate uses Techem's roughly 48% group-level margin and ista's 40%-plus evidence; Techem's submetering revenue is disclosed separately, so this is a normalized proxy rather than a segment margin. The 60% meter-owner estimate sits between SMS and Calisen.	MEDIUM
Maintenance capex / revenue	Building **9%** Meter owner **30%**	The 9% building-platform estimate reflects device, battery, communications and IT refresh. The 30% meter-owner estimate reflects the heavier replacement capital required by owned estates; rollout and growth capital are not cleanly separated from sustaining spend.	MEDIUM-LOW
Customer tenor INITIAL / EFFECTIVE	Building **8 / 14 yrs** Meter owner **3 / 13 yrs**	These are modeled terms, not disclosed weighted-average cohorts. Building installation and data migration support an 8-year initial and 14-year effective relationship; meter-owner legal forms can be short, while compensation can survive supplier switching for roughly 13 years.	MEDIUM-LOW
Utilization exposure	Building **Low** Meter owner **Moderate**	Building-platform exposure is Low because installed meters continue earning regardless of daily consumption. Meter-owner exposure is Moderate because installation throughput, supplier rollout decisions, meter removals and field-team productivity still matter.	MEDIUM
New-unit / customer payback	Building **4.0 yrs** Meter owner **6.0 yrs**	The modeled four-year building estimate centers the lower-cost device, billing and data cases. The modeled six-year meter-owner estimate is bounded by disclosed meter capex and annual rent evidence across SMS and Calisen.	MEDIUM-LOW
B2B vs B2C	Building **B2B; residential end users** Meter owner **Predominantly B2B**	Building owners and managers contract for submetering even when residents consume the utility and receive bills. Meter owners contract with energy suppliers or retailers, making both archetypes economically B2B.	HIGH

Directional framework shown where a point estimate is unsupported.

Submetering Operating Benchmark

SUBSECTOR CATEGORIES	Building submetering & billing	Meter asset ownership	Mixed metering & utility services
	ista / Techem / CARMA	Smart Metering Systems / Calisen	Energy Assets

ista

BUILDING SUBMETERING & BILLING

Reference data. 2025 sales were EUR1,279.0 million versus EUR1,216.1 million, up 5.2%; EBITDA margin evidence was above 40%. Total investment was EUR260 million, or 20.3% of sales, and S&P cited total capex at 12%-15% of revenue; neither measure is sustaining capex.

Business. European platform installing apartment meters and allocators, then providing recurring consumption allocation, billing, data, and sustainability workflows to owners and managers.

Economics. Unit: managed dwelling/device footprint. Regulation-shaped demand and dense estates support margin; access, hardware, resident communication, and data migration deter switching.

Techem

BUILDING SUBMETERING & BILLING

Reference data. 2025 submetering revenue was EUR1,041.5 million versus EUR933.5 million, up 11.6%; adjusted EBITDA was EUR557 million on EUR1,173.4 million group revenue, or 47.5%. Fitch expected about EUR190 million of annual capex in 2023-2025, roughly 16% of 2025 group revenue.

Business. European provider installing apartment devices and monetizing recurring heating/water allocation, billing, data, and digital building services for owners and managers.

Economics. Unit: serviced dwelling/device set. Regulation-shaped demand, radio infrastructure, and embedded billing deter switching; field service and procurement remain costs.

CARMA

BUILDING SUBMETERING & BILLING

Reference data. CARMA meters 135,000-plus units in 1,000-plus buildings under long-term inflation-protected contracts and processes 11 million-plus daily reads; revenue, EBITDA, capex and churn are undisclosed.

Business. Canadian suite-level metering and billing platform helping owners and managers allocate utilities, provide consumption visibility, and administer resident accounts.

Economics. Unit: contracted metered suite. Devices, data, billing, and move workflows likely deter switching; property decisions or service failures are plausible churn vectors. Capital funds meters, communications, IT, and reinstallation.

Smart Metering Systems

METER ASSET OWNERSHIP

Reference data. FY2022 ILARR rose 13% to GBP97.1 million; pre-exceptional EBITDA was GBP63.8 million on GBP135.5 million revenue, or 47.1%. SMS installed 480,000 meters; its 2.17 million-meter pipeline implied about GBP50 million ILARR, or GBP23 per meter annually. Sustaining capex is undisclosed.

Business. UK platform funding, installing, and managing supplier-facing smart meters for recurring rent and data income; batteries and EV-charging are excluded.

Economics. Unit: owned contracted meter. Ownership and workflow integration support tenure; rollout and supplier procurement drive growth and service utilization. Hardware-plus-installation makes replacement capital heavy.

Calisen

METER ASSET OWNERSHIP

Reference data. 2023 MAP revenue: GBP293.9m vs GBP236.2m; smart meters: 12.0m vs 8.2m. Group EBITDA/revenue: GBP265.4m/GBP358.2m (74.1%). Total capex: GBP355.5m; meter depreciation: GBP144.8m; capex/rent per meter: GBP219/GBP23.8 annually. Retailer orders: month-to-month; arrangements: 15 years; meter lives: 10-20 years.

Business. UK asset owner funding meters for energy retailers; the benchmark centers on MAP Services and excludes EV charging, heat pumps, and other transition activities.

Economics. Unit: owned revenue-generating meter. Policy drives installs; charges can survive supplier switching.

Energy Assets

MIXED METERING & UTILITY SERVICES

Reference data. Energy Assets manages more than 1.5 million assets. Clean segment revenue, EBITDA, sustaining capex, churn and contract-term disclosures are unavailable, so its company data do not anchor the numerical estimates.

Business. UK metering-and-data provider inside a wider network construction, ownership, water, fibre, and utility-services platform; customers need compliant metering, data, and field service.

Economics. Unit: installed asset with attached data/service. The offering spans recurring metering/data, installation, exchanges, and maintenance; public sources do not quantify revenue mix.

Liquid Environmental Services Operating Benchmark

SUBVERTICAL LENS	
	Two models should remain separate: installed onsite oil systems and route-based collection, treatment and processing. Onsite embedding supports tenure; route density and owned treatment support margin, but increase fleet and plant utilization exposure.

METRIC	ESTIMATE	ASSERTION	CONFIDENCE
Revenue growth	Onsite **8%** Route **9%**	The 8% onsite estimate reflects installed-kitchen wins, pricing and throughput across RTI's 45,000-plus sites. The 9% route estimate reflects modestly greater upside from density, cross-sell and treatment throughput.	MEDIUM
EBITDA margin	Onsite **21%** Route **28%**	The 21% onsite estimate is mechanism-based because RTI has no standalone margin disclosure. The 28% route estimate is set below historical S&P commentary citing LES margins above 30%, allowing for dated evidence, mixed services and third-party disposal.	MEDIUM-LOW
Maintenance capex / revenue	Onsite **Not estimated** Route **5.0%**	Onsite sustaining capital should rise with provider ownership of tanks, pumps, filters and monitoring hardware, shorter replacement cycles and a dispersed installed base. It should fall with long-lived standardized systems, customer-funded installation and dense service routes; route processing remains more exposed to truck and treatment-plant renewal, supporting the retained 5% estimate.	MEDIUM-LOW
Customer tenor INITIAL / EFFECTIVE	Onsite **Not est. / 9 yrs** Route **Not est. / 7 yrs**	Onsite contracts should be longer when the provider funds embedded equipment and bundles monitoring, compliance and maintenance, and shorter when customers own equipment or can switch without removal cost. Route contracts should be longer with dedicated capacity and bundled treatment, but shorter where work is bid per pickup; effective tenure tends to exceed the legal term when reliability and compliance make switching disruptive.	MEDIUM
Utilization exposure	Onsite **Moderate** Route **Moderate-High**	Onsite exposure is Moderate because deployed equipment remains in place, but gallons per kitchen and depot density affect economics. Route exposure is Moderate-High because stops per route, truck productivity and plant throughput determine contribution.	MEDIUM
New-unit / customer payback	Onsite **2.5 yrs** Route **1.25 yrs**	The 2.5-year onsite estimate allows recovery of hardware, installation and route support from the full customer relationship. The 1.25-year route estimate applies only to an incremental customer added within an existing dense market.	MEDIUM-LOW
B2B vs B2C	Onsite **Predominantly B2B** Route **Predominantly B2B**	Both archetypes are predominantly B2B because RTI serves commercial kitchens and institutions, while LES and WRM focus on commercial and industrial generators. Wind River's residential exposure does not change the route-processing center.	HIGH

Directional framework shown where a point estimate is unsupported.

Liquid Environmental Services Operating Benchmark

SUBSECTOR CATEGORIES			
	Installed onsite oil systems Restaurant Technologies	**Commercial liquid-waste routes** Liquid Environmental Solutions / Waste Resources Management	**Diversified wastewater services** Wind River Environmental

Restaurant Technologies

INSTALLED ONSITE OIL SYSTEMS

Reference data. RTI serves 45,000-plus kitchens through 41 depots; about 60% of 2022 revenue was QSR-related, and S&P expected low-teens growth in 2018. RTI advertises average customer savings of $250 and 9 labor hours per week; current margin, maintenance capex, contract terms and provider contribution are undisclosed.

Business. Installs automated cooking-oil systems and provides fresh-oil delivery, filtration, monitoring, used-oil removal, and recycling, solving kitchen labor, safety, quality, and disposal problems.

Economics. Unit: installed location; gallons measure usage and depot routes service capacity. Revenue mixes product, service, equipment, and recovery value. Equipment/workflow embedding retains customers; closures, operator changes, low throughput, service failures, or repricing cause churn.

Liquid Environmental Solutions

COMMERCIAL LIQUID-WASTE ROUTES

Reference data. LES expanded from 49 branches/20 recycling facilities historically and 54 branches/24 treatment plants in 2019 to 95 facility or partner sites and 129,500 customer locations by 2026. Historical S&P commentary cited ratings-adjusted EBITDA margins above 30%; current scale includes 670 field vehicles, versus 300-plus vacuum trucks in earlier releases.

Business. Collects, transports, treats, recycles, and disposes of grease, used oil, wastewater, grit, and related streams for commercial/industrial customers needing recurring sanitation and compliance.

Economics. Units: route customer for payback, truck/route for utilization, treatment plant for margin. Revenue mixes service, hauling, treatment/disposal, and recovery value. Plants and permits capture spread; recurring need, reliability, and processing access retain accounts.

Waste Resources Management

COMMERCIAL LIQUID-WASTE ROUTES

Reference data. WRM serves 10,000-plus clients in four states; a 2026 transaction adds 13 metros, and the company has 300-plus employees. Revenue, EBITDA, maintenance capex and contract terms are undisclosed; the McDonald Farms combination broadens the service mix beyond pure grease routes.

Business. Grease-centered platform collecting, processing, transporting, and recycling liquid waste while cross-selling jetting, trap repair, plumbing, septic, sludge, and remediation to commercial/institutional sites.

Economics. Unit: route customer linked to shared processing. Revenue mixes collection, treatment, recovery, transport, and field work. Vertical integration and bundled reliability retain accounts; switching remains easier than onsite equipment. Capital includes vacuum trucks, specialty units, pumps, tanks, and processing assets.

Wind River Environmental

DIVERSIFIED WASTEWATER SERVICES

Reference data. Wind River has 1,000 vehicles and 16 plants processing 200 million-plus gallons annually across 16 states, plus 90-plus acquisitions and a 75-plus-year operating legacy. It explicitly serves residential and commercial customers; revenue, EBITDA, maintenance capex and contract terms are undisclosed.

Business. East Coast consolidator spanning residential septic, commercial grease, plumbing/drains, municipal-industrial wastewater, specialty services, and treatment management, solving backup, compliance, and downtime risks.

Economics. Unit: route/service customer; plants drive network margin. Revenue mixes routes, septic cycles, emergencies, and projects. Local trust, recurring need, disposal access, and response retain customers; service inconsistency, competition, and integration cause churn.

Transportation Solutions Operating Benchmark

SUBVERTICAL LENS	Four models require separate estimates: railcar leasing, locomotive leasing, airport GSE leasing and outsourced yard logistics. Asset lessors earn higher margins with heavier reinvestment; yard logistics is labor-led, lower-margin and much lighter-capital.

METRIC	ESTIMATE	ASSERTION	CONFIDENCE
Revenue growth	Railcar **4%**; Loco **6%**; GSE **8%**; Yard **7%**	Railcar growth is held to 4% because mature fleets depend mainly on rate, utilization and measured additions; locomotives receive 6% for modernization demand. GSE and yard services receive 8% and 7% for outsourcing, electrification and new-site wins.	MEDIUM
EBITDA margin	Railcar **58%**; Loco **62%**; GSE **30%**; Yard **14%**	Railcar and locomotive estimates remain high because lease revenue sits on long-lived assets; the 62% locomotive estimate is normalized below Akiem's roughly 68% result. GSE carries workshop costs, while yard services are labor-led.	MEDIUM
Maintenance capex / revenue	Railcar **18%**; Loco **18%**; GSE **14%**; Yard **4%**	Railcar and locomotive estimates reflect overhaul, replacement, compliance and ERTMS work; GSE's 14% reflects refurbishment, batteries and workshop support. Yard services require only tractors and technology at 4%.	MEDIUM
Customer tenor INITIAL / EFFECTIVE	Railcar **4 / 9 yrs**; Loco **7 / 12 yrs**; GSE **5 / 9 yrs**; Yard **3 / 7 yrs**	Initial terms lengthen with asset value, approval burden and mobilization complexity, making locomotives longest and yard contracts shortest. Effective tenure is longer because fleets, airport packages and embedded yard operations are commonly renewed or refreshed.	MEDIUM
Utilization exposure	Railcar **Moderate**; Loco **Moderate**; GSE **Moderate-High**; Yard **High**	Rail exposure is Moderate because leased deployment matters but assets can be redeployed across long economic lives. GSE adds station activity and spare-fleet risk, while yard services are High because contribution depends on productive labor hours.	MEDIUM
New-unit / customer payback	Railcar **9 yrs**; Loco **10 yrs**; GSE **5.5 yrs**; Yard **1.5 yrs**	Rail assets receive nine-to-ten-year paybacks because high upfront cost must be recovered over multiple placements and residual value matters. GSE is cheaper but service-heavy at 5.5 years, while yard launch spend can recover in about 1.5 years.	MEDIUM
B2B vs B2C	All models **Predominantly B2B**	All four models are predominantly B2B because contractual buyers are rail operators, public authorities, airports, ground handlers, industrial firms and logistics providers. The benchmarked operations have no meaningful direct consumer revenue.	HIGH

Directional framework shown where a point estimate is unsupported.

Transportation Solutions Operating Benchmark

SUBSECTOR CATEGORIES	Railcar leasing	Locomotive leasing	Airport GSE leasing	Yard logistics
	VTG	Akiem	Air Rail / TCR	Lazer Logistics

VTG

RAILCAR LEASING

Reference data. 2024 operating cash flow was EUR405m and fixed-asset investment EUR536m, mainly new wagons and maintenance; tank-wagon utilization stayed high, intermodal utilization softened and rental pricing was positive. Customers hire wagons medium-to-long term, including a cited 10-year agreement, and are industrial shippers, rail operators and logistics firms. Standalone leasing EBITDA, sustaining capex and unit payback are undisclosed.

Business. Leases and manages European freight wagons for industrial shippers and rail operators.

Economics. Unit: earning wagon; revenue: recurring lease plus services. Customers avoid capital and secure compliant, cargo-specific capacity.

Akiem

LOCOMOTIVE LEASING

Reference data. 2021: 600-plus locomotives, 46 passenger trains, EUR220m revenue and EUR150m EBITDA, or roughly 68%. A EUR1.52bn 2026 refinancing, major OEM orders and ERTMS upgrades on about 250 locomotives support expansion. Customers include carriers, manufacturers and local authorities; weighted tenor, utilization, sustaining capex and payback are undisclosed.

Business. Leases and maintains locomotives and some passenger rolling stock across Europe.

Economics. Unit: locomotive under lease/full service. Operators outsource scarce, compliant traction.

Lazer Logistics

YARD LOGISTICS

Reference data. EQT reports 550-plus sites, 5,000 employees, 6,000 fleet assets and 9m annual service hours. Lazer expanded through acquisitions and serves logistics, manufacturing, food, retail and consumer-goods companies. Its labor-led site programs are embedded but rebiddable; organic growth, EBITDA, contract tenure, labor utilization, sustaining capex and site-launch payback are undisclosed.

Business. Runs warehouse and industrial yards using drivers, equipment, processes and technology.

Economics. Unit: site contract/service hour. Recurring site and labor revenue solves trailer-flow bottlenecks.

Air Rail

AIRPORT GSE LEASING

Reference data. Air Rail expanded from 180 units in 2014 to 4,700-plus by March 2026 across about 60 airports and eight countries; 68% of motorized units were electric. It serves airports, airlines and handlers through rental, maintenance, parts, resale and reconditioning. Segment revenue, EBITDA, sustaining capex, contract tenure, utilization and payback are undisclosed.

Business. AIR RAIL, S.L. leases, maintains and refurbishes airport GSE; it also has rail activities.

Economics. Unit: deployed GSE/station package. Rental, maintenance, parts and resale solve airline/handler capex and uptime needs. Airport service embeds relationships; rebids and transfer friction cause churn.

TCR

AIRPORT GSE LEASING

Reference data. TCR has 44,000-plus assets, 100-plus workshops, about 1,000 technicians and 200-plus airports. GIP describes long-term customer relationships; TCR provides multi-year fleet planning, refurbishment, electrification, fleet right-sizing and used-equipment resale for handlers, airlines and airports. EBITDA, sustaining capex, weighted tenor, paid utilization and unit payback are undisclosed.

Business. Rents, maintains, refurbishes and electrifies GSE for airlines, handlers and airports.

Economics. Unit: GSE/airport fleet package. Full-service rent and lifecycle services solve turnaround uptime and capital needs.

Business profiles grouped by subsector category.

Specialty Rental Operating Benchmark

SUBVERTICAL LENS	
	Consumer storage, B2B storage, modular space and trench safety differ in rental duration, service intensity and capital recovery. Simple standardized assets generally support stronger margins and conversion, while modular and project-driven fleets carry greater reset or utilization risk.

METRIC	ESTIMATE	ASSERTION	CONFIDENCE
Revenue growth	Consumer **Not estimated**; B2B storage **5%**; Modular **6%**; Trench **6%**	Consumer storage growth should rise with housing turnover, renovation, household dislocation, pricing and local container capacity; slower moves, higher acquisition cost and delivery constraints reduce it. Retained B2B, modular and trench estimates reflect rate, fleet additions and project demand.	MEDIUM
EBITDA margin	Consumer **Not estimated**; B2B storage **40%**; Modular **34%**; Trench **33%**	Consumer margins should rise with container occupancy, rental extensions, route density and pricing that covers delivery; paid acquisition, empty miles, repeated handling, yard expense and damage reduce them. Retained B2B margin is anchored to Mobile Mini, with modular and trench lower because installation, engineering and hauling add cost.	MEDIUM
Maintenance capex / revenue	Consumer **Not estimated**; B2B storage **6%**; Modular **8%**; Trench **7%**	Consumer sustaining capital should rise with asset age, damage, corrosion, road miles and repositioning; durable standardized containers, disciplined refurbishment and high recovery values reduce it. Retained B2B, modular and trench values reflect long asset lives, residuals and differing reset needs.	MEDIUM
Customer tenor INITIAL / EFFECTIVE	Consumer **1 mo / Not est.**; B2B storage **3 mos / 4 yrs**; Modular **18 mos / 7 yrs**; Trench **2 mos / 5 yrs**	PODS supports a one-month initial term. Effective consumer tenure should lengthen with uncertain move dates, extended renovations and passive monthly renewal, and shorten with defined moves, price pressure or easy substitution. B2B storage, modular and trench relationships persist through projects and repeat use.	MEDIUM
Utilization exposure	Consumer **Not estimated**; B2B storage **Moderate**; Modular **Moderate**; Trench **High**	Consumer utilization exposure should rise with fixed local fleets, seasonal moving demand, depot imbalance and limited redeployment, and fall with broad coverage, flexible transfers and balanced demand. B2B storage and modular are Moderate; trench is High because inventory rolls off with projects.	MEDIUM
New-unit / customer payback	Consumer **Not estimated**; B2B storage **4.5 yrs**; Modular **6.0 yrs**; Trench **3.5 yrs**	Consumer payback should shorten with high occupancy, longer rentals, strong delivery economics and repeated reuse, and lengthen with acquisition cost, empty transport, yard expense, damage and idle time. Retained B2B, modular and trench values use General Finance's four-year revenue recovery evidence and allow for cost, maintenance and residual value.	MEDIUM
B2B vs B2C	Consumer **Mixed; B2C lean**; B2B storage **Predominantly B2B**; Modular **Predominantly B2B**; Trench **Predominantly B2B**	Consumer storage is mixed with a B2C lean because household moves are a material demand source. Storage, modular and trench customers are principally contractors, companies, schools, institutions, government bodies and utilities.	MEDIUM

Directional framework shown where a point estimate is unsupported.

Specialty Rental Operating Benchmark

SUBSECTOR CATEGORIES	Consumer moving & storage	B2B portable storage	Modular space	Trench safety rental
	PODS	Mobile Mini / General Finance Corporation	Modulaire Group / McGrath RentCorp	Trench Plate Rental

PODS

CONSUMER MOVING & STORAGE

Reference data. PODS reports 2 million-plus long-distance moves and 7 million-plus deliveries; rentals renew monthly and serve residential and commercial moving/storage. Q3 2024 sales fell 7.5% and adjusted EBITDA fell 24%. Normalized margin, fleet count, sustaining capex, utilization, effective tenure and unit payback are not publicly disclosed.

Business. Delivers containers for residential and commercial moving/storage.

Economics. Unit: container-month plus move event; revenue is rent and logistics fees. Customers solve moves or renovations. PODS runs drivers, yards and linehaul; life-event completion causes churn.

Mobile Mini

B2B PORTABLE STORAGE

Reference data. In 2019, Storage rental revenue grew 4.0%, North America grew 6.3%, yield rose 3.5%, EBITDA margin was 41.0% and unit utilization was 79.1%. Month-based rentals served about 74,000 predominantly commercial customers. Assets had 30-year lives and 55% residual values; direct sustaining capex and cash payback were undisclosed.

Business. Rented storage containers and ground-level offices to commercial users.

Economics. Unit: box on rent; revenue was mostly rental. Branches deliver and refurbish simple assets solving temporary jobsite needs. Projects end placements, but accounts repeat.

General Finance Corporation

B2B PORTABLE STORAGE

Reference data. Typical leases averaged over 12 months and historical fleet utilization was 70%-85%. Management said revenue recovered original equipment cost within four years on average. Assets had 20-30-year lives and residuals up to 70%. North American leasing was commercial; segment EBITDA, sustaining capex and churn cohorts were undisclosed.

Business. Rented containers, mobile offices and modular units through regional brands.

Economics. Unit: fleet asset; revenue is lease plus sales. Customers need temporary space. GFC delivered, maintained and re-leased units; projects cause churn, while accounts repeat.

Modulaire Group

MODULAR SPACE

Reference data. In 1Q22, revenue was EUR370m and EBITDA EUR116m; revenue per unit rose 5%, units on rent increased and utilization was 86%. Modulaire serves 52,000-plus public, infrastructure and enterprise customers and emphasizes fleet refurbishment. Weighted contract terms, sustaining capex, customer tenure and direct payback are undisclosed.

Business. Rents and installs modular space for public, infrastructure and enterprise users.

Economics. Unit: module on rent; revenue is lease/service. Customers need temporary space. Modulaire transports, installs and refurbishes units; installation and repeat projects support retention.

McGrath RentCorp

MODULAR SPACE

Reference data. In 2025, Portable Storage rental revenue fell 3%; utilization was 73.0% for modular and 60.8% for storage. Initial terms are 12-24 months and 1-12 months; storage rents at 2%-4% of original cost monthly. Modular and storage assets use 18-year/50% and 25-year/62.5% life/residual assumptions and serve business and institutional customers.

Business. Rents modular buildings and storage containers to businesses and institutions.

Economics. Unit: module/container; revenue is rent plus delivery/service. Customers need classrooms, jobsite space or storage. McGrath installs and refurbishes; repeat accounts outlast placements.

Trench Plate Rental

TRENCH SAFETY RENTAL

Reference data. Before its 2021 merger, Trench Plate Rental had 37 branches and 5,000-plus customers. Rentals to contractors and utilities were short and extendable; durable steel plates, shoring and safety systems were inspected and redeployed between projects. Revenue growth, EBITDA, sustaining capex, utilization, churn and payback were not disclosed.

Business. Rented trench plates, shoring and safety systems to contractors and utilities.

Economics. Unit: jobsite equipment set; revenue is rental/support. Branches solve mandatory excavation safety, then inspect and redeploy steel. Projects end rentals, but contractors repeat.

Business profiles grouped by subsector category.